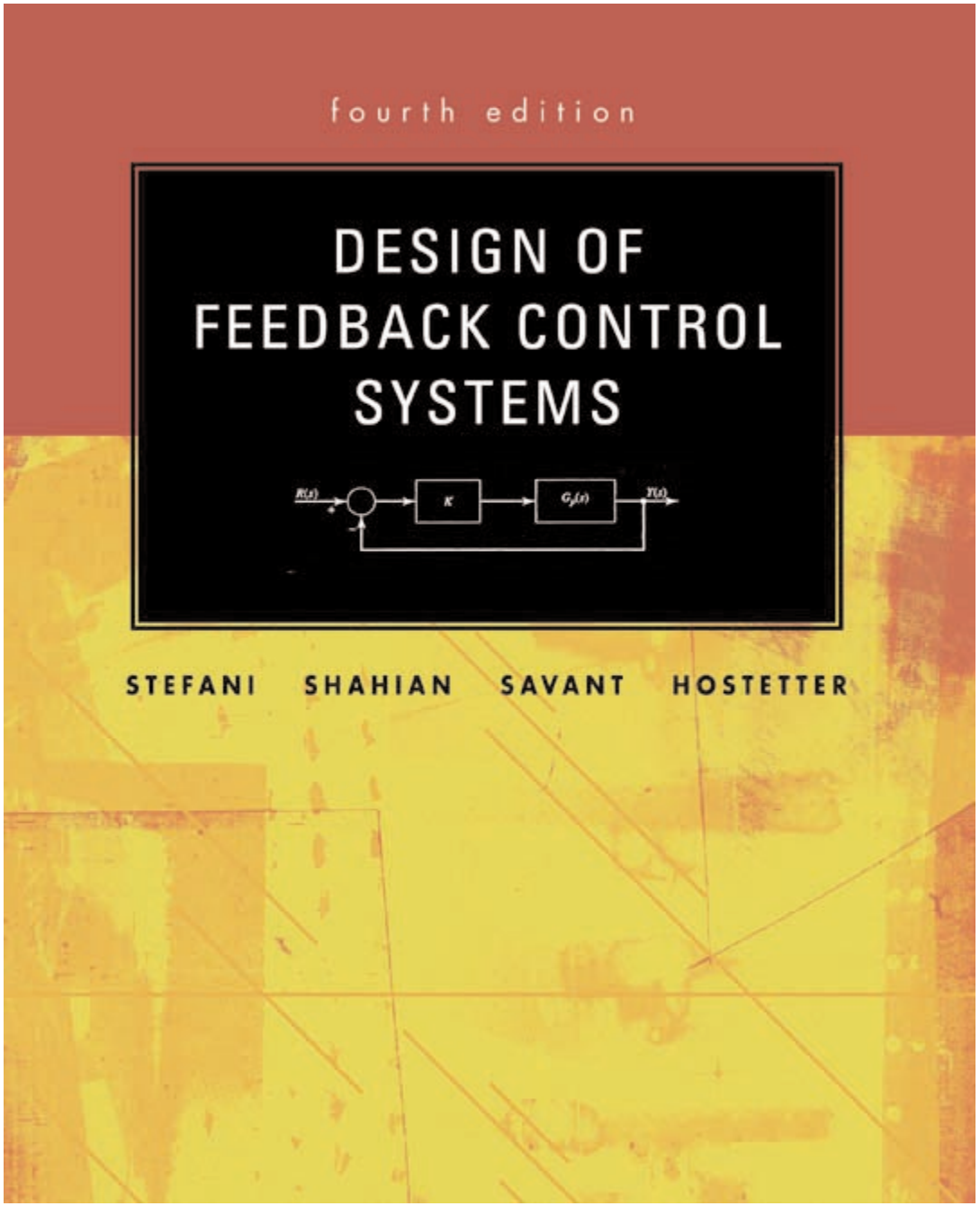


Instructor's Solutions Manual
to Accompany

Design of Feedback Control Systems

Fourth Edition

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CHAPTER 1

1.1 a. $x_0 = 1$

$y_0 = 1$

$m = \left.\frac{dy}{dx}\right|_{x=x_0} = 3x^2|_{x=1} = 3$

$y \approx 1 + 3(x-1)$

b) $x_0 = 1$

$y_0 = 2$

$m = \left.\frac{dy}{dx}\right|_{x=x_0} = \frac{2}{3}\left.\frac{1}{x^{1/3}}\right|_{x=1} = \frac{2}{3}$

$y \approx 2 + \frac{2}{3}(x-1)$

c) $x_0 = \frac{\pi}{4}$

$y_0 = 2\cos 2\left(\frac{\pi}{4}\right) = 2\cos\frac{\pi}{2} = 0$

$m = \left.\frac{dy}{dx}\right|_{x=x_0} = 2[-2\sin 2x]|_{x=\frac{\pi}{4}}$

$= -4\sin 2x|_{x=\frac{\pi}{2}} = -4\sin\frac{\pi}{2} = -4$

$y \approx -4\left(x - \frac{\pi}{4}\right)$

1.2(a). Write two loop equations directly from Fig P1.2a as follows:

$$\left(6s + 4 + \frac{1}{2s}\right) I_1 - \frac{1}{2s} I_2 = \frac{5}{s^2+1} - \frac{12}{s}$$

$$-\frac{1}{2s} I_1 + \left(\frac{1}{2s} + \frac{1}{3s}\right) I_2 = \frac{12}{s}$$

(b). Write three loop equations directly from Fig P1.2b as follows:

$$(4s+9) I_1 - 3 I_2 - 4s I_3 = -\frac{10}{s}$$

$$-3 I_1 + (3s+3) I_2 - 3s I_3 = -\frac{6s}{s^2+49}$$

$$-4s I_1 - 3s I_2 + 7s I_3 = \frac{6s}{s^2+49}$$

(c). Write two loop equations directly from Fig P1.2c. Note, the presence of a dependent voltage source which gives another equation as follows:

$$7V(s) = 7(6s)[I_1 - I_2]$$

Loop 1: $I_1(6s+11+\frac{1}{5s}) + I_2(-6s-7) = 7(6s)[I_1 - I_2]$

Loop 2: $I_1(-6s-7) + I_2(6s+11) = -\frac{1}{s^2+1}$

1.3 (a). $$V(s) = \frac{\frac{12}{s}(7s)}{12+7s} = \frac{70s}{s(7s+12)} = \frac{70}{7(s+\frac{12}{7})}$$

$$v(t) = 10 e^{-\frac{12}{7}t}$$

(b) $$V(s) = \frac{\frac{6}{s+3} \cdot \frac{1}{6s}}{8 + \frac{1}{6s}} = \frac{6/(s+3)}{48s+1} = \frac{6}{48(s+\frac{1}{48})(s+3)}$$

$$= \frac{1}{8(s+\frac{1}{48})(s+3)} = \frac{.042}{s+\frac{1}{48}} - \frac{.042}{s+3}$$

$$v(t) = .042 e^{-t/48} - .042 e^{-3t}$$

1.4 (a). Write two node equations directly from Fig. P1.4a

Node 1: $(\frac{1}{2} + 5s) V_1 + (-\frac{1}{2}) V_2 = \frac{1}{s^2+1}$

Node 2: $(-\frac{1}{2}) V_1 + (\frac{1}{3s} + \frac{1}{4} + \frac{1}{2}) V_2 = 0$

1.4(b) Write three node equations directly from Fig P1.4(b) as follows:

Node 1: $(4s + 1/3s)V_1 + (0)V_2 + (-4s)V_3 = \frac{-1}{s^2+1}$

Node 2: $(0)V_1 + (1/6 + 1/3s)V_2 + (-\frac{1}{3s})V_3 = \frac{1}{s^2+1}$

Node 3: $(-4s)V_1 + (-1/3s)V_2 + (4s + 1/3s)V_3 = \frac{-s}{s^2+1} + \frac{4}{s}$

(c) Write two node equations and one for the dependent current source, as follows:

Node 1: $(\frac{1}{3} + \frac{1}{9} + \frac{1}{2s})V_1 + (-\frac{1}{3} - \frac{1}{2s})V_2 = \frac{6s}{s^2+49}$

Node 2: $(-\frac{1}{3} - \frac{1}{2s})V_1 + (\frac{1}{3} + \frac{1}{2s} + 6s)V_2 = -5V(s)$

D.C.S.: $V(s) = V_1 - V_2$

(d) Write four node-pair equations, as follows:

Node 1: $(\frac{1}{R_1} + \frac{1}{R_2} + C_1 s)V_1 + (-sC_1)V_2 + (\frac{-1}{R_2})V_3 + (0)V_4 = \frac{V}{R_1}$

Node 2: $(-sC_1)V_1 + (sC_1 + \frac{1}{R_3})V_2 + (0)V_3 + (-\frac{1}{R_3})V_4 = 0$

Node 3: $(-\frac{1}{R_2})V_1 + (0)V_2 + (\frac{1}{R_2} + \frac{1}{R_4} + \frac{1}{sL})V_3 + (-\frac{1}{R_4})V_4 = 0$

Node 4: $(0)V_1 + (\frac{-1}{R_3})V_2 + (-\frac{1}{R_4})V_3 + (\frac{1}{R_3} + \frac{1}{R_4} + sC_2)V_4 = 0$

(e) Write three node-pair equations, as follows:

Node 1: $[s(C_1+C_2) + \frac{1}{R}]V_1 + (-sC_1)V_2 + (-sC_2)V_3 = V/R$

Node 2: $(-sC_1)V_1 + (\frac{1}{R_1} + sC_1 + \frac{1}{sL})V_2 + (\frac{-1}{sL})V_3 = 0$

Node 3: $(-sC_2)V_1 + (-\frac{1}{sL})V_2 + (\frac{1}{R_2} + sC_2 + \frac{1}{sL})V_3 = 0$

1.5 (a) $I(s) = \frac{(6/s)(1/2s)}{\frac{1}{2s} + 4} = \frac{3}{s(4s + \frac{1}{2})} = \frac{3/4}{s(s + \frac{1}{8})}$

(b) $I(s) = \frac{(\frac{7}{s+2})4}{(5s+4)} = \frac{28/5}{(s+2)(s+\frac{4}{5})}$

1.6

(a) $2\frac{di_1}{dt} + \underbrace{\frac{1}{3}\int_{-\infty}^{t} i_1(t)\,dt}_{v_1(t)} + 4(i_1 - i_2) + 10 = 0$

$$-10 + 4(i_2 - i_1) + 8\cos 9t + 6i_2 + 8\frac{di_2}{dt} + \underbrace{\frac{1}{7}\int_{-\infty}^{t} i_2(t)\,dt}_{v_2(t)}$$

$$2\frac{di_1}{dt} = -4i_1 + 4i_2 - v_1 - 10$$

$$8\frac{di_2}{dt} = 4i_1 - 10i_2 - v_2 + 10 - 8\cos 9t$$

$$3\frac{dv_1}{dt} = i_1$$

$$7\frac{dv_2}{dt} = i_2$$

$$\frac{di_1}{dt} = -2i_1 + 2i_2 - 0.5v_1 - 5$$

$$\frac{di_2}{dt} = 0.5i_1 - 1.25i_2 - 0.125v_2 + 1.25 - \cos 9t$$

$$\frac{dv_1}{dt} = \frac{1}{3}i_1$$

$$\frac{dv_2}{dt} = \frac{1}{7}i_2$$

1.6 (b) $3\frac{di_1}{dt} + \frac{1}{5}\int_{-\infty}^{t} i_1(\tau)d\tau + 4(i_1 - i_3) + 5(i_1 - i_2) = 0$

$$-10\cos 3t + 2\frac{di_2}{dt} + 5(i_2 - i_1) + 7(i_2 - i_3) = 0$$

$$7(i_3 - i_2) + 4(i_3 - i_1) + 4\frac{di_3}{dt} = 0$$

$$3\frac{di_1}{dt} = -9i_1 + 5i_2 + 4i_3 - v$$

$$2\frac{di_2}{dt} = +5i_1 - 12i_2 + 7i_3 + 10\cos 3t$$

$$4\frac{di_3}{dt} = +4i_1 + 7i_2 - 11i_3$$

$$5\frac{dv}{dt} = i_1$$

$$\frac{di_1}{dt} = -3i_1 + \frac{5}{3}i_2 + \frac{4}{3}i_3 - \frac{1}{3}v$$

$$\frac{di_2}{dt} = 2.5i_1 - 6i_2 + 3.5i_3 + 5\cos 3t$$

$$\frac{di_3}{dt} = i_1 + 1.75i_2 - 2.75i_3$$

$$\frac{dv}{dt} = 0.2i_1$$

1.7 (a) integrator

$$\frac{-V_{OUT}(s)}{V_{IN}(s)} = \frac{1}{sR_1C} = \frac{1}{s\,10\times10^{3}\times10\times10^{-6}} = \frac{1}{s\times10^{-1}}$$

$$= \frac{10}{s}$$

(b) lag circuit

$$k = \frac{5k}{2k} = 2.5$$

$$b = \frac{1}{10\times10^{3}\times10^{-6}} = \frac{1}{10^{-2}} = 100$$

$$a = \left(1+\frac{10k}{5k}\right)b = (1+2)b = 300$$

$$\frac{-V_{OUT}(s)}{V_{IN}(s)} = \frac{2.5(s+300)}{s+100}$$

(c) lead circuit

$$k = \frac{4k}{2k} = 2$$

$$b = \frac{1}{10\times10^{3}\times0.1\times10^{-6}} = \frac{1}{10^{1+3-1-6}} = \frac{1}{10^{-3}} = 1000$$

$$a = \left(1+\frac{10k}{2k}\right)b = 6k = 6000$$

$$\frac{-V_{OUT}(s)}{V_{IN}(s)} = \frac{2(s+1000)}{s+6000}$$

(d) PI

$$K_p = \frac{2k}{8k} = 0.25 \qquad K_i = \frac{1}{8\times10^{3}\times4\times10^{-6}} = \frac{1000}{32} = 31.25$$

$$\frac{-V_{OUT}(s)}{V_{IN}(s)} = 0.25 + \frac{31.25}{s}$$

(e) PD

$$K_D = \frac{2\mu}{8\mu} = 0.25 \qquad K_d = \frac{1}{4\times10^3\times2\times10^{-6}} = \frac{10^3}{8}$$

$$= \frac{1000}{8} = 125$$

$$\frac{-V_{OUT}(s)}{V_{IN}(s)} = 0.25 + 125s$$

1.8 (a) differentiator, $R_f = 2k\Omega$

(a) $R_f C = 0.001$

$$C = \frac{0.001}{R_f} = \frac{10^{-3}}{2\times10^{-3}} = 0.5\times10^{-6} = 0.5\mu F$$

(b) Lead

$k=5,\ b=6,\ a=60,\ R_1 = 1k\Omega$

$R_f = R_1 k = 5k\Omega$

$a = 60 = (1 + R_2/R_1)\ 6$

$R_2/R_1 = 9$

$R_2 = 9R_1 = 9k\Omega$

$C_1 = 1/bR_2 = \frac{1}{6.9\times10^3} = \frac{1}{54\times10^3} = 0.0185\times10^{-3}$

$= 18.5\mu F$

(c) Lag, $k=4,\ b=3,\ a=21,\ R_1 = 10k\Omega$

$R_a = R_1/k = 10\times10^3/4 = 2.5k\Omega$

$a = 21 = (1 + R_2/R_1)3$

$R_2/R_1 = 6 \qquad R_2 = 6R_1 = 60k\Omega$

$C_1 = 1/bR_2 = \frac{1}{3\times60\times10^3} = \frac{1}{180\times10^3} = 0.0056\times10^{-3}$

$= 5.6\mu F$

(d) Lag-Lead, $k=10,\ b=4,\ a=40,\ d=50,\ c=200,\ R_1 = 10k.$

$R_3 = R_1/k = 10\times10^3/10 = 10^3 = 1k\Omega$

$a = 40 = (1 + R_2/R_1)4 \qquad R_2/R_1 = 9 \qquad R_2 = 9R_1 = 90k\Omega$

$C_1 = 1/\omega R_2 = \frac{1}{4\times 90\times 10^3} = \frac{1}{360\times 10^3} = 0.0028\times 10^{-3}$

$= 2.8\mu F$

$C = 200 = (1 + R_4/R_3)50 \quad R_4/R_3 = 3 \quad R_4 = 3R_3 = 3k\Omega$

$C_2 = \frac{1}{\omega R_4} = \frac{1}{50\times 3\times 10^3} = \frac{1}{150\times 10^3} = 0.0067\times 10^{-3}$

$= 6.7\mu F$

(e) PD, $k_p = 100$, $k_d = 20$, $R_1 = 1k\Omega$

$C_2 = \frac{R_1}{k_d} = \frac{10^3}{20} = 0.05\times 10^{-3} = 50\mu F$

$C_1 = \frac{C_2}{k_p} = \frac{50\times 10^{-6}}{100} = 0.5\mu F$

(f) PID, $k_p = 100$, $k_d = 10$, $k_i = 1000$, $C_2 = 1\mu F$, $R_a = 1k\Omega$

$C_1 = \frac{C_2}{k_p} = \frac{10^{-6}}{100} = 0.01\mu F$

$R_1 = \frac{k_d}{C_2} = \frac{10}{10^{-6}} = 10\times 10^6 = 10\,M\Omega$

$C = \frac{1}{R_a k_i} = \frac{1}{10^3\times 10^3} = 10^{-6} = 1\mu F$

1.9

(a) $[8s^2+2s+8]X_1(s) - 4X_2(s) = 0$

$-4X_1(s) + [7s^2+4]X_2(s) = \frac{3}{s^2+9}$

(b) $[4s^2+7s+4]X_1(s) - 7sX_2(s) = 0$

$-7X_1(s) + [12s^2+7s+9]X_2(s) - 9X_3(s) = 10/s$

$-9X_2(s) + [6s^2+9]X_3(s) = 0$

(c) $[7s^2+5s+8]X_1(s) + [-2s-4]X_2(s) = 0$

$[-2s-4]X_1(s) + [7s^2+6s+4]X_2(s) = \frac{18}{s3}$

1.10 (a)

$$\frac{d}{dt}\begin{bmatrix} dx_1/dt \\ x_1 \\ dx_2/dt \\ x_2 \end{bmatrix} = \begin{bmatrix} -0.25 & -1 & 0 & 0.5 \\ 1 & 0 & 0 & 0 \\ 0 & 4/7 & 0 & -4/7 \\ 0 & 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} dx_1/dt \\ x_1 \\ dx_2/dt \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ f=\sin 3t \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} s+0.25 & 1 & 0 & -0.5 \\ -1 & s & 0 & 0 \\ 0 & -4/7 & s & 4/7 \\ 0 & 0 & -1 & s \end{bmatrix}\begin{bmatrix} sX_1(s) \\ X_1(s) \\ sX_2(s) \\ X_2(s) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ F(s)/7 = \frac{17}{s^2+9} \\ 0 \end{bmatrix}$$

(b)

$$\frac{d}{dt}\begin{bmatrix} dx_1/dt \\ x_1 \\ dx_2/dt \\ x_2 \\ dx_3/dt \\ x_3 \end{bmatrix} = \begin{bmatrix} -7/4 & -1 & 7/4 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 7/12 & -7/12 & -0.75 & 0 & 0.75 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1.5 & 0 & -1.5 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} dx_1/dt \\ x_1 \\ dx_2/dt \\ x_2 \\ dx_3/dt \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 10/12 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} s+7/4 & 1 & -7/4 & 0 & 0 & 0 \\ -1 & s & 0 & 0 & 0 & 0 \\ 0 & -7/12 & (s+7/12) & 0.75 & 0 & -0.75 \\ 0 & 0 & -1 & s & 0 & 0 \\ 0 & 0 & 0 & -1.5 & s & 1.5 \\ 0 & 0 & 0 & 0 & -1 & s \end{bmatrix} \begin{bmatrix} sX_1(s) \\ X_1(s) \\ sX_2(s) \\ X_2(s) \\ sX_3(s) \\ X_3(s) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 10/12s \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

(c)

$$d/dt\begin{bmatrix} dx_1/dx \\ x_1 \\ dx_2/dt \\ x_2 \end{bmatrix} = \begin{bmatrix} -5/7 & -8/7 & 2/7 & 4/7 \\ 1 & 0 & 0 & 0 \\ 2/7 & 4/7 & -6/7 & -4/7 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} dx_1/dt \\ x_1 \\ dx_2/dt \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ (9/7)t^2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} s+5/7 & 8/7 & -2/7 & -4/7 \\ -1 & 0 & 0 & 0 \\ -2/7 & -4/7 & s+6/7 & 4/7 \\ 0 & 0 & -1 & s \end{bmatrix} \begin{bmatrix} sX_1(s) \\ X_1(s) \\ sX_2(s) \\ X_2(s) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 18/7s^3 \\ 0 \end{bmatrix}$$

//

1.11 (a) $10\frac{d^2x}{dt^2} + 4x = 6e^{-2t}$

$(10s^2+4)X(s) = \frac{6}{s+2}$

$X(s) = \frac{6/10}{(s^2+2/5)(s+2)} = \frac{.23\angle 107^\circ}{s+j6.3} + \frac{.23\angle -107^\circ}{s-j6.3} + \frac{.14}{s+2}$

$x(t) = \left[.46\cos(.63t - 107^\circ) + .14e^{-2t}\right]u(t)$

(b) $10\frac{d^2x}{dt^2} + 7\frac{dx}{dt} = \frac{10}{s}$

$(10s^2+7s)X(s) = 10/s$

$X(s) = \frac{1}{s^2(s+7/10)} = \frac{10/7}{s^2} + \frac{-2.04}{s} + \frac{2.04}{s+7/10}$

$x(t) = \left[1.43t - 2.04 + 2.04e^{-.7t}\right]u(t)$

1.12 Write four node equations

X_1 $\quad X_1[M_1s^2 + B_1s + K_1] + X_2[-B_1s - K_1] + X_3(0) + X_4(0) = 0$

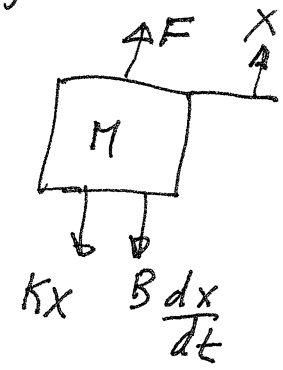
X_2 $\quad X_1[-B_1s - K_1] + X_2[M_2s^2 + (B_1+B_2+B_3)s + K_1+K_2+K_3]$
$+ X_3[-B_2s - K_2] + X_4(-B_3s - K_3) = 0$

X_3 $\quad X_1(0) + X_2[-B_2s - K_2] + X_3[M_3s^2 + B_2s + K_2] + X_4(0) = 0$

X_4 $\quad X_1(0) + X_2[B_3s - K_3] + X_3(0) + X_4[M_4s^2 + B_3s + K_3] = F$

1.13

Free body diagram



$$F = M\frac{d^2x}{dt^2} + B\frac{dx}{dt} + Kx$$

$$F(s) = [Ms^2 + Bs + K]X(s) \qquad F(s) = \frac{250}{s}$$

$$\frac{X(s)}{F(s)} = \frac{1}{Ms^2+Bs+K} = \frac{1}{M\left[s^2 + \frac{B}{M}s + \frac{K}{M}\right]}$$

$M = 10$

a) $s^2 + \frac{B}{10}s + \frac{K}{10} = (s+10)(s+2.5) = s^2 + 12.5s + 25$

$\frac{K}{10} = 25$ $\boxed{K = 250}$ $\frac{B}{10} = 12.5$ $\boxed{B = 125}$

$$X(s) = \frac{250/s}{10[s^2+12.5s+25]} = \frac{25}{s[s^2+12.5s+25]}$$

b) $s^2 + \frac{B}{10}s + \frac{K}{10} = (s+5)(s+5) = s^2 + 10s + 25$

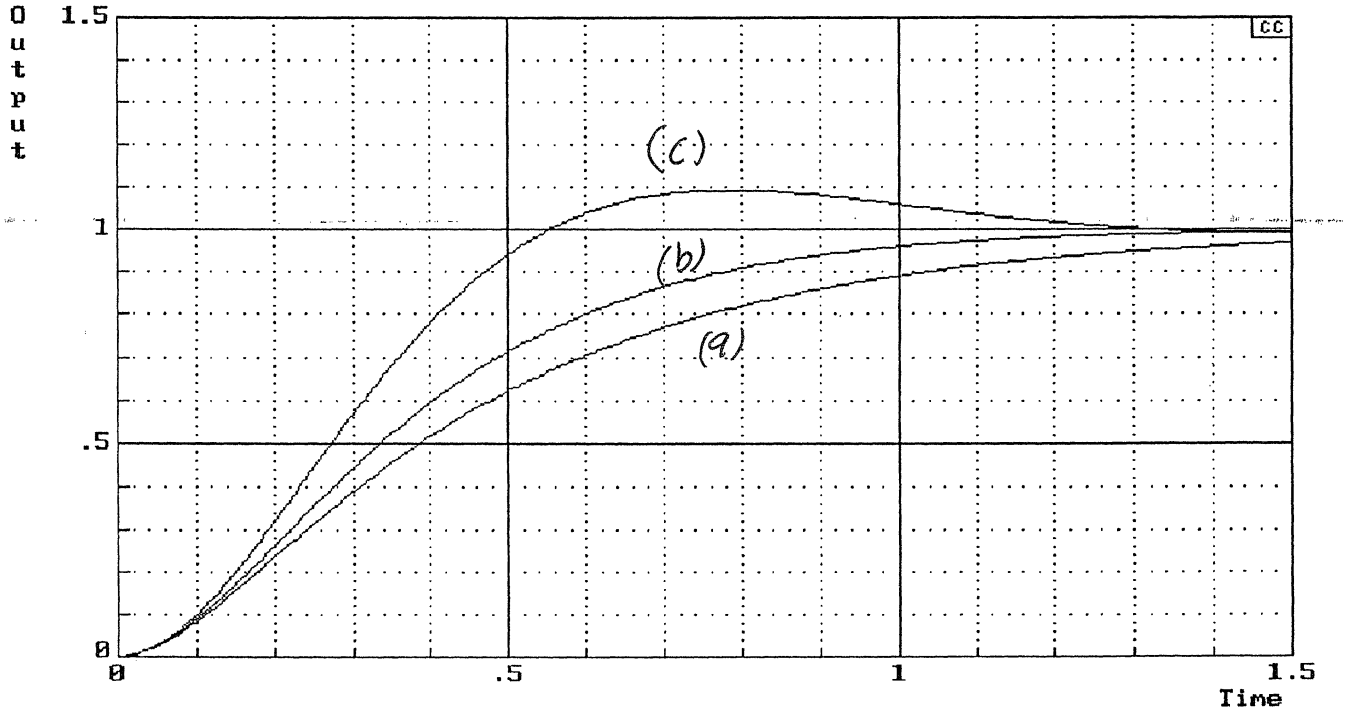
$\frac{K}{10} = 25$ $\boxed{K = 250}$ $\frac{B}{10} = 10$ $\boxed{B = 100}$

$$X(s) = \frac{250/s}{10[s^2+10s+25]} = \frac{25}{s[s^2+10s+25]}$$

c) $s^2 + \frac{B}{10}s + \frac{K}{10} = (s+3+j4)(s+3-j4) = s^2 + 6s + 25$

$\frac{K}{10} = 25$ $\boxed{K = 250}$ $\frac{B}{10} = 6$ $\boxed{B = 60}$

$$X(s) = \frac{250/s}{10[s^2+6s+25]} = \frac{25}{s[s^2+6s+25]}$$



Step response for Problem 1.13

1.14 Force is proportional to distance

$$M\frac{d^2x}{dt^2} + B\frac{dx}{dt} + Kx = -\left(\frac{L_1}{L_2}\right)F(t)$$

$$T(s) = \frac{-L_1/L_2}{Ms^2 + Bs + K}$$

1.15 (a) Write two equations

$$(5s^2+7)\oplus_1 - 7\oplus_2 = \frac{6}{s^2+49}$$

$$-7\oplus_1 + (10s^2+7)\oplus_2 = 0$$

(b) Write three equations

$$(8s^2+4s+6)\oplus_1 + (-6)\oplus_2 + (0)\oplus_3 = 0$$

$$(-6)\oplus_1 + (9s^2+6s+6)\oplus_2 + (-6s)\oplus_3 = 0$$

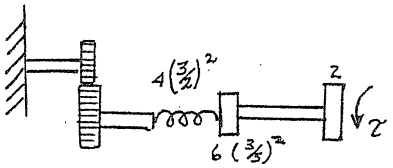
$$(0)\oplus_1 + (-6s)\oplus_2 + (3s^2+6s)\oplus_3 = \frac{7}{s+1}$$

(c) Write Two equations

$$\oplus_1[4s^2+9] + \oplus_2[-9] = 10/s$$

$$\oplus_1[-9) + \oplus_2[3s^2+4s+17] = 0$$

(d) Move the Spring (4) and the inertia (6) to the opposite side of the gear box. Only one equation is needed



$$\left[6\left(\tfrac{3}{5}\right)^2+2\right]s^2\oplus_2 + 4\left(\tfrac{3}{5}\right)^2\oplus_2 = \frac{40}{s^2+5}$$

and $\oplus_2 = \frac{3}{5}\oplus_1 = \dfrac{40}{(s^2+25)(4.16s^2+1.44)}$

1.16(a) $6\frac{d^2\theta}{dt^2} + 3\theta = 10$

$(6s^2+3)\Theta = \frac{10}{s} \Rightarrow \Theta = \frac{10/6}{s(s^2+\frac{1}{2})}$

$\Theta(s) = \frac{1.67\angle 180^\circ}{s+j.707} + \frac{1.67\angle -180^\circ}{s-j.707} + \frac{3.33}{s}$

$\theta(t) = 3.33 + 3.34\cos(.707t + 180^\circ)$

(b) $8\frac{d^2\theta}{dt^2} + 2\frac{d\theta}{dt} + 10\theta = 4\delta(t)$

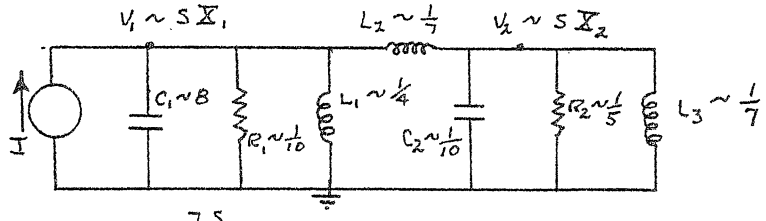
$\Theta(s) = \frac{4}{8s^2+2s+10} = \frac{1/2}{s^2+1/4s+5/4}$

$\Theta(s) = \frac{.23\angle 90^\circ}{s+.125+j1.1} + \frac{.23\angle -90^\circ}{s+.125-j1.1}$

$\theta(t) = .46e^{-.125t}\sin(1.1t)$

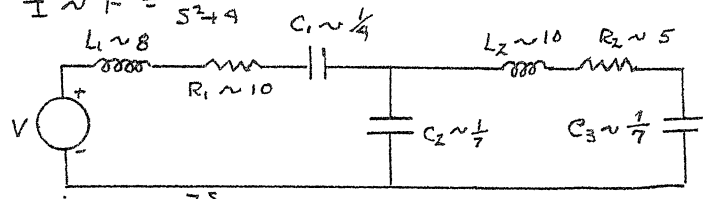
1.17 Two networks, nodal and loop, are analogous to the mechanical networks of Fig P1.17 as follows:

Nodal



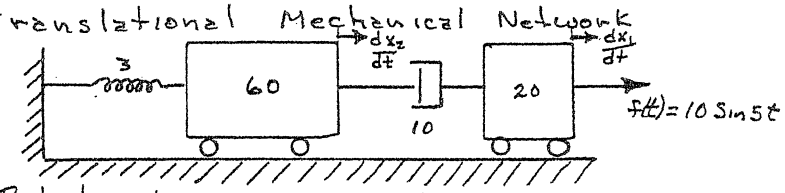
$I \sim F = \frac{7s}{s^2+4}$

Loop

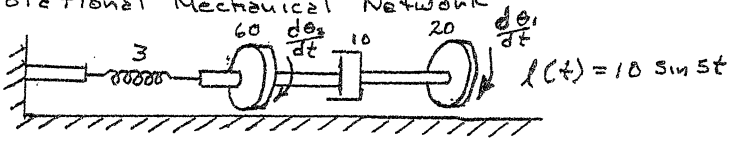


$V \sim F = \frac{7s}{s^2+4}$

1.18(a) Translational Mechanical Network



(b) Rotational Mechanical Network



1.19 Characteristic polynomial is

$$s^2 + s(-M_g - Z_d) + M_g Z_d - M_d(1 + Z_g)$$

(a) $s^2 + s(-(-20) - (-30)) + (-20)(-30) - 100(1+5)$

$= s^2 + 50s + 600 - 600 = s(s+50)$

marginally stable due to simple IA root

(b) $s^2 + (-(-30) - (-20)) + (-30)(-20) - (-200)(1+3)$

$= s^2 + 50s + 600 + 800 = s^2 + 50s + 1400$

roots are $\frac{-50 \pm \sqrt{50^2 - 4 \cdot 1400}}{2} = \frac{-50 \pm \sqrt{2500 - 5600}}{2}$

$= \frac{-50 \pm j\sqrt{3100}}{2} = -25 \pm j\,27.8$

stable due to 2 LHP roots

(c) $s^2 + s(-(-10) - (-40)) + (-10)(-40) - 200(1+2)$

$= s^2 + 50s + 400 - 600 = s^2 + 50s - 200$

roots are $\frac{-50 \pm \sqrt{50^2 - 4(-200)}}{2} = \frac{-50 \pm \sqrt{3300}}{2}$

$= -25 \pm 28.7 = -53.7,\ 3.7$

unstable due to RHP root

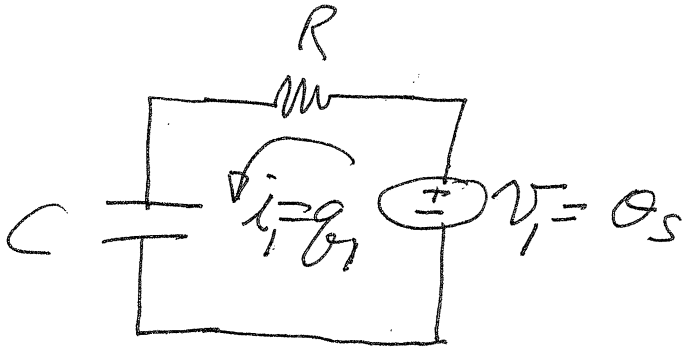
1.20 Room 1: Thermal capacitance C and resistance R

$$\theta_1(t) = \theta_1(0^-)e^{-t/RC} + \theta_s(1-e^{-t/RC})$$

$$\theta_1(RC) = 70e^{-1} + 100(1-e^{-1})$$

$$= 70(.368) + 100(.632) = 89^\circ$$

Electrical equivalent circuit



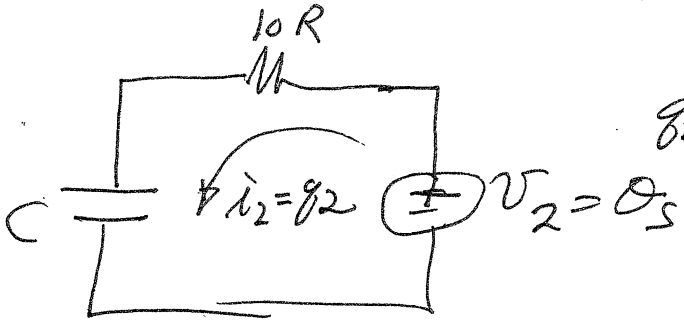
q_1 = heat flow 1

Room 2: Thermal capacitance C and resistance 10R

$$\theta_2(t) = \theta_1(0^-)e^{-t/10RC} + \theta_s(1-e^{-t/10RC})$$

$$\theta_2(RC) = 70e^{-.1} + 100(1-e^{-.1})$$

$$= 70(.905) + 100(0.005) = 72.9^\circ$$



q_2 = heat flow 2

1.21

a) Asymptotically marginally stable and BIBO marginally stable due to only LHP poles

b) Asymptotically unstable due to repeated poles at $s=0$ but BIBO stable because those poles are cancelled and remaining poles are LHP

c) Asymptotically unstable due to repeate IA poles at $\pm j2$ but BIBO marginally stable due to only simple uncancelled IA poles at $\pm j2$ (one pair are cancelled)

d) Asymptotically unstable and BIBO unstable due to RHP pole

e) Asymptotically unstable due to the RHP pole at $s=6$ but BIBO stable because the RHP pole is cancelled and remaining poles are LHP.

1.22 a, e marginally stable since $y_{zi}(t)$ goes neither to zero nor to ∞.

b, c, f stable since $y_{zi}(t)$ approaches 0 as t goes to ∞

d unstable because $y_{zi}(t)$ approaches ∞ (is unbounded)

1.23 a) $T(s) = \dfrac{\frac{K(s+4)}{s(s-2)}}{1+\frac{K(s+4)}{s(s-2)}} = \dfrac{K(s+4)}{s(s-2)+K(s+4)}$

$$T(s) = \frac{4(s+4)}{s^2+(K-2)s+4K} = \frac{4(s+4)}{P(s)}$$

If $K=1$, $P(s) = s^2-s+4$

roots of $P(s) = \dfrac{1 \pm \sqrt{1-16}}{2} = \dfrac{1 \pm j\sqrt{15}}{2} = 0.5 \pm j1.94$

Unstable due to RHP roots

b) $P(s) = s^2+8$ for $K=2$

roots are $\pm j2\sqrt{2} = \pm j2.828$

Marginally stable due to $\pm\infty$ roots

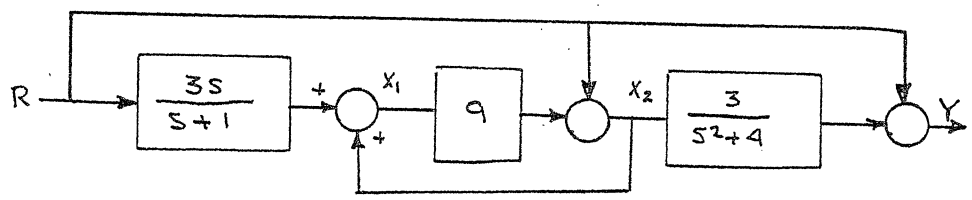
c) For $K=20$, $P(s) = s^2+18s+80$

$$\text{roots} = \frac{-18 \pm \sqrt{18^2-320}}{2} = \frac{-18 \pm \sqrt{36}}{2}$$

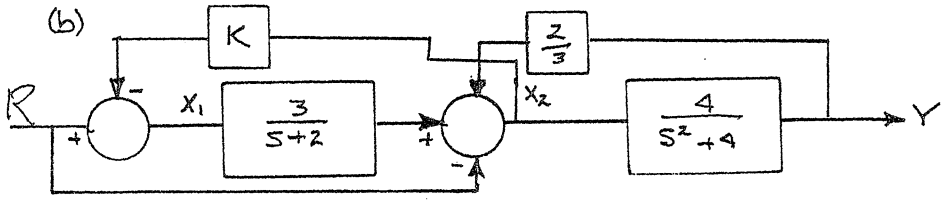
$$= -9 \pm 3 = -12, -6$$

Stable due to LHP roots

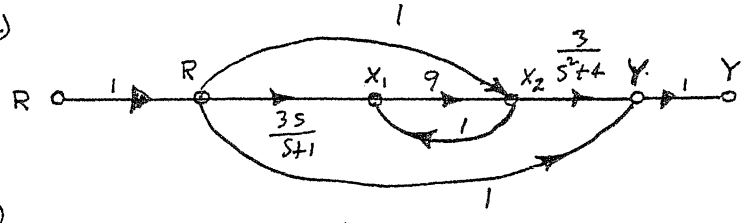
1.24 (a)



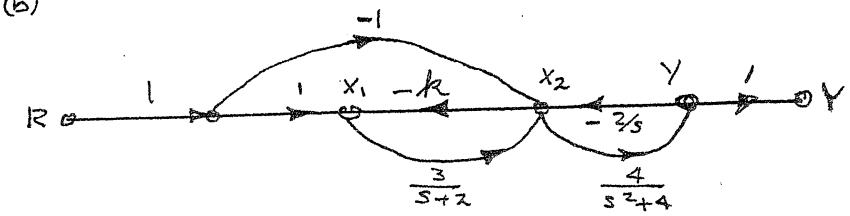
(b)



1.25 (a)



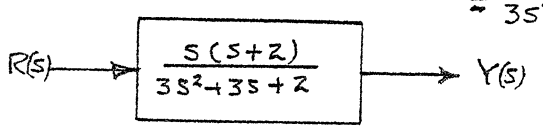
(b)



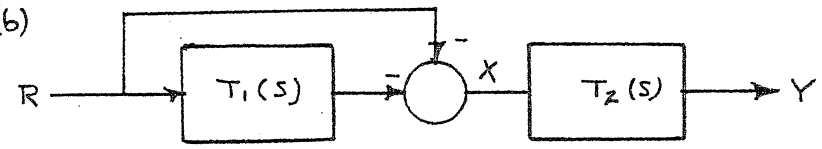
1.26 (a) $G(s) = \frac{5}{s+1}$ $\quad H(s) = \frac{2s}{s+2}$

$$T = \frac{G}{1+GH} = \frac{\frac{5}{s+1}}{1+\left(\frac{5}{s+1}\right)\left(\frac{2s}{s+2}\right)} = \frac{5(s+2)}{(s+1)(s+2)+2s^2}$$

$$= \frac{5(s+2)}{3s^2+3s+2}$$



1.26(b)



$$T_1 = \frac{\frac{5}{s+2}}{1-\frac{5}{s+2}} = \frac{5}{2}$$

$$T_2 = \frac{\frac{3s}{s+1}}{1+\frac{3s}{s+1}} = \frac{3s}{s+1+3s} = \frac{3s}{4s+1}$$

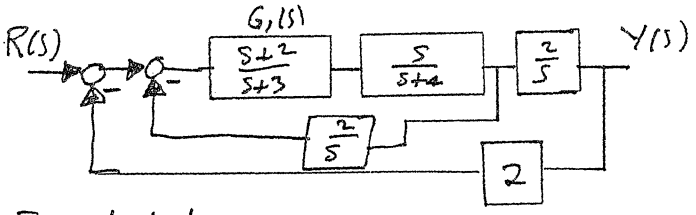
$X = -R - T_1 R$ and $Y = T_2 X = T_2[-R - T_1 R]$

and $Y = -R[T_2(1+T_1)]$

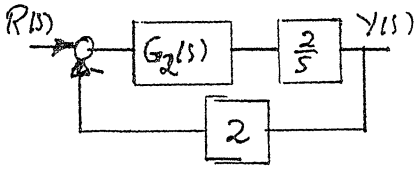
$$T = \frac{Y}{R} = \frac{-3s}{4s+1}\left(1+\frac{5}{2}\right) = \frac{-3s(2+s)}{2(4s+1)}$$

(C) From Block Diagram P1.26C

$$G_1(s) = \frac{1}{1+\frac{1}{s+2}} = \frac{s+2}{s+3}$$



Equivalent to



$$G_2(s) = \frac{\frac{(s+2)5}{(s+3)(s+4)}}{1+\frac{(s+2)8}{(s+3)(s+4)}\frac{2}{8}} = \frac{5(s+2)}{s^2+9s+16}$$

$$T(s) = \frac{\frac{2}{5}G_2(s)}{1+\frac{4}{5}G_2(s)} = \frac{\frac{2(s+2)}{s^2+9s+16}}{1+\frac{4(s+2)}{s^2+9s+16}} = \frac{2(s+2)}{s^2+13s+24}$$

1.26(d) Each equation is written directly from Fig P1.26d

$$Y_{11} = \frac{\frac{2}{s^2+4}}{1 + \frac{2}{s^2+4} \cdot \frac{5}{s+1}} = \frac{2(s+1)}{s^3+s^2+6s+4}$$

$$Y_{22} = \frac{\frac{5}{s+1}}{1 + \frac{2}{s^2+4} \cdot \frac{5}{s+1}} = \frac{5(s^2+4)}{s^3+s^2+6s+4}$$

$$Y_{12} = \frac{\frac{-5}{s+1} \cdot \frac{2}{s^2+4}}{1 + \frac{2}{s^2+4} \cdot \frac{5}{s+1}} = \frac{-2s}{s^3+s^2+6s+4}$$

$$Y_{21} = \frac{\frac{2}{s^2+4} \cdot \frac{5}{s+1}}{1 + \frac{2}{s^2+4} \cdot \frac{5}{s+1}} = \frac{2s}{s^3+s^2+6s+4}$$

1.27(a) $P_1 = 4s^2$ $\quad L_1 = \frac{4}{s^2+1}$ $\quad \Delta_1 = \Delta_2 = 1$

$P_2 = \frac{8}{s+1}$ $\quad \Delta = 1 - \frac{4}{s^2+1}$

$$T(s) = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta} = \frac{4s^2 + \frac{8}{s+1}}{1 - \frac{4}{s^2+1}}$$

$$= \frac{(4s^3+4s^2+8)(s^2+1)}{(s^2-3)(s+1)}$$

(b) $P_1 = \frac{1}{s^3}$ $\quad \Delta = 1 + \frac{8}{s} + \frac{2}{s^2} + \frac{4}{s^3}$

$L_1 = -\frac{8}{s}$ $\quad \Delta_1 = 1$

$L_2 = -\frac{2}{s^2}$

$L_3 = -\frac{4}{s^3}$

$$T(s) = \frac{P_1\Delta_1}{\Delta} = \frac{1/s^3}{1 + \frac{8}{s} + \frac{2}{s^2} + \frac{4}{s^3}}$$

$$T(s) = \frac{1}{s^3+8s^2+2s+4}$$

1.27(c) $P_1 = 1 \cdot \frac{1}{s+1}(-6)\frac{3s}{s+3}(1) = \frac{-18s}{(s+1)(s+3)}$

$P_2 = 1 \cdot \frac{1}{s+1} \cdot \frac{4}{s^2+1}(3)(1) = \frac{12}{(s+1)(s^2+1)}$

$P_3 = 1 \cdot \frac{1}{s+2}\left(\frac{-1}{s}\right)(3)(1) = \frac{-3}{s(s+2)}$

$P_4 = 1 \cdot \frac{1}{s+2}(s)\frac{4}{s^2+1}(3)(1) = \frac{12s}{(s+2)(s^2+1)}$

$P_5 = 1 \cdot \frac{1}{s+2}(s)(-6)\left(\frac{3s}{s+3}\right)(1) = \frac{-18s^2}{(s+2)(s+3)}$

$L_1 = (-6)\left(\frac{1}{s^2}\right) = -\frac{6}{s^2}$

$L_2 = -3$

$\Delta = 1 + \frac{6}{s^2} + 3 + \frac{18}{s^2}$

$\Delta_1 = 1+3 \qquad \Delta_2 = 1 \qquad \Delta_3 = 1 + \frac{6}{s^2}$

$\Delta_4 = 1 \qquad \Delta_5 = 1+3$

$$T(s) = \frac{P_1\Delta_1 + P_2\Delta_2 + P_3\Delta_3 + P_4\Delta_4 + P_5\Delta_5}{\Delta}$$

(d) Δ for all systems is $\Delta = 1 - \left(\frac{4}{s^2+2}\right)(3)\left(\frac{4}{s+1}\right)(2)$

For $T_{11}(s) = \frac{Y_1(s)}{R_1(s)}$ $\quad P_1 = (1)\left(\frac{s}{s+2}\right)\left(\frac{4}{s+2}\right)(3)\left(\frac{4}{s+1}\right)(1)$

$\Delta_1 = \Delta_2 = 1$ $\quad P_2 = (1)\,s\,(6)3\left(\frac{4}{s+1}\right)(1)$

For $T_{21}(s) = \frac{Y_2(s)}{R_1(s)}$ $\quad P_1 = (1)(s)(6)3(1) = 18s$

$P_2 = (1)\left(\frac{3}{s+2}\right)\left(\frac{4}{s+2}\right)(3)(1)$

$\Delta_1 = \Delta_2 = 1$

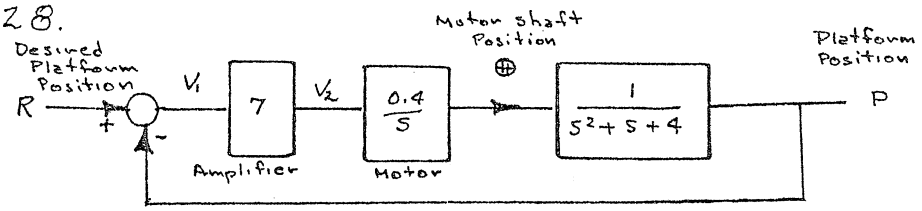
For $T_{12}(s) = \frac{Y_1(s)}{R_2(s)}$ $\quad P_1 = (1)(6)(3)\left(\frac{4}{s+1}\right)(1)$

$\Delta_1 = 1$

For $T_{22}(s) = \frac{Y_2(s)}{R_2(s)}$ $\quad P_1 = (1)(6)(s)(1)$

$\Delta_1 = 1$

1.28.



$$(S^2+S+4)P = \oplus$$

$$V_1 = R - P$$

$$S\oplus = 0.4 V_2$$

$$V_2 = 7 V_1$$

System Transfer function $T(S) = \frac{P(S)}{R(S)}$

$$T = \frac{\frac{2.8}{S(S^2+S+4)}}{1 + \frac{2.8}{S(S^2+S+4)}} = \frac{2.8}{S^3+S^2+4S+2.8}$$

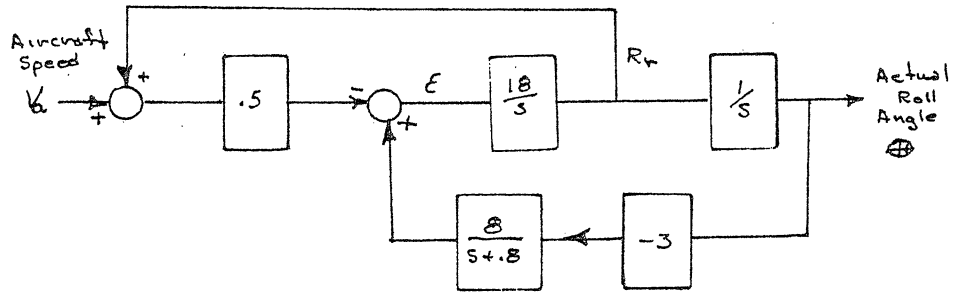
1.29 $T_1(S) = \frac{\text{Actual Roll Angle}}{\text{Input Proportional to Roll Angle}}$

$$T_2(S) = \frac{\text{Actual Roll Angle}}{\text{Air Speed}}$$

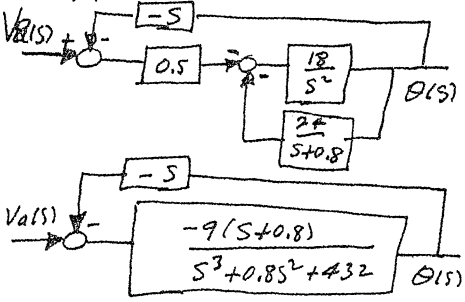
For $T_1(S)$: Aileron loop $= \frac{6\left(\frac{1}{3}\right)(3)}{1 + (.5)(6)\left(\frac{1}{5}\right)(3)} = \frac{18}{S+9}$

$$T_1(S) = \frac{\left(\frac{8}{S+.8}\right)\left(\frac{18}{S+9}\right)\cdot\frac{1}{S}}{1 + \frac{8(18)(3)}{S(S+.8)(S+9)}}$$

For $T_2(S)$: $T_1(S) = \frac{144}{S^3+9.8S^2+7.2S+432}$



1.29 (continued)



$$\frac{\frac{18}{s^2}}{1+\frac{18\times 24}{s^2(s+0.8)}} = \frac{18(s+0.8)}{s^3+0.85^2+432}$$

$$T_2(s) = \frac{\Theta(s)}{Va(s)} = \frac{\frac{-9(s+0.8)}{s^3+0.85^2+432}}{1+\frac{9s(s+0.8)}{s^3+0.85^2+432}} = \frac{-9(s+0.8)}{s^3+9.85^2+7.2s+432}$$

1.30 $$T(s) = \frac{\frac{5}{(s+a)^2}}{1-\frac{Ks}{(s+a)^2}} = \frac{5}{s^2+(2a+K)s+a^2}$$

Characteristic Equation is

$$s^2+(2a-K)s+a^2=0$$

for roots to be on $j\omega$ axis, $(2a-K)=0$

so $K=2a$

and the frequency is $f=\frac{\omega}{2\pi}=\frac{a}{2\pi}$

CHAPTER 2

2.1

The procedure is to reconstruct the diff. eqn, convert to Laplace to include initial conditions, and then solve for $y(t)$.

(a) $(S+3)Y(S) = 6R(S)$

$$\frac{dy}{dt} + 3y = 6r$$

$$SY(S) - y(0^-) + 3Y(S) = 6R(S)$$

$$(S+3)Y(S) = y(0^-) + 6R(S) \qquad R(S) = 1, \quad y(0^-) = -2$$

$$Y(S) = \frac{-2}{S+3} + \frac{6}{S+3} = \frac{4}{S+3}$$

$$y(t) = y_{not}(t) = 4e^{-3t}u(t) \qquad \tau = 1/3$$

forced response would be the step change from $y(0^-) = -2$ to $y(0^+) = 4$

(b) $(3S+2)Y(S) = -4S\,R(S)$

$$\frac{3dy}{dt} + 2y = -4\frac{dr}{dt}$$

$$3[SY(S) - y(0^-)] + 2Y(S) = -4[SR(S) - r(0^-)]$$

$$(3S+2)Y(S) = -\underbrace{4SR(S)}_{\frac{1}{S}} + 3\underbrace{y(0^-)}_{3} + 4\underbrace{r(0^-)}_{0}$$

$$Y(S) = \frac{-4}{3S+2} + \frac{9}{3S+2} = \frac{5}{3S+2} = \frac{5/3}{S+2/3}$$

$y_{nat}(t) = \frac{5}{3} e^{-\frac{3}{2}t} u(t) \qquad \tau = \frac{3}{2} = 1.5$

forced response would be the step change from $y(0^-) = 3$ to $y(0^+) = \frac{5}{3}$

(c) $(2s+3)Y(s) = (s+1)R(s)$

$2\frac{dy}{dt} + 3y = \frac{dr}{dt} + r$

$2[sY(s) - y(0^-)] + 3Y(s) = sR(s) - r(0^-) + R(s)$

$(2s+3)Y(s) = (s+1)\underbrace{R(s)}_{\frac{1}{s}} + 2\underbrace{y(0^-)}_{4} - \underbrace{r(0^-)}_{0}$

$Y(s) = \frac{s+1}{s(2s+3)} + \frac{8}{2s+3} = \underbrace{\frac{0.5(s+1)}{s(s+1.5)}}_{Y_{zs}(s)} + \frac{4}{s+1.5}$

$Y_{zs}(s) = \frac{K_1}{s} + \frac{K_2}{s+1.5}$

$K_2 = \left[(s+1.5)Y_{zs}(s)\right]\Big|_{s=-1.5} = \frac{0.5(s+1)}{s}\Big|_{s=-1.5} = 0.167$

$K_1 = \left[sY_{zs}(s)\right]\Big|_{s=0} = \frac{0.5(s+1)}{s+1.5}\Big|_{s=0} = \frac{0.5}{1.5} = \frac{1}{3}$

$Y(s) = \frac{1/3}{s} + \frac{0.167}{s+1.5} + \frac{4}{s+1.5} = \frac{1/3}{s} + \frac{4.167}{s+1.5}$

For $t > 0$

$y(t) = \underbrace{\frac{1}{3}}_{y_{forced}(t)} + \underbrace{4.167e^{-1.5t}}_{y_{nat}(t)} \qquad \tau = \frac{1}{1.5} = 0.67$

(d) $(s+7)Y(s) = 3R(s)$

$\frac{dy}{dt} + 7y = 3r$

$sY(s) - y(0^-) + 7Y(s) = 3R(s)$

$$(s+7)\,Y(s) = 3\underbrace{\frac{R(s)}{\frac{2s}{s^2+16}}} + \frac{y(0^-)}{2}$$

$$Y(s) = \frac{6s}{(s+7)(s^2+16)} + \frac{2}{s+7}$$

$$= \frac{K_1}{s+7} + \frac{K_2 s + K_3}{s^2+16} + \frac{2}{s+7}$$

$$K_1 = \left[(s+7)\,Y_{zs}(s)\right]\Big|_{s=-7} = \frac{6s}{s^2+16}\Big|_{s=-7} = -\frac{42}{65} = -0.646$$

$$\frac{K_2 s + K_3}{s^2+16} = Y_{zs}(s) - \frac{(-0.646)}{s+7} = \frac{6s}{(s+7)(s^2+16)} + \frac{0.646}{s+7}$$

$$= \frac{6s + 0.646(s^2+16)}{(s+7)(s^2+16)} = \frac{0.646[s^2 + 9.29s + 16]}{(s+7)(s^2+16)} \qquad (s+7)(s+2.29)$$

$$= \frac{\overbrace{0.646}^{K_2}\, s + \overbrace{1.48}^{K_3}}{s^2+16}$$

$$Y(s) = \frac{-0.646}{s+7} + \frac{0.646s + 1.48}{s^2+16} + \frac{2}{s+7}$$

$$= \frac{1.354}{s+7} + \frac{0.646s + 1.48}{s^2+16} \qquad 4 \times 0.37$$

$b^2 = 16$ so $b = 4$

For $t > 0$

$$y(t) = \underbrace{1.354e^{-7t}}_{y_{not}(t)} + \underbrace{0.646\cos 4t + 0.37\sin 4t}_{y_{forced}(t)}$$

$$\tau = \frac{1}{7} = 0.143$$

(e) $(2s+1)\,Y(s) = 2R(s)$

$$2\frac{dy}{dt} + y = 2r$$

$$2\left[sY(s) - y(0^-)\right] + Y(s) = 2R(s)$$

$$\underbrace{(2s+1)}_{2(s+\frac{1}{2})} Y(s) = \underbrace{2R(s)}_{\frac{6}{s+\frac{1}{2}}} + \underbrace{2y(0^-)}_{3}$$

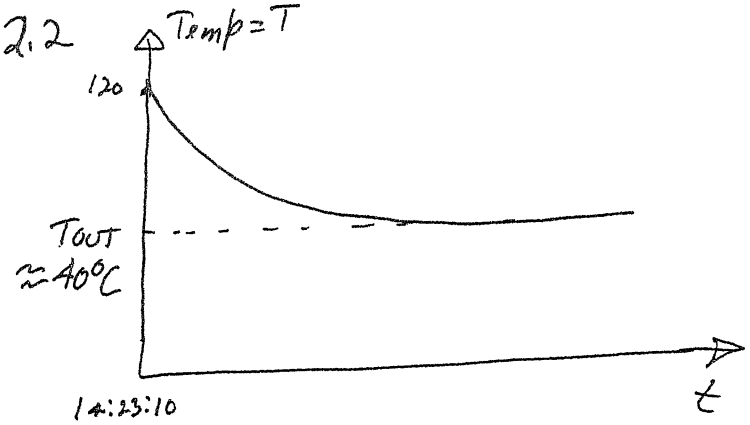
$$Y(s) = \frac{2\cdot 6}{2(s+\frac{1}{2})^2} + \frac{6}{2(s+\frac{1}{2})} = \frac{6}{(s+1/2)^2} + \frac{3}{s+1/2}$$

In cases like this, the forced response is usually taken to be the particular solution to the ~~diff~~ eqn, which is the inverse transform of $\frac{6}{(s+\frac{1}{2})^2}$. Thus, the forced response becomes

$y_{ss}(t)$ and, for $t>0$,

$$y(t) = \underbrace{6te^{-\frac{1}{2}t}}_{y_{forced}(t)} + \underbrace{3e^{-\frac{1}{2}t}}_{y_{nat}(t)} \qquad \tau = 2$$

2.2



With $T_{OUT} \simeq 40°C$, for $t \geq 14:23:10$

$$T(t) = T_{OUT} + \left[120° - T_{OUT}\right] e^{-t/\tau}$$

$T(t) = 40° + [120° - 40°]\, e^{-t/\tau} = 40° + 80°\, e^{-t/\tau}$

$T(\tau) = 40° + 80° e^{-1} = 40° + 80°(0.368) = 69.4°$

interpolate, subtracting 14:23:10 from each time:

t	Temb
25:25 = 25.42	80.5
(35.8)	(69.4)
41:50 = 41.83	63.0

2.3

Here, each complex term is written as

$$Y(s) = \frac{num}{(s^2+as+b)(other)} = \frac{\overset{K}{\widetilde{M\angle\theta}}}{s+a-jb} + \frac{\overset{K^*}{\widetilde{M\angle-\theta}}}{s+a+jb} + other$$

$$y(t) = \left[2M e^{-at}\cos(bt+\theta) + other\right] u(t)$$

Also, $Y(s) = Y_{zs}(s) = T(s)\,R(s)$.

(a) $$Y(s) = \frac{2}{s^2+4s+16} = \frac{K}{s+2-j3.46} + \frac{K^*}{s+2+j3.46}$$

$$K = \left[(s+2-j3.46)\,Y(s)\right]\Big|_{s=-2+j3.46}$$

$$= \left[\frac{2}{s+2+j3.46}\right]\Big|_{s=-2+j3.46} = \frac{2}{-2+j3.46+2+j3.46}$$

$$= \frac{2}{j6.72} = 0.3\angle -90°$$

$$y_{nat}(t) = 0.6\, e^{-2t}\cos(3.46t - 90°)\, u(t)$$

The forced response is the change from $y(0^-) = 0$ to $y(0^+) = 0.6\cos(-90°) = 0$ (no forced)

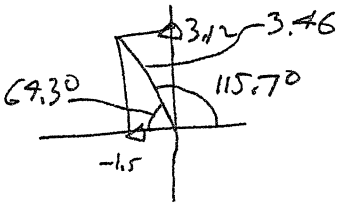
(b) $Y(s) = \frac{s-2}{s(s^2+3s+12)} = \frac{s-2}{s(s+1.5+j3.12)(s+1.5-j3.12)}$

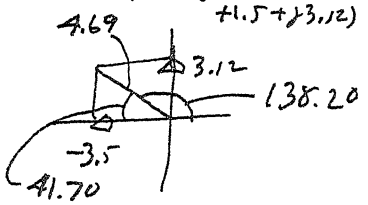
$= \frac{K_1}{s} + \frac{K_2}{s+1.5-j3.12} + \frac{K_2^*}{s+1.5+j3.12}$

$K_1 = sY(s)|_{s=0} = \frac{s-2}{s^2+3s+12}\Big|_{s=0} = -\frac{2}{12} = -0.167$

$K_2 = \frac{s-2}{s(s+1.5+j3.12)}\Big|_{s=-1.5+j3.12} = \frac{-1.5+j3.12-2}{(-1.5+j3.12)(-1.5+j3.12+1.5+j3.12)}$

$= \frac{-3.5+j3.12}{(-1.5+j3.14)(j6.24)}$





$K_2 = \frac{4.69\angle 138.2^\circ}{(3.46\angle 115.70)(6.24\angle 90^\circ)} = 0.22\angle -67.5^\circ$

For $t \geq 0$

$y(t) = \underbrace{-0.167}_{y_{forced}(t)} + \underbrace{0.44\, e^{-1.5t}\cos(3.12t - 67.5^\circ)}_{y_{nat}(t)}$

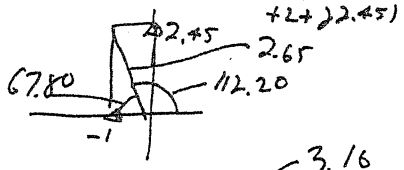
(c) $Y(s) = \frac{s+1}{s(s^2+4s+10)} = \frac{s+1}{s(s+2-j2.45)(s+2+j2.45)}$

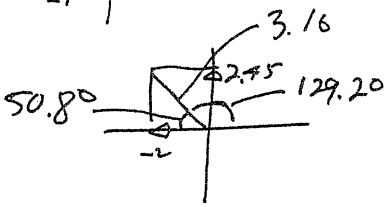
$= \frac{K_1}{s} + \frac{K_2}{s+2-j2.45} + \frac{K_2^*}{s+2+j2.45}$

$K_1 = sY(s)|_{s=0} = \frac{s+1}{s^2+4s+10}\Big|_{s=0} = \frac{1}{10} = 0.1$

$$K_2 = \left.\frac{s+1}{s(s+2+j2.45)}\right|_{s=-2+j2.45} = \frac{-2+j2.45+1}{(-2+j2.45)(-2+j2.45+2+j2.45)}$$

$$= \frac{-1+j2.45}{(-2+j2.45)(j4.90)}$$





$$K_2 = \frac{2.65\angle 112.2^\circ}{(3.16\angle 129.20)(4.9\angle 90^\circ)} = 0.17\angle -107^\circ$$

For $t > 0$

$$y(t) = \underbrace{0.1}_{y_{forced}(t)} + \underbrace{0.34e^{-2t}\cos(2.45t - 107^\circ)}_{y_{nat}(t)}$$

(d) $$Y(s) = \frac{2}{s^2(s^2+s+6)} = \frac{2}{s^2(s+0.5-j2.4)(s+0.5+j2.4)}$$

$$= \frac{K_2}{s} + \frac{K_1}{s^2} + \frac{K_3}{s+0.5-j2.4} + \frac{K_3^*}{s+0.5+j2.4}$$

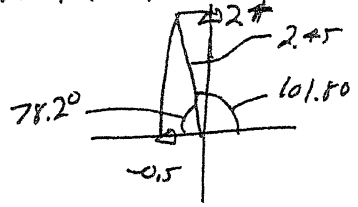
$$K_2 = s^2Y(s)\big|_{s=0} = \left.\frac{2}{s^2+s+6}\right|_{s=0} = \frac{2}{6} = \frac{1}{3} = 0.33$$

$$K_1 = \left[\frac{d}{ds}\, s^2Y(s)\right]\Big|_{s=0} = \left[\frac{d}{ds}\, 2(s^2+s+6)^{-1}\right]\Big|_{s=0}$$

$$= -\frac{2}{(s^2+s+6)^2}(2s+1)\Big|_{s=0} = -\frac{2}{36} = -\frac{1}{18} = -0.056$$

$$K_3 = \frac{2}{s^2(s+0.5+j2.4)}\Big|_{s=-0.5+j2.4}$$

$$= \frac{2}{(-0.5+j2.4)^2(-0.5+j2.4+0.5+j2.4)}$$



$$K_3 = \frac{2}{(2.45\angle 101.8)^2 j4.8}$$

$$= 0.069\angle -293.6° = 0.069\angle 66.4°$$

For

$$Y(s) = \frac{0.33}{s^2} + \frac{-0.056}{s} +$$

For $t \geq 0$

$$y(t) = \underbrace{0.33t - 0.056}_{y_{forced}(t)} + \underbrace{0.138e^{-0.5t}\cos(2.4t + 66.4°)}_{y_{natural}(t)}$$

2.4

For each part of this problem, the denom of T(s), the charact. polynomial, is compared with the standard form $s^2+2\zeta\omega_n s+\omega_n^2$ and $\omega = \omega_n\sqrt{1-\zeta^2}$

(a) $\omega_n^2 = 25$ $\boxed{\omega_n = 5}$

$2\zeta\omega_n = 4$ $10\zeta = 4$ $\boxed{\zeta = 0.4}$ $\boxed{a = 2}$

$\omega = 5\sqrt{1-0.4^2} = \boxed{4.58}$

(b) $4s^2+2s+50 = 4(s^2+0.5s+12.5)$

$\omega_n^2 = 12.5$ $\boxed{\omega_n = 3.54}$ $\boxed{a = 0.25}$

$2\zeta\omega_n = 0.5$ $(2)(3.54)\zeta = 0.5$ $\boxed{\zeta = 0.07}$

$\omega = \sqrt{1-0.07^2}\times 3.54 = \boxed{3.54}$

(c) $3s^2+s+10 = 3\left(s^2+\frac{1}{3}s+\frac{10}{3}\right)$

$\omega_n^2 = \frac{10}{3}$ $\boxed{\omega_n = 1.82}$

$2\zeta\omega_n = \frac{1}{3}$ $(2)(1.82)\zeta = \frac{1}{3}$ $\boxed{\zeta = 0.09}$ $\boxed{a = 0.167}$

$\omega = 1.82\sqrt{1-(.09)^2} = \boxed{1.81}$

(d) $\boxed{\omega_n = \sqrt{16} = 4}$ $\boxed{\zeta = 0}$ $\boxed{\omega = \omega_n = 4}$

$\boxed{a = 0}$

(e) $2s^2+6s+20 = 2(s^2+3s+10)$ $\boxed{\omega_n = \sqrt{10} = 3.16}$

$2\zeta\omega_n = 3$ $2(3.16)\zeta = 3$ $\boxed{a = 1.5}$ $\boxed{\zeta = 0.47}$ $\omega = 3.16\sqrt{1-(.47)^2} = \boxed{2.78}$

2.5

Let $G(s) = \frac{N(s)}{D(s)}$ $\quad T(s) = \dfrac{\frac{N(s)}{D(s)(s+2)}}{1+\frac{N(s)}{D(s)(s+2)}}$

$T(s) = \dfrac{N(s)}{D(s)(s+2)+N(s)}$ $\quad$ If $\begin{cases} D(s)=s \\ N(s)=1 \end{cases}$

$T(s) = \dfrac{1}{s(s+2)+1} = \dfrac{1}{s^2+2s+1}$ $\quad \begin{cases} \omega_n = 1 \\ \zeta = 1 \end{cases}$

So system is crit. damped. Answer is no unique.

2.6

$$T(s) = \frac{\frac{4K}{s^2+12s+2}}{1+\frac{4K}{s^2+12s+2}} = \frac{4K}{s^2+12s+2+4K}$$

$P(s) = s^2+12s+2+4K = s^2+2\zeta\omega_n s+\omega_n^2$

If $\zeta = 0.7$ then $2(.7)\omega_n = 12$ $\quad \omega_n = 8.57$

$\omega_n^2 = 73.47 = 2+4K$ $\quad 4K = 71.47$ $\quad \boxed{K = 17.87}$

2.7

$$T(s) = \frac{\frac{Ks}{s^2+2s+300}}{1+\frac{4Ks}{s^2+2s+300}} = \frac{Ks}{s^2+(2+K4)s+300}$$

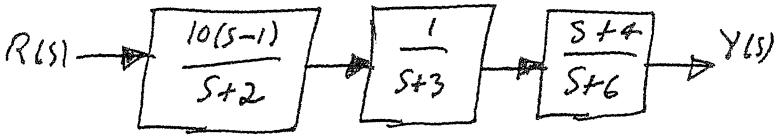
$s^2+2\underbrace{\zeta\omega_n}_{a}s+\omega_n^2 = s^2+(2+4K)s+300$

For $a \geq 10$ $\quad 2+4K \geq 2a = 20$ $\quad 1+2K \geq 10$

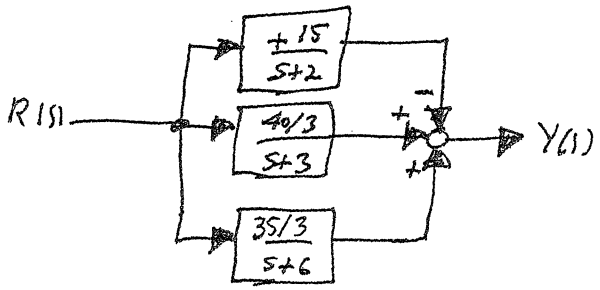
$\boxed{K \geq \frac{9}{2} = 4.5}$

2.8 $T(s) = \dfrac{10(s-1)(s+4)}{(s+2)(s+3)(s+6)}$

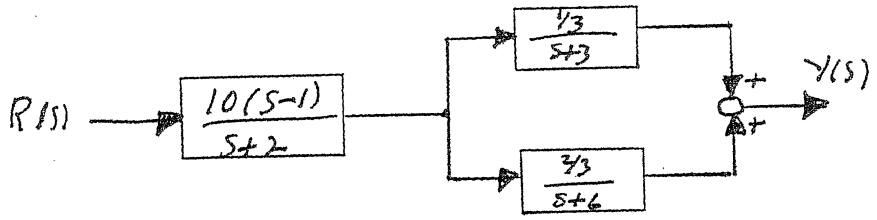
(a)



(b) $T(s) = \dfrac{-15}{s+2} + \dfrac{40/3}{s+3} + \dfrac{35/3}{s+6}$



(c) $T(s) = \dfrac{10(s-1)}{s+2}\left[\dfrac{s+4}{(s+3)(s+6)}\right] = \dfrac{10(s-1)}{s+2}\left[\dfrac{1/3}{s+3} + \dfrac{3/3}{s+6}\right]$



2.9 Each charact. polynomial should be compared to $s^2 + 2\zeta\omega_n s + \omega_n^2$

(a) $P(s) = s^2 + 8s + 8$ $\omega_n = \sqrt{8} = 2.83$

$2\zeta\omega_n = 8$ $2\zeta(2.83) = 8$ $\zeta = 1.41$

$\zeta > 1 \Rightarrow$ **overdamped**

(b) $P(s) = s^2 + s + 7$ $\omega_n = \sqrt{7} = 2.65$

$2\zeta\omega_n = 1$ $2\zeta(2.65) = 1$ $\zeta = 0.19$

$\zeta < 1 \Rightarrow$ **underdamped**

(c) $2s^2 + 9s + 3 = 2(s^2 + 4.5s + 1.5)$ $\omega_n = \sqrt{1.5} = 1.22$

$2\zeta\omega_n = 4.5$ $2\zeta(1.22) = 4.5$ $\zeta = 1.84$

$\zeta > 1 \Rightarrow$ **overdamped**

(d) $P(s) = s^2 + s + 6$ $\omega_n = \sqrt{6} = 2.45$

$2\zeta\omega_n = 1$ $2\zeta(2.45) = 1$ $\zeta = 0.20$

$\zeta < 1 \Rightarrow$ **underdamped**

(e) $P(s) = s^2 + 6s + 9$ $\omega_n = \sqrt{9} = 3$

$2\zeta\omega_n = 6$ $2\zeta(3) = 6$ $\zeta = 1$

$\zeta = 1 \Rightarrow$ **critically damped**

2.10 Routh array

s^3	1	a_1
s^2	a_2	a_0
s^1	$\frac{a_1 a_2 - a_0}{a_0}$	
s^0	a_0	

For no left column sign changes when $a_0 > 0$

$a_1 a_2 - a_0 > 0$

$\boxed{a_1 a_2 > a_0}$

Also $\begin{cases} a_1 > 0 \\ a_2 > 0 \end{cases}$ not $\begin{cases} a_1 < 0 \\ a_2 < 0 \end{cases}$

2.11

(a) $$T(s) = \frac{\frac{K}{s^2+4s+8}}{1+\frac{3K}{(s^2+4s+8)(s^2+2s+12)}}$$

$$= \frac{K(s^2+2s+12)}{\underbrace{s^4+6s^3+28s^2+64s+(96+3K)}_{P(s)}}$$

Routh Array is

s^4	1	28	$96+3K$
s^3	6	64	
s^2	17.33	$96+3K$	
s^1	$\frac{1109.33-576-18K}{17.33} = \frac{533-18K}{17.33}$		
s^0	$96+3K$		

From s^1 $\quad 533-18K > 0 \quad K < 29.6$

From s^0 $\quad 96+3K > 0 \quad K > -32$

For stability $\boxed{-32 < K < 29.6}$

(b) Any characteristic polynomial will do.

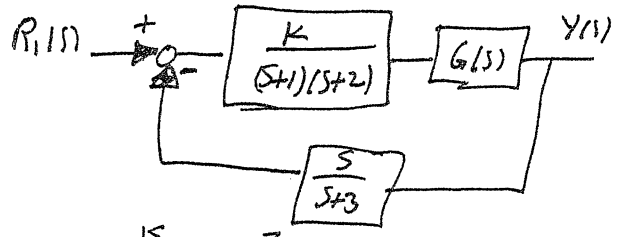
For example, take the denominator of

$$T_1(s) = \left.\frac{Y(s)}{R_1(s)}\right|_{R_2(s)=0}$$

Let

$$G(s) = \frac{\frac{7}{s^2+3s+3}}{1+\frac{7}{s^2+3s+3}} = \frac{7}{s^2+3s+10}$$

(b) Continued



$$T(s) = \frac{\frac{K}{(s+1)(s+2)}\,\frac{7}{s^2+3s+10}}{1+\frac{7K5}{(s+1)(s+2)(s^2+3s+10)(s+3)}}$$

$$P(s) = (s+1)(s+2)(s^2+3s+10)(s+3)+7K5$$

$$= s^5+9s^4+39s^3+99s^2+(128+7K)s+60$$

s^5	1	39	$128+7K$
s^4	9	99	60
s^3	28	$121.33+7K$	
s^2	$60-2.25K$	60	
s^1	$\frac{-15.75(K-24.09)(K+14.76)}{60-2.25K}$		
s^0	60		

For stability, we want no left column sign changes.

$60-2.25K > 0 \quad K < 26.67$

2.11 (b) continued

$-15.75(K-24.09)(K+14.76) > 0$ If $60-2.25K > 0$

$(K-24.09)(K+14.76) < 0$

Either

$K-24.09 < 0$, $K+14.76 > 0$ so $\{K < 24.09\}$, $\{K > -14.76\}$ $\Rightarrow$ $-14.76 < K < 24.09$

OR

$K-24.09 > 0$, $K+14.76 < 0$ so $\{K > 24.09\}$, $\{K < -14.76\}$ $\Rightarrow$ null set

Solution is intersection of

$\{-14.76 < K < 24.2\}$ and $\{K < 26.67\}$

Result

$\boxed{-14.76 < K < 24.2}$

(c) $P_1 = \left(\frac{3}{s+2}\right)\left(\frac{3}{s+9}\right)$, using Masons Rule

$L_1 = -\frac{15}{s+2}$ $L_2 = -\frac{9}{s+9}$ $L_3 = -\frac{9k}{(s+2)(s+9)}$

$\Delta = 1-(L_1+L_2+L_3) = 1+\left[\frac{15}{s+2}+\frac{9}{s+9}+\frac{9k}{(s+2)(s+9)}\right]$

$= \frac{(s+2)(s+9)+15(s+9)+9(s+2)+9k}{(s+2)(s+9)}$

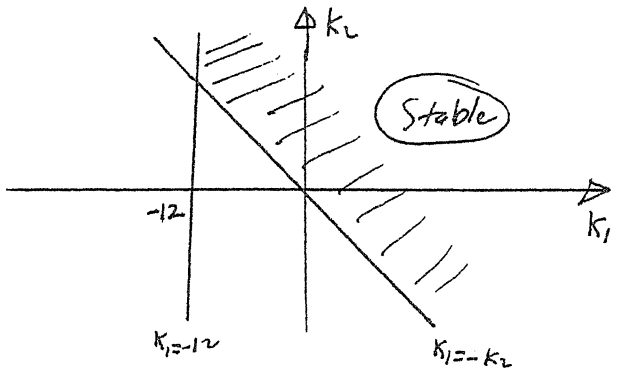
$P(s) = \frac{}{s^2+35s+171+9k}$ $\begin{cases} \text{Stable } 171+9k > 0 \\ k > -\frac{171}{9} \\ \boxed{k > -19} \end{cases}$

2.12

For stability of $P(s) = s^2 + (6+0.5K_1)s + 3(K_1+K_2)$

$6+.5K_1 > 0 \quad .5K_1 > -6 \quad K_1 > -12$

$K_1 + K_2 > 0 \quad K_1 > -K_2$



2.13

Using 2.10, we need

$a_0, a_1, a_2 > 0$

$a_1 a_2 > a_0$ for $P(s) = s^3 + a_2 s^2 + a_1 s + a_0$

(a) $P(s) = K\left[s^3 + \frac{1}{K}s^2 + \frac{6}{K}s + \frac{12}{K}\right]$ $\begin{cases} a_0 = \frac{12}{K} \\ a_1 = \frac{6}{K} \\ a_2 = \frac{1}{K} \end{cases}$

$K > 0 \Rightarrow a_0, a_1, a_2 > 0$

$\left(\frac{6}{K}\right)\left(\frac{1}{K}\right) > \frac{12}{K} \quad 6 > 12K \quad 12K < 6 \quad \boxed{K < \frac{1}{2}} \quad \boxed{K > 0}$

$0 < K < \frac{1}{2}$

(b) $a_0 = 12 \quad a_1 = 3 \quad a_2 = K \quad K > 0$ and

$3K > 12 \quad \boxed{K > 4}$

(c) $a_0 = K \quad a_1 = K \quad a_1 = 7 \quad K > 0$ and

$7K > K$

$7 > 1$ so $\boxed{K > 0}$ works

2.14

(a) $T(s) = \dfrac{\dfrac{2K}{s^2+3s+15}}{1+\dfrac{2Ks}{(s^2+3s+15)(s+10)}} = \dfrac{2K(s+10)}{(s^2+3s+15)(s+10)+2Ks}$

$P(s) = s^3 + 13s^2 + (45+2K)s + 150$

Using problem 2.10, $a_0 = 150$, $a_1 = 45+2K$, $a_2 = 13$

$45+2K > 0 \quad 2K > -45 \quad K > -22.5$

$(13)(45+2K) > 150 \quad 2K > \frac{150}{13} - 45 = -33.46$

$K > -16.73$ so $\boxed{K > -16.73}$

(b) $T(s) = \dfrac{\dfrac{2}{s^2+Ks+15}}{1+\dfrac{2}{(s^2+Ks+15)(s+3)}} = \dfrac{2(s+3)}{(s^2+Ks+15)(s+3)+2}$

$P(s) = s^3 + (3+K)s^2 + (15+3K)s + 47$

$a_0 = 47$, $a_1 = 15+3K$, $a_2 = 3+K$

$15+3K > 0 \quad 3K > 15 \quad (K > 5)$

$3+K > 0 \quad (K > -3)$

$(3+K)(15+3K) > 47$

$3K^2 + 24K + 45 > 47$

$K^2 + 8K - \frac{2}{3} > 0$ roots $\dfrac{-8 \pm \sqrt{64 + \frac{8}{3}}}{2}$

$(K + 8.082)(K - 0.082) > 0 \quad = -8.082, 0.082$

either $\left\{\begin{matrix} K+8.082 > 0 \\ K-0.082 > 0 \end{matrix}\right\} \left\{\begin{matrix} K > -8.082 \\ K > 0.082 \end{matrix}\right\}$ $(K > 0.082)$

or $\left\{\begin{matrix} K+8.082 < 0 \\ K-0.082 < 0 \end{matrix}\right\} \left\{\begin{matrix} K < -8.082 \\ K < 0.082 \end{matrix}\right\}$ $(K < -8.082)$

So must be $\boxed{K > 0.082}$

(c) Using Masons Rule: $P = \frac{3K}{(s+K)(s+3)} \quad L_1 = -\frac{6}{s+5}$

$L_2 = -\left(\frac{3}{5}\right)\frac{3K}{(s+K)(s+3)} \quad \Delta = 1 - (L_1 + L_2) = 1 + \frac{6}{s+5} + \frac{9K}{s(s+K)(s+5)}$

2.14(c) continued

$P(s) = s(s+k)(s+5) + 6s(s+k) + 9k$

$= s^3 + (k+\ )s^2 +$

Using Problem 2.10, $a_0 = 9k$, $a_1 = 11k$, $a_2 = k+11$

$a_0 > 0 \Rightarrow k > 0$; $a_1 > 0 \Rightarrow k > 0$; $a_2 > 0 \Rightarrow k > -11$

$a_1 a_2 > a_0 \Rightarrow (k+11)(11k) > 9k$

$k + 11 > \frac{9}{11}$ $\quad k > -11 + \frac{9}{11} = -10.18$

$k > -10.18$

So we must have $\boxed{k > 0}$

2.15 $\frac{1}{a} = 7200 \Rightarrow a = 1.39 \times 10^{-4}$

$\frac{A}{a^2} = 0.35 \Rightarrow A = 6.75 \times 10^{-9}$

$\tau = 5$; $K = 7.3 \times 10^{-6}$

$R(s) = \left(\frac{A}{K}\right)\frac{\tau s + 1}{(s+a)^2} = \frac{4.62 \times 10^{-3} s + 9.25 \times 10^{-4}}{(s + 1.39 \times 10^{-4})^2}$

$= \frac{K_1}{s + 1.39 \times 10^{-4}} + \frac{K_2}{(s + 1.39 \times 10^{-4})^2}$

$K_1 = 4.62 \times 10^{-3}$ $\quad K_2 = 7.86 \times 10^{-4}$

$r(t) = 4.62 \times 10^{-3} e^{-1.39 \times 10^{-4} t}$

$+ 7.86 \times 10^{-4} t\, e^{-1.39 \times 10^{-4} t}$

2.16

K_1	M
.02	990
.74	992
.93	994
1.24	996
1.87	998
3.75	1000

2.17

K_1	M
.74	19.8
.42	19.84
.54	19.88
.75	19.92
1.25	19.96
3.75	20.00

2.18

$$Ae^{an} = 1000(1.007)^n$$

$$\ln[Ae^{an}] = \ln[1000(1.007)^n]$$

$$\ln A + an = \ln 1000 + n\underbrace{\ln(1.007)}_{0.0070}$$

$A = 1000 \quad a = 0.0070$

2.19

$$T(S) = \frac{\frac{180K_1(0.23)}{(S+6)(S+30)}}{1 + \frac{180K_1K_2}{(S+6)(S+30)}} = \frac{\overbrace{41.4}}{\frac{0.23(180)K_1}{S^2+36S+180+180K_1K_2}}$$

For unit step input $Y(S) = T(S)R(S) = T(S)\left(\frac{1}{S}\right)$

$$Y(S) = \underbrace{\frac{A}{S}}_{\text{forced}} + \underbrace{\frac{B}{S+a} + \frac{C}{S+b}}_{\text{natural}}$$

$$A = [S\,Y(S)]|_{S=0} = \left[S\,T(S)\left(\frac{1}{S}\right)\right]\Big|_{S=0}$$

$$= T(0)$$

(a) $$A = \frac{41.4K_1}{180+180K_1K_2} = 0.4$$

For critical damping $P(S) = S^2+36S+180+180K_1K_2$

$$= S^2+2\zeta\omega_n S+\omega_n^2$$

$$= S^2+2\omega_n S+\omega_n^2$$

2.19(a) continued

$\omega_n = \sqrt{180+180K_1K_2}$ $\quad 2\omega_n = 2\sqrt{180+180K_1K_2} = 36$

from forced response

$41.4K_1 = 0.4(180+180K_1K_2)$

$\left(\frac{36}{2}\right)^2 = 324$

$41.4K_1 = 0.4(324) = 129.6$ $\quad \boxed{K_1 = 3.13}$

$180(3.13)K_2 = 324-180 = 144$ $\quad \boxed{K_2 = 0.72}$

(b) $A = \dfrac{41.4K_1}{180+180K_1K_2} = 0.23$

$s^2+2\zeta\omega_n s+\omega_n^2 = s^2+1.414\omega_n s+\omega_n^2$

$= s^2+36s+180+180K_1K_2$

from forced response

$41.4K_1 = 0.23(180+180K_1K_2)$

$\omega_n = \sqrt{180+180K_1K_2}$

$2\zeta\omega_n = 1.414\sqrt{180+180K_1K_2} = 36$

$180+180K_1K_2 = \left(\frac{36}{1.414}\right)^2 = 648.20$

$41.4K_1 = 0.23(648.20) = 149.1$; $\boxed{K_1 = 3.6}$

$(180)(3.6)K_2 = 648.2-180 = 468.2$; $\boxed{K_2 = 0.72}$

2.20 $T(s) = \dfrac{\left(\frac{5K_1+K_2}{s}\right)\left(\frac{2}{s+4}\right)}{1+\frac{2(5K_1+K_2)}{s(s+4)}} = \dfrac{2sK_1+2K_2}{s^2+(4+2K_1)s+2K_2}$

$\omega_n^2 = 2K_2 = 36$ $\quad \boxed{K_2 = 18}$

$2\zeta\omega_n = 4+2K_1$

$2(0.7)(6) = 4+2K_1$ $\quad 4+2K_1 = 8.4$ $\quad 2K_1 = 4.4$ $\quad \boxed{K_1 = 2.2}$

2.21 $T_1(s) = \left.\frac{\theta(s)}{R(s)}\right|_{\substack{F(s)=0 \\ \text{Init cond}=0}} = \dfrac{\dfrac{2K}{(s^2+7s+5)(s+6)}}{1+\dfrac{2K}{(s^2+7s+5)(s+6)}}$

$$= \frac{2K}{s^3+6.7s^2+4.7s+3+2K}$$

$$T_2(s) = \left.\frac{\theta(s)}{F(s)}\right|_{\substack{R(s)=0 \\ \text{Init cond}=0}} = \frac{\dfrac{2}{s^2+7s+5}}{1+\dfrac{2K}{(s^2+7s+5)(s+6)}}$$

$$= \frac{2(s+6)}{s^3+6.7s^2+4.7s+3+2K}$$

$P(s) = s^3+6.7s^2+4.7s+3+2K$

Using Problem 2.10, $q_0 = 3+2K$, $q_1 = 4.7$, $q_2 = 6.7$

$q_0 > 0$ so $3+2K > 0$ $\quad 2K > -3$ $\quad K > -1.5$

$q_1 q_2 > q_0$ so $(4.7)(6.7) > 3+2K$

$3+2K < 31.49$ $\quad 2K < 28.49$ $\quad K < 14.25$

Stable for $\boxed{-1.5 < K < 14.25}$

2.22

In general (for any K_1, K_2, a_1, a_2, a_3)

$$T(s) = \frac{\frac{K_1}{(s+a_1)} \frac{a_3}{s^2+a_2 s+a_3}}{1+\left(\frac{K_1}{s+a_1}\right)\left(\frac{a_3}{s^2+a_2 s+a_3}\right)\left(\frac{K_2}{s+a_4}\right)}$$

$$= \frac{K_1 a_3 (s+a_4)}{s^4 + s^3\underbrace{(a_1+a_2+a_4)}_{\alpha_3} + s^2\underbrace{(a_1a_4+a_1a_2+a_2a_4+a_3)}_{\alpha_2} + s\underbrace{(a_1a_2a_4+a_1a_3+a_3a_4)}_{\alpha_1} + \underbrace{a_1a_3a_4+K_1K_2a_3}_{\alpha_0}}$$

Next, $\alpha_0, \alpha_1, \alpha_2, \alpha_3$ can be evaluated for some set of K_1, K_2, a_1, a_2, a_3 and the Routh array is formed

s^4	1	α_2	α_0
s^3	α_3	α_1	
s^2	β_1	α_0	
s^1	γ_1		
s^0	α_0		

$$\beta_1 = \frac{\alpha_2\alpha_3 - \alpha_1}{\alpha_3} \qquad \gamma_1 = \frac{\alpha_1\beta_1 - \alpha_0\alpha_3}{\beta_1}$$

2.23

$$T(s) = \frac{\frac{100K}{(s+10)(s+100)}}{1+\frac{100K(10^{-3})(10+s)}{(s+10)(s+100)}10} = \frac{100K}{s^2+(110+.01K)+1000+.1K}$$

For $K>0$, coefficients >0 so second order system is stable

2.24

$$T(s) = \frac{\frac{10^4}{s(s+2)(s+10)}}{1 + \frac{10^4 \, 10^{-4} K}{s(s+2)(s+10)}} = \frac{10^4}{s^3 + 12s^2 + 20s + K}$$

For Problem 2.10, $a_0 = K$, $a_1 = 20$, $a_2 = 12$. For $K > 0$ $a_0, a_1, a_2 > 0$ so for stability

$a_1 a_2 > a_0$ $\quad (20)(12) > K$ $\quad K < 240$

Stable for $\boxed{0 < K < 240}$

2.25

$$T(s) = \frac{\frac{K(s+8)}{s(s+1)(s^2+2s+1)}}{1 + \frac{K(s+8)}{s(s+1)(s^2+2s+1)}} = \frac{K(s+8)}{s^4 + 3s^3 + 3s^2 + (1+K)s + 8K}$$

s^4	1	3	$8K$
s^3	3	$1+K$	
s^2	$\frac{8-K}{3}$	$8K$	
s^1	$\frac{3(-K^2-65K+8)}{8-K}$		
s^0	$8K$		

For marg. stab. a row of zeros is needed. Here, we need

$-K^2 - 65K + 8 = 0$

$-(K^2 + 65K - 8) = 0$

$K = \frac{-65 \pm \sqrt{65^2 + 32}}{2}$

$= -65.12, \; 0.13$

Choose value for $K > 0$ (so s^0 row is > 0).

$\boxed{K = 0.13}$

$E(s) = \frac{8-K}{3} s^2 + 8K = 2.625 s^2 + 1.04$

roots $\pm j\sqrt{\frac{1.04}{2.62}} = \boxed{\pm j \, 0.63}$

~~2.25~~ SO

2.26

$$T(s) = \frac{\frac{0.1A\,K_p}{S(0.07S+1)}}{1+\frac{0.1A\,K_p}{S(0.07S+1)}} = \frac{0.1AK_p}{.07S^2+S+0.1AK_p}$$

$$= \frac{0.1AK_p}{.07\left[S^2+\frac{1}{.07}S+\frac{0.1AK_p}{0.7}\right]} \qquad \text{Since } K_p = 3.18$$

$$T(s) = \frac{4.54A}{S^2+14.29S+4.54A} = \frac{4.54A}{S^2+2\zeta\omega_n S+\omega_n^2}$$

$$= \frac{4.54A}{S^2+0.2\omega_n S+\omega_n^2}$$

$0.2\omega_n = 14.29 \qquad \omega_n = 71.45$

$\omega_n^2 = 4.54A \qquad 5105 = 4.54A \qquad \boxed{A = 1124}$

2.27

$$T(s) = \frac{\frac{K}{(S+1)S(S+5)}}{1+\frac{K}{(S+1)(S)(S+5)(S+2)}} = \frac{K(S+2)}{\underbrace{S^4+8S^3+17S^2+10S+K}_{P(s)}}$$

(a)

S^4	1	17	K
S^3	8	10	
S^2	15.75	K	
S^1	$\frac{157.5-8K}{15.75}$		
S^0	K		

For stability

$K > 0$

$157.5 - 8K > 0$

$8K < 157.5$

$K < 19.69$

Stable for $\boxed{0 < K < 19.69}$

(b) There is a row of zeros for $K = 0$ (root at $S = 0$)

(c) and $K = 19.69$ for which $E(s) = 15.75S^2 + K$

$E(s) = 0 = 15.75S^2 + 19.69$ for $S = \pm j\sqrt{\frac{19.69}{15.75}} = \boxed{\pm j\,1.12}$

CHAPTER 3

3.1

(a) $T(s) = \dfrac{\frac{7}{s+7}}{1+\frac{7}{s+7}}$

$= \dfrac{7}{s+14}$

Rel stab = $\boxed{14}$

(b) Use Masonis rule

$P_1 = \frac{1}{2s} \quad P_2 = \frac{7}{s^2}$

$L_1 = -\frac{4}{s} \quad L_2 = -\frac{20}{s^2}$

$\Delta = 1 - (L_1 + L_2)$

$= 1 + \frac{4}{s} + \frac{20}{s^2}$

$= \dfrac{s^2+4s+20}{s^2}$

$\Delta_1 = 1 \quad \Delta_2 = 1$

$T(s) = \dfrac{P_1\Delta_1 + P_2\Delta_2}{\Delta}$

$= \dfrac{\frac{1}{2s} + \frac{7}{s^2}}{\frac{s^2+4s+20}{s^2}}$

$= \dfrac{\frac{1}{2}s+7}{s^2+4s+20}$

$= \dfrac{0.5(s+14)}{(s+2+j4)(s+2-j4)}$

Rel. stab = $\boxed{2}$

(c) $T(s) = \dfrac{\frac{15}{s+2}}{1+\frac{15}{s+2}\frac{2}{s+2}}$

$= \dfrac{15(s+2)}{(s+2)^2+30}$

$= \dfrac{15(s+2)}{s^2+4s+34}$

roots $\dfrac{-4 \pm j\sqrt{16-136}}{2}$

$= -2 \pm j\,5.48$

Rel. Stab. = $\boxed{2}$

3.2

$T(s) = \dfrac{\frac{k(s-2)}{s^2+3s+7}}{1+\frac{k(s-2)}{s^2+3s+7}}$

$= \dfrac{k(s-2)}{s^2+(3+k)s+7-2k}$

Using the charact. poly.

$s^2+(3+k)s+7-2k$

$= (s+\sigma+j\omega)(s+\sigma-j\omega)$

$= s^2+2\sigma s+\sigma^2+\omega^2$

$= s^2+2\zeta\omega_n s+\omega_n^2$

3.2 continued

For rel. stab. ≥ 2

$2\sigma \geq 4$

$3+k \geq 4$

$k \geq 1$

Choose $k=1$

Since $\omega_n = \sqrt{7-2k}$

$= \sqrt{5}$

$2\zeta\omega_n = 2\sigma = 4$

$\zeta = \frac{4}{2\omega_n} = \frac{2}{\omega_n} = \frac{2}{\sqrt{5}}$

$= 0.89 > 0.5$

So $\boxed{k=1}$ works

3.3 For each problem

$y(0) = \lim_{s\to\infty} sY(s)$

So take only the highest powers of s in each numerator and denom.

(a) $y(0) = \lim_{s\to\infty} s\left(\frac{-5s^3}{s^3}\right)$

$= \boxed{\infty}$

(b) $y(0) = \lim_{s\to\infty} s\left(\frac{2s+0}{s^2}\right)$

$= \lim_{s\to\infty} 2 = \boxed{2}$

(c) $y(0) = \lim_{s\to\infty} s\left(\frac{4s^2}{11s^4}\right)$

$= \boxed{0}$

(d) $y(0) = \lim_{s\to\infty} s\left(\frac{4s^2}{5s}\right)$

$= \boxed{0}$

3.4 For each problem,

$y(\infty) = \lim_{s\to 0} sY(s)$

So take only the lowest powers of s in each numerator and denom., if all poles of $sY(s)$ are LHP.

(a) Using Prob 2.10

$a_0 = 2, a_1 = 8, a_2 = 7$

$a_0, a_1, a_2 > 0$

$a_1 a_2 = 56 > a_0 = 2$

(Stable) So exists

$y(\infty) = \lim_{s\to 0} s\left(\frac{7}{2}\right) = \boxed{0}$

(b), (c), (d) use Routh

(b)

s^4	1	4	18
s^3	2	9	
s^2	-0.5	18	
s^1	81		
s^0	18		

3.4(b) continued
unstable so FV does not exist

(c)

s^4	4	2	9
s^3	6	2	
s^2	2/3	9	
s^1	-79		
s^0	9		

Unstable so FV does not exist

(d) $SY(s) = \dfrac{10}{s^4+3s^3+7s^2+4s+10}$

s^4	1	7	4
s^3	3	4	
s^2	17/3	4	
s^1	1.88		
s^0	4		

Stable so FV exists.

$y(\infty) = \lim_{s\to 0} s\left(\frac{10}{4s}\right) = \boxed{\frac{10}{4} = 2.5}$

3.5

(a) Unity feedback and stable due to positive coeff in $T(s)$.

$K_0 = G_E(0) = \frac{12}{10} = 1.2$

$e_{STEP}(\infty) = \frac{1}{1+K_0} = \frac{1}{1+1.2}$

$= \boxed{0.45}$

(b) $T(s) = \dfrac{\frac{1}{s^2+9}}{1+\frac{3}{s+4}\frac{1}{s^2+9}}$

$= \dfrac{s+4}{(s+4)(s^2+9)+3}$

$= \dfrac{s+4}{s^3+4s^2+9s+39}$

From Prob 2.10

$a_0, a_1, a_2 > 0$

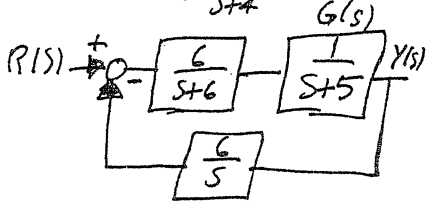
$a_0 = 3, a_1 = 9, a_2 = 4$

$a_1 a_2 = 36 < a_0 = 39$

unstable so FV does not exists.

(c) Simplify right feed. loop

$G(s) = \dfrac{\frac{1}{s+4}}{1+\frac{1}{s+4}} = \frac{1}{s+5}$



3.5 (d) continued

$$T(s) = \frac{\frac{6}{s+6}\frac{1}{s+5}}{1+\frac{6}{5}\frac{6}{s+6}\frac{1}{s+5}}$$

$$= \frac{6s}{s(s+6)(s+5)+36}$$

$$= \frac{6s}{s^3+11s^2+30s+36}$$

$a_0=36 \quad a_1=30 \quad a_2=11$

$a_0, a_1, a_2 > 0$

$a_1 a_2 = 330 > a_0 = 36$

Stable so $\exists v$ exists

$$T_E(s) = 1 - \frac{6s}{s^3+11s^2+30s+36}$$

$$= \frac{s^3+11s^2+24s+36}{s^3+11s^2+30s+36}$$

$e_{STEP}(\infty) = T_E(0) = \boxed{1}$

3.6

(a) Type given by num. of $T_E(s)$ or denom of $G_E(s)$

$$T_E(s) = 1 - \frac{7}{s^2+4s+7}$$

$$= \frac{s(s+4)}{s^2+4s+7}$$

Stable due to pos. Second order coeff

Type 1

$e_{STEP}(\infty) = T_E(0) = \boxed{0}$

$$e_{RAMP}(\infty) = \lim_{s\to 0} \frac{T_E(s)}{s} = \lim_{s\to 0} \frac{s+4}{s^2+4s+7} = \boxed{\frac{4}{7}}$$

$$e_{PAR}(\infty) = \lim_{s\to 0} \frac{T_E(s)}{s^2} = \lim_{s\to 0} \frac{s+4}{s(s^2+4s+7)} = \boxed{\infty}$$

(b) Stable due to second order pos. coeff. as in (a).

$$T_E(s) = 1 - \frac{4s+7}{s^2+4s+7}$$

$$= \frac{s^2}{s^2+4s+7}$$

Type 2

$e_{STEP}(\infty) = e_{RAMP}(\infty) = 0$

$$e_{PAR}(\infty) = \lim_{s\to 0} \frac{T_E(s)}{s^2} = \lim_{s\to 0} \frac{1}{s^2+4s+7} = \boxed{\frac{1}{7}}$$

(c) Stable for pos second order coeff.

$$T_E(s) = \frac{s^2+4s+7}{s^2+4s+12}$$

Type 0

$e_{STEP}(\infty) = T_E(0) = \boxed{\frac{7}{12}}$

$e_{RAMP}(\infty) = e_{PAR}(\infty) = \boxed{\infty}$

3.6 (d)

$$T_E(s) = 1 - \frac{s+4}{s^3+4s^2+5s+4}$$

$$= \frac{s^2(s+4)}{s^3+4s^2+5s+4}$$

type [2]

$a_0 = 4 \quad a_1 = 1 \quad a_2 = 4$

$a_0, a_1, a_2 > 0$ but

$a_1 a_2 = a_0$

$4 = 4$

System is marg. stable so no s.s. values

(e)

$$T_E(s) = \frac{s^2(s^2+3s+8)}{s^4+3s^3+10s^2+5s+4}$$

type [2]

Use the Routh array

s^4	1	10	4
s^3	3	1	
s^2	$\frac{29}{3}$	4	
s^1	-0.14		
s^0	4		

System is unstable so FVs do not exist.

3.7

For each of these

$K_0 = G_E(0)$

$K_1 = \lim_{s \to 0} s\,G_E(s)$

$K_2 = \lim_{s \to 0} s^2 G_E(s)$

use lowest powers of s

The steady state errors depend on the stability.

(a) $K_0 =$ [∞] type [1]

$K_1 = \lim_{s \to 0} s\left(\frac{7}{7s}\right) =$ [1]

$K_2 = \lim_{s \to 0} s^2\left(\frac{7}{7s}\right) =$ [0]

$$T(s) = \frac{\frac{-3s+7}{s^4+3s^3+2s^2+7s}}{1 + \frac{-3s+7}{s^4+3s^3+2s^2+7s}}$$

$$= \frac{-3s+7}{s^4+3s^3+2s^2+4s+7}$$

Use Routh Array

s^4	1	2	7
s^3	3	4	
s^2	$\frac{2}{3}$	7	
s^1	-27.5		
s^0	7		

Unstable

3.7 (b)

type [1]

$K_0 = G(s)|_{s=0} = \boxed{\infty}$

$K_1 = \lim_{s\to 0} sG(s)$

$= \lim_{s\to 0} s\left(\frac{2}{7s}\right) = \boxed{\frac{2}{7}}$

$K_2 = \lim_{s\to 0} s^2\left(\frac{2}{7s}\right) = \boxed{0}$

$$T(s) = \frac{\frac{2}{s^3+2s^2+7s}}{1+\frac{2}{s^3+2s^2+7s}}$$

$$= \frac{2}{s^3+2s^2+7s+2}$$

$a_0=2,\ a_1=7,\ a_2=2$

$a_1 a_2 = 14 > a_0 = 2$

So the system is stable

$e_{STEP}(\infty) = \frac{1}{1+K_0} = \boxed{0}$

$e_{RAMP}(\infty) = \frac{1}{K_1} = \boxed{\frac{7}{2}}$

$e_{PAR}(\infty) = \frac{1}{K_2} = \boxed{\infty}$

(c) Type [4]

$K_0 = K_1 = K_2 = \infty$

$$T(s) = \frac{\frac{6s^2-3s+6}{s^4}}{1+\frac{6s^2-3s+6}{s^4}}$$

$$T(s) = \frac{6s^2-3s+6}{s^4+6s^2-3s+6}$$

Due to 0 coef and negative coef., the system is not stable.

(d) type [3]

$K_0 = K_1 = K_2 = 0$

$$T(s) = \frac{\frac{4s^2+9s+11}{s^4+8s^3}}{1+\frac{4s^2+9s+11}{s^4+8s^3}}$$

$$= \frac{4s^2+9s+11}{s^4+8s^3+4s^2+9s+11}$$

Use the Routh array

s^4	1	4	11
s^3	8	9	
s^2	$\frac{23}{8}$	11	
s^1	-21.6		
s^0	11		

The system is unstable due to the left column sign changes

3.8 Let $G(s) = \frac{N(s)}{D(s)}$

$$T(s) = \frac{\frac{N(s)}{D(s)}\frac{1}{s+2}}{1+s\frac{N(s)}{D(s)}\frac{1}{s+2}\frac{1}{s+4}}$$

3.8 (continued)

$$T(s) = \frac{(s+4)N(s)}{(s+2)(s+4)D(s)+5N(s)}$$

$$= \frac{(s+4)N(s)}{(s^2+6s+8)D(s)+5N(s)}$$

Let $D(s) = s+1$

$$T(s) = \frac{(s+4)N(s)}{s^3+7s^2+14s+8+5N(s)}$$

$$T_E(s) = 1 - T(s)$$

$$= \frac{s^3+7s^2+14s+8-4N(s)}{s^3+7s^2+14s+8+5N(s)}$$

For type 2 behavior

$$14s+8-4N(s)=0$$

$$N(s) = \frac{14s+8}{4} = 3.5s+2$$

$$\boxed{G(s) = \frac{3.5s+2}{s+1}}$$

$$T_E(s) = \frac{s^2(s+7)}{s^3+10.5s^2+16s+8}$$

$a_0 = 8, a_1 = 16, a_2 = 10.5$

$a_0, a_1, a_2 > 0$

$a_1 a_2 = 168 > a_0 = 8$

Stable

3.9 (a) $S_a = \frac{a}{T}\frac{\partial T}{\partial a}$

$$\frac{\partial T}{\partial a} = \frac{(s+a)7 - 7a\overbrace{\frac{\partial}{\partial a}(s+a)}^{1}}{(s+a)^2}$$

$$= \frac{7s+7a-7a}{(s+a)^2} = \frac{7s}{(s+a)^2}$$

$$S_a = \frac{a}{\frac{7a}{s+a}}\frac{7s}{(s+a)^2} = \frac{s}{s+a}$$

$$= \boxed{\frac{s}{s+3}}$$

(b) $\frac{\partial T}{\partial a} = \frac{\partial}{\partial a} 6(s^2+9s+2)^{-1}$

$$= -6(s^2+9s+2)^{-2}\underbrace{\frac{\partial}{\partial a}(s^2+9s+2)}_{s}$$

$$= \frac{-6s}{(s^2+9s+2)^2}$$

$$S_a = \frac{a}{\frac{6}{s^2+9s+2}}\frac{-6s}{(s^2+9s+2)^2}$$

$$= -\frac{as}{s^2+9s+2}$$

$$= \boxed{-\frac{2s}{s^2+2s+2}}$$

(c) $\frac{\partial T}{\partial a} = \frac{\left[(s^2+9s+10)\overbrace{\frac{\partial}{\partial a}(9s+10)}^{s} - (9s+10)\overbrace{\frac{\partial}{\partial a}(s^2+9s+10)}^{s}\right]}{(s^2+9s+10)^2}$

$$= \frac{(s^2+9s+10)s - (9s+10)s}{(s^2+9s+10)^2}$$

$$= \frac{s^3}{(s^2+9s+10)^2}$$

$$S_a = \frac{a}{\left(\frac{9s+10}{s^2+9s+10}\right)}\frac{s^3}{(s^2+9s+10)^2}$$

$$= \frac{9s^3}{(9s+10)(s^2+9s+10)} = \frac{4s^3}{(4s+10)(s^2+4s+10)}$$

3.10

Let $G(s) = \frac{N(s)}{D(s)}$

$$T(s) = \frac{1}{s+6} + \frac{N(s)\,s}{(s+9)D(s)}$$

$$= \frac{(s+9)D(s) + sN(s)(s+6)}{(s+6)(s+9)\,D(s)}$$

$$= \frac{(s+9)D(s) + s(s+6)N(s)}{(s^2+15s+54)\,D(s)}$$

Let $D(s)(s+9) = -s(s+6)N(s)$

$$\boxed{\frac{N(s)}{D(s)} = -\frac{s+9}{s(s+6)}}$$

3.11

$$T(s) = \frac{\frac{4}{s}}{1+4\frac{k}{s}} = \frac{4}{s+4k}$$

$$T_E(s) = 1 - T(s) = \frac{s+4k-4}{s+4}$$

$$E(s) = \frac{T_E(s)}{s}$$

For $S = \int_0^\infty e^2(t)\,dt$ to be other than infinite, then $e_{STEP}(\infty) = 0$ and the system must be type 1. Thus

$4k - 4 = 0$ $\boxed{k = 1}$

3.12

Table 3.4 may be used to minimize J

(a) $N_0 = 2$, $N_1 = k$

$D_0 = 10$, $D_1 = 7$, $D_2 = 1$

$$J = \frac{N_1^2 D_0 + N_0^2 D_2}{D_0 D_1 D_2}$$

$$= \frac{10k^2 + 4}{140}$$

$$\frac{\partial J}{\partial k} = 0 = \frac{20k}{140}$$

$\boxed{k = 0}$

(b) $N_0 = k$, $N_1 = -6$

$D_0 = 6$, $D_1 = 5$, $D_2 = 1$

$$J = \frac{N_1^2 D_0 + N_0^2 D_2}{D_0 D_1 D_2}$$

$$= \frac{36(6) + k^2}{60}$$

$$\frac{\partial J}{\partial k} = 0 = \frac{2k}{60}$$

$\boxed{k = 0}$

3.13

$$T_D(s) = \frac{1}{1 + \frac{9K}{s+5}} = \frac{s+5}{s+5+9K}$$

$$Y(s) = T_D(s) D(s) \frac{1}{s} = \frac{s+5}{s(s+5+9K)}$$

$$y(\infty) = \lim_{s\to 0} sY(s) = \lim_{s\to 0}\left(\frac{s+5}{s+5+9K}\right) = \frac{5}{5+9K} < 0.1$$

$$5 < 0.1(5+9K)$$

$$50 < 5 + 9K$$

$$9K > 45 \quad \boxed{K > 5}$$

3.14

For the velocity control loop

$$G_V(s) = \frac{\frac{0.0211}{s}\frac{10}{s+.4}}{1 + \frac{0.0211}{s}\frac{10}{s+0.4}3.24(s+0.309)} = \frac{(0.0211)(10)}{s(s+0.4) + 0.684(s+0.309)} = \frac{0.211}{s^2 + 1.085 + 0.211}$$

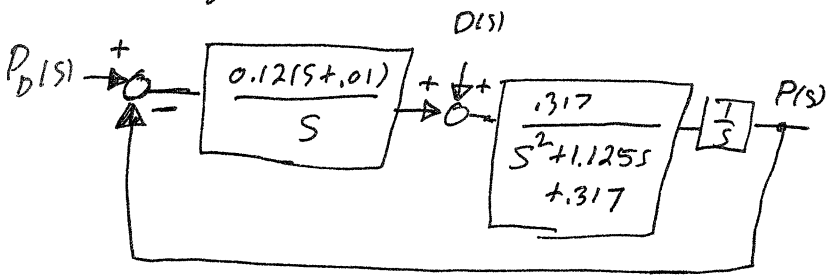
The forward path is

$$G_E(s) = 0.12\frac{(s+0.01)}{s} G_V(s) \frac{1}{s}$$

System is type 2, so $\boxed{e_{step}(\infty) = 0}$

3.15

The system is



$$T_D(s) = \frac{P(s)}{D(s)}\bigg|_{P_D(s)=0} = \frac{\frac{0.317}{s(s^2+1.125s+0.317)}}{1 + \frac{0.12(s+.01)(.317)}{s^2(s^2+1.125s+0.317)}} = \frac{0.317\,s}{s^4+1.125s^3+0.317s^2+0.038s+0.00038}$$

$$T_E(s) = 1 - T(s) = \frac{s^4+1.125s^3+0.317s^2-0.279s+0.00038}{s^4+1.125s^2+0.317s^2+0.038s+0.00038}$$

$$e_{STEP}(\infty) = T_E(0) = \boxed{1}$$

3.16

$$T(s) = \frac{2\pi K}{s^2 + 9s + 2\pi K} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

For $\zeta = 1$

$$T(s) = \frac{\omega_n^2}{s^2 + 2\omega_n s + \omega_n^2}$$

For $\boxed{T_r = 0.1}$ and $\zeta = 1$

$$\omega_n T_r = 3.25$$

$$\omega_n = \frac{3.25}{0.1} = 32.5$$

$$\boxed{a = 2 \times 32.5 = 65}$$

$$2\pi K = \omega_n^2$$

$$K = \frac{\omega_n^2}{2\pi} = \frac{32.5^2}{2\pi}$$

$$= \boxed{168}$$

3.17

Since the design is

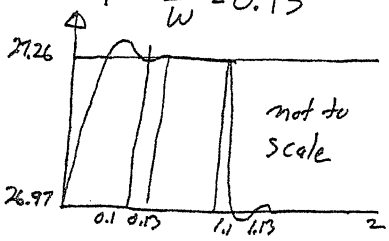
$\zeta = 0.5$, $\omega_n = 53$

$T_s = 0.1$

$$\omega = \omega_n\sqrt{1-\zeta^2}$$

$$= 53\sqrt{1-.5^2} = 45.9$$

$$T = \frac{2\pi}{\omega} = 0.13$$



3.18

$$T_R(s) = \left.\frac{V(s)}{R(s)}\right|_{L(s)=0}$$

$$= \frac{K\left(\frac{20}{s+12}\right)\left(\frac{1}{s+0.5}\right)\left(\frac{6}{s+2}\right)}{1 + 8K\left(\frac{20}{s+12}\right)\left(\frac{1}{s+0.5}\right)\left(\frac{6}{s+2}\right)}$$

$$= \frac{120K}{s^3 + 14.5s^2 + 31s + 12 + 960K}$$

$$T_L(s) = \left.\frac{V(s)}{L(s)}\right|_{R(s)=0}$$

$$= \frac{\frac{6}{s+2}}{1 + 8K\left(\frac{20}{s+12}\right)\left(\frac{1}{s+0.5}\right)\left(\frac{6}{s+2}\right)}$$

$$= \frac{6(s+12)(s+0.5)}{s^3 + 14.5s^2 + 31s + 12 + 960K}$$

For stability

$a_0 = 12 + 960K$, $a_1 = 31$, $a_2 = 14.5$

$a_0 = 12 + 960K > 0$ $\quad K > -\frac{12}{960}$

$K > -0.125$

$a_1 a_2 = (31)(14.5) > 12 + 960K$

$960K < 437.5$

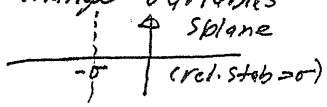
$K < 0.456$

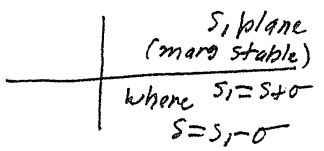
Stable

$-0.125 < K < 0.456$

For relative stability,

change variables





For the s plane, the charact. poly.

$$P(s) = s^3 + 14.5s^2 + 31s + 12 + 960K$$

with change of var

$$P(s_1) = (s-\sigma)^3 + 14.5(s-\sigma)^2 + 31(s-\sigma) + 12 + 960K$$

$$= s_1^3 + s_1^2(14.5 - 3\sigma) + s_1(3\sigma^2 - 29\sigma + 31) + (12 + 960K + (-\sigma^3 + 14.5\sigma^2 - 31\sigma))$$

a_2' a_1' a_0'

The $P(s_1)$ poly must indicate marginal stab. if $P(s)$ poly. has rel. stab. $= \sigma$

Therefore

$$a_0' = 0$$

$$\boxed{K = \frac{\sigma^3 - 14.5\sigma^2 + 31\sigma - 12}{960}}$$

For marginal stability (not unst.)

$a_2' \geq 0 \quad a_1' \geq 0 \quad a_0' = 0$

So $a_2' = 14.5 - 3\sigma \geq 0$

$\sigma \leq 4.83$

$a_1' = 3(\sigma - 8.44)(\sigma - 1.22) \geq 0$ for $\sigma \leq 4.83$ then

$\sigma \leq 8.44$ and $\sigma \leq 1.22$

So $\sigma_{max} = 1.22$

When $K = 0.006306$

At max rel stability in s_1 plane, roots are for

$$P(s_1) = s_1^3 + 10.8s_1^2$$

roots $= 0, 0, -10.8$

In s plane subtract 1.22

so roots are

$-1.22, -1.22, -12$ and

$$\boxed{y_{nat}(t) = K_1 t e^{-1.22t} + K_2 e^{-1.22t} + K_3 e^{-12t}}$$

Design for max. stability for safety.

$$T_{E_R}(s) = 1 - T_R(s) = \frac{s^3 + 14.5s^2 + 31s + 12 + 840K}{s^3 + 14.5s^2 + 31s + 12 + 960}$$

type 0 so steb, ramp, and parab. errors are non zero

$$T_{E_L}(s) = 1 - T_L(s) = \frac{s^3 + 8.55s^2 - 44s + 960K - 24}{s^3 + 14.5s^2 + 31s + 12 + 960}$$

type 0 so steb, ramp, and parab. errors are non zero.

3.19

$$T(s)=\frac{\left(\frac{k_1s^2+k_2s+k_3}{s}\right)\left(\frac{4}{s+1}\right)\left(\frac{0.5}{s^2+s+3}\right)}{1+\left(\frac{k_1s^2+k_2s+k_3}{s}\right)\left(\frac{4}{s+1}\right)\left(\quad\right)\left(\frac{0.2}{s+6}\right)}=\frac{(k_1s^2+k_2s+k_3)(2)(s+6)}{(s+6)s(s+1)(s^2+s+3)+(k_1s^2+k_2s+k_3)0.4}$$

$$=\frac{2(s+6)(k_1s^2+k_2s+k_3)}{s^5+8s^4+16s^3+(27+0.4k_1)s^2+(18+0.4k_2)s+0.4k_3}$$

As in Prob 3.18, shift to S_1 plane using $S_1=(S-0.5)$. Splane system has rel. stab. of 0.5 if S_1 system is marginally stable. The next step is to get $P(S_1)$ from denom of $T(S)$,

$$P(S_1)=(S_1-0.5)^5+8(S_1-0.5)^4+16(S_1-0.5)^3+27(S_1-0.5)^2$$
$$+18(S_1-0.5)+0.4k_1(S-0.5)^2+0.4k_2(S_1-0.5)+0.4k_3$$

$$=S_1^5+5.5S_1^4+2.5S_1^3+S_1^2\underbrace{(13.75+0.4k_1)}_{a_2'}$$
$$+S_1\underbrace{(-0.6875-0.4k_1+0.4k_2)}_{a_1'}+\underbrace{(-3.7813+0.1k_1-0.2k_2+0.4k_3)}_{a_0'}$$

If $a_2'=a_1'=a_0'$ then $P(S_1)$ is marg. stable with 3 roots at $S_1=0$ so $P(S)$ system has 3 roots at $S=-0.5$.

$a_2'=13.75+0.4k_1=0$ $\boxed{k_1=-34.375}$

$a_1'=-0.6875-0.4k_1+0.4k_2=0$ $k_2=\frac{0.6875+0.4k_1}{0.4}$

$a_0'=0=-3.7813+0.1k_1-0.2k_2+0.4k_3=\boxed{-32.656}$

$\boxed{k_3=1.719}$

$P(S_1)=S_1^5+5.5S_1^4+2.5S_1^3$ roots are $0,0,0,-5,-0.5$

Roots of $P(S)$ must be $-0.5,-0.5,-0.5,-5.5,-1$

$P(S)=S^5+8S^4+16S^3+13.25S^2+4.9375S+0.6875$

3.20

(a) This block represents system dynamics, the integration of water flow rate to obtain total weight. Thus, this block is fixed and not adjustable, whereas an amplifier would be adjustable

(b) $$T(s)=\frac{\frac{1000}{s}\frac{10^4}{s^3+3\times10^3 s+14\times10^6}}{1+\frac{1000}{s}\frac{10^4}{s^3+3\times10^3 s+14\times10^6}}=\frac{10^4}{s^3+3\times10^3 s^2+14\times10^6 s+10^7}$$

$a_0=10^7,\ a_1=14\times10^6,\ a_2=3\times10^3$

$a_0,a_1,a_2>0 \qquad a_1a_2=42\times10^9>a_0=10^7$ so stable.

(c)

$$y(\infty)=\lim_{s\to0} sY(s)=\lim_{s\to0} sT(s)\frac{1}{s}=T(0)=\boxed{1}$$

(d) $$T_w(s)=\frac{\frac{10^4}{s^2+3\times10^3 s+14\times10^6}}{1+\frac{1000}{s}\frac{10^4}{s^2+3\times10^3 s+14\times10^6}}=\frac{10^4 s}{s^3+3\times10^3 s^2+14\times10^6 s+10^7}$$

$$y(\infty)=\lim_{s\to0} sY(s)=\lim_{s\to0} sT_w(s)\frac{1}{s}=T_w(0)=\boxed{0}$$

3.21

$$T(s)=\frac{\frac{(k_1s+k_2)}{s+1000}\frac{k_3}{s+k_3}\frac{50}{s^2+10s+50}}{1+\frac{(k_1s+k_2)}{s+1000}\frac{k_3}{s+k_3}\frac{50}{s^2+10s+50}}$$

$$=\frac{50k_3(k_1s+k_2)}{(s+1000)(s+k_3)(s^2+10s+50)+(k_1s+k_2)50k_3}$$

3.21 (continued) Let $k_3 = 0.2$

$$T(s) = \frac{10(k_1 s + k_2)}{s^4 + 1010.2s^3 + 10252s^2 + (52010 + 10k_1)s + 10000 + 10k_2}$$

As in problem 3.18, change variables on the charact. polynomial where $s = s_1 - \sigma$, σ is the rel. stab. and the system of $P(s_1)$ is marg. stable causing the system of $P(s)$ to have rel. stab. = σ. Here, let $\sigma = 3$

$$P(s_1) = (s_1-3)^4 + 1010.2(s_1-3)^3 + 1025(s_1-3)^2 + 52010(s_1-3) + 10000 + 10k_1(s_1-3) + 10k_2$$

$$= s_1^4 + 998.2s_1^3 + 1214.2s_1^2 + s_1\underbrace{(17665 + 10k_1)}_{a_1'} + \underbrace{(10k_2 - 80956 - 30k_1)}_{a_0'}$$

For $a_1' = 0 = 17665 + 10k_1$ $\quad k_1 = -1766.5$

For $a_0' = 0 = 10k_2 - 80956 - 30(-1766.5)$

$$k_2 = \frac{80956 + 30(-1766.5)}{10} = 2796.1$$

$P(s_1) = s_1^4 + 998.2s_1^3 + 1214.2s_1^2$ roots are $0, 0, -998, -1.22$

Then in Splane, roots are found by subtracting 3

roots are $-3, -3, -1000, -4.2$

$$P(s) = s^4 + 1010.2s^3 + 10252s^2 + 34345s + 37961$$

$$T(s) = \frac{17665s + 27961}{P(s)} \qquad T_E(s) = 1 - T(s)$$

$$T_E(s) = \frac{s^4 + 1010.2s^3 + 10252s^2 + 16680s + 10000}{s^4 + 1010.2s^3 + 10252s^2 + 34345s + 37961}$$

$$e_{step(\infty)} = T_E(0) = \frac{10000}{37961} = 0.263 = 26.3\%$$

which meets the design goal

3.22

For a first order model

let $y(\infty) = 3.3$. Then

$y(t) = 3.3(1 - e^{-t/\tau})u(t)$

For $t = \tau$

$y(\tau) = 3.3(1 - 0.368)$

$= 3.3(0.632) = 2.1$

From the table $\tau \simeq 0.29$

$y(t) = 3.3(1 - e^{-3.45t})u(t)$

(comparing data

t	actual y	y first order model	y second order model
0	0	0	0
0.15	1.1	1.3	0.69
0.3	2.2	2.1	1.9
0.45	2.8	2.6	2.8
0.6	3.3	2.9	3.3
0.75	3.5	3.1	3.5
0.9	3.5	3.2	3.5
1.05	3.4	3.2	3.4
1.2	3.3	3.2	3.3
1.35	3.3	3.3	3.3

$Y(s) = 3.3\left[\frac{1}{s} - \frac{1}{s+3.45}\right]$

$= 3.3\,\frac{3.45}{s(s+3.45)}$

$= T(s)\,R(s)$

$= T(s)\left(\frac{6.3}{s}\right)$

$T(s) = \frac{Y(s)}{R(s)} = \frac{3.3}{6.3}\,\frac{3.45}{s+3.45}$

$= \boxed{\frac{1.807}{s+3.45}}$

For a second order model

$y(\infty) = 3.3$.

$\%\text{overshoot} = \frac{0.2}{3.3} \times 100 = 6\%$

From Fig 2.10(b), $\xi \simeq 0.65$

Peak time is at about

$\frac{1}{2}$ period $\simeq 0.8$ sec

$T \simeq 1.6$ sec

$w = \frac{2\pi}{T} = \frac{2\pi}{1.6} = 3.93$

$w_n = \frac{w}{\sqrt{1-\xi^2}} = 5.17$

Using Equation (2.4) with

$z = \sqrt{1-\xi^2} = 0.76$

$\xi w_n = 3.36$

$y(t) = 3.3[1 + 1.316e^{-3.36t}\cos(3.93t + 139.5^\circ)]$

$y(t)$ data is shown to the left. Then

$T(s) = \frac{w_n^2}{s^2 + 2\xi w_n s + w_n^2}$

$= \boxed{\frac{26.7}{s^2 + 6.72s + 26.7}}$

3.23

In this problem

$K_0 = G_E(0)$

$K_1 = \lim_{s\to 0} sG_E(s)$

$K_2 = \lim_{s\to 0} s^2 G_E(s)$.

For a stable system

$e_{STEP}(\infty) = \frac{1}{1+K_0}$

$e_{RAMP}(\infty) = \frac{1}{K_1}$

$e_{PAR}(\infty) = \frac{1}{K_2}$

(a) type $\boxed{0}$

$K_0 = G_E(0) = \boxed{\frac{K}{480}}$

$K_1 = K_2 = \boxed{0}$

$e_{STEP}(\infty) = \frac{1}{1+\frac{K}{480}} = \boxed{\frac{480}{480+K}}$

$e_{RAMP}(\infty) = e_{PAR}(\infty) = \boxed{\infty}$

(Stable due to second order positive coef.)

(b) Assume K is selected for stability.

type $\boxed{1}$

$K_0 = \boxed{\infty}$

$K_1 = \lim_{s\to 0} s\,\frac{K}{s(s+6)(s+80)} = \boxed{\frac{K}{480}}$

$K_2 = \boxed{0}$

$e_{STEP}(\infty) = \boxed{0}$

$e_{RAMP}(\infty) = \boxed{\frac{480}{K}}$

$e_{PAR}(\infty) = \boxed{\infty}$

(c) Assume K is selected for stability.

Due to s in numerator of $G_E(s)$, type number is $\boxed{=1}$ so obviously

$K_0 = K_1 = K_2 = \boxed{0}$

$e_{STEP}(\infty) = e_{RAMP}(\infty) = e_{PAR}(\infty) = \boxed{\infty}$

(d) Assume K is adjusted for stability.

Type $\boxed{2}$

$K_0 = K_1 = \infty$

$K_2 = \lim_{s\to 0} s^2 G_E(s) = \frac{K}{\omega_n^2}$

$e_{STEP}(\infty) = e_{RAMP}(\infty) = \boxed{0}$

$e_{PAR}(\infty) = \boxed{\frac{\omega_n^2}{K}}$

(e) System is stable due to second order positive coef.

3.23(e) continued

type 0

$K_0 = G_{E(0)} = \boxed{\frac{100}{\omega_n^2}}$

$K_1 = K_2 = \boxed{0}$

$e_{STEP(\omega)} = \frac{1}{1+\frac{100}{\omega_n^2}}$

$= \boxed{\frac{\omega_n^2}{\omega_n^2+100}}$

$e_{RAMP}(\omega) = e_{PAR}(\omega) = \boxed{\infty}$

(f) type 3

$$T(s) = \frac{\frac{70(s^2+2s+150)}{s^5+2s^4+10s^3}}{1+\frac{70(s^2+2s+150)}{s^5+2s^4+10s^3}}$$

$$= \frac{70(s^2+2s+150)}{s^5+2s^4+10s^3+70s^2+1750s+10500}$$

Use Routh array

s^5	1	10	1750
s^4	2	70	10500
s^3	-25		unstable

3.24

$$T(s) = \frac{K_1 \frac{K}{s(\tau s+1)}}{1+\frac{K(\alpha\tau s+1)}{s(\tau s+1)^2}}$$

$$= \frac{K_1 K(\tau s+1)}{s(\tau s+1)^2 + K(\alpha\tau s+1)}$$

To Get $G_{E(s)}$

$$T(s) = \frac{G_{E(s)}}{1+G_{E(s)}}$$

$$T(s)[1+G_{E(s)}] = G_{E(s)}$$

$$G_{E(s)}[1-T(s)] = T(s)$$

$$G_{E(s)} = \frac{T(s)}{1-T(s)}$$

$$T(s) = \frac{K_1 K\tau s + K_1 K}{s^3\tau^2 + 2\tau s^2 + (1+K\alpha\tau)s + K}$$

$$1-T(s) = s^3\tau^2 + 2\tau s^2 + (1+K\alpha\tau - K_1 K\tau)s + K - K_1 K$$

$$G_{E(s)} = \frac{K_1 K\tau s + K_1 K}{s^3\tau^2 + 2\tau s^2 + (1+K\alpha\tau - K_1 K\tau)s + K - K_1 K}$$

type 0 so

$K_0 = G_{E(0)} = \boxed{\frac{K_1 K}{K - K_1 K}}$

$K_1 = K_2 = \boxed{0}$

3.25

The inner loop (with $T_d(s)=0$)

(a) has transmittance

$$G_1(s)=\frac{\frac{1}{Js}}{1+\frac{K_2}{Js}}=\frac{1}{Js+K_2}$$

$$G_E(s)=A_1\left(\frac{\tau_1 s+1}{\tau_2 s+1}\right)A_2K_1G_1(s)\left(\frac{1}{s}\right)$$

The system is type 1 with

$K_0=\infty$

$$K_1=\lim_{s\to 0} sG_E(s)=\frac{A_1A_2K_1}{K_2}$$

$e_{STEP}(\infty)=0$ so $\boxed{y_{ss}=1}$ for $r(t)=u(t)$

$$e_{RAMP}(\infty)=\frac{1}{K_1}=\frac{K_2}{A_1A_2K_1}=r_{ss}-y_{ss}$$

So $y_{ss}=r_{ss}-e_{RAMP}(\infty)$

$$=\boxed{t-\left(\frac{K_2}{A_1A_2K_1}\right)}$$

(b)

$$T_D(s)=\frac{\frac{1}{Js}\frac{1}{s}}{1+\frac{K_2}{Js}+\frac{1}{Js^2}A_1A_2K_1\left(\frac{\tau_1 s+1}{\tau_2 s+1}\right)}$$

$$T_D(s)=\frac{\tau_2 s+1}{J\tau_2 s^3+s^2(J+K_2\tau_2)+s(K_2+A_1A_2K_1\tau_1)+A_1A_2K_1}$$

$$Y(s)=T_D(s)\frac{T_0}{s}$$

$$y(\infty)=\lim_{s\to 0} sT_D(s)\frac{T_0}{s}=T_D(0)T_0=\boxed{\frac{T_0}{A_1A_2K_1}}$$

3.26

$$Y(s)=T_D(s)\frac{D}{s^2}$$

$$y(\infty)=\lim_{s\to 0} sT_D(s)\frac{D}{s^2}=\lim_{s\to 0}\frac{DT_D(s)}{s}=\boxed{\infty}$$

3.27

$$T_D(s)=\frac{\tau_2 s+1}{J\tau_2 s^3+s^2(J+K_2\tau_2)+s(K_2+A_1\frac{K}{s}K_1\tau_1)+A_1\frac{K}{s}K_1}$$

$$=\frac{s(\tau_2 s+1)}{J\tau_2 s^4+s^3(J+K_2\tau_2)+s^2K_2+sA_1KK_1\tau_1+AK_1K}$$

$$y(\infty)=\lim_{s\to 0}\frac{bT_D(s)}{s}=\boxed{\frac{b}{AK_1K}}$$

3.28

$$T_D(s) = \frac{\tau_2 s + 1}{J\tau_2 s^3 + s^2(J + K_2\tau_2) + s(K_2 + A_1 K_s k_1 \tau_1) + A_1 K_s k_1}$$

$$= \frac{\tau_2 s + 1}{s[J\tau_2 s^2 + s(J + K_2\tau_2 + A_1 K k_1 \tau_1) + K_2 + A_1 K k_1]}$$

$$Y(s) = \frac{b}{s^2} T_D(s)$$

$$y(\infty) = \lim_{s\to 0} s \frac{b}{s^2} T_D(s)$$

$$= \lim_{s\to 0} \frac{b T_D(s)}{s}$$

$$= \boxed{\infty}$$

3.29

(a) type $\boxed{2}$

(b) $K_0 = K_1 = \boxed{\infty}$

$$K_2 = \lim_{s\to 0} s^2 G(s)$$

$$= \boxed{\frac{K s_1}{s_2 s_4}}$$

3.30 (a)

$$T(s) = \frac{\frac{A K_m K_p}{s(\tau s + 1)}}{1 + \frac{A K_m K_p (\tau_1 s + 1)}{s(\tau s + 1)}}$$

$$T(s) = \frac{A K_m K_p}{s(\tau s + 1) + A K_m K_p (\tau_1 s + 1)}$$

$$= \frac{A K_m K_p / \tau}{s^2 + s\left(\frac{1 + A K_m K_p \tau_1}{\tau}\right) + \frac{A K_m K_p}{\tau}}$$

$$= \frac{A K_m K_p / \tau}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\boxed{\omega_n = \sqrt{\frac{A K_m K_p}{\tau}}}$$

$$2\zeta\omega_n = \frac{1 + A K_m K_p \tau_1}{\tau}$$

$$\boxed{\zeta = \left(\frac{1 + A K_m K_p \tau_1}{2\tau}\right)\sqrt{\frac{\tau}{A K_m K_p}}}$$

(b) $T_E(s) = 1 - T(s)$

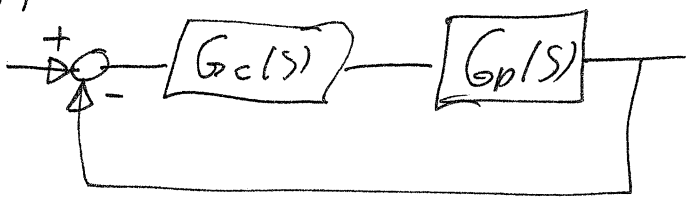
$$= \frac{s\left(s + \frac{1 + A K_m K_p \tau_1}{\tau}\right)}{s^2 + s\left(\frac{1 + A K_m K_p \tau_1}{\tau}\right) + \frac{A K_m K_p}{\tau}}$$

$$e_{RAMP}(\infty) = \lim_{s\to 0} \frac{T_E(s)}{s}$$

$$= \left(\frac{1 + A K_m K_p \tau_1}{\tau}\right)\left(\frac{\tau}{A K_m K_p}\right)$$

$$= \boxed{\frac{1}{A K_m K_p} + \tau_1}$$

3.31



For Ziegler-Nichols, start with $G_c(s) = K_p$.

Select $K_p = K_{po}$ for marginal stability

$$\frac{G_c(s)G_p(s)}{1+G_c(s)G_p(s)} = \frac{K_p \frac{4000}{(s+40)(s+10)^2}}{1+\frac{K_p 4000}{(s+40)(s+10)^2}}$$

$$= \frac{4000K_p}{(s+40)\underbrace{(s+10)^2}_{s^2+20s+100} + 4000K_p} = \frac{4000K_p}{s^3+60s^2+900s+4000+4000K_p}$$

Get Routh array

s^3	1	900
s^2	60	$4000+4000K_p$
s^1	$\frac{54000-4000-4000K_p}{60}$	$= \frac{50000-4000K_p}{60}$
s^0	$4000+4000K_p$	

Set s^1 coeff to zero. $50000 - 4000K_p = 0$

$$K_{po} = K_p = \frac{50000}{4000} = 12.5$$

To find w_o take even divisor from s^2 row

$60s^2 + 4000 + 4000K_p = 60s^2 + 54000$

roots $s^2 = -\frac{54000}{60} = -900$ $\quad s = \pm j30$ $\quad w_o = 30$

$T_o = \frac{2\pi}{w_o} = \frac{2\pi}{30} = 0.209$ $\quad T_i = .5 \times T_o = \boxed{.1045}$

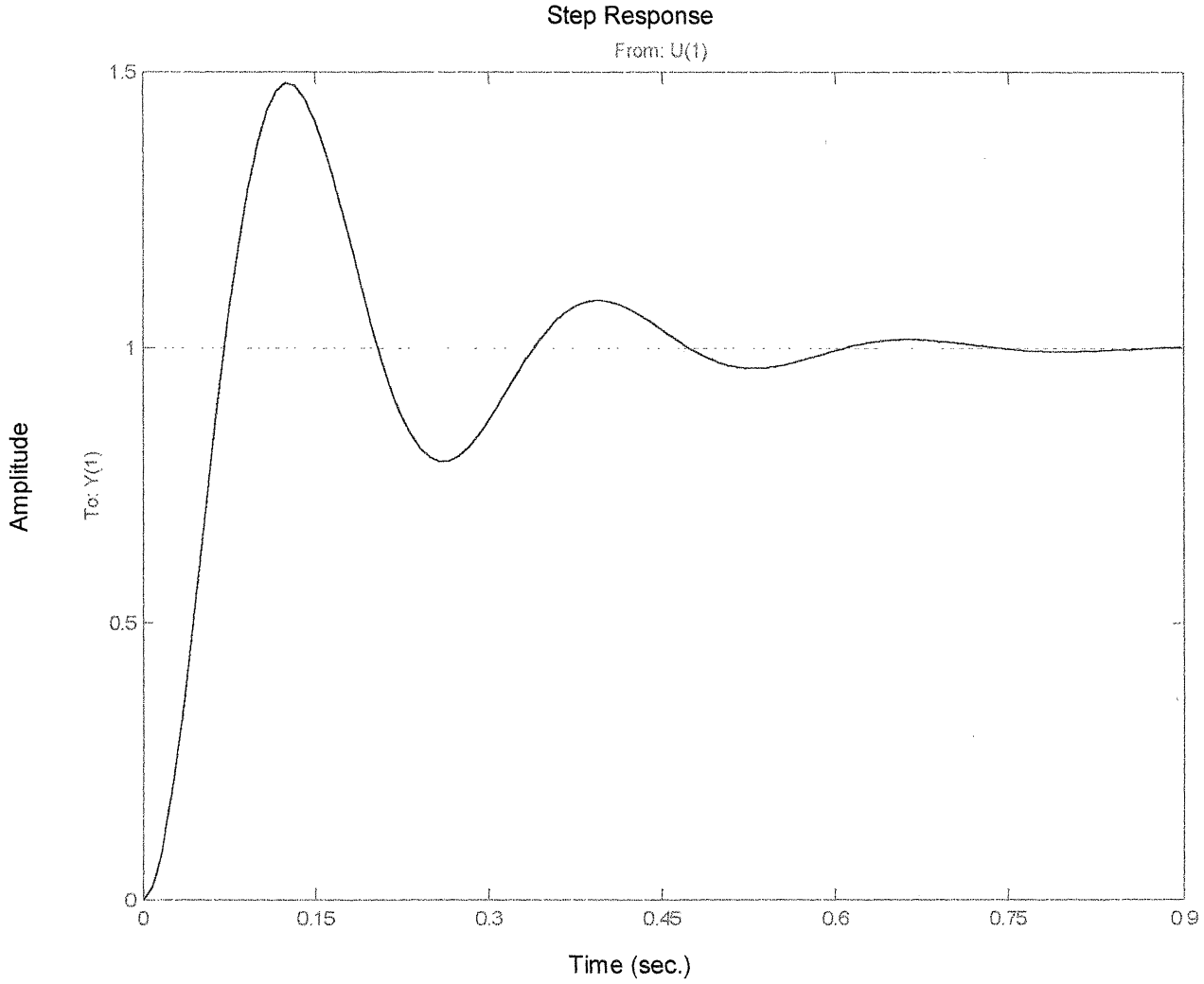
$K_p = .6 \times K_{po} = .6 \times 12.5 = \boxed{7.5}$ $\quad T_d = .125 * T_o = \boxed{.026}$

$$G_c(s) = \frac{K_p T_i T_d s^2 + K_p T_i s + K_p}{T_i s}$$

Use MATLAB and values above

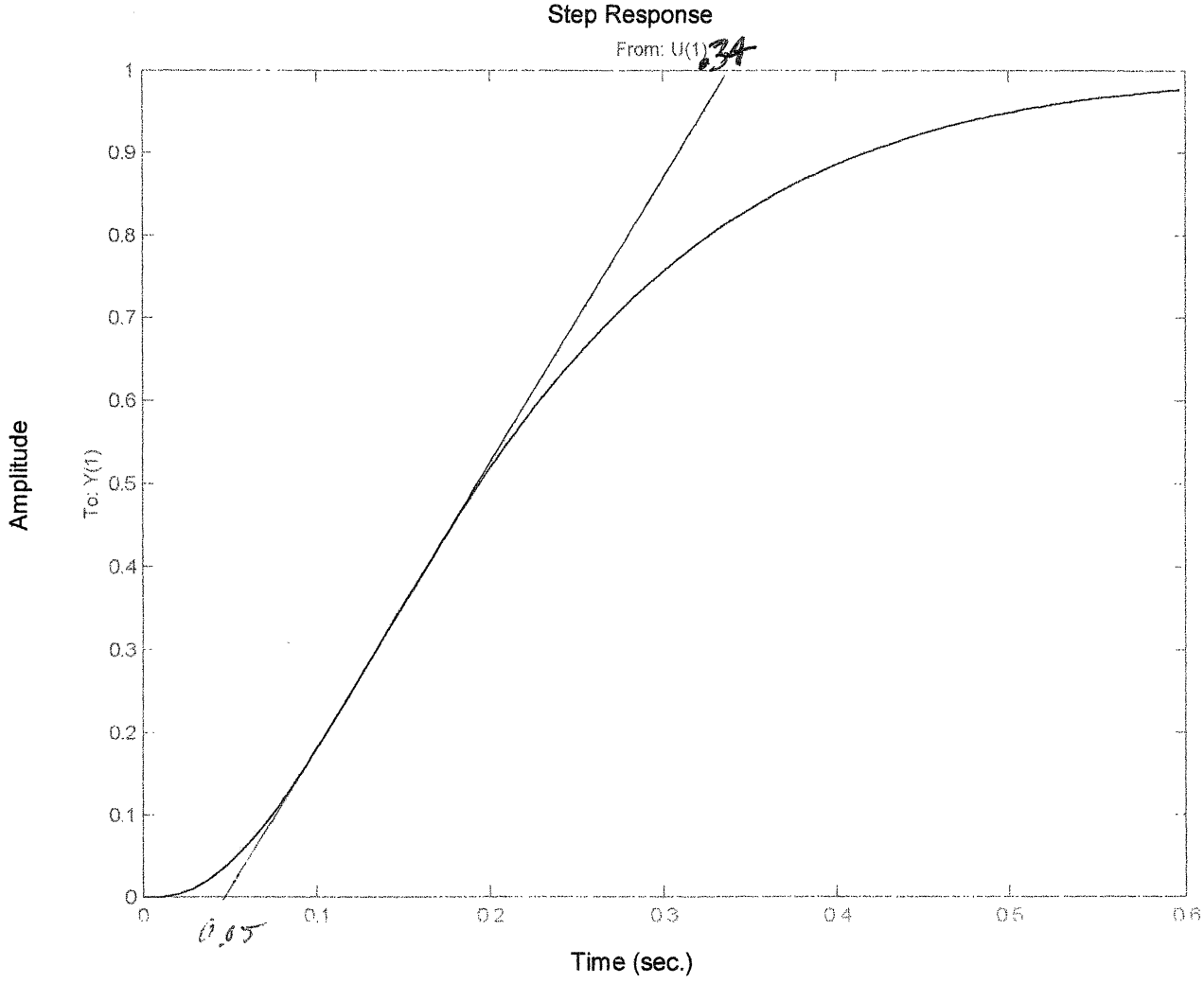
```
numgc = Kp*[Td*Ti  Ti  1]
dengc = [Ti  0]
numgp = [4000]
dengp = [1  60  900  4000]
```

```
n = conv(numgc, numgp)
d = conv(dengc, dengp)
g = tf(n, d)
t = g/(1+g)
step(t)
```



3.32

For CHR, start by getting the step response of the plant to find R



$T_U = 0.05$ $T_g = 0.29$ $R = T_g/T_U = .29/.05 = 5.8$

From Table 3.6, choose a PID compensator since $3 < R < 7.5$

overdamped

$K_p = .8R/1 = .8 \times 5.8 = \boxed{3.48}$

$T_i = T_g = \boxed{0.29}$

$T_d = 0.5 T_U = 0.5 * 0.05 = \boxed{0.025}$

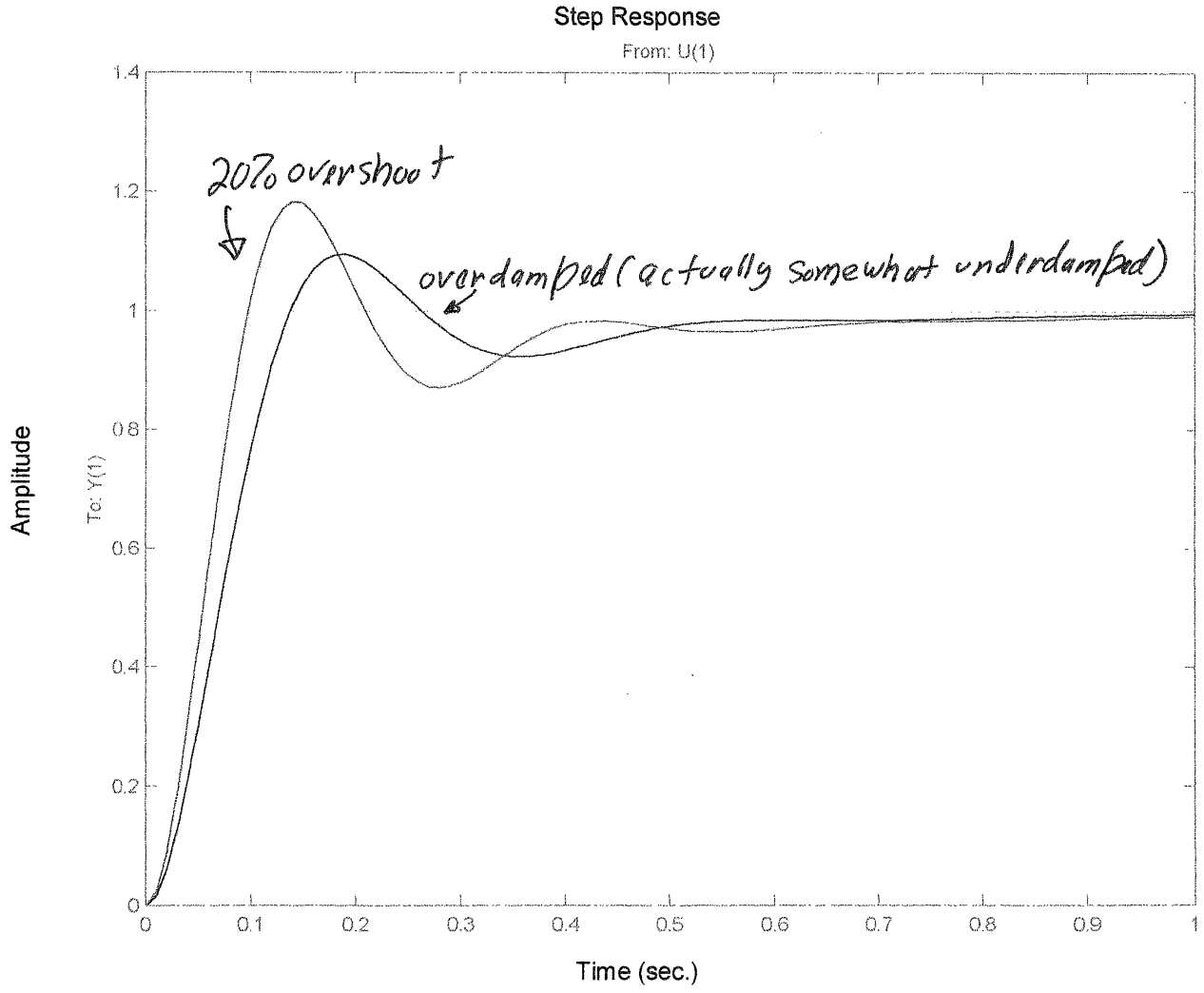
20% overshoot

$K_p = .95R/1 = .95 * 5.8 = \boxed{5.51}$

$T_i = 1.35\,T_g = 1.35 \times .29 = \boxed{0.3915}$

$T_d = .47 * T_U = .47 \times .05 = \boxed{.0235}$

For step response, use MATLAB coding for 3.31



3.33

$$P(s) = (s+8)(s+10+j10)(s+10-j10)$$

$$= (s+8)(s^2+20s+200)$$

$$= s^3+28s^2+360s+1600$$

$$= s^3+bs^2+Kbs+Kba$$

$b=20 \quad Kb=360 \quad K=\frac{360}{28}=\frac{90}{7}=14.3$

$Kba=1600 \quad a=\frac{1600}{Kb}=\frac{1600}{360}=\frac{40}{9}=5.5$

3.34 $r(t)=tu(t)=$ unit ramp

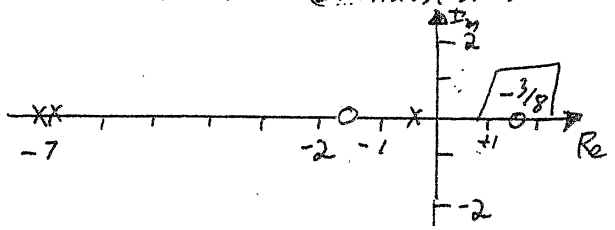
$$K_1=\lim_{s\to 0} sG_c(s)G_p(s)=\lim_{s\to 0} s\frac{Kb}{s(s+b)}=K=10$$

$$e_{Ramp}(\infty)=\frac{A}{K_1}=\frac{1}{10}=0.1 \quad (r(t)=At\,u(t) \text{ so } A=1)$$

CHAPTER 4

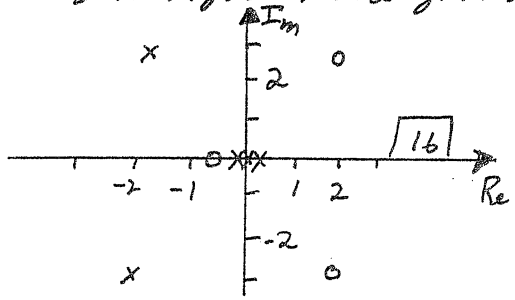
4.1

(a) $F(s) = \dfrac{-3(s^2 - \frac{8}{3})}{8(s+\frac{1}{8})(s+7)^2} = \dfrac{(-\frac{3}{8})(s+1.63)(s-1.63)}{(s+0.125)(s+7)^2}$

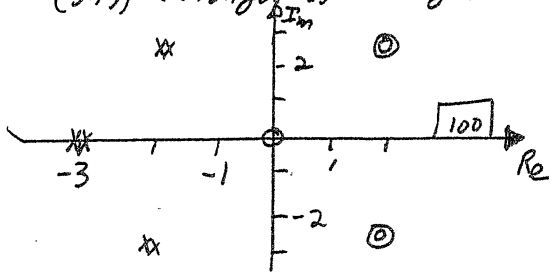


(b) $F(s) = \dfrac{8(2)(s^2 - 4s + 10)(s+\frac{1}{2})}{s^3(s^2+4s+10)}$

$= \dfrac{16(s-2+j2.45)(s-2-j2.45)(s+0.50)}{s^3(s+2+j2.45)(s+2-j2.45)}$

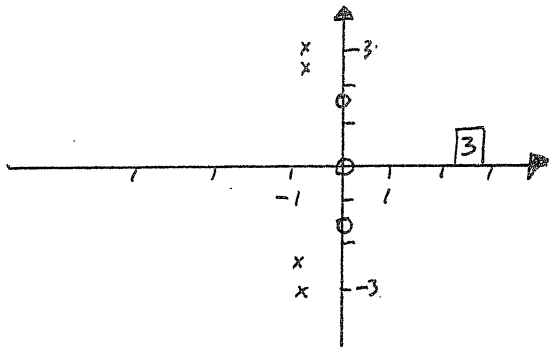


(c) $F(s) = \dfrac{100\,s(s-2+j2.45)^2(s-2-j2.45)^2}{(s+3)^2(s+2+j2.45)^2(s+2-j2.45)^2}$



4.1 (continued)

(d) $F(s) = \dfrac{3s(s+j1.732)(s-j1.732)}{(s+1+j3)(s+1-j3)(s+1+j2.65)(s+1-j2.65)}$



4.2 To determine the range of K for stability, write $KG(s)H(s) = K\frac{N(s)}{D(s)}$ so that

$$1 + KG(s)H(s) = 1 + K\frac{N(s)}{D(s)} = \frac{D(s)+KN(s)}{D(s)}$$

The charact. polynomial. $P(s) = D(s) + KN(s)$

(a) $G(s)H(s) = \dfrac{Y_4}{s(s+1/4)} = \dfrac{N(s)}{D(s)}$

$P(s) = s(s+0.25) + 0.25K$

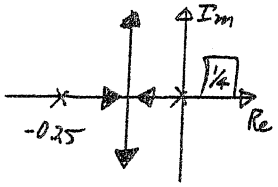
$= s^2 + 0.25s + 0.25K$

For stability, coeff > 0

so $\boxed{K > 0}$

$n = 2 \quad m = 0 \quad n - m = 2$

$\sigma = 0.125, \theta_L = \pm 90°$



(b) $G(s)H(s) = \dfrac{3(s+1/3)}{4s(s+1/4)} = \dfrac{0.75(s+1/3)}{s(s+0.25)}$

$P(s) = s^2 + s(0.25 + 0.75K) + 0.25K$

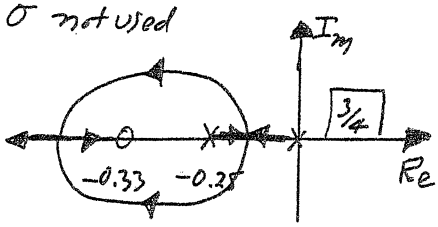
For positive coeff

$0.25 + 0.75K > 0 \quad K > -\frac{1}{3}$

$\boxed{K > 0}$ ← choose

$n = 2, m = 1, n - m = 1, \theta_L = 180°$

σ not used



4.2 (continued)

(c) $G(s)H(s) = \dfrac{s+1}{4s(s+1/4)} = \dfrac{0.25(s+1)}{s(s+0.25)}$

$P(s) = s^2 + 0.25s + K(0.25)(s+1)$

$= s^2 + s(0.25 + 0.25K) + 0.25K$

For coeff >0

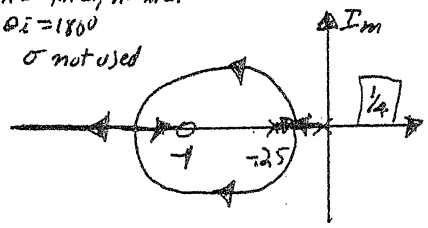
$0.25 + 0.25K > 0 \quad K > -1$

$\boxed{K>0}$ ← Choose

$n=2, m=1, n-m=1$

$\theta_i = 180°$

σ not used



(d) $G(s)H(s) = \dfrac{0.25}{s(s+1)(s+0.25)}$

$D(s) + KN(s) = P(s)$

$= s^3 + 1.25s^2 + 0.25s + 0.25K$

$a_0 = 0.25K \quad a_1 = 0.25$

$a_2 = 1.25$

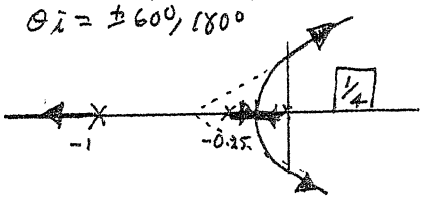
From prob 2.10

$a_1 a_2 = (1.25)(0.25) > 0.25K$

$\boxed{K < 1.25}$

$n=3, m=0, n-m=3, \sigma = -0.42$

$\theta_i = \pm 60°, 180°$



(e) $G(s)H(s) = \dfrac{1}{(s+5+j3.87)(s+5-j3.87)}$

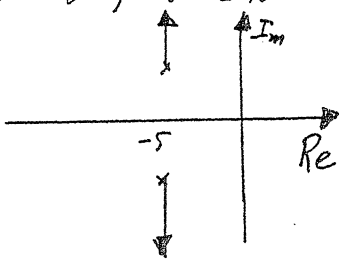
$P(s) = D(s) + KN(s) = s^2 + 10s + 40 + K$

For coeff >0

$40 + K > 0 \quad \boxed{K > -40}$

For $K>0$, $n=2, m=0, n-m=2$

$\sigma = -5, \quad \theta_i = \pm 90°$



(f) $G(s)H(s) = \dfrac{s+4}{(s+0.5+j2.4)(s+0.5-j2.4)}$

$P(s) = D(s) + KN(s)$

$= s^2 + s(1+K) + 6 + 4K$

For coeff >0

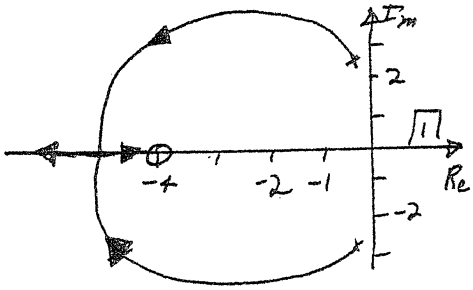
$1+K > 0 \quad \boxed{K > -1}$ ← Choose

$6 + 4K > 0 \quad K > -1.5$

For $K>0$ root locus

$n=2, m=1, n-m=1$

$\theta_i = 180°$, σ not used



4.2 (continued)

(g) $G(s)H(s) = \frac{5}{s(s+2)(s+10)}$

$P(s) = D(s) + KN(s) = s^2 + (12+K)s + 20$

For roots > 0 : $12 + K > 0$

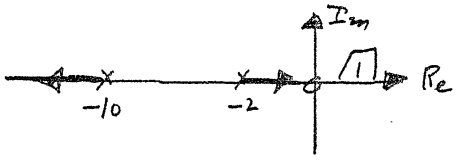
$\boxed{K > -12}$

For $K > 0$

$n = 2,\ m = 1,\ n - m = 0$

$\theta_i = 180°$

σ not used



(h) $G(s)H(s) = \frac{1}{s(s+2)(s+1+j7)(s+1-j7)}$

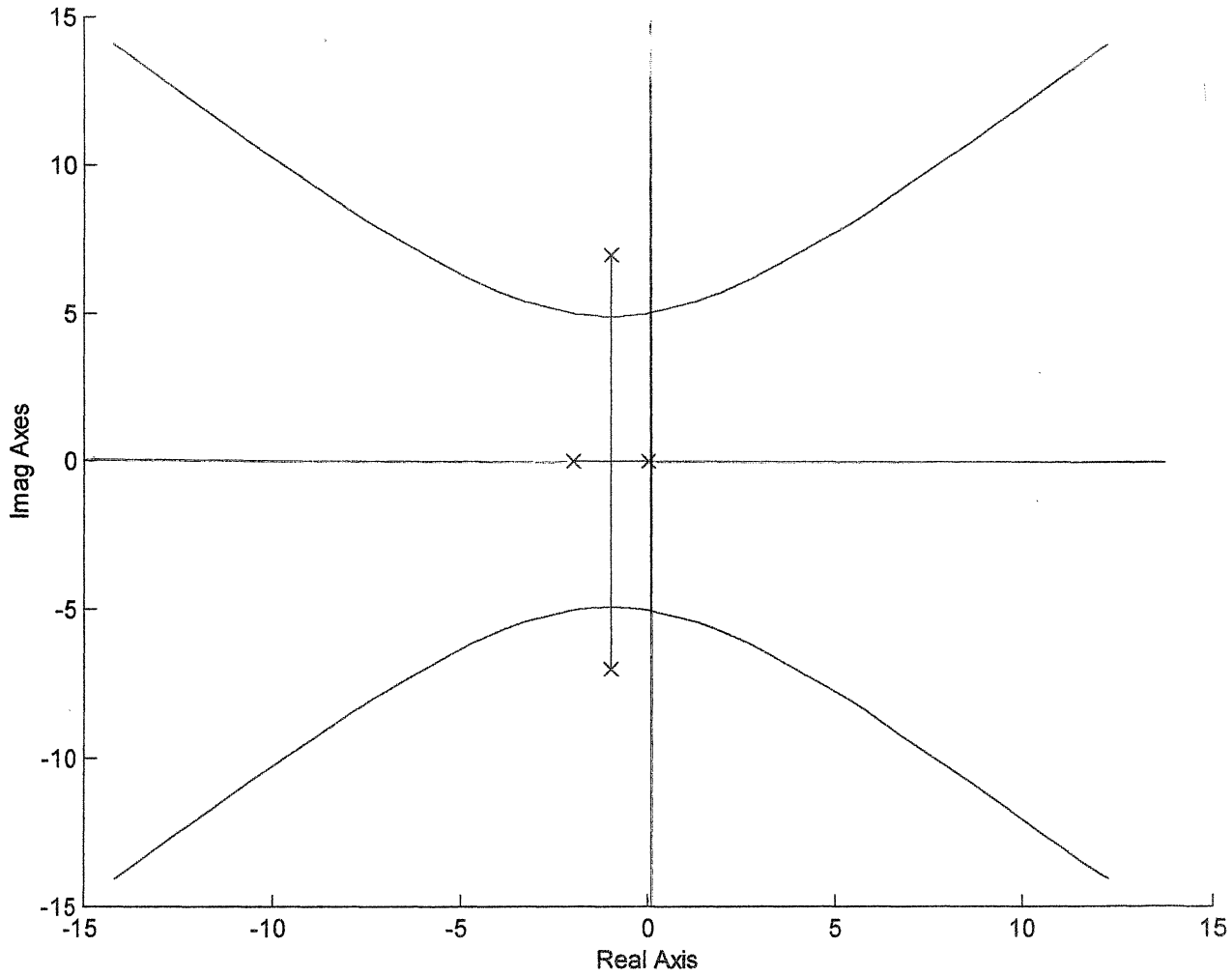
$P(s) = D(s) + KN(s) = s^4 + 4s^3 + 54s^2 + 100s + K$

s^4	1	54	K
s^3	4	100	
s^2	29	K	
s^1	$(2900-4K)/29$		
s^0	K		

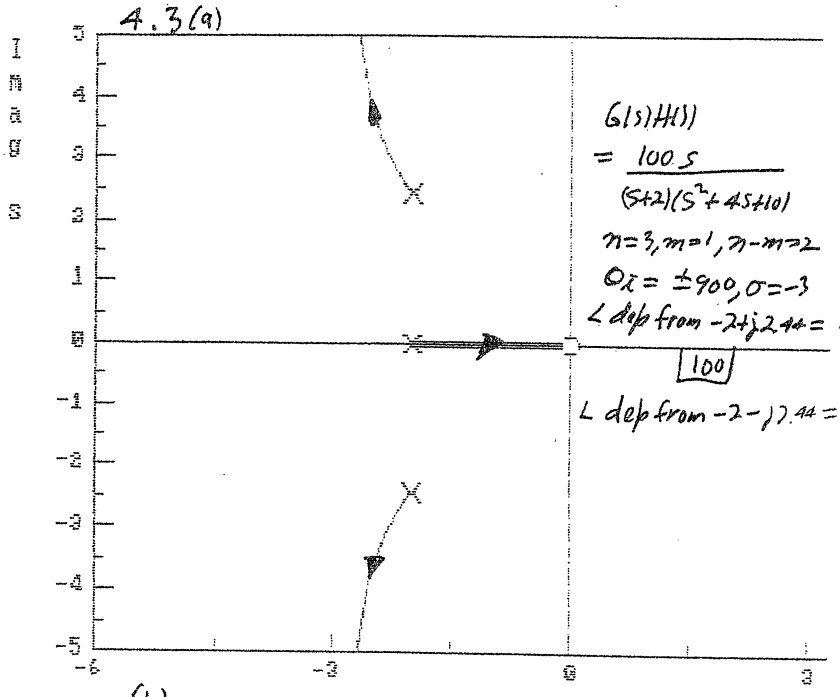
$K > 0$

$2900 - 4K > 0 \quad K < 725$

$\boxed{0 < K < 725}$



4.3(a)



$G(s)H(s)$

$= \dfrac{100\,s}{(s+2)(s^2+4s+101)}$

$n=3, m=1, n-m=2$

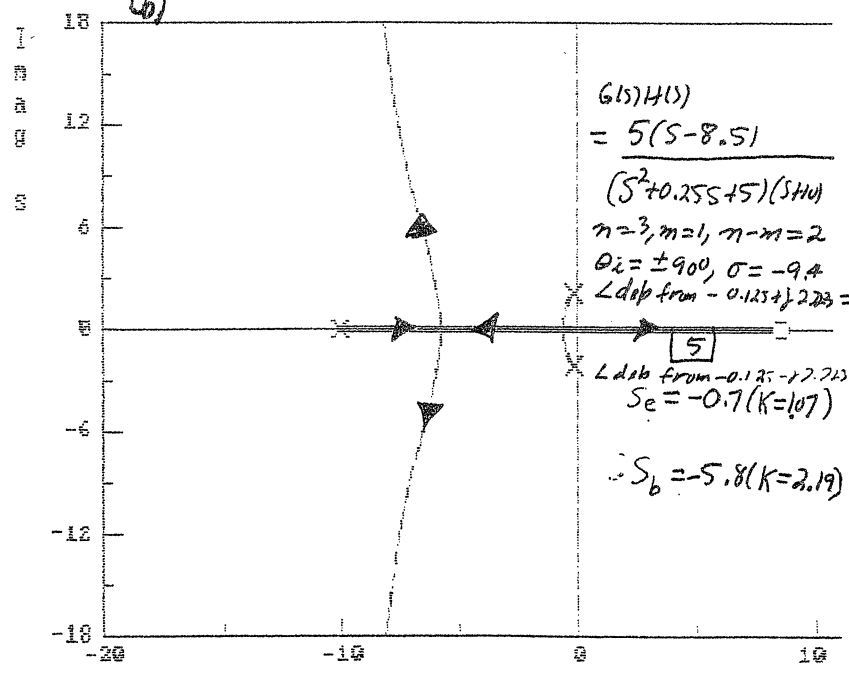
$\theta_i = \pm 90°, \sigma = -3$

$\angle$ dep from $-2+j2.44 = 129.2°$

100

$\angle$ dep from $-2-j2.44 = -129.2°$

(b)



$G(s)H(s)$

$= \dfrac{5(s-8.5)}{(s^2+0.25s+5)(s+10)}$

$n=3, m=1, n-m=2$

$\theta_i = \pm 90°, \sigma = -9.4$

$\angle$ dep from $-0.125+j2.23 = -112.3°$

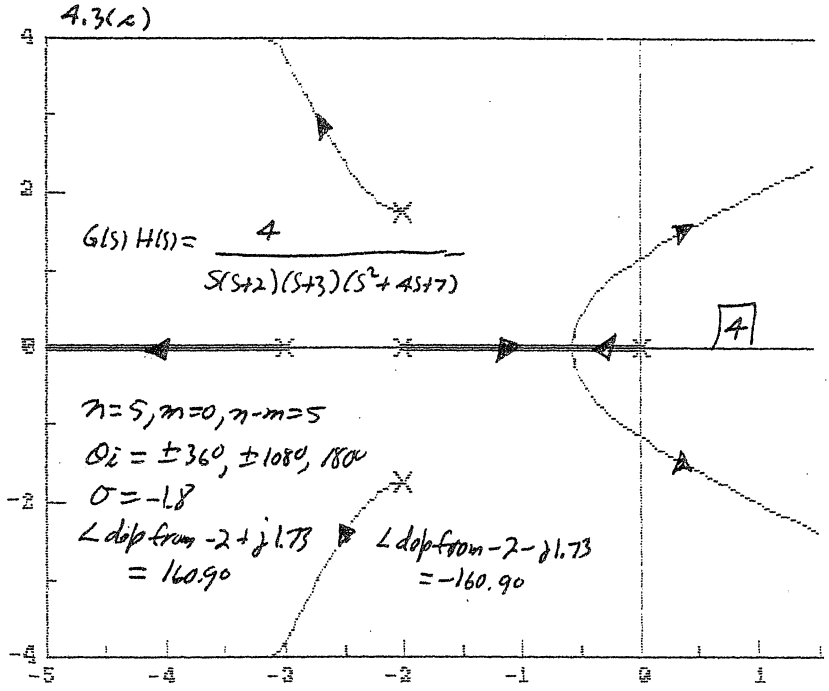
5

$\angle$ dep from $-0.125-j2.23 = +112.3°$

$S_e = -0.7 \; (K=107)$

$\therefore S_b = -5.8 \; (K=2.19)$

4.3(c)



$G(s)H(s) = \dfrac{4}{s(s+2)(s+3)(s^2+4s+7)}$

4

$n=5, m=0, n-m=5$

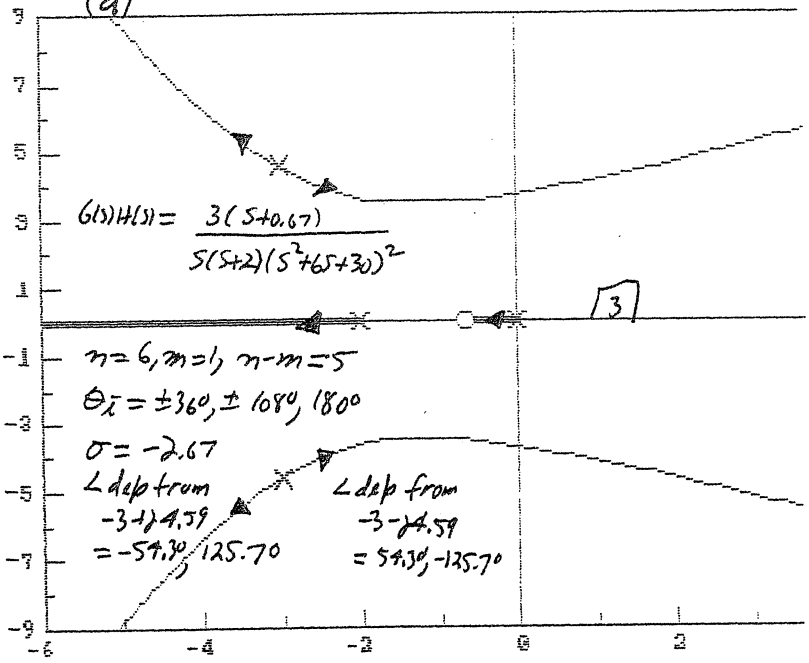
$\theta_i = \pm 36^\circ, \pm 108^\circ, 180^\circ$

$\sigma = -1.8$

$\angle$ dep from $-2+j1.73$ $= 160.90^\circ$

$\angle$ dep from $-2-j1.73$ $= -160.90$

(d)



$G(s)H(s) = \dfrac{3(s+0.67)}{s(s+2)(s^2+6s+30)^2}$

3

$n=6, m=1, n-m=5$

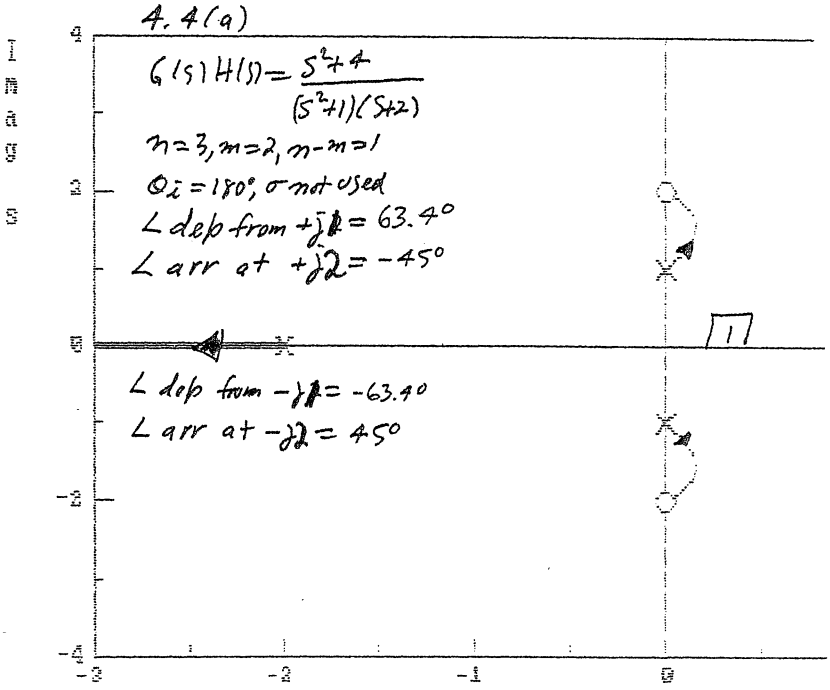
$\theta_i = \pm 36^\circ, \pm 108^\circ, 180^\circ$

$\sigma = -2.67$

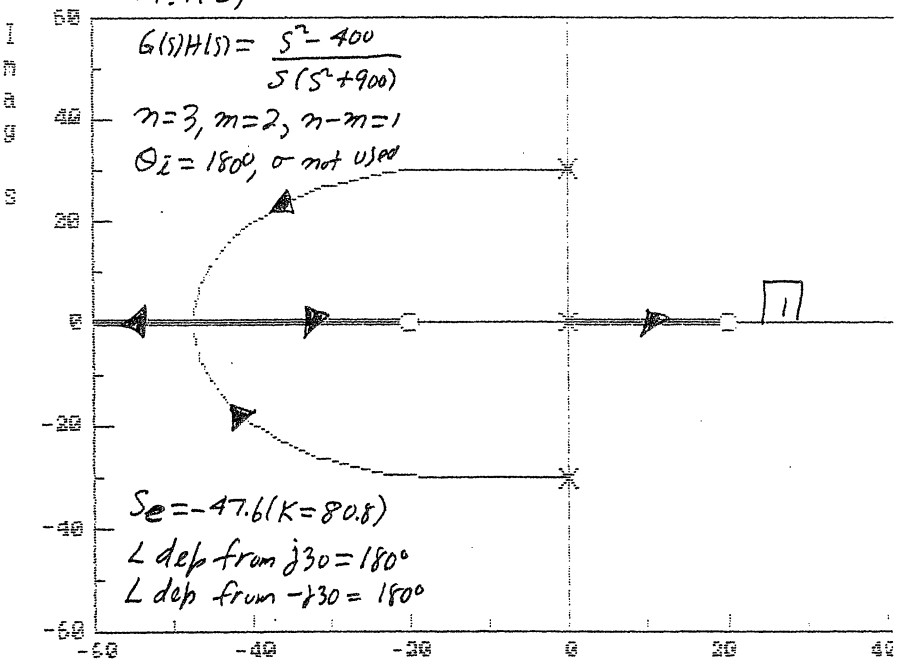
$\angle$ dep from $-3+j4.59$ $= -54.30^\circ, 125.70^\circ$

$\angle$ dep from $-3-j4.59$ $= 54.30^\circ, -125.70^\circ$

4.4(a)


Imag
4
2
0
-2
-4
-3
-2
-1
0
1
G(s)H(s) = s²+4 / (s²+1)(s+2)
n=3, m=2, n-m=1
Θi = 180°, σ not used
∠ dep from +j1 = 63.4°
∠ arr at +j2 = -45°
∠ dep from -j1 = -63.4°
∠ arr at -j2 = 45°

4.4(c)


Imag
60
40
20
0
-20
-40
-60
-60
-40
-30
0
20
40
1
G(s)H(s) = s²-400 / s(s²+900)
n=3, m=2, n-m=1
Θi = 180°, σ not used
Se = -47.6 (K=80.8)
∠ dep from j30 = 180°
∠ dep from -j30 = 180°

4.4(d)

$G(s)H(s) = \dfrac{10(s^2+4s+20)}{(s+2)(s+6)}$

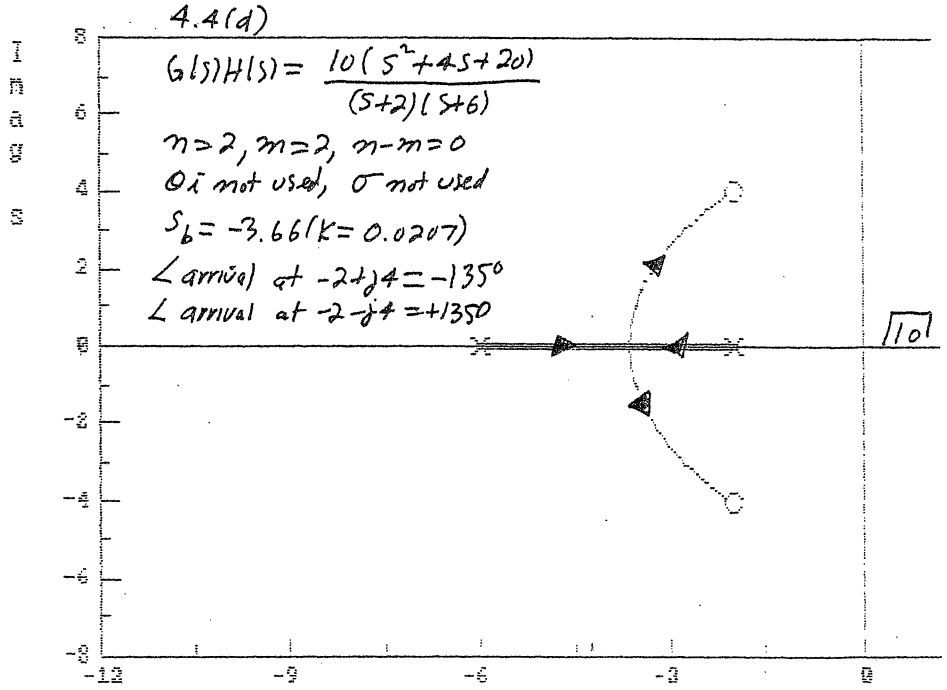
$n=2, m=2, n-m=0$

θ_i not used, σ not used

$s_b = -3.66 (K = 0.0207)$

$\angle$ arrival at $-2+j4 = -135^\circ$

$\angle$ arrival at $-2-j4 = +135^\circ$



4.4(f)

$G(s)H(s) = \dfrac{4s^2}{(s^2+400s+130000)^2}$

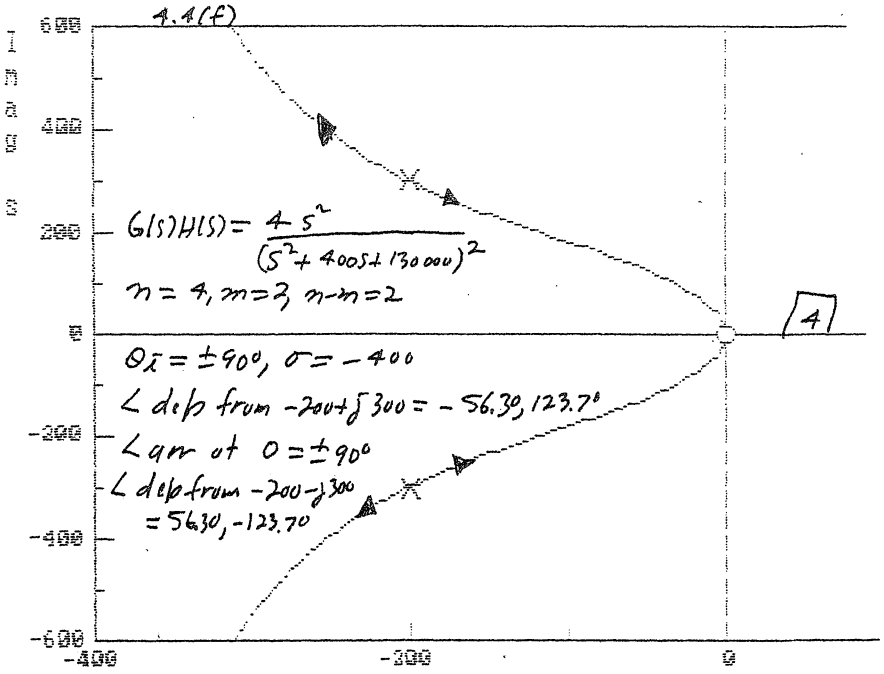
$n=4, m=2, n-m=2$

$\theta_i = \pm 90^\circ, \sigma = -400$

$\angle$ dep from $-200+j300 = -56.30, 123.7^\circ$

$\angle$ arr at $0 = \pm 90^\circ$

$\angle$ dep from $-200-j300 = 56.30, -123.70$



4.4 (g)

$$G(s)H(s) = \frac{75(s^2+9)}{(s+2)(s+3)(s^2+2s+5)}$$

$n = 4,\ m = 3,\ n-m = 1$

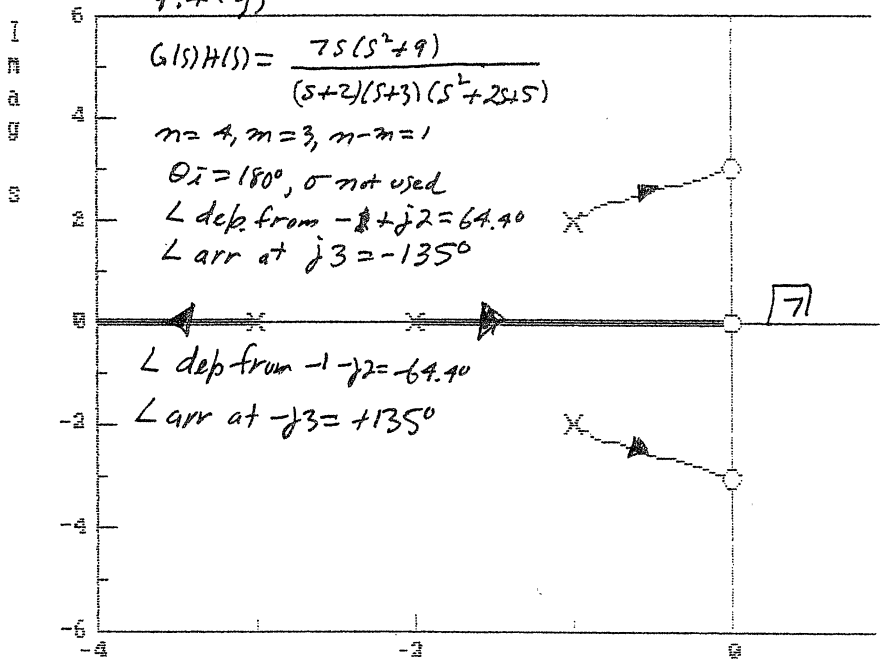
$\theta_i = 180°$, σ not used

$\angle$ dep. from $-1+j2 = 64.40$

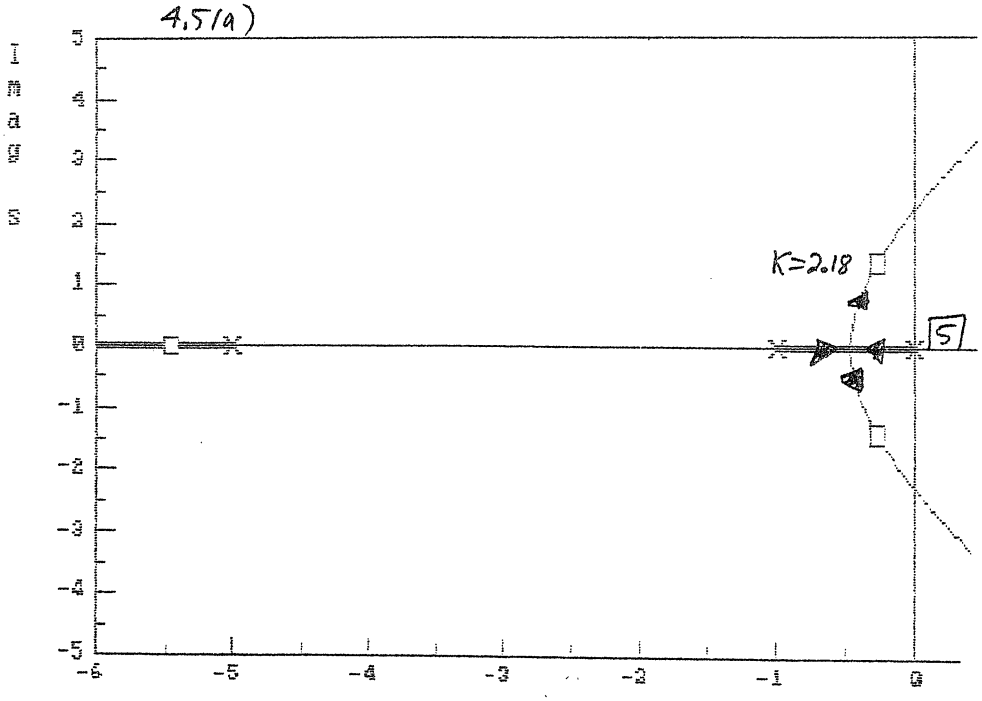
$\angle$ arr at $j3 = -135°$

$\angle$ dep from $-1-j2 = -64.40$

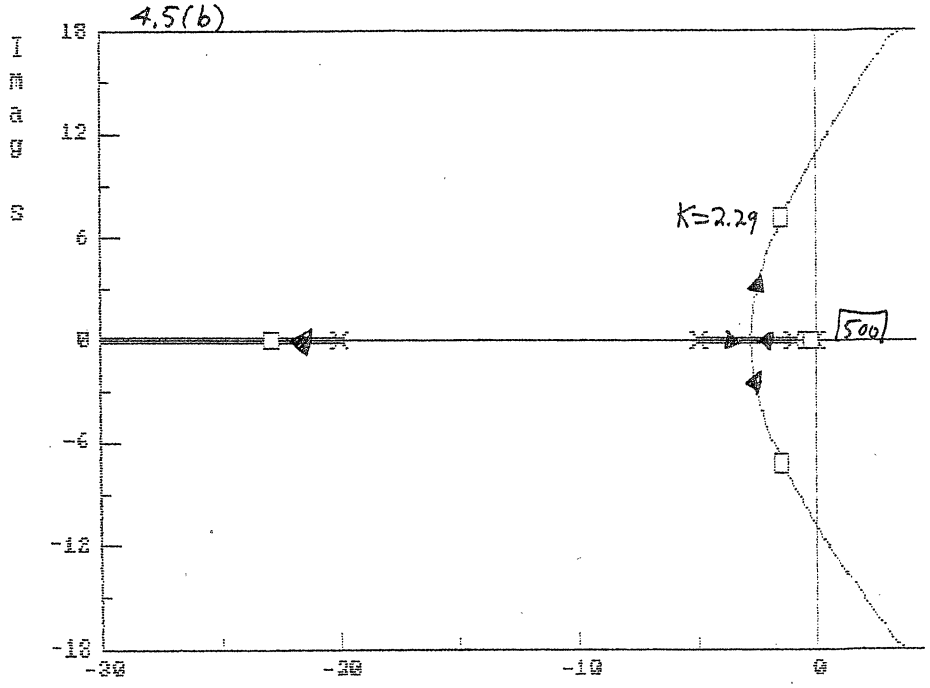
$\angle$ arr at $-j3 = +135°$



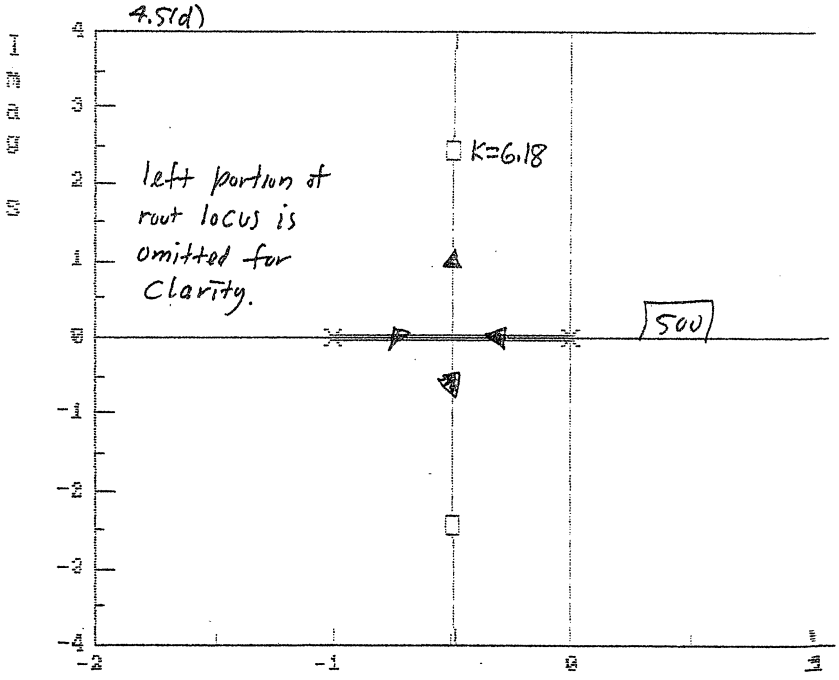
4.51 a)



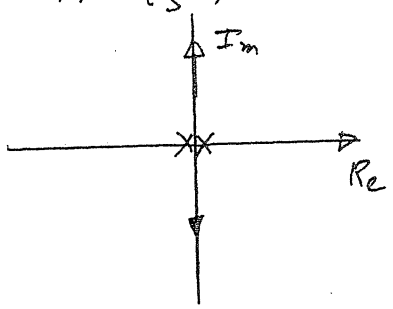
4.5(b)


K=2.29
500

4.5(d)


K=6.18
left portion of root locus is omitted for clarity.
500

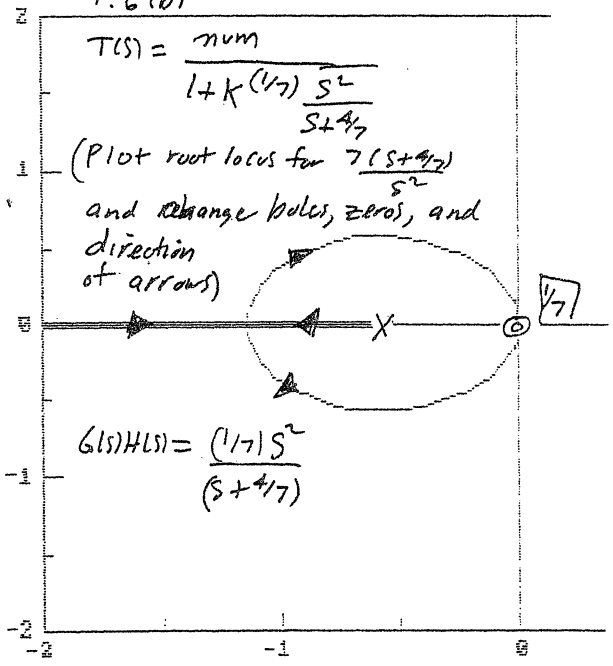
4.6 (a) $T(s) = \dfrac{num}{1 + K\left(\frac{1}{s^2}\right)}$ $\quad G(s)H(s) = \dfrac{1}{s^2}$



4.6 (b)

$T(s) = \dfrac{num}{1 + K(1/7)\frac{s^2}{s+4/7}}$

(Plot root locus for $\frac{7(s+4/7)}{s^2}$ and change poles, zeros, and direction of arrows)

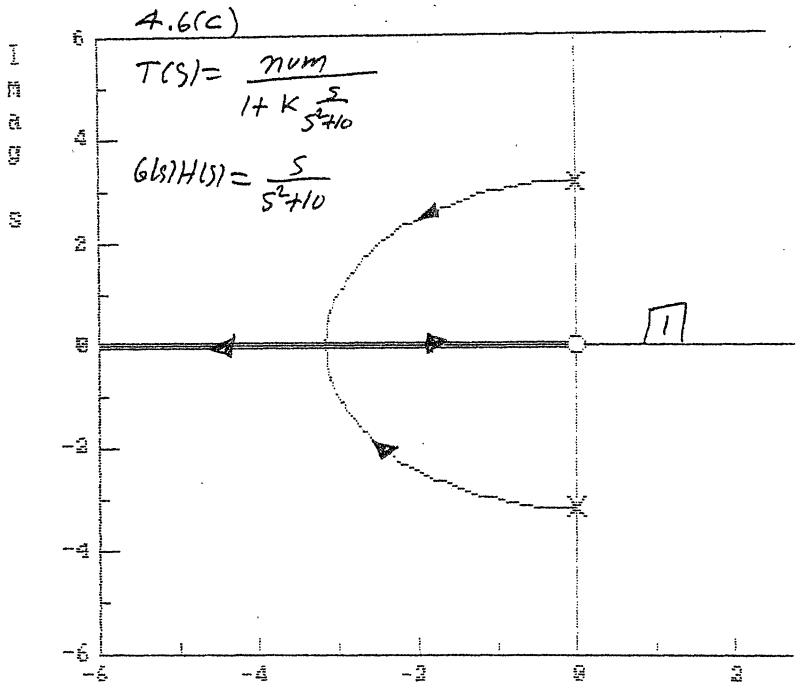


$G(s)H(s) = \dfrac{(1/7)s^2}{(s+4/7)}$

4.6(c)

$$T(s) = \frac{num}{1 + K\frac{s}{s^2+10}}$$

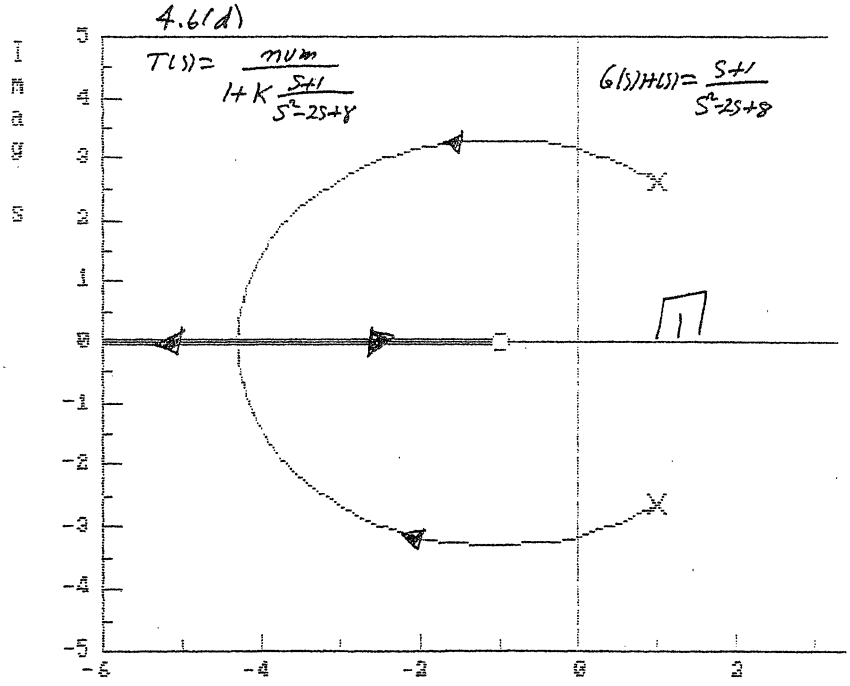
$$G(s)H(s) = \frac{s}{s^2+10}$$



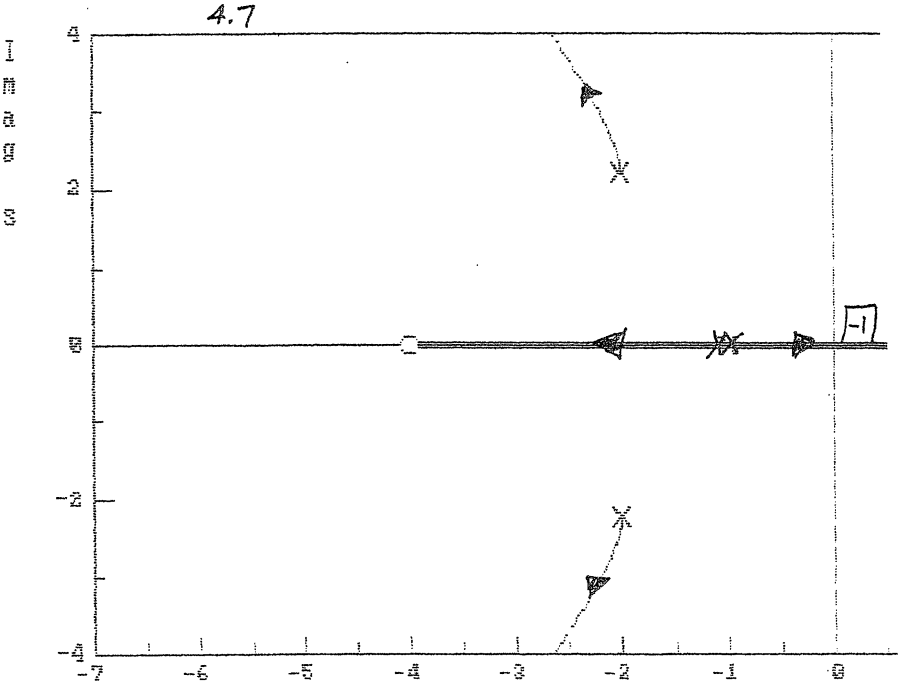
4.6(d)

$$T(s) = \frac{num}{1 + K\frac{s+1}{s^2-2s+8}}$$

$$G(s)H(s) = \frac{s+1}{s^2-2s+8}$$



4.7

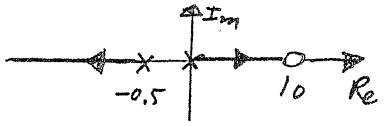


4.8 (See text)

4.9

$$T(s) = \frac{num}{1 + \frac{K(s-10)}{4s(s+0.5)}}$$

$$G(s)H(s) = \frac{(0.25)(s-10)}{s(s+0.5)}$$



4.10

For compensator $G_c(s) = \frac{s+20+K_1 s}{s+20}$

$$T(s) = \frac{\left(\frac{s+20+K_1 s}{s+20}\right) K_2 \left(\frac{0.1}{s+2}\right) \frac{1}{s}}{1 + \left(\frac{s+20+K_1 s}{s+20}\right) K_2 \left(\frac{0.1}{s+2}\right) + \frac{1}{s}}$$

$$= \frac{num}{s^3 + 22s^2 + (40+0.1K_2)s + 2K_2 + K_1 s(0.1K_2)}$$

82

If K_2 is fixed and K_1 is variable

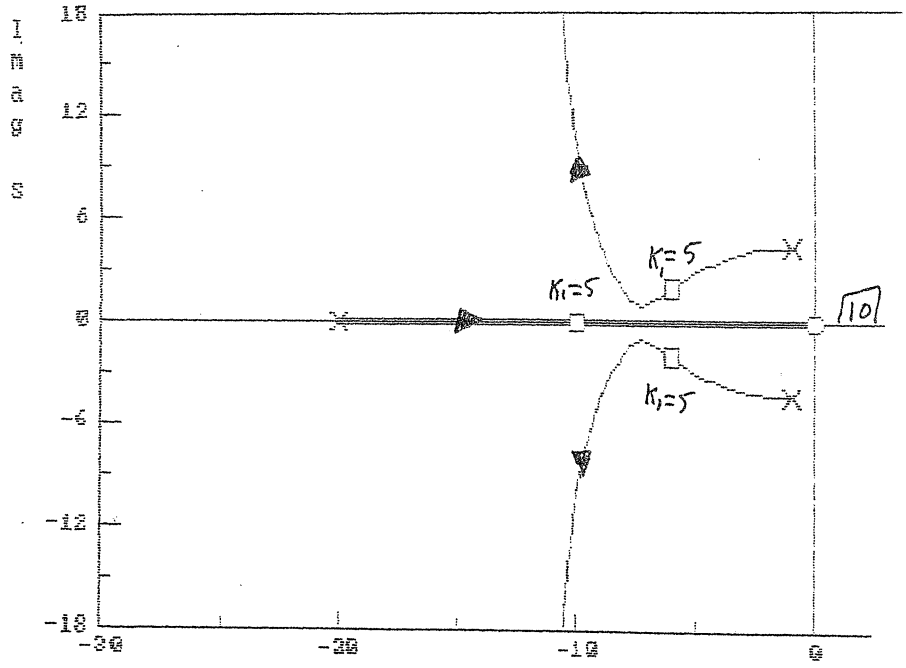
$$T(s) = \frac{num'}{1 + K_1 \dfrac{0.1K_2 s}{s^3 + 22.5s^2 + (40 + 0.1K_2)s + 2K_1}}$$

If $K_2 = 100$ and we want a root locus as K_1 varies

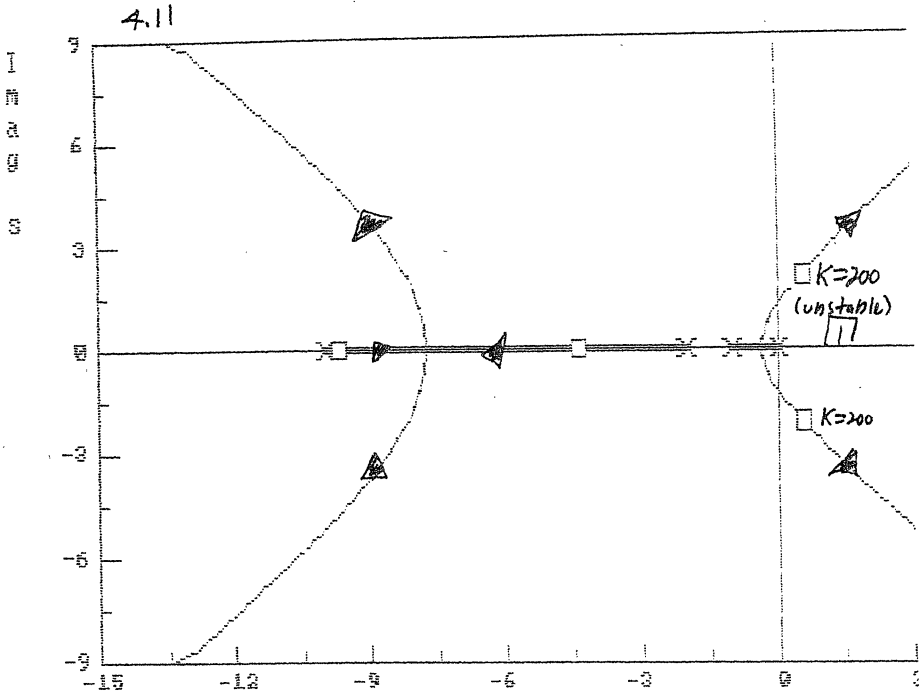
$$G(s)H(s) = \frac{10s}{s^3 + 22.5s^2 + 50s + 200}$$

OL poles are $-1 \pm j3, -20$

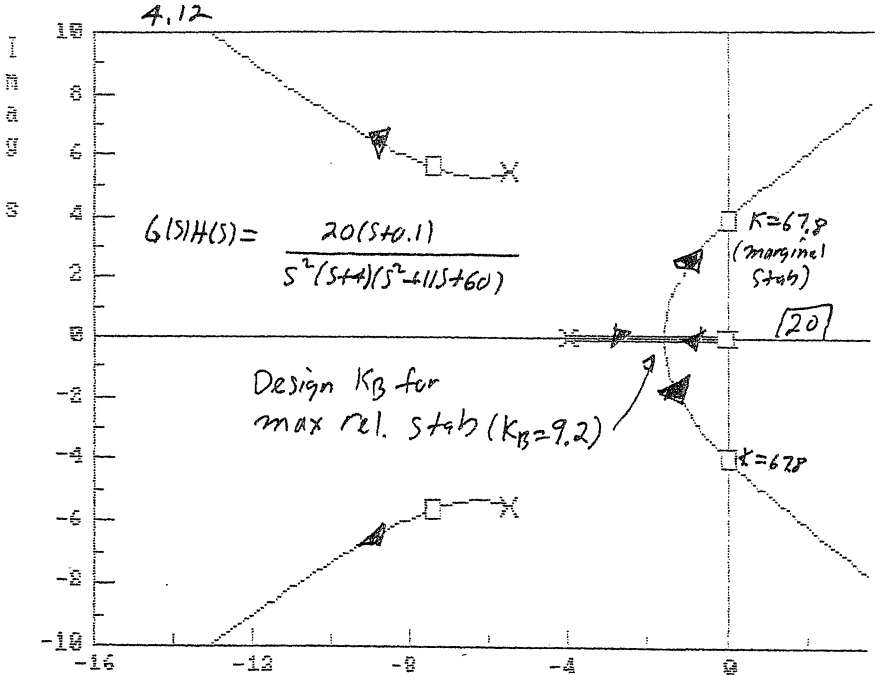
On the root locus below, if $K_1 = 5$ the CL poles are -10 and $-6 \pm j2$ so the relative stab = 6, thus meeting the requirement.



4.11


Imag
9
6
3
0
-3
-6
-9
-15
-12
-9
-6
-3
0
K=∞
(unstable)
11
K=∞

4.12

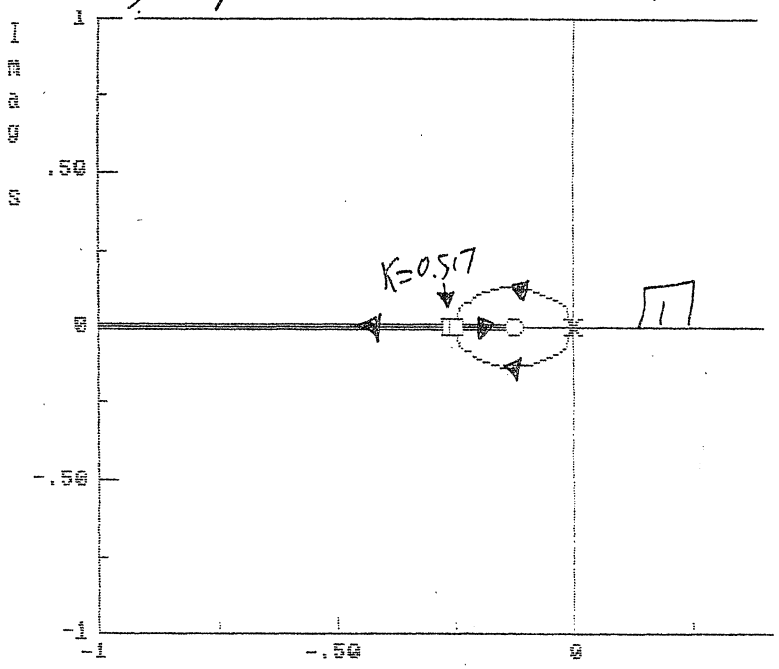

Imag
10
8
6
4
2
0
-2
-4
-6
-8
-10
-16
-12
-8
-4
0
G(s)H(s) = 20(s+0.1) / s²(s+4)(s²+11s+60)
K=67.8
(marginal stab)
20
Design KB for
max rel. stab (KB=9.2)
K=67.8

4.14

$$T(s) = \frac{\frac{K}{S(S+8)(S+K)}}{1+\frac{K}{S(S+8)(S+K)}} = \frac{K}{S(S+8)(S+K)+K} = \frac{K}{S^2(S+8)+K[S^2+8S+1]}$$

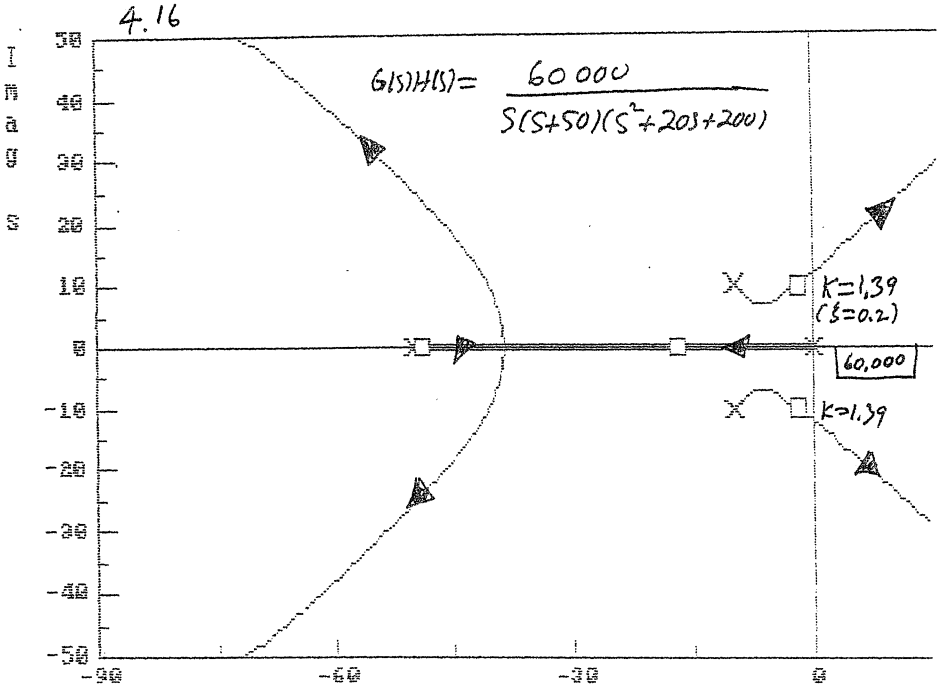
$$= \frac{num}{1+K\,\frac{S^2+8S+1}{S^2(S+8)}} \qquad G(s)H(s) = \frac{S^2+8S+1}{S^2(S+8)}$$

Focusing on part of RL nearest to origin, K=0.517



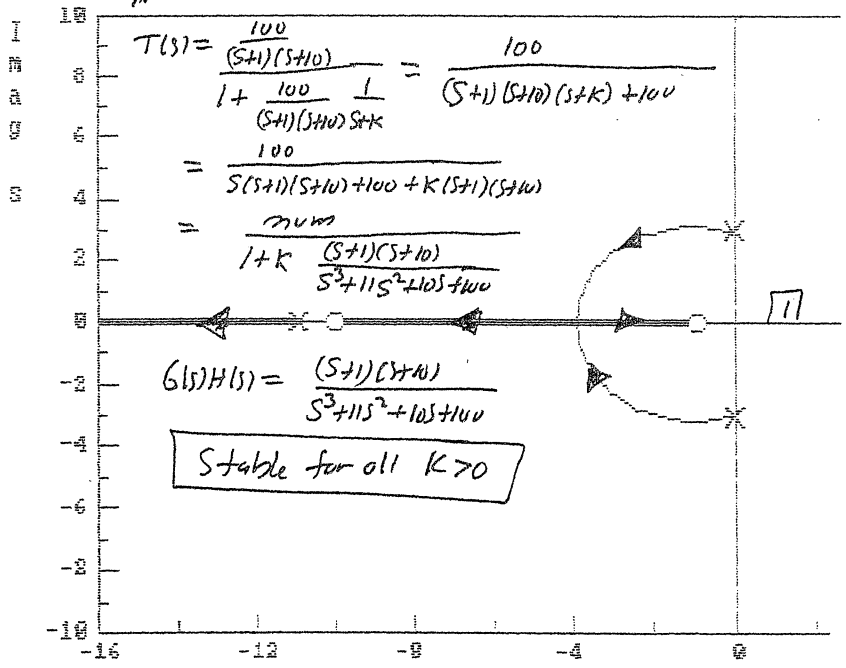
4.15 See text

4.16



$$G(s)H(s) = \frac{60\,000}{S(S+50)(S^2+20S+200)}$$

4.17



$$T(s) = \frac{\frac{100}{(S+1)(S+10)}}{1 + \frac{100}{(S+1)(S+10)}\frac{1}{S+K}} = \frac{100}{(S+1)(S+10)(S+K)+100}$$

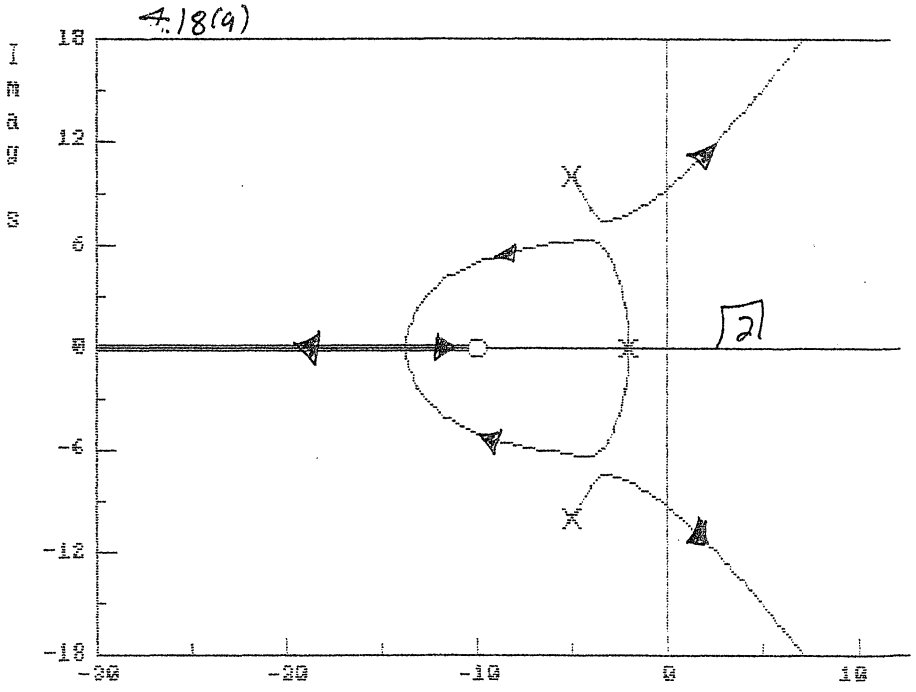
$$= \frac{100}{S(S+1)(S+10)+100+K(S+1)(S+10)}$$

$$= \frac{num}{1+K\frac{(S+1)(S+10)}{S^3+11S^2+10S+100}}$$

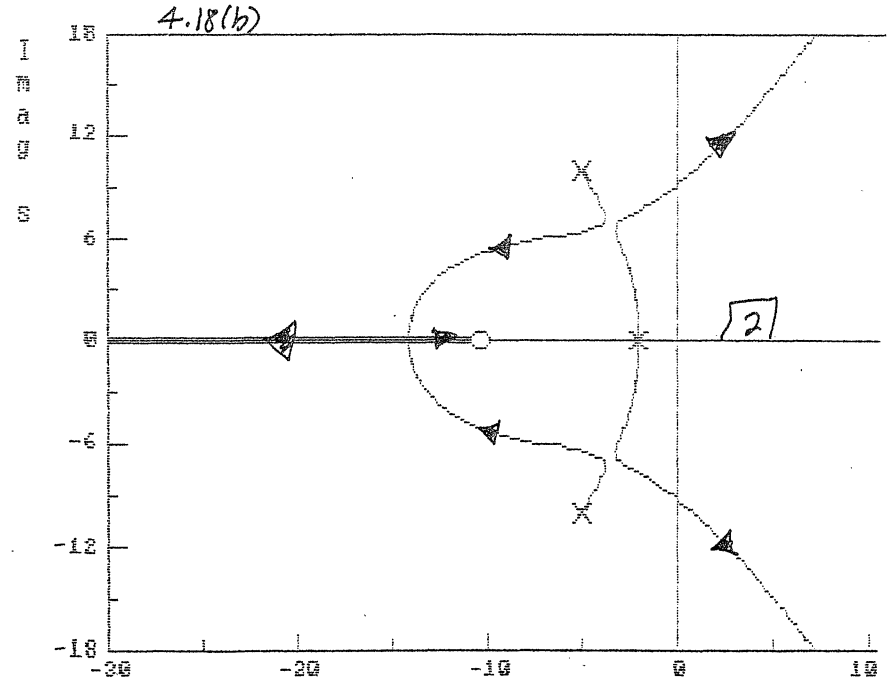
$$G(s)H(s) = \frac{(S+1)(S+10)}{S^3+11S^2+10S+100}$$

Stable for all $K>0$

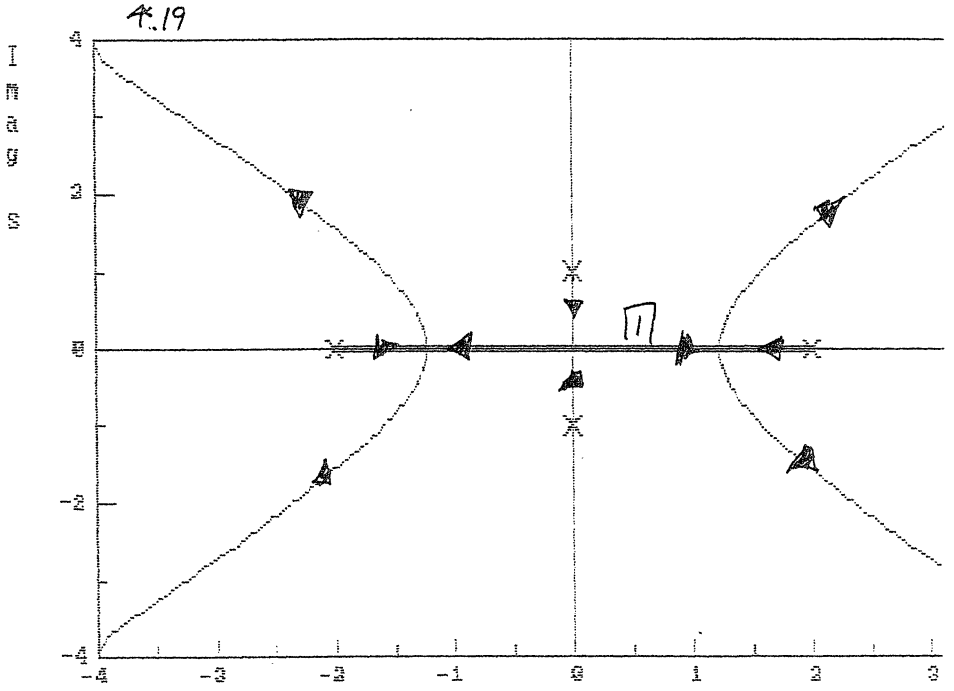
4.18(a)


Imag
18
12
6
0
-6
-12
-18
-30
-20
-10
0
10
2

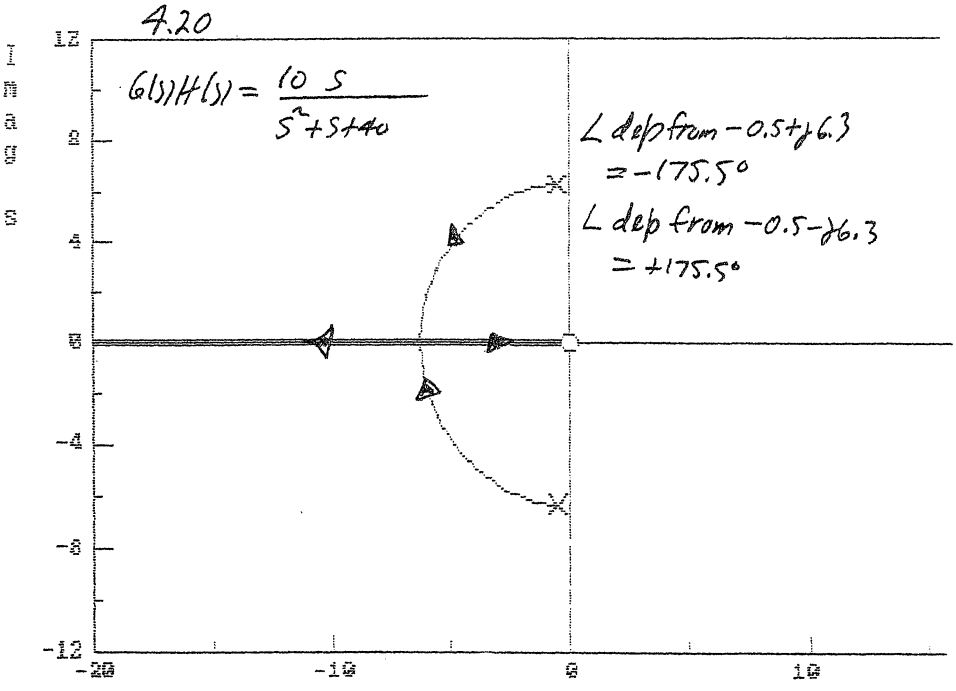
4.18(b)


Imag
18
12
6
0
-6
-12
-18
-30
-20
-10
0
10
2

4.19


Imag
4
3
0
-2
-4
-4
-3
-2
-1
0
1
2
3

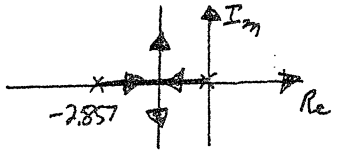
4.20


Imag
12
8
4
0
-4
-8
-12
-20
-10
0
10
$G(s)H(s) = \frac{10s}{s^2+s+40}$
$\angle$ dep from $-0.5+j6.3$
$= -175.5°$
$\angle$ dep from $-0.5-j6.3$
$= +175.5°$

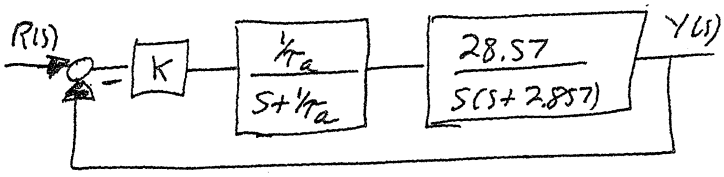
4.21

(a) $G(s)H(s) = \dfrac{10}{s(0.35)(s+1/0.35)} = \dfrac{10/0.35}{s(s+1/0.35)}$

$= \dfrac{28.57}{s(s+2.857)}$



(b)



$$1 + K\frac{N(s)}{D(s)} = 1 + \frac{K\,28.57(1/T_a)}{s(s+\frac{1}{T_a})(s+2.857)}$$

$P(s) = D(s) + KN(s) = s(s+\frac{1}{T_a})(s+28.57) + K(28.57)(1/T_a)$

Let $K = 30$

$P(s) = s^3 + s^2\underbrace{(2.857+\frac{1}{T_a})}_{a_2} + s\underbrace{\frac{2.857}{T_a}}_{a_1} + \underbrace{30(28.57)(\frac{1}{T_a})}_{a_0}$

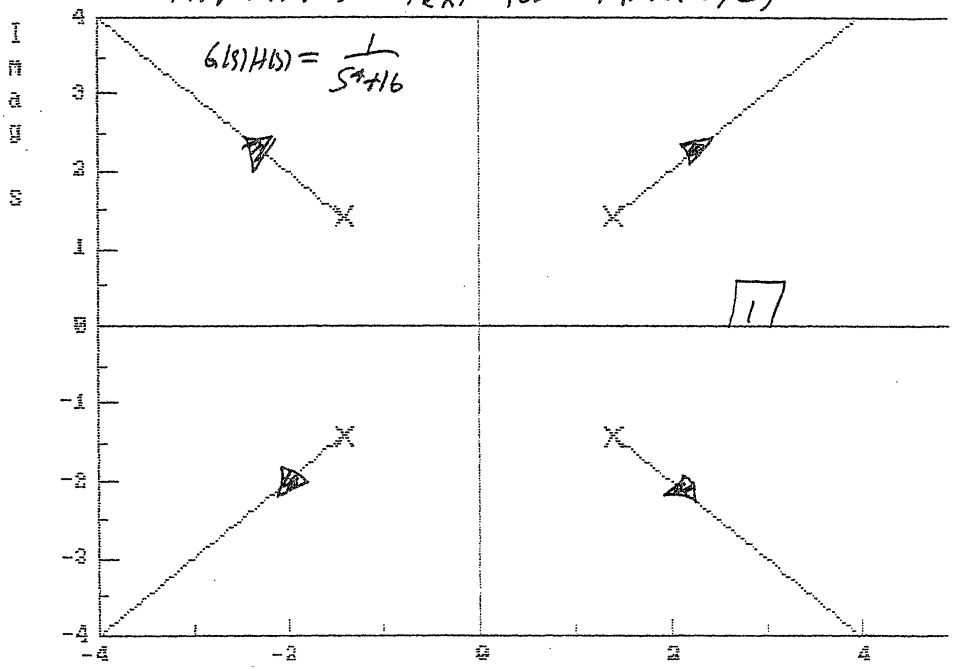
For marginal stab (Prob 2.10) $a_2 a_1 = a_0$

$$\left(\frac{2.857T_a+1}{T_a}\right)\left(\frac{2.857}{T_a}\right) = \frac{30(28.57)}{T_a}$$

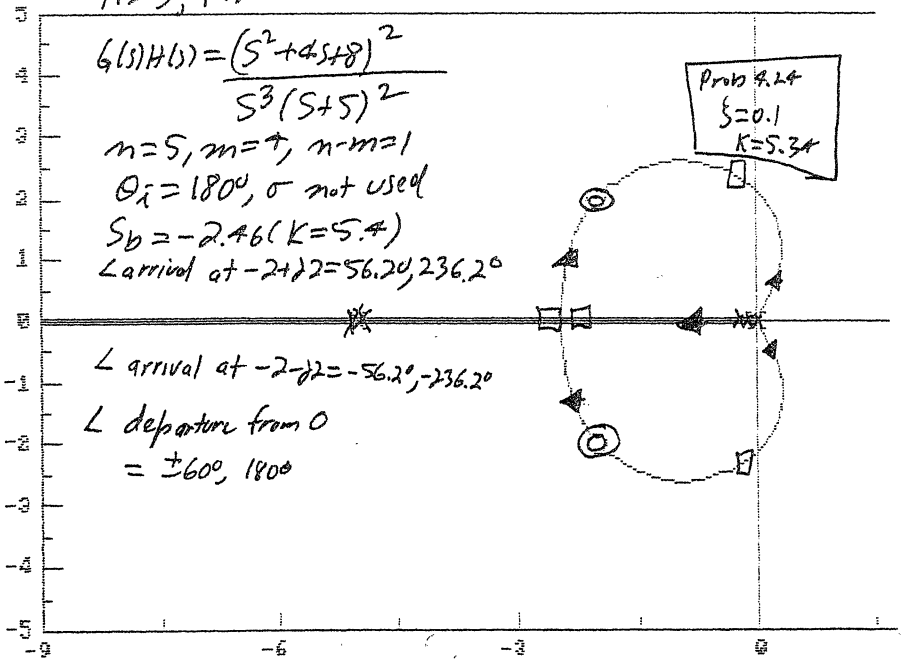
$$2.857^2 T_a + 2.857 = 30(28.57)T_a$$

$$T_a = \frac{2.857}{30(28.57) - 2.857^2} = \boxed{0.00336}$$

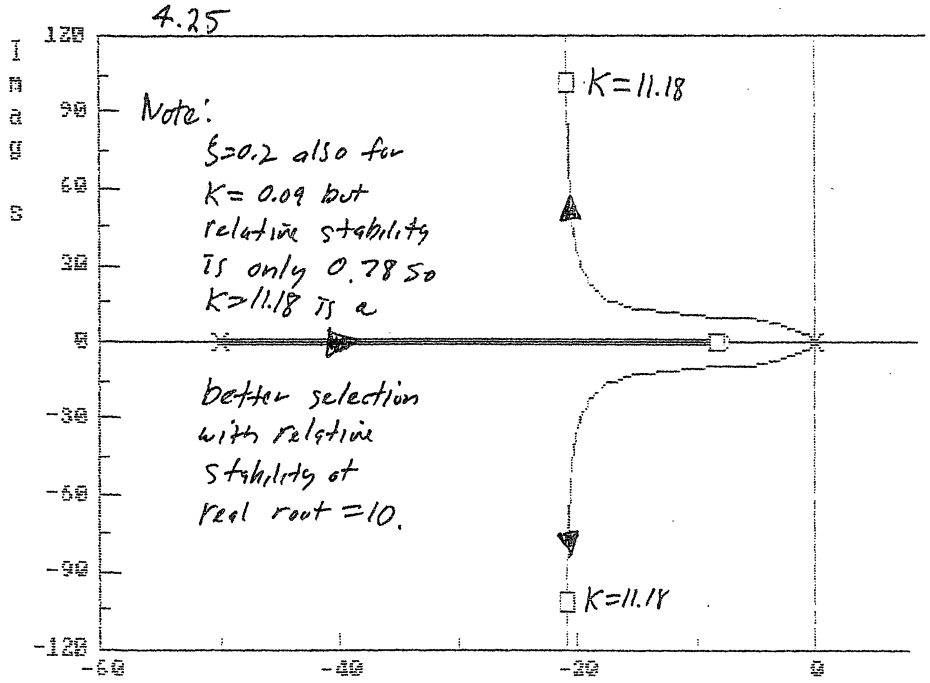
4.22(a). See Text for 4.22(b,c)



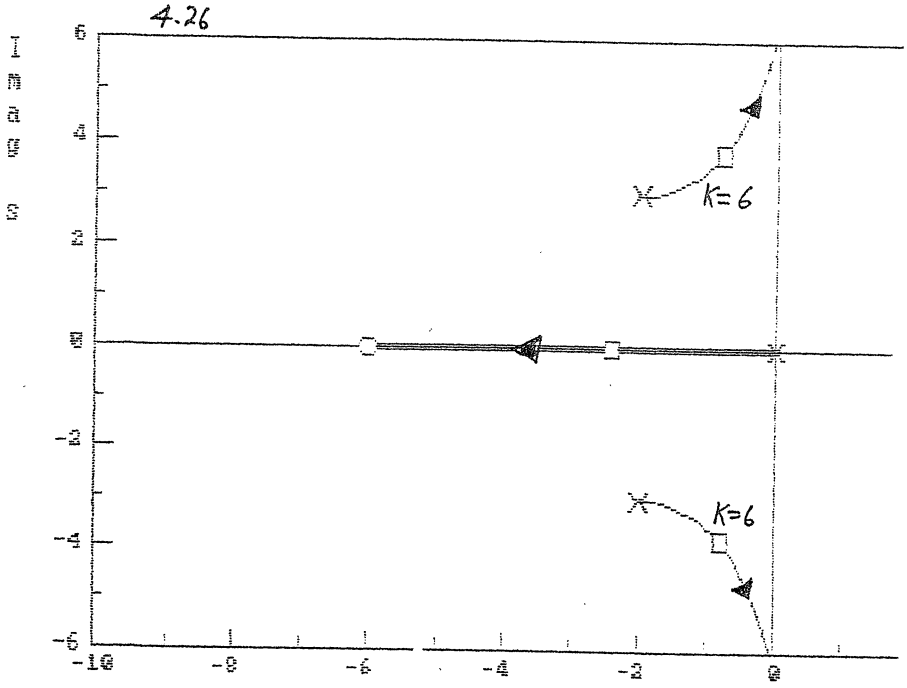
4.23, 4.24



4.25

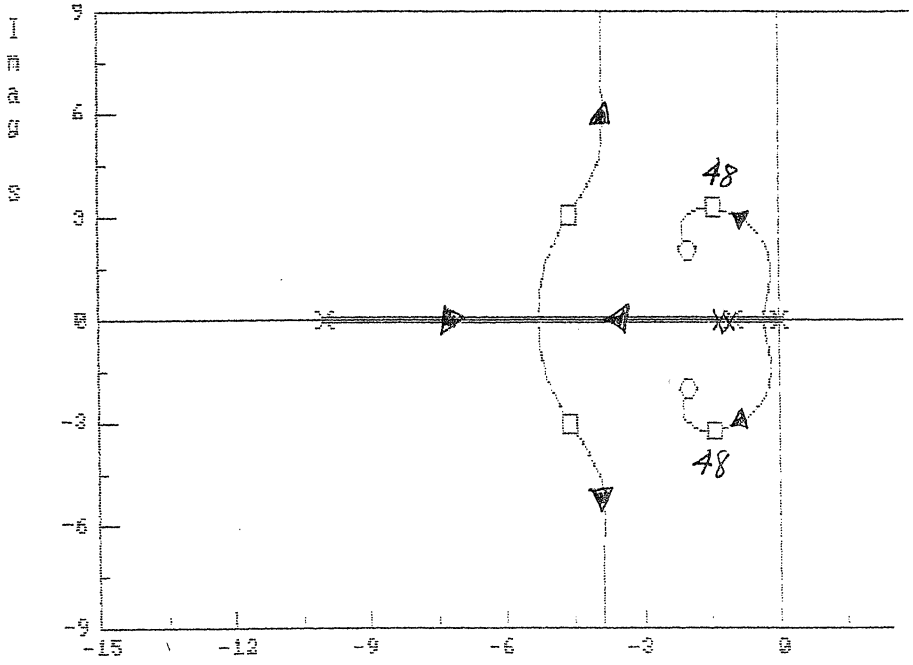

Note:
ζ=0.2 also for
K= 0.09 but
relative stability
is only 0.785o
K=11.18 is a
better selection
with relative
stability of
real root =10.
K=11.18
K=11.18

4.26


K=6
K=6

4.27

(a) $G(s)H(s) = \dfrac{s^2+4s+8}{s(s+1)^2(s+10)}$



$n=4$

$m=2$

$n-m=2$

$\theta_z = \pm 90^\circ$

$\sigma = -4$

$\angle$ arrival at $-2+j2 = 112.2^\circ$

$\angle$ arrival at $-2-j2 = -112.2^\circ$

$s_b = -0.36 \; (K=0.213)$

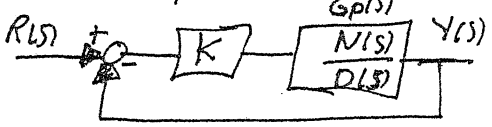
$s_b = -5.3 \; (K=30.9)$

(b) $K=48$ for $\zeta = 0.4$ (dominant roots)

CHAPTER 5

5.1

For this problem, write the system as



$$T(s) = \frac{\frac{KN(s)}{D(s)}}{1+\frac{KN(s)}{D(s)}} = \frac{KN(s)}{\underbrace{D(s)+KN(s)}_{P(s),\text{ the charact. poly.}}}$$

(a) $P(s) = (s+10)+\frac{10}{K}(s+4) = s+10+10s+40$

$= 11s+410$ CL pole $= -\frac{410}{11} = \boxed{-37.3}$

$K_0 = K\,G_p(0) = 10\left(\frac{4}{10}\right) = \boxed{4}$

(b) $P(s) = (s+4)+\frac{10}{K}(s+10) = s+4+10s+100$

$= 11s+104$ CL pole $= -\frac{104}{11} = \boxed{-9.45}$

$K_0 = K\,G_p(0) = 10\left(\frac{10}{4}\right) = \boxed{25}$

(c) $P(s) = (s+2)^2+\frac{10}{K}(s+10) = s^2+4s+4+10s+100$

$= s^2+14s+104$

CL poles $-7 \pm j\,7.42$ $K_0 = \frac{10(10)}{2^2} = \boxed{25}$

(d) $P(s) = (s+10)(s+4)+\frac{58}{K} = s^2+14s+98$

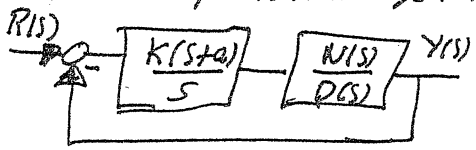
CL poles $= -7 \pm j7$ $K_0 = \frac{58}{(10)(4)} = 1.45$

5.2

As in 5.1, for the uncompensated system,

$P(s) = D(s)+K_0N(s)$.

5.2 (continued)
For the PI compensated system:



$$T(s) = \frac{\frac{K(s+a)N(s)}{sD(s)}}{1+\frac{K(s+a)N(s)}{sD(s)}} = \frac{K(s+a)N(s)}{s[D(s)+KN(s)]+aKN(s)}$$

$$= \frac{num}{1+a\frac{KN(s)}{s[D(s)+KN(s)]}} \qquad G(s)H(s) = \frac{KN(s)}{s[D(s)+KN(s)]}$$

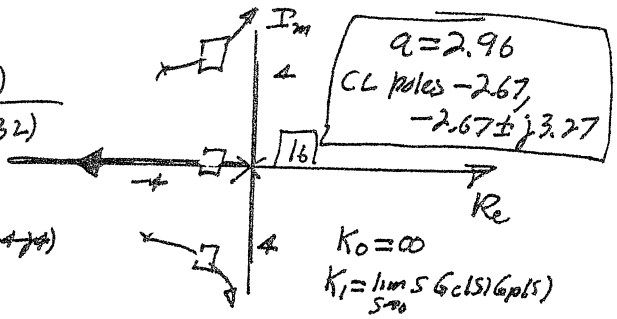
(a) Uncomp: $P(s) = (s+4)^2 + \overbrace{K_u}^{2.67}6 = s^2+8s+\overbrace{16+16}^{32}$

$= s^2+2\zeta\omega_n s+\omega_n^2 \qquad \omega_n = \sqrt{32} = 5.657$

$2\zeta\omega_n = 2\zeta(5.657) = 8 \qquad \zeta = 0.707 \qquad K_0 = 1$

$K = K_u = 2.67$

$$G(s)H(s) = \frac{\overbrace{(2.67)}^{K}(6)}{s(s^2+8s+32)} = \frac{6K}{s(s+4+j4)(s+4-j4)}$$



$K_0 = \infty$

$K_1 = \lim_{s\to 0} sG_c(s)G_{pl}(s)$

$= Ka\frac{N(a)}{D(a)}$

$= (2.67)(2.96)\frac{6}{16}$

$= \boxed{2.96}$

Try increasing K above 2.67

$G_c(s)G_{pl}(s) = \frac{K6(s+2.96)}{s(s+4)^2}$

K	2.67	4	5	6
T_s	1.1	1.02	0.92	1.18

(choose $K = 4$)

(b) $P(s) = (s+4)^3 + \overbrace{K_u}^{40} = s^3+12s^2+48s+104$

CL poles -7.42, $\underbrace{-2.29 \pm j2.96}_{\text{dominant}}$

5.2(b) (continued)

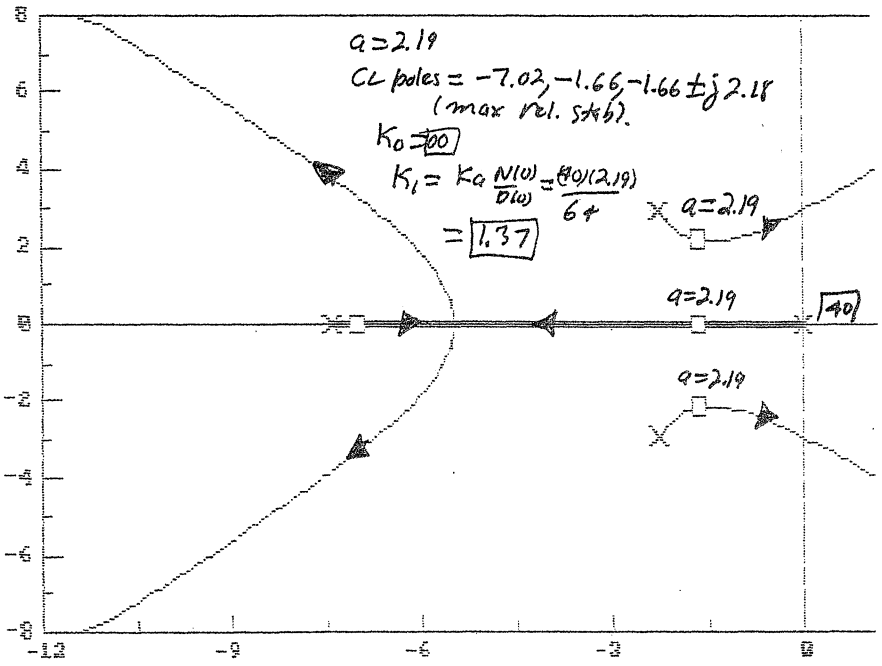
$\omega_n = \sqrt{2.29^2 + 2.46^2} = 3.74$ $\quad \zeta = \frac{\sigma}{\omega_n} = \frac{2.29}{3.74} = 0.61$

$K_0 = K_U \frac{N(0)}{D(0)} = \frac{40}{64} = 0.625$

$K = K_U = 40$

~~[illegible]~~

$$G(s)H(s) = \frac{\overbrace{40}^{K}}{s(s^3 + 12s^2 + 48s + 104)}$$



$$G_c(s)G_p(s) = \frac{K(s+2.19)}{s(s+4)^3}$$

Try increasing K_c

Select $\boxed{K = 43}$

K	40	43	45
T_s	1.3	1.2	1.85

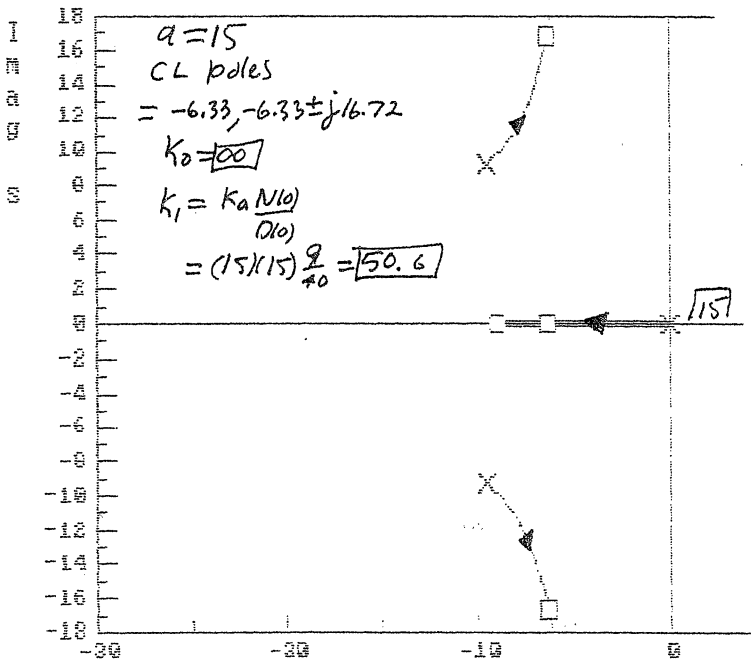
(C) $D(s) = s^2 + 4s + 40 + \frac{15}{K_0}(s+9) = s^2 + 19s + 175$

CL poles $-9.5 \pm j\,9.21$ $\omega_n = \sqrt{175} = 13.22$

$\zeta = \frac{9.5}{13.22} = 0.718$ $K_0 = K_0 \frac{N(0)}{D(0)} = \frac{(15)(9)}{40} = 3.37$

With a variable, $K = 15$

Effective $G(s)H(s) = \dfrac{15[s+9)}{s[s^2 + 19s + 175]}$



With $q = 15$, try increasing K

$G_c(s)G_p(s) = \dfrac{K(s+15)(s+9)}{s(s^2 + 4s + 40)}$

K	15	~~20~~	30	50
T_s	0.40	0.35	0.27	0.12

We can increase K without limit. One CL pole moves to $-\infty$ while other CL poles are cancelled by CL zeros.

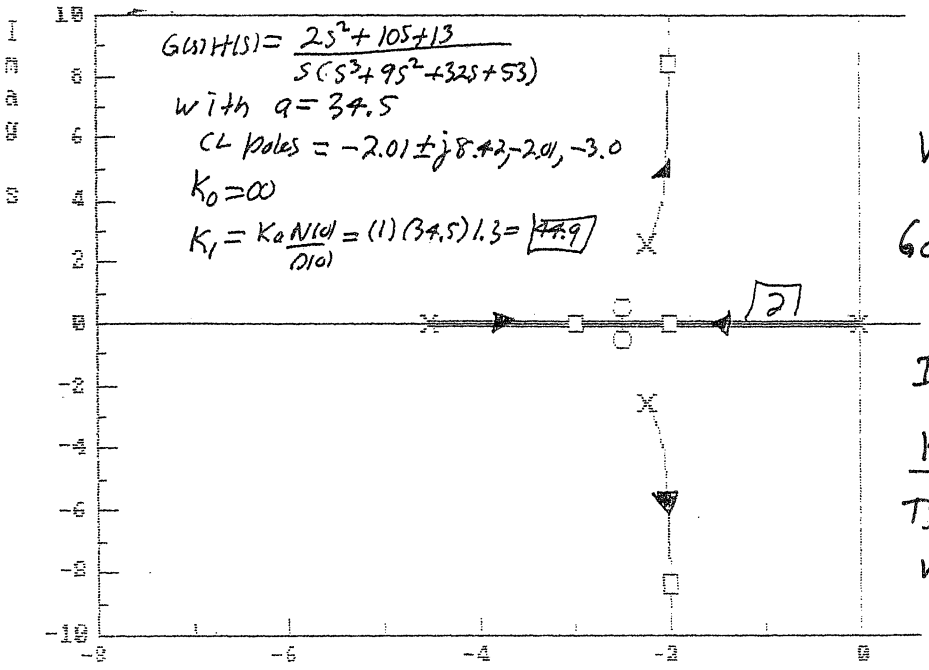
(d)

$$P(s) = (s^3+7s^2+22s+40) + K_0(2s^2+10s+13)$$
$$= s^3+9s^2+32s+53$$

CL poles $-2.24 \pm j2.59$ (dominant), -4.51

$K_0 = \frac{K_0 N(0)}{D(0)} = 1.3$ $\quad \omega_n = \sqrt{2.24^2+2.59^2} = 3.424$ $\quad \zeta = \frac{2.24}{3.424} = 0.65$

$K = K_0 = 1$



With $a = 34.5$

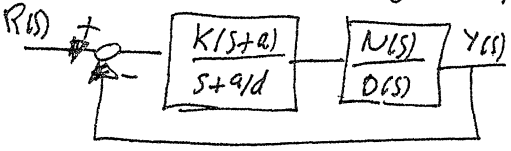
$$G_{cl}(s)G_p(s) = \frac{K(s+34.5)(2s^2+10s+13)}{s(s+4)(s^2+3s+10)}$$

If we increase K with a = 34.5

K	1	5	10
TS	1.53	.42	.27

We can increase K without limit as in 5.2(c).

5.3 The uncomp. system is handled as in 5.2. For a lag comp.



$$T(s) = \frac{\frac{K(s+a)N(s)}{(s+\frac{a}{d})D(s)}}{1+\frac{K(s+a)N(s)}{(s+\frac{a}{d})D(s)}} = \frac{K(s+a)N(s)}{s[D(s)+KN(s)]+\frac{a}{d}[D(s)+KdN(s)]}$$

$$= \frac{num}{1+\frac{a(\frac{1}{d})[D(s)+KdN(s)]}{s[D(s)+KN(s)]}} \qquad G(s)H(s) = \frac{(\frac{1}{d})[D(s)+KdN(s)]}{s[D(s)+KN(s)]}$$

For large d, this becomes a PI comp.

5.3 (continued)

(a) $P(s) = s^2+4s+10+\frac{10}{K_0}(10) = s^2+4s+110$

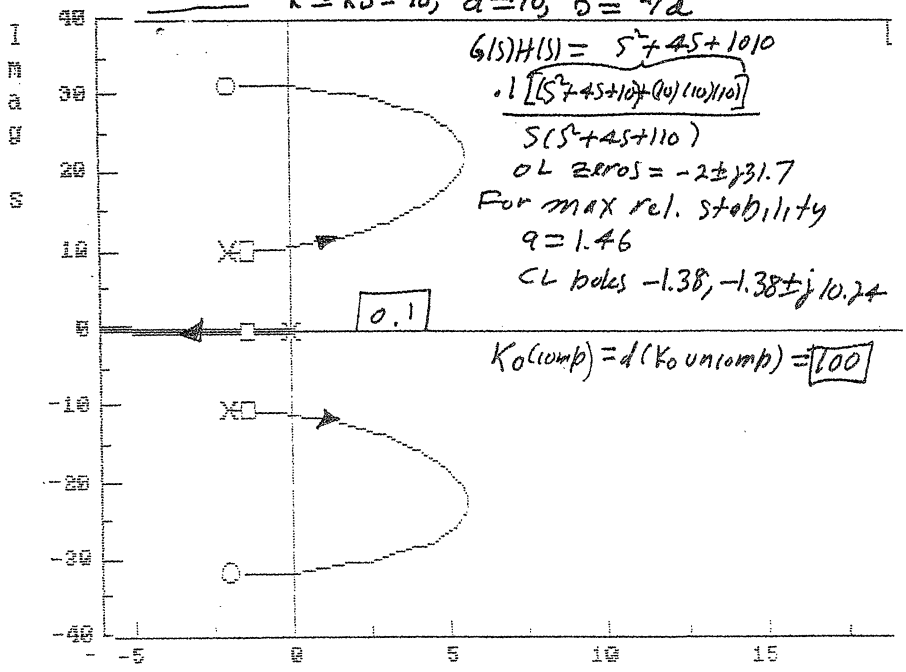
CL poles $= -2 \pm j10.3$

$\omega_n = \sqrt{110} = 10.49, \quad \zeta = \frac{2}{10.49} = 0.191, \quad K_0 = KG_p(0) = 10$

$K = K_0 = 10$

~~$K = \frac{1}{3}(\quad)$~~

$K = K_0 = 10, \; a = 10, \; b = 9/a$



$G(s)H(s) = s^2+4s+1010$

$\cdot 1\,[(s^2+4s+10)+(10)(10)(10)]$

$s(s^2+4s+110)$

OL zeros $= -2 \pm j31.7$

For max rel. stability

$a = 1.46$

CL poles $-1.38, -1.38 \pm j10.24$

$K_0(\text{comp}) = d(K_0 \text{ uncomp}) = \boxed{100}$

with $a = 1.46$, try increasing K

$$G_c(s)G_p(s) = \frac{10K(s+1.46)}{(s+0.146)(s^2+4s+10)}$$

K	10	15	20
T_s	1.92	2.06	202

No improvement for $K > 10$. Select $\boxed{K=10}$

b)

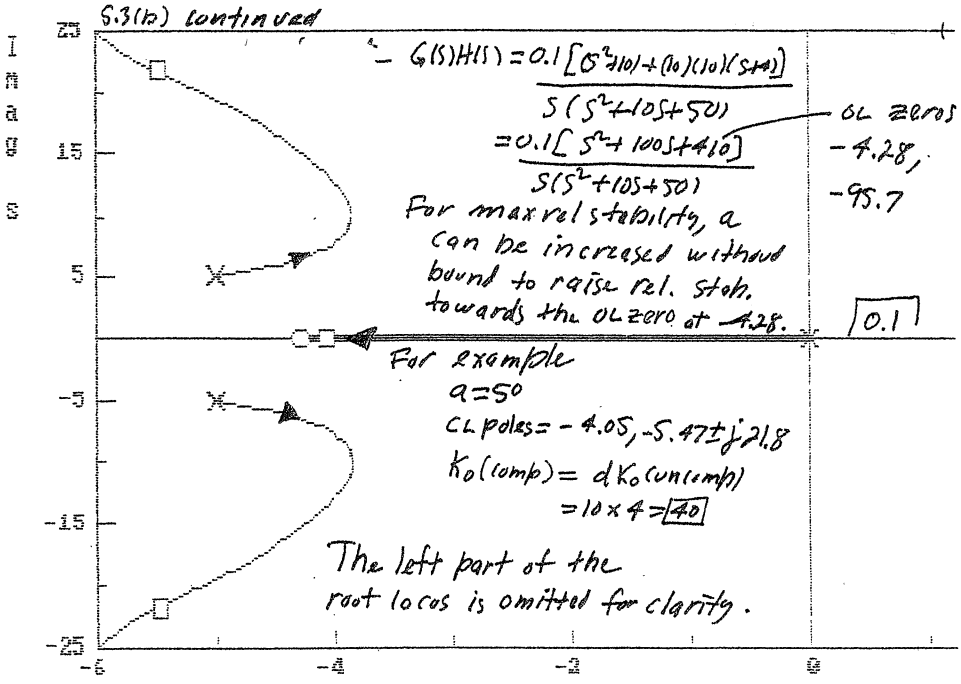
$P(s) = (s^2+10)+\frac{10}{K_0}(s+4) = s^2+10s+50$

CL poles $= -5 \pm j5, \quad \omega_n = \sqrt{50} = 7.07$

$\zeta = \frac{5}{7.07} = 0.707 \quad K_0 = K_0 G_p(0) = 10\left(\frac{4}{10}\right) = 4$

$K = K_U = 10,\ d = 10,\ b = 9/10$

5.3(b) continued



$G(s)H(s) = \dfrac{0.1[s^2+10]+(10)(10)(s+9)]}{s(s^2+10s+50)} = \dfrac{0.1[s^2+100s+910]}{s(s^2+10s+50)}$

OL zeros $-4.28,\ -95.7$

For max rel stability, a can be increased without bound to raise rel. stab. towards the OL zero at -4.28.

0.1

For example

$a = 50$

CL poles $= -4.05,\ -5.47 \pm j21.8$

$K_0(\text{comp}) = dK_0(\text{uncomp})$

$= 10 \times 4 = \boxed{40}$

The left part of the root locus is omitted for clarity.

[c] $P(s) = (s^2+9) + \dfrac{10}{K_U}(2s+20) = s^2+20s+209$

CL poles $= -10 \pm j10.44.\quad \omega_n = \sqrt{209} = 14.47\quad \zeta = \dfrac{10}{14.47} = 0.692$

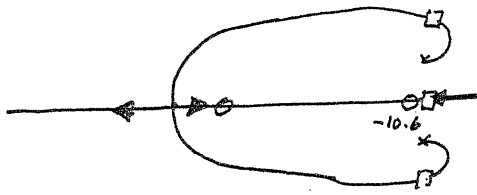
$K_0 = K_U G_p(0) = 10\left(\dfrac{20}{9}\right) = 22.2$

[redacted]

$K = K_U = 10,\ d = 10,\ b = 9/d$

OL zeros $= -10.6,\ -18.9$

$G(s)H(s) = \dfrac{0.1[(s^2+9)+(10)(10)(2s+20)]}{s[s^2+20s+209]} = \dfrac{0.1[s^2+200s+209]}{s(s^2+20s+209)}$



a can be increased without limit to raise rel. stab. towards the OL zero at -10.6.

For $a = 100$

CL poles $-10, -10 \pm j43.7$

$K_0 = 10(22.2) = \boxed{222}$

5.3(d) $P(s) = (s+5)(s^2+4s+10) + \frac{10}{K_0}\,6$

$P(s) = s^3+9s^2+30s+110$ CL poles $\underbrace{-1.02 \pm j3.84}_{\text{dominant}}, -7$

$\omega_n = \sqrt{1.02^2+3.84^2} = 3.97 \quad \zeta = \frac{1.02}{3.97} = 0.257 \quad K_0 = 10\left(\frac{6}{50}\right) = 1.2$

$K = K_0 = 10, \; d = 10, \; \theta = 9/10$

$$G(s)H(s) = \frac{0.1\left[(s^3+9s^2+30s+50)+(10)(10)\,6\right]}{s(s^3+9s^2+30s+110)}$$

$$= \frac{0.1\left[s^3+9s^2+30s+650\right]}{s(s^3+9s^2+30s+110)}$$

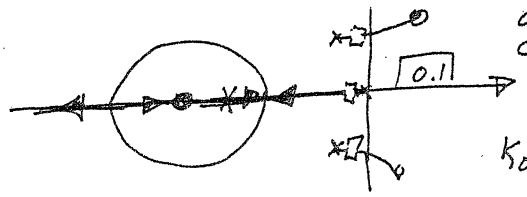
OL zeros $-11.38, +1.19 \pm j7.46$

max rel stab

$a = 1.13$

CL poles $-0.78, -0.78 \pm j3.65, -6.7$

$K_0 = 10 \times 1.2 = \boxed{12}$



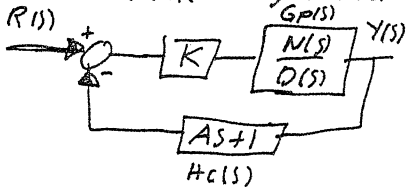
With $a = 1.13$, try increasing K

$$G_c(s)G_p(s) = \frac{6K(s+1.13)}{(s+0.113)(s+5)(s^2+4s+10)}$$

K	10	11	12	15
T_s	3.76	3.73	3.73	5.00

No improvement for $K > 10$. Select $\boxed{K = 10}$

5.4 The uncomp. system is handled as in 5.2. For rate feedback to get root locus for variable A



$$T(s) = \frac{K\frac{N(s)}{D(s)}}{1+K\frac{N(s)}{D(s)}(As+1)} = \frac{KN(s)}{D(s)+KN(s)+AsKN(s)} = \frac{num}{1+A\frac{sKN(s)}{D(s)+KN(s)}}$$

$$G(s)H(s) = \frac{sKN(s)}{D(s)+KN(s)}$$

$P(s) = (s^3+4s^2+4s) + \frac{20}{K_0}(s+8)$

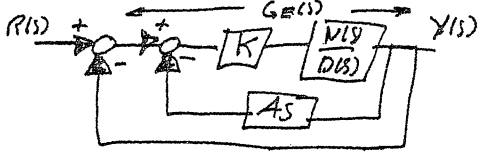
$= s^3+4s^2+6s+160$

CL poles $= -0.589 \pm j7.51, -2.82$

$K_0 = \infty \quad K_1 = \lim_{s\to 0} s\,K_0\,G_p(s) = 20\left(\frac{8}{40}\right) = 4$

5.4 (continued)

For the error coefficient



$$G_E(s) = \frac{KN(s)/D(s)}{1 + KN(s)As/D(s)} = \frac{KN(s)}{D(s) + KAsN(s)} = \frac{K(s+8)}{s(s^2+4s+40) + KAs(s+8)}$$

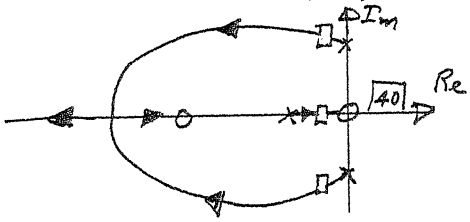
$$= \frac{K(s+8)}{s[s^2 + s(4+KA) + 40 + 8KA]}$$

$$K_1(\text{comp}) = \lim_{s\to 0} s\,G_E(s) = \frac{K8}{40+8KA} = \frac{K\frac{8}{40}}{1+\frac{8}{40}KA}$$

Choose $K = 2K_U = 40$ to maintain K_1 (comp)

$$G(s)H(s) = \frac{s(40)(s+8)}{s^3+4s^2+40s+40(s+8)} = \frac{40s(s+8)}{s^3+4s^2+80s+320}$$

OL poles $= -4, \pm j\,8.9$



For max. rel. stab.

$A = 0.1325$

CL poles

$-3.10, -3.10 \pm j\,9.67$

$$K_1(\text{comp}) = \frac{40\frac{8}{40}}{1+\frac{8}{40}(40)(.1325)} = 3.85$$

rel. stab. increases from 0.59 to 3.1

K_1 is nearly the same.

5.5

The uncomp. system may be handled as in Prob 5.2

$$P(s) = (s+4)(s^2+10s+50) + \overbrace{K_U}^{600} = s^3+14s^2+90s+\overbrace{200}^{800}+600$$

CL poles $= \underbrace{-0.98 \pm j\,8.09}_{\text{dominant}}, -12$ $\quad \omega_n = \sqrt{.98^2+8.09^2} = 8.15$

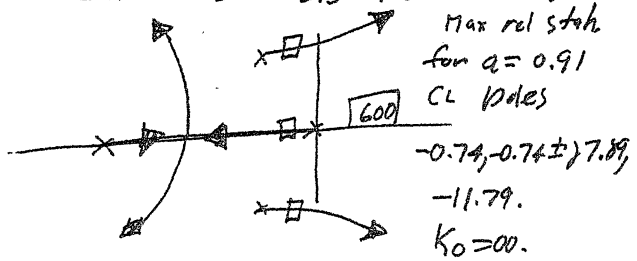
$$\zeta = \frac{0.98}{8.15} = 0.120 \qquad K_0 = K_U G_p(0) = \frac{600}{200} = 3$$

5.5 (continued)

$K = K_0 = 600$, $A = 0$

To get a root locus for variable a, see prob 5.2

$$G(s)H(s) = \frac{KN(s)}{s[D(s)+KN(s)]} = \frac{600}{s(s^3+14s^2+9as+800)}$$



Max rel stab for $a = 0.91$

CL poles

$-0.74, -0.74 \pm j\,7.89$,

-11.79.

$K_0 = 00$.

Now rel. stab. drops to 0.74 but $K_0 = \infty$

We must be careful in increasing K.

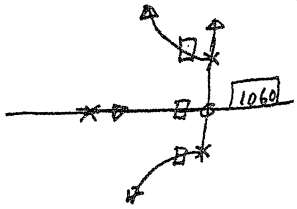
$P(s) = s^3 + \underbrace{14}_{a_2}s^2 + \underbrace{90}_{a_1}s + \underbrace{200}_{a_0} + K$, for uncomp system.

If we use the result of prob 5.4 to get $G(s)H(s)$ for a root locus as A varies,

$$G(s)H(s) = \frac{s\,KN(s)}{D(s)+KN(s)}$$

Now at marg. stab. $9a_2 = a_0$ so $1260 = 200 + K$

$K = 1060$ so try $G(s)H(s) = \dfrac{s\,1060}{s^3+14s^2+90s+1260}$



Max rel. stab. for CL poles $\pm j9.5, -14$

$A = 0.186$, CL poles $-4.66 \pm j\,15.7, -4.66$.

rel stab is now 4.66.

$q=0.91,\ K=1060,\ A=0.186$ — CL zeros $-1.16, -4.22$

$$G(s) = \frac{KA\left[s^2+\frac{1}{A}s+\frac{q}{A}\right]}{s} = \frac{K(1.186)\left[s^2+5.385+4.89\right]}{s}$$

with $K=1060$ { CL zeros $-1.16, -4.22$; CL poles $-0.97, -4.18, -4.42 \pm j14.7$ } $T_s = 1.28\,s$

with $K=2120$ { CL zeros $-1.16, -4.22$; CL poles $-1.05, -4.20, -4.37 \pm j20$ } $T_s = 0.7\,s$

With $K=2120$ there are CL pole-zero cancellations leaving dominant poles at $-4.37 \pm j20$. Compared to uncomp case, rel. stab rises from 0.74 to 4.37 and K_0 goes from 3 to ∞. Higher K does not reduce T_s.

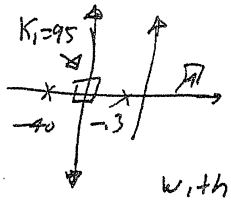
5.6

$$T(s) = \frac{40\left(\frac{.1}{s+.3}\right)\left(\frac{1}{s}\right)}{1+40\left(\frac{.1}{s+.3}\right)\left(\frac{K_1}{s+40}\right)+40\left(\frac{.1}{s+.3}\right)\frac{1}{s}\frac{K_2(s+6)}{s+4}}$$

$$= \frac{4(.1)(s+4)(s+40)}{s(s+.3)(s+4)(s+40)+4K_1s(s+4)+4K_2(s+6)(s+40)}$$

For $K_2=0$, variable K_1

$$G(s)H(s) = \frac{4s(s+4)}{s(s+.3)(s+4)(s+40)} = \frac{4}{(s+.3)(s+40)}$$

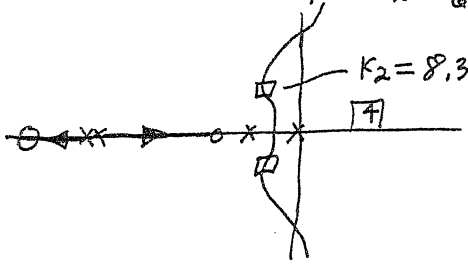


If $K_1 = 98.5$, CL poles are $-20.15 \pm j20.15$ (max rel. stab)

For Fixed K_1, variable K_2

$$G(s)H(s) = \frac{4(s+6)(s+40)}{s[s+4][(s+.3)(s+4)+4K_1]}$$

with $K_1 = 98.5$ $\quad G(s)H(s) = \frac{4(s+6)(s+40)}{s(s+4)(s+20.15)^2}$



$K_2 = 8.32$, CL poles

$-3.6 \pm j3.6$ ($\zeta = 0.7$)

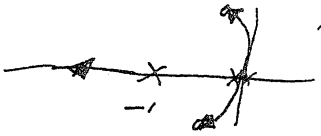
$-12.7, -24$

$\boxed{K_1 = 98.5,\ K_2 = 8.32}$

5.7

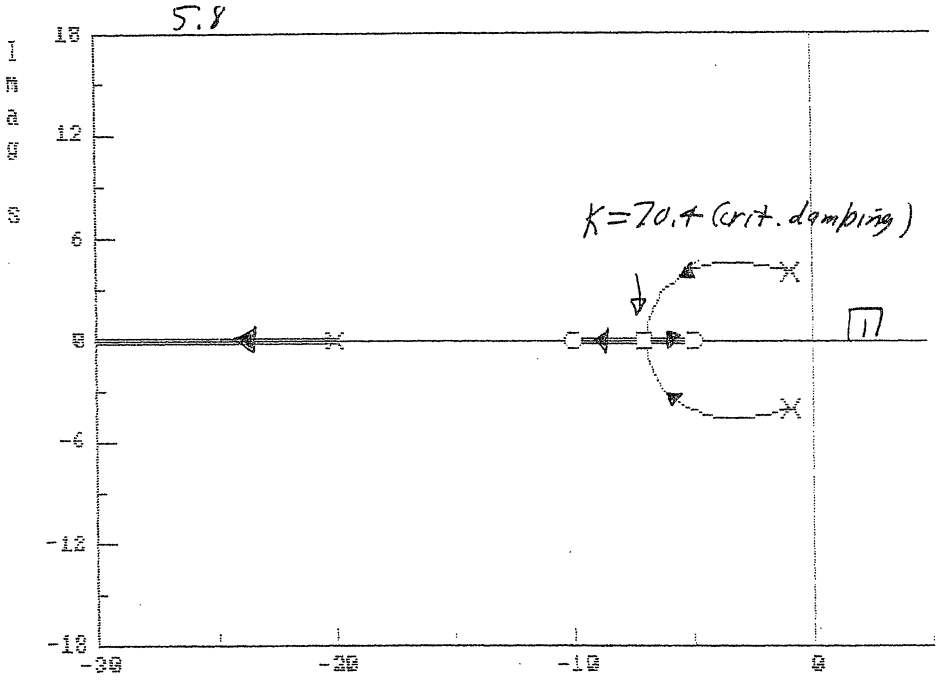
$$T(s) = \frac{I(s)}{R(s)} = \frac{\frac{1}{s}\frac{K}{(s+1)(s+1)}}{1+\frac{1}{s}\frac{K}{(s+1)(s+1)}} = \frac{K}{s^2(s+1)+s(s+1)K+K}$$

$$= \frac{K}{s^2(s+1)+K(s^2+s+1)} \qquad G(s)H(s) = \frac{s^2+s+1}{s^2(s+1)}$$

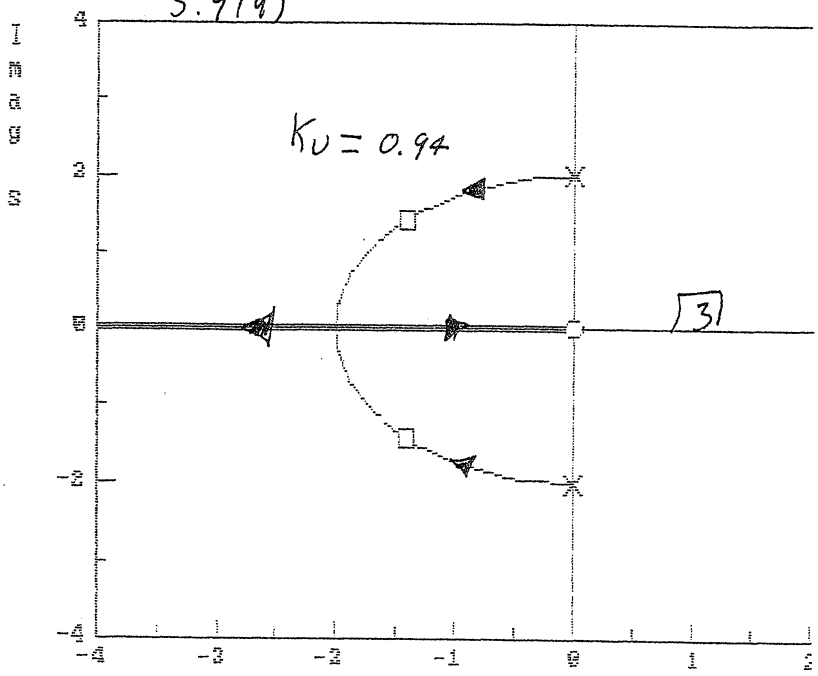


for variable K
max rel stability occurs for K=∞.
Max allowable K would be $\frac{1}{\tau_{min}}$
$\tau_{min} = 1$ Max $\boxed{K=1}$

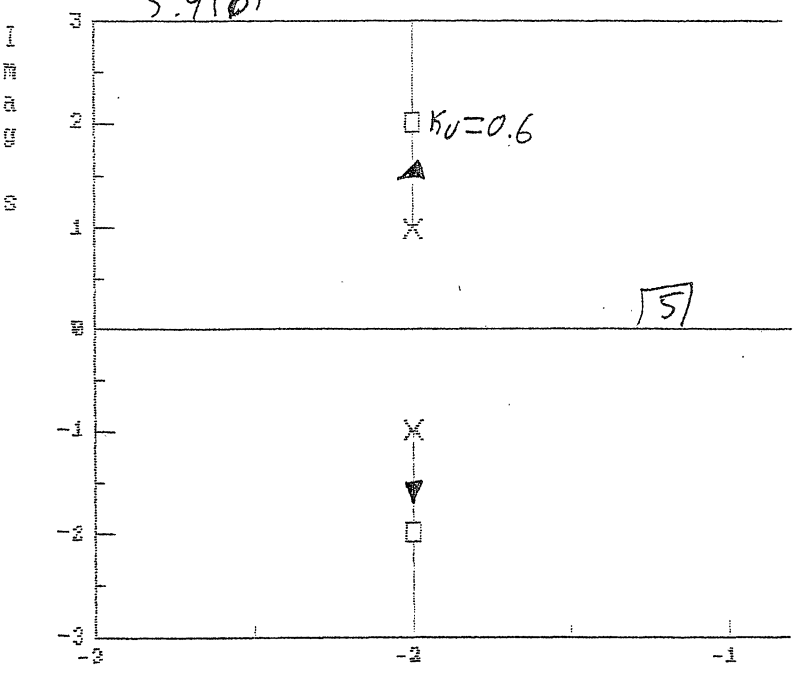
5.8



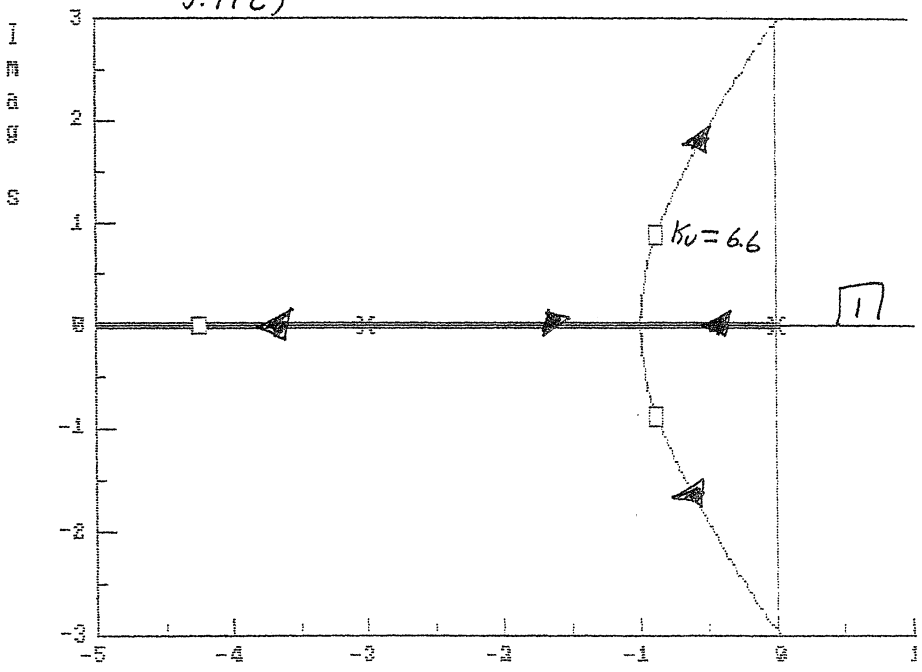
5.9(a)



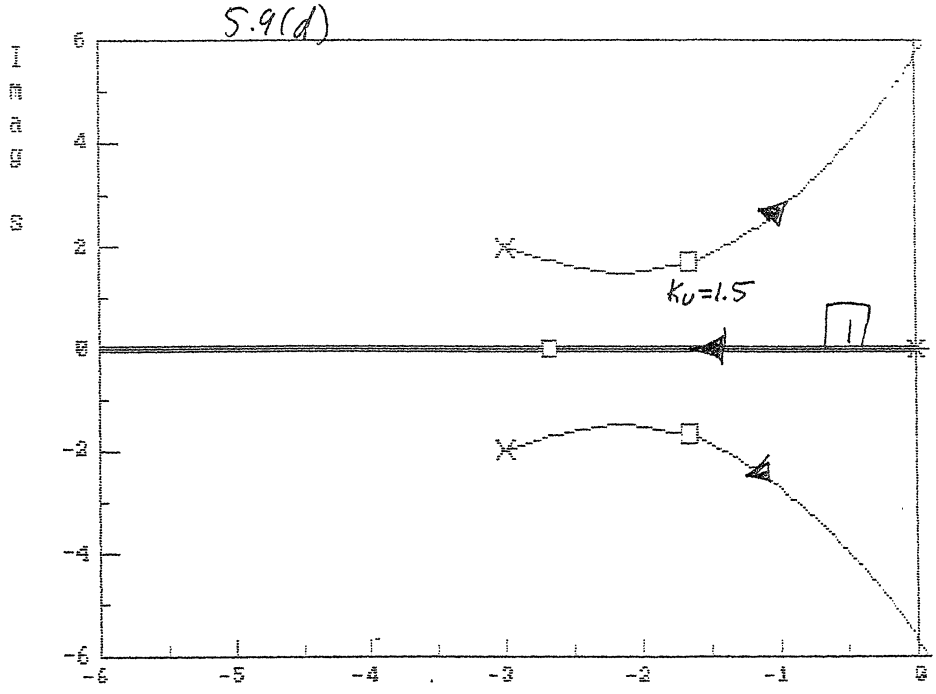
5.9(b)



5.91(c)



5.9(d)



5.10 $G_p(s) = \dfrac{s+3}{(s+2)(s+2+j0.5)(s+2-j0.5)} = \dfrac{s+3}{(s+2)(s^2+4s+4.25)}$

$K_0 = K G_p(0) = K\left(\frac{3}{2\times 4.25}\right) = 0.35K$ $\quad e_{step}(\infty) = \frac{1}{1+K_0} = \frac{1}{1+.35K}$

$e_{step}(\infty) \le 0.5$ $\quad \frac{1}{1+.35K} \le 0.5$ $\quad 1+0.35K \ge 2$ $\quad .35K \ge 1$ $\quad K \ge 2.83$

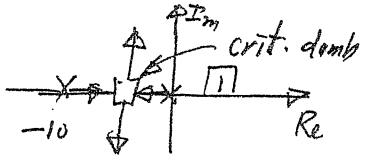
$T(s) = \frac{KG_p(s)}{1+KG_p(s)}$ For each K, get damping ratio using MATLAB command damp(t).

K	2.83	4	6	6.6
ζ	0.65	0.59	0.52	0.5

Need $K \ge 2.83$, $K \le 6.6$ $\Rightarrow$ $\boxed{2.83 \le K \le 6.6}$

5.11

$G(s)H(s) =$



For crit damping, CL poles are at -5

$P(s) = (s+5)^2 = s(s+10)+K$

$s^2+10s+25 = s^2+10s+K$ $\quad \boxed{K=25}$

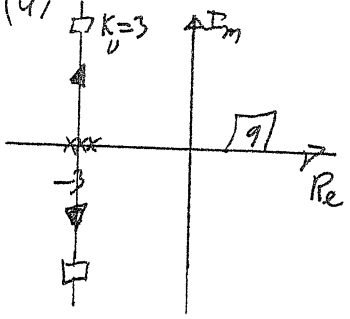
Since system is type 1, $K_0 = \infty$, $e_{STEP}(\infty) = 0$

$K_1 = \lim_{s\to 0} s\left(\frac{K}{s+10}\right)\left(\frac{1}{s}\right) = \frac{K}{10}$

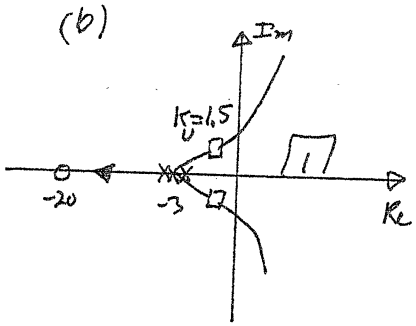
$e_{ramp}(\infty) = \frac{0.2}{K_1} = 0.2\left(\frac{10}{K}\right) = \boxed{\frac{2}{K}}$

5.12

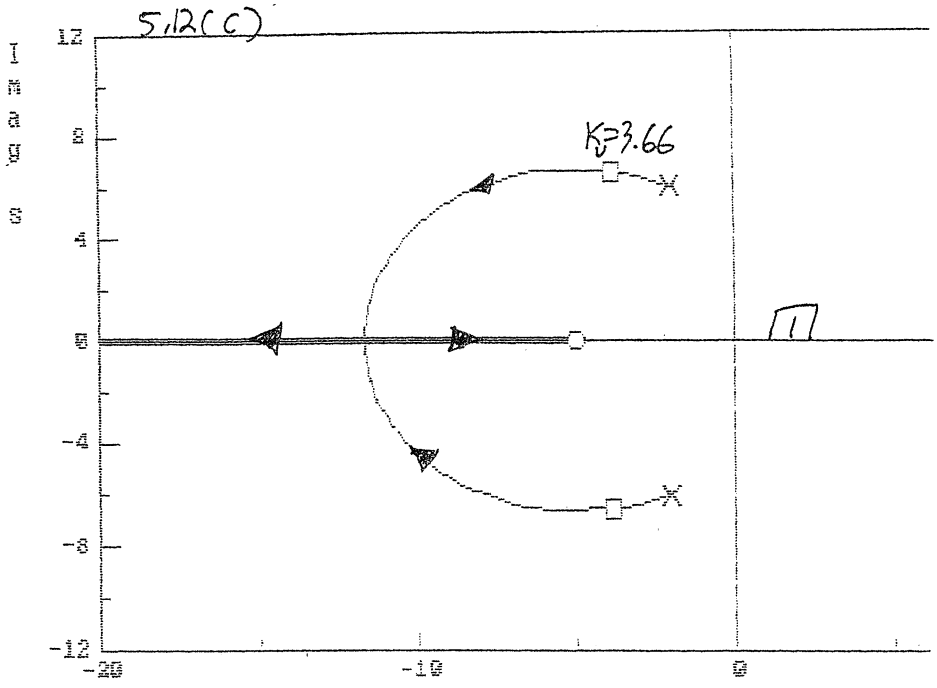
(a)



(b)



5.12(c)



5.13

See Problem 5.2 for the nomenclature.

(a) $P(s) = s^2+7s+10+K_U(10) = s^2+7s+11$

CL poles $-2.38, -4.62$ $\omega_n = \sqrt{11} = 3.32$, $2\zeta\omega_n = 7$

$2\zeta(3.32) = 7$ $\zeta = 1.05$

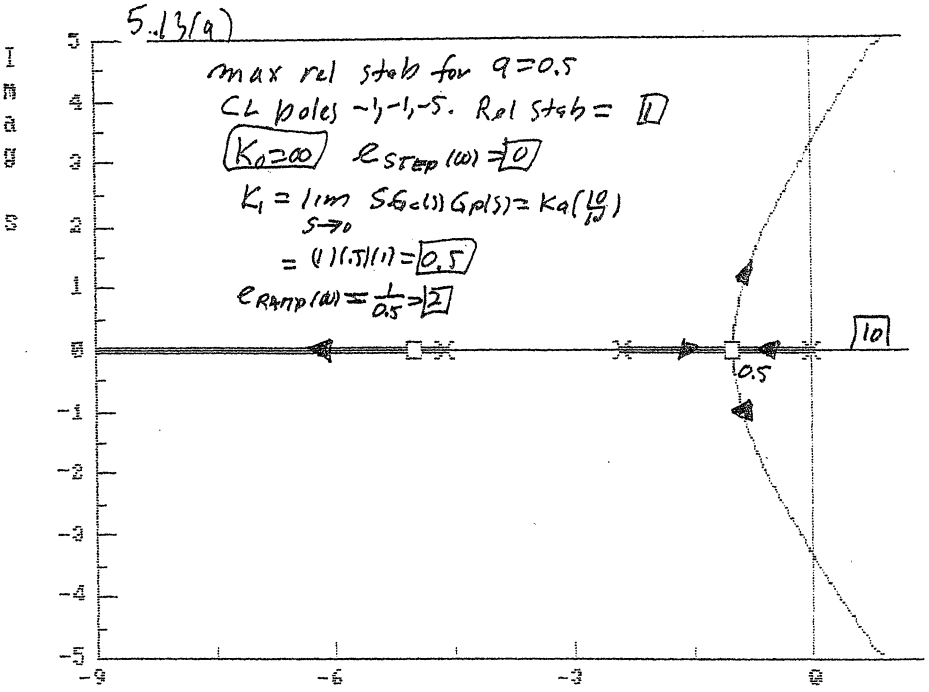
~~The value for K_U was determined for ζ but ω_n will not be correct here for $\zeta = 1.05$~~

$K = K_U = 1$

$$G(s)H(s) = \frac{KN(s)}{s[D(s)+KN(s)]} = \frac{10}{s(s^2+7s+11)}$$

OL poles $-2.38, -4.62$

5.13(a)



max rel stab for $q=0.5$

CL poles $-1, -1, -5$. Rel stab = 11

$K_0 = \infty$ $e_{STEP}(\infty) = 0$

$K_1 = \lim_{s\to 0} sG_c(s)G_p(s) = K_a\left(\frac{10}{10}\right)$

$= (1)(.5)(1) = 0.5$

$e_{RAMP}(\infty) = \frac{1}{0.5} = 2$

(b)

$P(s) = s(s^2+4s+3) + K_U(s+2) = s^3+4s^2+5s+4$

CL poles $= \underbrace{-0.65 \pm j\,1.03}_{\text{dominant}}, -2.70$ $\omega_n = \sqrt{.65^2+1.03^2} = 1.23$

$\zeta = \frac{0.65}{1.23} = 0.526$ type 1 so $K_0 = \infty$

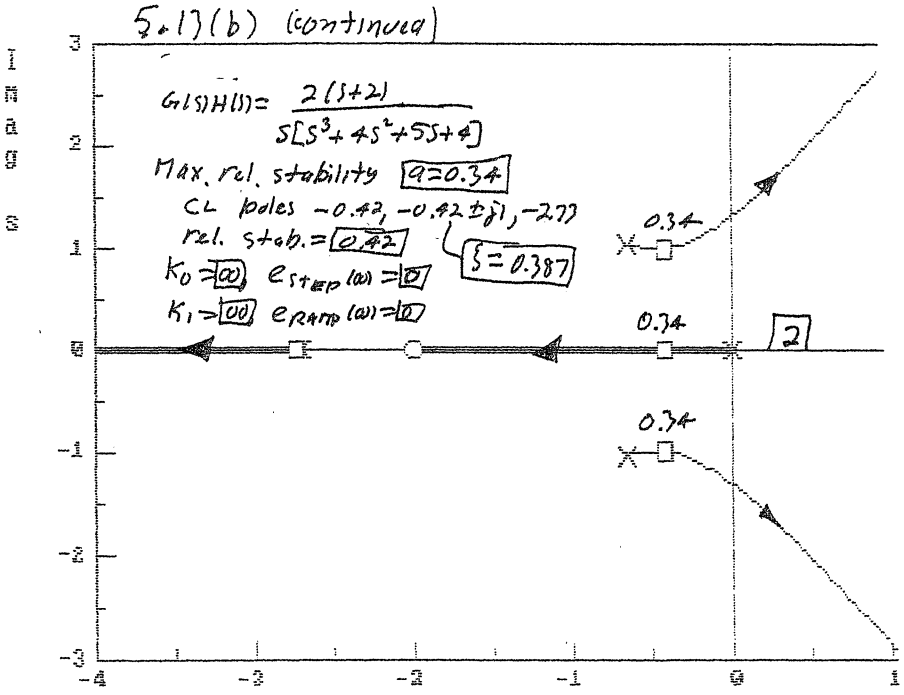
$K_1 = \lim_{s\to 0} K_U G_p(s) = (2)\left(\frac{2}{3}\right)$

$= 1.33$

$K = K_U = 2$

[redacted]

5.17(b) (continued)



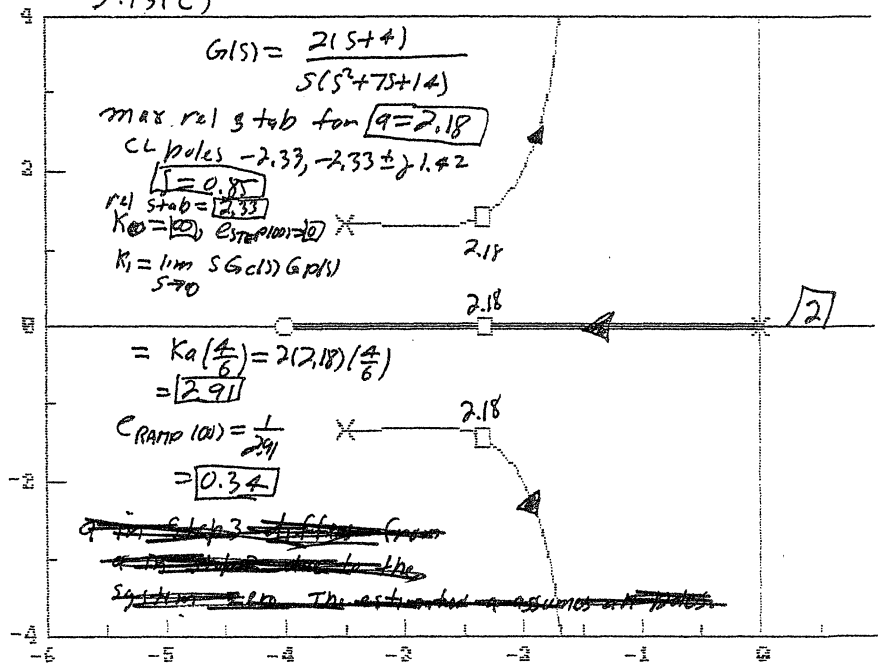
(c) $P(s) = s^2+5s+6+K_v(s+4) = s^2+7s+14$

CL poles $= -3.5 \pm j1.32$ $\omega_n = \sqrt{14} = 3.74$

$\zeta = \frac{3.5}{3.74} = 0.936$ $K_0 = K_v(6p10) = 2(\frac{4}{6}) = 1.33$

$K = K_v = 2$

5.13(c)



(d)

$P(s) = s^2 + 4s + 13 + K_0(6s+30) = s^2 + 10s + 43$

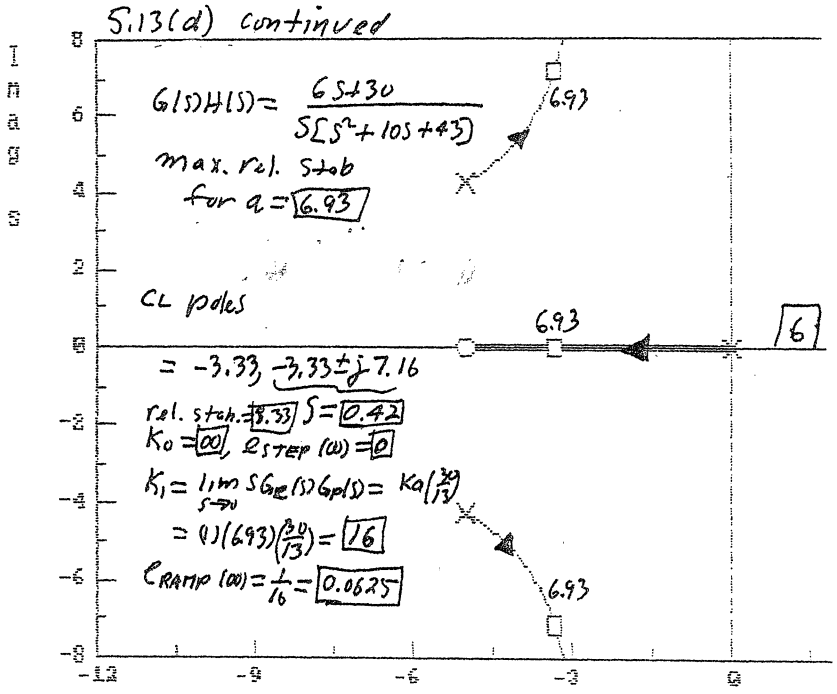
CL poles $= -5 \pm j\,4.24$, $\omega_n = \sqrt{43} = 6.56$

$\zeta = \frac{5}{6.56} = 0.914$

$K_0 = K_0\, G_{p(0)} = 1\,\frac{30}{13} = 2.31$

$K = K_0 = 1$

5.13(d) continued



5.14

See Problems 5.1 and 5.3 for nomenclature

(a) $P(s) = s(s^2+4s+13) + \frac{4}{K_U}(3s+30) = s^3+4s^2+25s+120$

CL poles $= 0.22 \pm j\,5.19, -4.45$ $\quad K_0 = \infty$

$K_1 = \lim_{s\to 0} s\,K_U G_{OL}(s) = 4\left(\frac{30}{13}\right) = 9.23$

the uncompensated system is unstable, therefore a lag-comp system would also be unstable so a lag comp would not be used here.

5.14 (b)

$$P(s) = s^2 + 10s + \overbrace{K_0}^{20}(3) = s^2 + 10s + 60$$

CL poles $= -5 \pm j\,5.92$. $\omega_n = \sqrt{60} = 7.78$

$\zeta = \frac{5}{7.78} = 0.6425$ $\quad K_0 = \infty$ (type 1)

$K_1 = \lim_{s \to 0} s\,K_0\,G_p(s) = 20\left(\frac{3}{10}\right) = 6$

~~$\alpha = \zeta\omega_n$ cos θ = 0.5 ... = 2.2~~

$d = 10$, $K = K_0 = 20$, $b = 9/10$

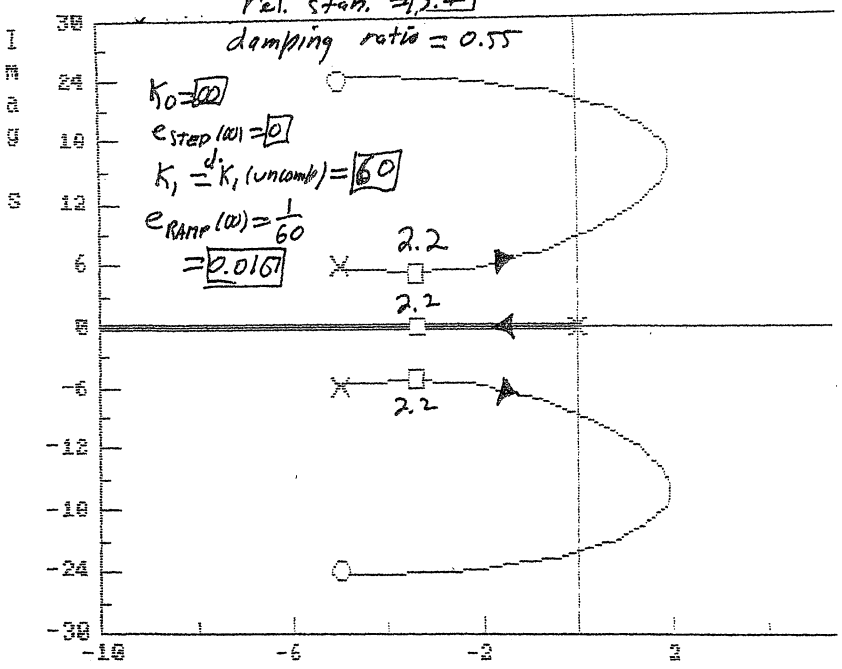
OL zeros $-5 \pm j2.4$

$$G(s)H(s) = \frac{0.1\left[s^2 + 10s + (20)(10)(3)\right]}{s\left(s^2 + 10s + 60\right)} = \frac{0.1\left(s^2 + 10s + 600\right)}{s\left[s^2 + 10s + 60\right]}$$

For max. rel. stability $\alpha = 2.2$

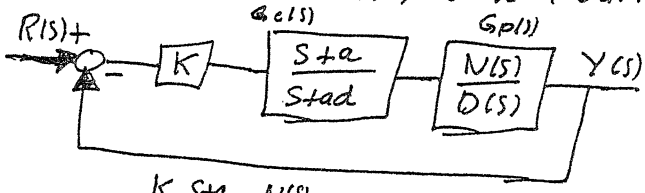
CL poles $-3.4 \pm j\,5.19, -3.4$

rel. stab. $= \boxed{3.4}$



5.15

For the root locus with a as a variable



$d > 1$

$$T(s) = \frac{K\frac{s+a}{s+ad}\frac{N(s)}{D(s)}}{1+K\frac{s+a}{s+ad}\frac{N(s)}{D(s)}} = \frac{K(s+a)N(s)}{s[D(s)+KN(s)]+ad[D(s)+\frac{1}{d}KN(s)]}$$

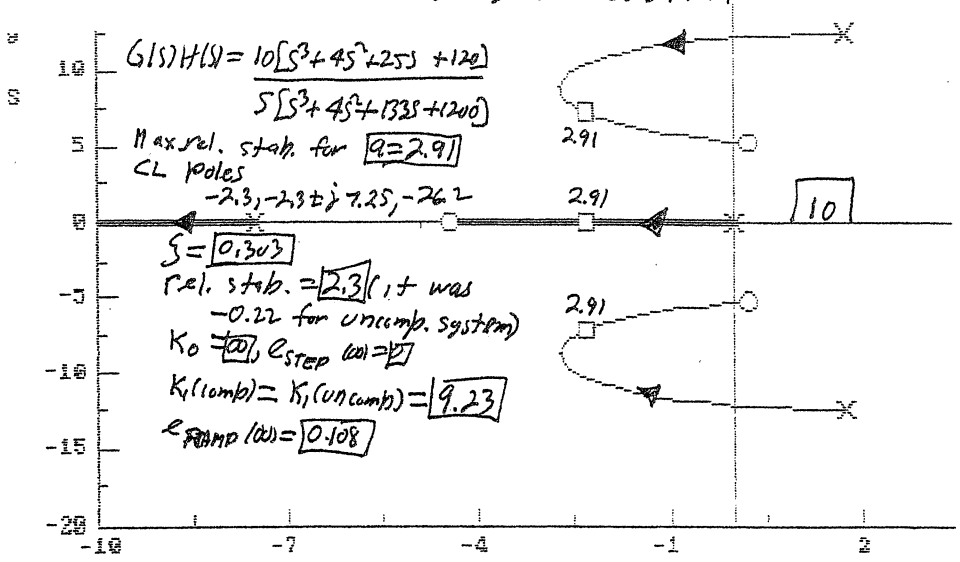
$$= \frac{num}{1+a\frac{d[D(s)+\frac{1}{d}KN(s)]}{s[D(s)+KN(s)]}} \qquad G(s)H(s) = \frac{d[D(s)+\frac{1}{d}KN(s)]}{s[D(s)+KN(s)]}$$

When $d=0$, we have PI comp.

(a) See Prob 5.14 (a)

$d = 10$ $\quad K = K_{cd} = K = (4)10 = 40$

$$G(s)H(s) = \frac{10[s(s^2+4s+13)+0.1(40)(3s+30)]}{s(s^3+4s^2+13s+40(3s+30))}$$



5.15 (continued)

(b) See Problem 5.14(b).

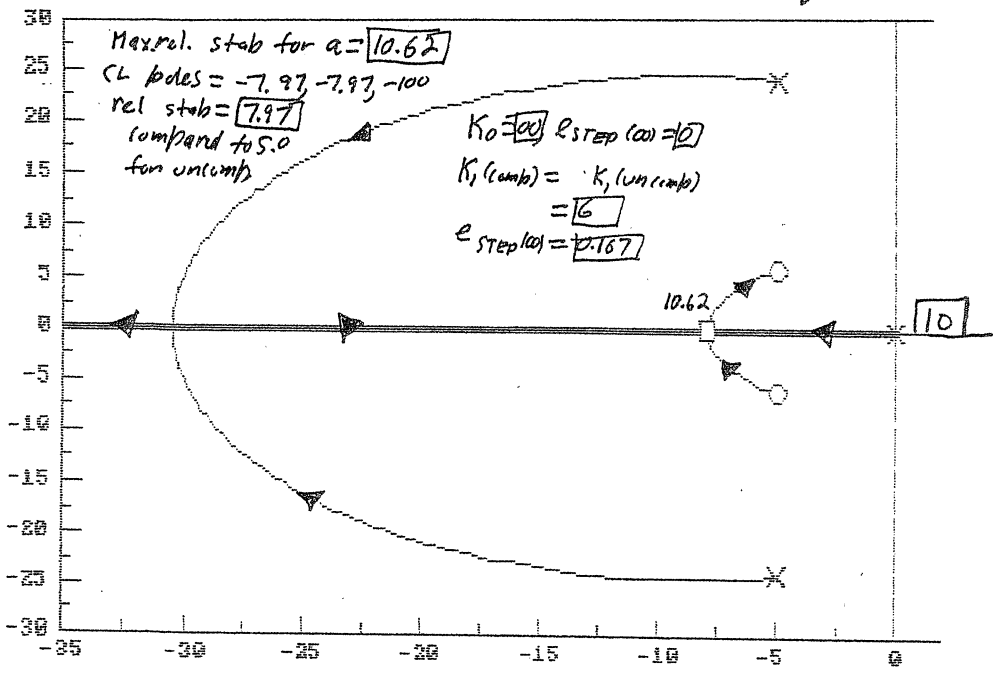
$d = 10$ $\quad K = K_0 d = 20(10) = 200$

$$G(s)H(s) = \frac{10[s^2 + 10s + (0.1)(200)(3)]}{s^2 + 10s + 200(3)}$$

$$= \frac{10[s^2 + 10s + 60]}{s^2 + 10s + 600}$$

OL zeros $-5 \pm j5.92$

OL poles $-5 \pm j24$



5.16

(a) A lag-lead comp would not be used here due to uncomp. system being unstable.

(b) After

$$G_{cl}(s) = K \frac{(s+2.2)}{(s+0.22)_{lag}} \frac{(s+10.62)}{(s+106.2)_{lead}}$$

115

5.16(b) (continued)

$$G_c(s)G_p(s) = K\frac{(s+2.2)(s+10.62)}{(s+0.22)(s+106.2)}\cdot\frac{3}{s(s+10)}$$

CL zeros $-2.2, -10.62$

Gain	CL Poles	T_s
200	$-3.35 \pm j\,1.93,\ -9.92,\ -100.4$	1.12
400	$-2.75,\ -9.95 \pm j\,3.09,\ -93.8$	0.70
1000	$-2.4,\ -11,\ -51.5 \pm j\,5$	0.19
2000	$-2.3,\ -10.8,\ -51.7 \pm j\,55$	0.08

CL poles at $-2.3, -10.8$ cancelled by CL zeros

dominant response $-51.7 \pm j55$ has about 0.7 damping

$K_0 = \infty$ $e_{STEP}(\infty) = 0$ $K_1 = 3K/10 = 600$

$e_{RAMP}(\infty) = \frac{1}{600}$

5.17

See Problems 5.1 and 5.4 for nomenclature.

(a) $P(s) = s(s+10)^2 + \frac{250}{K_U} = s^3 + 20s^2 + 100s + 250$

CL poles $-2.9 \pm j3.03, -14.2$

$K_0 = \infty$ $K_1 = \lim_{s\to 0} s K_U G_p(s) = \frac{250}{100} = 2.5$

$$G_e(s) = \frac{KN(s)}{D(s)+KAsN(s)} = \frac{K}{s(s+10)^2 + KAs}$$

$$= \frac{K}{s[s^2+20s+100+KA]} \qquad K_0 = \infty$$

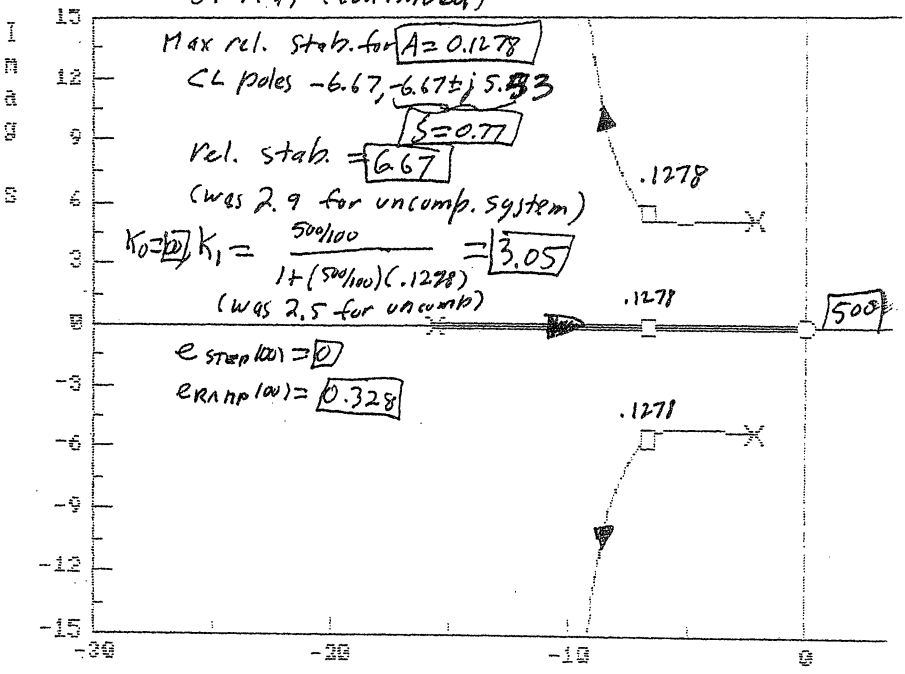
$$K_1 = \lim_{s\to 0} sG_e(s) = \frac{K}{100+KA} = \frac{K/100}{1+(K/100)A}$$

Try $K = 2K_U = 500$

$$G(s)H(s) = \frac{sKN(s)}{D(s)+KN(s)} = \frac{500s}{s^3+20s^2+100s+500}$$

CL poles $-15.65, -2.17 \pm j\,5.2$

5.17(a) (continued)



Max rel. stab. for $A = 0.1278$

CL poles $-6.67, -6.67 \pm j\,5.33$

$\zeta = 0.77$

rel. stab. $= 6.67$

(was 2.9 for uncomp. system)

$K_0 = 10$, $K_1 = \dfrac{500/100}{1+(500/100)(.1278)} = 3.05$

(was 2.5 for uncomp)

$e_{STEP(100)} = 0$

$e_{RAMP(100)} = 0.328$

(b)

$$P(s) = s^2+7s+10 + \frac{1}{K_U}(10) = s^2+7s+20$$

CL poles $-3.5 \pm j\,2.78$

$K_0 = K_U G_p(0) = 1\left(\frac{10}{10}\right) = 1 \qquad K_1 = 0$

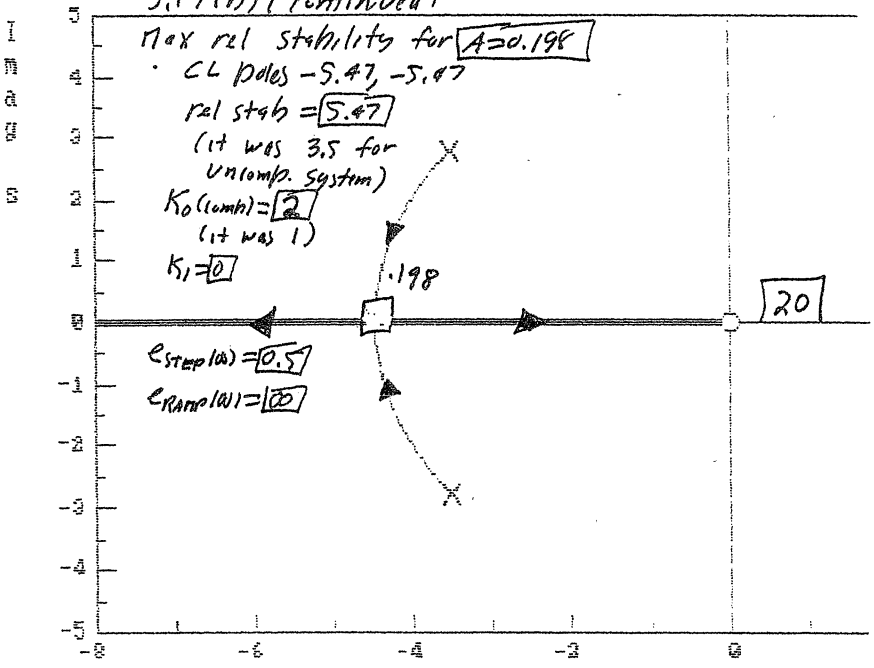
$$G_{cl}(s) = \frac{KN(s)}{D(s)+KA_s N(s)} = \frac{10K}{s^2+7s+10+10KA_s}$$

$$= \frac{10K}{s^2+s(7+10KA)+10} \qquad K_0 = \frac{10K}{10} = K$$

Try $K = 2K_U = 2$ so that $K_0 = 2$

$$G(s)H(s) = \frac{5KN(s)}{D(s)+KN(s)} = \frac{10K}{s^2+7s+10+10K} = \frac{20}{s^2+7s+30}$$

5.17 (b) (continued)



5.18

The uncompensated and PD parts of PID compensation were performed in prob 5.17. the PI comp. remains to be added to get a PID comp.

(a)

CL poles = $\underbrace{-2.9 \pm j3.03}_{\text{dominant}}, -14.2$

$K_0 = \infty$, $K_1 = 2.5$, $\omega_n = \sqrt{2.9^2 + 3.03^2} = 4.19$

$\zeta = \frac{2.9}{4.17} = 0.69$

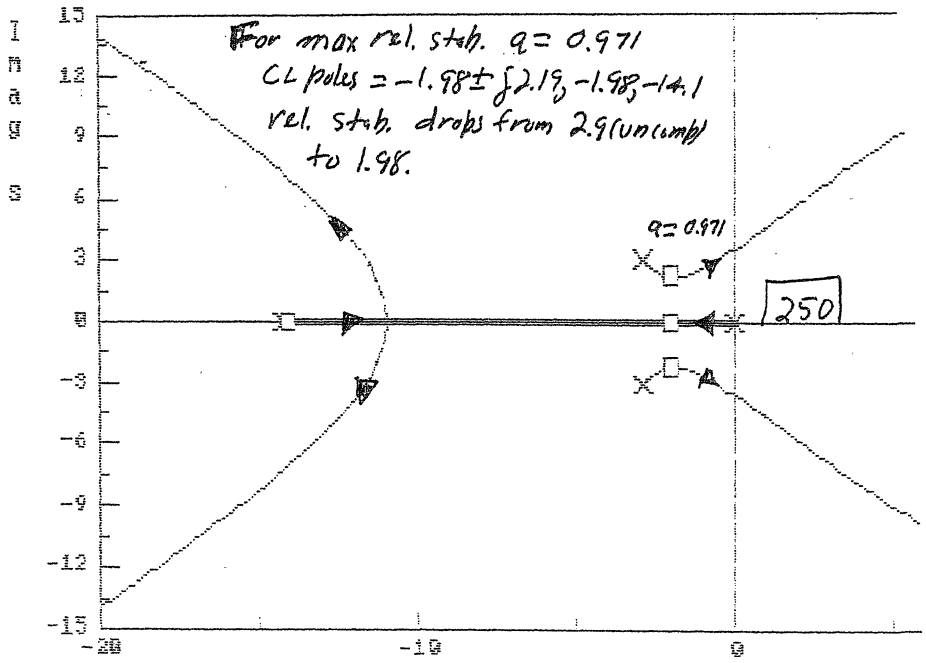
~~$\frac{K_0}{K_1}$ (2.5) = 0.69~~

5.18 (a) (continued)

For variable a, $G(s)H(s) = \frac{KN(s)}{s[D(s)+KN(s)]}$

as in problem 5.2. $K = K_v = 250$

$G(s)H(s) = \frac{250}{s[s^3+20s^2+100s+250]}$



From Prob 5.17(a), $A = 0.1278$. $q = 0.971$

$G_c(s) = \frac{KA[s^2+\frac{1}{A}s+\frac{q}{A}]}{s} = K(0.1278)\frac{[s^2+7.82s+7.59]}{s}$ (0.1278 ≈ 1/7.82; 7.59 ≈ 0.971/0.1278; 7.82 ≈ −6.68 − 1.135)

With $K = 500$ as for PD

CL zeros = −1.14, −6.68

CL poles = −1.94, −2.97, −7.5 ± j5.2 $T_S = 1.8$

$K_D = \infty$, $e_{STEP}(\infty) = 0$, $K_p = \infty$, $e_{RAMP}(\infty) = 0$

With $K = 1500$

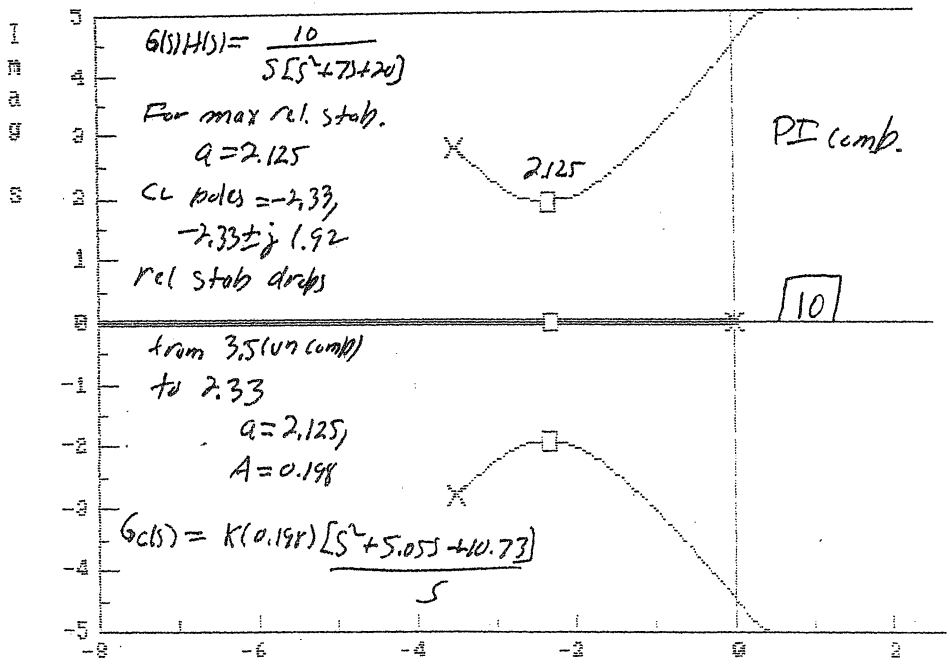
CL poles = −1.25, −6.09, −6.33 ± j12.3 $T_S = 0.41$

there is better cancellation of CL poles with CL zeros moving dominant roots to −6.33 ± j12.3 and reducing T_S

5.18(b)

CL poles $= -3.5 \pm j2.78$, $\omega_n = \sqrt{20} = 4.47$

$\zeta = \frac{3.5}{4.47} = 0.78$, $K_0 = 0$, $K_1 = 1$



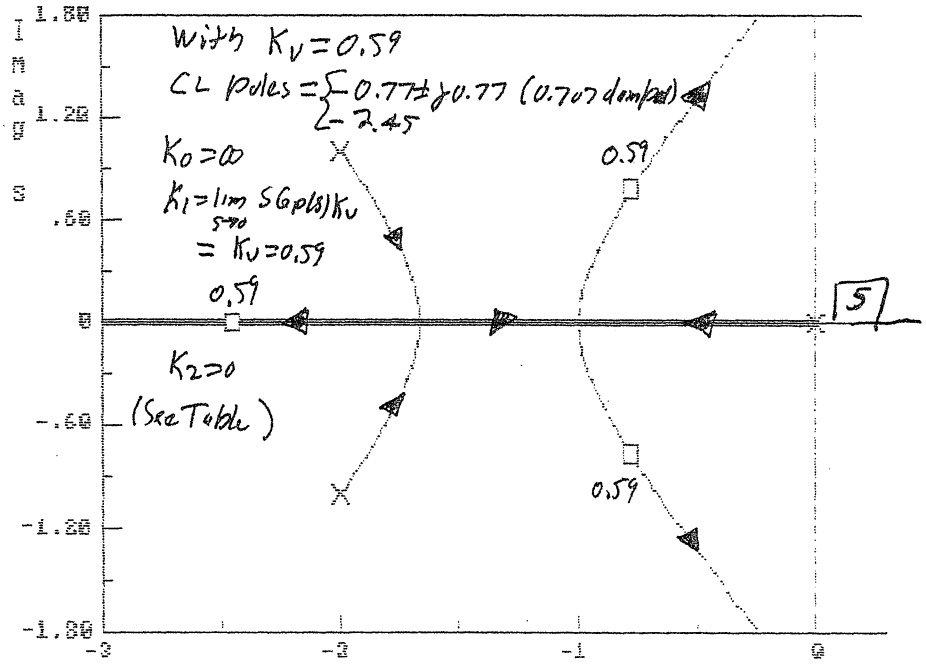
	CL poles	CL zeros	
$T_S = 0.75$ — $K=2$	$-1.57 \pm j1.72, -7.82$	$-2.52 \pm j2.09$	Let $K=6$ with CL pole-zero cancellation dominant pole is -14.7
$T_S = 0.46$ — $K=4$	$-1.91 \pm j2, -11.1$	"	
$T_S = 0.35$ — $K=6$	$-2.08 \pm j2.08, -14.7$	"	

Rel. stab $\Rightarrow \boxed{14.7}$. $K_0 = \infty$, $e_{STEP}(\infty) = 0$. $K_1 = \lim_{s \to 0} sG_c(s)H(s)$

$K_1 = \underbrace{K}_{6}(.198)(10.73)\left(\frac{19}{10}\right) = 12.71$. $e_{RAMP}(\infty) = 0.078$

5.19

(a)



(b)

For lag compensator nomenclature, see Prob 5.3.

See part (a) CL poles $-0.77 \pm j0.77$, -2.45

$\omega_n = \sqrt{(.77)^2 + .77^2} = 1.09$ $\quad \frac{2.95}{0.59}$

$\zeta = 0.707$, $P(s) = s^3 + 4s^2 + 5s + 5K_v$

$K = K_v = 0.59$

[redacted]

$K = K_v = 0.59 \quad d = 10 \quad b = a/d$

5.19 (b) (continued)

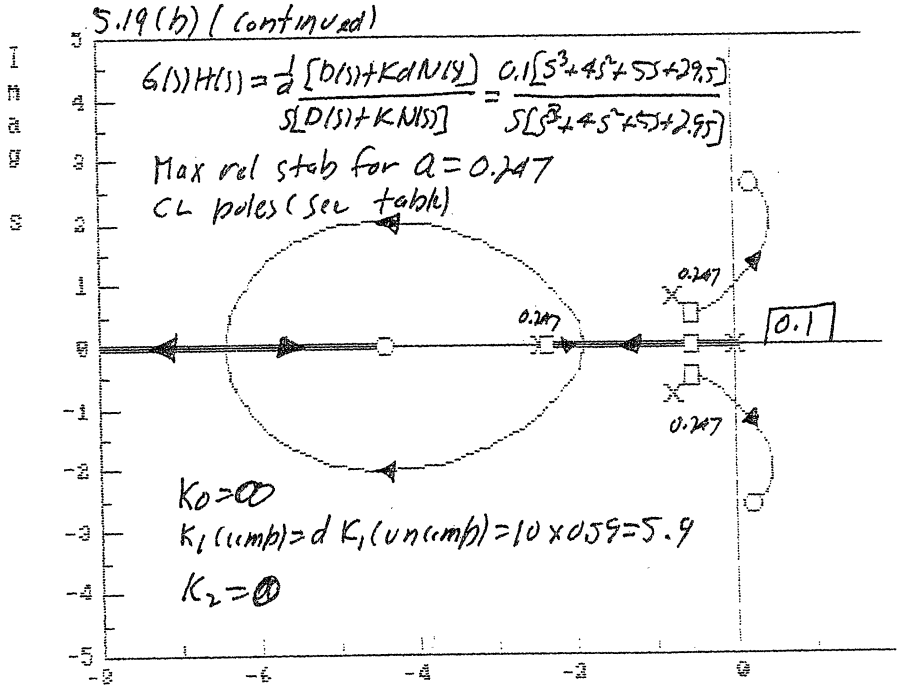


Table of results for Problem 5.19

Comp	CL Poles	Rel Stab	ζ	$e_{STEP}(\infty)$	$e_{RAMP}(\infty)$	$e_{PAR}(\infty)$
Uncomp $K_U = 0.59$	$-0.7 \pm j0.77$ -2.45	0.77	0.707	0	1.69	∞
Lag $G_c(s) = \frac{0.59(s+.247)}{s+0.0247}$	$-0.59 \pm j0.49$ -2.36	0.56	0.75	0	0.169	∞
Rate feed $K = 1.18$ $H_c(s) = 0.505(s+1.96)$	-1.33 $-1.33 \pm j1.63$	1.33	0.63	0	1.35	∞
Lead $G_c(s) = \frac{5.9(s+3.578)}{s+35.78}$	-1.33 $-1.33 \pm j0.68$ -35.8	1.33	0.87	0	1.69	∞

(c) Rate feedback. See Prob 5.4 for nomenclature. (See part (a))

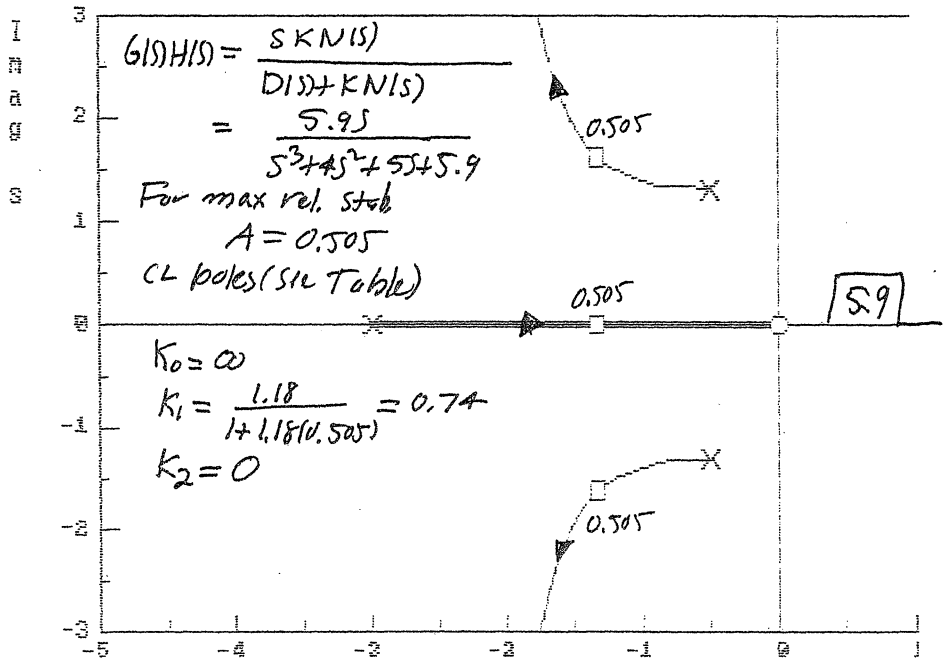
5.19(c) (continued)

$$G_E(s) = \frac{KN(s)}{D(s)+KN(s)A_s} = \frac{K(s)}{s^3+4s^2+5s+K(s)A_s}$$

$$G_E(s) = \frac{5K}{s[s^3+4s+5+5KA]}\ ;\ K_1 = \lim_{s\to 0} G_E(s) = \frac{5K}{5+KA}$$

$$K_1 = \frac{K}{1+KA}$$

Let $K = 2K_U = 1.18$



(d) (See part (g))

$d = 10$

$K = K_U d = (0.59)(10) = 5.9$,

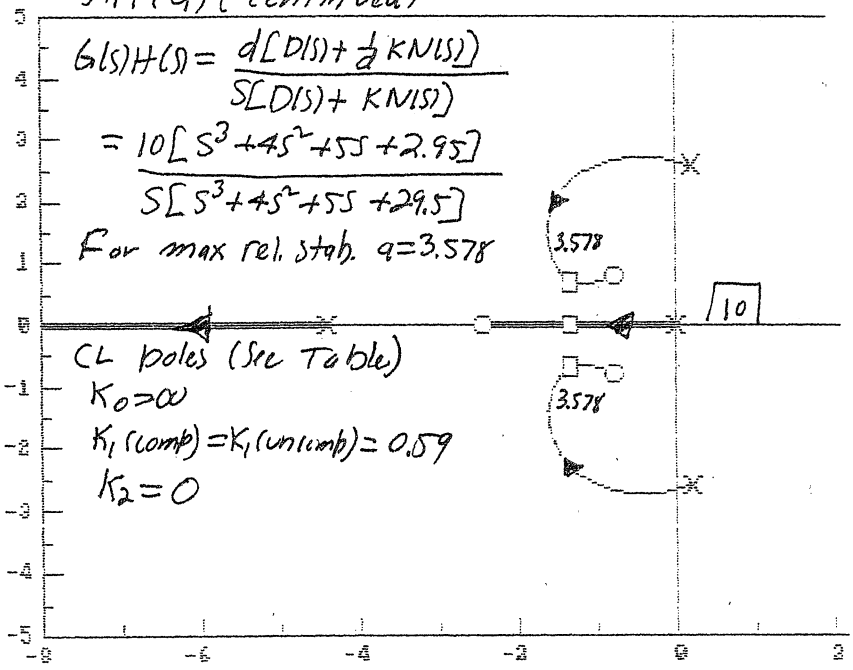
See problem 5.15 for nomenclature

5.19 (d) (continued)

$$G(s)H(s) = \frac{d[D(s) + \frac{1}{d}KN(s)]}{S[D(s) + KN(s)]}$$

$$= \frac{10[s^3 + 4s^2 + 5s + 2.95]}{S[s^3 + 4s^2 + 5s + 29.5]}$$

For max rel. stab. $q = 3.578$



CL poles (See Table)

$K_0 = \infty$

$K_1(comp) = K_1(uncomp) = 0.59$

$K_2 = 0$

5.20

$$G_p(s) = \frac{1}{(s+1)(s^2+4s+5)} = \frac{1}{s^3+5s^2+9s+5}$$

$$K_0(uncomp) = K_U G_p(0) = \frac{K_U}{5}$$

For comp. system

$$e_{STEP}(\infty) = \frac{1}{1+K_0(comp)} < 0.05$$

$1 + K_0(comp) > 20 \quad K_0(comp) > 19$

For lag comp.

$$K_0(comp) = d\, K_0(uncomp) = d\,\frac{K_U}{5}$$

$\frac{K_U}{5} > 19 \quad d = 10 \quad K_U > 9.5$

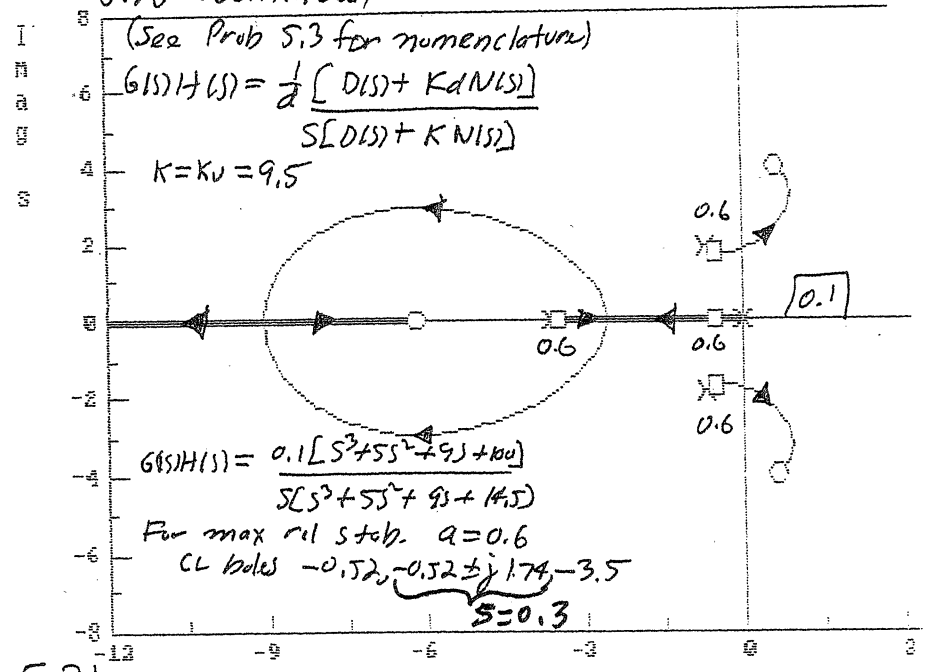
Try $K_U = 9.5$

5.20 (continued)

(See Prob 5.3 for nomenclature)

$$G(s)H(s) = \frac{1}{a}\,\frac{[D(s) + K_d N(s)]}{s[D(s) + K N(s)]}$$

$K = K_v = 9.5$



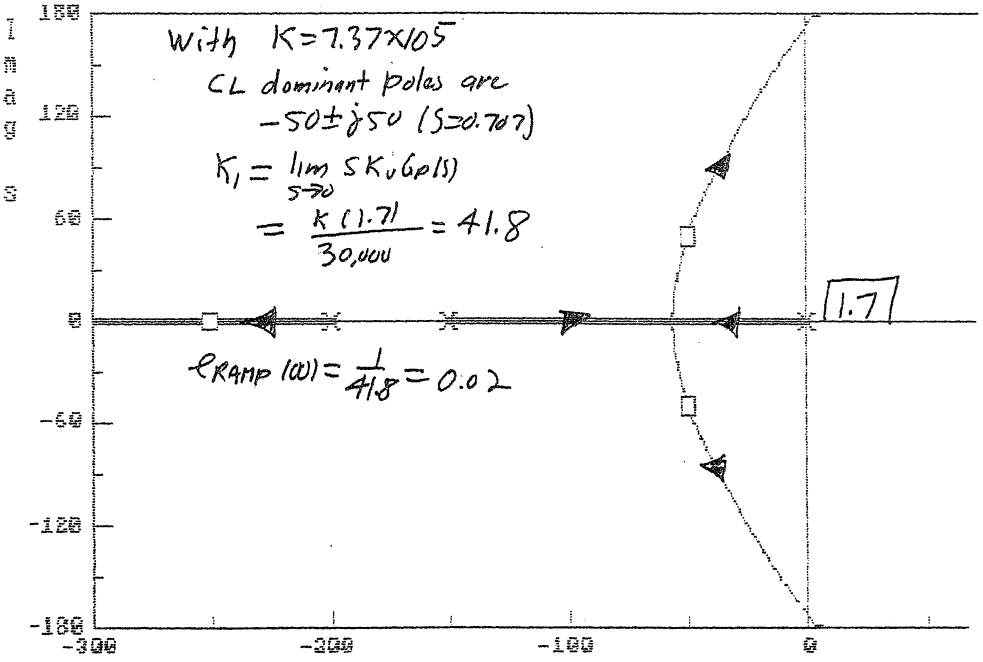
$$G(s)H(s) = \frac{0.1[s^3 + 5s^2 + 9s + 100]}{s[s^3 + 5s^2 + 9s + 14.5]}$$

For max rel stab. $a = 0.6$

CL poles $-0.52, -0.52 \pm j1.74, -3.5$

$\zeta = 0.3$

5.21



With $K = 7.37 \times 10^5$

CL dominant poles are $-50 \pm j50$ ($\zeta = 0.707$)

$$K_1 = \lim_{s \to 0} sK_v G_p(s) = \frac{K(1.7)}{30{,}000} = 41.8$$

1.7

$$e_{RAMP}(\infty) = \frac{1}{41.8} = 0.02$$

5.22

See Prob 5.4 for nomenclature.

$$P(s) = (s+0.5)(s^2 - .4s + .20) + \hat{K}_U'(10)(s+0.05)$$

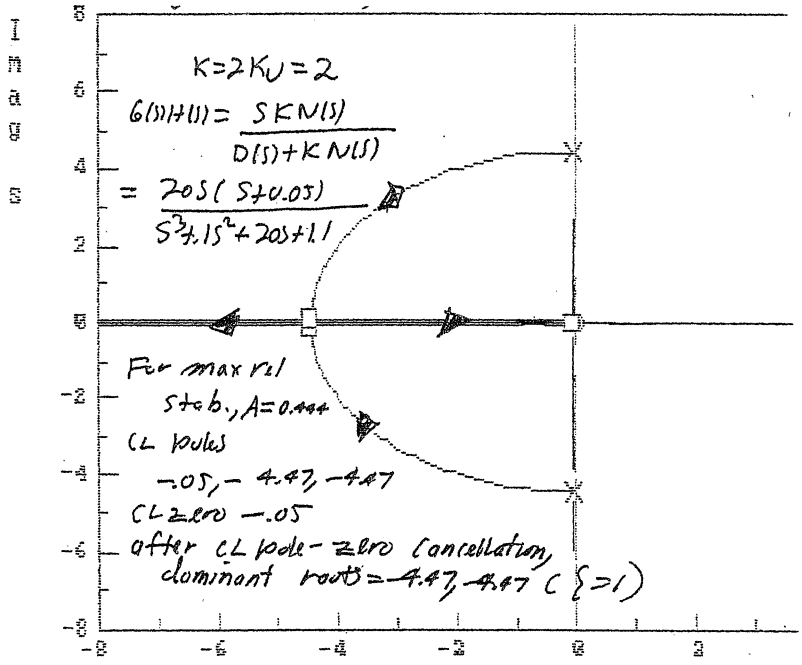
$$= s^3 + 0.1s^2 + 10s + 0.6$$

CL poles $= -.02 \pm j3.16, -.06$

$$G_{\epsilon}(s) = \frac{KN(s)}{D(s) + KA_s N(s)}$$

$$K_0 = G_{\epsilon}(0) = \frac{KN(0)}{D(0) + \underbrace{KA(0)N(0)}_{0}} = \frac{KN(0)}{D(0)} = \frac{K_U(10)(.05)}{(.5)(.20)}$$

$$= 5K_U$$



5.23. (a) The characteristic polynomial is

$$P(s) = s^2 + 2\zeta\omega_n s + \omega_n^2$$

$\omega_n = 4$

$\zeta = 0.7$

For zero steady state step error, select $b_0 = a_0$ as in Section 5.12.2. Then $b_0 = 16$

$$T(s) = \frac{16}{s^2 + 2(.7)(4)s + 4^2} = \boxed{\frac{16}{s^2 + 5.6s + 16}}$$

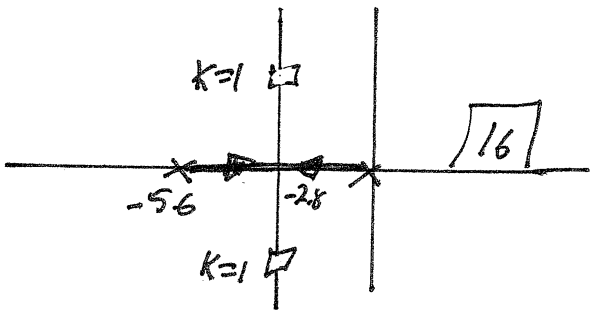
(b) $$G_p(s) = \frac{1}{(s+2)^2} = \frac{1}{s^2+4s+4} = \frac{N_p(s)}{D_p(s)}$$

$$T(s) = \frac{16}{s^2+5.6s+16} = \frac{N_t(s)}{D_t(s)}$$

$$G_c(s) = \frac{N_t(s)\, D_p(s)}{N_p(s)\,[D_t(s) - N_t(s)]}$$

$$= \frac{16\,(s+2)^2}{(1)\,[s^2+5.6s+16-16]} = \boxed{\frac{16(s+2)^2}{s(s+5.6)}}$$

(c) $$G_c(s)\,G_p(s) = \frac{KT(s)}{1-T(s)} = \frac{K\,N_t(s)}{D_t(s) - N_t(s)} = \frac{K(16)}{s(s+5.6)}$$



For K=1 closed-loop poles are $-2.8 \pm j\,2.86$

5.24. $P(s) = D_c(s) D_p(s) + N_c(s) N_p(s)$

$$G_c(s) = \frac{K_d s^2 + K_p s + K_i}{s} = \frac{N_c(s)}{D_c(s)}$$

$$G_p(s) = \frac{1}{(s+2)^2} = \frac{1}{s^2+4s+4} = \frac{N_p(s)}{D_p(s)}$$

$$P(s) = (s^2+5.6s+16)(s+20) = s^3+25.6s^2+128s+320$$
$$= s(s^2+4s+4) + (K_d s^2 + K_p s + K_i)(1)$$
$$= s^3 + (4+K_d)s^2 + (4+K_p)s + K_i$$

$K_i = 320$ $\quad 4+K_p = 128$ $\quad 4+K_d = 25.6$

$K_p = 124$ $\quad K_d = 21.5$

$$\boxed{G_c(s) = \frac{21.6s^2+124s+320}{s}}$$

5.25 (a) For zero steady state error, select $b_0 = q_0$ as in Section 5.12.2. Here

$$P(s) = (s+5)^2(s+50) = (s^2+10s+25)(s+50)$$
$$= s^3+60s^2+525s+1250$$

$q_0 = 1250$ so $b_0 = 1250$

$$\boxed{T(s) = \frac{1250}{(s+5)^2(s+50)}}$$

5.25 (b) $$G_p(s) = \frac{1}{(s+1)(s^2+4)} = \frac{1}{s^3+s^2+4s+4} = \frac{N_p(s)}{D_p(s)}$$

$$T(s) = \frac{1250}{(s+5)^2(s+50)} = \frac{1250}{s^3+60s^2+525s+1250} = \frac{N_t(s)}{D_t(s)}$$

$$G_c(s) = \frac{N_t(s) D_p(s)}{N_p(s)[D_t(s) - N_t(s)]}$$

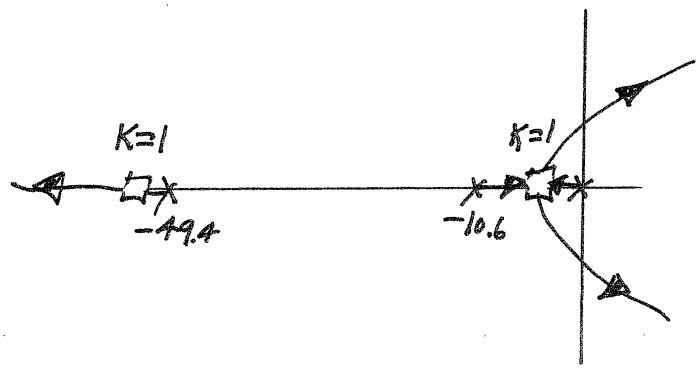
5.25 (b) continued

$$G_c(s) = \frac{1250(s+1)(s^2+4)}{(1)\left[s^3+60s^2+525s+1250-1250\right]}$$

$$\boxed{G_c(s) = \frac{1250(s+1)(s^2+4)}{s(s^2+60s+525)}}$$

(c) $G_c(s)G_p(s) = \frac{KT(s)}{1-T(s)} = \frac{KN_{CL}(s)}{D_{CL}(s)-N_{CL}(s)}$

$$= \frac{K(1250)}{s(s^2+60s+525)} = \frac{K(1250)}{s(s+10.6)(s+49.4)}$$



5.26

(a) $G_p'(s) = \frac{1}{(s+2)(s^2+4)}$

$$T'(s) = \frac{G_p'(s)G_c(s)}{1+G_p'(s)G_c(s)}$$

$$= \frac{\frac{1}{(s+2)(s^2+4)} \cdot \frac{1250(s+1)(s^2+4)}{s(s^2+60s+525)}}{1+\frac{1}{(s+2)(s^2+4)} \cdot \frac{1250(s+1)(s^2+4)}{s(s^2+60s+525)}}$$

$$= \frac{1250(s+1)}{s(s+2)(s^2+60s+525)+1250(s+1)}$$

$$= \frac{1250(s+1)}{s^4+62s^3+645s^2+2300s+1250}$$

$$= \boxed{\frac{1250(s+1)}{(s+0.657)(s+5.66+j2.44)(s+5.66-j2.44)(s+50.01)}}$$

The closed-loop poles all move from $-5, -5, -50$ and one new closed-loop pole is at -0.657

5.26 continued

(b) $H_1(s) = T_2(s) = \boxed{\frac{25}{(s+5)^2}}$

$$G_{c2}(s) = \frac{T_2(s)}{G_{p2}(s)} = \frac{25(s^2+4)}{(s+5)^2}$$

$$G_{c1}(s) = \frac{T_1(s)}{G_{p1}(s)[1-T_1(s)]} = \frac{\frac{50}{s+50}}{\left(\frac{1}{s+1}\right)\left[1-\frac{50}{s+50}\right]}$$

$$= \boxed{\frac{50(s+1)}{s}}$$

(c) $$T_1'(s) = \frac{T_1(s)\, G_{p1}'(s)}{G_{p1}(s) + T_1(s)\left[G_{p1}'(s) - G_{p1}(s)\right]}$$

$$= \frac{\frac{50}{s+50}\left(\frac{1}{s+2}\right)\left(\frac{s+1}{s+1}\right)}{\left(\frac{s+2}{s+2}\right)\left(\frac{s+50}{s+50}\right)\left(\frac{1}{s+1}\right) + \frac{50}{s+50}\left[\frac{1}{s+2} - \frac{1}{s+1}\right]}$$

$$\frac{s+1-s-2}{(s+1)(s+2)} = -\frac{1}{(s+1)(s+2)}$$

$$= \frac{50(s+1)}{(s+2)(s+50)-50} = \frac{50(s+1)}{s^2+52s+100-50}$$

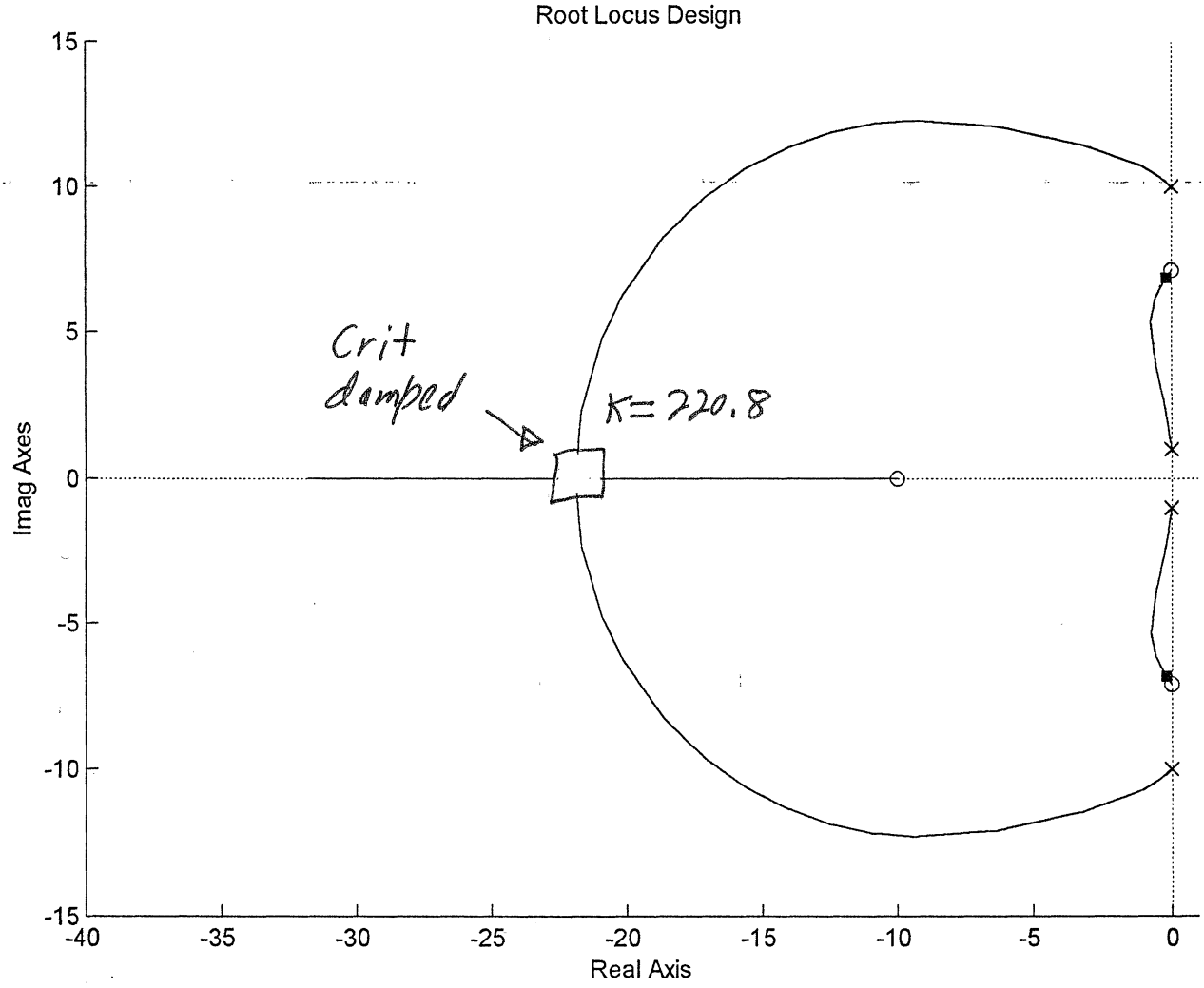
$$= \boxed{\frac{50(s+1)}{s^2+52s+50}}$$

5.27

$$G_p(s) = \frac{1}{s^2+1} + \frac{1}{s^2+100} = \frac{s^2+100+s^2+1}{(s^2+1)(s^2+100)} = \frac{2s^2+101}{(s^2+1)(s^2+100)}$$

$$= \frac{2(s^2+50.5)}{(s^2+1)(s^2+100)} \qquad H_c(s) = 0.1(s+10)$$

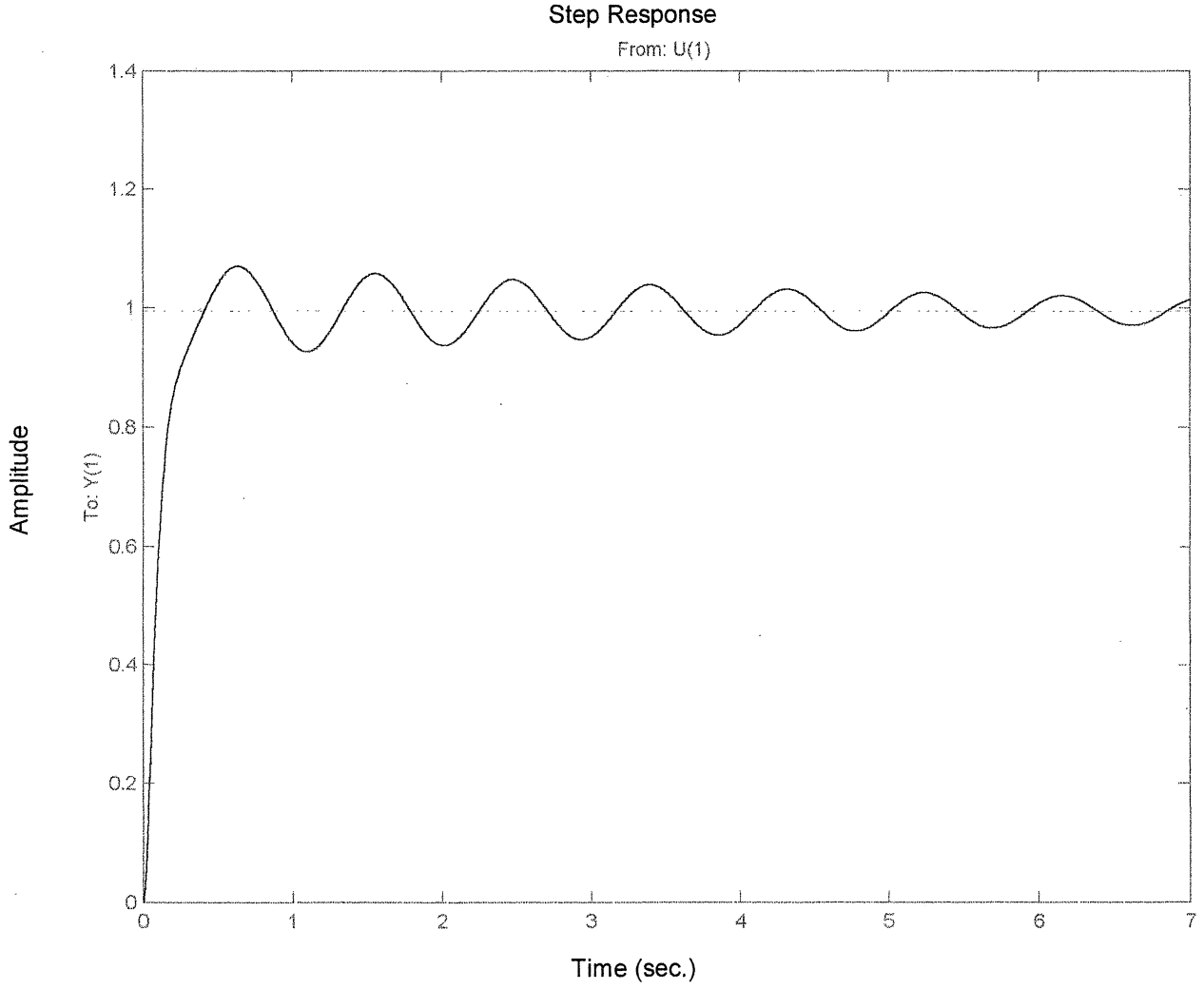
$$G_p(s)H_c(s) = \frac{0.2(s+10)(s^2+50.5)}{(s^2+1)(s^2+100)}$$



For $K = 220.8$, closed-loop poles

$-0.19 \pm j\,6.8, -22, -22$

$$T(s) = \frac{KG_p(s)}{1 + KG_p(s)H_c(s)}$$



CHAPTER 6

6.1

(a) $F(j\omega) = \frac{1}{j\omega+8}$ $A(\omega) = \frac{1}{\sqrt{\omega^2+64}}$ $\phi(\omega) = -\tan^{-1}\frac{\omega}{8}$

$\omega = 5$ $A(\omega) = 0.106$ $\phi(\omega) = -\tan^{-1}\frac{5}{8} = -32^0$

$y(t) = 5(.106)\cos(5t-32^0) = \boxed{0.53\cos(5t-32^0)}$

(b) $F(j\omega) = \frac{(j\omega)^2}{j\omega+4}$ $A(\omega) = \frac{\omega^2}{\sqrt{\omega^2+16}}$ $\phi(\omega) = 180^0 - \tan^{-1}\frac{\omega}{4}$

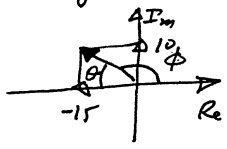
$\omega = 4$ $A(\omega) = \frac{4^2}{\sqrt{4^2+16}} = 2.83$ $\phi(\omega) = 180^0 - \tan^{-1}1 = 135^0$

$y(t) = 100(2.83)\cos(4t+40^0+135^0) = \boxed{283\cos(4t+175^0)}$

(c) $F(j\omega) = \frac{10}{(j\omega)^2+j\omega+10} = \frac{10}{10-\omega^2+j(2\omega)} \overset{\omega=5}{=} \frac{10}{-15+j10}$

$A(\omega) = \frac{10}{\sqrt{(-15)^2+10^2}}$

$= 0.555$



$\phi(\omega) = -[180^0-\theta] = -[180^0-\tan^{-1}\frac{10}{15}] = -146.3^0$

$y(t) = 4(0.555)\cos(5t-70^0-146.3^0) = 2.22\cos(5t-216.3^0)$

$= \boxed{2.22\cos(5t+143.7^0}$

(d) $F(j\omega) = \frac{j\omega}{(j\omega+4)(j\omega+8)}$ $A(\omega) = \frac{\omega}{\sqrt{\omega^2+16}\sqrt{\omega^2+64}}$

$\phi(\omega) = 90^0 - \tan^{-1}\frac{\omega}{4} - \tan^{-1}\frac{\omega}{8}$ $\omega = 4$

$A(\omega) = \frac{4}{\sqrt{16+16}\sqrt{16+64}} = 0.079$ $\phi(\omega) = 90^0 - \tan^{-1}1 - \tan^{-1}0.5$

$= 18.4^0$

$y(t) = 10(0.079)\cos(4t+120^0+18.4^0)$

$= \boxed{0.79\cos(4t+138.4^0)}$

6.1(e)

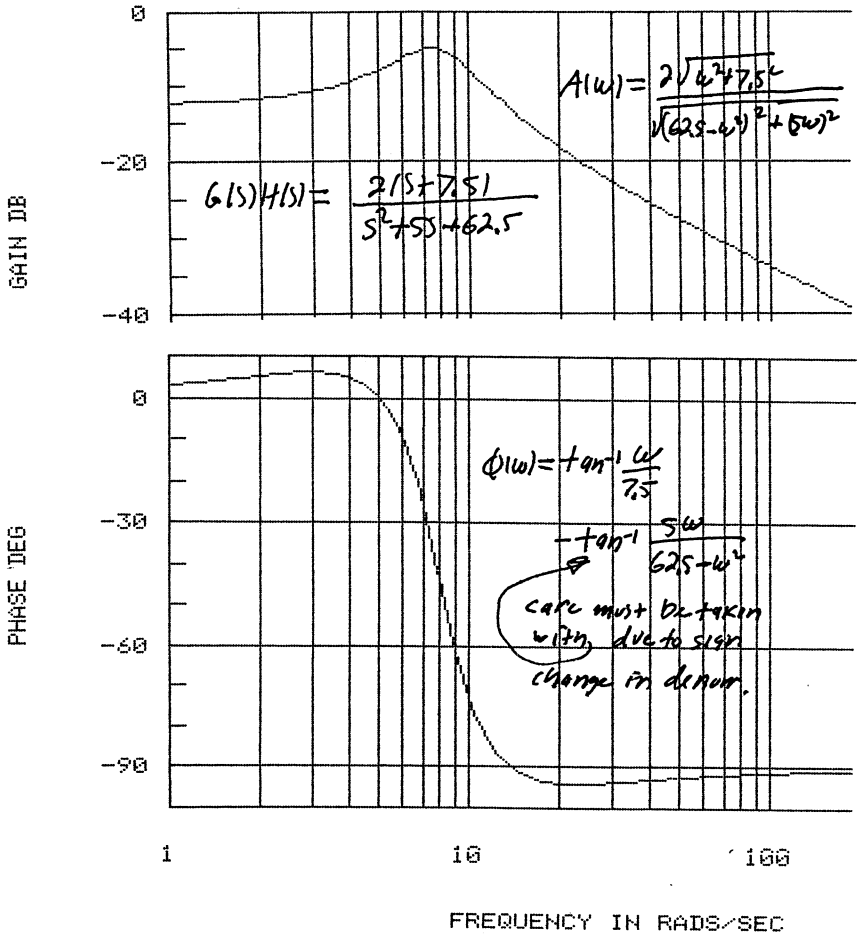
$$F(j\omega) = \frac{10}{(j\omega+6)^2} \quad A(\omega) = \frac{10}{(\sqrt{\omega^2+36})^2} = \frac{10}{\omega^2+36}$$

$$\phi(\omega) = -2\tan^{-1}\frac{\omega}{6} \quad \omega = 6 \quad A(\omega) = \frac{10}{36+36} = 0.139$$

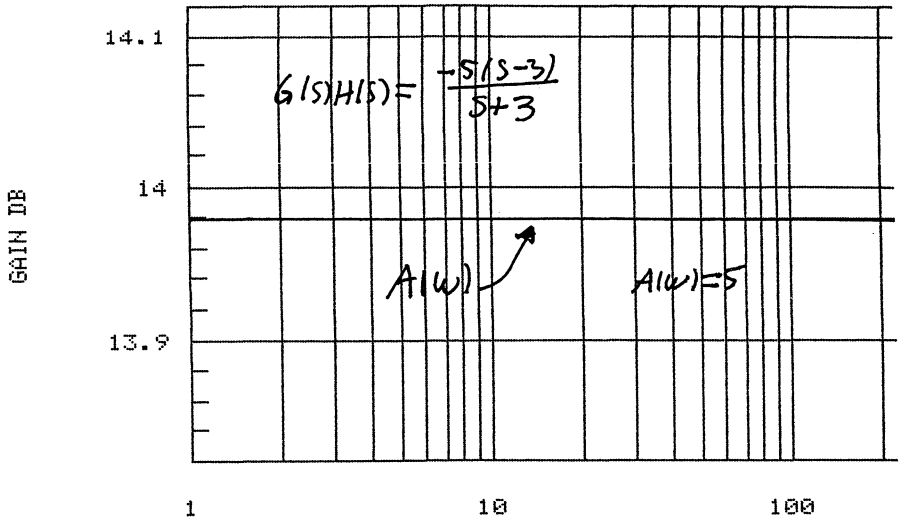
$$\phi(\omega) = -2\tan^{-1}1 = -90^\circ$$

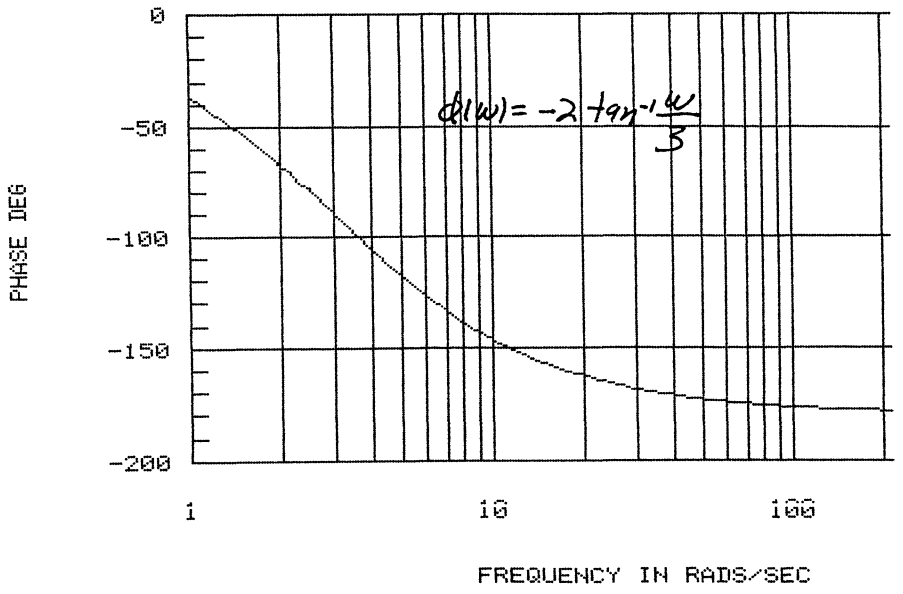
$$y(t) = 5(0.139)\cos(6t-90^\circ) = \boxed{0.7\cos(6t-90^\circ)}$$

6.21(a)



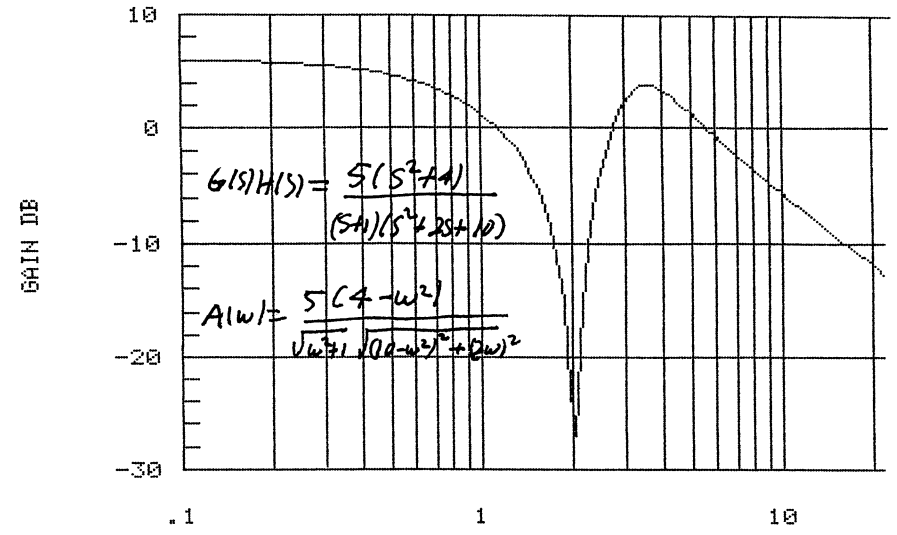
6.2(b)

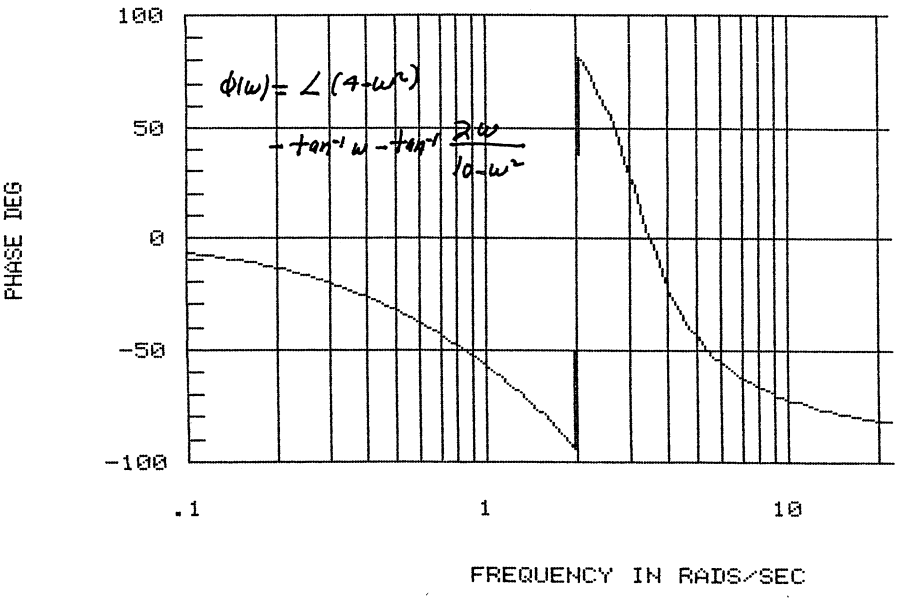




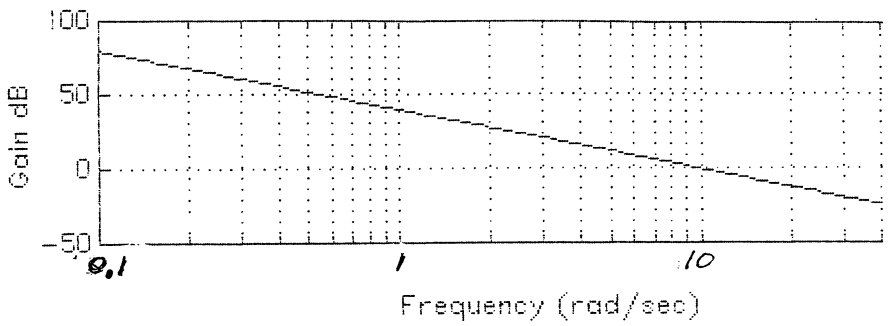
6.2(c) see text.

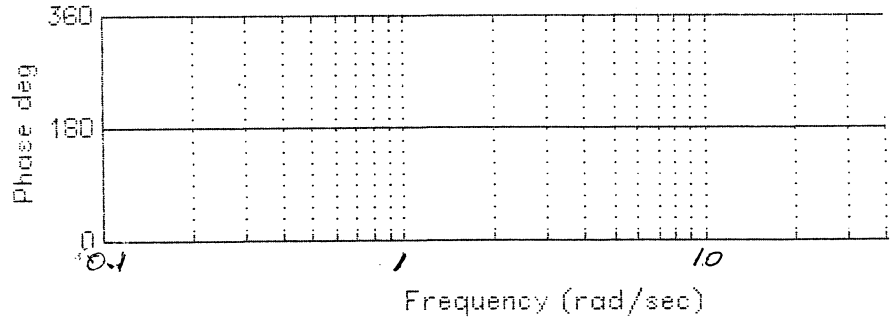
6.2(d)


10
0
-10
-20
-30
.1
1
10
GAIN DB
$G(s)H(s) = \frac{5(s^2+4)}{(s+1)(s^2+2s+10)}$
$A(\omega) = \frac{5|4-\omega^2|}{\sqrt{\omega^2+1}\sqrt{(10-\omega^2)^2+(2\omega)^2}}$

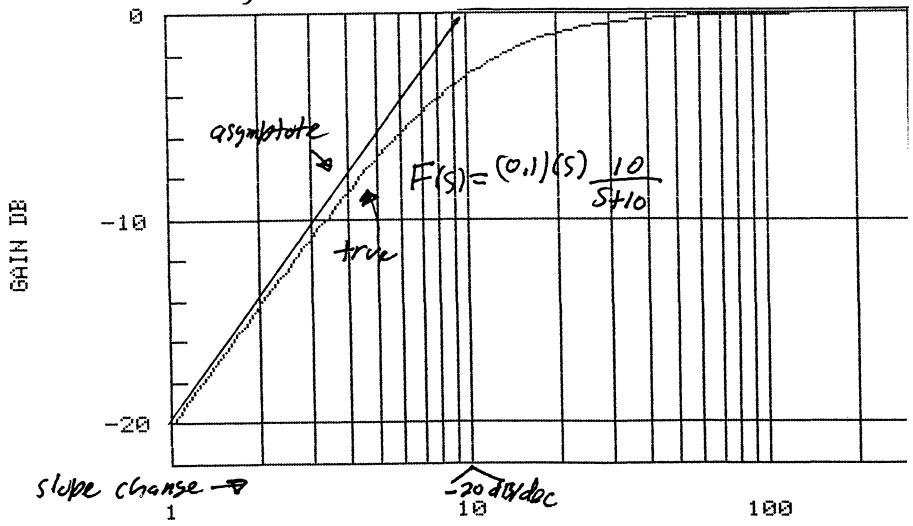

100
50
0
-50
-100
.1
1
10
PHASE DEG
FREQUENCY IN RADS/SEC
$\phi(\omega) = \angle(4+\omega^2)$
$-\tan^{-1}\omega - \tan^{-1}\frac{2\omega}{10-\omega^2}$

6.3(a)

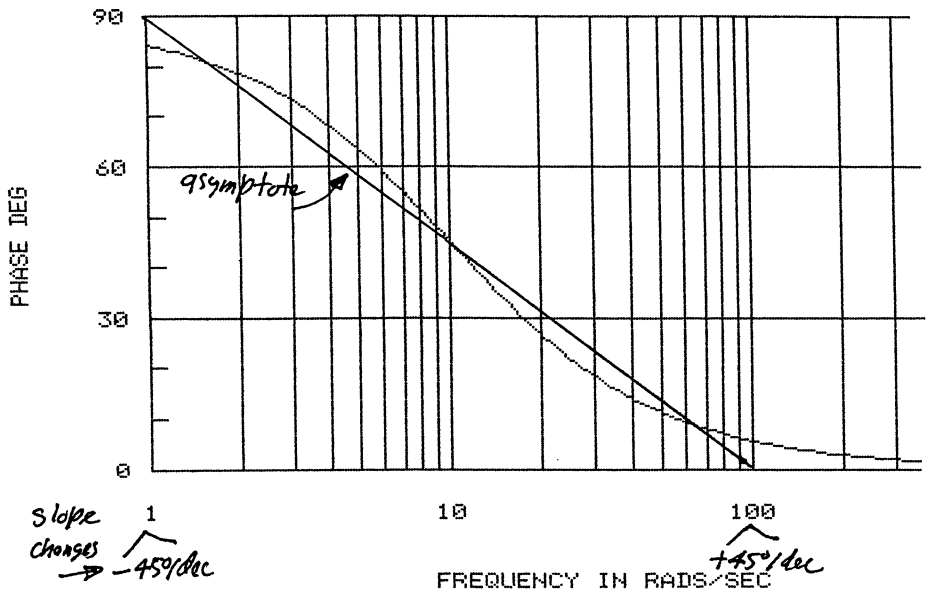

100
50
0
-50
Gain dB
0.1
1
10
Frequency (rad/sec)


360
180
0
Phase deg
0.1
1
10
Frequency (rad/sec)

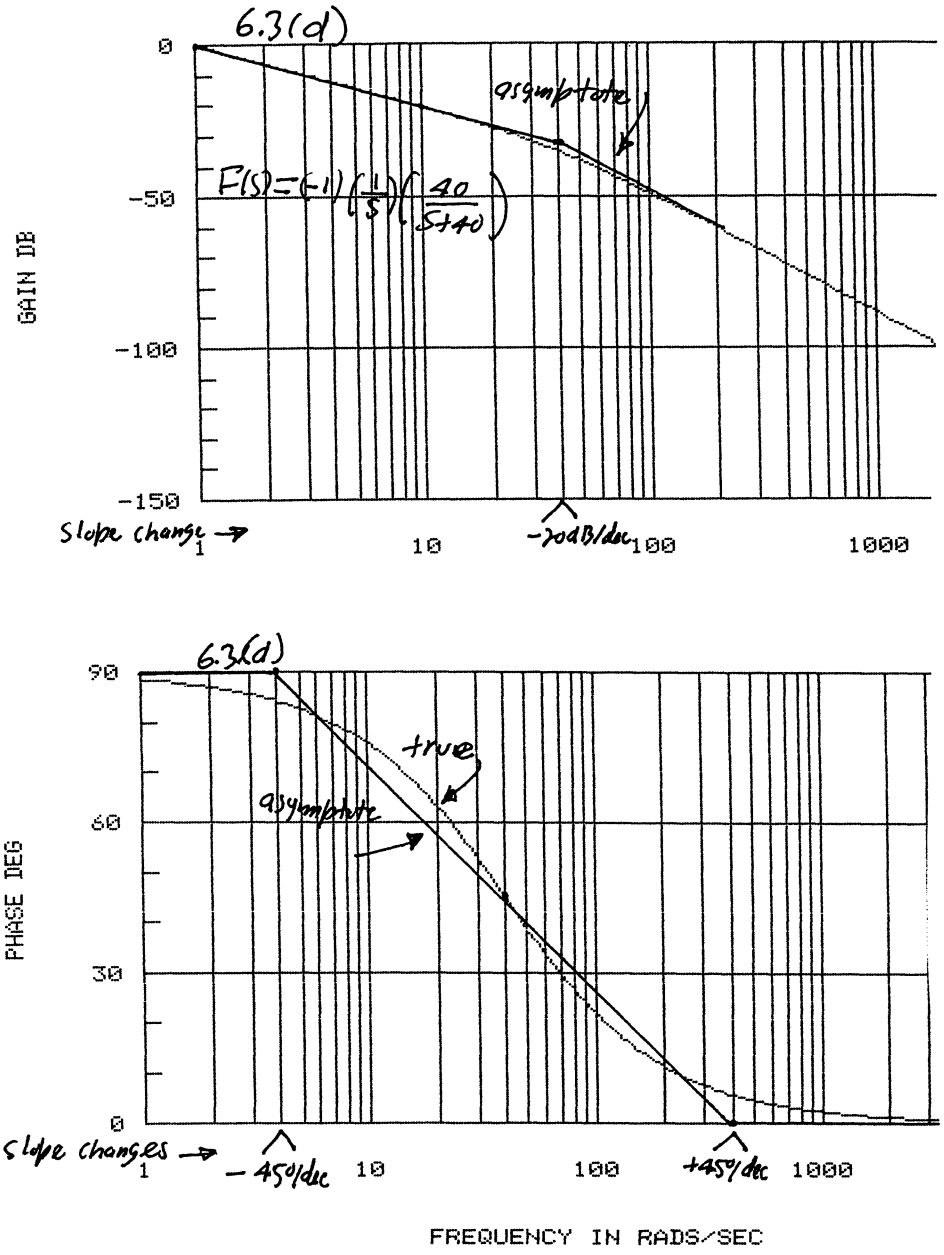
6.3(b)


0
-10
-20
GAIN DB
asymptote
true
F(s) = (0.1)(s) 10/(s+10)
slope change
-20 dB/dec
1
10
100

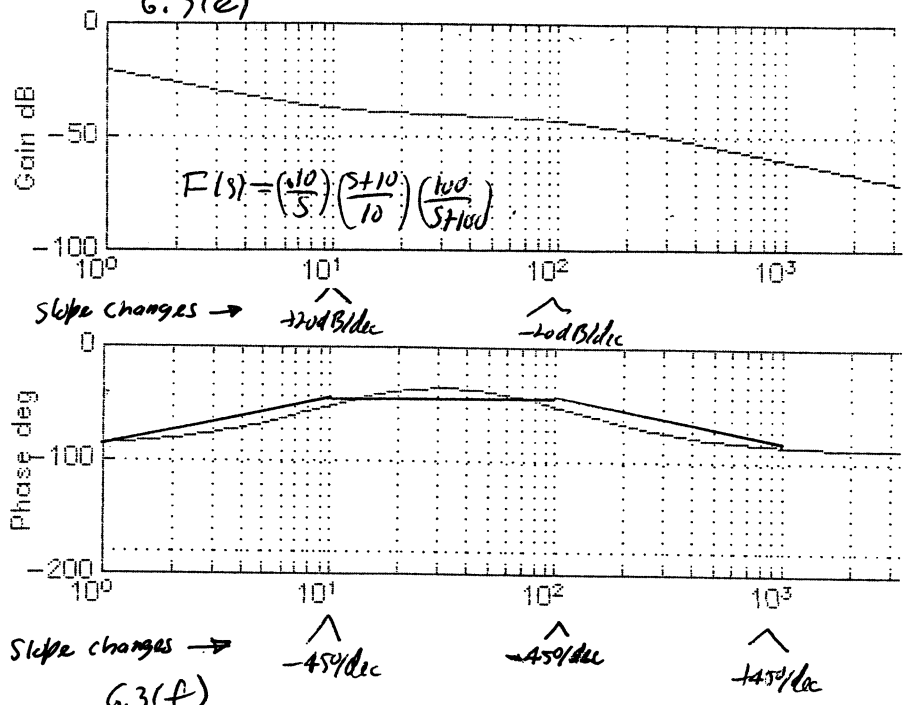
6.3(b) (continued)



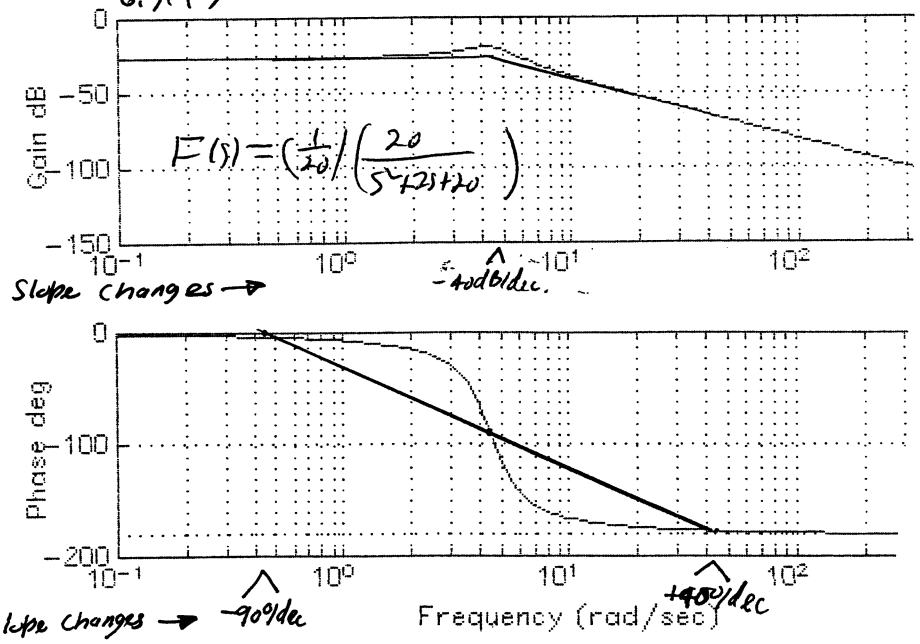
6.3 (c) (See text)


6.3(d)
0
-50
-100
-150
GAIN DB
asymptote
F(S)=(-1)(1/S)(40/S+40)
Slope change
1
10
-20dB/dec
100
1000
6.3(d)
90
60
30
0
PHASE DEG
true
asymptote
Slope changes
1
-45°/dec
10
100
+45°/dec
1000
FREQUENCY IN RADS/SEC

6.3(e)

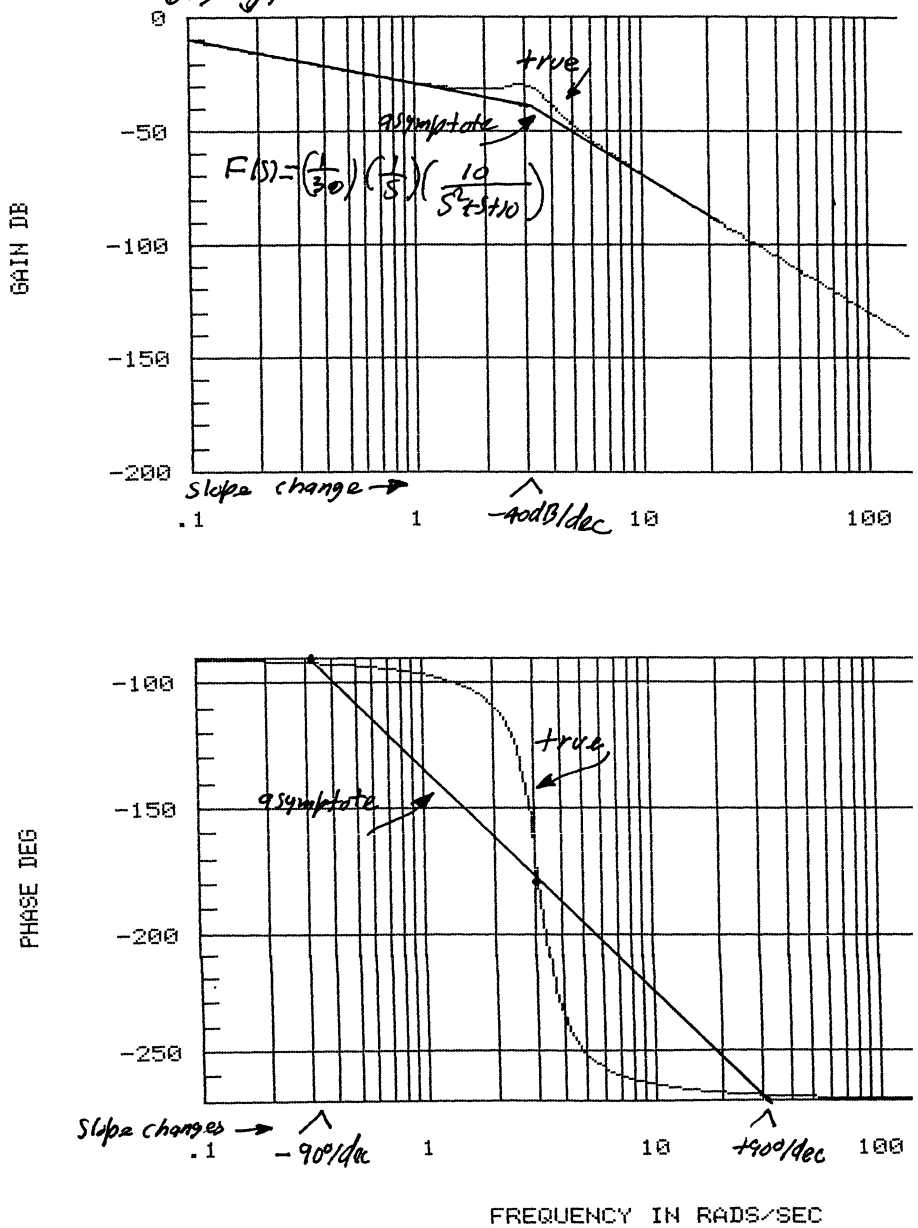

Gain dB
0
−50
−100
10^0
10^1
10^2
10^3
$F(s) = \left(\frac{10}{s}\right)\left(\frac{s+10}{10}\right)\left(\frac{100}{s+100}\right)$
Slope changes →
+20dB/dec
−20dB/dec
Phase deg
0
−100
−200
Slope changes →
−45°/dec
−45°/dec
+45°/dec

6.3(f)

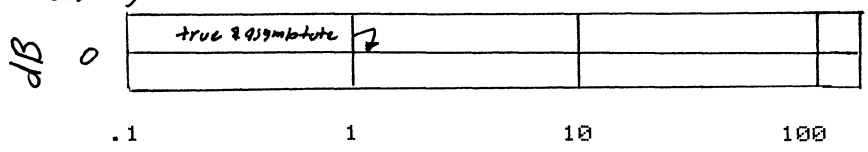

Gain dB
0
−50
−100
−150
10^{-1}
10^0
10^1
10^2
$F(s) = \left(\frac{1}{20}\right)\left(\frac{20}{s^2+2s+20}\right)$
Slope changes →
−40dB/dec.
Phase deg
0
−100
−200
Slope changes →
−90°/dec
+90°/dec
Frequency (rad/sec)

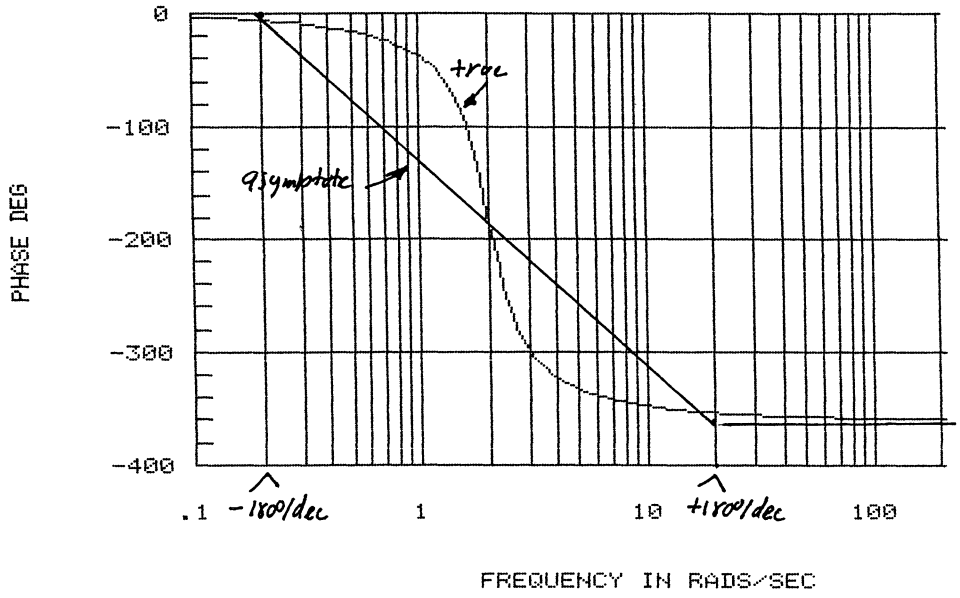
6.3(f) (See text)

6.3 (g)



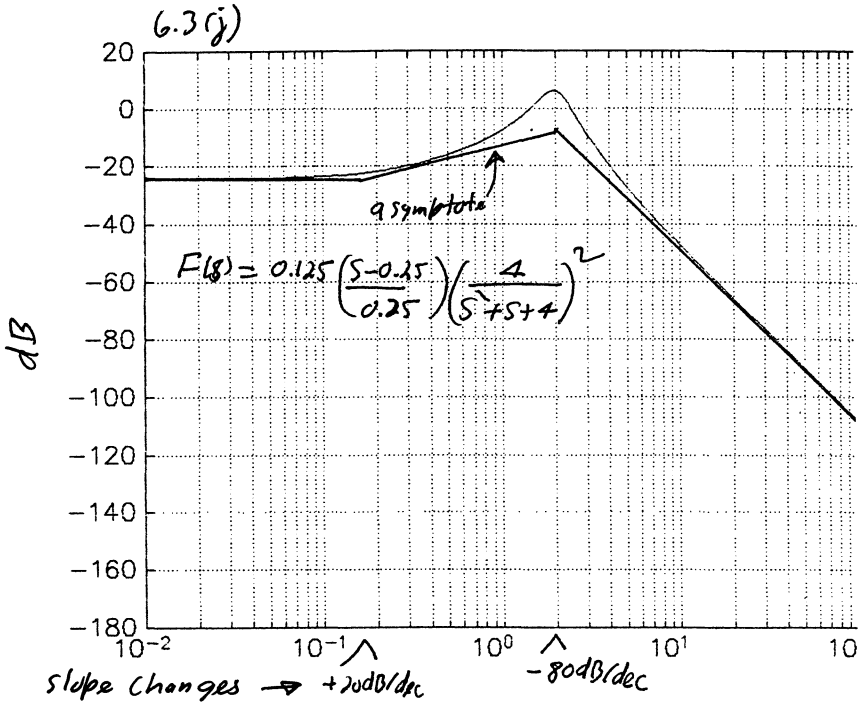
6.3(h)

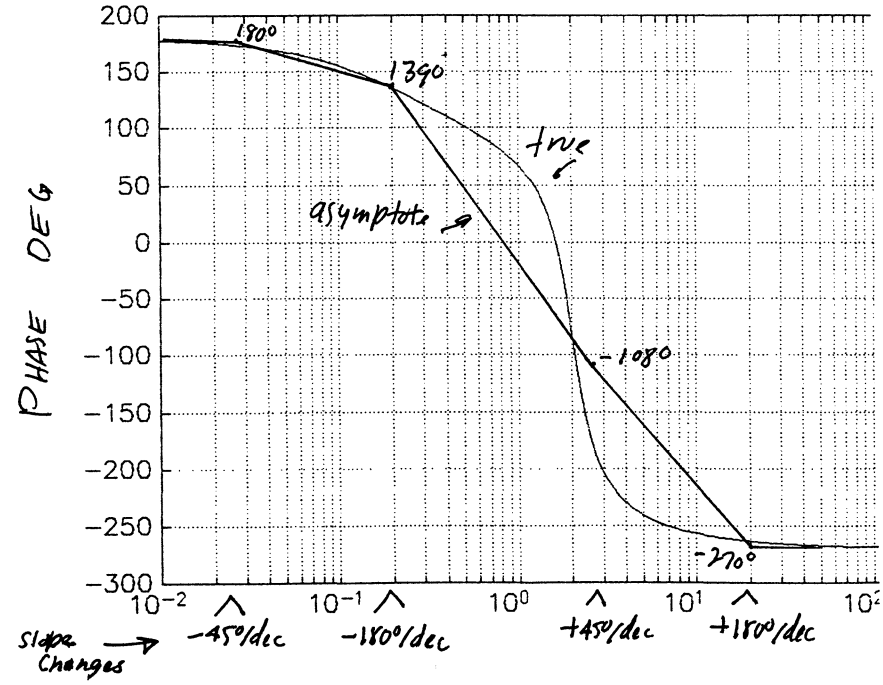




6.3(i) (See text)

6.3(j)

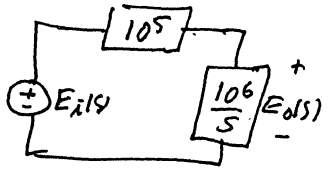

dB
asymptote
$F(s) = 0.125\left(\frac{s-0.25}{0.25}\right)\left(\frac{4}{s^2+s+4}\right)^2$
slope changes → +20dB/dec
-80dB/dec


PHASE DEG
180°
139°
true
asymptote
-108°
-270°
slope changes →
-45°/dec
-180°/dec
+45°/dec
+180°/dec

6.4

convert each circuit to an impedance equivalent

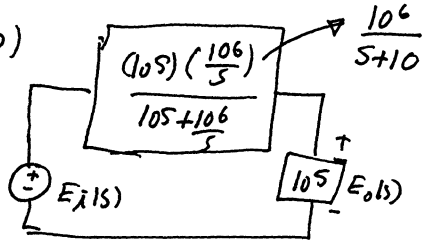
(a)



$$F(s) = \frac{E_o(s)}{E_i(s)} = \frac{\frac{106}{s}}{105 + \frac{106}{s}}$$

$$= \frac{106}{105s + 106} = \boxed{\frac{10}{s+10}}$$

(b)

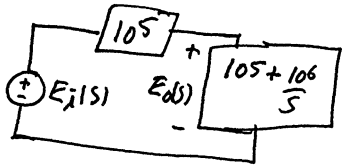


$$F(s) = \frac{E_o(s)}{E_i(s)} = \frac{10^5}{10^5 + \frac{10^6}{s+10}}$$

$$= \frac{10^5(s+10)}{10^5[s+10]+10^6}$$

$$= \boxed{\frac{s+10}{s+20}}$$

(c)



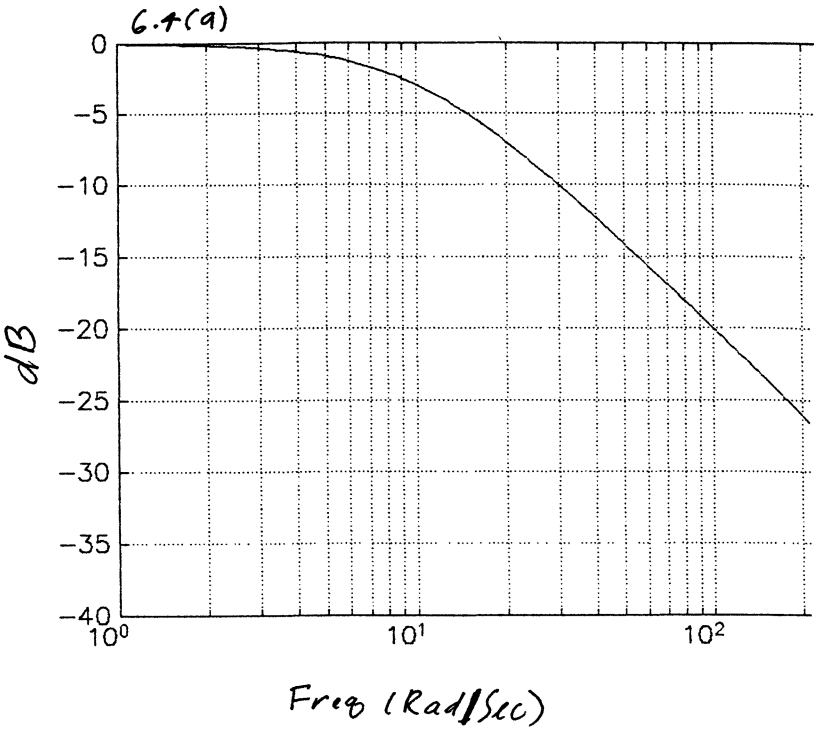
$$F(s) = \frac{E_o(s)}{E_i(s)}$$

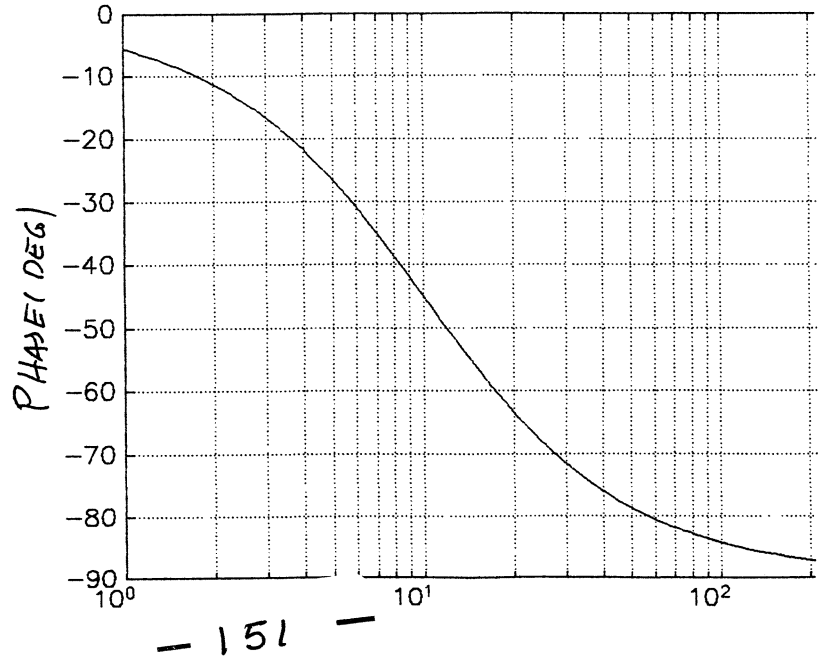
$$= \frac{105 + \frac{10^6}{s}}{105 + 105 + \frac{10^6}{s}}$$

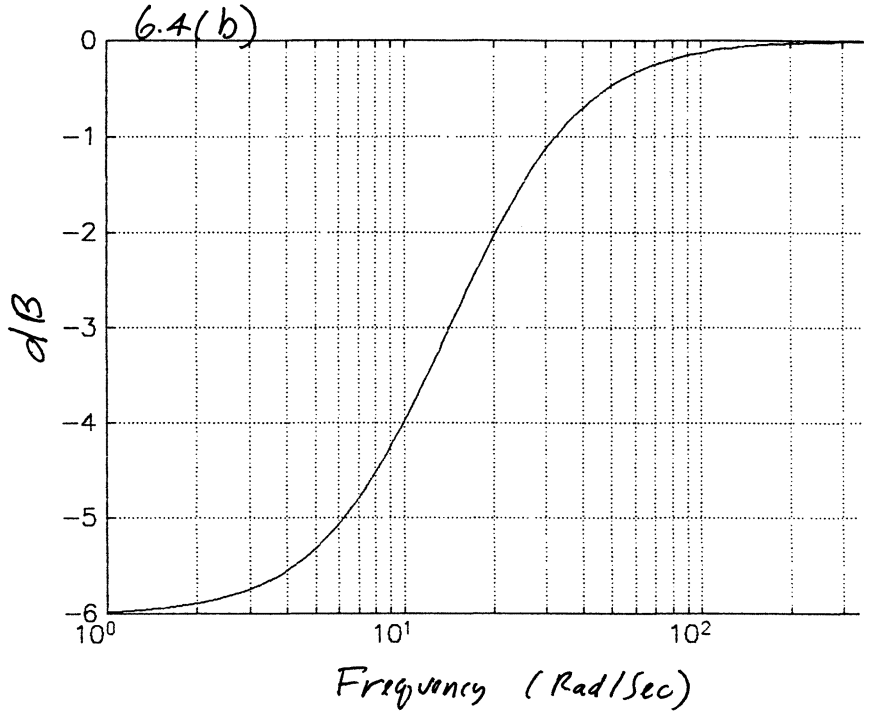
$$= \frac{105s + 10^6}{2 \times 105 s + 10^6}$$

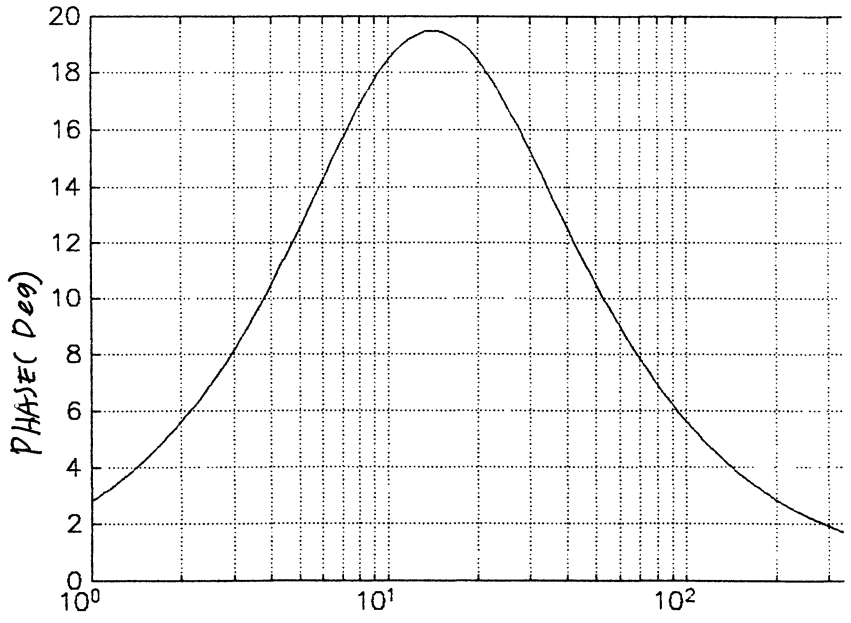
$$= \boxed{\frac{s+10}{2(s+5)}}$$

Bode plots follow for 6.4(a) and 6.4(b). The Bode for 6.4(c) is in the text.


6.4(a)
0
-5
-10
-15
-20
-25
-30
-35
-40
dB
10^0
10^1
10^2
Freq (Rad/Sec)


0
-10
-20
-30
-40
-50
-60
-70
-80
-90
PHASE (DEG)
10^0
10^1
10^2


6.4(b)
0
−1
−2
−3
−4
−5
−6
dB
10^0
10^1
10^2
Frequency (Rad/Sec)


20
18
16
14
12
10
8
6
4
2
0
PHASE (Deg)
10^0
10^1
10^2

6.5

(a) $f(t) = 10\frac{d^2x}{dt^2} + 7\frac{dx}{dt}$

With zero init. conditions

$F(s) = 10s^2X(s) + 7sX(s)$

$T(s) = \frac{F(s)}{X(s)} = \boxed{\frac{1}{s(10s+7)}}$

(b) $f(t) = 10\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 5x$

With zero init cond.

$F(s) = 10s^2X(s) + 2sX(s) + 5X(s)$

$T(s) = \frac{F(s)}{X(s)} = \boxed{\frac{1}{10s^2+2s+5}}$

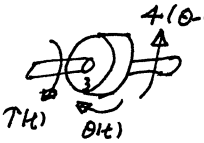
Bode plots follow on the next page

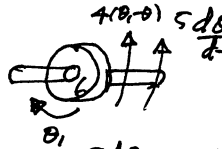
6.6

(a) $T(t) = 10\frac{d^2\theta}{dt^2} + 8\frac{d\theta}{dt} + 5\theta$ with zero init cond

$T(s) = \theta(s)[10s^2+8s+5]$ $F(s) = \frac{\theta(s)}{T(s)} = \boxed{\frac{1}{10s^2+8s+5}}$

(b) Take free body diagrams for each inertial mass





$T(t) - 4(\theta - \theta_1) = 3\frac{d^2\theta}{dt^2}$

$3\frac{d^2\theta}{dt^2} + 4\theta - 4\theta_1 = T$

With zero Ics: $T(s) = \theta(s)[3s^2+4] - 4\theta_1(s)$

$-5\frac{d\theta_1}{dt} - 4\theta_1 + 4\theta = 6\frac{d^2\theta_1}{dt^2}$

$6\frac{d^2\theta_1}{dt^2} + 5\frac{d\theta_1}{dt} + 4\theta_1 = 4\theta$

With zero initial conditions

$\theta_1(s)[6s^2+5s+4] = 4\theta(s)$

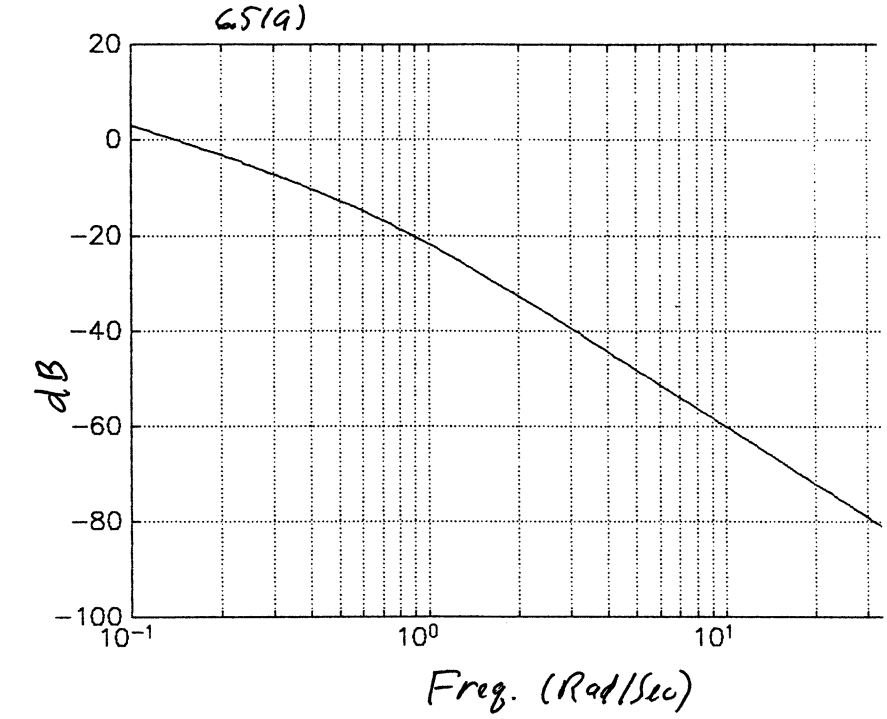
$\theta_1(s) = \frac{4\theta(s)}{6s^2+5s+4}$

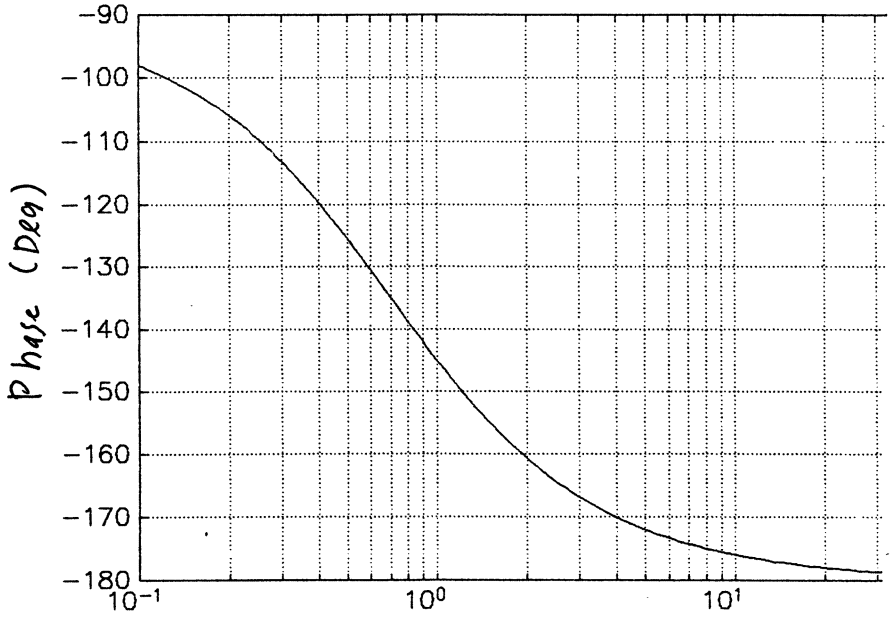
$T(s) = \theta(s)\left[3s^2+4 - \frac{16}{6s^2+5s+4}\right]$

$= \frac{18s^4+15s^3+36s^2+20s}{6s^2+5s+4}$

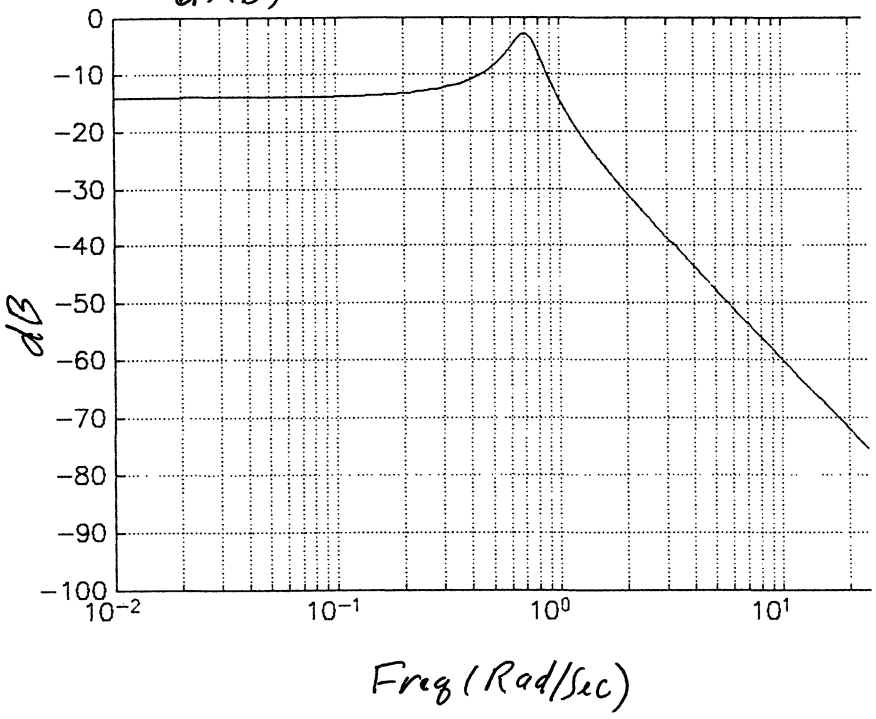
$F(s) = \frac{\theta(s)}{T(s)} = \boxed{\frac{6s^2+5s+4}{10s^4+15s^3+36s^2+20s}}$

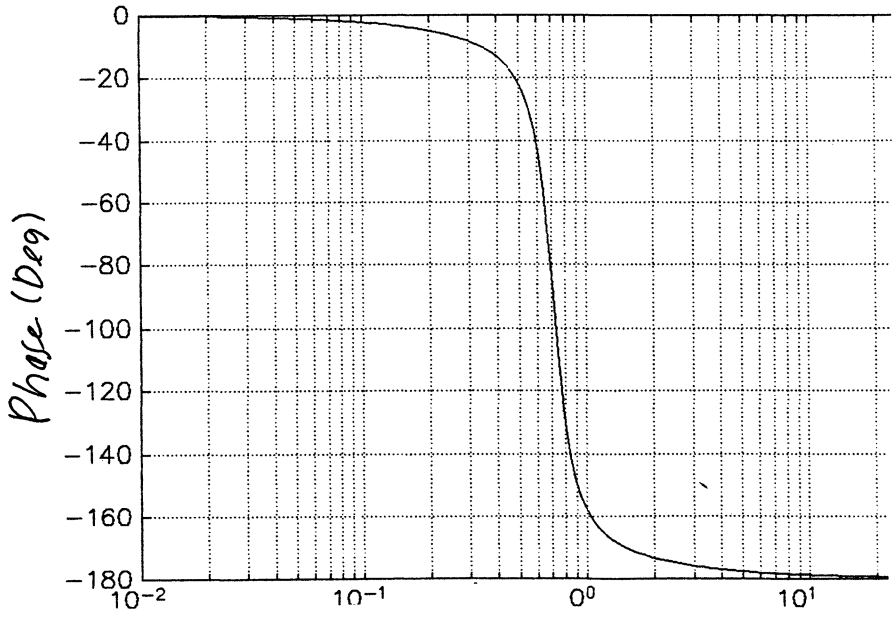
* Bode plots follow those of Prob 6.6


G5(s)
20
0
-20
-40
-60
-80
-100
dB
10^{-1}
10^{0}
10^{1}
Freq. (Rad/Sec)

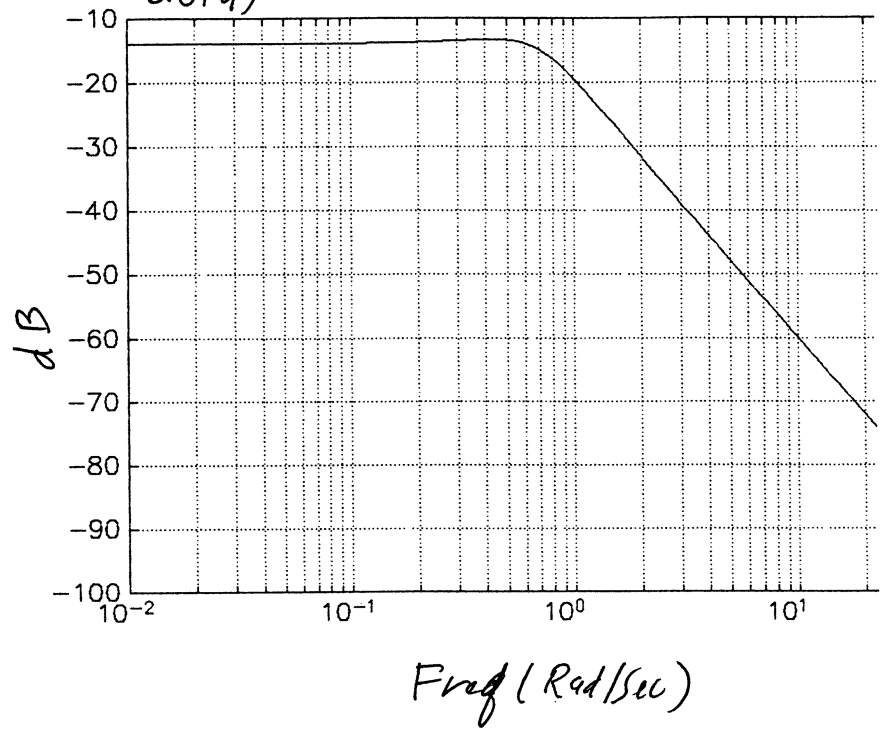

-90
-100
-110
-120
-130
-140
-150
-160
-170
-180
Phase (Deg)
10^{-1}
10^{0}
10^{1}

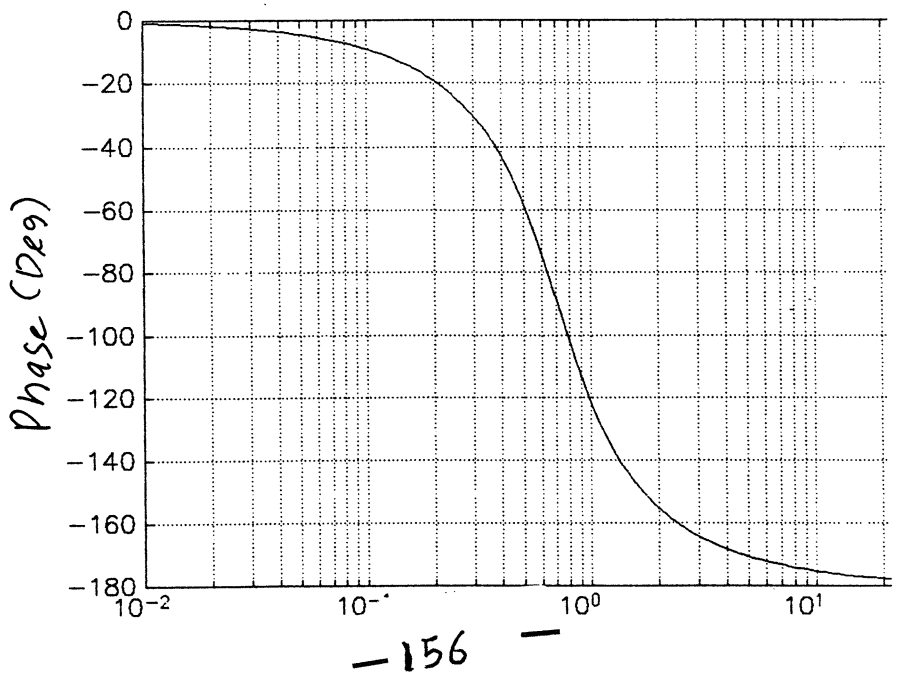
6.5(b)


0
−10
−20
−30
−40
−50
−60
−70
−80
−90
−100
dB
10^{-2}
10^{-1}
10^{0}
10^{1}
Freq (Rad/Sec)

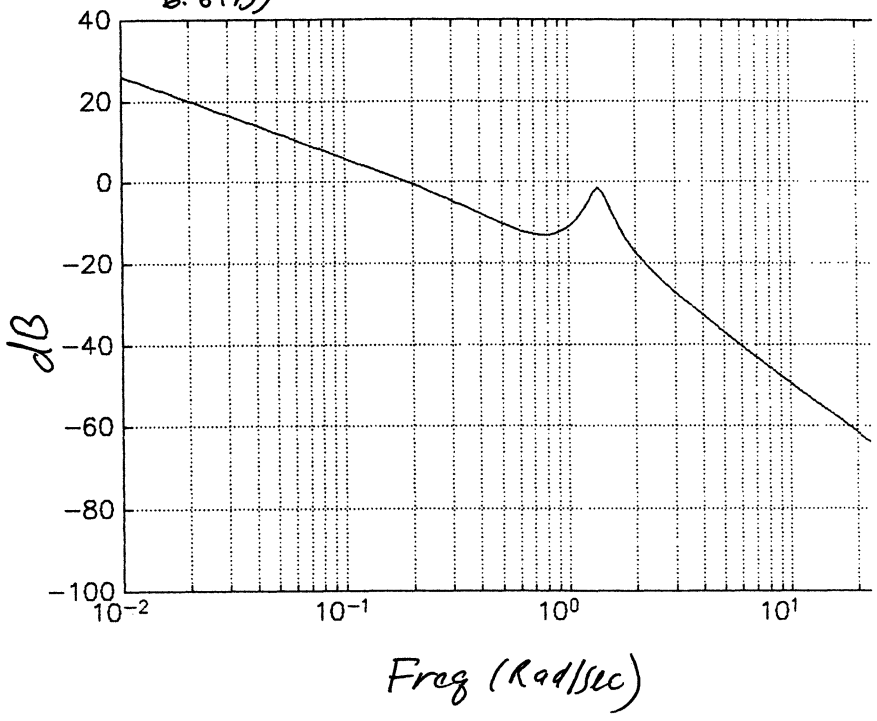

0
−20
−40
−60
−80
−100
−120
−140
−160
−180
Phase (Deg)
10^{-2}
10^{-1}
0^{0}
10^{1}

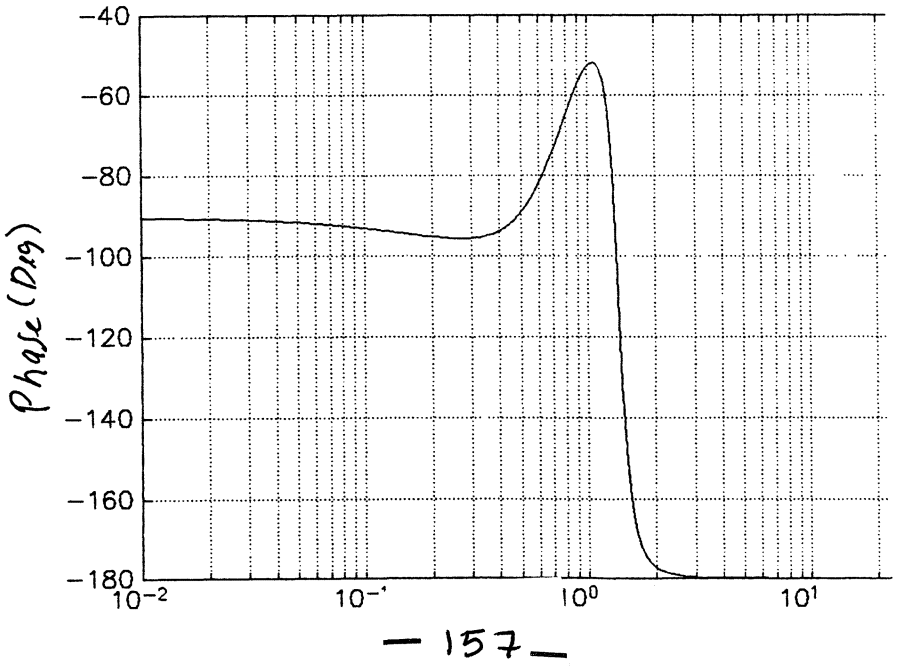
6.6(a)


-10
-20
-30
-40
-50
-60
-70
-80
-90
-100
dB
10^{-2}
10^{-1}
10^{0}
10^{1}
Freq (Rad/Sec)


0
-20
-40
-60
-80
-100
-120
-140
-160
-180
Phase (Deg)
10^{-2}
10^{-1}
10^{0}
10^{1}

6.6(b)


40
20
0
−20
−40
−60
−80
−100
dB
10^{-2}
10^{-1}
10^{0}
10^{1}
Freq (Rad/sec)


−40
−60
−80
−100
−120
−140
−160
−180
Phase (Deg)
10^{-2}
10^{-1}
10^{0}
10^{1}

6.7 Corner freq at 100, 1000, 10,000

$$F(s) = \left(\frac{s+100}{100}\right)\left(\frac{1000}{s+1000}\right)\left(\frac{10{,}000}{s+10{,}000}\right)^2$$

6.8 Slope changes indicate corner freq at 1, 10

$$F(s) = \left(\frac{1}{s+1}\right)^2\left(\frac{s+10}{10}\right)^2$$

6.9

(a)

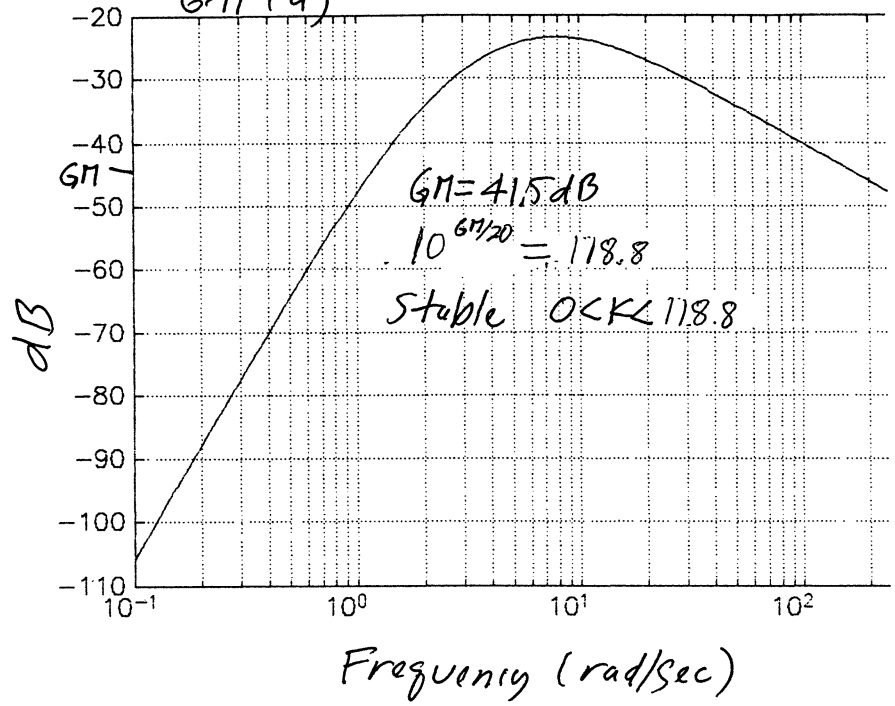
ω	$A(\omega)$	dB	$\angle$
0.1	20	26	-90°
0.2	10.08	20.7	-97°
0.4	5.06	14	-99°
1	1.8	5	-109°
2	0.72	-2.8	-123°
4	0.28	-11	-135°
8	0.05	-26	-171°
16	0.006	-44	$+111.5^\circ$

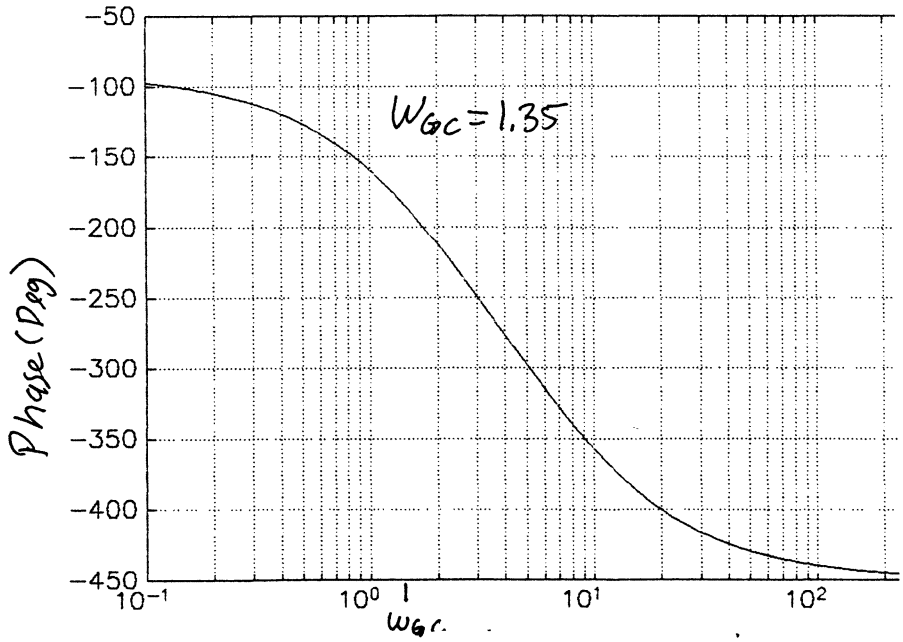
$$F(s) = \left(\frac{2}{s}\right)\left(\frac{4}{s+4}\right)\left(\frac{10}{s+10}\right)$$

6.10

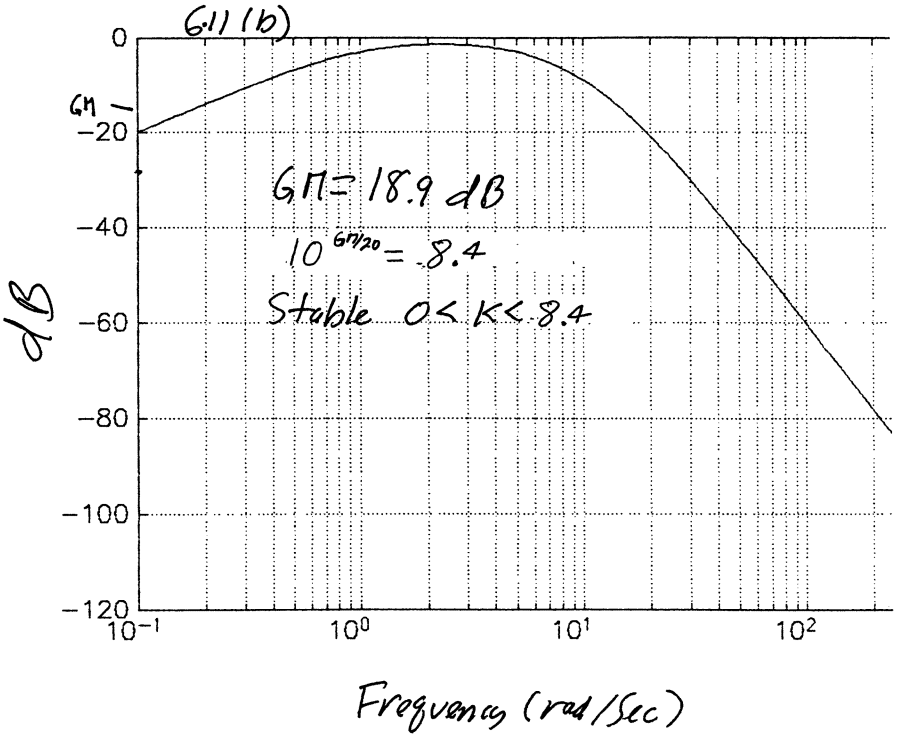
Part	ω_{gc}	PM	ω_{pc}	GM(dB)
(a)	—	∞	12.13	30
(b)	0.99	78.7	10.0	26
(c)	—	∞	6.93	54
(d)	27.6	79	109.6	17.6
(e)	0.01	89.9°	10.0	66.

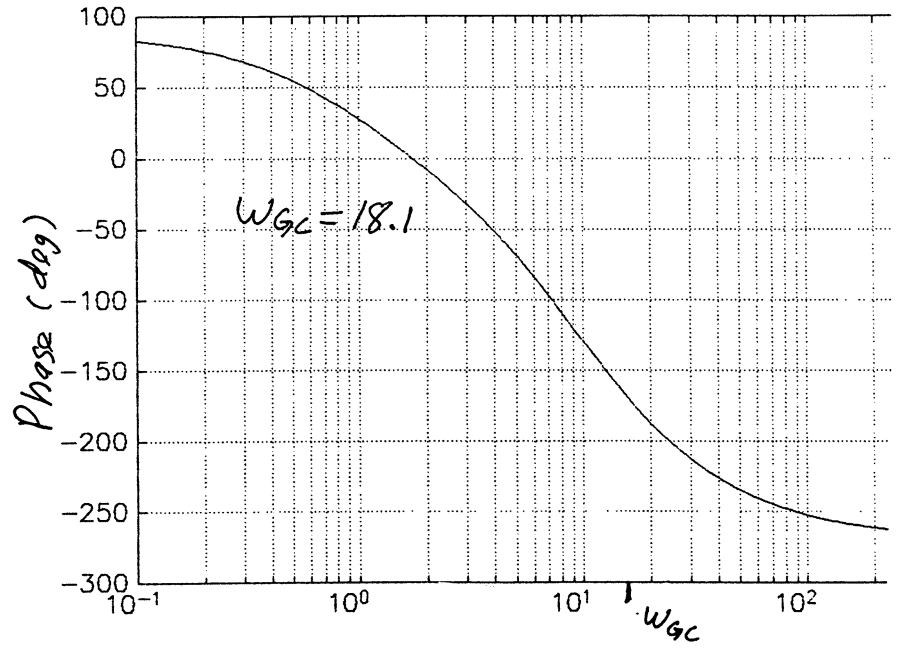
G.11 (a)


GM
GM = 41.5 dB
$10^{GM/20} = 118.8$
Stable $0 < K < 118.8$
dB
Frequency (rad/sec)

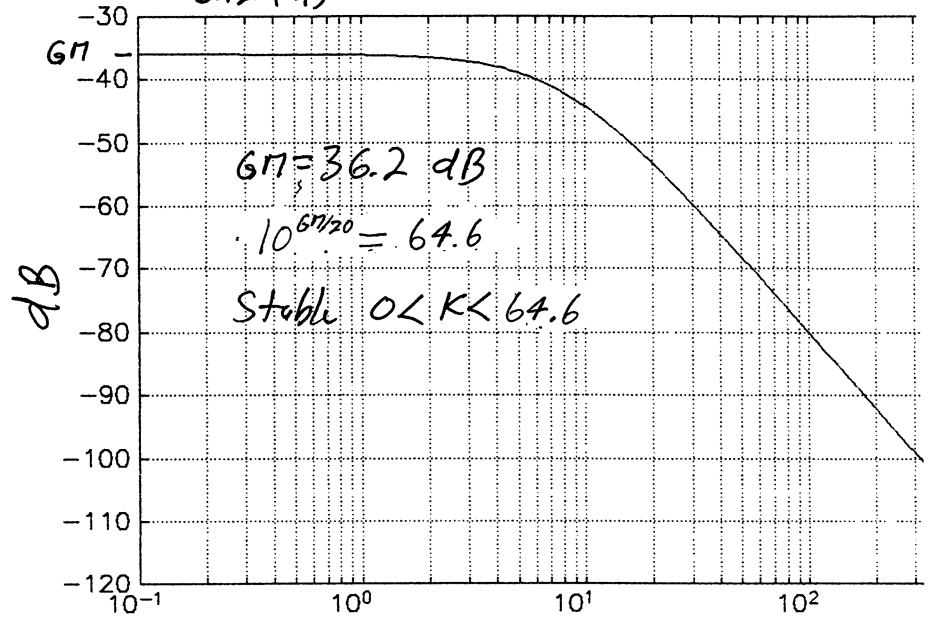

$W_{GC} = 1.35$
Phase (Deg)
W_{GC}

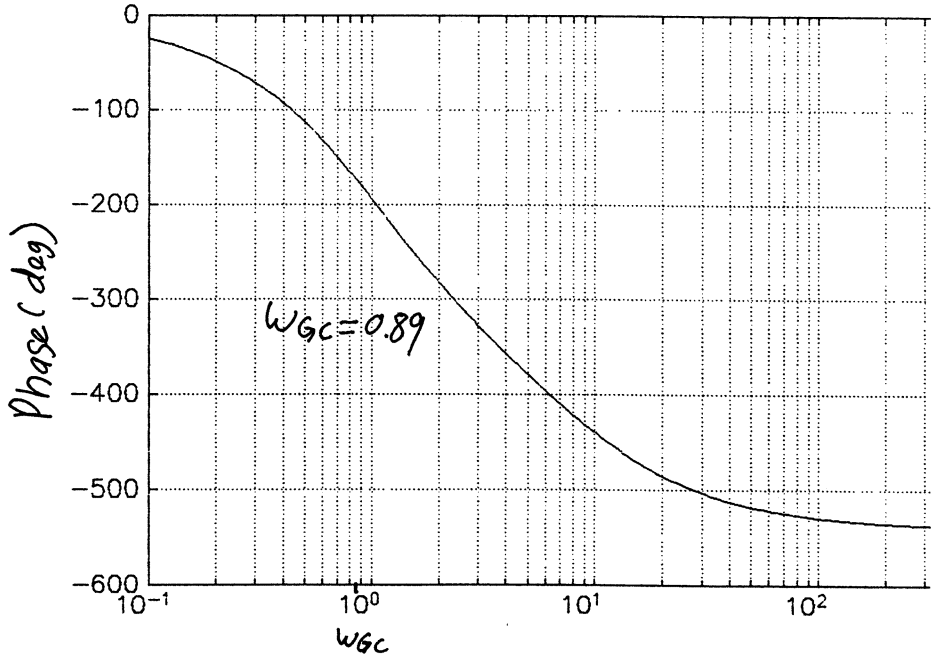
1


G11(b)
GM
GM = 18.9 dB
$10^{GM/20} = 8.4$
Stable $0 < K < 8.4$
dB
0
−20
−40
−60
−80
−100
−120
10^{-1}
10^{0}
10^{1}
10^{2}
Frequency (rad/Sec)


$W_{GC} = 18.1$
Phase (deg)
100
50
0
−50
−100
−150
−200
−250
−300
10^{-1}
10^{0}
10^{1}
10^{2}
W_{GC}

6.12 (a)


GM
GM=36.2 dB
$10^{GM/20} = 64.6$
Stable $0 < K < 64.6$
dB
Frequency (rad/sec)


$\omega_{GC} = 0.89$
Phase (deg)
ω_{GC}

6.12(b)

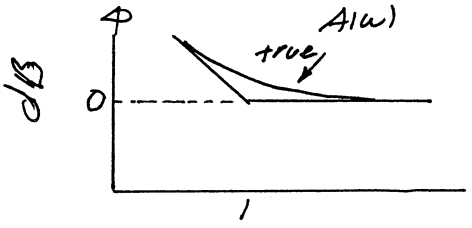
$$G(s) = \left(\frac{s-K}{s+K}\right)^2 \qquad G(j\omega) = \left(\frac{j\omega - K}{j\omega + K}\right)^2$$

$$G(j\omega) = \left[\frac{\sqrt{\omega^2+K^2}\angle 180^\circ - \tan^{-1}\frac{\omega}{K}}{\sqrt{\omega^2+K^2}\angle \tan^{-1}\frac{\omega}{K}}\right]^2$$

$$= 1\angle\left(360^\circ - 4\tan^{-1}\frac{\omega}{K}\right) = 1\angle -4\tan^{-1}\frac{\omega}{K}$$

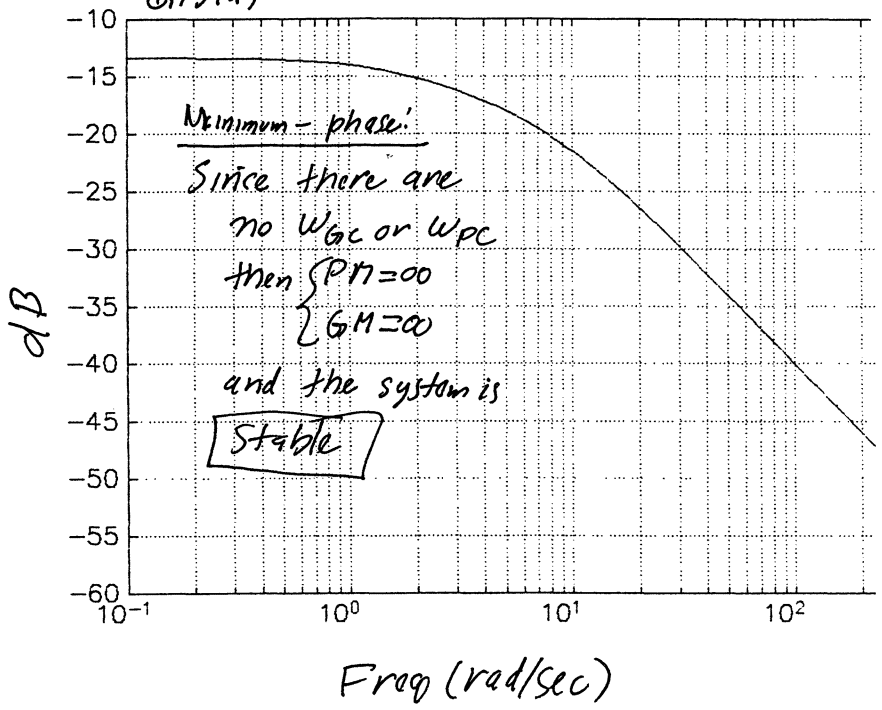
$$H(s) = \frac{s+1}{5} \qquad H(j\omega) = \frac{\sqrt{1+\omega^2}}{\omega}\angle \tan^{-1}\omega - 90^\circ$$

$$G(j\omega)H(j\omega) = \underbrace{\frac{\sqrt{1+\omega^2}}{\omega}}_{A(\omega)}\angle \overbrace{\tan^{-1}\omega - 90^\circ - 4\tan^{-1}\frac{\omega}{K}}^{\phi(\omega)}$$



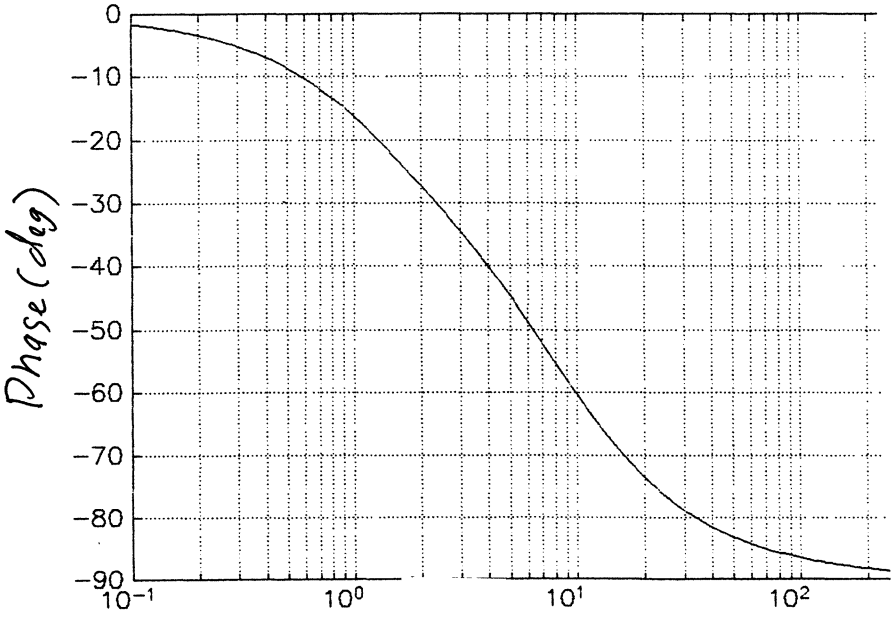
For any K, $\phi(\omega) = 180^\circ$ at a freq where $A(\omega) > 0$ dB hence the system is unstable

6.13(a)

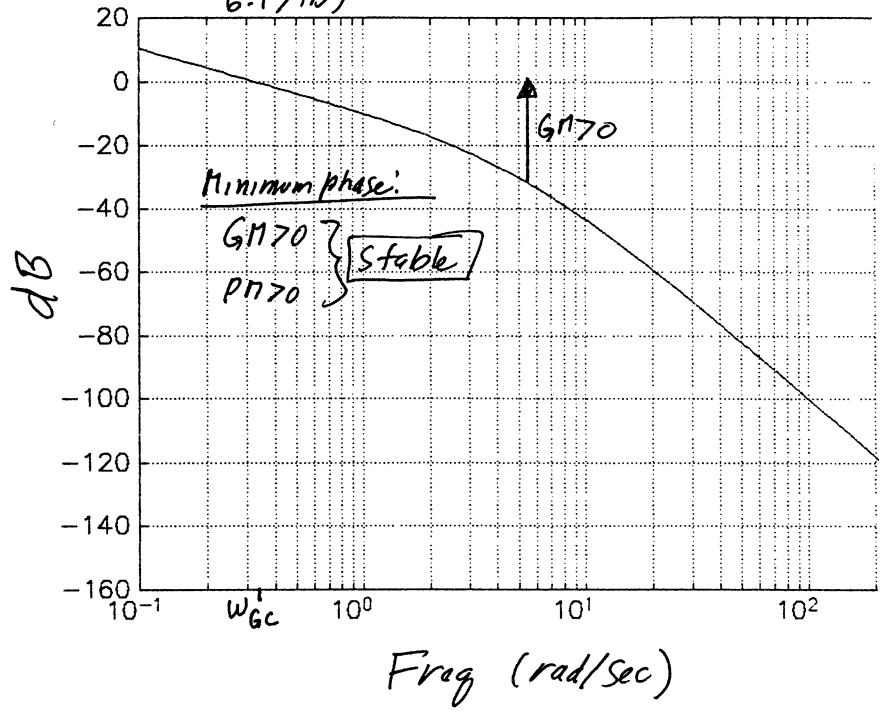


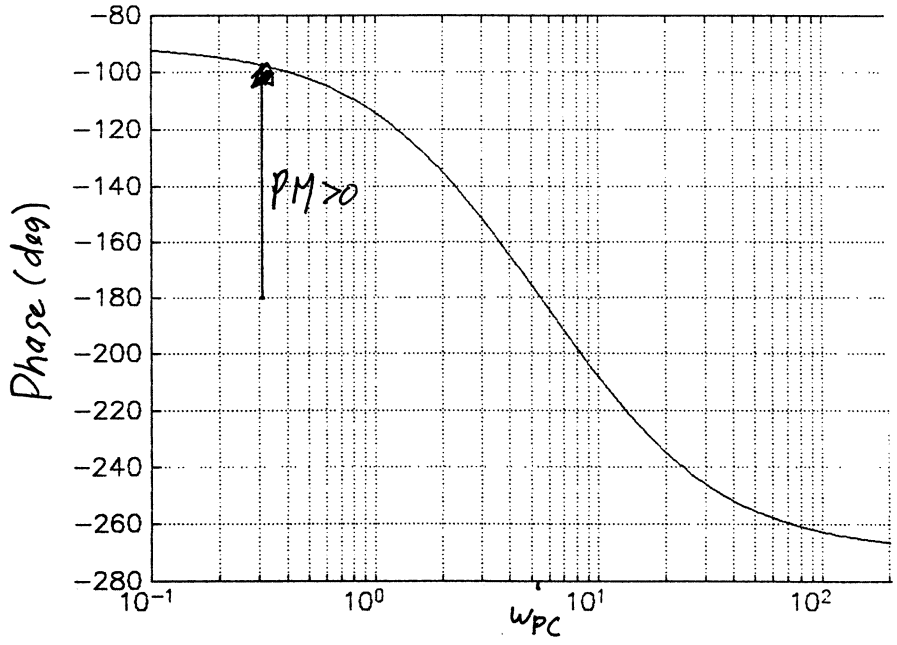
Minimum-phase:

Since there are no ω_{gc} or ω_{pc} then $\begin{cases} PM = \infty \\ GM = \infty \end{cases}$ and the system is **Stable**

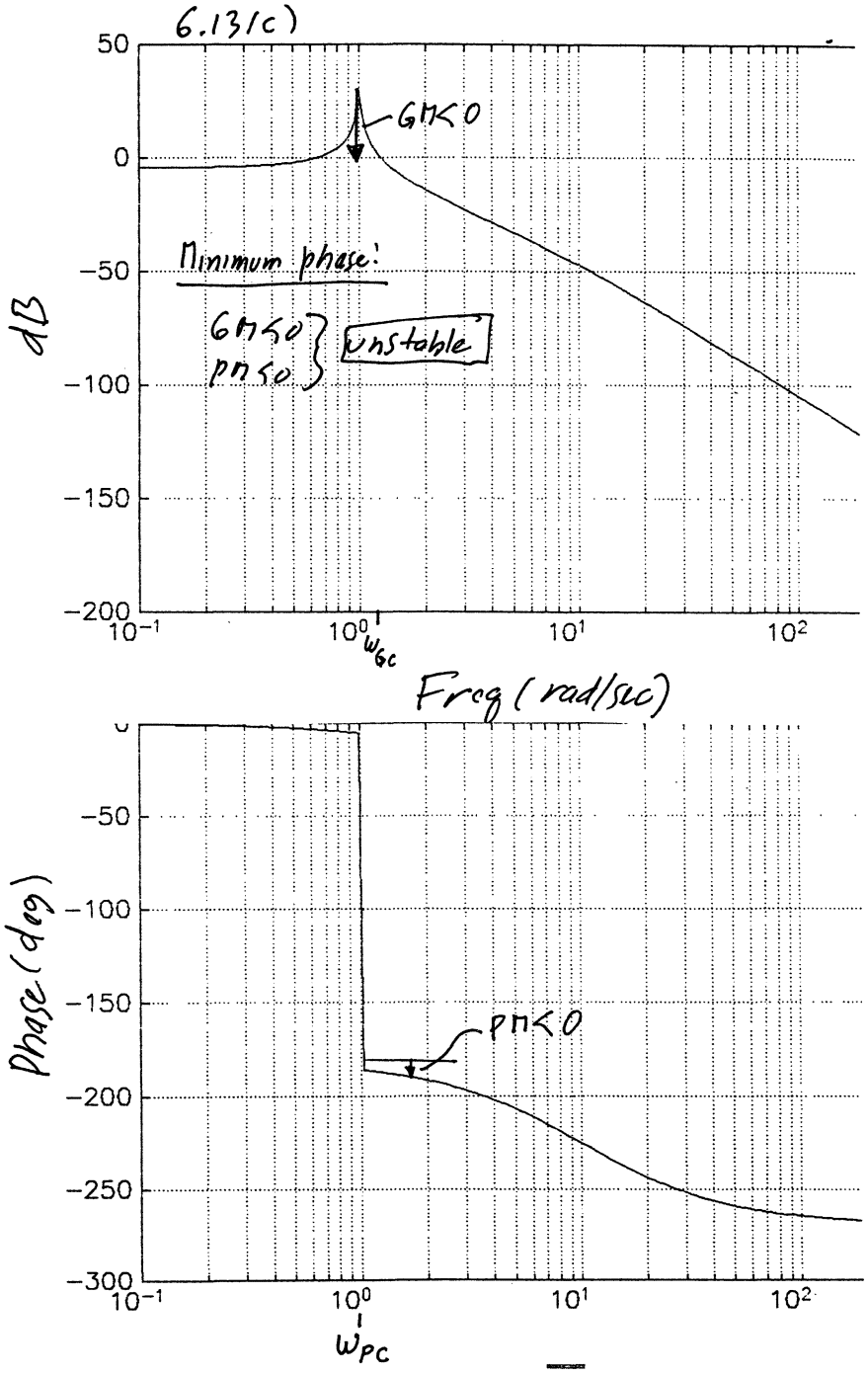


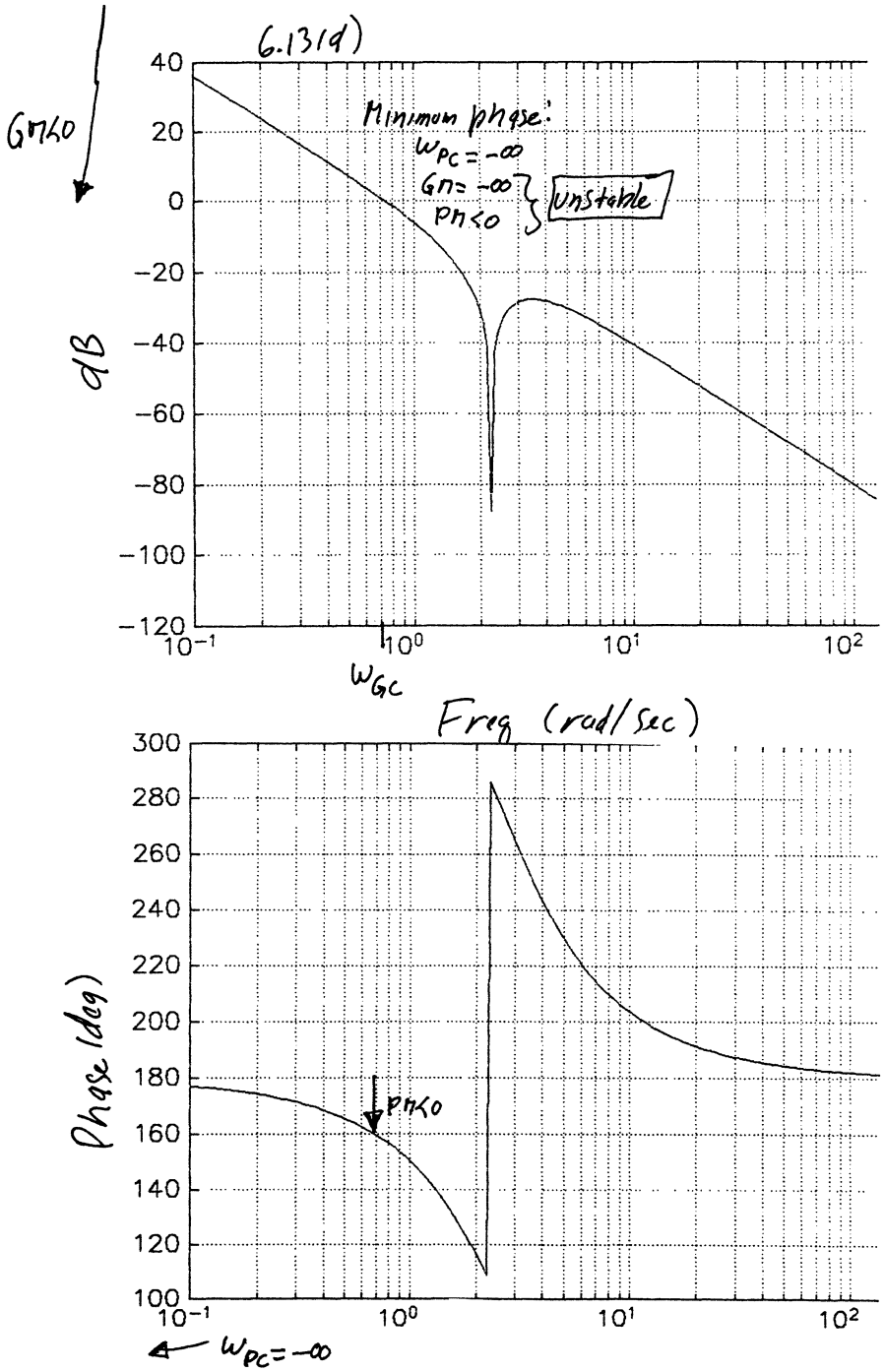
6.13(b)


20
0
−20
−40
−60
−80
−100
−120
−140
−160
dB
GM>0
Minimum phase:
GM>0
PM>0
Stable
10^{-1}
ω_{GC}
10^{0}
10^{1}
10^{2}
Freq (rad/sec)


−80
−100
−120
−140
−160
−180
−200
−220
−240
−260
−280
Phase (deg)
PM>0
10^{-1}
10^{0}
ω_{PC}
10^{1}
10^{2}

1


6.13(c)
50
0
−50
−100
−150
−200
dB
GM<0
Minimum phase:
GM<0
PM<0
unstable
10^{-1}
10^{0}
ω_{Gc}
10^{1}
10^{2}
Freq (rad/sec)
−50
−100
−150
−200
−250
−300
Phase (deg)
PM<0
10^{-1}
10^{0}
ω_{Pc}
10^{1}
10^{2}

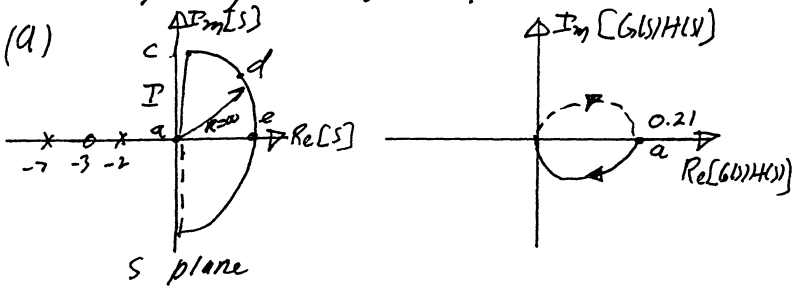

6.13(d)
GM<0
Minimum phase:
$w_{pc} = -\infty$
$G_m = -\infty$
$pm < 0$
unstable
dB
40
20
0
−20
−40
−60
−80
−100
−120
10^{-1}
10^{0}
10^{1}
10^{2}
w_{GC}
Freq (rad/sec)
Phase (deg)
300
280
260
240
220
200
180
160
140
120
100
pm<0
$w_{pc} = -\infty$

1

6.14

The answers to 6.14(b,d) are in the text. For 6.14(a,c), the Bode plots from problem 6.13(a,c) may be used for the mapping of the positive imaginary axis (I).

(a)

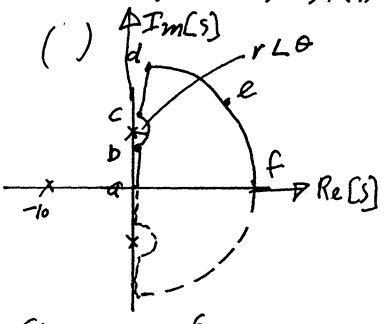


$$\begin{pmatrix}\text{Number of RHP}\\ \text{CL poles}\end{pmatrix} = \begin{pmatrix}\text{Number of CW}\\ \text{encirc. of -1}\end{pmatrix} + \begin{pmatrix}\text{Number of}\\ \text{RHP OL poles}\end{pmatrix}$$

$$= 0+0 = 0 \quad \boxed{\text{Stable}}.$$

System is stable for all $K>0$ also

( )



$G(s)H(s) = \frac{6}{(s^2+1)(s+10)}$

at $s=0$ (point a)

$G(0)H(0) = \frac{6}{10}$

Near $s=j1$, let $s = j1 + r\angle\theta$

$-90° \le \theta \le +90°$, r small

$G(s)H(s) = \frac{6}{(s+j1)(s-j1)(s+10)}$

$s = j1 + r\angle\theta$

$G(s)H(s) = \frac{6}{(j1+j1)(r\angle\theta)(j1+10)}$

$\sqrt{11}\angle\tan^{-1}\frac{1}{10}$

$= \frac{6}{(2\angle 90°)(r\angle\theta)(\sqrt{11}\angle 5.7°)}$

$= R\angle -\theta - 95.7°$

R large

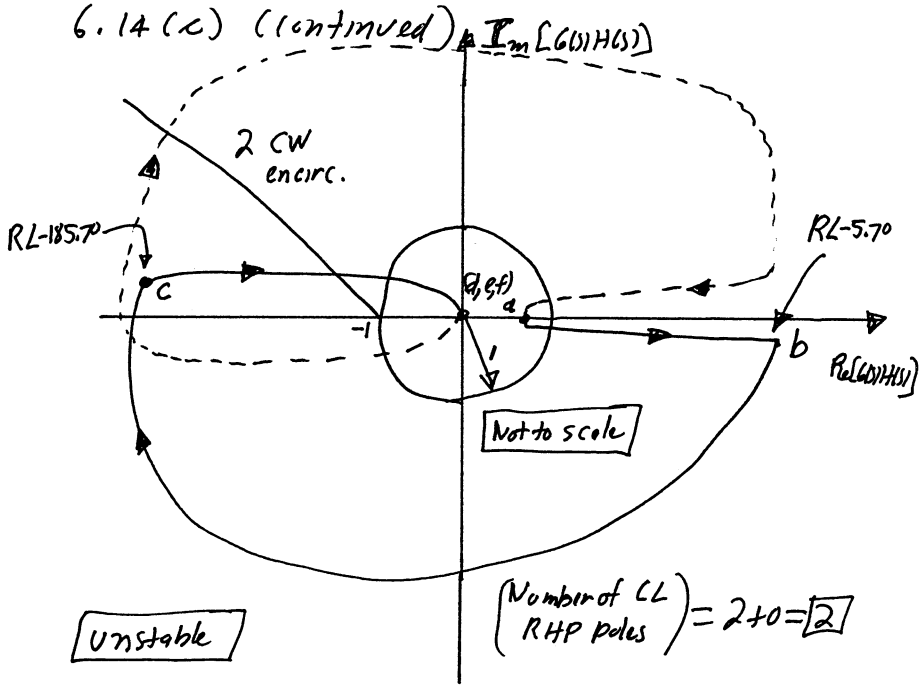
In G(s)H(s) plane, for R large, since $-90° \le \theta \le +90°$, then

$90° \ge -\theta \ge -90°$

$-5.7° \ge -\theta - 95.7° \ge -185.7°$

So mapping of [b,c] goes from $-5.7°$ to $-185.7°$.

6.14 (c) (continued)



Unstable

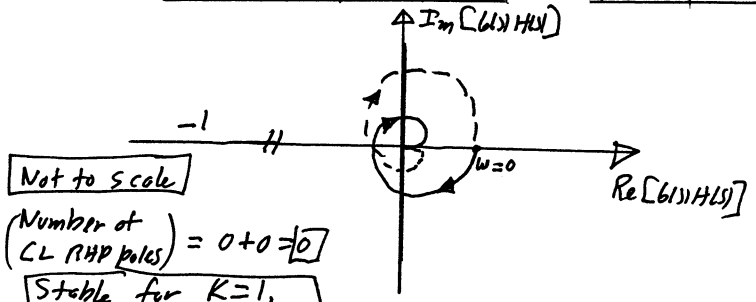
(Number of CL RHP poles) = 2+0 = 2

6.15 (a)

$G(0)H(0) = 0.02$

Data for I

ω	0	0.1	1	10	100	∞
$A(\omega)$	0.02	0.02	0.02	1.5×10^{-3}	3×10^{-7}	0
$\phi(\omega)$	0°	-3.90	-38.40	-233°	-345.70	-360°



Not to scale

(Number of CL RHP poles) = 0+0 = 0

Stable for K=1, unstable for large K

6.15(b)

$G(0)H(0) = 0.03$

Data for plotting I

ω	0	.1	1	10	100	∞
$A(\omega)$	.03	.03	.03	6×10^{-4}	1.2×10^{-7}	0
$\phi(\omega)$	0	-4.7°	-48°	-264°	-347°	-360°

The shape of the Nyquist plot is generally the same as for problem 6.15(a). The system is stable for K=1 and unstable for large K.

6.15(c)

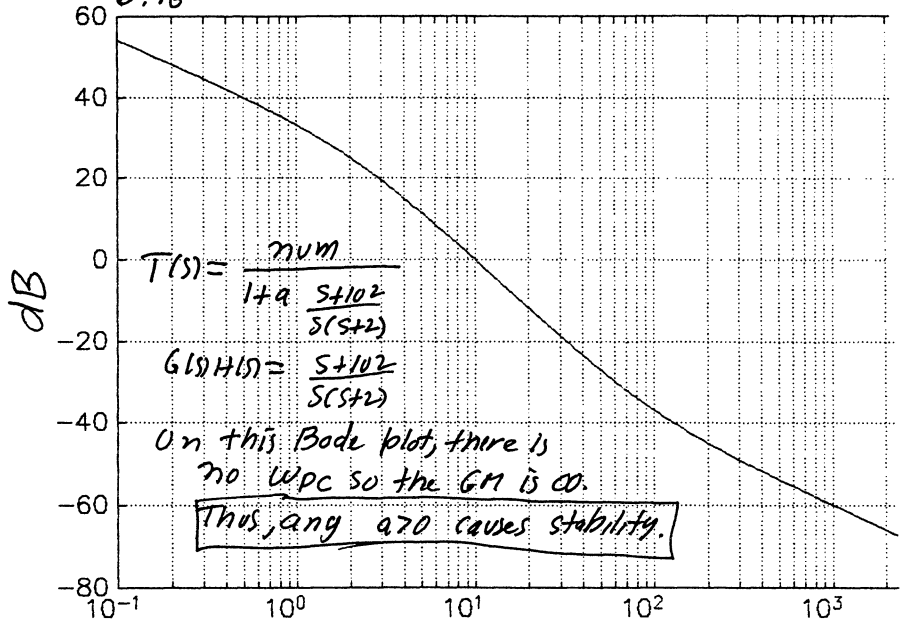
$G(0)H(0) = 0.04$

Data for plotting I

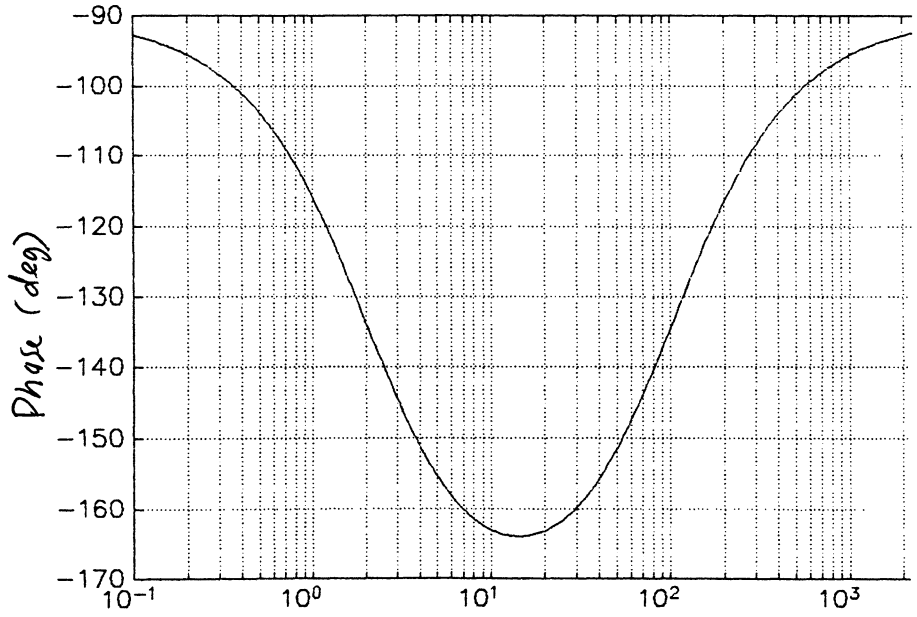
ω	0	0.1	1	3	10	100	∞
$A(\omega)$	.04	.04	.043	.07	.018	.02	0
$\phi(\omega)$	0°	-1.2°	-13.7°	-84.3°	-187°	-257°	-360°

The shape ~~of~~ the Nyquist plot is generally the same as for problem 6.15(a), although the magnitude increases between 0° and 90°. The stability conclusion is the same. The system is stable for K=1 and unstable for large K.

6.16



Frequency (rad/sec)



−170

1

6.17

Let $G(jw) = M \angle \theta$

Define $GDB = 20 \log_{10} M$

$GP = \theta$

Then for $T(jw)$

$$T(jw) = \frac{G(jw)}{1+G(jw)} = \frac{M\angle\theta}{1+R+jI}$$

$R = M\cos\theta \quad I = M\sin\theta$

$$T(jw) = \frac{M\angle\theta}{M_1\angle\theta_1} \begin{cases} M_1 = \sqrt{(1+R)^2+I^2} \\ \theta_1 = \tan^{-1}\frac{I}{1+R} \end{cases}$$

$$T(jw) = \frac{M}{M_1}\angle\theta-\theta_1$$

The following set of instructions must be coded to get from GDB and GP to TDB (dB value of $T(jw)$) and TP (the $\angle T(jw)$)

Instructions:

$M = 10^{(GDB/20)}$

$R = M\cos(GP)$

$I = M\sin(GP)$

$$MT = \frac{M}{\sqrt{(1+R)^2+I^2}}$$

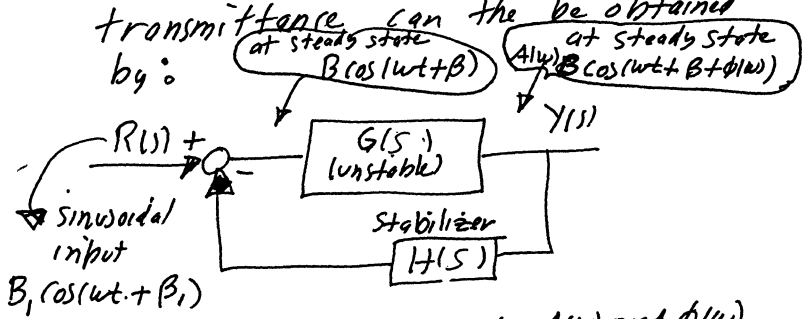
$TDB = 20\log_{10} MT$

$TP = GP - \tan^{-1}\left(\frac{I}{1+R}\right)$

6.18

To obtain a frequency response by exciting a system with a sinusoidal input, the sinusoidal steady state output is required. This steady state result will not occur unless the closed-loop system is stable.

If the closed-loop system is stable, the frequency response of an unstable transmittance can the be obtained by:



For each ω, we can get $A(\omega)$ and $\phi(\omega)$.

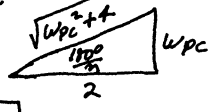
6.19

The largest value of K would cause the system to be marginally stable at the phase crossover frequency ω_{pc}. At ω_{pc}

$$G_E(j\omega) = \frac{K}{(j\omega+2)^n} = \frac{K}{(\sqrt{\omega_{pc}^2+4})^n \angle n\tan^{-1}\frac{\omega_{pc}}{2}}$$

$$= \frac{K}{\sqrt{\omega_{pc}^2+4}} \angle \underbrace{-n\tan^{-1}\frac{\omega_{pc}}{2}}_{-180^\circ}$$

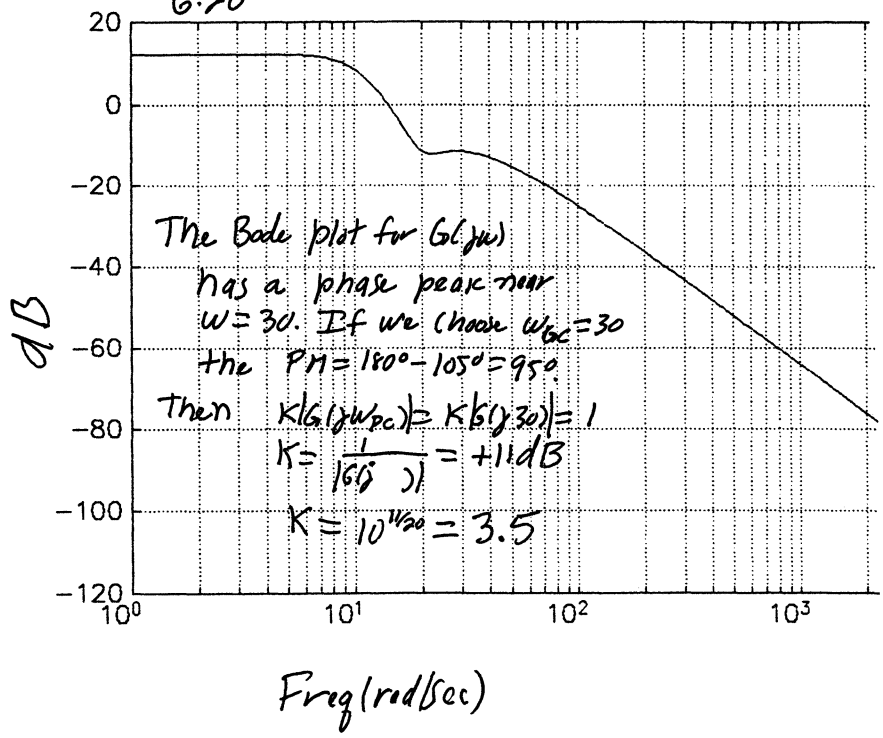
$$\tan^{-1}\frac{\omega_{pc}}{2} = \frac{180^\circ}{n}$$



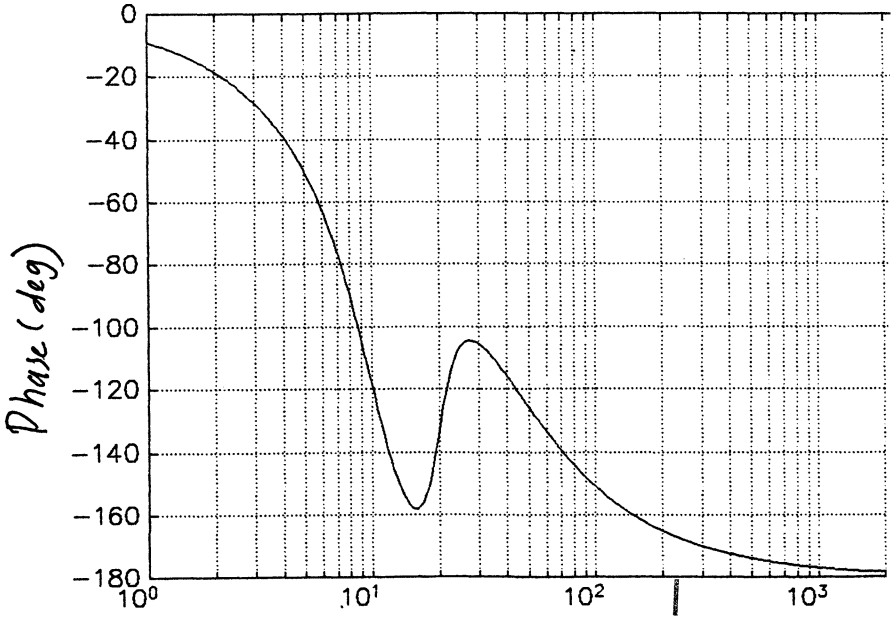
$$\cos\left(\frac{180^\circ}{n}\right) = \frac{2}{\sqrt{\omega_{pc}^2+4}}$$

$$\boxed{0 < K < (\omega_{pc}^2+4)^n = \frac{2^n}{\cos^n\left(\frac{180^\circ}{n}\right)}}$$

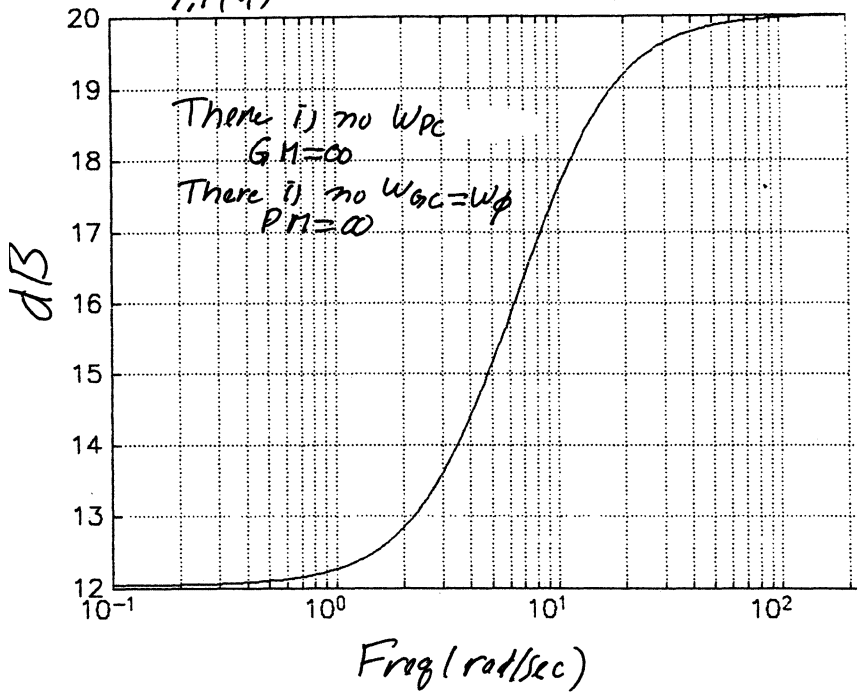
6.20

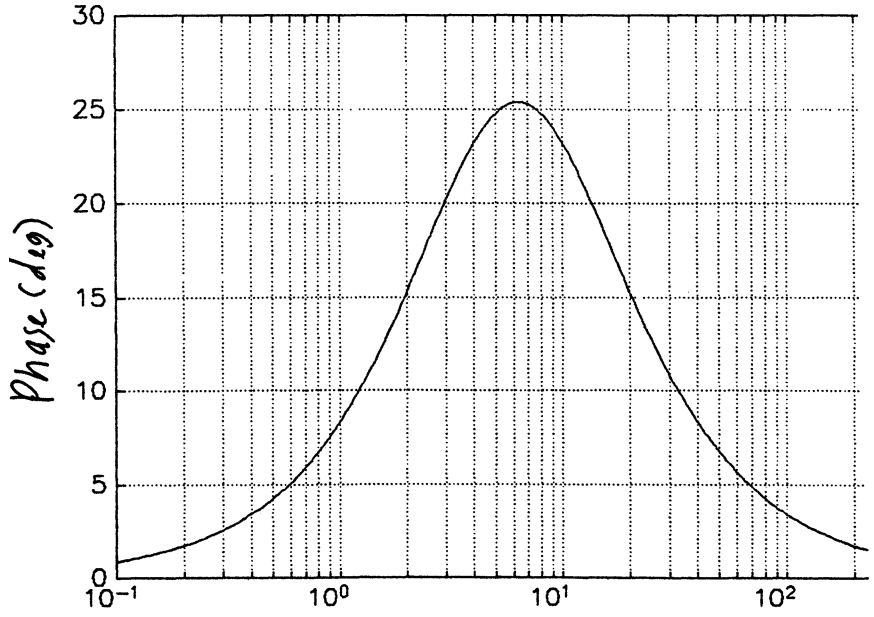


Freq (rad/sec)

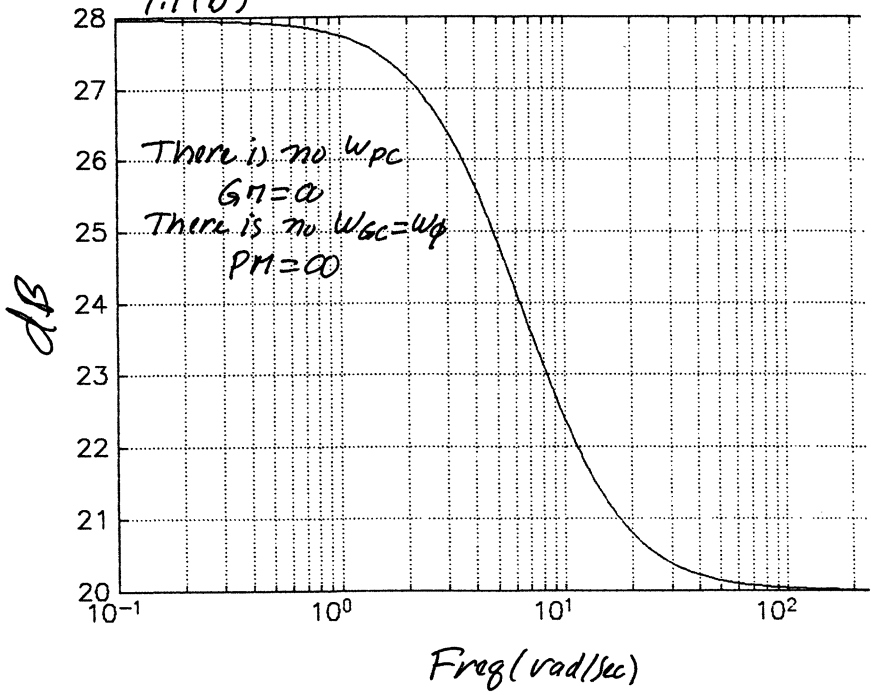


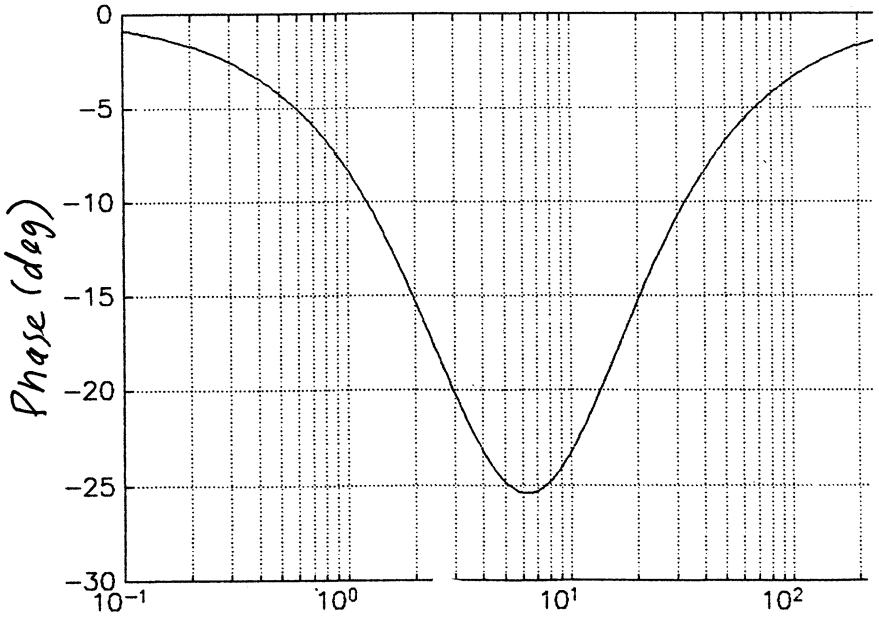
7.1(a)


There is no ω_{pc}
$GM=\infty$
There is no $\omega_{gc}=\omega_p$
$PM=\infty$
dB
Freq (rad/sec)

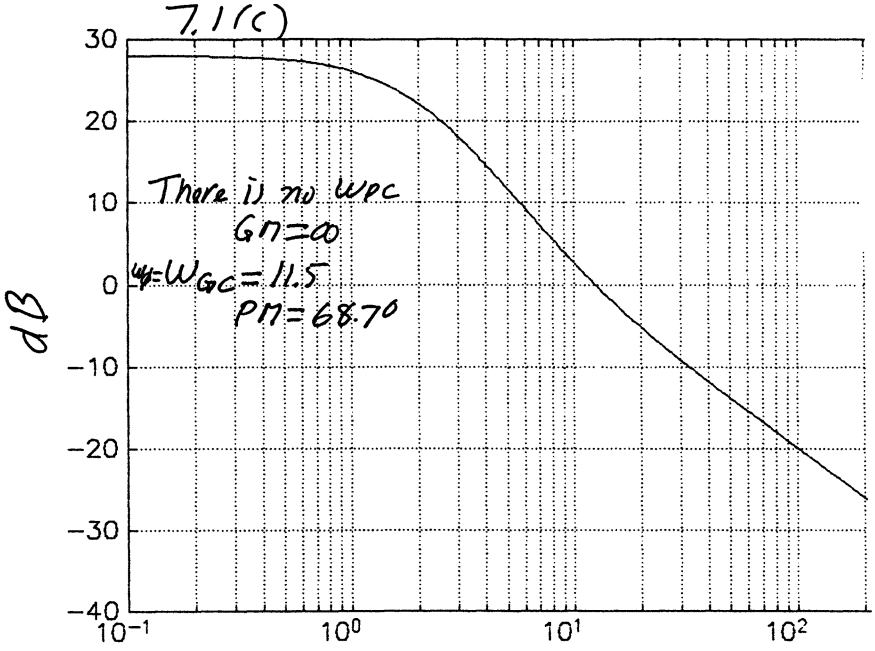

Phase (deg)

7.1(b)

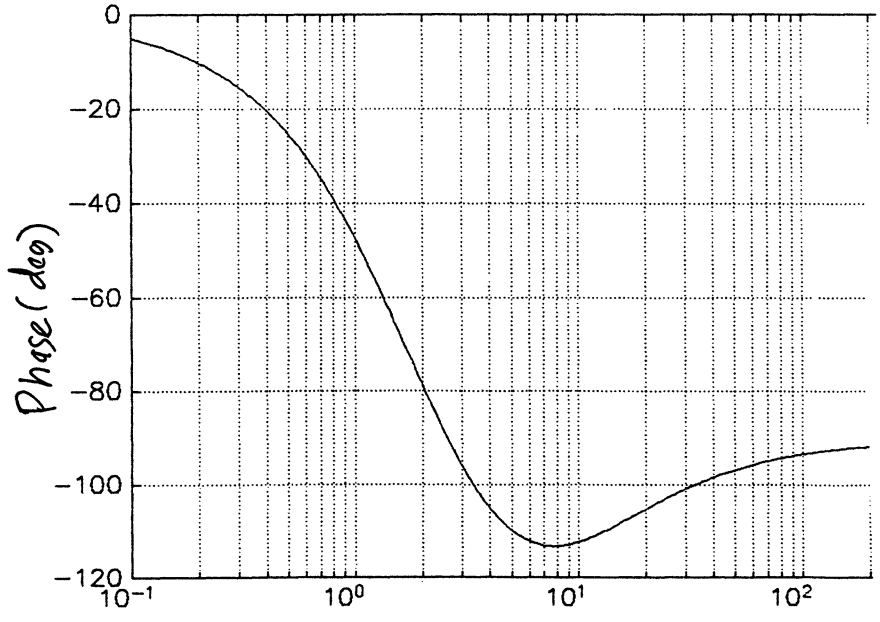

There is no ω_{pc}
$G_m = \infty$
There is no $\omega_{gc} = \omega_\phi$
$P_M = \infty$
dB
Freq (rad/sec)


Phase (deg)

7.1(c)

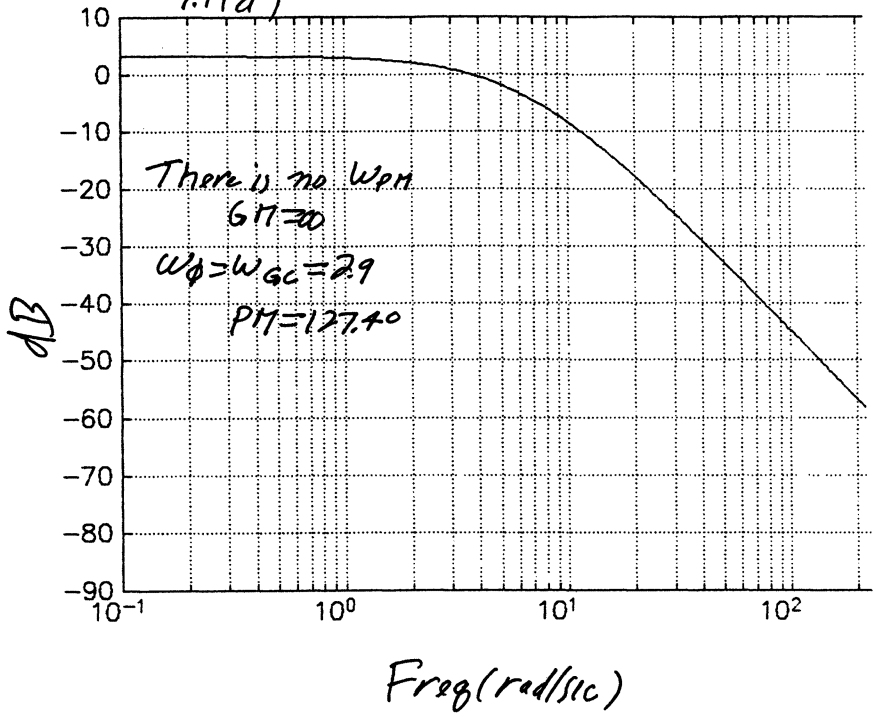


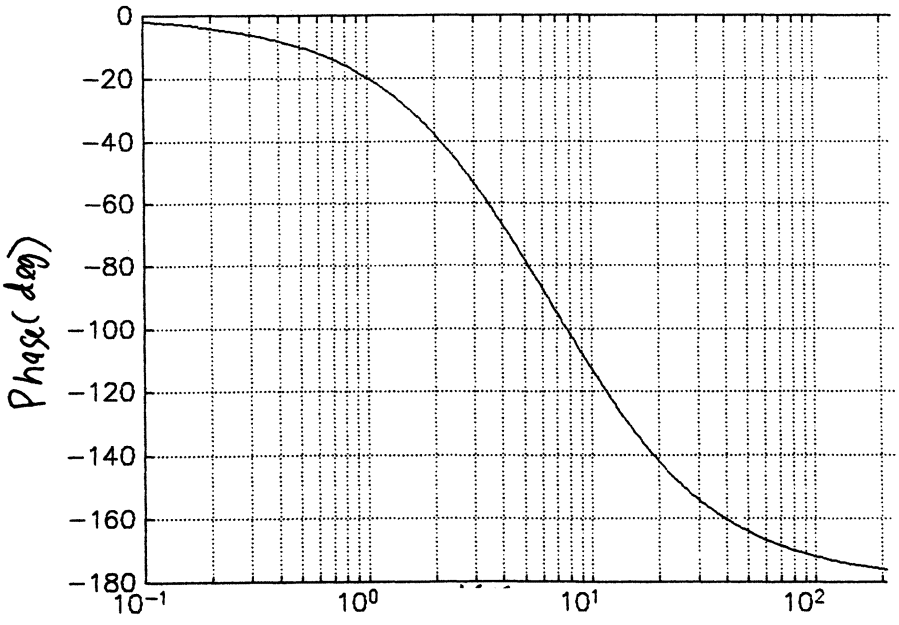
Freq(rad/sec)



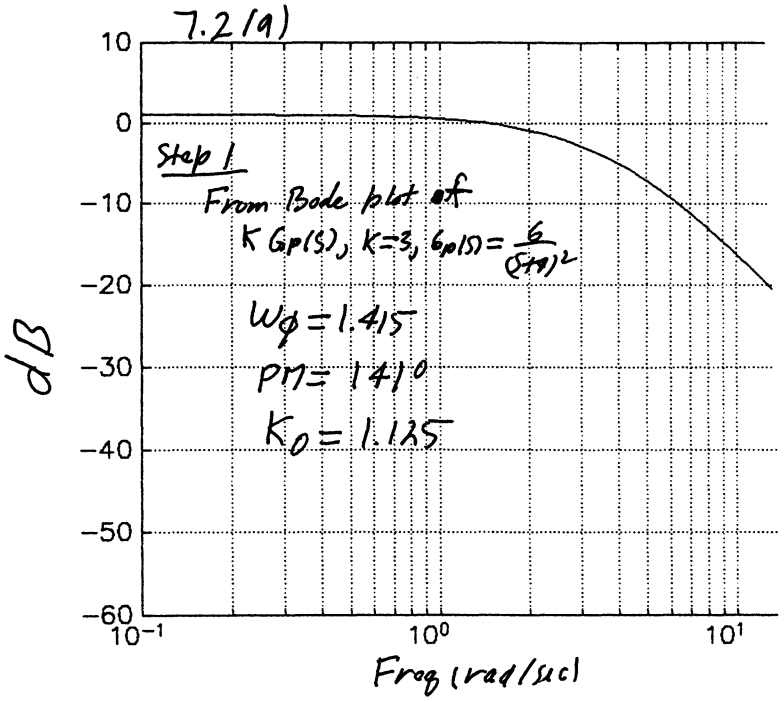
1

7.1(d)

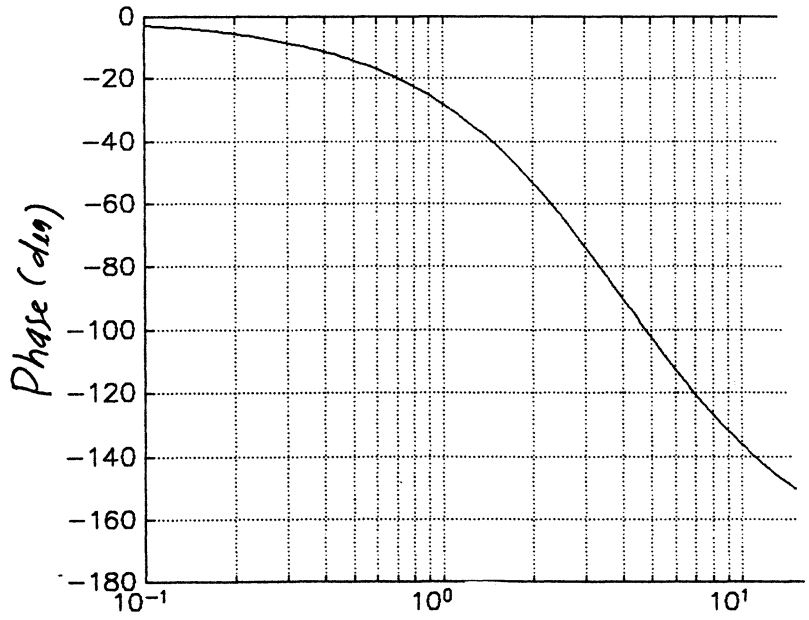

There is no ω_{pm}
GM = ∞
$\omega_\phi = \omega_{gc} = 29$
PM = 127.4°
dB
Freq (rad/sec)


Phase (deg)

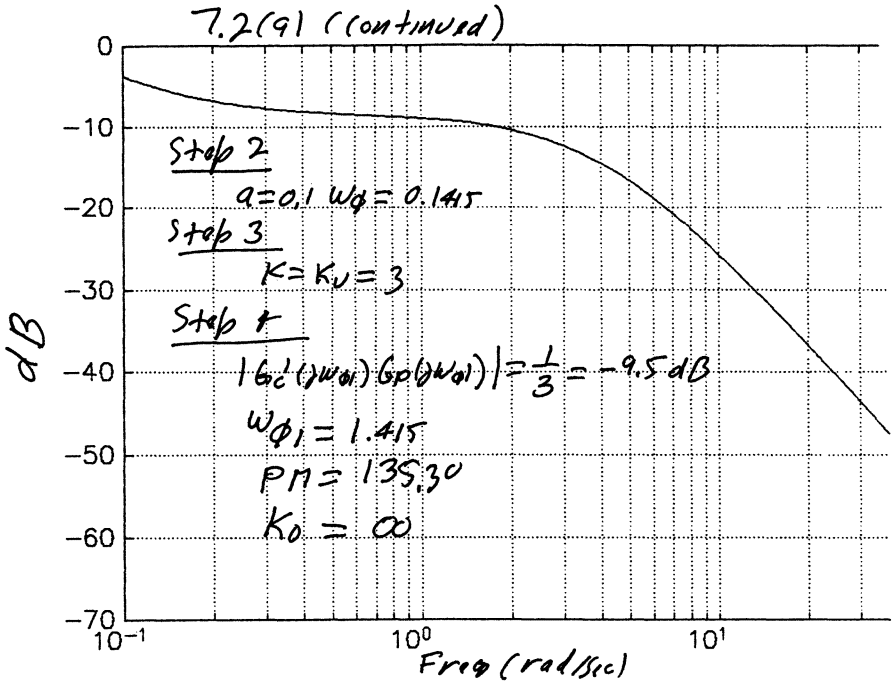
7.2 (a)



Bode plot of uncomp system



7.2(9) (continued)



Step 2

$q = 0.1 \quad w_q = 0.1415$

Step 3

$K = K_v = 3$

Step 4

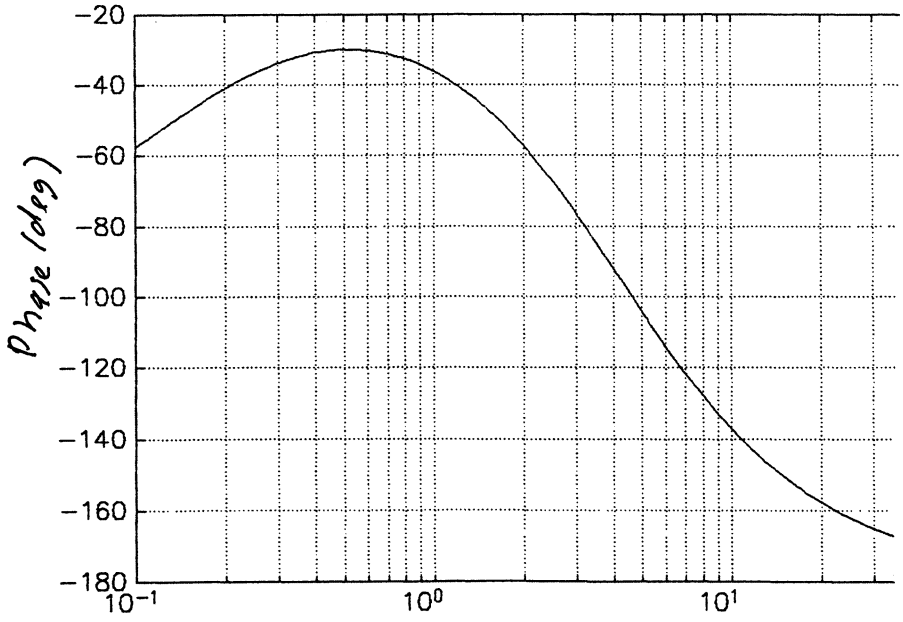
$|G_c'(jw_{\phi_1}) G_p(jw_{\phi_1})| = \frac{1}{3} = -9.5\,dB$

$w_{\phi_1} = 1.415$

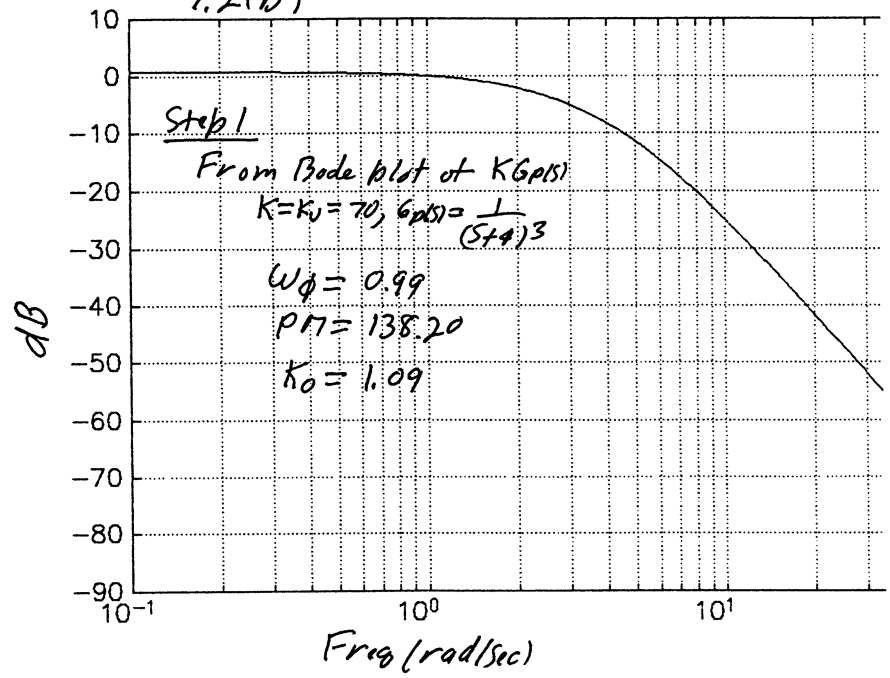
$PM = 135.3^\circ$

$K_0 = \infty$

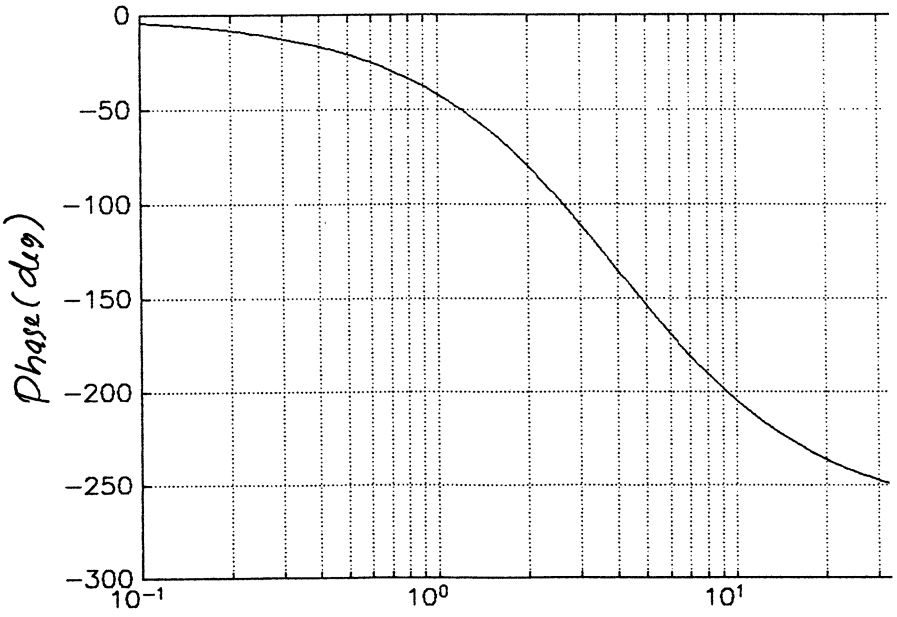
Bode plot of $G_c'(s) G_p(s) = \frac{s+0.1415}{s} \frac{3}{(s+1)^2}$



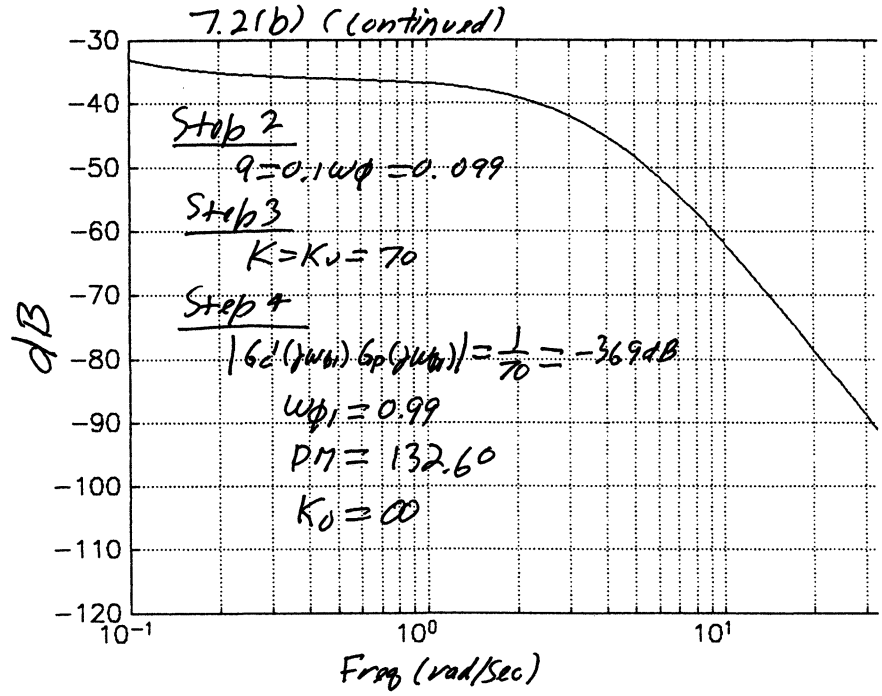
7.2(b)



Bode plot of uncomp system



7.2(b) (continued)



Step 2

$q = 0.1\omega_\phi = 0.099$

Step 3

$K = K_v = 70$

Step 4

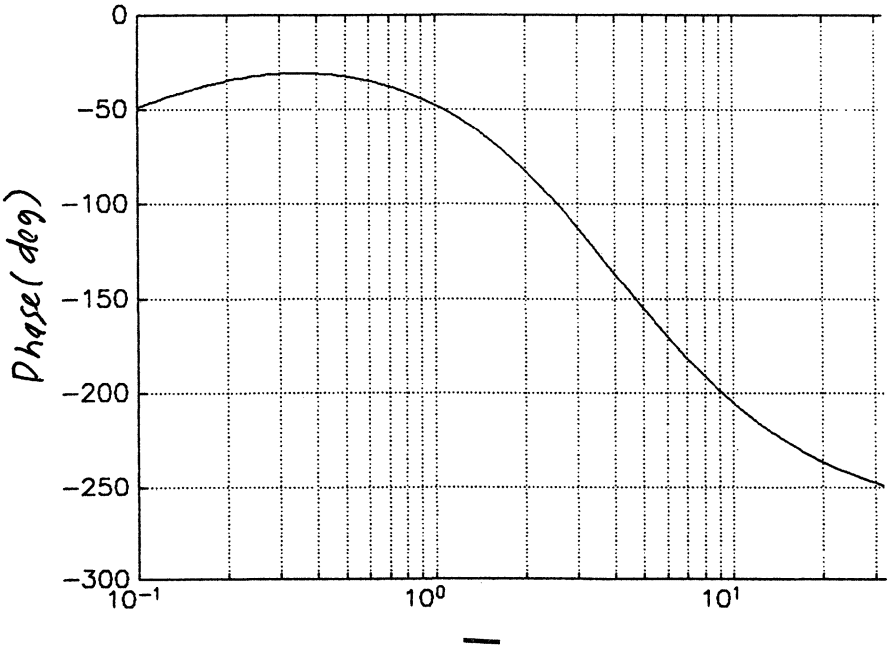
$|G_c'(j\omega_{bi})\,G_p(j\omega_{bi})| = \frac{1}{70} = -36.9\,dB$

$\omega_{\phi 1} = 0.99$

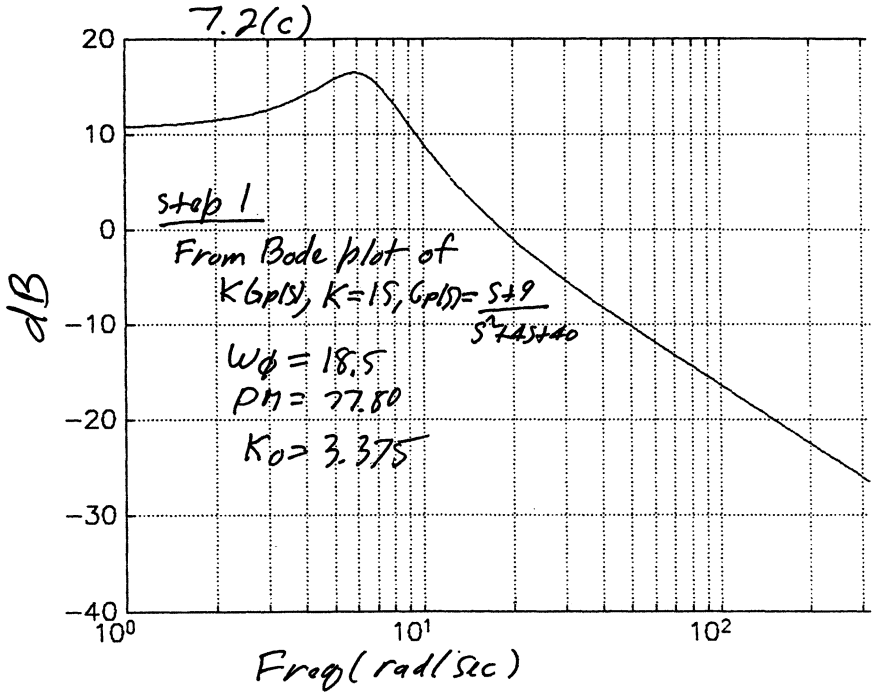
$PM = 132.6^\circ$

$K_0 = \infty$

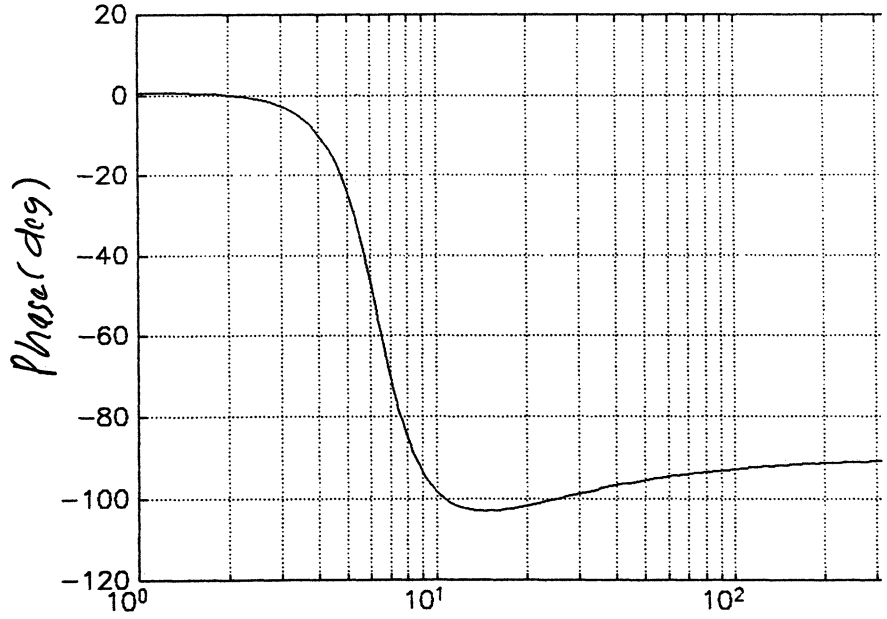
Bode plot of $G_c'(s)G_p(s) = \frac{s+0.099}{s}\,\frac{1}{(s+1)^3}$



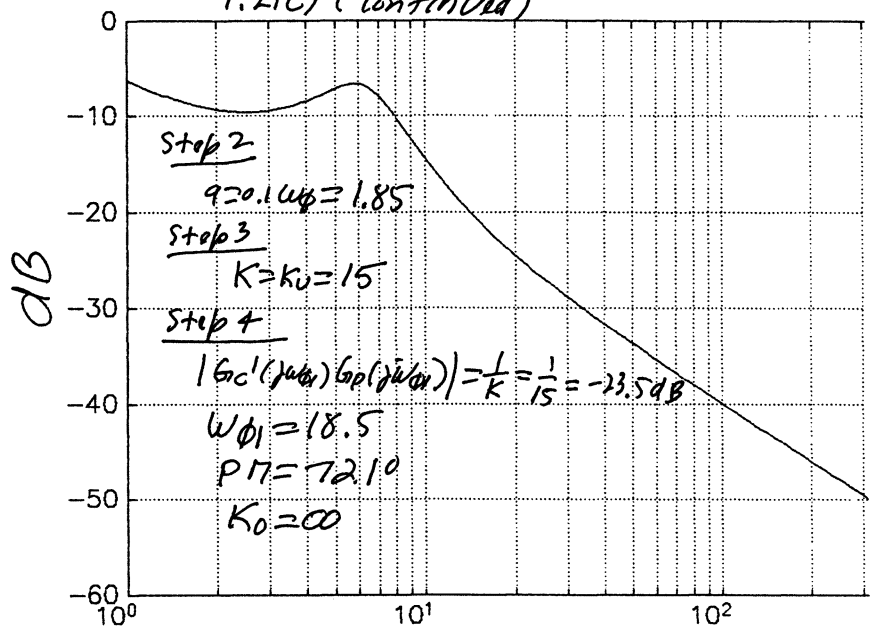
7.2(c)



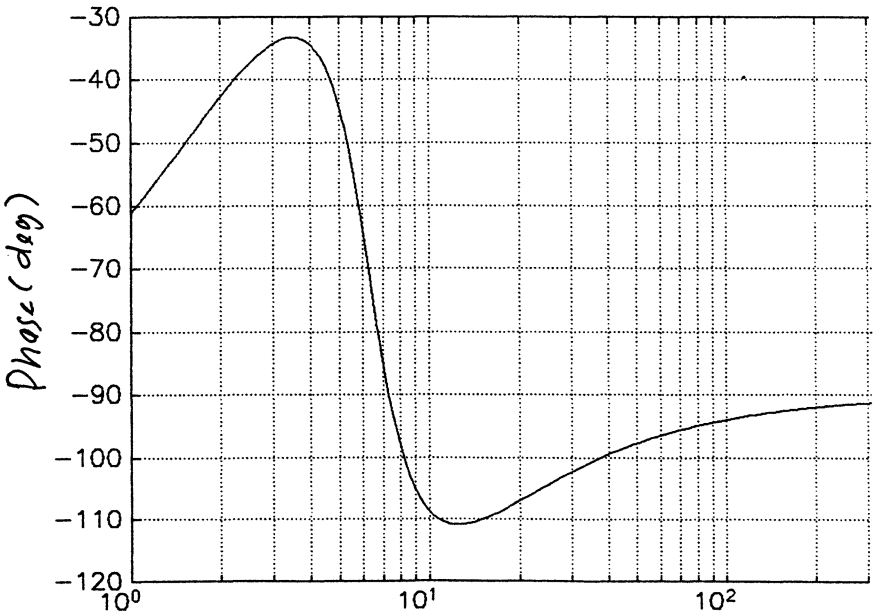
Bode plot of uncomp. system



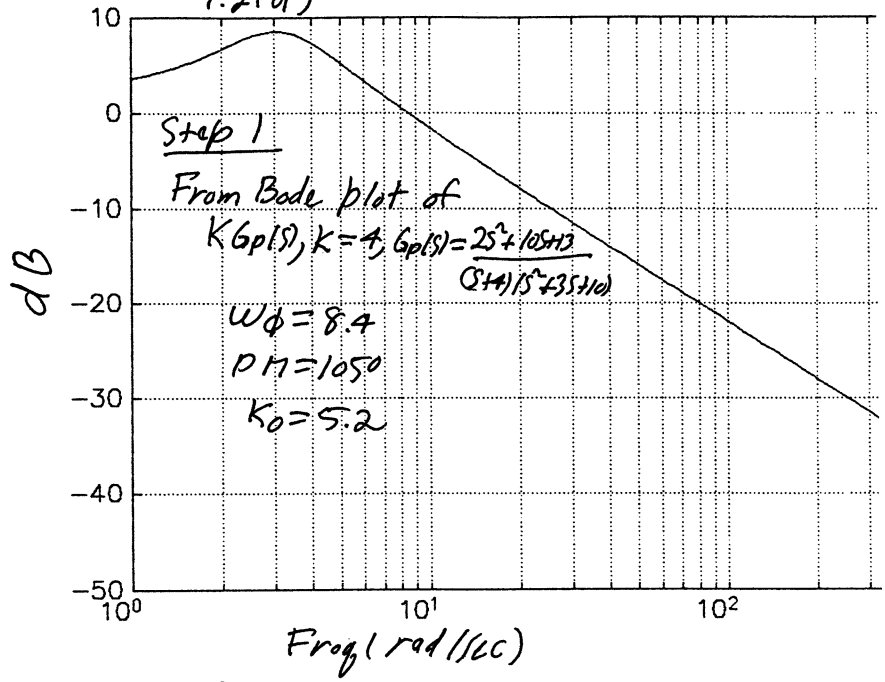
7.21c) (continued)



Bode plot of $G_c'(s)G_p(s) = \frac{s+1.85}{s}\,\frac{s+9}{s^2+4s+40}$



7.2(d)



Step 1

From Bode plot of

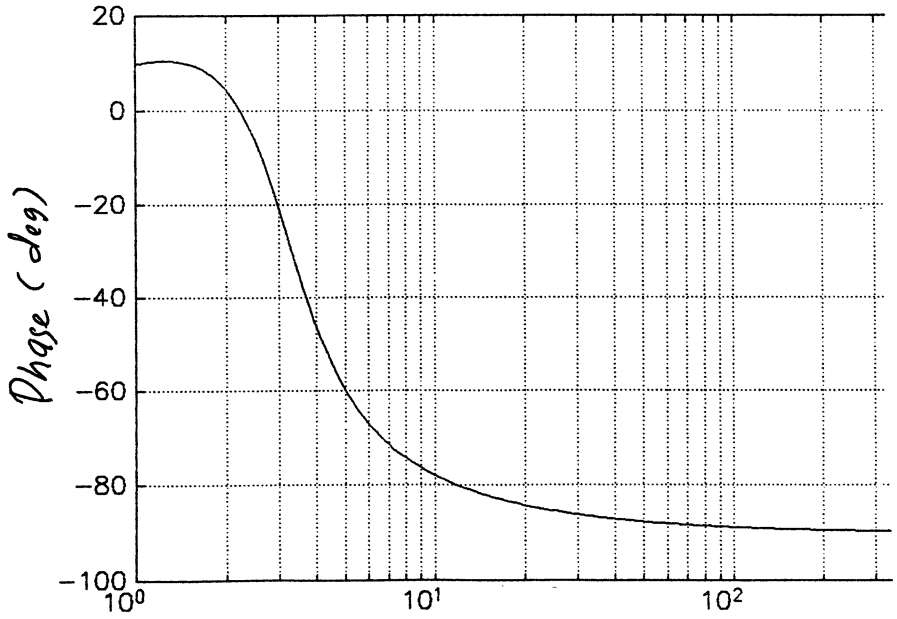
$KG_p(s)$, $K=4$, $G_p(s)=\frac{2s^2+10s+13}{(s+1)(s^2+3s+10)}$

$\omega_\phi = 8.4$

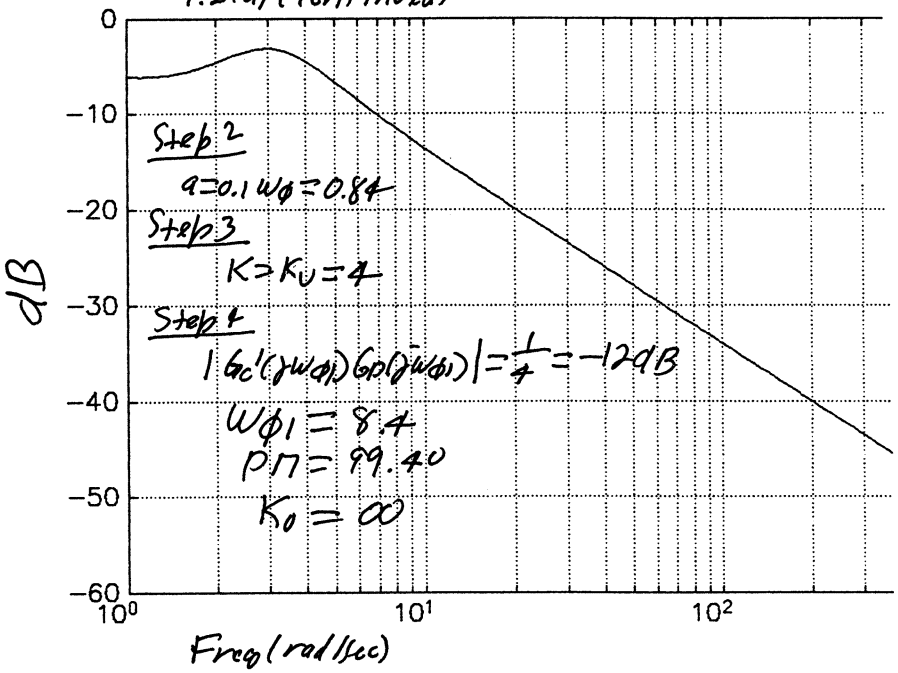
$PM = 105^0$

$K_0 = 5.2$

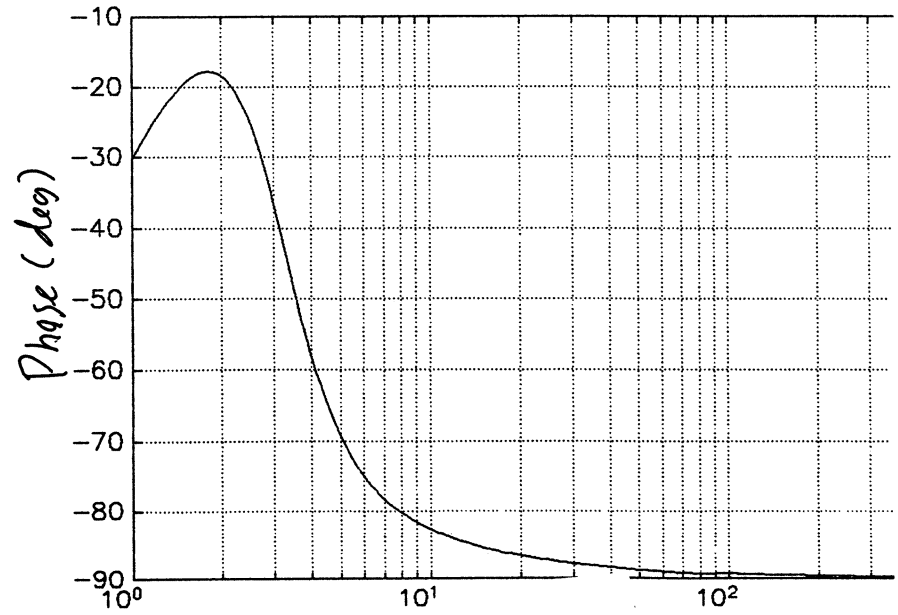
Bode plot of uncomp. system



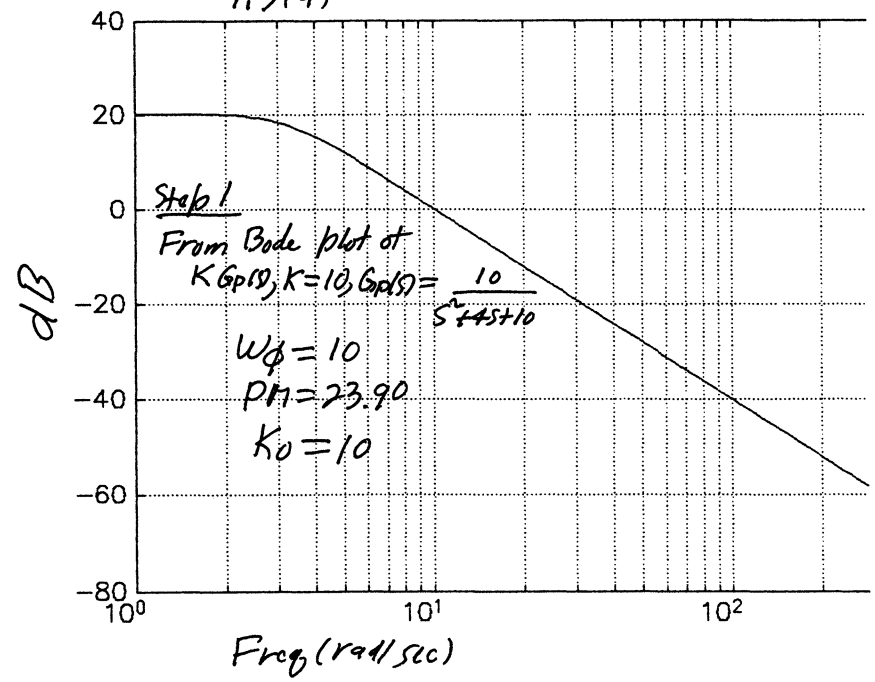
7.2(d) (continued)



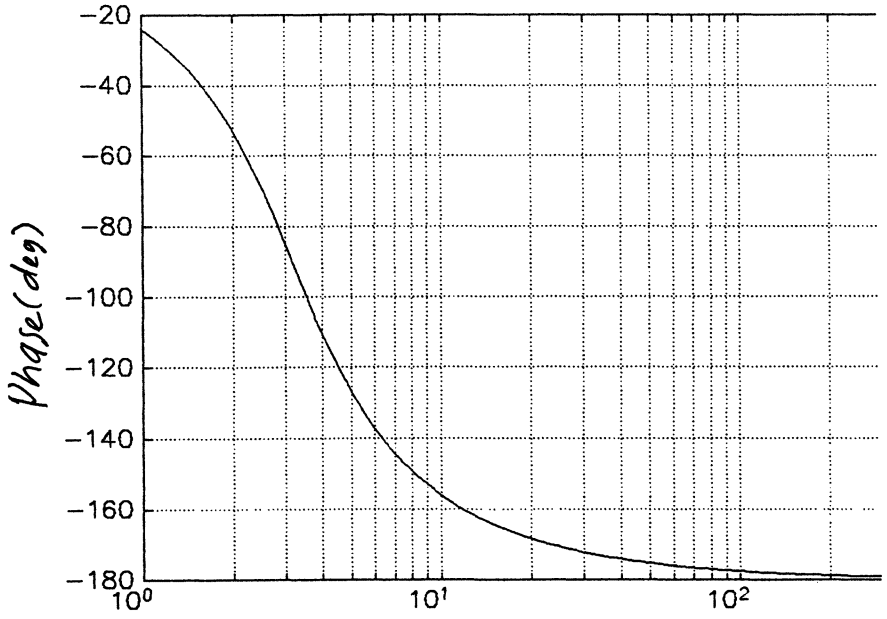
Bode plot of $G_c'(s)\,G_p(s) = \frac{s+0.84}{s}\,\frac{(25^\wedge+10s+13)}{(s+4)(s^\wedge+3s+10)}$



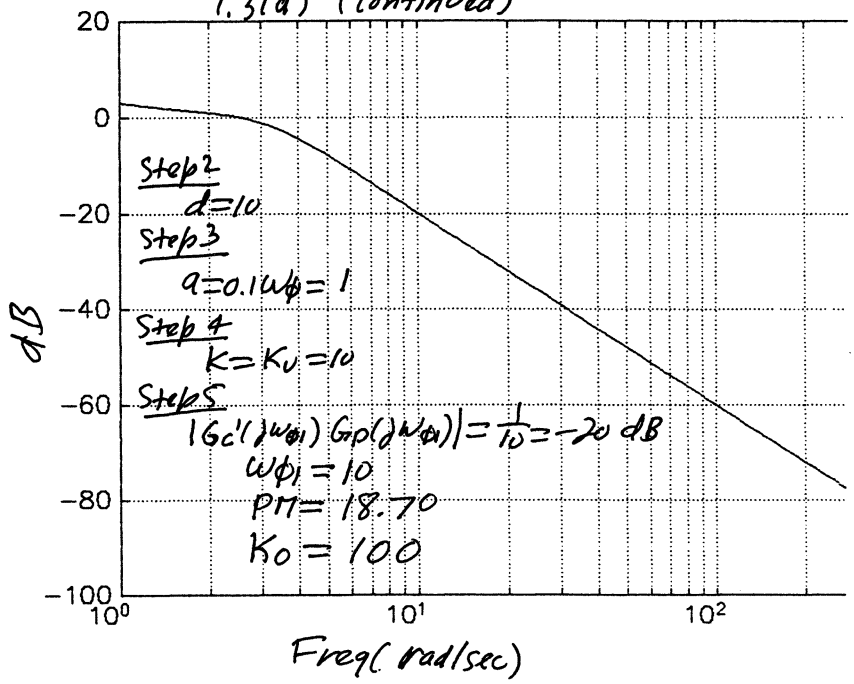
7.3(g)



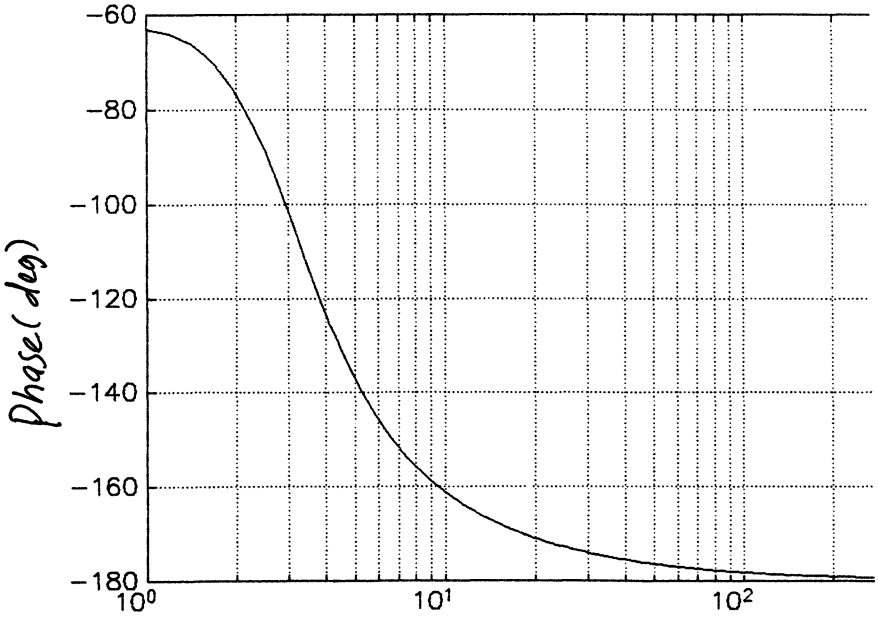
Bode of Uncompensated System



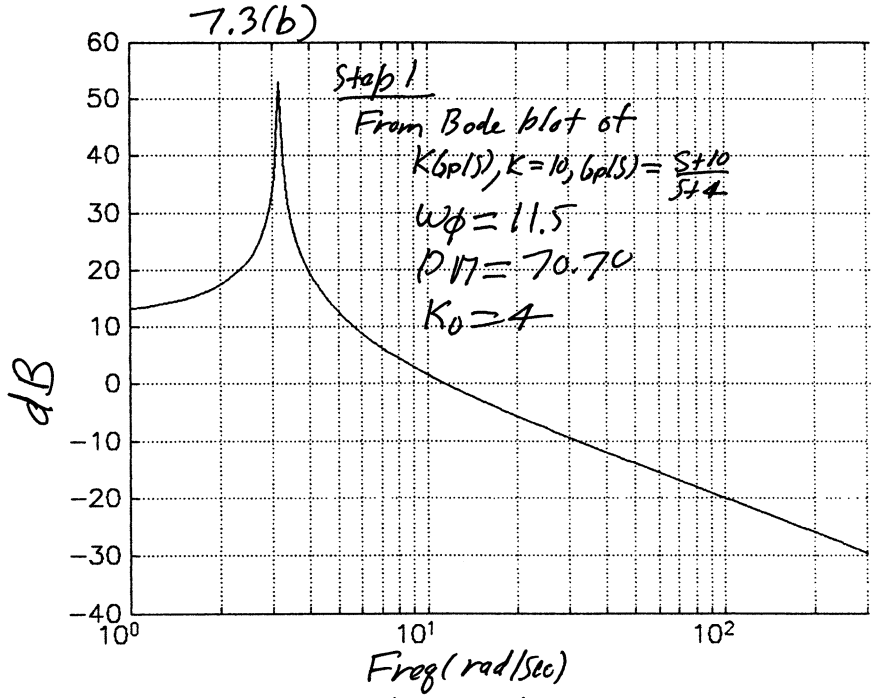
7.3(a) (continued)



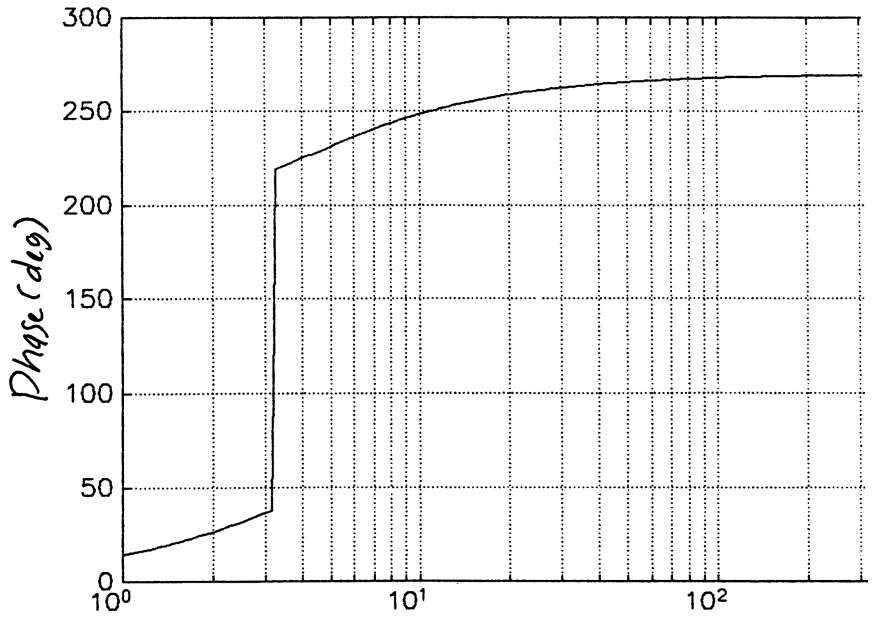
Bode plot of $G_c'(s)G_p(s) = \left(\frac{s+1}{s+.1}\right)\frac{10}{s^2+4s+10}$



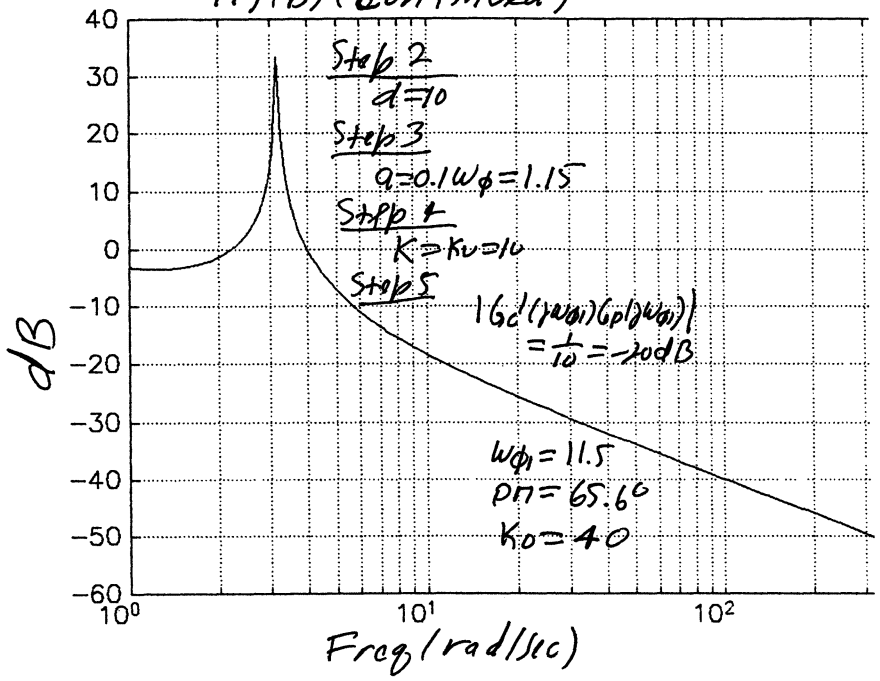
7.3(b)



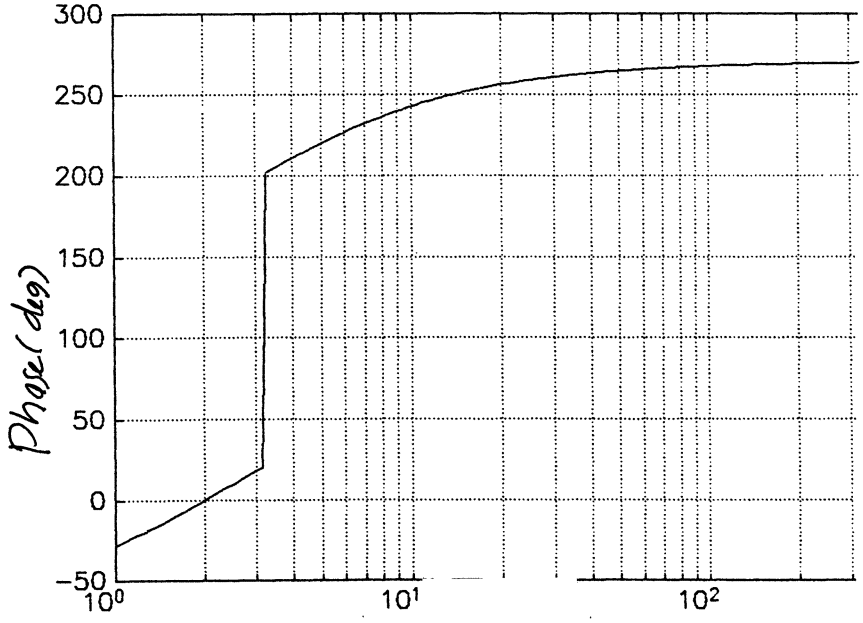
Bode plot of uncomp. system.



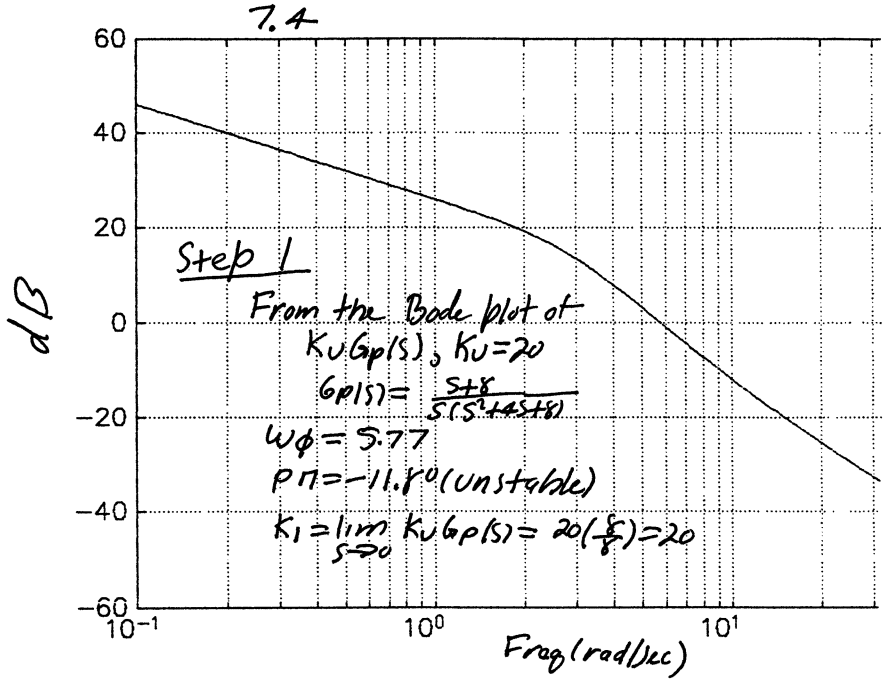
7.3(b) (continued)



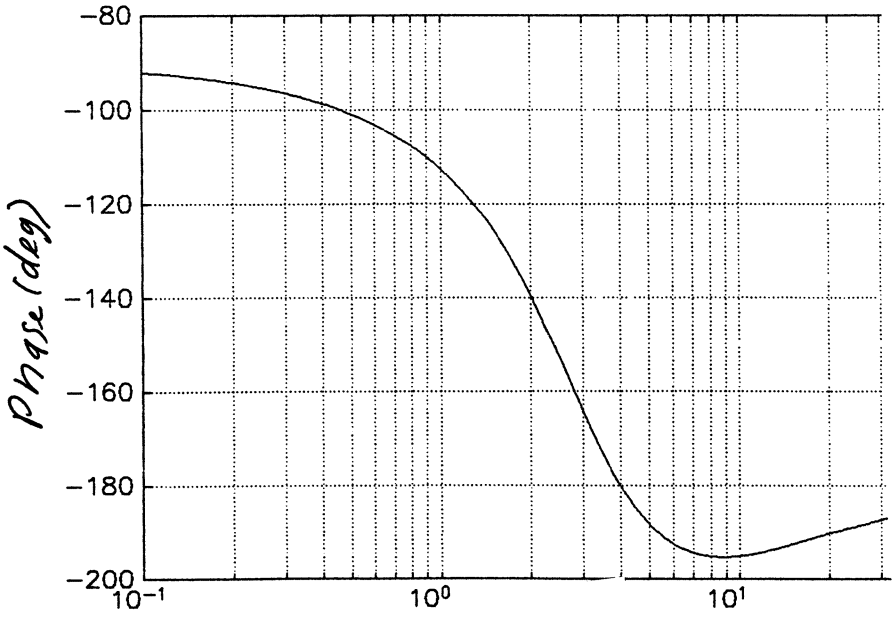
Bode plot of $G_c'(s)G_p(s)=\left(\frac{s+1.15}{s+.115}\right)\frac{s+4}{s^2+10}$



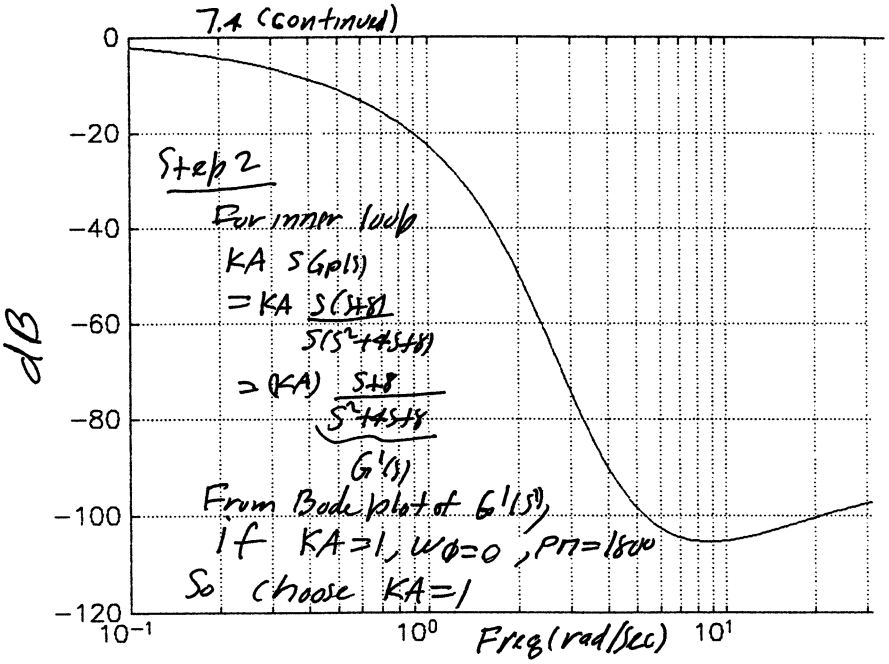
7.4



Bode plot of Uncomp. System



7.4 (continued)



Step 2

For inner loop

$KA\ G_p(s)$

$= KA \dfrac{s(s+8)}{s(s^2+4s+8)}$

$= (KA) \dfrac{s+8}{s^2+4s+8}$

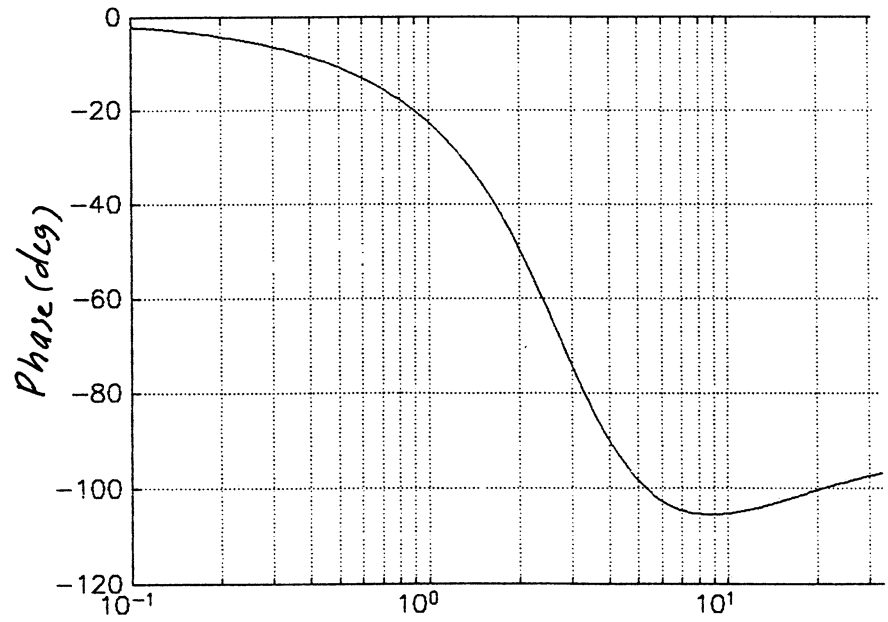
$G'(s)$

From Bode plot of $G'(s)$,

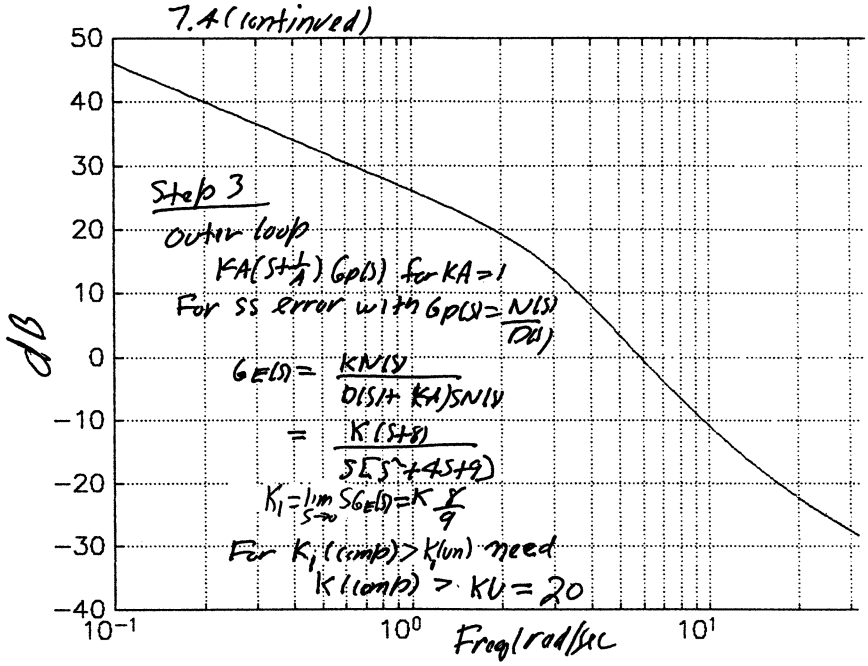
If $KA=1$, $\omega\phi=0$, $PM=180°$

So choose $KA=1$

Bode Plot of $\dfrac{s+8}{s^2+4s+8}$



7.4 (continued)



Step 3

Outer loop

$KA(s+\frac{1}{A})\,Gp(s)$ for $KA=1$

For SS error with $Gp(s)=\frac{N(s)}{D(s)}$

$$G_E(s)=\frac{K N(s)}{D(s)(s+KA)+KN(s)}$$

$$=\frac{K(s+8)}{s(s^2+4s+9)}$$

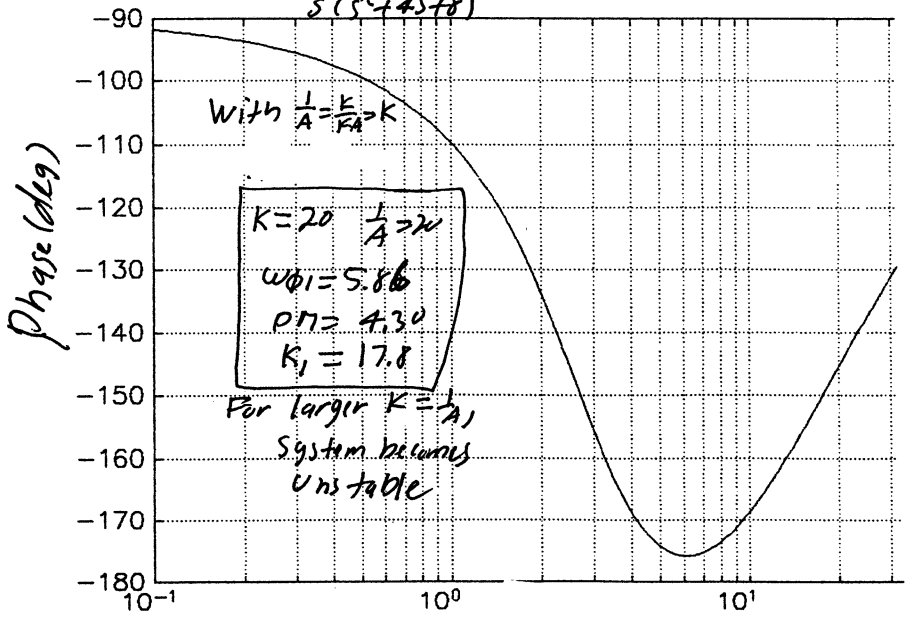
$$K_1=\lim_{s\to 0} sG_E(s)=K\frac{8}{9}$$

For K_1 (comp) $>$ K_1 (un) need

K (comp) $>$ $KU=20$

Bode Plot of $KA(s+\frac{1}{A})\,Gp(s)$

$$=(1)\frac{(s+20)(s+8)}{s(s^2+4s+8)}$$



With $\frac{1}{A}=\frac{K}{KA}=K$

$K=20$ $\frac{1}{A}=20$

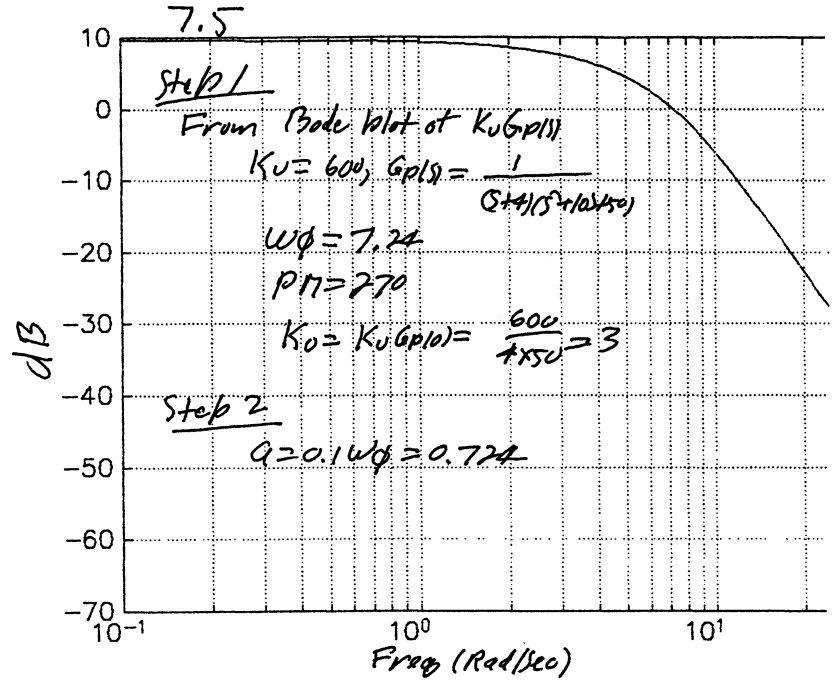
$\omega\phi_1=5.86$

$PM=4.30$

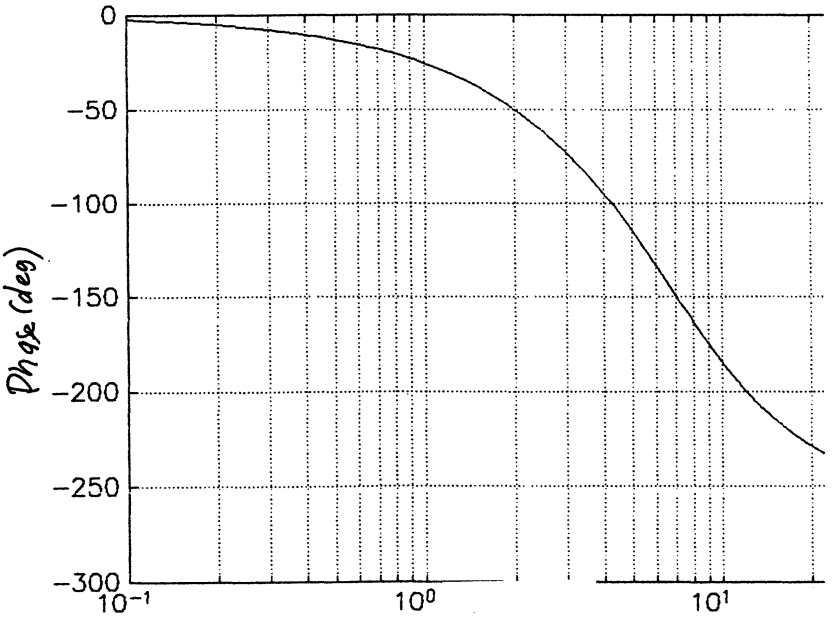
$K_1=17.8$

For larger $K=\frac{1}{A}$, System becomes unstable

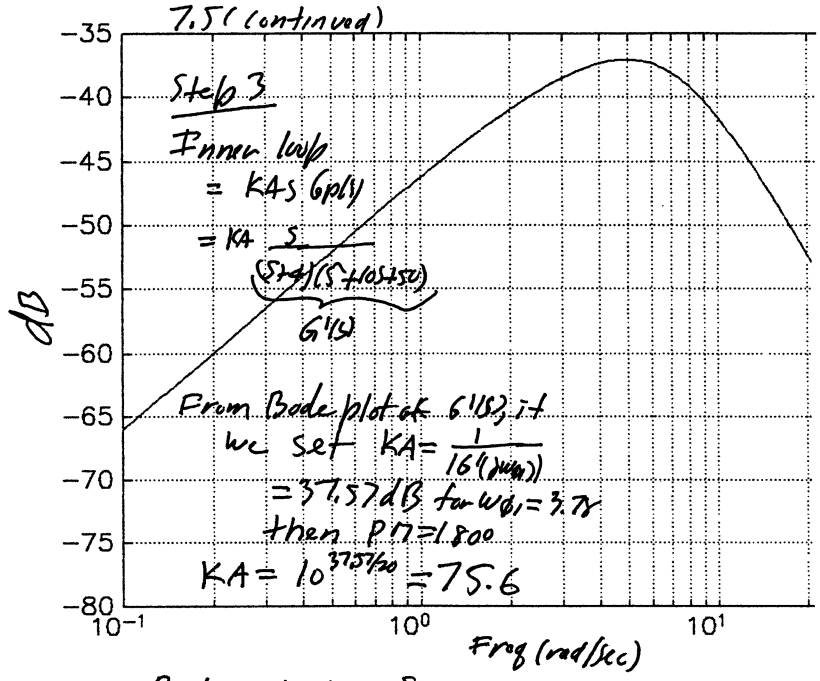
7.5



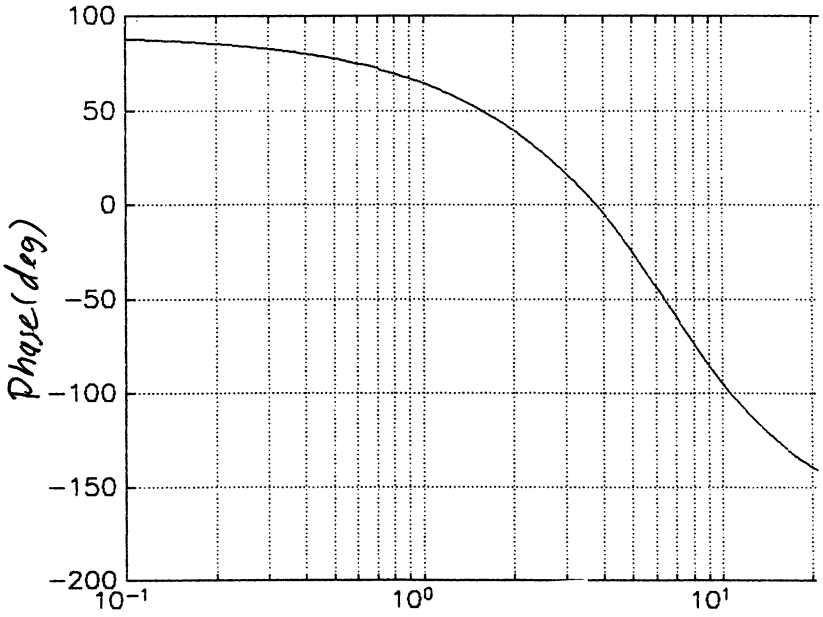
Bode Plot of Uncomp System



7.5 (continued)



Bode Plot of $\frac{S}{(S+4)(S+10S+50)}$



7.5 (continued)

Step 4 If $G_p(s) = \frac{N(s)}{D(s)}$ $G_E(s) = \frac{KN(s)}{D(s) + KA\,N(s)}$

$$G_E(s) = \frac{K}{(s+4)(s^2+10s+50) + KA\,s} \qquad K_0 = G_E(0) = \frac{K}{200}$$

Outer loop transfer function

$$= (1+A)(s+\tfrac{1}{A})\,G_p(s) = \frac{\frac{75.6}{KA}(s+\frac{1}{A})}{(s+4)(s^2+10s+50)}$$

For various $\frac{1}{A} = \frac{K}{KA} = \frac{K}{75.6}$

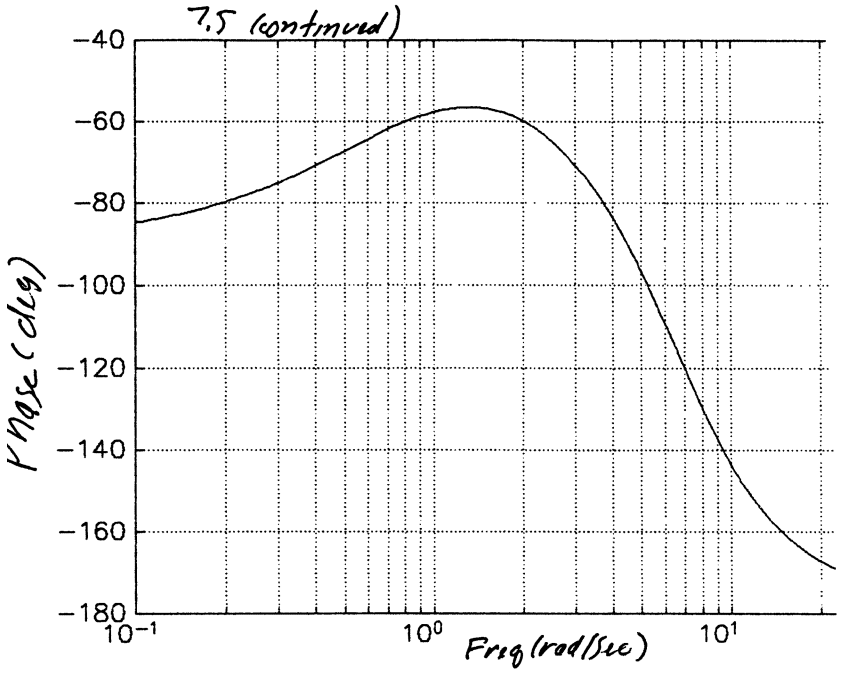
$\frac{1}{A}$	K	ω_ϕ	PM	$K_0 = \frac{K}{200}$	
7.94	600	8.82	54.90	3	
10.58	800	9.54	41.50	4	good trade off
13.22	1000	10.09	31.80	5	

Step 5

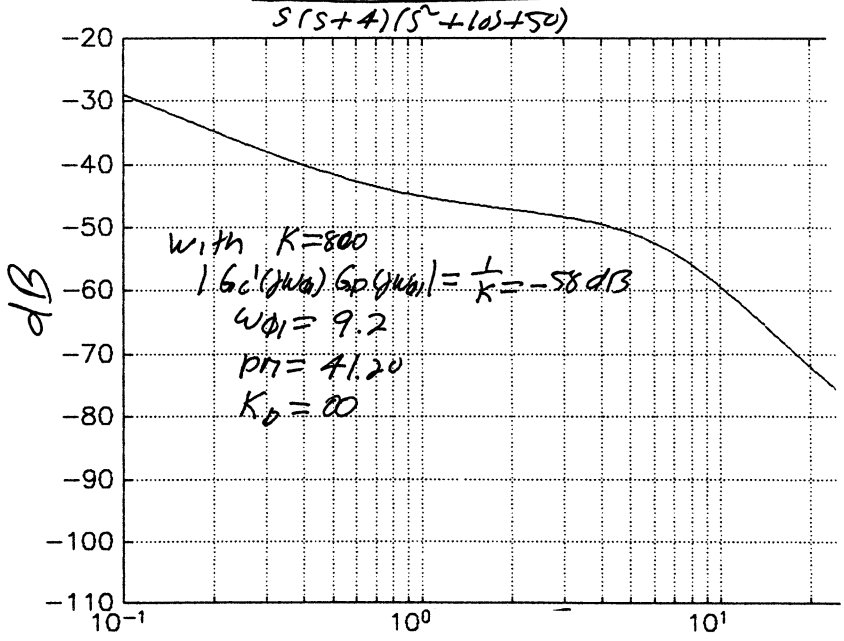
$K = 800$, $A = \frac{1}{10.58} = 0.0945$, $a = 0.724$

$$G_c'(s) = \frac{A\left[s^2 + \frac{1}{A}s + \frac{a}{A}\right]}{s} = \frac{0.0945\left[s^2 + 10.58s + 7.66\right]}{s}$$

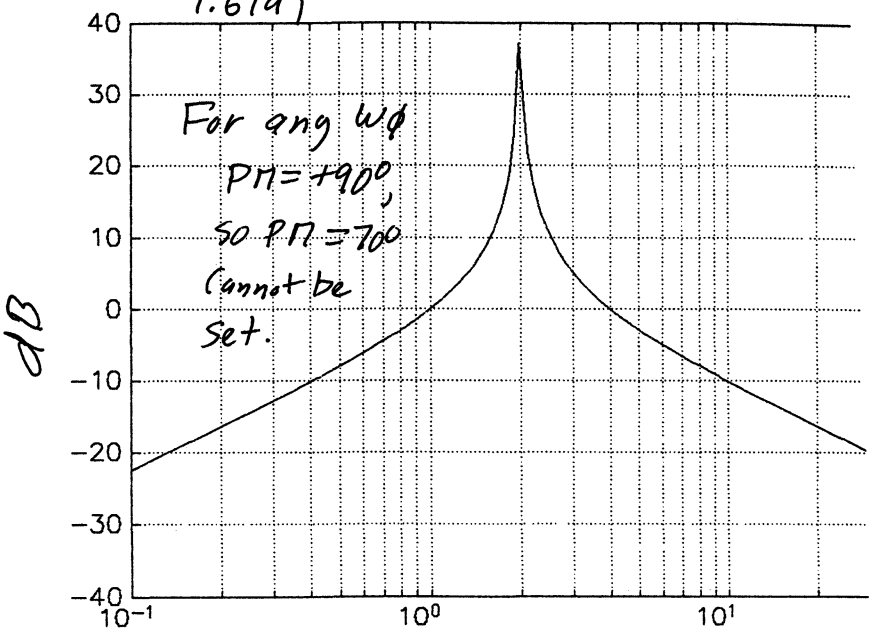
7.5 (continued)



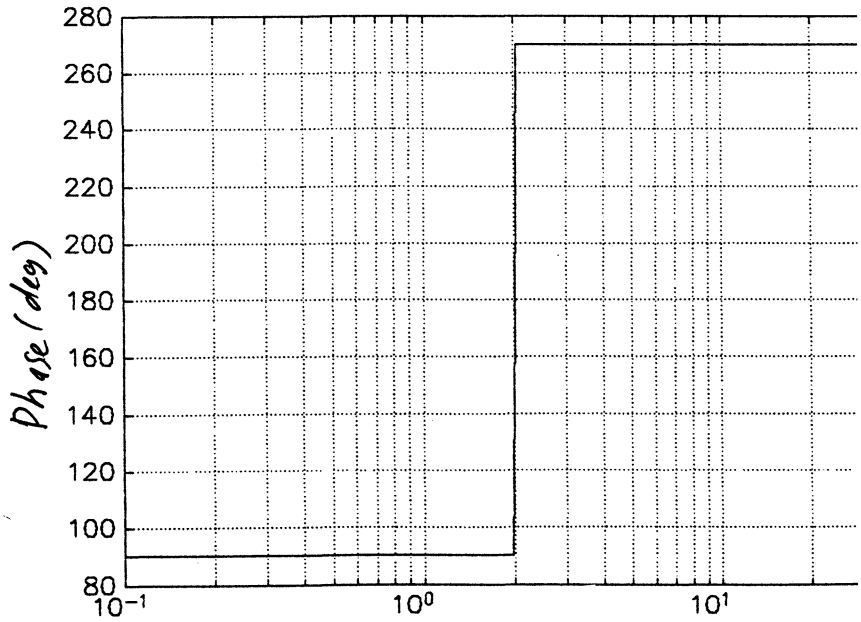
Bode Plot of $G_c'(s)\,G_p(s) = \dfrac{0.0945[s^2+10.58s+7.60]}{s(s+4)(s^2+10s+50)}$



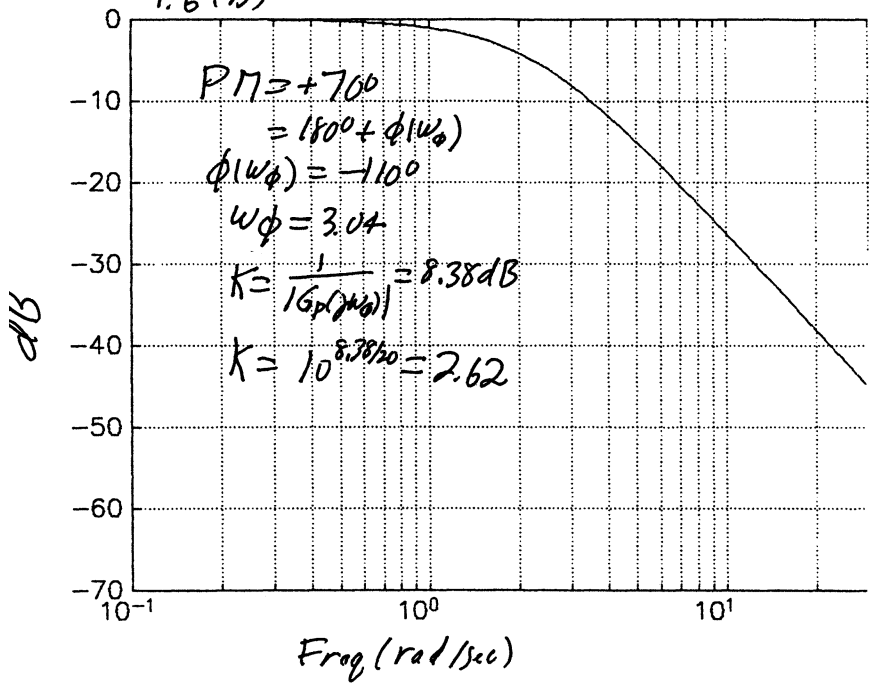
7.6(a)



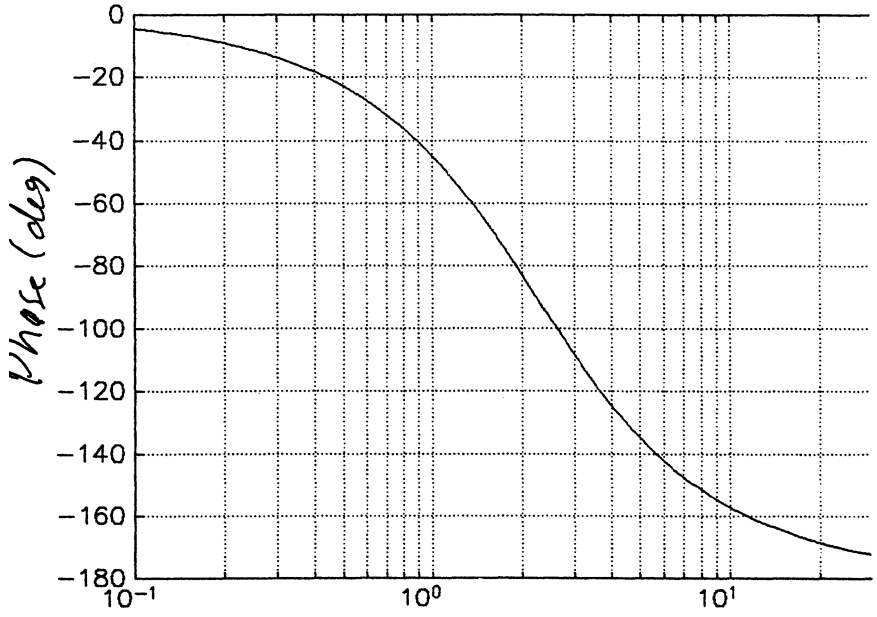
Bode plot of $G_p(s)$



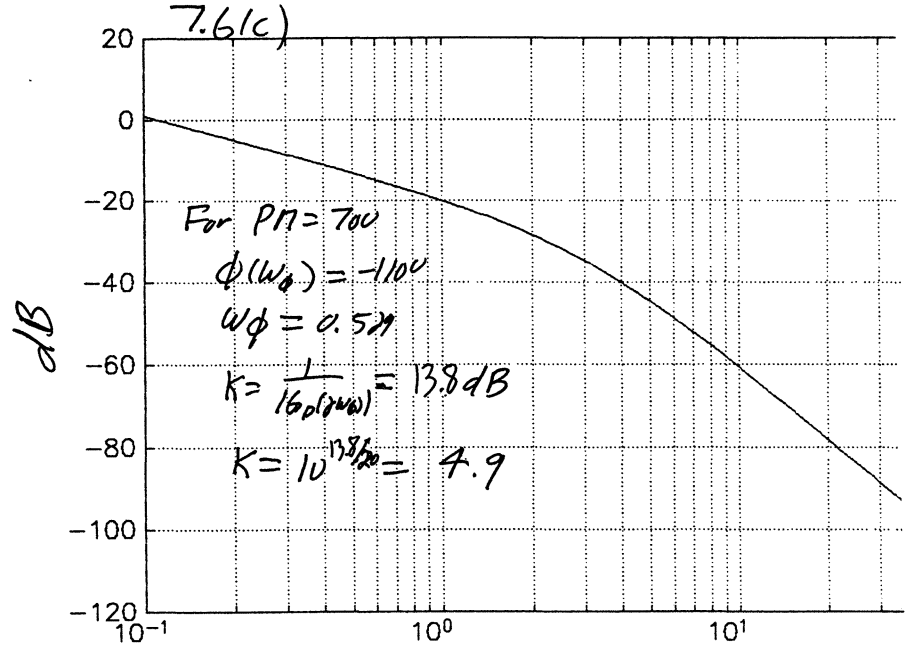
7.6(b)



Bode plot of $G_p(s)$

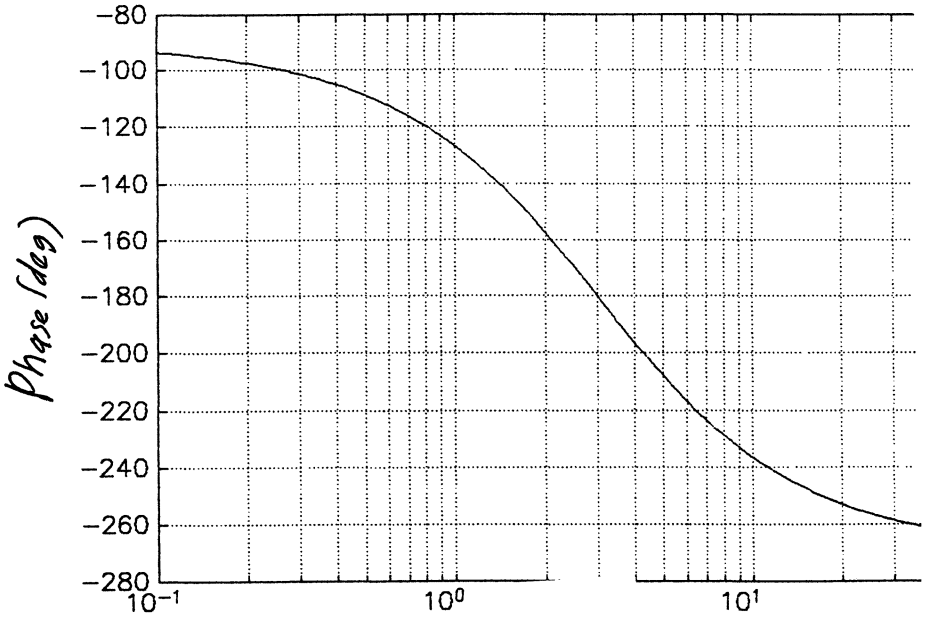


7.61c)

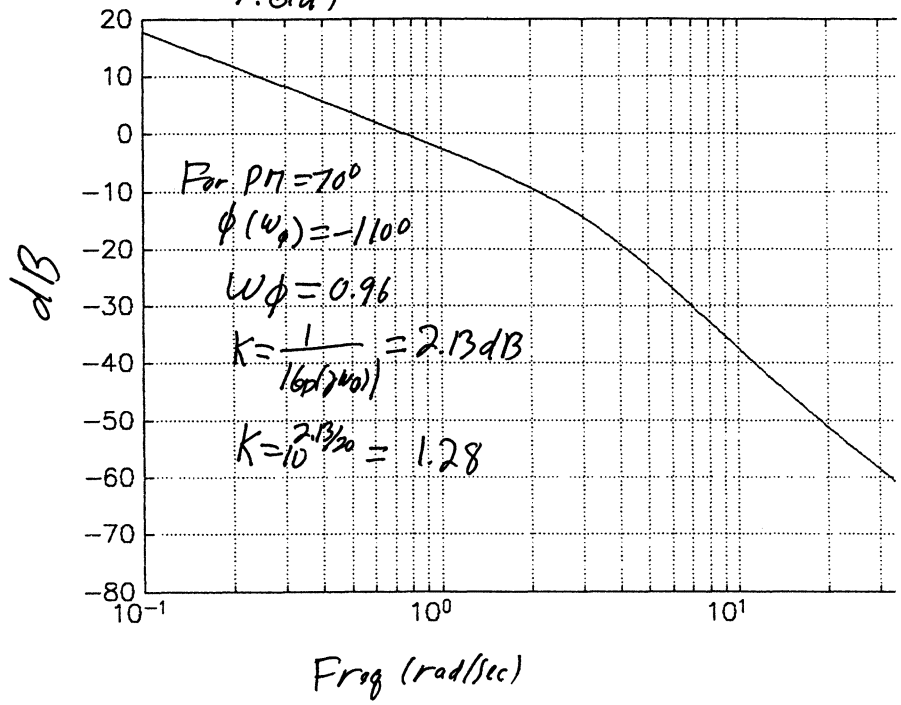


Freq (rad/sec)

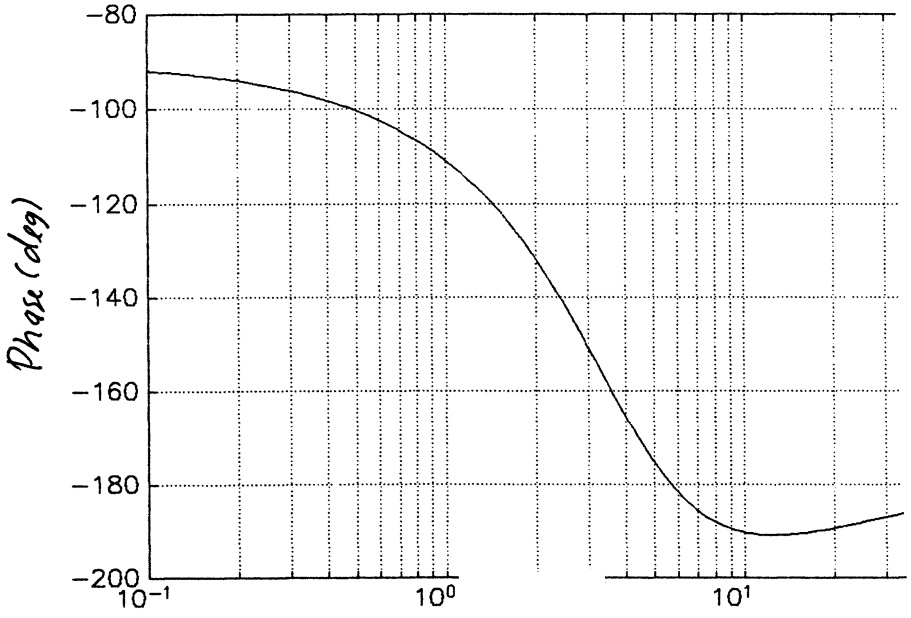
Bode plot of G(s)



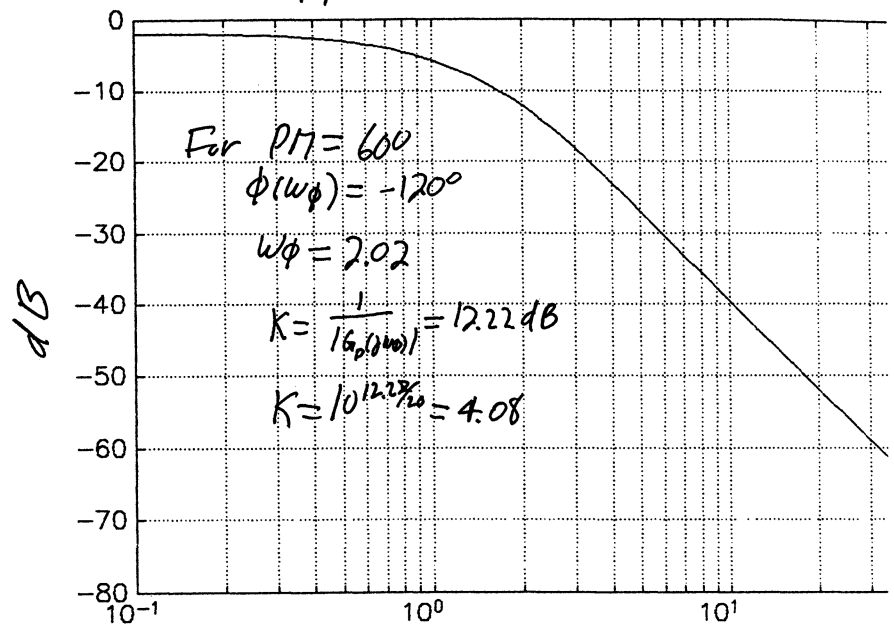
7.6(d)



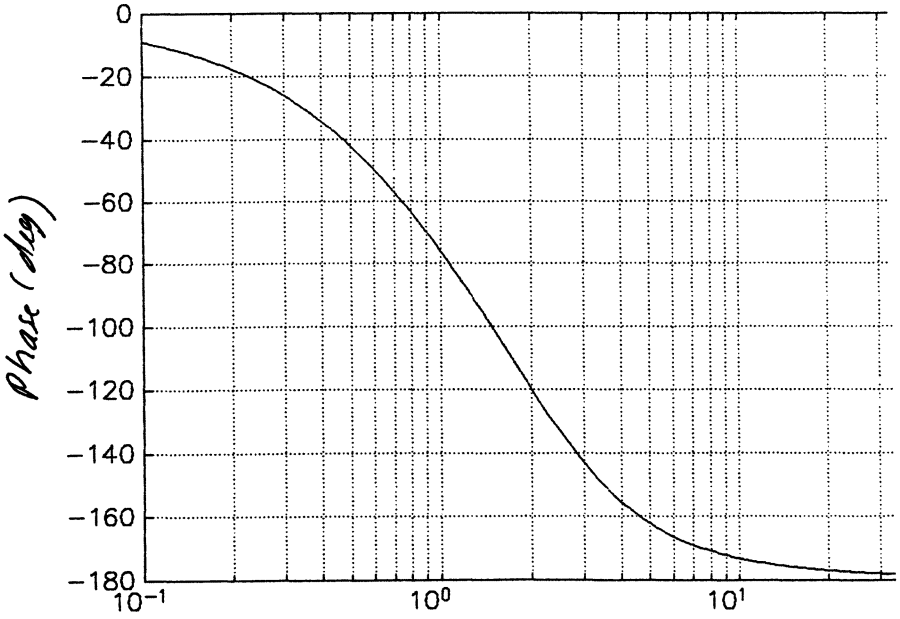
Bode plot of $G_{pl}(s)$



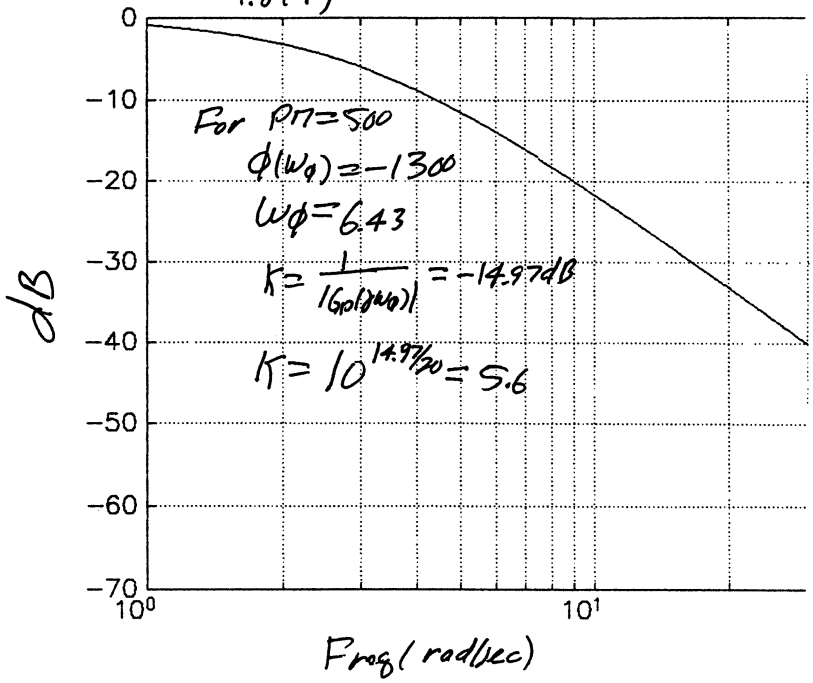
7.7



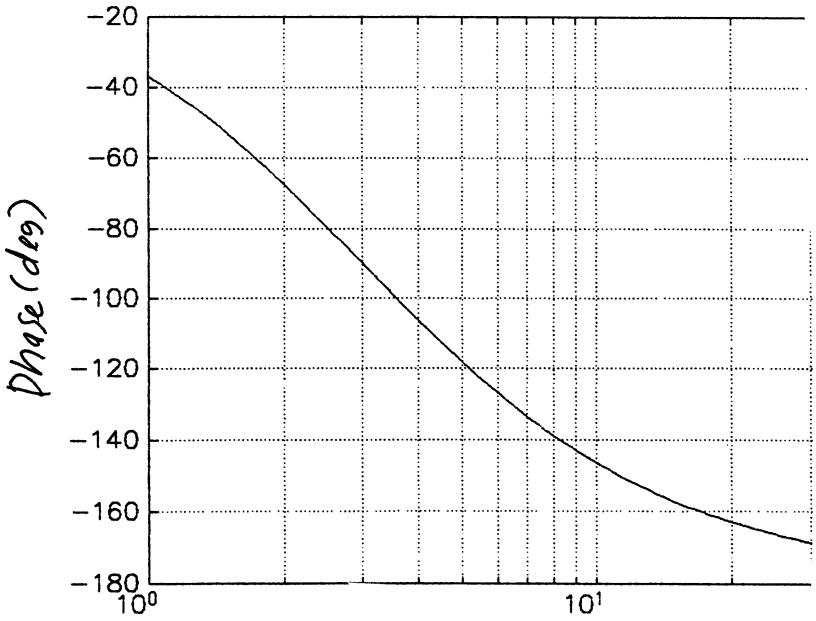
Bode plot of $G_p(s)$



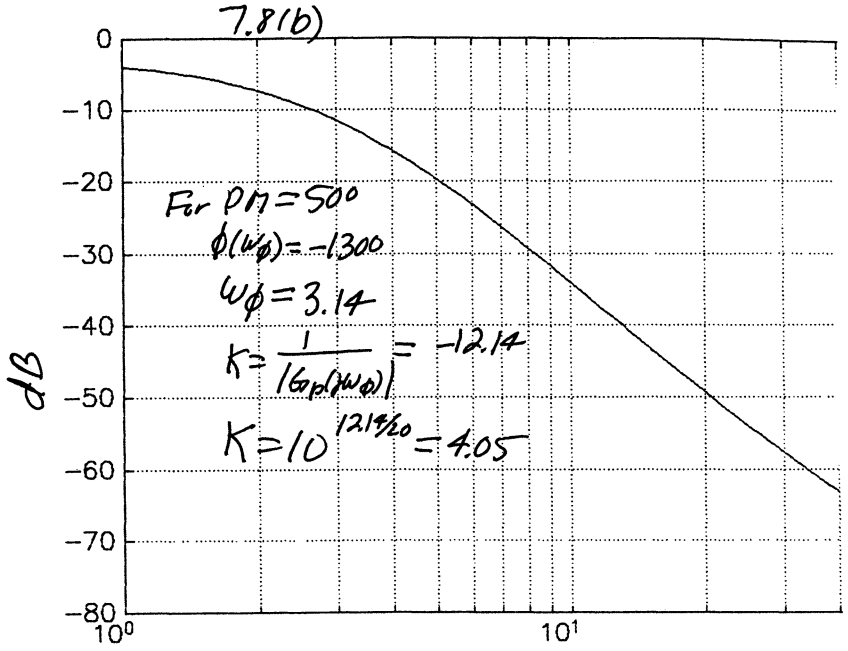
7.8(a)



Bode plot of $G_{OL}(s)$

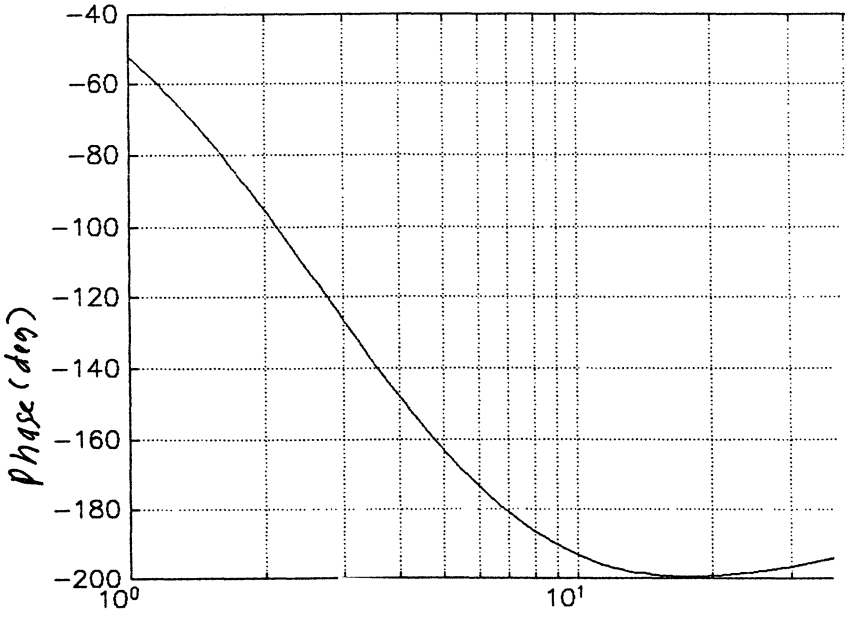


7.8(b)

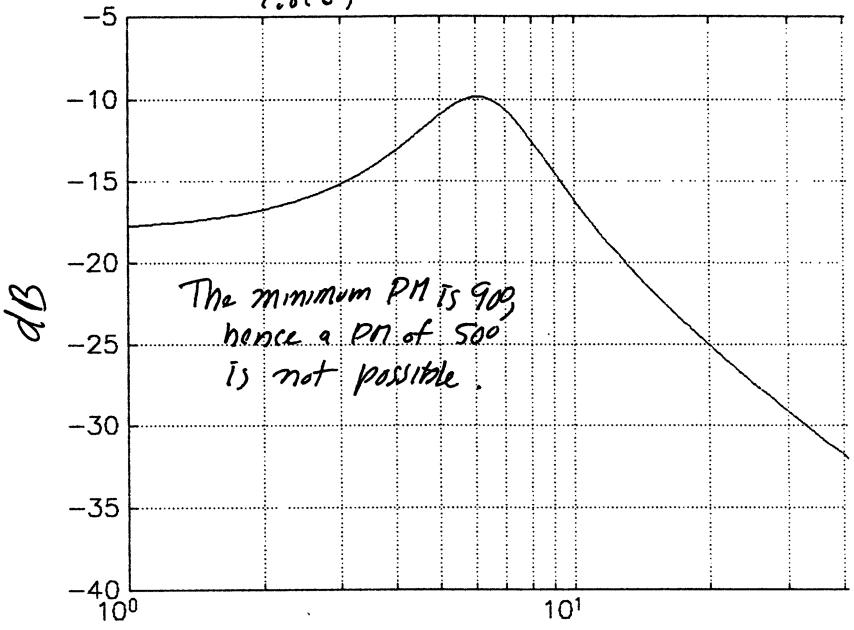


Freq (rad/sec)

Bode plot of Gp(s)

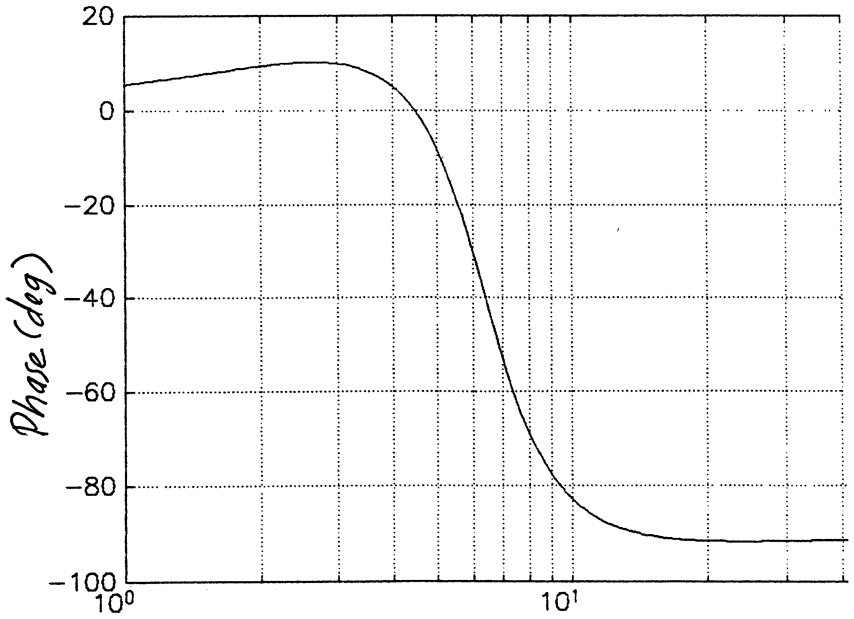


7.8(c)

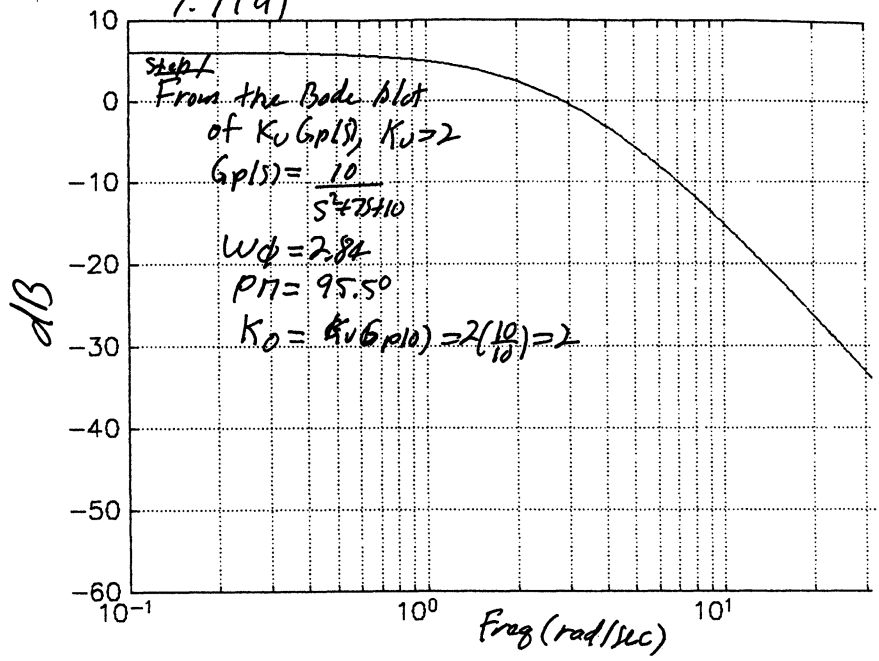


Frrq (rad/sec)

Bode plot of Gp(s)



7.9(a)



Step 1
From the Bode plot of $K_0 G_p(s)$, $K_0 = 2$

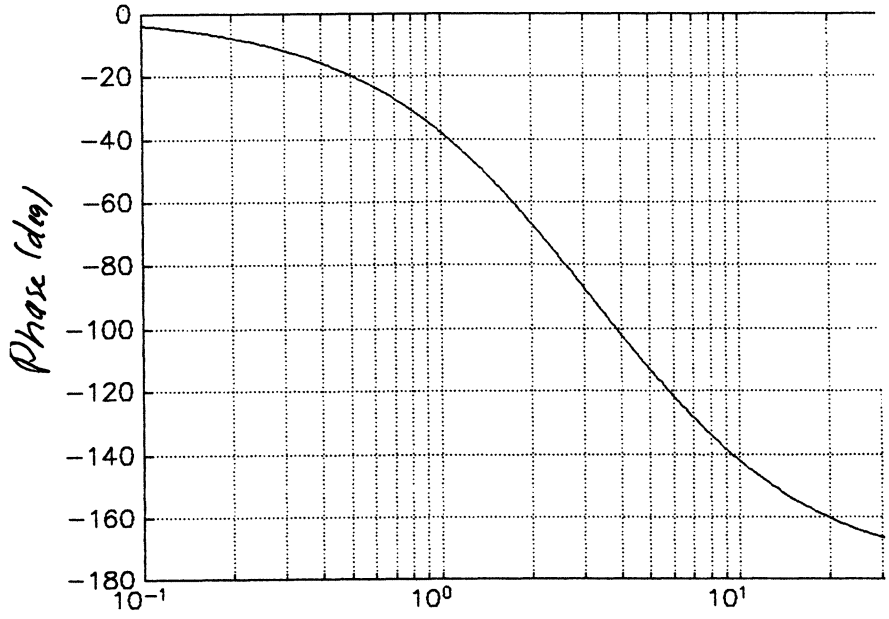
$G_p(s) = \frac{10}{s^2+7s+10}$

$\omega_\phi = 2.84$

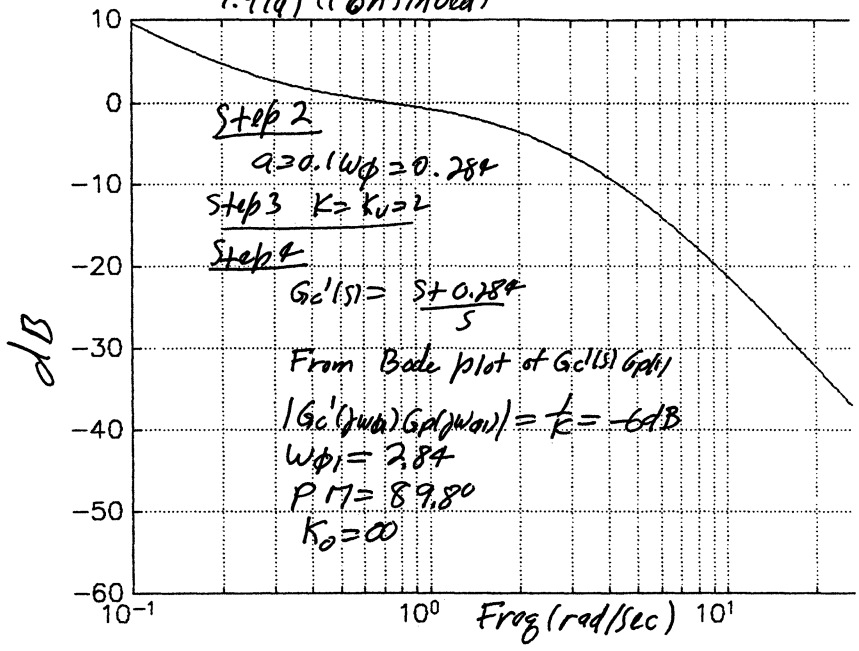
$PM = 95.5°$

$K_0 = \frac{1}{|G_p(j\omega)|} = 2\left(\frac{10}{10}\right) = 2$

Bode Plot of Uncomp. System



7.9(q) (continued)



Step 2

$a = 0.1\omega_\phi = 0.284$

Step 3 $K = K_u = 2$

Step 4

$G_c'(s) = \dfrac{s + 0.284}{s}$

From Bode plot of $G_c'(s)G_p(s)$

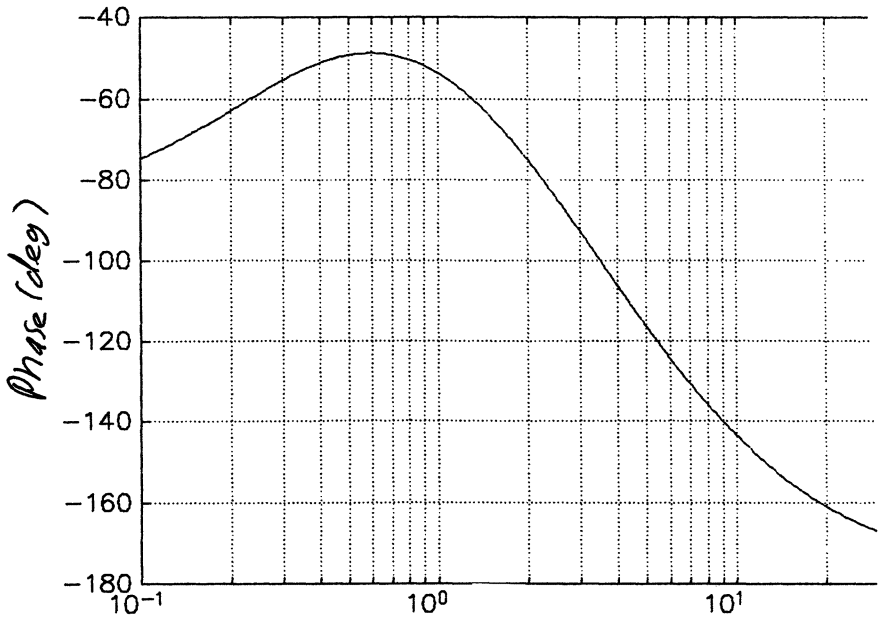
$|G_c'(j\omega_a)G_p(j\omega_a)| = \frac{1}{2} = -6\,dB$

$\omega_{\phi_1} = 2.84$

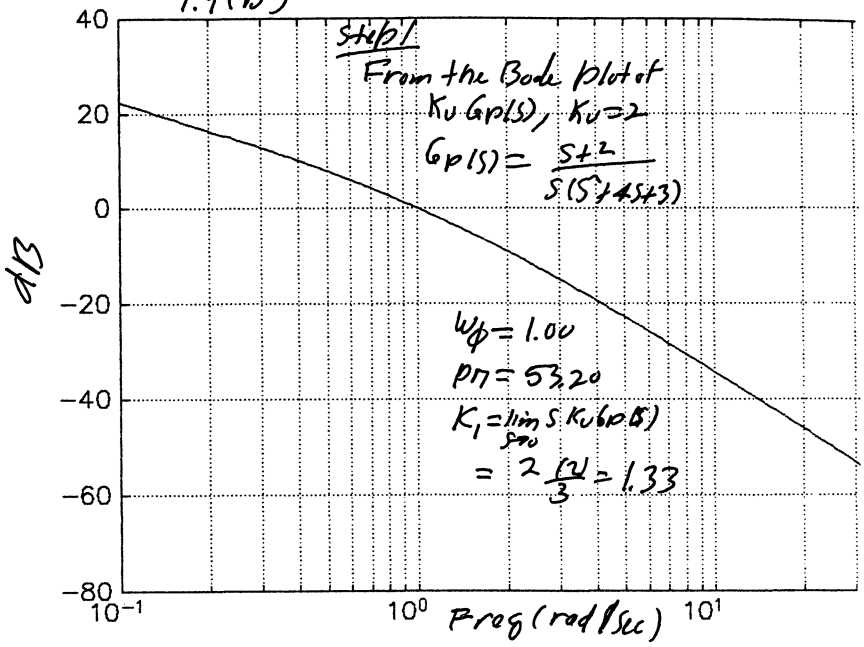
$PM = 89.8^\circ$

$K_0 = \infty$

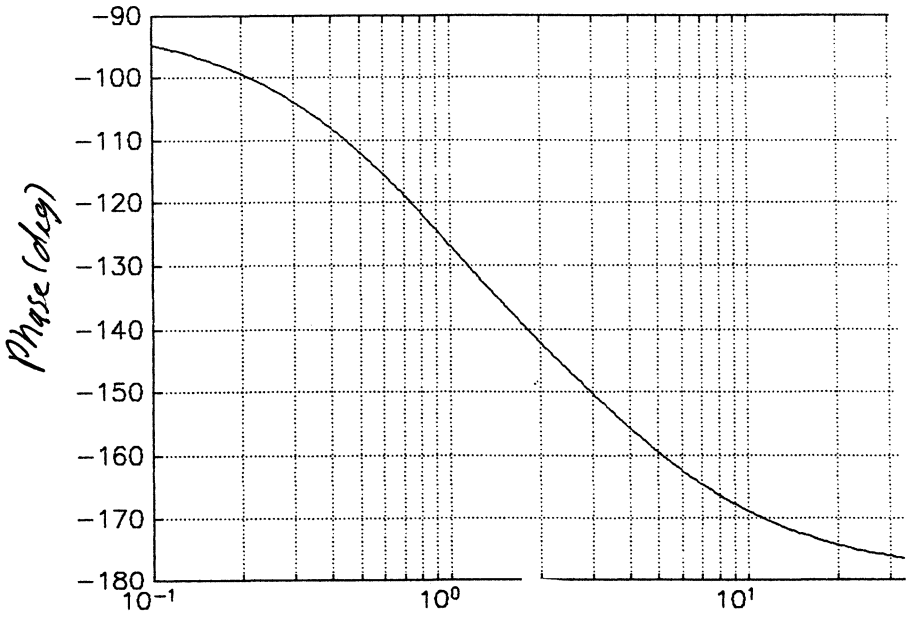
Bode Plot of $G_c'(s)G_p(s) = \dfrac{(s+.284)10}{s(s^2+.7s+10)}$



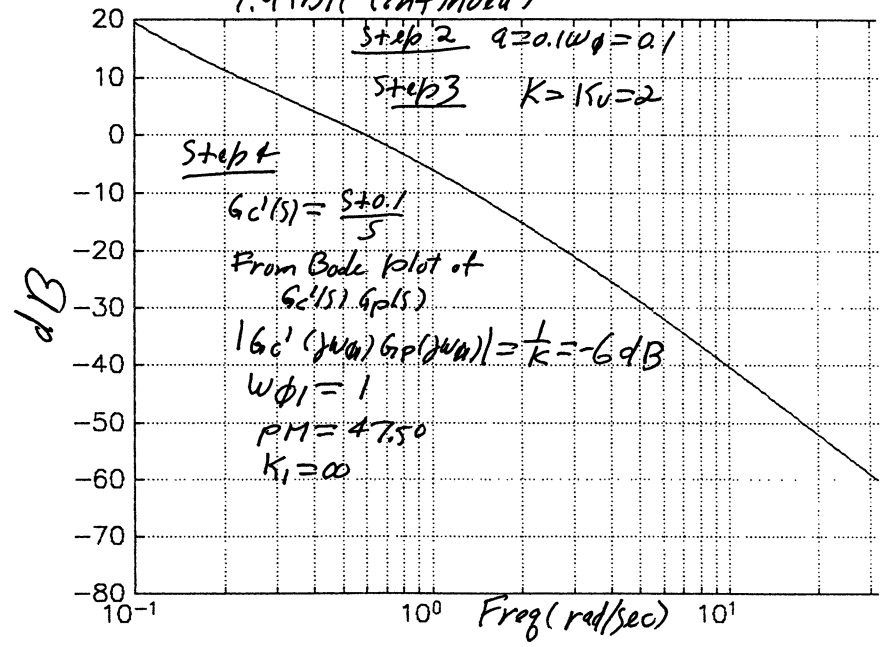
7.9(b)



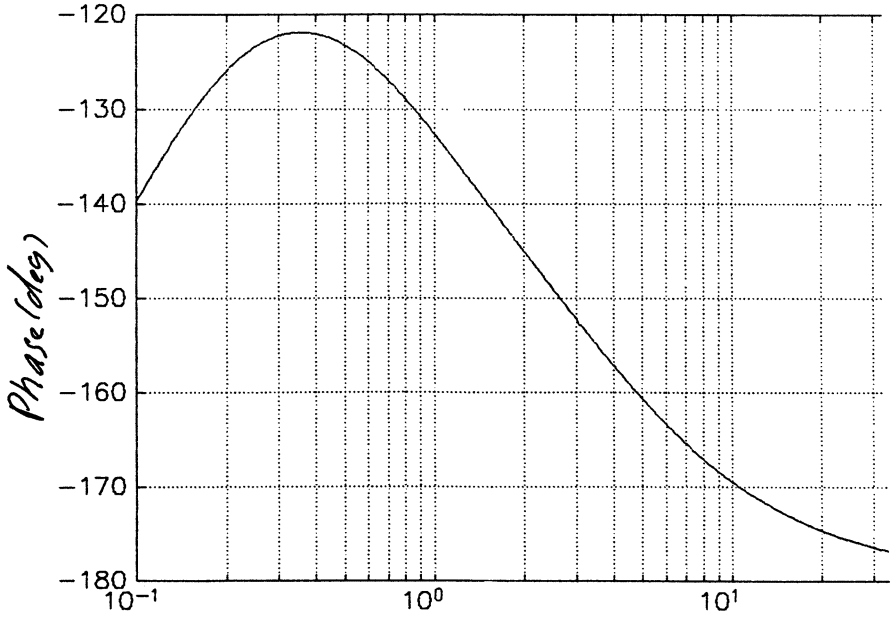
Bode Plot of Uncomp. System



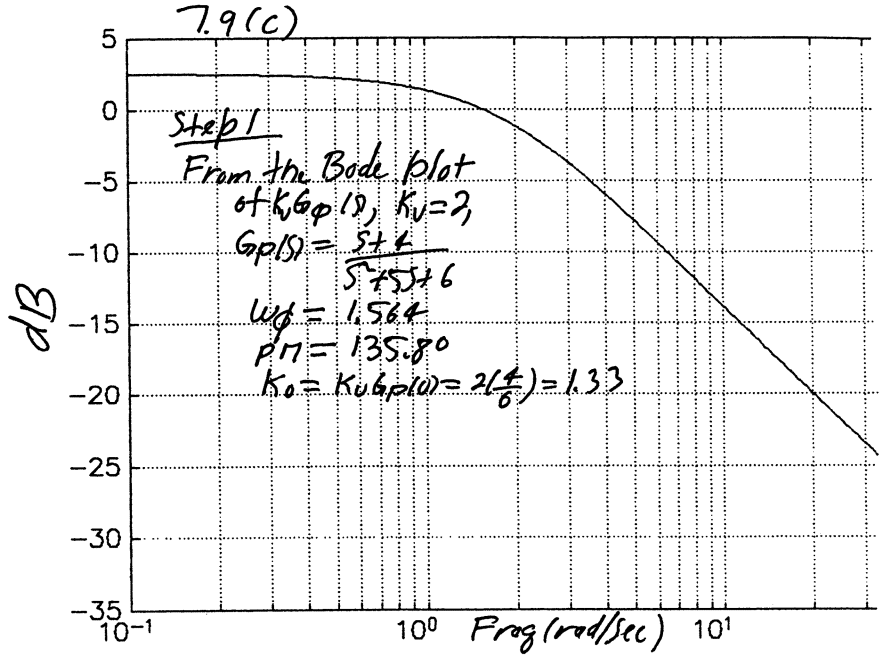
7.9(b)(continued)



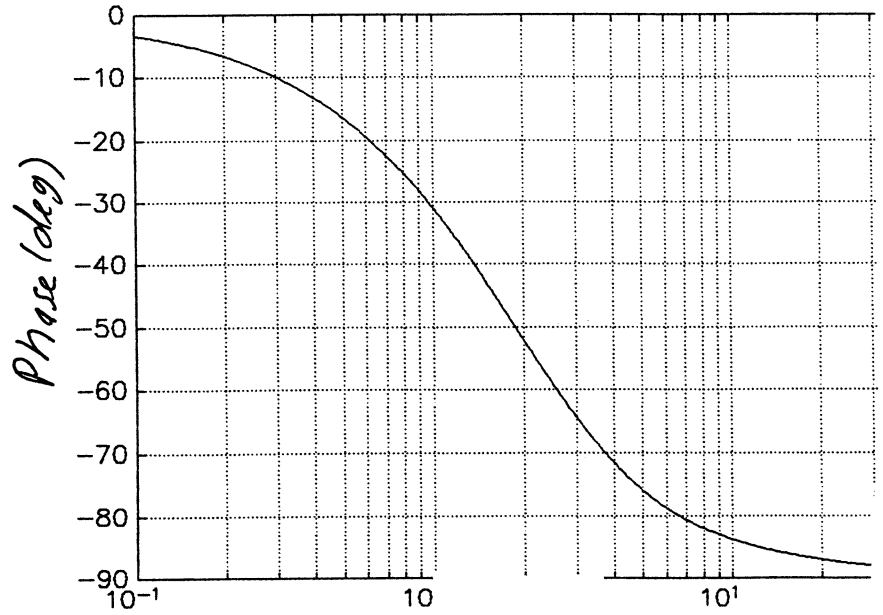
Bode Plot of $G_c'(s)G_p(s)=\frac{(s+0.1)(s+2)}{s^2(s^2+4s+3)}$



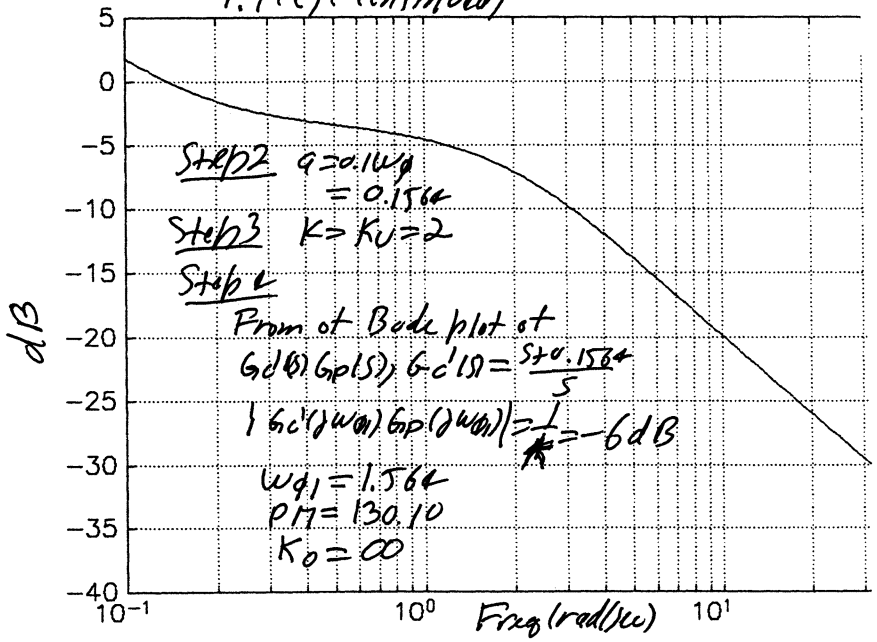
7.9(c)



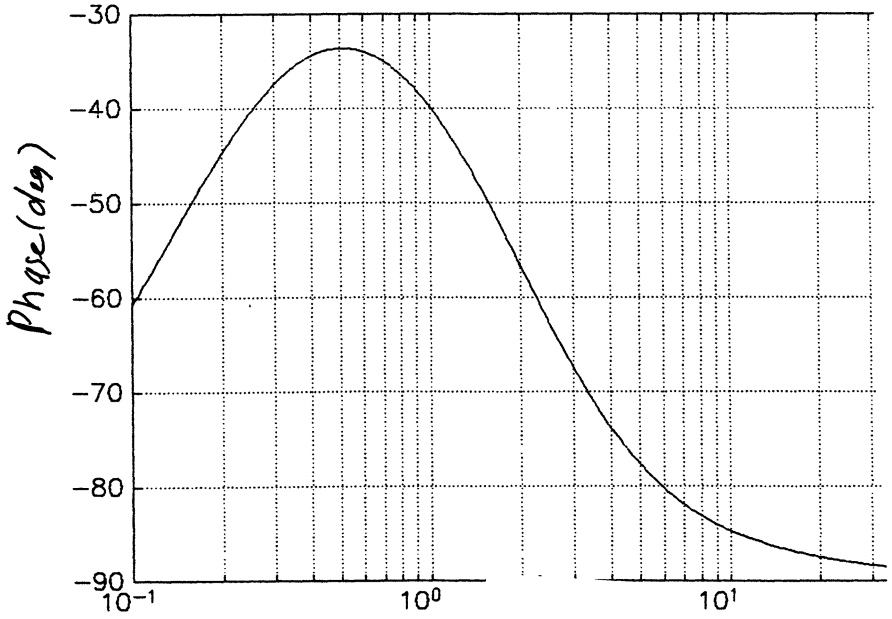
Bode Plot of Uncomp. System



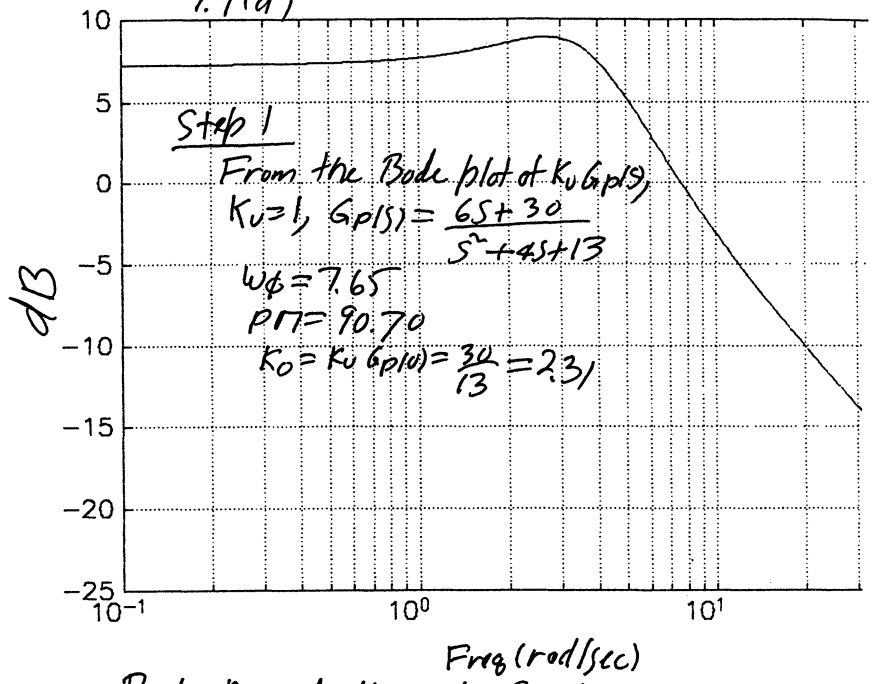
7.9(c) (continued)



Bode Plot of $G_c'(s)G_p(s) = \left(\frac{s+0.156}{5}\right)\left(\frac{s+4}{s^2+5s+6}\right)$



7.9(d)



Step 1

From the Bode plot of $K_u G_p(s)$,

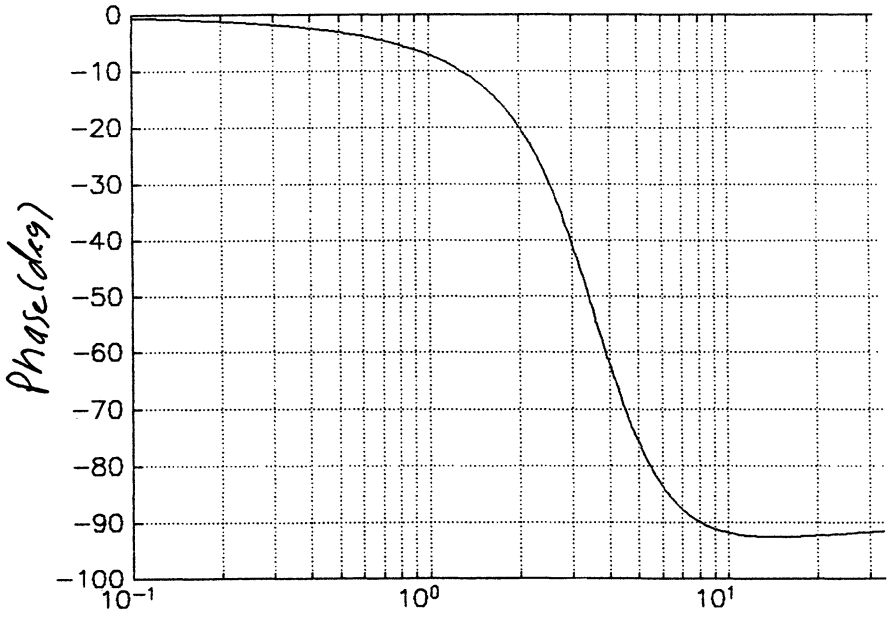
$K_u = 1$, $G_p(s) = \frac{6s+30}{s^2+4s+13}$

$\omega_\phi = 7.65$

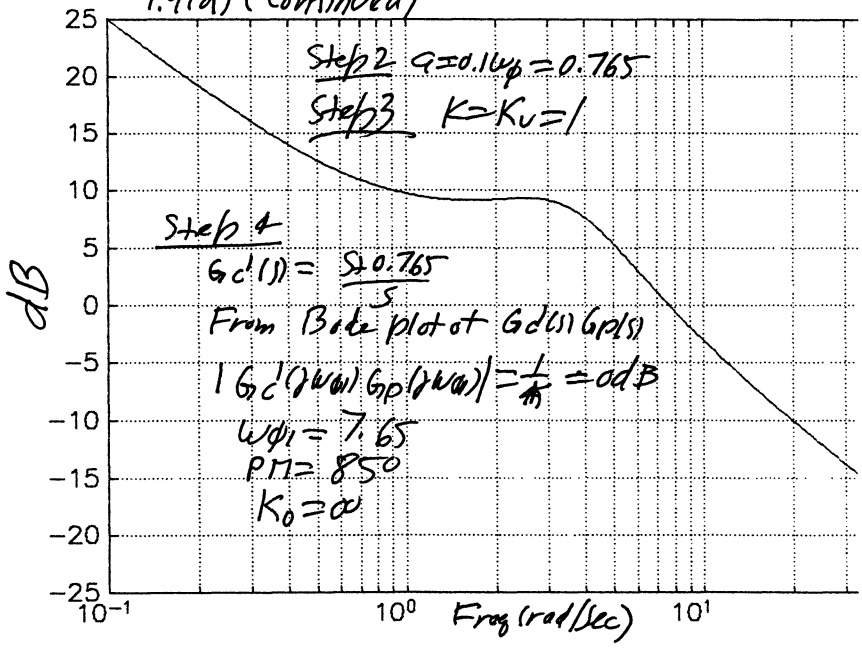
$PM = 90.70$

$K_0 = K_u G_p(0) = \frac{30}{13} = 2.31$

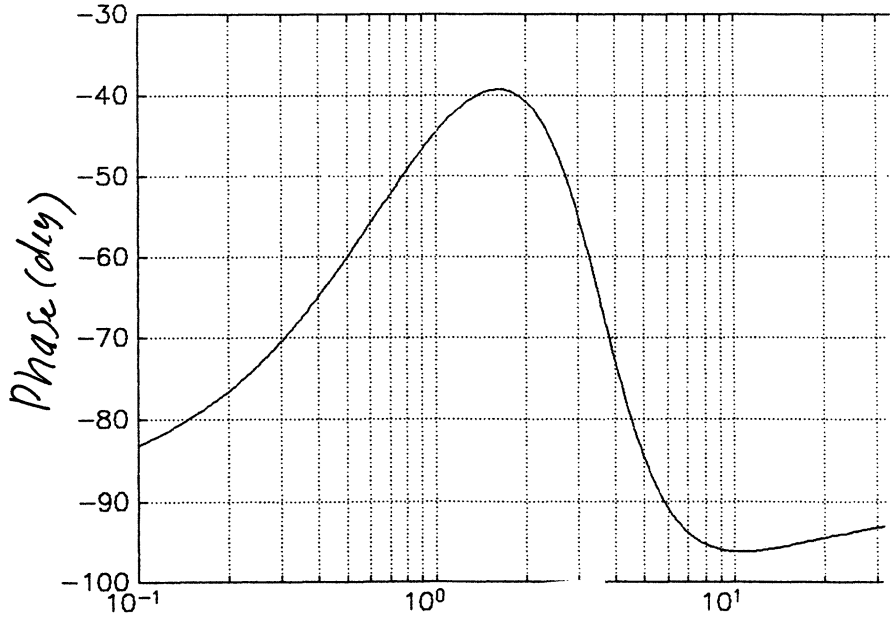
Bode Plot of Uncomp. System



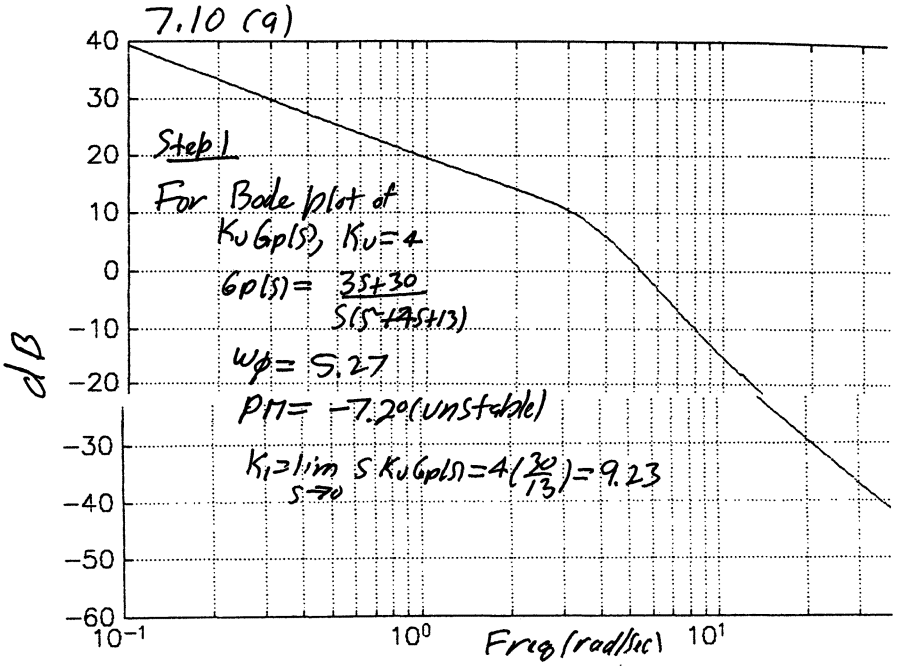
7.9(d) (continued)



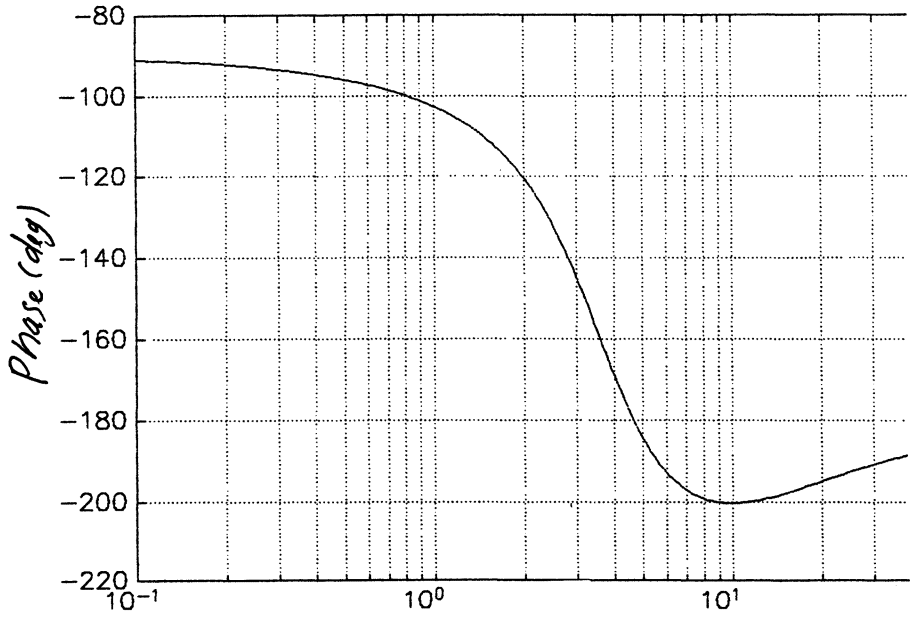
Bode Plot of $G_c'(s)G_p(s) = \frac{s+0.765}{s} \frac{6s+30}{s^2+4s+13}$



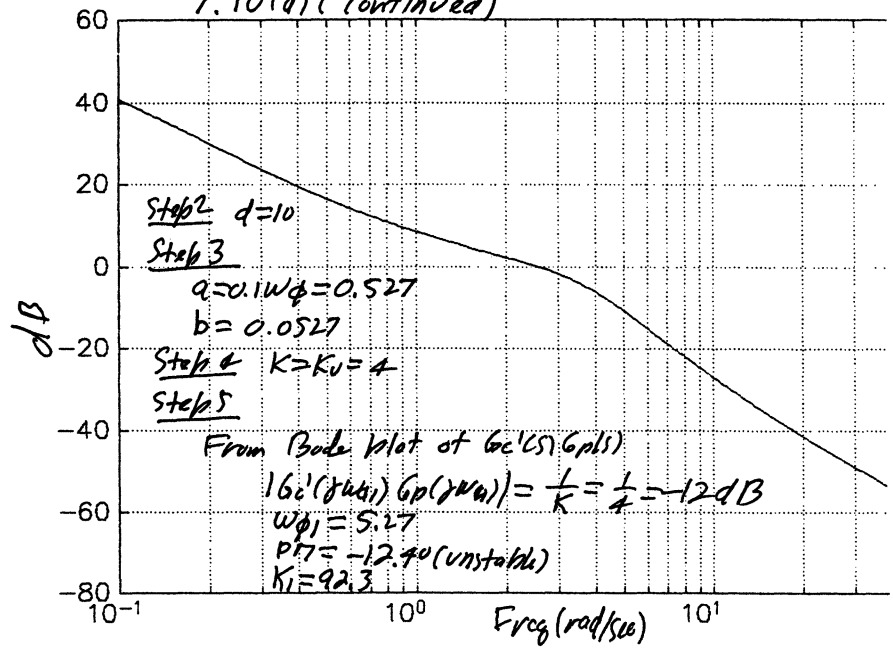
7.10 (a)



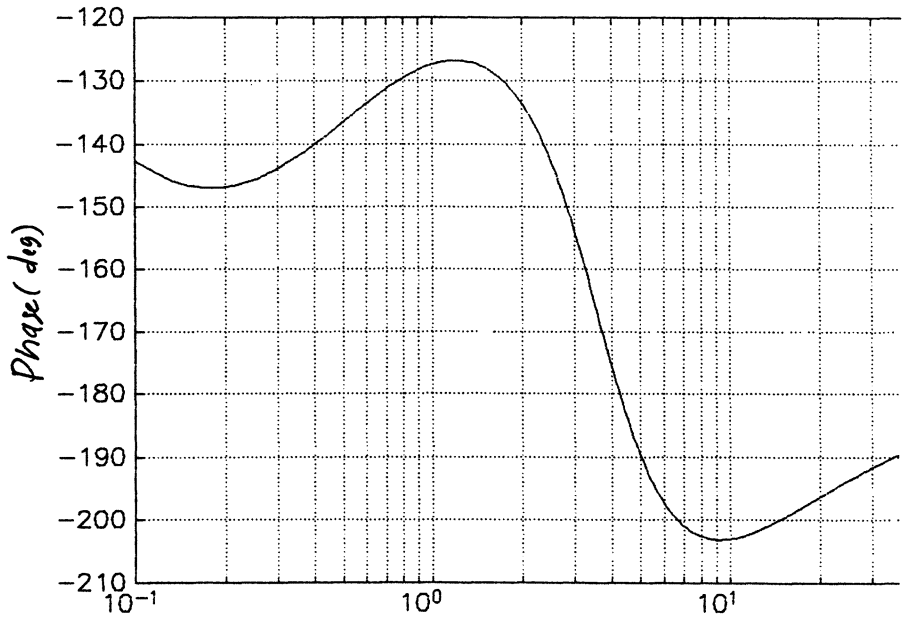
Bode Plot of Uncomp. System



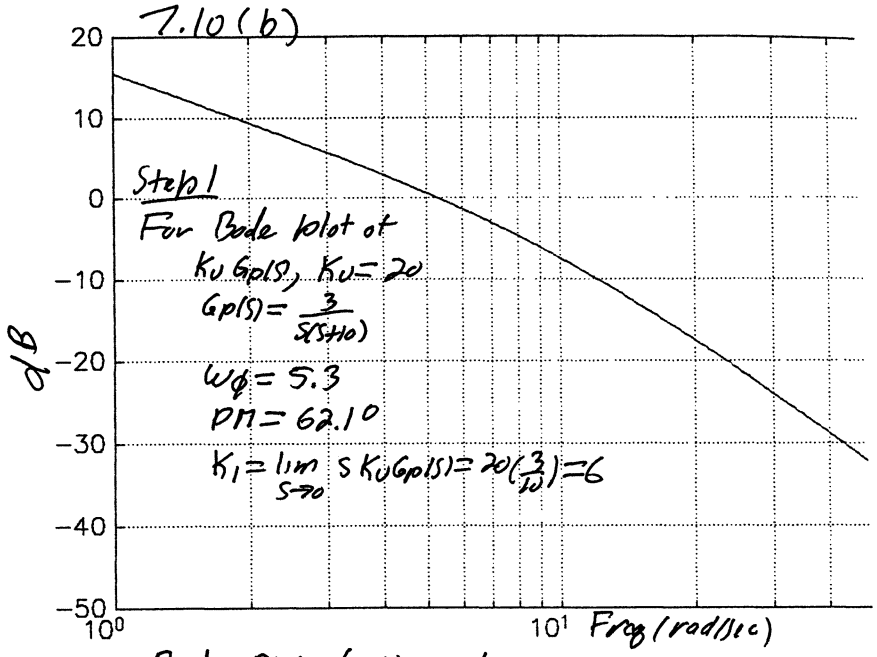
7.10(a) (continued)



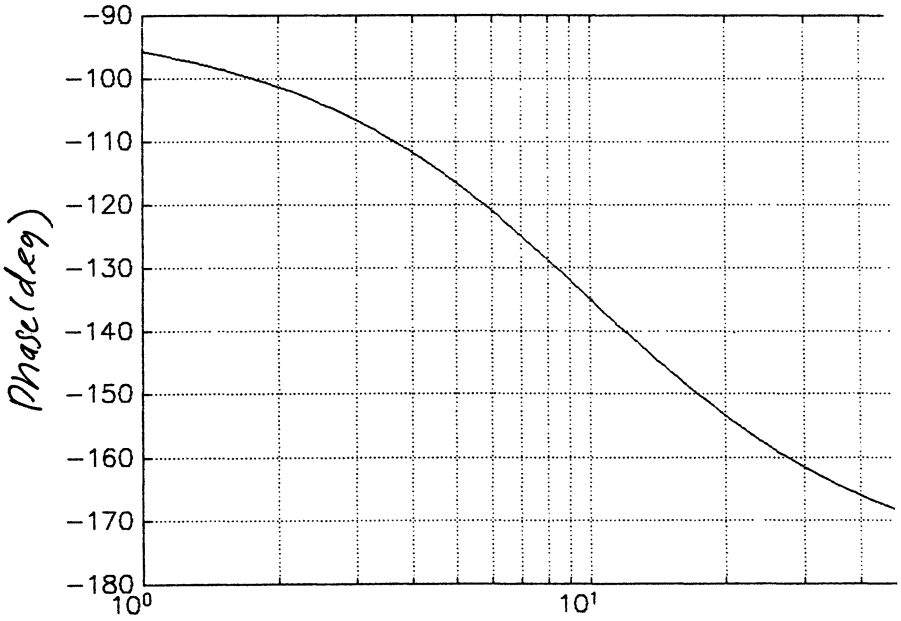
Bode Plot of $G_c'(s)G_p(s) = \left(\frac{s+0.527}{s+0.0527}\right)\left(\frac{35+30}{s(s^2+4s+3)}\right)$



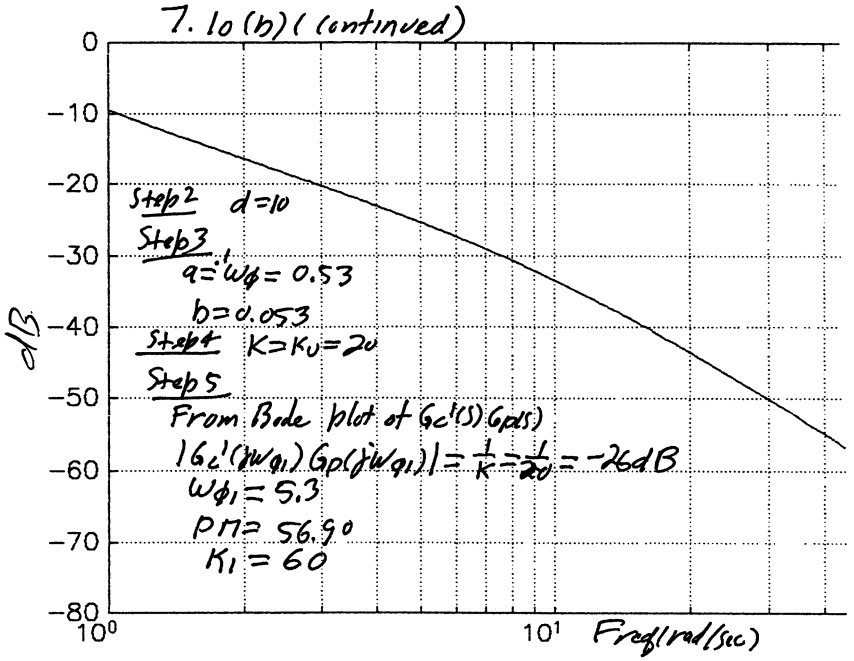
7.10 (b)



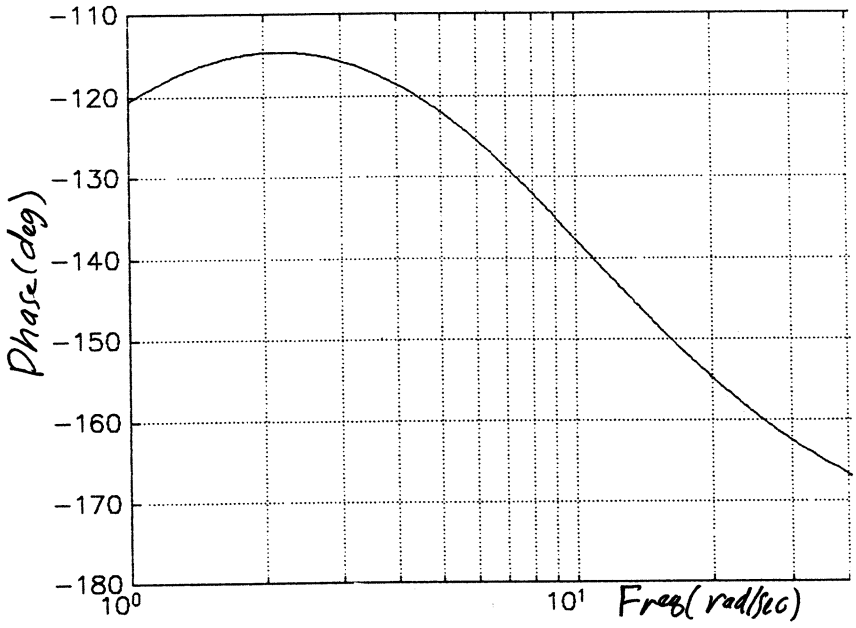
Bode Plot of Uncomp. System



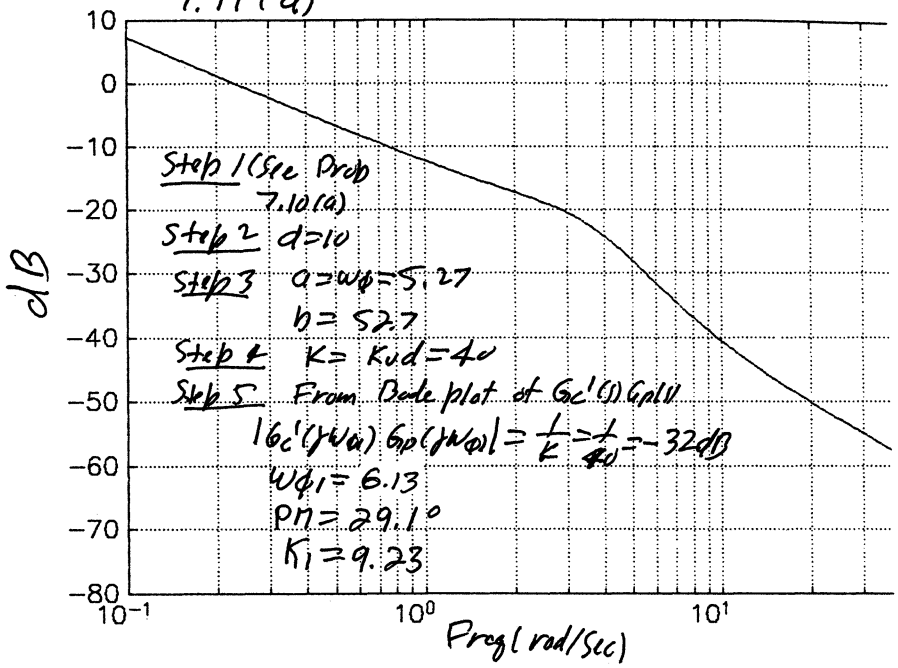
7.10(b) (continued)



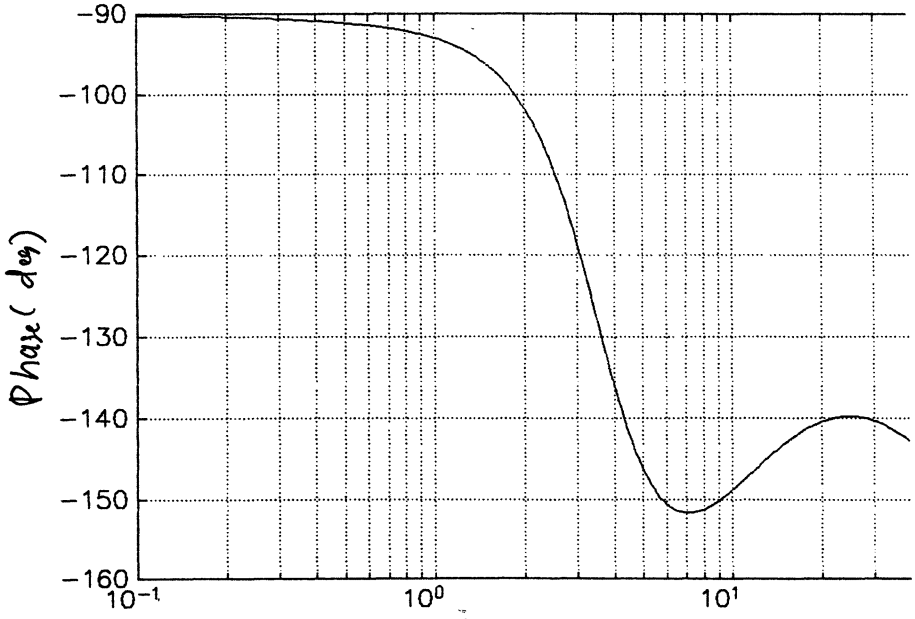
Bode Plot of $G_c'(s)G_p(s) = \left(\frac{s+0.53}{s+5.3}\right)\left(\frac{3}{s(s+10)}\right)$



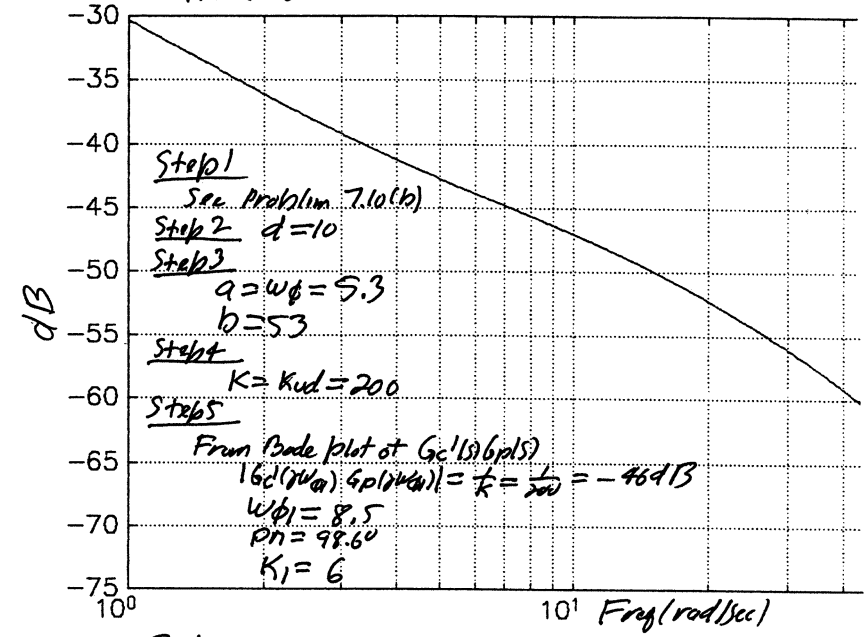
7.11 (a)



Bode Plot of $G_c'(s)G_p(s) = \left(\frac{s+5.27}{s+52.7}\right)\left(\frac{3s+30}{s(s^2+4s+13}\right)$

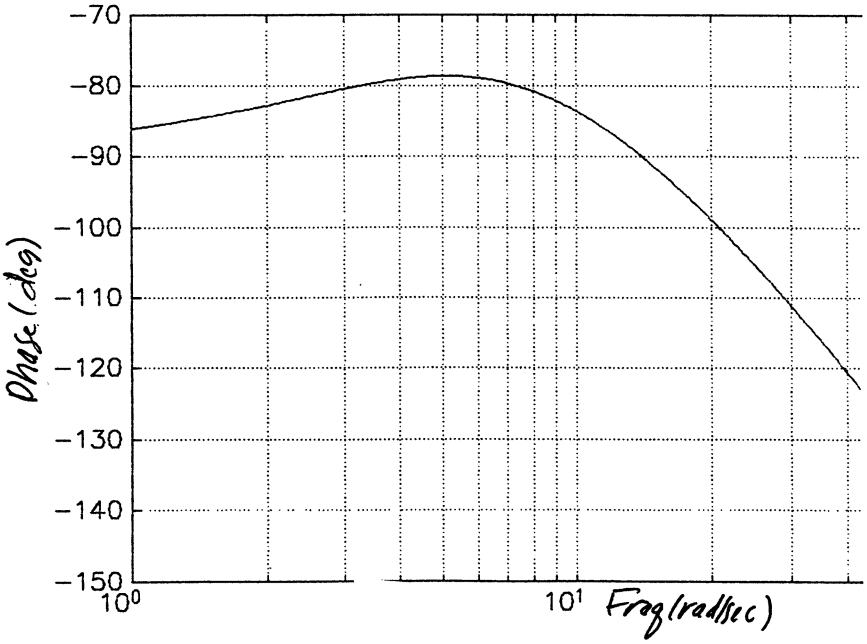


7.11(b)

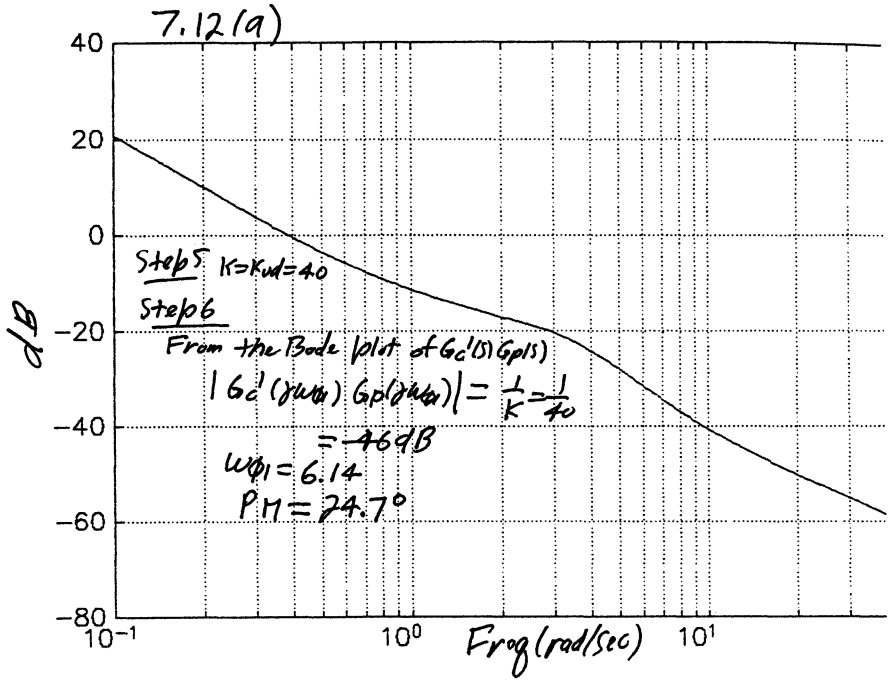


Step 1
See Problem 7.10(b)
Step 2 $d = 10$
Step 3
$a = \omega_\phi = 5.3$
$b = 53$
Step 4
$K = K_{ud} = 200$
Step 5
From Bode plot of $G_c'(s)G_p(s)$
$|G_c'(j\omega_{co})G_p(j\omega_{co})| = \frac{1}{K} = \frac{1}{200} = -46\,dB$
$\omega_{\phi 1} = 8.5$
$pm = 98.6^\circ$
$K_1 = 6$

Bode Plot of $G_c'(s)G_p(s) = \left(\frac{s+53}{s+53}\right)\left(\frac{3}{s(s+10)}\right)$



7.12 (a)



Step 5 $K = K_{ud} = 40$

Step 6

From the Bode plot of $G_c'(s)G_p(s)$

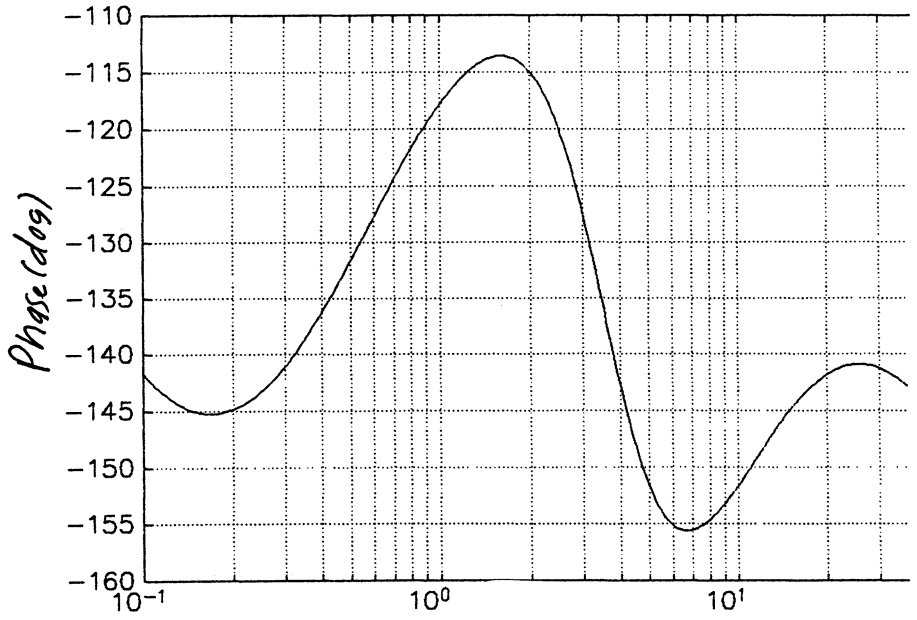
$|G_c'(j\omega_{co})\,G_p(j\omega_{co})| = \frac{1}{K} = \frac{1}{40}$

$= -46\,dB$

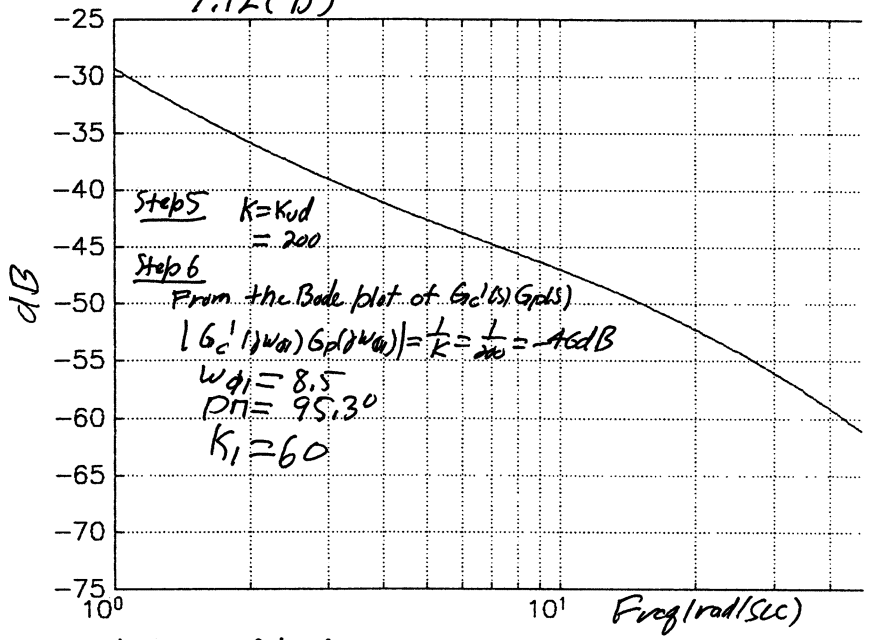
$\omega_{\phi 1} = 6.14$

$PM = 24.7^\circ$

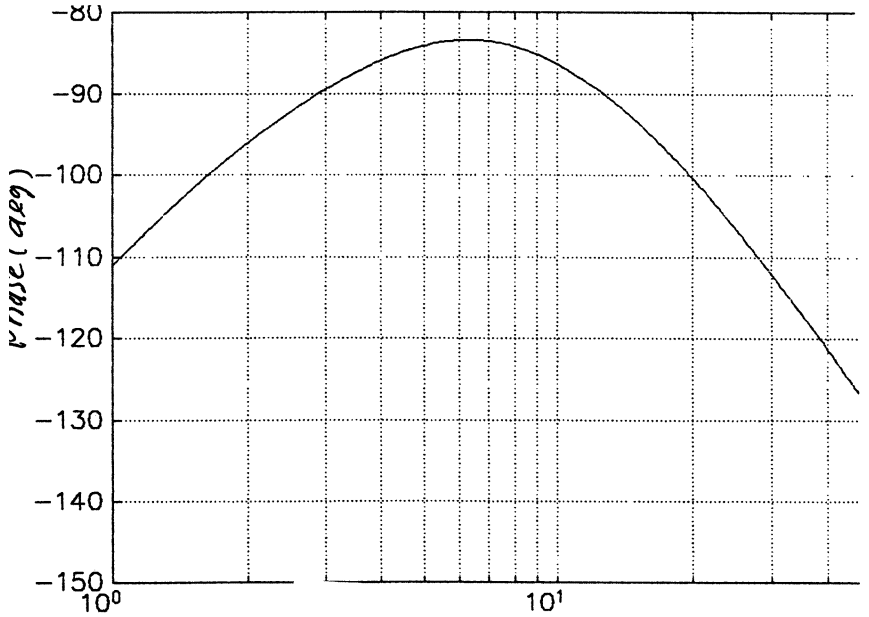
Bode Plot of $G_c'(s)G_p(s) = \left(\frac{s+.527}{s+.0527}\right)\left(\frac{s+5.27}{s+52.7}\right)\left(\frac{3s+30}{s(s^2+4s+3)}\right)$



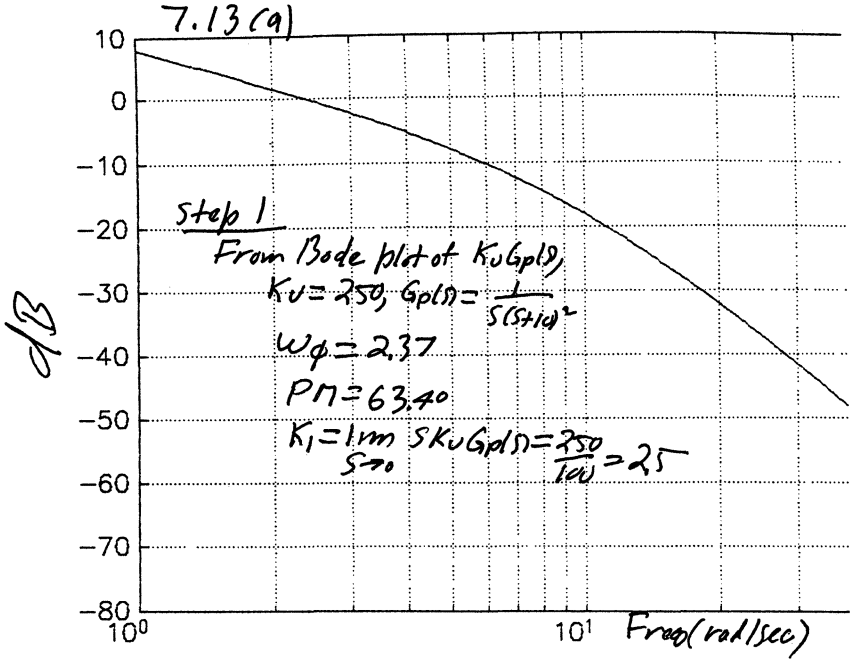
7.12(b)



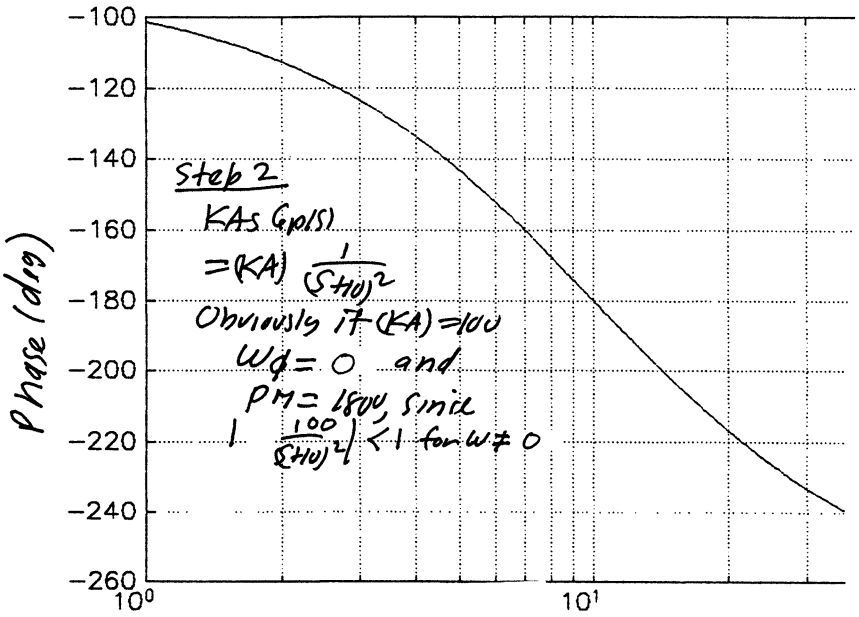
Bode Plot of $G_c'(s)G_p(s) = \left(\frac{s+0.53}{s+0.053}\right)\left(\frac{s+5.3}{s+53}\right)\left(\frac{3}{s(s+10)}\right)$



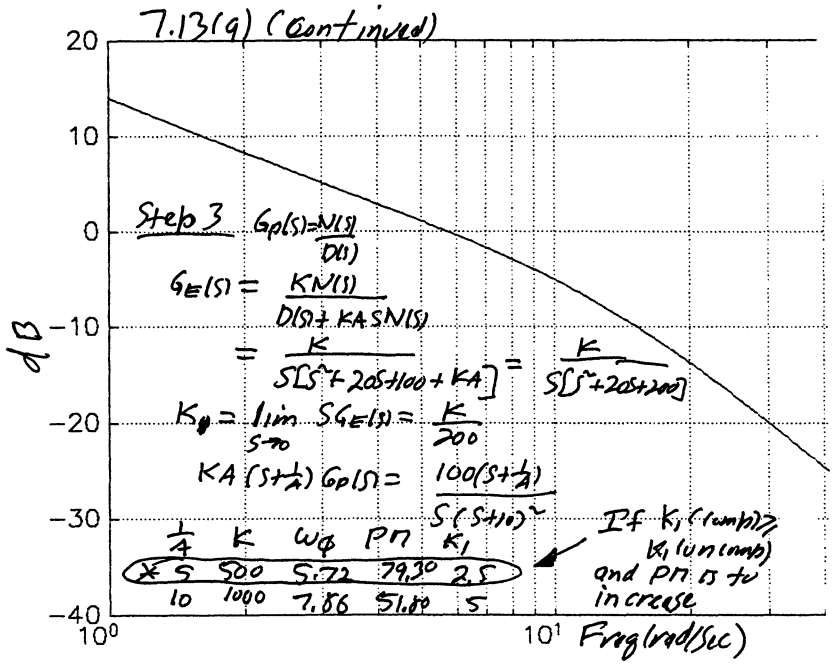
7.13 (a)



Bode Plot of Uncomp. System



7.13(g) (continued)



Step 3 $G_p(s) = \frac{N(s)}{D(s)}$

$$G_{\epsilon}(s) = \frac{KN(s)}{D(s) + KA\,N(s)} = \frac{K}{s[s^2+20s+100+KA]} = \frac{K}{s[s^2+20s+200]}$$

$$K_p = \lim_{s\to 0} sG_{\epsilon}(s) = \frac{K}{200}$$

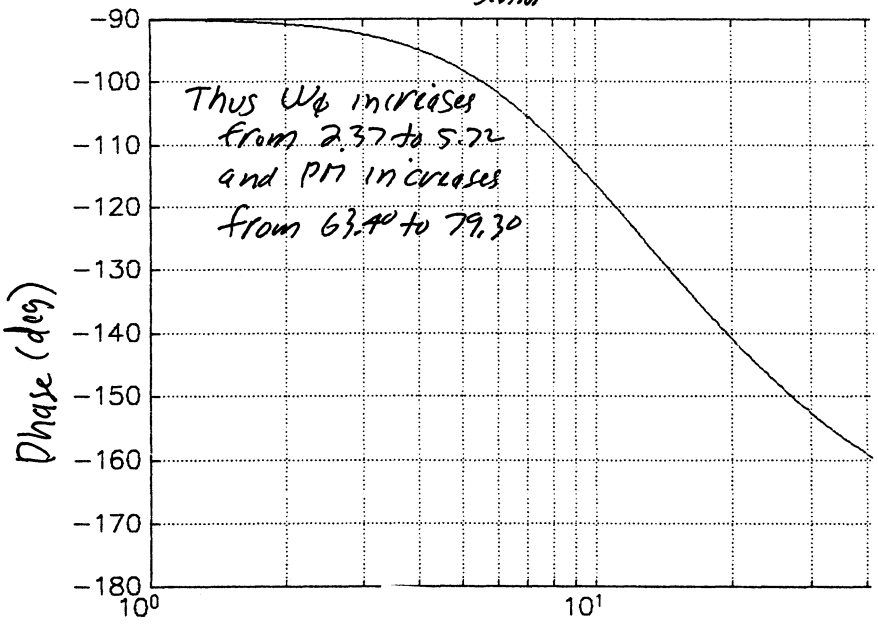
$$KA\left(s+\tfrac{1}{2}\right)G_p(s) = \frac{100\left(s+\frac{1}{2}\right)}{s(s+10)^2}$$

$\frac{1}{4}$	K	ω_ϕ	PM	K_1
5	500	5.72	79.30	2.5
10	1000	7.86	51.89	5

If K_1 (ramp) $\nearrow$ K_1 (unramp) and PM is to increase

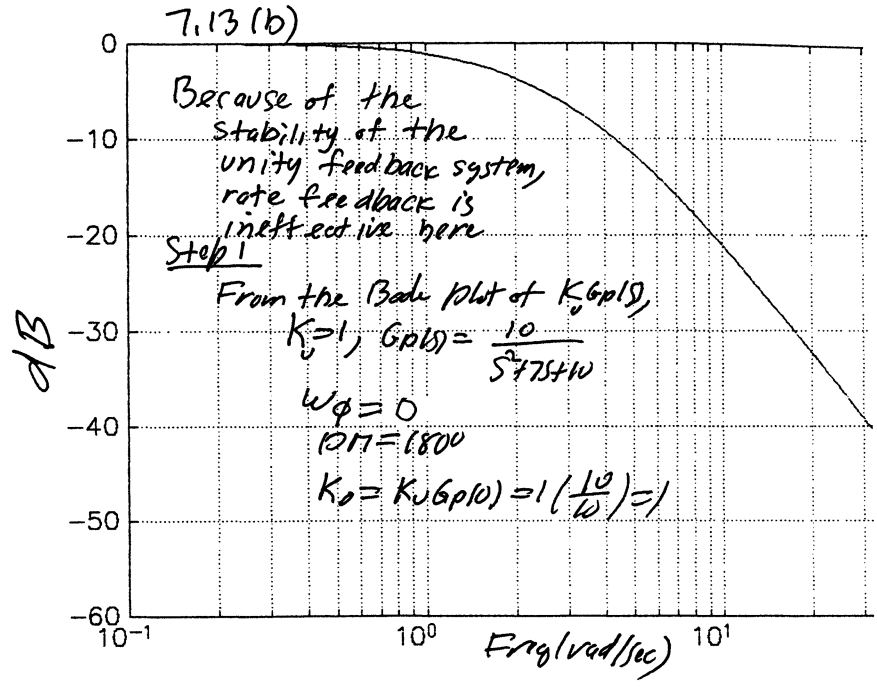
Bode Plot of $(KA)\left(s+\frac{1}{2}\right)G_p(s)$

$$= 100(s+5)\frac{1}{s(s+10)^2}$$



Thus ω_ϕ increases from 2.37 to 5.72 and PM increases from 63.4° to 79.30

7.13 (b)



Because of the stability of the unity feedback system, rate feedback is ineffective here

Step 1

From the Bode plot of $K_v G_p(s)$,

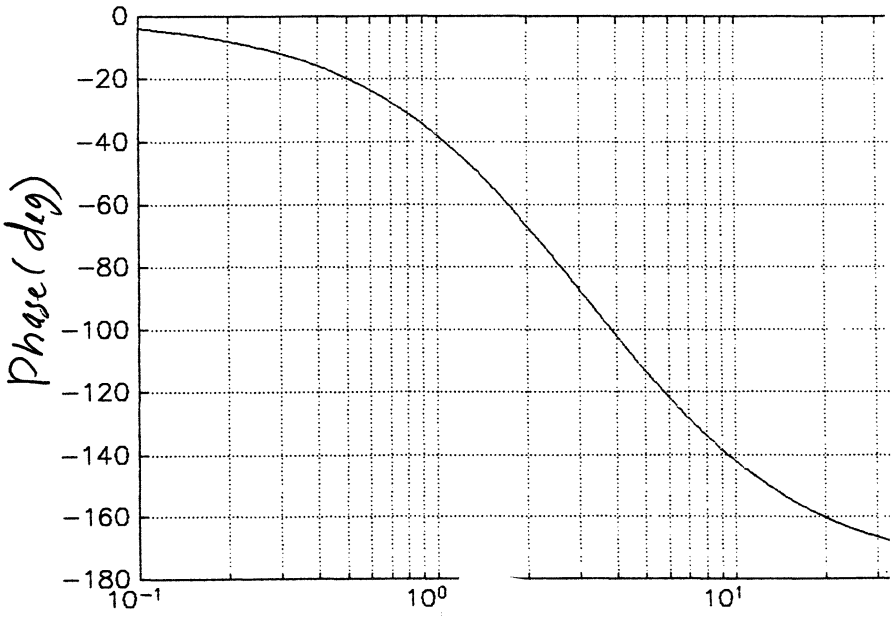
$K_v = 1$, $G_p(s) = \frac{10}{s^2 + 7s + 10}$

$\omega_\phi = 0$

$PM = 180^\circ$

$K_p = K_v G_p(0) = 1\left(\frac{10}{10}\right) = 1$

Bode Plot of Unramp. System



7.13(b) (continued)

Step 2

$(KA)\,S\,G_p(s)$

$= (KA)\,\dfrac{10s}{s^2+7s+10}$

$G'(s)$

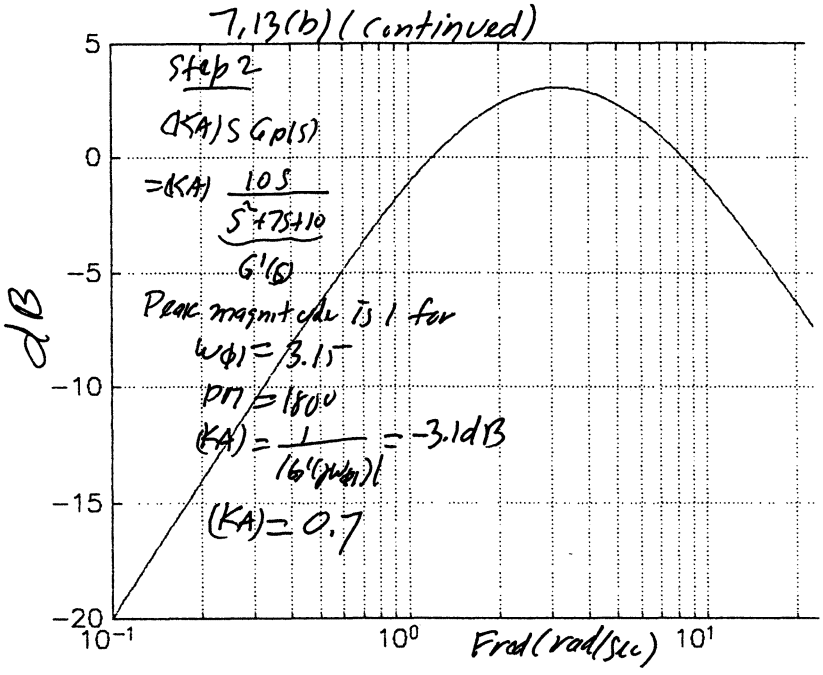
Peak magnitude is 1 for

$\omega_{\phi 1} = 3.1$

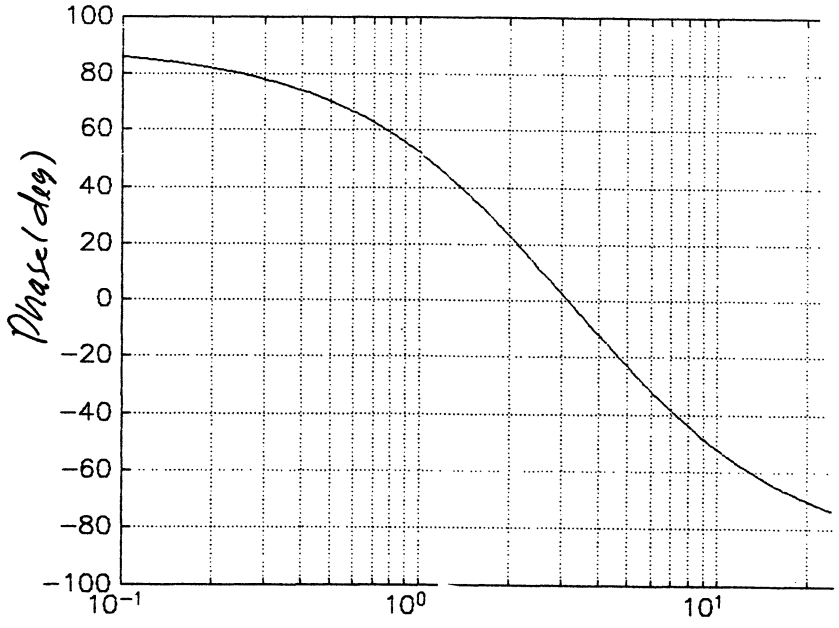
$PM = 180°$

$(KA) = \dfrac{1}{|G'(j\omega_{\phi 1})|} = -3.1\,dB$

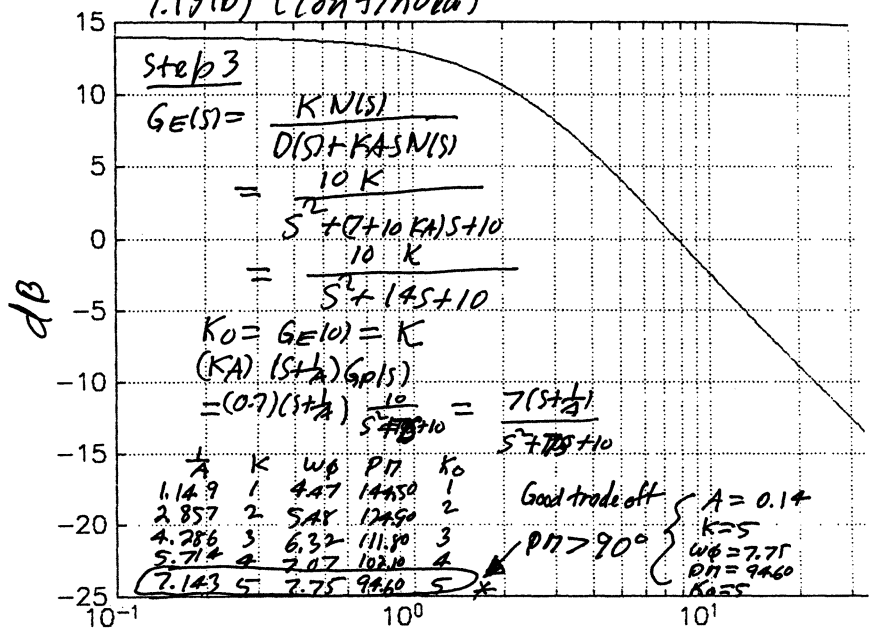
$(KA) = 0.7$



Bode Plot of $G'(s) = \dfrac{10s}{s^2+7s+10}$



7.13(b) (continued)



Step 3

$$G_E(s) = \frac{K N(s)}{D(s) + KA\,s\,N(s)} = \frac{10K}{s^2 + (7 + 10KA)s + 10} = \frac{10K}{s^2 + 14s + 10}$$

$K_0 = G_E(0) = K$

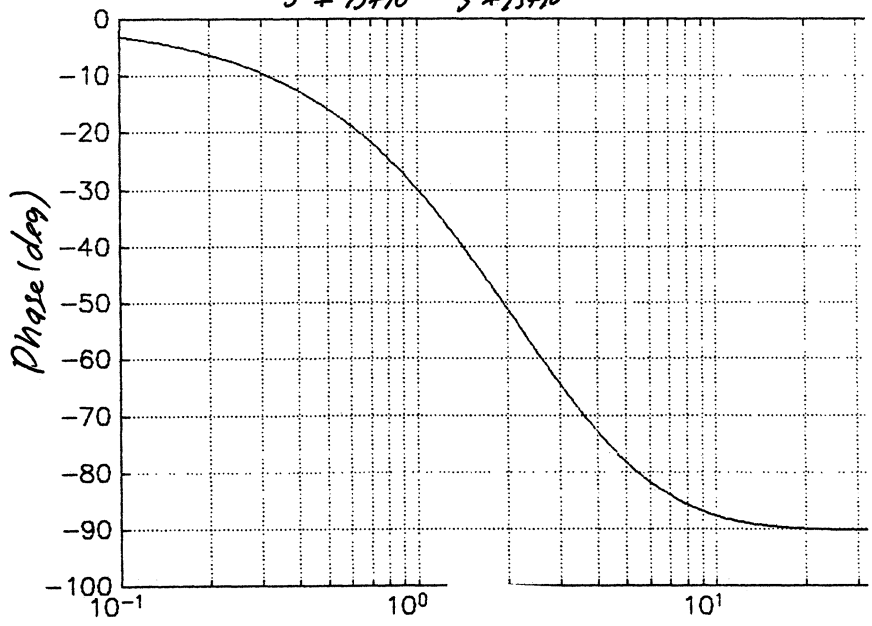
$$(KA)\left(s + \tfrac{1}{A}\right)G_p(s) = (0.7)\left(s + \tfrac{1}{A}\right)\frac{10}{s^2 + 7s + 10} = \frac{7\left(s + \frac{1}{A}\right)}{s^2 + 7s + 10}$$

$\frac{1}{A}$	K	ω_ϕ	PM	K_0	
1.149	1	4.47	144.50	1	
2.857	2	5.48	124.50	2	Good trade off
4.286	3	6.32	111.80	3	PM > 90°
5.714	4	7.07	102.10	4	
7.143	5	7.75	94.60	5	★

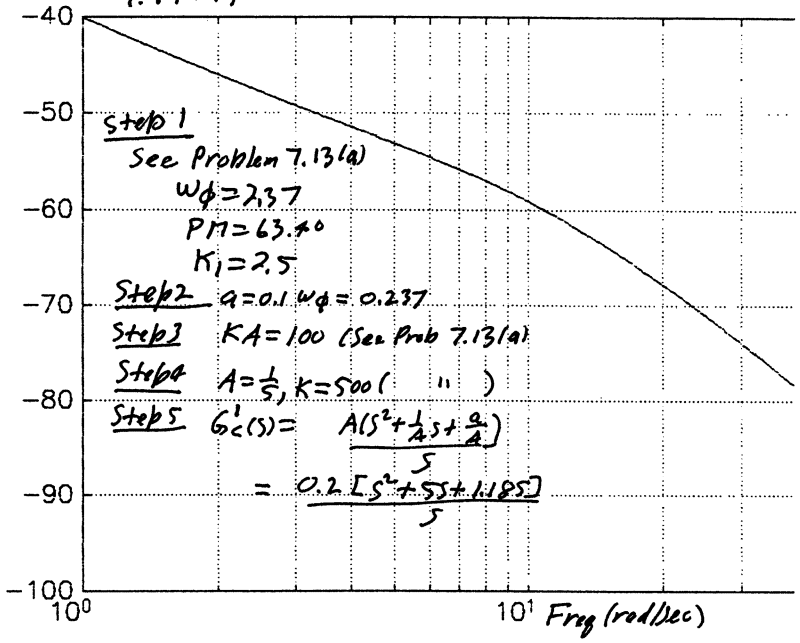
$A = 0.14$, $K = 5$, $\omega_\phi = 7.75$, PM = 94.60, $K_0 = 5$

Bode Plot of $(KA)\left(s + \frac{1}{A}\right)G_p(s)$

$$= \frac{7\left(s + \frac{1}{A}\right)}{s^2 + 7s + 10} = \frac{7(s + 7.143)}{s^2 + 7s + 10}$$



7.14(a)



Step 1
See Problem 7.13(a)
$\omega_\phi = 2.37$
$PM = 63.4°$
$K_1 = 2.5$

Step 2 $a = 0.1$ $\omega_\phi = 0.237$

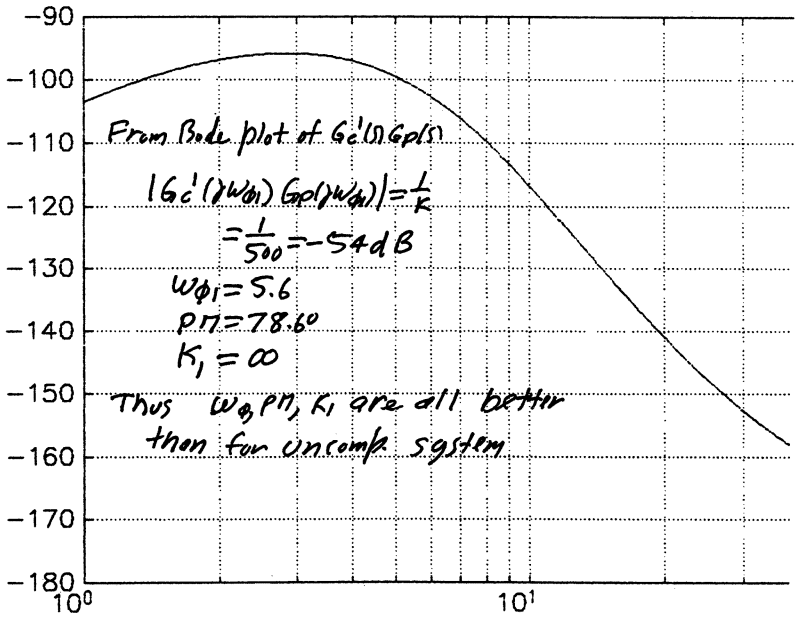
Step 3 $KA = 100$ (See Prob 7.13(a))

Step 4 $A = \frac{1}{5}$, $K = 500$ (")

Step 5 $G_c'(s) = \frac{A(s^2 + \frac{1}{A}s + \frac{a}{A})}{s}$

$= \frac{0.2[s^2 + 5s + 1.185]}{s}$

Bode Plot of $G_c'(s)G_p(s) = \frac{0.2[s^2+5s+1.185]}{s^2(s+10)^2}$



From Bode plot of $G_c'(s)G_p(s)$

$|G_c'(j\omega_{\phi_1})G_p(j\omega_{\phi_1})| = \frac{1}{K}$

$= \frac{1}{500} = -54\,dB$

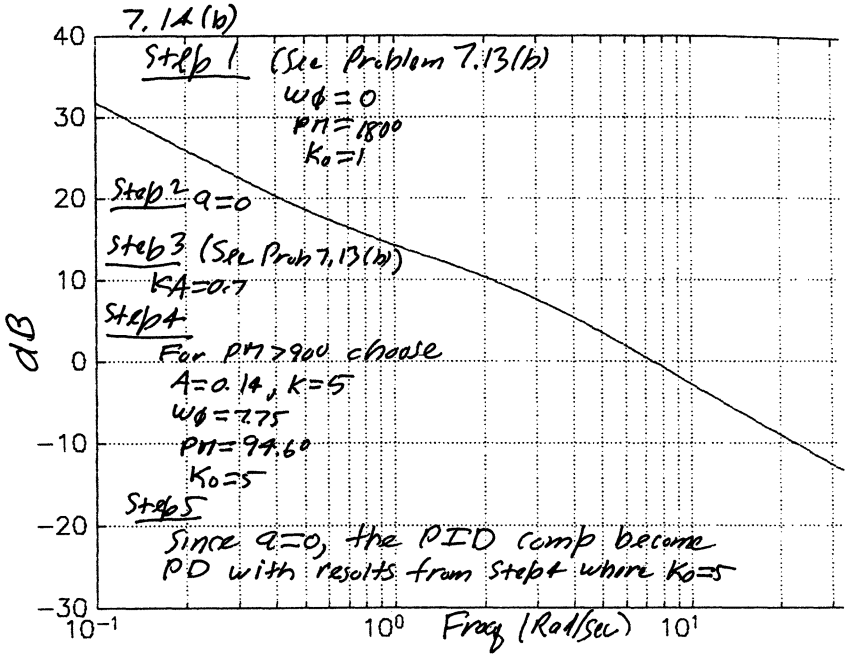
$\omega_{\phi_1} = 5.6$

$PM = 78.6°$

$K_1 = \infty$

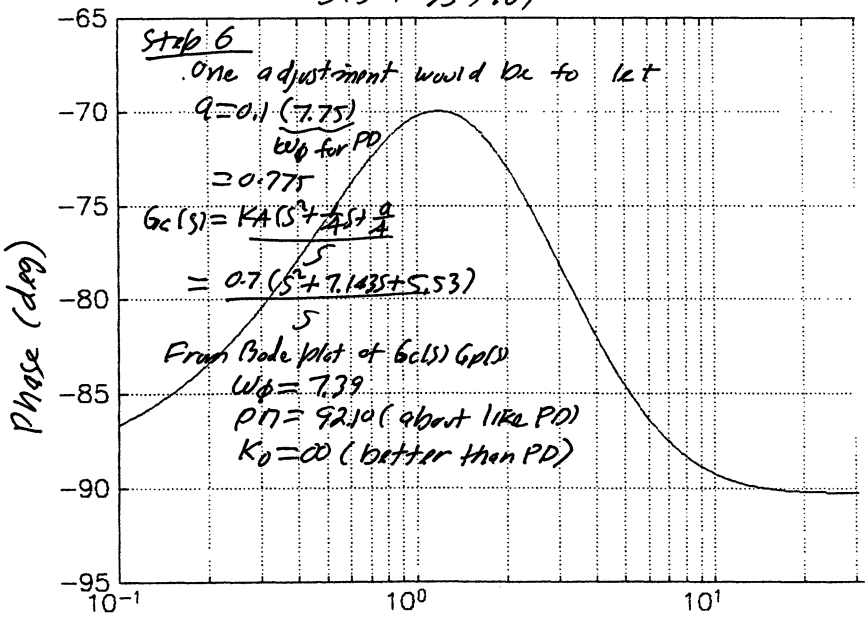
Thus ω_ϕ, PM, K_1 are all better than for uncomp system

7.14 (b)

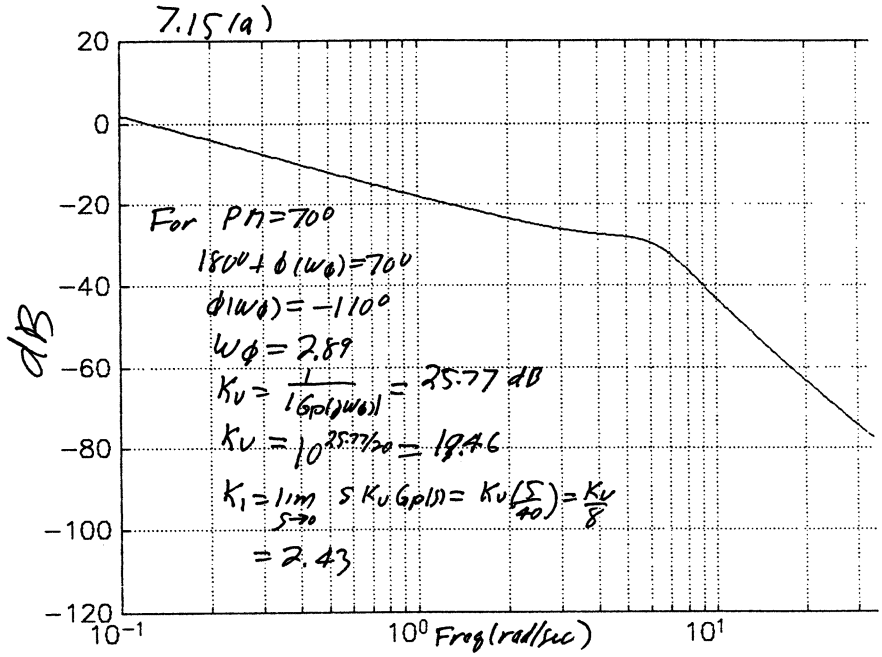


Bode Plot of $G_c(s)G_p(s)$

$$= \frac{7D(s^2 + 7.143s + 5.53)}{s(s^2 + 7s + 10)}$$



7.15(a)



For $PM = 70°$

$180° + \phi(\omega_\phi) = 70°$

$\phi(\omega_\phi) = -110°$

$\omega_\phi = 2.89$

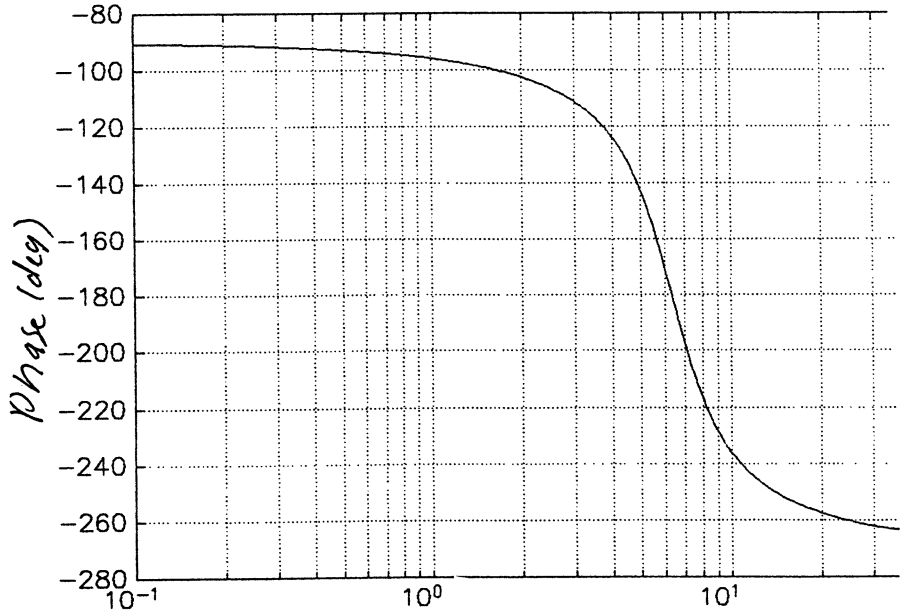
$K_v = \frac{1}{|G_p(j\omega_\phi)|} = 25.77\ dB$

$K_v = 10^{25.77/20} = 19.46$

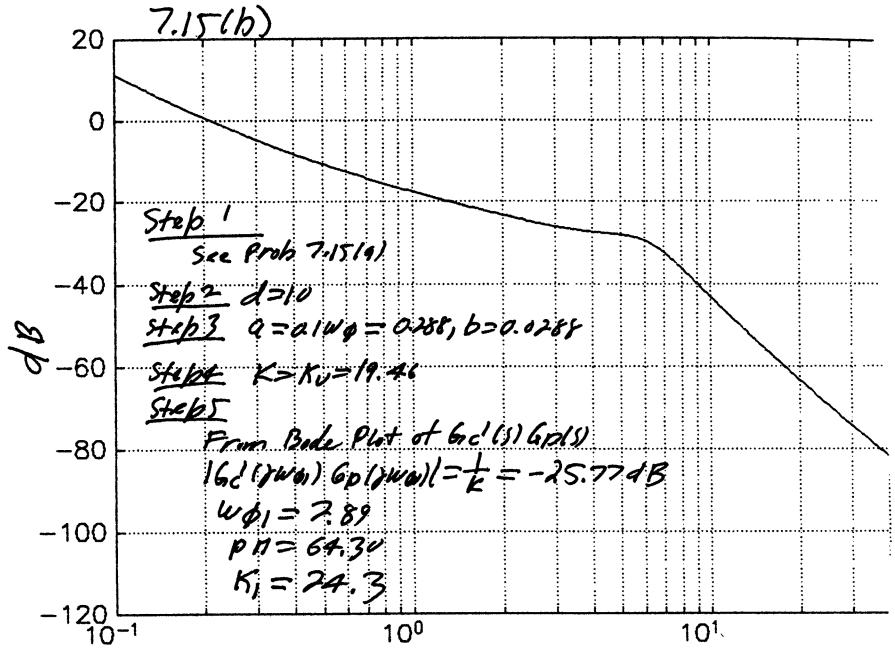
$K_1 = \lim_{s \to 0} s K_v G_p(s) = K_v\left(\frac{5}{40}\right) = \frac{K_v}{8}$

$= 2.43$

Bode Plot of $G_p(s) = \frac{5}{s(s^2 + 4s + 40)}$

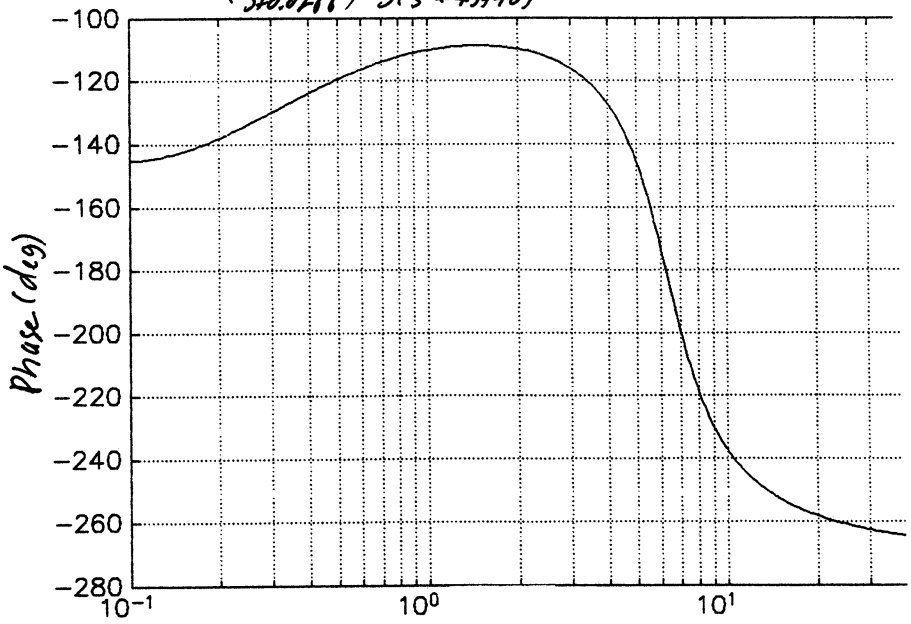


7.15(b)

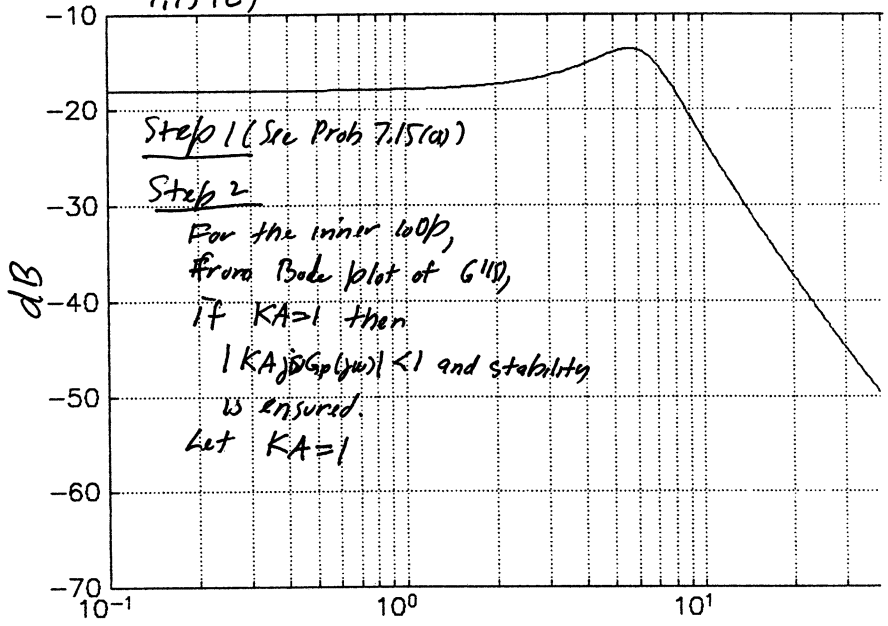


Bode Plot of $G_c'(s)\,G_p(s)$

$= \left(\frac{s+0.288}{s+0.0288}\right)\frac{5}{s(s^2+4s+40)}$



7.15(c)



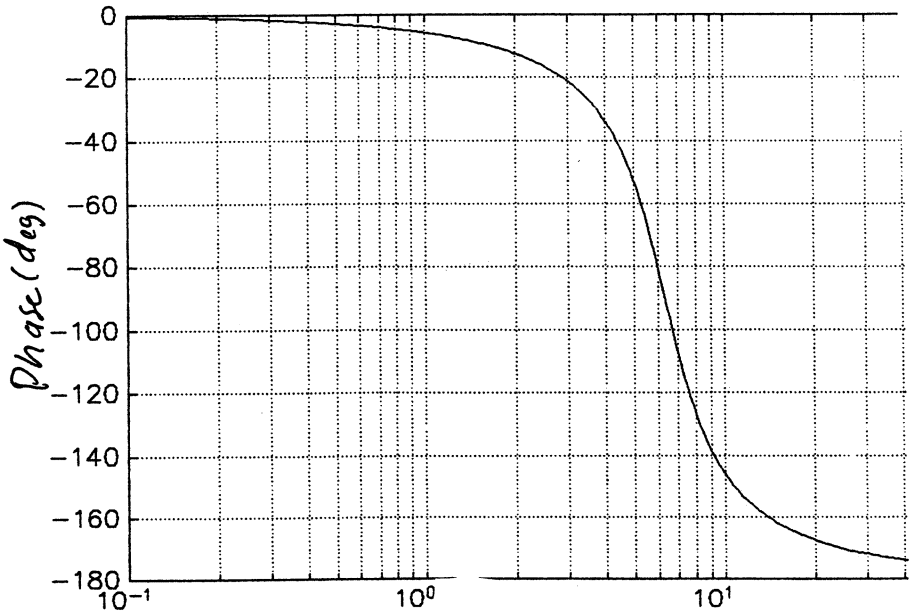
Step 1 (See Prob 7.15(a))

Step 2

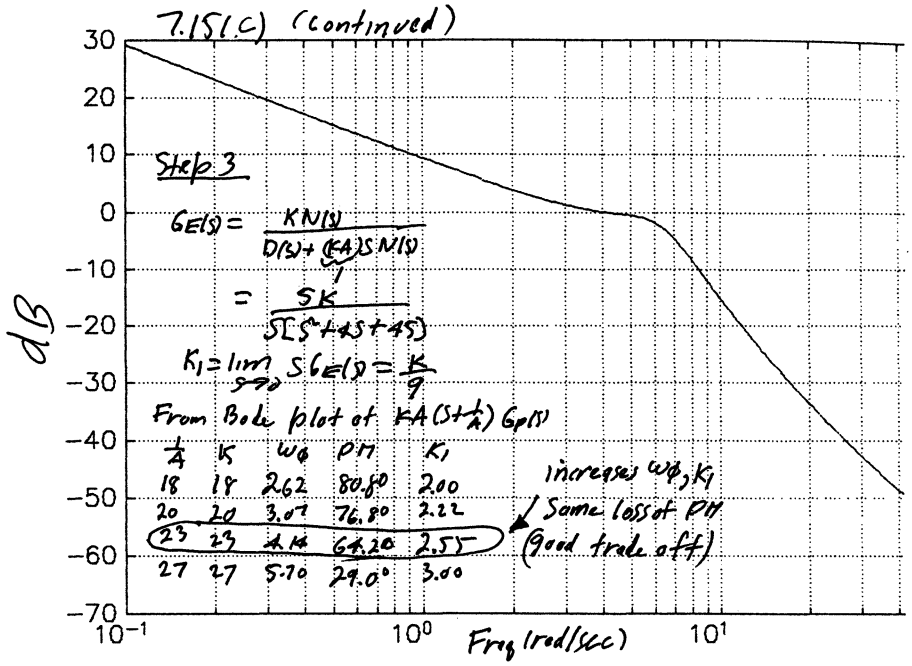
For the inner loop, from Bode plot of $G'(s)$, if $K_A=1$ then $|K_A G_p(j\omega)| < 1$ and stability is ensured.

Let $K_A=1$

Bode Plot of $SG_p(s) = \dfrac{5}{s^2+4s+40} = G'(s)$



7.15(c) (continued)



Step 3

$$G_E(s) = \frac{KN(s)}{D(s)+(KA)sN(s)} = \frac{5K}{s(s^2+4s+45)}$$

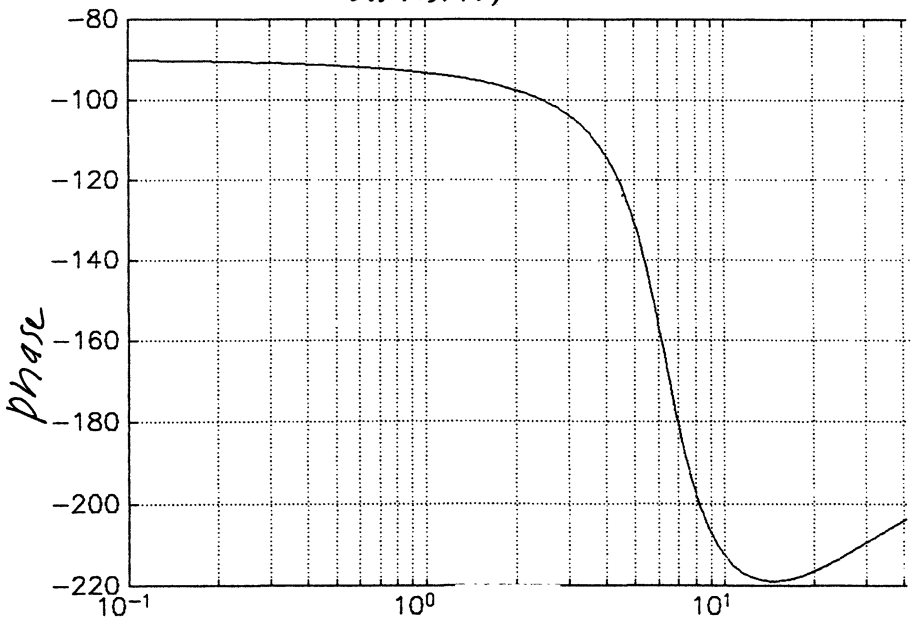
$$K_1 = \lim_{s\to 0} sG_E(s) = \frac{K}{9}$$

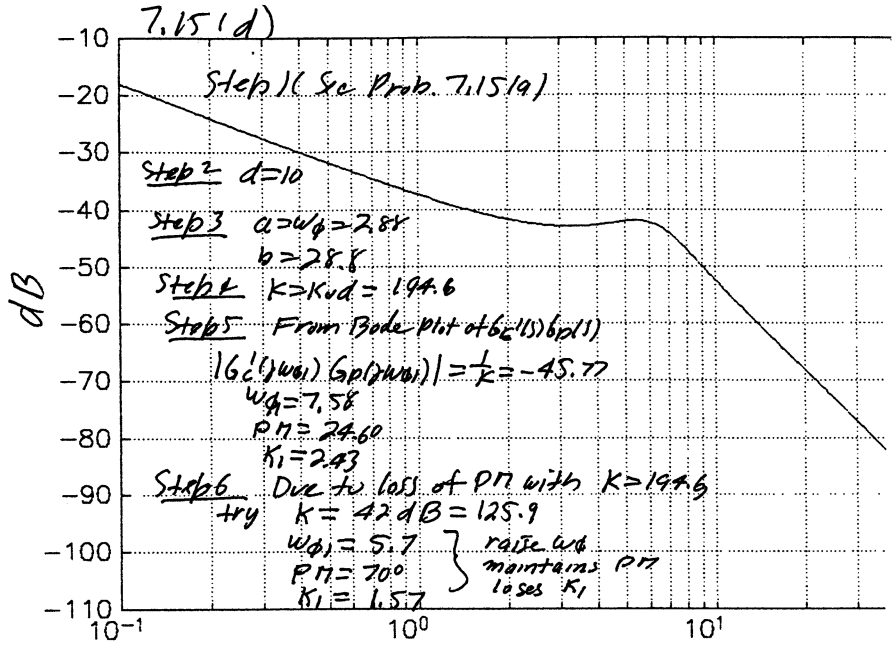
From Bode plot of $KA(s+\frac{1}{A})G_p(s)$

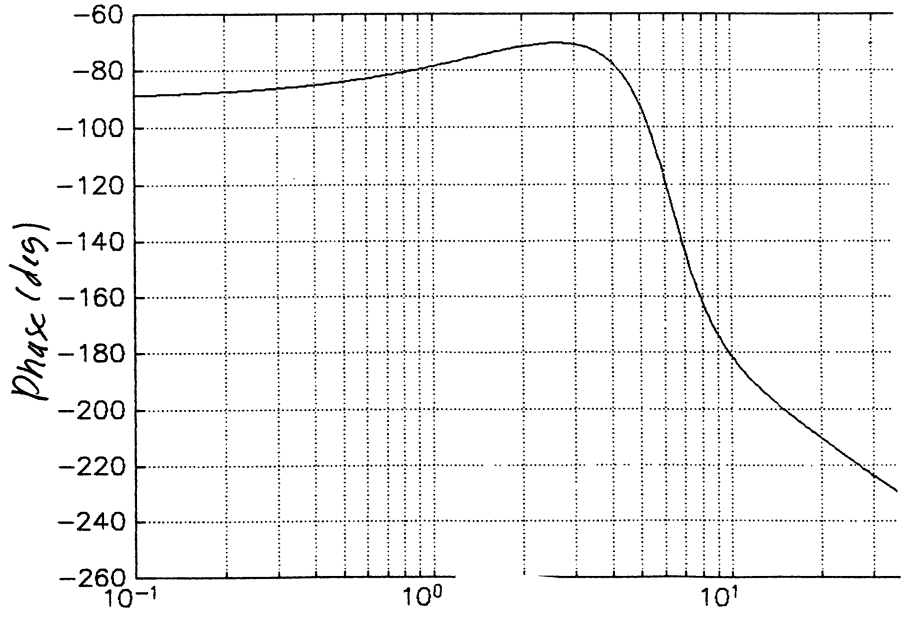
$\frac{1}{A}$	K	ω_ϕ	PM	K_1	
18	18	2.62	80.6°	2.00	
20	20	3.07	76.8°	2.22	
23	23	4.14	64.2°	2.55	← Same loss of PM (good trade off)
27	27	5.70	29.0°	3.00	

Increases ω_ϕ, K_1

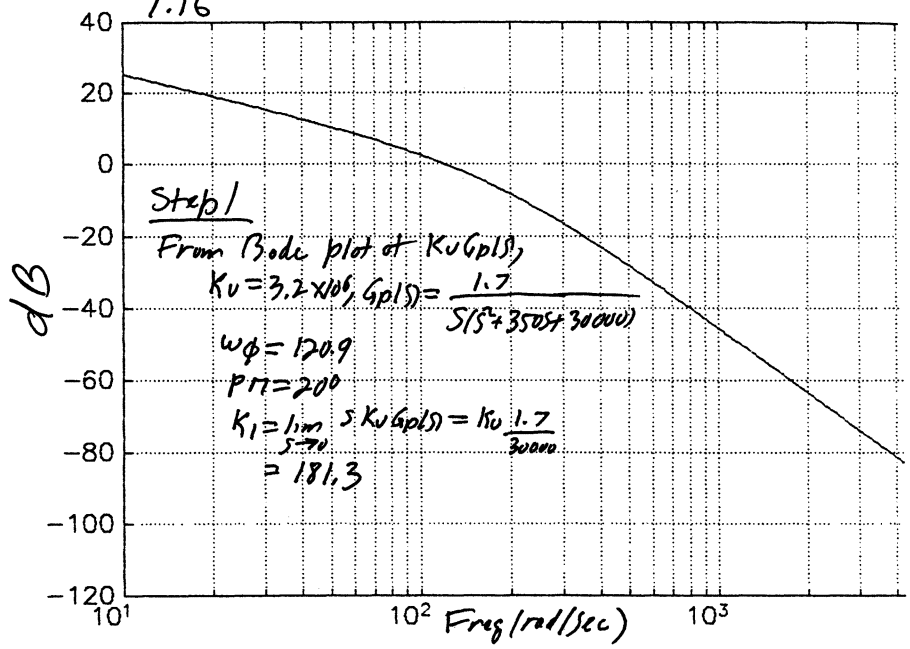
Bode Plot of $KA(s+\frac{1}{A})G_p(s) = (s+23)\frac{5}{s(s^2+4s+40)}$




7.15(d)
Step 1 (See Prob. 7.15(a))
Step 2 d=10
Step 3 a=ωφ=2.85
b=28.8
Step 4 K=Kvd=194.6
Step 5 From Bode plot of Gc'(s)Gp(s)
|Gc'(jω01)Gp(jω01)|=1/K=-45.77
ωφ=7.58
PM=24.60
K1=2.43
Step 6 Due to loss of PM with K=194.6
try K=42 dB=125.9
ωφ1=5.7
PM=70°
K1=1.57
raise ωφ maintains PM loses K1
dB
-10
-20
-30
-40
-50
-60
-70
-80
-90
-100
-110
10^-1
10^0
10^1


Phase (deg)
-60
-80
-100
-120
-140
-160
-180
-200
-220
-240
-260
10^-1
10^0
10^1

7.16



Step 1

From Bode plot of $K_v G_p(s)$,

$K_v = 3.2 \times 10^6$, $G_p(s) = \dfrac{1.7}{s(s^2+350s+30000)}$

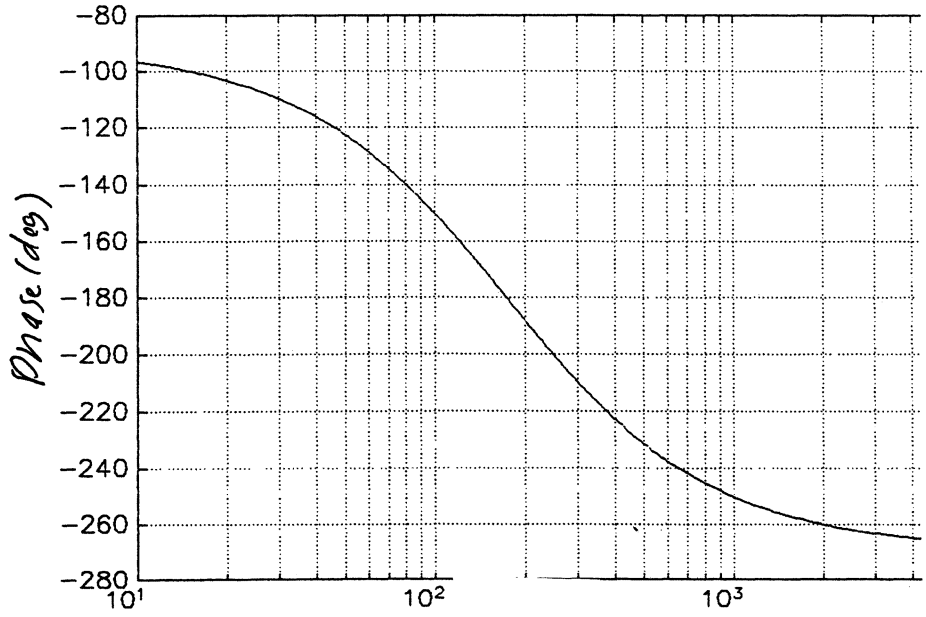
$\omega_\phi = 120.9$

$PM = 20°$

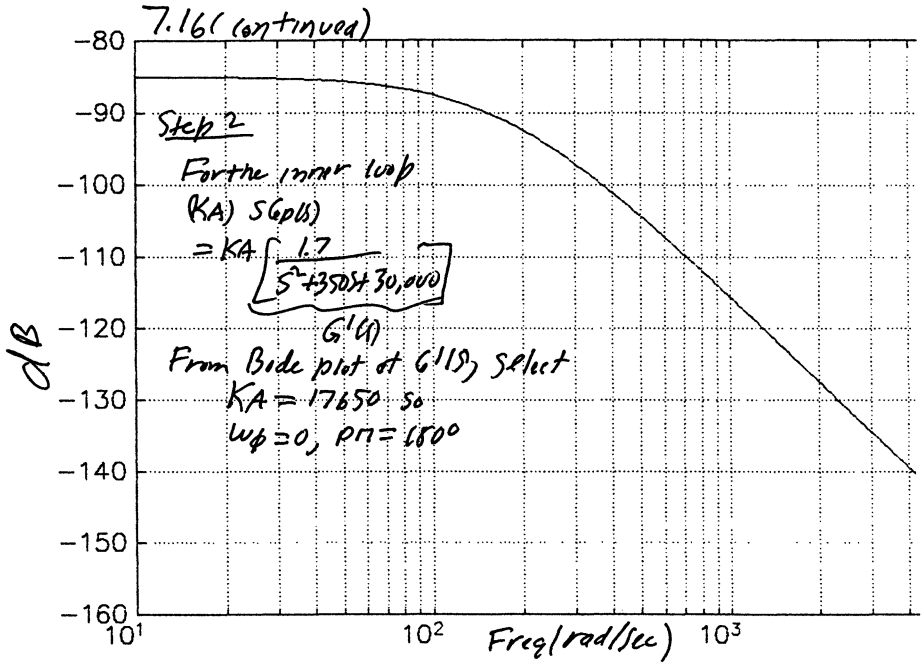
$K_1 = \lim_{s \to 0} s K_v G_p(s) = K_v \dfrac{1.7}{30000}$

$= 181.3$

Bode Plot of $K_v G_p(s)$



7.16 (continued)



Step 2

For the inner loop

$(K_A)\ SG_p(s)$

$= K_A \left[\frac{1.7}{s^2+350s+30{,}000} \right]$

$G'(s)$

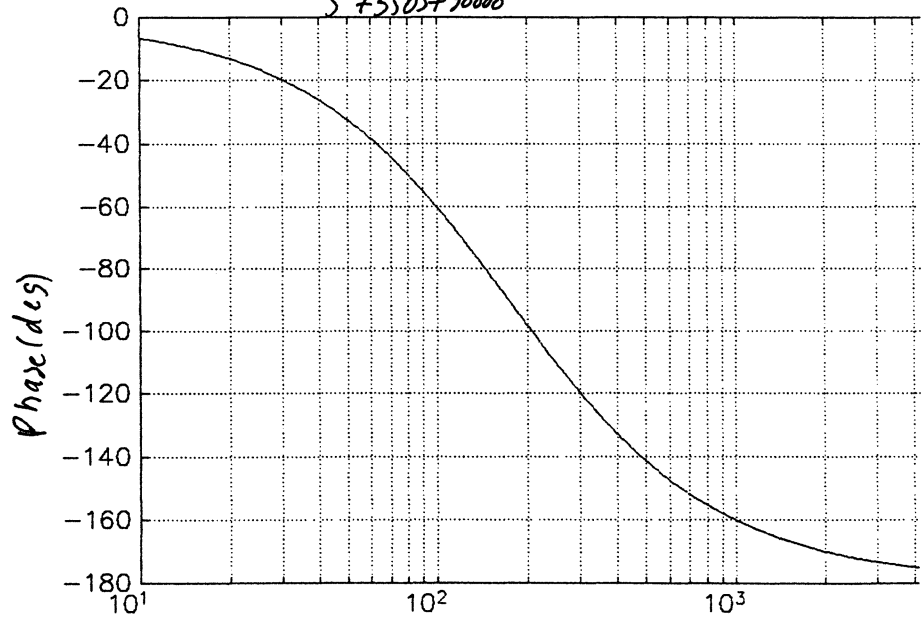
From Bode plot of $G'(s)$ select

$K_A = 17650$ so

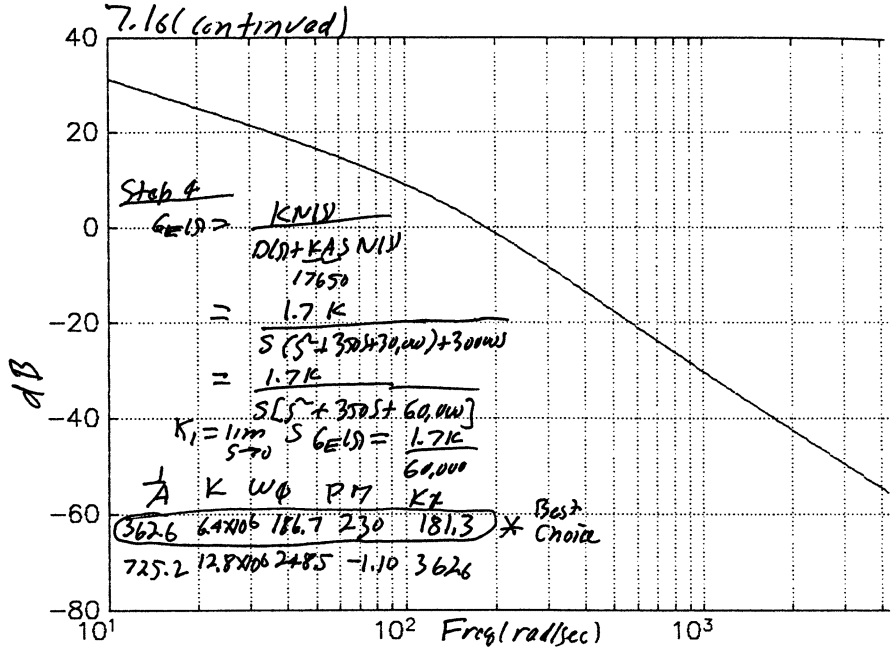
$\omega_\phi = 0$, $pm = 180^\circ$

Bode Plot of $SG_p(s)$

$= \frac{1.7}{s^2+350s+30000}$



7.16 (continued)



Stab ϕ

$G_E(s) = \dfrac{KN(s)}{D(s)+KA_1 N(s)}$

$= \dfrac{1.7K}{S(S^2+350S+30{,}000)+300{,}000}$

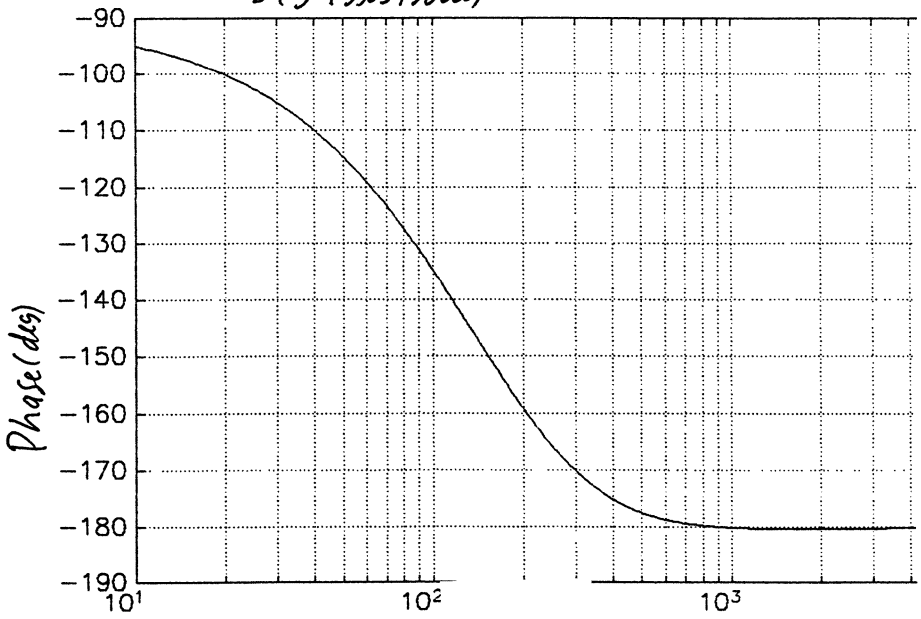
$= \dfrac{1.7K}{S[S^2+350S+60{,}000]}$

$K_1 = \lim_{S\to 0} S\,G_E(s) = \dfrac{1.7K}{60{,}000}$

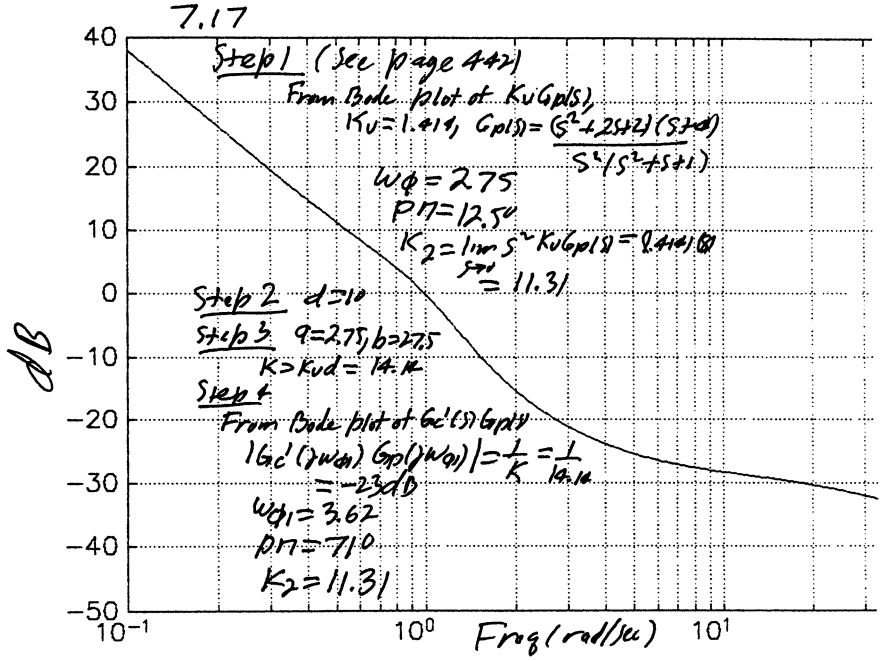
$\frac{1}{A}$	K	ω_ϕ	PM	K_1	
362.6	6.4×10^6	186.7	230	181.3	✱ Best Choice
725.2	12.8×10^6	248.5	−1.10	3626	

Bode Plot of $KA(S+\frac{1}{A})G_p(s)$

$= \dfrac{30000(S+362.6)}{S(S^2+350S+30000)}$



7.17



Step 1 (See page 442)

From Bode plot of $K_u G_{p(s)}$,

$K_u = 1.414$, $G_{p(s)} = \frac{(s^2+2s+2)(s+4)}{s^2(s^2+s+1)}$

$\omega_\phi = 2.75$

$PM = 12.5^\circ$

$K_2 = \lim_{s\to 1} s^2 K_u G_{p(s)} = (1.414)(8)$

$= 11.31$

Step 2 $d = 10$

Step 3 $a = 2.75$, $b = 27.5$

$K > K_{ud} = 14.14$

Step 4

From Bode plot of $G_c'(s) G_{p(s)}$

$|G_c'(j\omega_{\phi 1}) G_{p(j\omega_{\phi 1})}| = \frac{1}{K} = \frac{1}{14.14}$

$= -23\,dB$

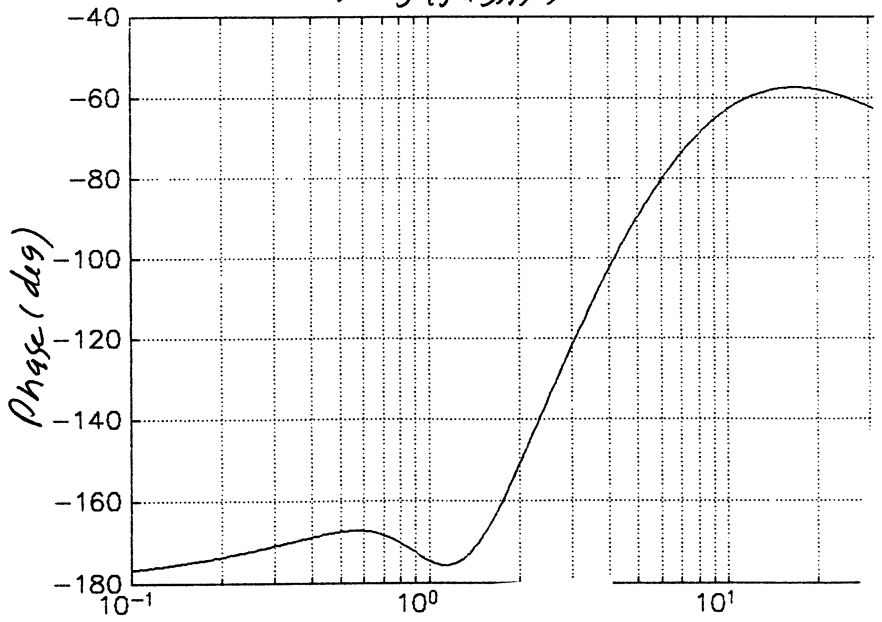
$\omega_{\phi 1} = 3.62$

$PM = 71^\circ$

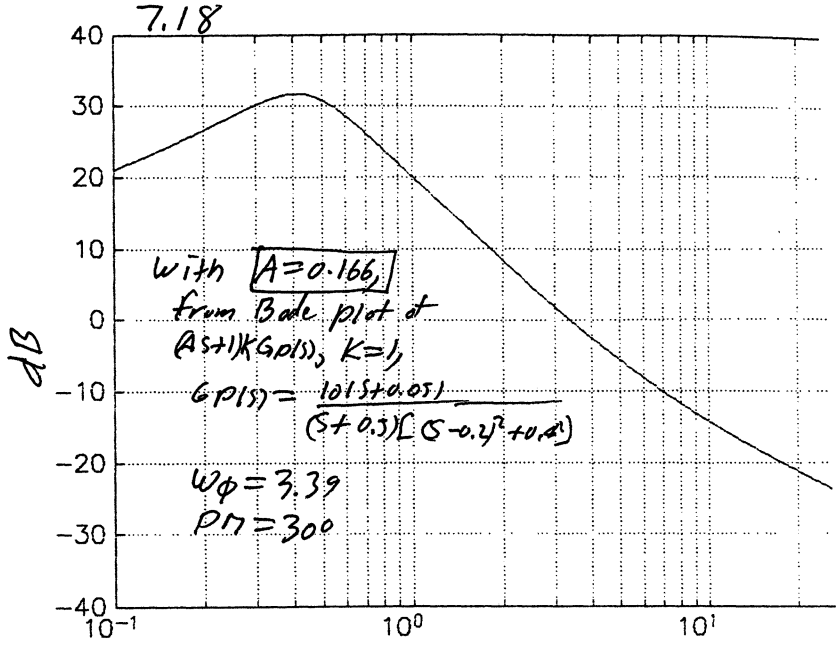
$K_2 = 11.31$

Bode Plot of $G_c'(s) G_{p(s)}$

$= \left(\frac{s+2.75}{s+27.5}\right)\left(\frac{(s^2+2s+2)(s+4)}{s^2(s^2+s+1)}\right)$

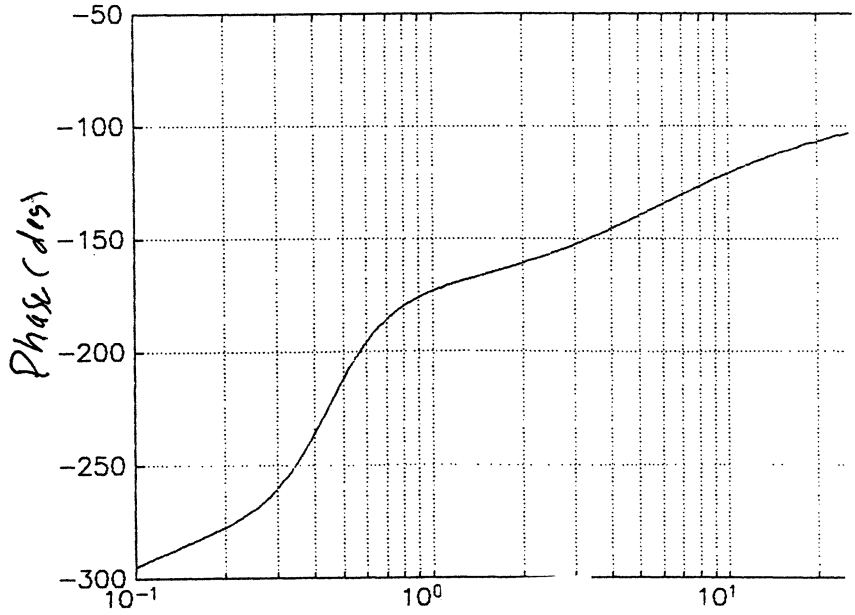


7.18



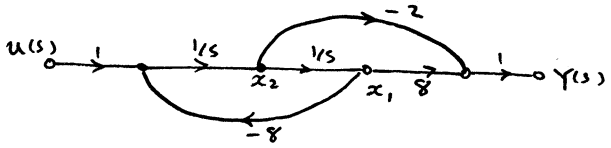
Bode Plot of $(As+1)\frac{10(s+0.05)}{(s+0.5)[(s-0.2)^2+0.4^2]}$,

$A=0.166$



Chapter 8

8.1 a) $T(s) = \dfrac{-2s+8}{s^2+8} = \dfrac{-\frac{2}{s}+\frac{8}{s^2}}{1-\left(-\frac{8}{s^2}\right)}$

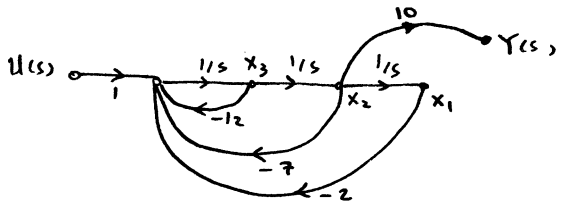


$\dot{x}_1 = x_2$

$\dot{x}_2 = -8x_1 + u \quad \Rightarrow \quad \dot{x} = \begin{pmatrix} 0 & 1 \\ -8 & 0 \end{pmatrix} x + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u$

$y = 8x_1 - 2x_2 \qquad y = \begin{pmatrix} 8 & -2 \end{pmatrix} x$

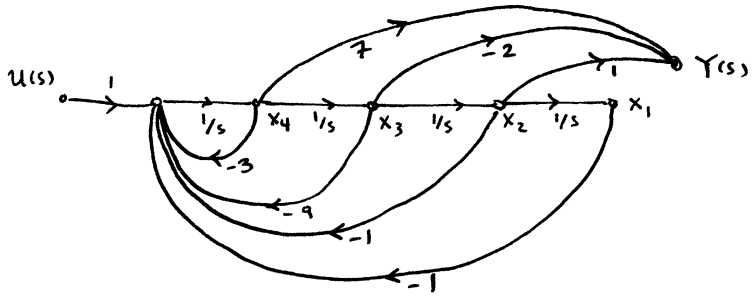
b) $T(s) = \dfrac{10s}{s^3+12s^2+7s+2} = \dfrac{10/s^2}{1-\left(\frac{-12}{s}-\frac{7}{s^2}-\frac{2}{s^3}\right)}$



$$\dot{x} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -7 & -12 \end{pmatrix} x + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} u$$

$$y = \begin{pmatrix} 0 & 10 & 0 \end{pmatrix} x$$

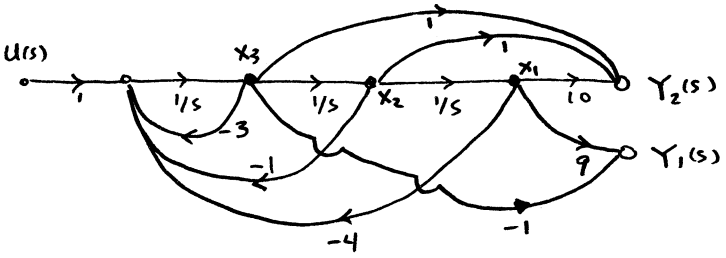
c) $T(s) = \dfrac{7s^3-2s^2+s}{s^4+3s^3+9s^2+s+1} = \dfrac{\frac{7}{s}-\frac{2}{s^2}+\frac{1}{s^3}}{1-\left(-\frac{3}{s}-\frac{9}{s^2}-\frac{1}{s^3}-\frac{1}{s^4}\right)}$



$$\dot{x} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & -1 & -9 & -3 \end{pmatrix} x + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} u$$

$$y = \begin{pmatrix} 0 & 1 & -2 & 7 \end{pmatrix} x$$

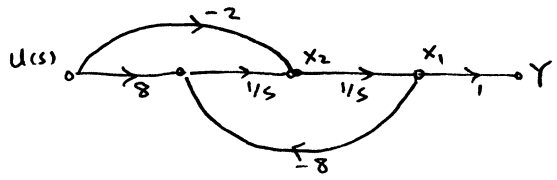
d) $$T(s) = \frac{1}{s^3 + 3s^2 + s + 4} \begin{bmatrix} -s^2 + 9 \\ s^2 + s + 10 \end{bmatrix}$$



$$\dot{x} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -4 & -1 & -3 \end{pmatrix} x + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} u$$

$$Y = \begin{pmatrix} 9 & 0 & -1 \\ 10 & 1 & 1 \end{pmatrix} x$$

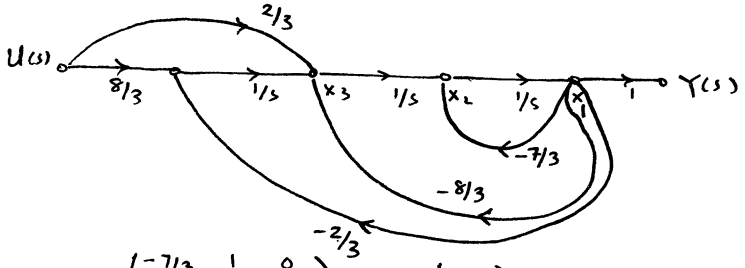
8.2 a) $$T(s) = \frac{-2s + 8}{s^2 + 8}$$



$$\dot{x} = \begin{pmatrix} 0 & 1 \\ -8 & 0 \end{pmatrix} x + \begin{pmatrix} -2 \\ 8 \end{pmatrix} u$$

$$Y = \begin{pmatrix} 1 & 0 \end{pmatrix} x$$

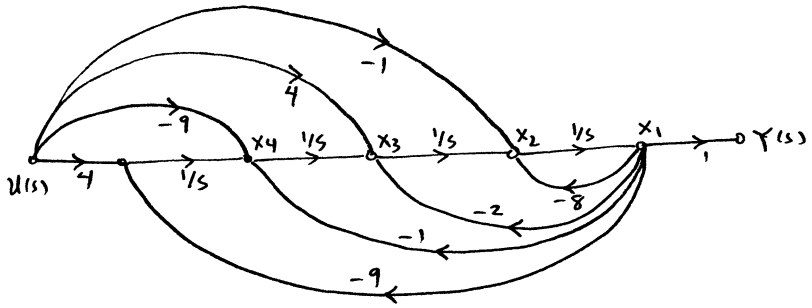
b) $T(s) = \dfrac{2s+8}{3s^3+7s^2+8s+2} = \dfrac{\frac{2}{3s^2}+\frac{8}{3s^3}}{1-\left(-\frac{7}{3s}-\frac{8}{3s^2}-\frac{2}{3s^3}\right)}$



$$\dot{x} = \begin{pmatrix} -7/3 & 1 & 0 \\ -8/3 & 0 & 1 \\ -2/3 & 0 & 0 \end{pmatrix} x + \begin{pmatrix} 0 \\ 2/3 \\ 8/3 \end{pmatrix} u$$

$$Y = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} x$$

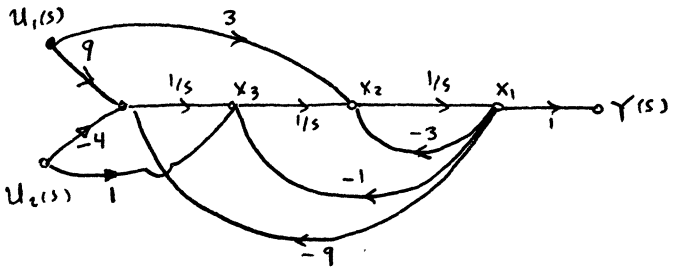
c) $T(s) = \dfrac{-s^3+4s^2-9s+4}{s^4+8s^3+2s^2+s+9} = \dfrac{-\frac{1}{s}+\frac{4}{s^2}-\frac{9}{s^3}+\frac{4}{s^4}}{1-\left(-\frac{8}{s}-\frac{2}{s^2}-\frac{1}{s^3}-\frac{9}{s^4}\right)}$



$$\dot{x} = \begin{bmatrix} -8 & 1 & 0 & 0 \\ -2 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ -9 & 0 & 0 & 0 \end{bmatrix} x + \begin{bmatrix} -1 \\ 4 \\ -9 \\ 4 \end{bmatrix} u$$

$$Y = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} x .$$

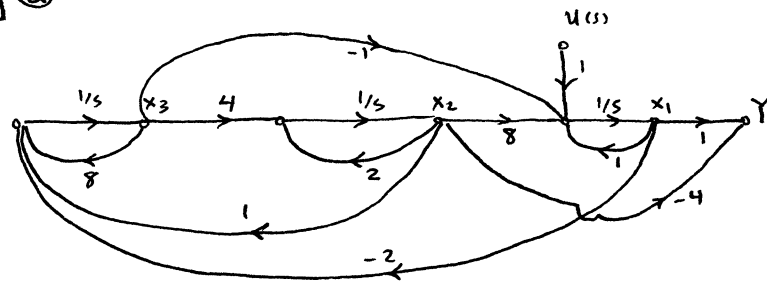
d) $T(s) = \dfrac{1}{s^3+3s^2+s+9}\begin{bmatrix} 3s^2+9 & s-4 \end{bmatrix}$



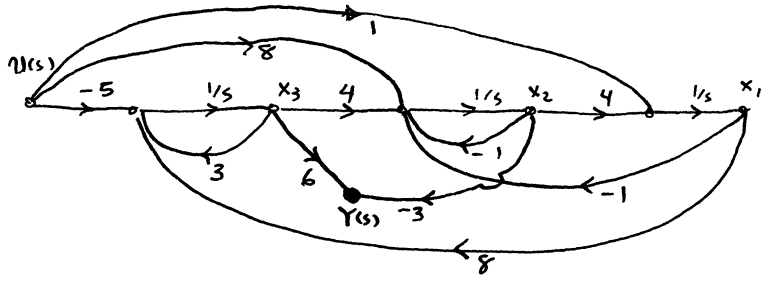
$$\dot{x} = \begin{bmatrix} -3 & 1 & 0 \\ -1 & 0 & 1 \\ -9 & 0 & 0 \end{bmatrix} x + \begin{bmatrix} 3 & 0 \\ 0 & 1 \\ 9 & -4 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} x$$

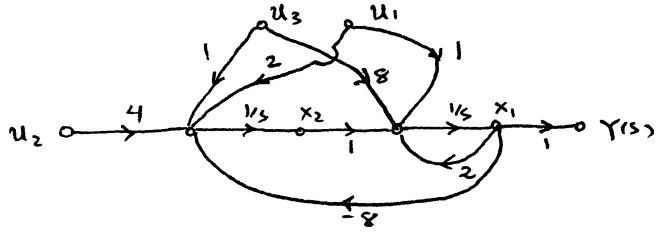
8.3 ⓐ



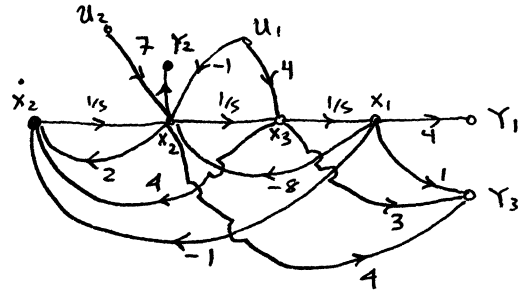
b)



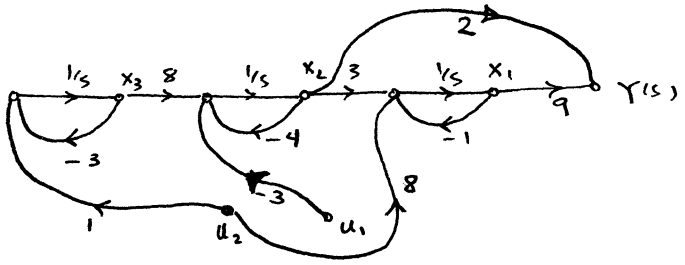
c)



d)



8.4



8.5

a) $T(s) = C(sI-A)^{-1}B+D = \frac{1}{s^2+2s+8}\begin{bmatrix} 6s+20 \\ -16s-96 \end{bmatrix}$

b) $T(s) = \begin{bmatrix} 16s-4 & -29s+3 \end{bmatrix}\frac{1}{s^2-8s+3}$

c) $T(s) = \frac{1}{s^3+4s^2-30s+32}\begin{bmatrix} 4s^2+19s-80 & 2s^2+4s-168 \\ -24s^2+112s-128 & 68s^2-144s+8 \end{bmatrix}$

8.6

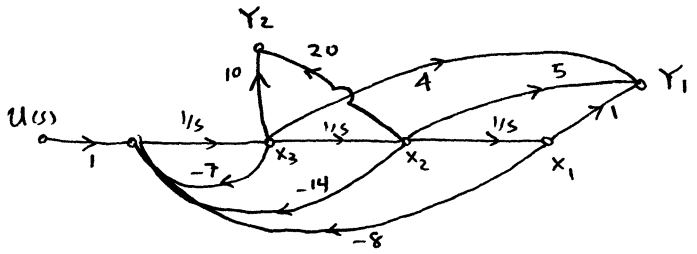
a) Ch. eq. $= |sI-A| = s^2-7s+1$ → unstable

b) " $= s^3+11s^2+25s$ a pole at $s=0$, marg. stable.

c) " $= s^3+5s^2-2s-16$ unstable.

8.7 $T(s) = \dfrac{1}{(s+1)(s+2)(s+4)} \begin{bmatrix} (4s+1)(s+1) \\ 10s\,(s+2) \end{bmatrix}$

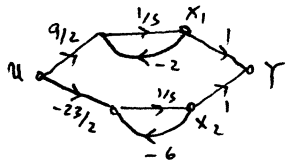
$$T(s) = \frac{1}{1-\left(-\frac{7}{s}-\frac{14}{s^2}-\frac{8}{s^3}\right)} \begin{bmatrix} \frac{4}{s}+\frac{5}{s^2}+\frac{1}{s^3} \\ \frac{10}{s}+\frac{20}{s^2} \end{bmatrix}$$



$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -8 & -14 & -7 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 5 & 4 \\ 0 & 20 & 10 \end{bmatrix} x$$

8.8 a) $T(s) = \dfrac{9/2}{s+2} + \dfrac{-23/2}{s+6}$



$$\dot{x} = \begin{bmatrix} -2 & 0 \\ 0 & -6 \end{bmatrix} + \begin{bmatrix} 9 \\ -23 \end{bmatrix} \frac{u}{2}$$

$$y = [1 \quad 1]\, x$$

b) $T(s) = \dfrac{-5/18}{s+2} + \dfrac{97/18}{s+8} + \dfrac{-28/9}{s+3}$

$$\dot{x} = \begin{pmatrix} -2 & 0 & 0 \\ 0 & -8 & 0 \\ 0 & 0 & -3 \end{pmatrix} x + \begin{pmatrix} -5 \\ 97 \\ -56 \end{pmatrix} \frac{1}{18}\, u$$

$$Y = (1 \quad 1 \quad 1)\, x$$

c) $T(s) = \frac{10/15}{s} + \frac{-10/6}{s+3} + \frac{1}{s+5} \Rightarrow \dot{x} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -5 \end{pmatrix} x + \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} u$

$y = \begin{pmatrix} \frac{10}{15} & -\frac{10}{6} & 1 \end{pmatrix} x$

8.9 a) $T(s) = \frac{45}{(s+1+j2.45)(s+1-j2.45)} = \frac{2-0.81j}{s+1+j2.45} + \frac{2+0.81j}{s+1-j2.45}$

$\dot{x} = \begin{pmatrix} -1+j2.45 & 0 \\ 0 & -1-j2.45 \end{pmatrix} x + \begin{pmatrix} 1 \\ 1 \end{pmatrix} u$

$y = (2-0.81j \quad 2+0.81j)\, x$

and $\dot{x} = \begin{pmatrix} 0 & 1 \\ -7 & -2 \end{pmatrix} x + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u$

$y = (0 \quad 4)\, x$

b) $T(s) = \frac{0.23s - 2.53}{(s+3)^2+1} + \frac{1.23}{s+8}$ (using Program CC)

$\dot{x} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -80 & -58 & -14 \end{pmatrix} x + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} u$

$y = (-8 \quad 3 \quad 1)\, x$

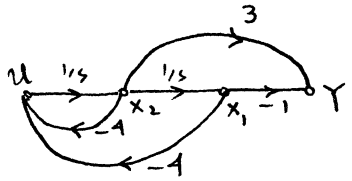
c) $T(s) = \frac{0.23}{s+2} - \frac{0.23s}{(s+1)^2+16}$

$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -34 & -21 & -4 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$, $y = (4 \quad 0 \quad 0)\, x$

8.10 a) $T = \frac{3s-1}{(s+2)^2}$

$\dot{x} = \begin{pmatrix} 0 & 1 \\ -4 & -4 \end{pmatrix} x + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u$

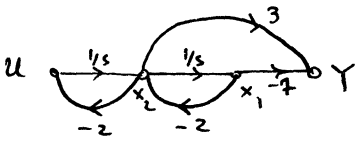
$y = (-1 \quad 3)\, x$



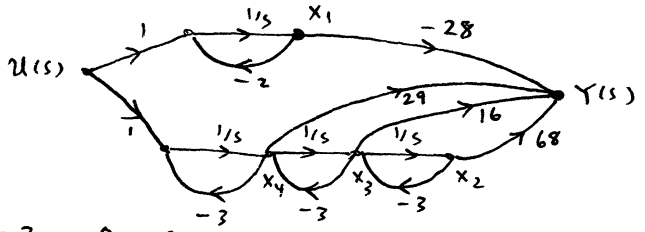
The previous diagram gives a phase variable realization. For Jordan form, we need another diagram. $T = \frac{3}{s+2} - \frac{7}{(s+2)^2}$

$$\dot{x} = \begin{bmatrix} -2 & 1 \\ 0 & -2 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = [-7 \quad 3]\, x$$



b) $T(s) = \frac{-28}{s+2} + \frac{68}{(s+3)^3} + \frac{16}{(s+3)^2} + \frac{29}{s+3}$

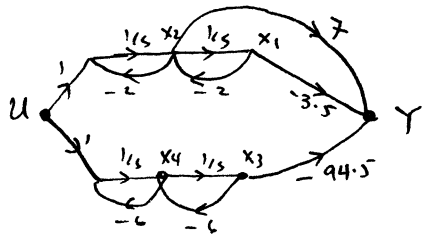


$$\dot{x} = \begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & -3 & 1 & 0 \\ 0 & 0 & -3 & 1 \\ 0 & 0 & 0 & -3 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$Y = [-28 \quad 68 \quad 16 \quad 29]\, x$$

c) $T(s) = \frac{-3.5}{(s+2)^2} + \frac{7}{s+2} + \frac{0}{s+6} - \frac{94.5}{(s+6)^2}$

$$\dot{x} = \begin{bmatrix} -2 & 1 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -6 & 1 \\ 0 & 0 & 0 & -6 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} u$$



$$Y = [-3.5 \quad 7 \quad -94.5 \quad 0]\, x$$

8.11 ⓐ $\lambda(A) = -4, -5$, $P = \begin{bmatrix} 1 & 1 \\ 5 & 4 \end{bmatrix}$

$\dot{z} = (P^{-1}AP)z + P^{-1}Bu$
$y = CPz$
$\Rightarrow \begin{cases} \dot{z} = \begin{pmatrix} -4 & 0 \\ 0 & -5 \end{pmatrix} z + \begin{pmatrix} 1 \\ 0 \end{pmatrix} u \\ y = (-10 \quad -13)\, z \end{cases}$

b) $\lambda(A) = -2, -3$, $P = \begin{bmatrix} 1 & 1 \\ -2 & -3 \end{bmatrix}$

$\dot{z} = \begin{pmatrix} -2 & 0 \\ 0 & -3 \end{pmatrix} z + \begin{pmatrix} 3 & 1 \\ -2 & -1 \end{pmatrix} u$

$y = (-2 \quad -3)\, z$

c) $\lambda(A) = -1, -4$, $P = \begin{bmatrix} 1 & 1 \\ 1 & 4 \end{bmatrix}$

$\dot{z} = \begin{pmatrix} -4 & 0 \\ 0 & -1 \end{pmatrix} z + \begin{pmatrix} 1 & 1/3 \\ 0 & -1/3 \end{pmatrix} u$

$y = \begin{pmatrix} 2 & 5 \\ -1 & -1 \end{pmatrix} z$

d) $\lambda(A) = 0, -1, -2$ $P = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 2 & 1 \\ 2 & 0 & 0 \end{bmatrix}$

$\dot{z} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{pmatrix} z + \begin{pmatrix} 0.5 \\ -1 \\ 1.5 \end{pmatrix} u$

$y = \begin{pmatrix} 1 & 3 & 2 \\ 5 & 2 & 1 \end{pmatrix} z$

8.12 $\lambda(A) = 0, -1 \pm j4$ $P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1+4j & -1-4j \\ 0 & -15-8j & -15+8j \end{bmatrix}$

$\dot{x} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1-j4 & 0 \\ 0 & 0 & -1+4j \end{pmatrix} x + \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} u$

$y = \left(\frac{32}{17} \quad \frac{-15-j8}{34} \quad \frac{-15+j8}{34} \right) x$

Alternatively $T(s) = \frac{32/17}{s} - \frac{15/17\, s - 14.25}{s^2 + 2s + 17}$

$\dot{x} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -17 & -2 \end{pmatrix} x + \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} u$, $y = \left(\frac{32}{17} \quad -14.25 \quad \frac{-15}{17} \right) x$

8.13 $\lambda(A) = -1, -3, -3$, $P = \begin{bmatrix} 1 & 1 & 0 \\ -1 & -3 & 1 \\ 1 & 9 & -6 \end{bmatrix}$

$$\dot{z} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -3 & 1 \\ 0 & 0 & -3 \end{bmatrix} z + \begin{bmatrix} 15 \\ -15 \\ -22 \end{bmatrix} \frac{u}{4}$$

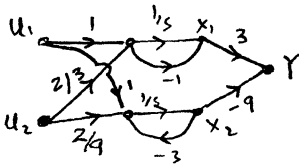
$$y = \begin{bmatrix} 0 & -2 & 1 \\ 0 & 6 & -5 \end{bmatrix} z$$

8.14 (a)

$$T(s) = \frac{1}{1-\left(\frac{-4}{s} - \frac{3}{s^2}\right)} \begin{bmatrix} -\frac{6}{s} & \frac{4}{s^2} \end{bmatrix}$$

$$= \frac{1}{(s+1)(s+3)} \begin{bmatrix} -6s & 4 \end{bmatrix} =$$

$$= \left[\frac{3}{s+1} + \frac{-9}{s+3} \;,\; \frac{2}{s+1} - \frac{2}{s+3} \right]$$



$$\dot{x} = \begin{pmatrix} -1 & 0 \\ 0 & -3 \end{pmatrix} x + \begin{pmatrix} 1 & 2/3 \\ 1 & \frac{2}{9} \end{pmatrix} u$$

$$y = (3 \quad -9)\, x$$

(b)

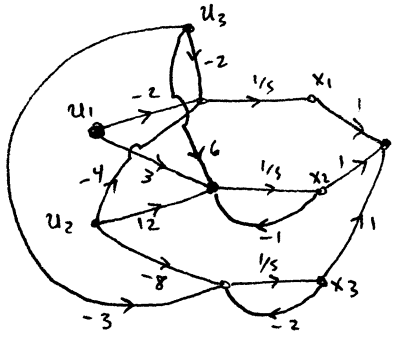
$$T(s) = \left[\frac{(s-2)(s+2)}{s(s+1)(s+2)} \quad \frac{4(s-2)}{s(s+1)(s+2)} \quad \frac{(s+4)(s-1)}{s(s+1)(s+2)} \right]$$

$$= \left[\frac{-2}{s} + \frac{3}{s+1} \quad \frac{-4}{s} + \frac{12}{s+1} + \frac{-8}{s+2} \quad \frac{-2}{s} + \frac{6}{s+1} + \frac{-3}{s+2} \right]$$

$$\dot{x} = \begin{bmatrix} 0 & 0 \\ -1 & 0 \\ 0 & -2 \end{bmatrix} x +$$

$$+ \begin{bmatrix} -2 & -4 & -2 \\ 3 & 12 & 6 \\ 0 & -8 & -3 \end{bmatrix} u$$

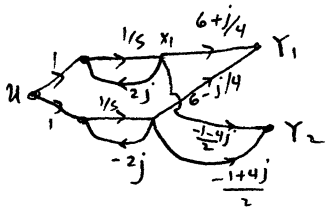
$$y = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} x$$



c) $T(s) = \frac{1}{s^2+4}\begin{bmatrix} 3s-1 \\ -s+8 \end{bmatrix}$

$$\dot{x} = \begin{bmatrix} 2j & 0 \\ 0 & -2j \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$

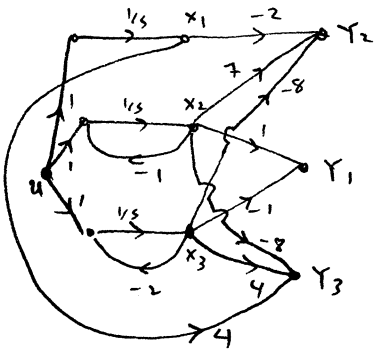
$$y = \begin{bmatrix} \frac{6+j}{4} & \frac{6-j}{4} \\ \frac{-1-4j}{2} & \frac{-1+4j}{2} \end{bmatrix} x$$



d) $T = \frac{1}{s(s+1)(s+2)}\begin{bmatrix} s \\ -3s^2-4 \\ 8 \end{bmatrix} = \begin{bmatrix} \frac{1}{s+1} + \frac{-1}{s+2} \\ \frac{-2}{s} + \frac{7}{s+1} + \frac{-8}{s+2} \\ \frac{4}{s} + \frac{-8}{s+1} + \frac{4}{s+2} \end{bmatrix}$

$$\dot{x} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u$$

$$Y = \begin{bmatrix} 0 & 1 & -1 \\ -2 & 7 & -8 \\ 4 & -8 & 4 \end{bmatrix} x$$



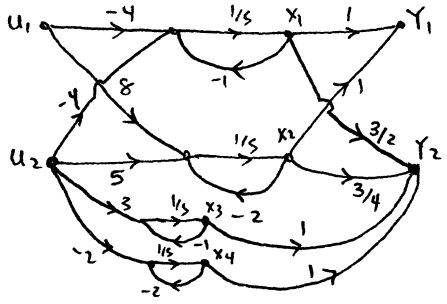
8.15

$$T(s) = \frac{1}{(s+1)(s+2)}\begin{bmatrix} 4s & s-3 \\ -6 & s+4 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{-4}{s+1} + \frac{8}{s+2} & \frac{-4}{s+1} + \frac{5}{s+2} \\ \frac{-6}{s+1} + \frac{6}{s+2} & \frac{3}{s+1} + \frac{-2}{s+2} \end{bmatrix}$$

$$\dot{x} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix} x + \begin{bmatrix} -4 & -4 \\ 8 & 5 \\ 0 & 3 \\ 0 & -2 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 3/2 & 3/4 & 1 & 1 \end{bmatrix} x$$



8.16 $T(s) = 3 + \dfrac{-7s-10}{s^2+3s+2}$

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \;, \quad y = \begin{bmatrix} -10 & -7 \end{bmatrix} x + 3u$$

8.17 a) $M_c = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$, $M_o = \begin{bmatrix} 1 & 1 \\ 2 & -3 \end{bmatrix}$

$|M_c| = 0$, $|M_o| \neq 0$ $\therefore$ observable, uncontrollable.

$T(s) = c(sI-A)^{-1}B + D = \dfrac{(s-1)}{(s-1)(s-2)} = \dfrac{1}{s-2}$

$\lambda(A) = 1, 2 \Rightarrow$ not asymp. stable, poles $= 2 \Rightarrow$ not BIBO stable.

b) $M_c = \begin{bmatrix} 1 & 0 & 3 & 5 & 9 & 5 \\ 2 & 0 & 0 & -5 & -6 & -5 \\ 0 & -1 & 0 & 2 & 0 & -4 \end{bmatrix}$ rank $= 3$ $\therefore$ Contr.

$M_o = \begin{bmatrix} 4 & 1 & -3 \\ 3 & 2 & -1 \\ 10 & 1 & -9 \\ 5 & 2 & -3 \\ 28 & 1 & -27 \\ 11 & 2 & -9 \end{bmatrix}$ rank $= 2$ $\therefore$ unobservable.

$\lambda(A) = 1, 3, -2$ $\therefore$ not asy. stable., not BIBO stable.

c) $M_c = \begin{bmatrix} 1 & -1 & 1 & -1 & 1 & -1 \\ 2 & 0 & -3 & -3 & 12 & 6 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$, rank $= 2$ $\therefore$ uncont.

$M_o = \begin{bmatrix} 0 & 1 & 1 \\ 0 & -2 & 3 \\ 12 & -12 & 1 \\ -6 & 6 & 3 \\ -24 & 36 & -23 \\ 12 & -18 & 15 \end{bmatrix}$, rank $= 3$ $\therefore$ observable.

$\lambda(A) = -3, 1, 1 \Rightarrow$ not asymp. stable.

8.18 a) $A = \begin{bmatrix} +2 & & \\ & -3 & \\ & & -4+j \\ & & & -4-j \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ - \\ - \\ - \end{bmatrix}$ → any non zero number

b) same A, $C = [-\ 0\ -\ -]$

c) same A, $B' = [0\ -\ -\ -]$, $C = [0\ -\ -\ -]$

d) same A, $B' = [-\ -\ 0\ 0]$

e) same A, $B' = [-\ 0\ -\ -]$, $C = [0\ -\ -\ -]$.

8.19 $T(s) = \dfrac{(s+2)}{(s-1)(s+2)} = \dfrac{1}{s-1}$

Since, pole is at $s=1 \Rightarrow$ sys. is not BIBO stable and it can be detected from input-output measurements.

$M_o = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, rank $=1$ ∴ not observable.

Note: Instability is detected because it is the stable mode that is cancelled out (we say the system is <u>detectable</u>).

Let $C = [-2\ \ 1]$, $T(s) = \dfrac{(s-1)}{(s-1)(s+2)} = \dfrac{1}{s+2}$

$\lambda(A) = 1, -2$, pole $= -2$ ∴ BIBO stable, but not asymp. stable.

$M_o = \begin{bmatrix} -2 & 1 \\ 4 & -2 \end{bmatrix}$, rank $=1 \Rightarrow$ unobservable.

In this case, the unstable mode cancels out and instability can not be detected from I/O measurements. The system is <u>not detectable</u>.

8.20 $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ a_1 & a_2 & a_3 \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$$M_c = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & a_3 \\ 1 & a_3 & \alpha \end{bmatrix} \quad \text{where } \alpha = a_2 + a_3^2$$

$\det M_c = -1 \neq 0$, therefore it's impossible. It can be shown in general, that the phase-variable form is always completely controllable.

8.21 We have $y = Cx$, if C is nonsingular, C^{-1} exists, and $x = C^{-1}y$. In particular, $x(0) = C^{-1}y(0)$. Because we can always find x_0 from $y(0)$, the sys. is observable. Another way to see this: If C is nonsingular $\Rightarrow C = n \times n$ and rank $C = n$, since $M_o = \begin{bmatrix} C \\ CA \\ \vdots \end{bmatrix} \Rightarrow$ rank $M_o \geq n \Rightarrow$ obs.

8.22 $$x(t) = 7e^{-2t} + \int_0^t e^{-2(t-\tau)} \cdot 3 \cdot e^{3\tau} d\tau$$

$$\int_0^t e^{-2t} \cdot e^{2\tau} e^{3\tau} d\tau = e^{-2t} \int_0^t e^{5\tau} d\tau = e^{-2t} \frac{e^{5\tau}}{5} \Big|_0^t = \frac{e^{-2t}}{5}\left[e^{5t} - 1\right]$$

$$\therefore x(t) = 7e^{-2t} + \frac{3}{5}\left[e^{3t} - e^{-2t}\right] = \underline{\frac{32}{5} e^{-2t} + \frac{3}{5} e^{3t}}$$

8.23 a)

$$sX(s) - x(0^-) = -3X(s) + 2\frac{5}{s}$$

$$\Rightarrow (s+3)X(s) = 7 + \frac{10}{5} = \frac{7s+10}{s} \Rightarrow X(s) = \frac{7s+10}{s(s+3)}$$

$$X(s) = \frac{10/3}{s} + \frac{11/3}{s+3} \Rightarrow x(t) = \frac{1}{3}\left[10 + 11e^{-3t}\right] u(t)$$

$$y(t) = \underline{\frac{4}{3}\left[10 + 11e^{-3t}\right] u(t)}$$

b) $sX(s) - x(0) = AX(s) + BU(s)$, $U(s) = 1$

$$X(s) = (sI - A)^{-1}[x(0) + B] = \begin{pmatrix} s+3 & -1 \\ -7 & s \end{pmatrix}^{-1} \begin{pmatrix} 4 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} s & 1 \\ 7 & s+3 \end{pmatrix} \begin{pmatrix} 4 \\ 0 \end{pmatrix} \frac{1}{s(s+3)-7} = \begin{bmatrix} 4s \\ 28 \end{bmatrix} \frac{1}{s^2+3s-7}$$

$$x(t) = \begin{bmatrix} 1.01\, e^{1.5t} + 2.98\, e^{-4.54t} \\ 4.6\, e^{1.5t} - 4.6\, e^{-4.54t} \end{bmatrix}$$

$$y(t) = x_1(t) = 1.01\, e^{1.5t} + 2.98\, e^{-4.54t}.$$

c) $$X(s) = \begin{pmatrix} s & -1 \\ 3 & s+8 \end{pmatrix}^{-1} \begin{bmatrix} 7 + \frac{20}{s} \\ -3 + \frac{1}{s-1} - \frac{5}{s} \end{bmatrix} = \frac{\begin{pmatrix} s+8 & 1 \\ -3 & s \end{pmatrix} \begin{bmatrix} - \\ - \end{bmatrix}}{s^2+8s+3}$$

$$= \frac{1}{s^2+8s+3} \begin{bmatrix} s+8 & 1 \\ -3 & s \end{bmatrix} \begin{bmatrix} \frac{7s+20}{s} \\ \frac{-3s^2+3s+s-5s+5}{s(s-1)} \end{bmatrix}$$

$$= \frac{1}{(s^2+8s+3)s(s-1)} \begin{bmatrix} 4s^3+68s^2+89s-160 \\ -3s^3-22s^2-34s+60 \end{bmatrix}$$

$$\Rightarrow x(t) = \begin{bmatrix} 53.3 - 46.5\, e^{-0.39t} + 8.3\times10^{-2}\, e^{t} - 2.8\, e^{-7.6t} \\ -20 + 17.6\, e^{-0.39t} + 8.3\times10^{-2}\, e^{t} - 0.7\, e^{-7.6t} \end{bmatrix}$$

$$y(t) = 33.3 - 28.8\, e^{-0.39t} + 0.16 e^{t} - 3.6\, e^{-7.6t}$$

NOTE : Most algebra done by Program CC.

d) $sX_1 = -3X_1 + X_2 + 1$

$sX_2 = -2X_1 + X_3 + \frac{2}{s}$

$sX_3 = 0 \Rightarrow \dot{x}_3 = 0 \Rightarrow x_3(t) = x_3(0) = 0$

$$\Rightarrow X_1 = \frac{1}{s+3}\left[X_2 + 1\right], \quad X_2 = \frac{1}{s}\left[-2X_1 + \frac{2}{s}\right]$$

$$\Rightarrow X_1 = \frac{1}{s+3}\left[-\frac{2}{s}X_1 + \frac{2}{s^2} + 1\right] \Rightarrow X_1\left(1 + \frac{2}{s(s+3)}\right) = \frac{1}{s+3} \cdot \frac{2+s^2}{s^2}$$

$$X_1\left(\frac{(s+1)(s+2)}{s(s+3)}\right) = \frac{s^2+2}{s^2(s+3)} \Rightarrow X_1 = \frac{s^2+2}{s(s+1)(s+2)}$$

$$X_2 = \frac{1}{s}\left[\frac{2}{s} - \frac{2(s^2+2)}{s(s+1)(s+2)}\right] = \frac{1}{s}\left[\frac{2s^2+6s+4-2s^2-4}{s(s+1)(s+2)}\right]$$

$$X_2 = \frac{6}{s(s+1)(s+2)}$$

$$\therefore \begin{cases} x_1(t) = -3e^{-t} + 3e^{-2t} + 1 \\ x_2(t) = -6e^{-t} + 3e^{-2t} + 3 \\ x_3(t) = 0 \end{cases} \qquad \begin{cases} Y_1 = X_1 + X_3 = X_1 \\ Y_2 = X_3 = 0 \end{cases}$$

8.24 $x(t) = \phi(t)\,x(0) = \begin{bmatrix} \frac{e^{-t}}{2} + \frac{e^{-2t}}{2} & 2(e^{-t} - e^{-2t}) \\ 3e^{-t} - 3e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}\begin{bmatrix} -7 \\ 2 \end{bmatrix}$

$$= \begin{bmatrix} \frac{e^{-t}}{2} - \frac{15}{2}e^{-2t} \\ -23e^{-t} + 25e^{-2t} \end{bmatrix}.$$

8.25 a) $\Phi(s) = (sI - A)^{-1} = \begin{pmatrix} s & -1 \\ 6 & s+4 \end{pmatrix}^{-1} = \frac{\begin{pmatrix} s+4 & 1 \\ -6 & s \end{pmatrix}}{s^2+4s+6}$

$$\phi(t) = \begin{bmatrix} 1.7e^{-2t}\sin(1.4t+0.6) & 0.7e^{-2t}\sin(1.4t) \\ -4.2e^{-2t}\sin(1.4t) & -1.7e^{-2t}\sin(1.4t-0.6) \end{bmatrix}$$

b) $\Phi(s) = \begin{pmatrix} s+1 & -4 \\ -8 & s+2 \end{pmatrix}^{-1} = \begin{pmatrix} s+2 & 4 \\ 8 & s+1 \end{pmatrix}\frac{1}{s^2+3s-30}$

$$\phi(t) = \begin{bmatrix} 0.5e^{4.1t} + 0.4e^{-7.1t} & 0.3e^{4.1t} - 0.3e^{-7.1t} \\ 0.7e^{4.1t} - 0.7e^{-7.1t} & 0.4e^{4.1t} + 0.5e^{-7.1t} \end{bmatrix}.$$

c) $\phi(s) = e^{At}$. It might be easier to find $e^{At} = Pe^{\Lambda t}P^{-1}$

where $e^{\Lambda t} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & e^{-t} & 0 \\ 0 & 0 & e^{-4t} \end{bmatrix}$, $\lambda(A) = 0, -1, -4$

P = matrix of eigenvectors $= \begin{bmatrix} 5 & 1 & 1 \\ 25 & 4 & 1 \\ 4 & 0 & 0 \end{bmatrix}$, $P^{-1} = \begin{bmatrix} 0 & 0 & 1/4 \\ -1/3 & 1/3 & -10/6 \\ 4/3 & -1/3 & 5/12 \end{bmatrix}$

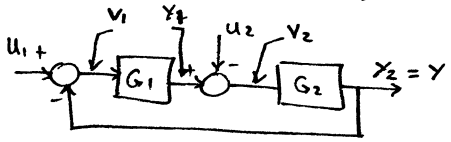
$$\Phi(t) = \begin{bmatrix} \frac{-e^{-t} + 4e^{-4t}}{3} & \frac{e^{-t} - e^{-4t}}{3} & \frac{5}{4} - \frac{5}{8}e^{-t} + \frac{5}{12}e^{-4t} \\ \frac{-4e^{-t} + 4e^{-4t}}{3} & \frac{4e^{-t} - e^{-4t}}{3} & \frac{25}{4} - \frac{20}{3}e^{-t} + \frac{5}{12}e^{-4t} \\ 0 & 0 & 1 \end{bmatrix}$$

8.26 diagonal system : typical state is $\dot{z}_i(t) = \lambda_i z_i(t)$

$$\Rightarrow z_i(t) = e^{\lambda_i t} z_i(0) \Rightarrow z(t) = \begin{pmatrix} e^{\lambda_1 t} & & 0 \\ & \ddots & \\ 0 & & e^{\lambda_n t} \end{pmatrix} z(0)$$

$z(t) = \Phi(t) z(0) \Rightarrow \Phi(t) =$ diagonal

8.27 a)



$$Y_2 = G_2 V_2 = \frac{10}{s^2+3s+10} \Rightarrow \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -10 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} V_2$$

$$Y_2 = \begin{bmatrix} 10 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$Y_1 = G_1 V_1 = \frac{2}{s+4}$$

$$\Rightarrow \dot{x}_3 = -4x_3 + V_1$$
$$Y_1 = 2x_3$$

Now $V_2 = Y_1 - u_2 = 2x_3 - u_2$

$$V_1 = u_1 - Y_2 = u_1 - 10x_1$$

Combine both systems (x_1, x_2, x_3), inputs (u_1, u_2), output (y)

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = -10x_1 - 3x_2 + V_2 = -10x_1 - 3x_2 + 2x_3 - u_2$$
$$\dot{x}_3 = -4x_3 + V_1 = -4x_3 + u_1 - 10x_1$$
$$y = y_2 = 10x_1$$

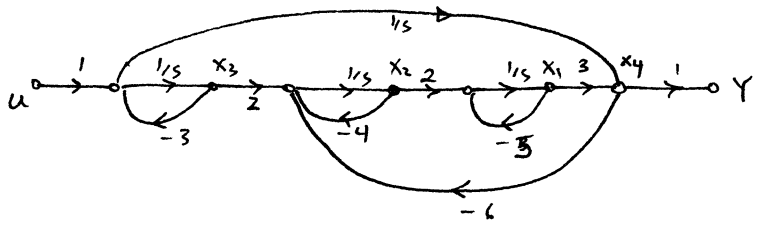
$$\Rightarrow \begin{cases} \dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ -10 & -3 & 2 \\ -10 & 0 & -4 \end{bmatrix} x + \begin{bmatrix} 0 & 0 \\ 0 & -1 \\ 1 & 0 \end{bmatrix} u \\ y = \begin{bmatrix} 10 & 0 & 0 \end{bmatrix} x \end{cases} \qquad u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$T(s) = C(sI-A)^{-1}B+D = \frac{1}{s^3+7s^2+22s+60}\begin{bmatrix} 20 & -10(s+4) \end{bmatrix}.$$

Using Mason's Rule : $T_{11} = \frac{Y}{u_1} = \frac{G_1G_2}{1+G_1G_2} = \frac{20}{s^3+7s^2+22s+60}$ ✓

$T_{12} = \frac{Y}{u_2} = \frac{-G_2}{1+G_1G_2} = \frac{-10(s+4)}{"}$ ✓ O.K.

b)



$Y = x_4 + 3x_1$

$\dot{x}_1 = -5x_1 + 2x_2$

$\dot{x}_2 = -4x_2 + 2x_3 - 6x_4 - 6(3x_1)$

$\dot{x}_3 = -3x_3 + u$

$\dot{x}_4 = u$

$$\Rightarrow \dot{x} = \begin{bmatrix} -5 & 2 & 0 & 0 \\ -18 & -4 & 2 & -6 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} u$$

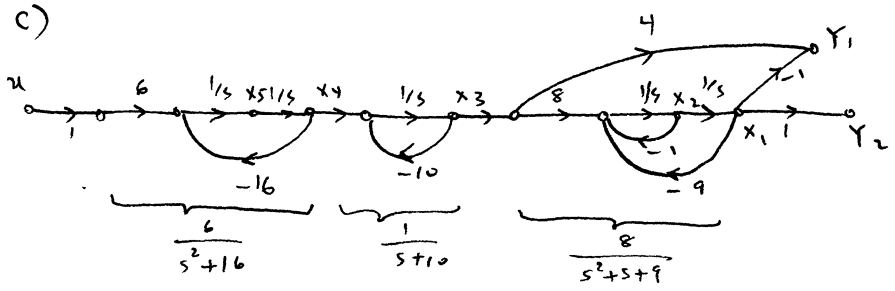
$$Y = \begin{bmatrix} 3 & 0 & 0 & 1 \end{bmatrix} x$$

$$T(s) = C\Phi(s)B = \frac{s^3+12s^2+59s+60}{s^4+12s^3+83s^2+168s}$$

Using Masons Rule : $T(s) = \frac{\frac{1}{s} + \left[\frac{2}{s+3}\cdot\frac{2}{s+4}\cdot\frac{3}{s+5}\right]}{1-\frac{6(-6)}{(s+4)(s+5)}} =$

They are equal !

c)



$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -9x_1 - x_2 + 8x_3$$

$$\dot{x}_3 = -10x_3 + x_4$$

$$\dot{x}_4 = x_5$$

$$\dot{x}_5 = -16x_4 + 6u$$

$$y_1 = -x_1 + 4x_3$$

$$y_2 = x_1$$

$$\Rightarrow \dot{x} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ -9 & -1 & 8 & 0 & 0 \\ 0 & 0 & -10 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -16 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 6 \\ 0 \\ 6 \end{bmatrix} u$$

$$y = \begin{bmatrix} -1 & 0 & 4 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 6 \end{bmatrix} u$$

$$T = C\Phi B + D = \frac{1}{s^5 + 11s^4 + 35s^3 + 26s^2 + 30s + 1440} \begin{bmatrix} 24s^2 + 24s + 168 \\ 48 \end{bmatrix}$$

Same result is obtained using Mason's Rule.

Chapter 9

9-1 a Desired Ch. eq $= \Delta_d(s) = s^4 + 18s^3 + 146s^2 + 595s + 1088$

Let $\Delta_J(s) = \det[sI - (A - BK)]$ & solve for k

$k = [1080 \quad 589 \quad 139 \quad 13]$

In steady state $\dot{x}_{ss} = 0 = (A - BK)x_{ss} + Br$

$x_{ss} = -(A - BK)^{-1} Br \Rightarrow y_{ss} = -C(A - BK)^{-1} Br$

$r = 1 \Rightarrow y_{ss} = \begin{bmatrix} 1.8 \\ 0.9 \end{bmatrix} \times 10^{-3}$

9-1 b $k = [70 \quad 47 \quad 10]$ for B use $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ only

i.e. $B = [B_1 \quad B_2]$ B_1 for control.

use B_2 to compute y_{ss} for r $\Rightarrow$

$$y_{ss} = -C(A - B_1K)^{-1} B_2 r = -0.5625$$

9-2 (a) Note That $C = [I_2 \; 0]$, Hence split A & B

$$A = \left[\begin{array}{cc|c} -2 & 1 & 0 \\ -4 & 0 & 1 \\ \hline -2 & 0 & 0 \end{array}\right], \quad B = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \quad A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$$

$D = A_{22} - LA_{12}$ is the observer error matrix. Choose L to place the pole of D at $s = -30$

$$D = 0 - L\begin{bmatrix} 0 \\ 1 \end{bmatrix} = -[l_1 \quad l_2]\begin{bmatrix} 0 \\ 1 \end{bmatrix} = -l_2 = -30$$

$\therefore l_2 = 30$, $l_1 =$ arbitrary (set $l_1 = 0$) $\Rightarrow L = [0 \quad 30]$

$$F = DL + A_{21} - LA_{11} = -30[0 \quad 30] + [-2 \quad 0] - [0 \quad 30]\begin{bmatrix} -2 & 1 \\ -4 & 0 \end{bmatrix}$$

$F = [118 \quad -900]$, $G = B_2 - LB_1 = -1 - 30 = -31$

$$\begin{cases} \dot{z} = -30z + [118 \quad -900]y - 31u \\ \hat{x}_3 = z + [0 \quad 30]y \end{cases}$$ observer

(b) $A_{11}=0 \quad A_{12}=1 \quad A_{21}=-4 \quad A_{22}=-4 \quad B_1=0, B_2=3$

$D = A_{22} - LA_{12} = -4 - l = -30 \Rightarrow l = 28$

$F = -30(28) + (-4) - 28(0) = -844$

$G = 3 - 28(0) = 3$

$$\Rightarrow \begin{cases} \dot{z} = -30z + 3u - 844y \\ \hat{x}_2 = z + 28y \end{cases}$$

9-3 (a) $A - LC = \begin{pmatrix} 0 & 1 \\ -4 & -4 \end{pmatrix} - \begin{pmatrix} l_1 \\ l_2 \end{pmatrix}(1 \quad 0) = \begin{pmatrix} 0-l_1 & 1 \\ -4-l_2 & -4 \end{pmatrix}$

$$\begin{vmatrix} s+l_1 & -1 \\ 4+l_2 & s+4 \end{vmatrix} = s^2 + (4+l_1)s + 4l_1 + 4 + l_2 = (s+20)^2 = s^2 + 40s + 400$$

$\Rightarrow 4 + l_1 = 40 \Rightarrow l_1 = 36$

$4l_1 + 4 + l_2 = 400 \Rightarrow l_2 = 396 - 4(36) = 252 \Rightarrow L = \begin{bmatrix} 36 \\ 252 \end{bmatrix}$

$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$

(b) $L = [32 \quad 141]$ using same procedure.

9-4 (a) $k = [12 \quad 3]$

Now find a reduced-order observer. $A_{22} = -4$, $A_{12} = 1$, $A_{21} = -4$, $A_{11} = 0$

$A_{22} - LA_{12} = -4 - l = -20 \Rightarrow l = 16$

$F = -20(16) + (-4) = -324$, $G = 3$

$$\begin{cases} \dot{z} = -20z - 324y + 3u \\ \hat{x}_2 = z + 16y \end{cases}, \quad y = x_1$$

Now let $u = -k\hat{x} = -12x_1 - 3\hat{x}_2$

$$\begin{bmatrix}\dot{x}\\ \dot{\tilde{x}}_u\end{bmatrix}=\left[\begin{array}{c|c} A-BK & BK_2 \\ \hline 0 & A_{22}-LA_{12}\end{array}\right]\begin{bmatrix}x\\ \tilde{x}_u\end{bmatrix}=\left[\begin{array}{cc|c} 0 & 1 & 0 \\ -40 & -13 & 9 \\ \hline 0 & 0 & -20\end{array}\right]\begin{bmatrix}x\\ \tilde{x}_u\end{bmatrix}$$

$\lambda = -5, -8, -20$

(b) $\ell = 12$, $F = -243$, $G = 2$, $D = -20$

$$\begin{cases}\dot{\hat{z}} = -20\hat{z} - 243y + 2u \\ \hat{x}_2 = \hat{z} + 12y\end{cases}$$

$K = [18.5 \quad 2.5]$

$$\lambda\left[\begin{array}{cc|c} 0 & 1 & 0 \\ -40 & -13 & 5 \\ \hline 0 & 0 & -20\end{array}\right] = -5, -8, -20$$

9-5

a) $M_c = \begin{bmatrix}1 & 2\\ -1 & -2\end{bmatrix}$, $\det M_c = 0$, not controllable.

b) $T(s) = (c_1 \quad c_2)\begin{pmatrix}s-2 & 0\\ 1 & s-1\end{pmatrix}^{-1}\begin{pmatrix}1\\ -1\end{pmatrix} = (c_1 \quad c_2)\underbrace{\begin{pmatrix}s-1 & 0\\ -1 & s-2\end{pmatrix}}_{(s-2)(s-1)}\begin{pmatrix}1\\ -1\end{pmatrix}$

$$= \frac{(c_1\ c_2)\begin{pmatrix}s-1\\ -s+1\end{pmatrix}}{(s-2)(s-1)} = \frac{c_1 s - c_1 - c_2 s + c_2}{(s-2)(s-1)} = \frac{(c_1-c_2)s + (c_2-c_1)}{(s-2)(s-1)}$$

$$= (c_1 - c_2)\frac{(s-1)}{(s-2)(s-1)}$$

For $c_1 \neq c_2$, $(s-1)$ cancels out, this must be an uncontrollable mode, since unstable $\Rightarrow$ not stabilizable.

(c) $T(s) = 0$

(d) $T(s) = \frac{2(s-1)}{(s-2)(s-1)} = \frac{2}{s-2}$

(e) NO, not stabilizable.

9-6 Because we want to determine stabilizability & detect... it is best to find the Transfer function.

$$T(s) = (0\ 0\ 1)(sI-A)^{-1}\begin{pmatrix}1\\0\\0\end{pmatrix} = (3,1)\text{ element of }(sI-A)^{-1}.$$

$$sI-A = \begin{bmatrix} s & -1 & 1 \\ 2 & s+3 & 0 \\ -p & -1 & s-1 \end{bmatrix}, \quad |sI-A| = -2+p(s+3)+(s-1)[s(s+3)+2]$$

$$|sI-A| = -2+ps+3p+s^3+2s^2-3s+2s-2 = s^3+2s^2+(p-1)s+(3p-4)$$

$$\text{Adj}(sI-A) = \begin{vmatrix} - & - & -2+p(s+3) \\ - & - & - \\ - & - & - \end{vmatrix}$$

$$\therefore \boxed{T(s) = \frac{ps+3p-2}{s^3+2s^2+(p-1)s+(3p-4)}}$$

a) Now Find $M_c = \begin{bmatrix} 1 & 0 & -2-p \\ 0 & -2 & 6 \\ 0 & p & -2+p \end{bmatrix}$,

$\det M_c = -2(-2+p)-6p = 0$ for $4-2p-6p=0 \Rightarrow p=\frac{1}{2}$

$\therefore \det M_c = 0$ when $p=\frac{1}{2} \Rightarrow$ Controllable if $p \neq \frac{1}{2}$

$$M_o = \begin{bmatrix} 0 & 0 & 1 \\ p & 1 & 1 \\ -2+p & p-2 & -p+1 \end{bmatrix}, \quad \det M_o = p(p-2)-(p-2) = (p-2)(p-1)$$

$\therefore \det M_o = 0$ when $p = 1, 2 \Rightarrow$ observable if $p \neq 1, 2$

b) see above

c) check for $p=\frac{1}{2}$, $T(s)\Big|_{p=\frac{1}{2}} = \dfrac{\frac{s}{2}-\frac{1}{2}}{s^3+2s^2+(-\frac{1}{2})s+(-\frac{5}{2})}$

$T(s)=0$ for $s=1$, $s^3+2s^2-\frac{s}{2}-\frac{5}{2}\Big|_{s=1} = 1+2-\frac{1}{2}-\frac{5}{2} = 0$

$\therefore T(s)$ has a factor of $(s-1)$ as a pole & zero.

Because $(s-1)$ is an unstable pole & it is uncontrollable $\Rightarrow$ system stabilizable if $p \neq \frac{1}{2}$.

For $p=1$, $T(s) = \left.\dfrac{s+1}{s^3+2s^2-1}\right|_{s=-1} = \dfrac{0}{-1+2-1} = \dfrac{0}{0}$

$\therefore (s+1)$ cancels out as expected (because system is not observable for $p=1$), But $(s+1)$ is stable $\Rightarrow$ detectable for $p=1$.

for $p=2$, $T(s) = \dfrac{2s+4}{s^3+2s^2+s+2} = \left.\dfrac{2(s+2)}{s^3+2s^2+s+2}\right|_{s=-2} = \dfrac{0}{-8+8-2+2}$

$\therefore (s+2)$ cancels out, because it is stable, it's detectable.

Conclusion { controllable for all $p \neq \frac{1}{2}$
stabilizable " " " "
observable " " $p \neq 1, 2$
detectable for all p.

9-7 a) $M_c = \begin{bmatrix} 1 & 5 & 13 \\ 3 & 3 & 3 \\ 2 & -2 & -10 \end{bmatrix}$, $\det M_c = 0 \Rightarrow$ not controllable

$M_o = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 5 & 2 & 1 \end{bmatrix}$, $\det M_o \neq 0 \Rightarrow$ observable.

b) $\lambda(A) = 1, -1, 2$, The eigenvectors are $\begin{bmatrix} 1 & 1 & 1 \\ -1 & -3 & 0 \\ -2 & 2 & -1 \end{bmatrix} = P$

$\dot{z} = \bar{P}^1 A P z + \bar{P}^1 B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix} z + \begin{pmatrix} -3 \\ 0 \\ 4 \end{pmatrix} u$

$y = CPz = (1 \;\; 1 \;\; 1)\, z$

c) since second component of $\bar{B}$ is zero $\Rightarrow$ mode $= -1$ is uncontrollable

d) $G(s) = \dfrac{(s+2)(s+1)}{(s+1)(s-1)(s+2)} = \dfrac{s^2+3s+2}{s^2-2s-s+2} = \dfrac{s+2}{(s-1)(s+2)}$

As expected $(s+1)$ cancels out (uncontrollable mode)

e) Since $(s+1)$ is stable and the sys is also observable we conclude the system is stabilizable & detectable.

9-8 $M_c = \begin{bmatrix} 0 & -1 & -4 \\ 0 & 0 & 0 \\ 1 & 3 & 8 \end{bmatrix}$, $\det M_c = 0 \Rightarrow$ uncont.

$M_o = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -3 & 2 \\ 4 & -7 & 4 \end{bmatrix}$, $\det M_o = 0 \Rightarrow$ unobser.

$G(s) = \dfrac{s^2-3s+2}{s^3-5s^2+8s-4} = \dfrac{(s-1)(s-2)}{(s-1)(s-2)^2} = \dfrac{1}{s-2}$

$\dot{z} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix} z + \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} u \Rightarrow s=1$ uncontrollable

$y = (1 \quad 0 \quad -1)\, z \Rightarrow s=+2$ unobservable

$(s-1)$, $(s-2)$ cancel out.

9-9 (a) $M_c = \begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix}$, $\det M_c = 0 \Rightarrow$ no.

(b) $M_o = \begin{bmatrix} c_1 & c_2 \\ -c_1 & 2c_2 \end{bmatrix}$, $\det M_o = 2c_2c_1 + c_1c_2$

$\therefore \det M_o = 0$ when $c_1c_2 = 0 \Rightarrow$ observable

as long as $c_1c_1 \neq 0$.

(c) $A - BK = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ k_1 & k_2 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ -k_1 & 2-k_2 \end{pmatrix}$

$$\det[(A-BK)-\lambda I] = \begin{bmatrix} -1-\lambda & 0 \\ -k_1 & 2-k_2-\lambda \end{bmatrix} = +(1+\lambda)$$

$(-2+k_2+\lambda) = \lambda^2 + \lambda(k_2-2+1) + (k_2-2)$

to place poles at $(-1,-2) \Rightarrow \lambda^2+3\lambda+2$

$\therefore$ $k_2-2+1=3$ and $k_2-2=2 \Rightarrow k_2=4$, k_1 = arbitrary

$\therefore$ $k = [\text{anything} \quad 4]$ does it.

(d) for poles at $(-2,-2) \Rightarrow \lambda^2+4\lambda+4$

$\begin{cases} k_2-2+1=4 \Rightarrow k_2=5 \\ k_2-2=4 \Rightarrow k_2=6 \end{cases}$ impossible.

(e) system <u>not controllable</u>, means we cannot necessarily move poles to any arbitrary place. For example, to $(-1,-2)$ is possible, but not to $(-2,-2)$. Note that

$\det[\lambda I-(A-BK)] = (\lambda+1)(\lambda+k_2-2)$

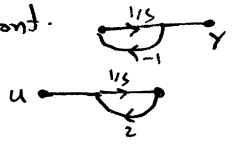
means that $(\lambda+1)$ is fixed (the uncontrollable mode)

$\therefore$ Any pole location, where $\lambda=-1$, λ = anywhere is possible, but $\lambda=-1$ cannot be moved. Hence, there is no discrepancy; the apparent discrepancy has to do with the notion of controllability.

(f) $c_1=1$, $c_2=0$ ($c_1c_2=0 \Rightarrow$ not observable)

$\dot{x} = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix} x + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u$ → uncont.

$y = (1 \quad 0)\, x$ → unobs.

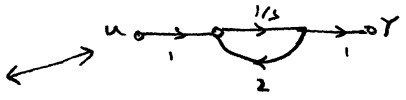


Because there is no path from the input to the output, $G(s) = 0$.



(g) $\dot{x} = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix} x + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u$

$y = (0 \quad 1)\, x$



Clearly $G(s) = \frac{1}{s-2}$

(h) The system must be at least detectable, i.e., we must measure x_2. $\Rightarrow C = [C_1 \quad C_2]$, $C_2 \neq 0$.

9-10 $A = \begin{pmatrix} 2 & 0 \\ 1 & 0 \end{pmatrix}$, $b = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $C = (0 \quad 1)$.

Let $T = [1 \quad 0] \Rightarrow P = \begin{bmatrix} C \\ T \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$,

$Q = P^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \Rightarrow$ new $C = CQ = (0 \quad 1)\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$= (1 \quad 0)$

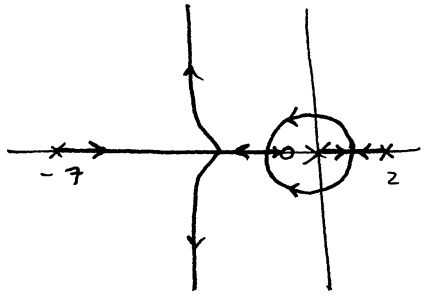
∴ New matrices are:

$A = \begin{pmatrix} 0 & 1 \\ 0 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $C = (1 \quad 0)$

$L = 4$, $D = -2$, $G = 1$, $F = -8$

$$\begin{cases} \dot{z} = -2z + u - 8y \\ \hat{x}_2 = z + 4y \end{cases}$$ ← observer

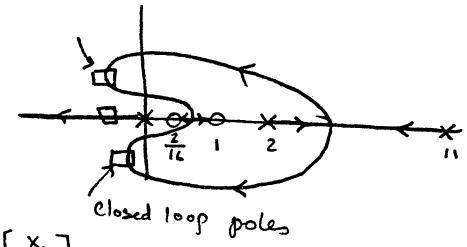
$G_1 H = \dfrac{225+4}{s(s+7)(s-2)}$



9-11 $k = [4 \quad 6]$, $F = -3$, $G = -2$, $L = 2$, $D = -1$

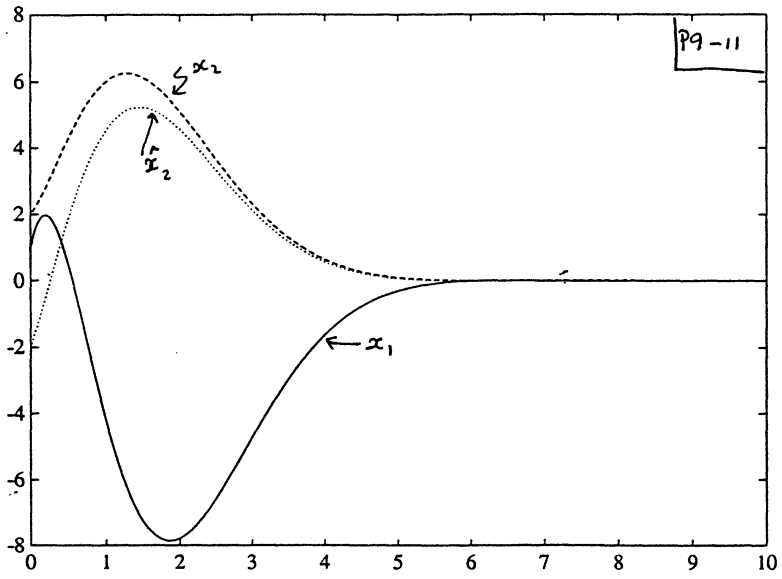
$$\begin{cases} \dot{z} = -z - 2u - 3y \\ \hat{x}_2 = z + 2y \end{cases}, \qquad H(s) = \frac{16s - 2}{s - 11}, \quad G = \frac{s-1}{s(s-2)}$$

$$GH = \frac{(16s-2)(s-1)}{s(s-2)(s-11)}$$



$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{\hat{x}}_2 \end{bmatrix} = \begin{bmatrix} -3 & 1 & -6 \\ 1 & 1 & 0 \\ 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \hat{x}_2 \end{bmatrix}$$

Let $\begin{bmatrix} x_1(0) \\ x_2(0) \\ \hat{x}_2(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}$, The zero-input response (ZIR) to these initial conditions is shown below in Figure P9-11



9-12 $k = [4 \quad 6]$, $L = \begin{bmatrix} 6 \\ 14 \end{bmatrix}$

$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$, $u = -k\hat{x}$

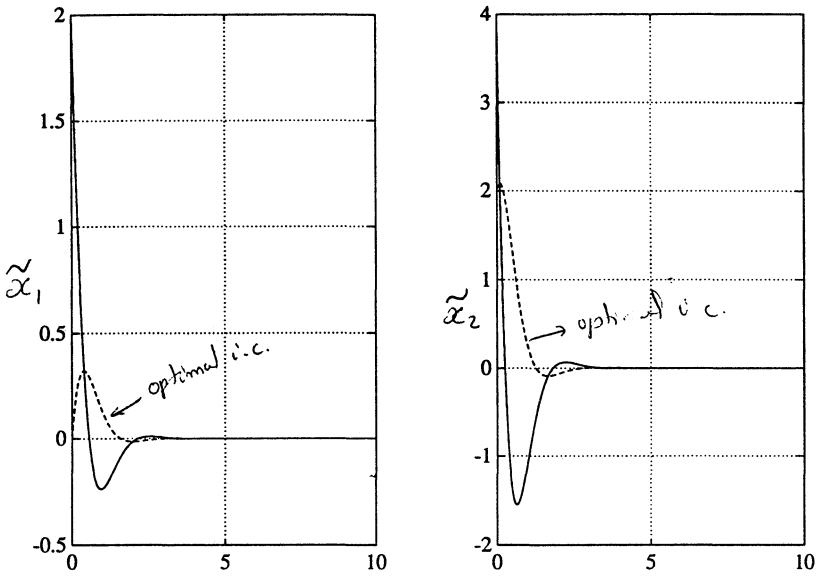
$$\begin{bmatrix} \dot{x} \\ \dot{\hat{x}} \end{bmatrix} = \begin{bmatrix} A & -BK \\ LC & A - BK - LC \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix} = \begin{bmatrix} 1 & 1 & -1 & -6 \\ 1 & 1 & 0 & 0 \\ 6 & 0 & -9 & -5 \\ 14 & 0 & -13 & 1 \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix}$$

using $x(0) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\hat{x}(0) = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$, see Figure P9-12A.

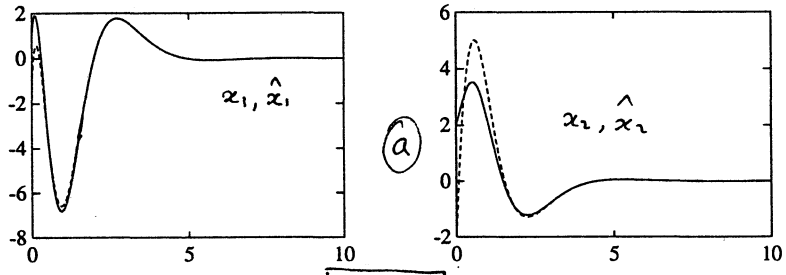
Using $\hat{x}(0) = C'(CC')^{-1} y(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} y(0)$

$y(0) = C\,x(0) = [1 \quad 0] \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1 \Rightarrow \hat{x}(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

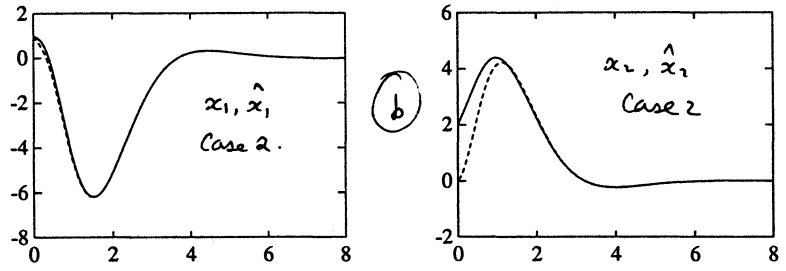
See Fig P9-12B for simulations. If you look at peak to peak values, they seem to be smaller in the 2nd case.



Plots of observer errors



P9-12



9-13 (a) $K = [22 \quad 22 \quad 11]\frac{1}{6}$

$H(s) = \frac{11(s+1\pm j)}{(s+1)(s+2)}$ state feedback case.

b) see below

c) $L = [47 \quad 305 \quad 977]'$

$H(s) = \frac{3081(s+1\pm j)}{(s+84.1)(s-27.9)(s+2.8)}$ full order obs.

d) $L' = [7 \quad 25]$

$H(s) = \frac{75.1(s+1\pm j)}{(s+2.8)(s-59)}$ reduced order obs.

e) $L' = [42 \quad 80 \quad 2]$

$H(s) = \frac{451(s+1\pm j)}{(s+1)(s+2)(s+51)}$ full order observ, robust

f) $L' = [2 \quad 0]$

$H(s) = \frac{11(s+1\pm j)}{(s+1)(s+2)}$ reduced order obs., robust.

Data:		GM	PM	% overshoot	peak time t_p	settling time t_s
1	SF	-21	∞	32	0.9	3
2	FOO	-1.7	20.1	63	2.8	0.58
3	ROO	-18.9	73.7	32	0.9	3
4	FOO, rob.	-1.7	36.4	62	0.55	2.8
5	ROO, rob	-20.8	∞	32	0.9	3

Compare 4 and 2, 4 has better PM & same GM.

" 5 " 3, 5 " 4 " " better GM.

% overshoots are identical, ROO has less overshoot. But FOO has faster response (smaller t_p).

Better margins imply robustness. Let's verify it by using these control & observer gains on the "true" A matrix.

9) $A_{true} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & 0 & -1 \end{bmatrix}$ call this A_t.

State feedback : $\lambda(A_t - BK) = -10.5,\ 0.29,\ -1.7$

unstable !

For, Case **c** : need to check closed loop stability first.

$$\dot{x} = A_t x + Bu$$
$$u = -k\hat{x}$$
$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$$

$$\Rightarrow \begin{cases} \dot{x} = A_t x - BK\hat{x} \\ \dot{\hat{x}} = A\hat{x} - BK\hat{x} + LCx - LC\hat{x} \end{cases}$$

$$\begin{bmatrix} \dot{x} \\ \dot{\hat{x}} \end{bmatrix} = \begin{bmatrix} A_t & -BK \\ LC & A - BK - LC \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix} \leftarrow \text{closed loop system}$$

use the model "A" matrix for observer

use "True A" matrix for actual plant.

Case **c** : has an eigenvalue of 0.03 => unstable.

Case e : $\lambda = -39.9, -10, -2, -1, -0.9 \pm j1.2$ Stable

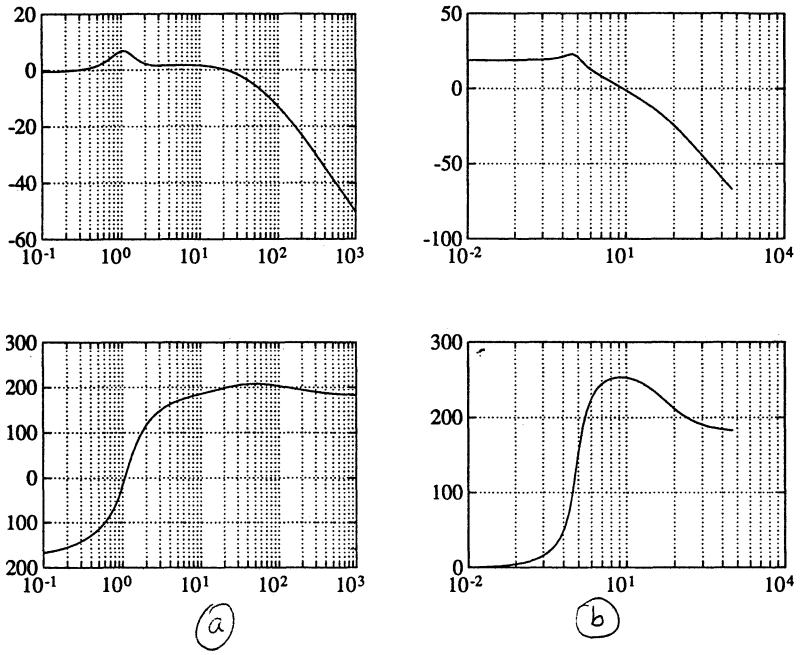
∴ The robust observer is able to tolerate model errors and remain stable.

For the reduced order observer case.

Case d : has an eigenvalue at $s = 0.057$ => unstable.

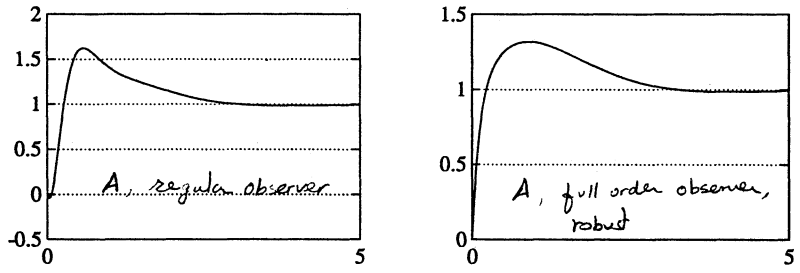
Case f : $\lambda = -10, -0.97 \pm j1.19, -2, -1$ Stable.

Therefore, the observers in (e, f) are stable even for the modified A matrix and hence are truly robust.

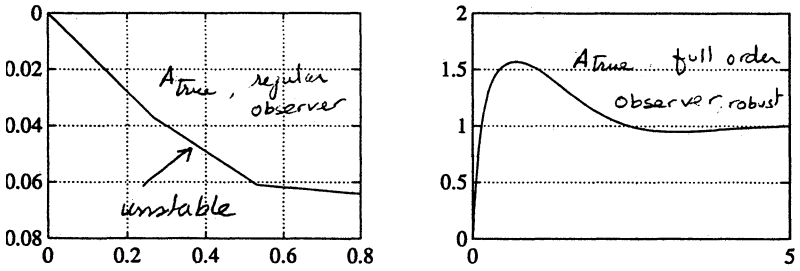


Bode plots using A_{true}. a) regular observer b) robust observer

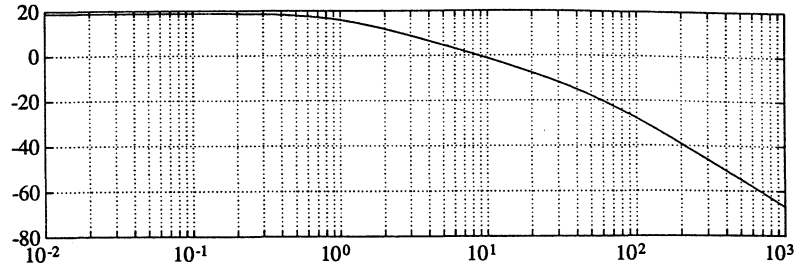
P 9-13

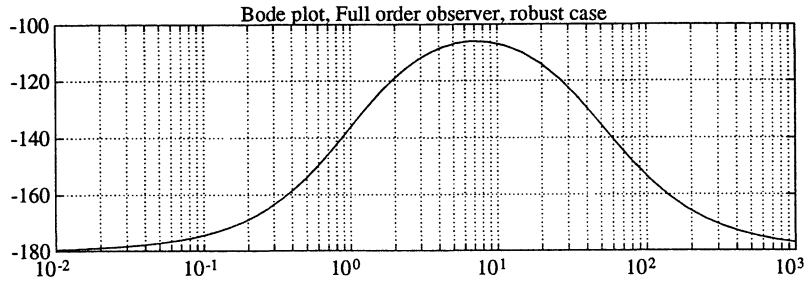


P9-13

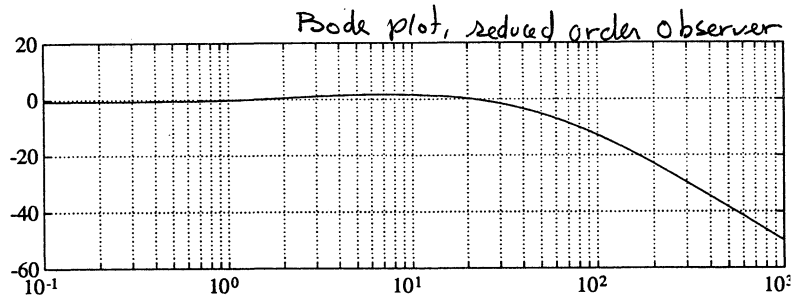


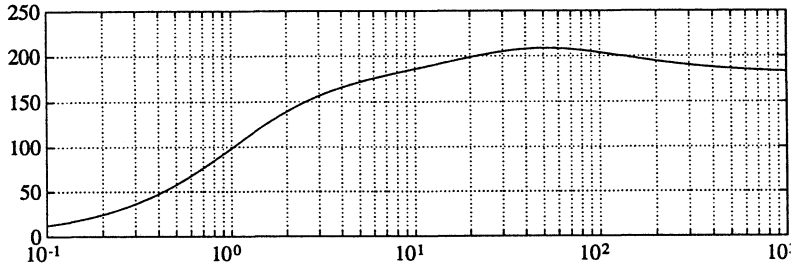
Step response using A_{true} and A. Full order observer case.


20
0
-20
-40
-60
-80
10-2
10-1
100
101
102
103


Bode plot, Full order observer, robust case
-100
-120
-140
-160
-180
10-2
10-1
100
101
102
103

Prob. 9-13


Bode plot, reduced order observer
20
0
-20
-40
-60
10-1
100
101
102


250
200
150
100
50
0
10-1
100
101
102

9-14 $K = [-256 \quad -150 \quad -7.5]$

Then $\lambda(A - B_1K) = -5, -5, -5$

note that $B = [B_1 \quad B_2] = \begin{bmatrix} 0 & 0 \\ 0 & -1 \\ 2 & 0 \end{bmatrix}$

B_1 is used for control. B_2 is for the disturbance input.

when $f = 1$, $d_{ss} = -C(A - B_1K)^{-1}B_2 f = -0.12$

when $f = 0$, $u = -kx + u_1$, u_1 = reference input.

$d_{ss} = -C(A - B_1K)^{-1}B_1 u_1 = -0.004$ for $u_1 = 1$

For $d_{ss} = 15 \Rightarrow u_1 = \dfrac{d_{ss}}{-0.004} = -3750$

Actually $d_{ss} = 15$ mm $\Rightarrow u_1 = \dfrac{15 \times 10^{-3}}{-0.004} = -3.75$

9-15 $h = 0.2 \sin(\pi t/10)$, $\dot{h} = 0.2(\pi/10)\cos(\pi t/10)$

$\dddot{h} = -0.2(\pi/10)^2 \sin(\pi t/10) = -(\pi/10)^2 h$

Let $x_4 = h \Rightarrow \dot{x}_4 = x_5$

$x_5 = \ddot{h} \Rightarrow \dot{x}_5 = \dddot{h} = -(\pi/10)^2 \ddot{h} = -(\pi/10)^2 x_4$

$\dot{x} = Ax + B_1 u_1 + B_2 f$, $f = \ddot{h} = x_5$, $B_2 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$

$\dot{x} = Ax + B_1 u_1 + B_2 x_5$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0.2 & 0 & -0.25 & 0 & -1 \\ 0 & 0.8 & -30 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -(\pi/10)^2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 2 \\ 0 \\ 0 \end{bmatrix} u_1$$

$$u = kx + u_1$$

$$\dot{x} = Ax + BKx + Bu_1$$

$$BK = \begin{bmatrix} 0 \\ 0 \\ 2 \\ 0 \\ 0 \end{bmatrix} [k_1 \cdots k_5] = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 2k_1 & 2k_2 & \cdots & & 2k_5 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$A+BK = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0.2 & 0 & -0.25 & 0 & -1 \\ 2k_1 & 0.8+2k_2 & -30+2k_3 & 2k_4 & 2k_5 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -(\pi/10)^2 & 0 \end{bmatrix}$$

Ch. 10

10.1

(a) $K = [0.3333 \quad 0.0809]$ (b) $K = [0 \quad 0.3333]$

(c) $K = [0.3333 \quad 0.3487]$

Note that when the weight on a certain state is reduced, the control gain component is also reduced. For example, in (a), $q_{22} = 0 \Rightarrow K_2 \ll K_1$. In (b), $q_{11} = 0$, $k_1 \ll k_2$. Hence, if you don't care about a particular state, the controller won't put much effort in controlling it.

10.2

(a) Let $C_p = [1 \quad 0] \Rightarrow Q = C_p' C_p$

$$G_p(s) G_p(-s) = [C_p (sI - A)^{-1} B][C_p (-sI - A)^{-1} B]$$

$$= \frac{3}{(s+2)(s+2)} \cdot \frac{3}{(-s+2)(-s+2)}$$

(b) Let $C_p = [0 \quad 1]$

$$G_p(s) = \frac{3s}{(s+2)(s+2)} \cdot \frac{-3s}{(-s+2)(-s+2)}$$

we need negative gain root locus here.

(c) There is no $C_p \ni Q = C_p' C_p$.

To obtain RSL, start from the Hamiltonian matrix.

$$H = \begin{bmatrix} A & -\frac{1}{\rho}BB' \\ -Q & -A' \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -4 & -4 & 0 & -9/\rho \\ -1 & 0 & 0 & 4 \\ 0 & -1 & -1 & 4 \end{bmatrix}$$

$$\det(sI - \mathcal{H}) = (s^4 - 8s^2 + 16) - \frac{9}{\rho}(1 - s^2) = 0$$

$$\Rightarrow 1 + \frac{1}{\rho}\underbrace{\frac{-9(1-s^2)}{s^4 - 8s^2 + 16}}_{G_1(s)\,G_1(-s)} = 0$$

$$G_1(s) = \frac{3(1+s)}{s^2+4s+4} \qquad G_1(-s) = \frac{-3(1-s)}{s^2-4s+4}$$

use negative root locus for stable solutions.

Plots are attached.

10.3 (a) $Q = [1\ 0]'[1\ 0]$, $C_q = [1\ \ 0]$

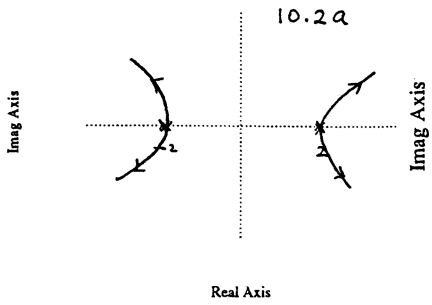
$$G_{1q}(s) = \frac{3}{s^2+8s+3} \Rightarrow G_{1q}(s)\,G_{1q}(-s) = \frac{9}{(s^2+8s+3)(s^2-8s+3)}$$

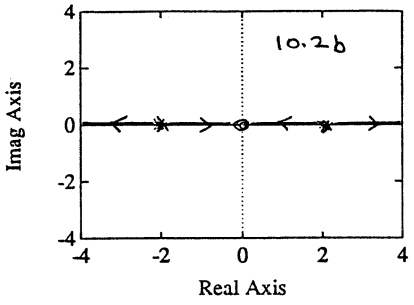
(b) Because Q can't be factored, use the Hamiltonian.

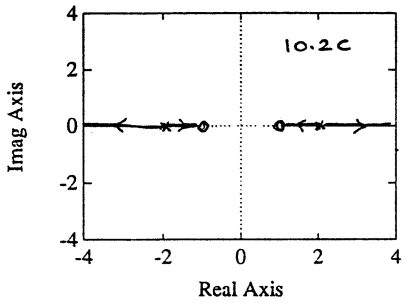
$$\det(sI - \mathcal{H}) = (s^4 - 58s^2 + 9) - \frac{16}{\rho}(1 - s^2) = 0$$

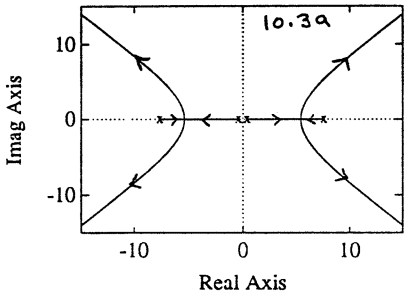
$$\therefore 1 + \frac{1}{\rho}\frac{4(1+s)(1-s)(-4)}{(s^2+8s+3)(s^2-8s+3)} = 0$$

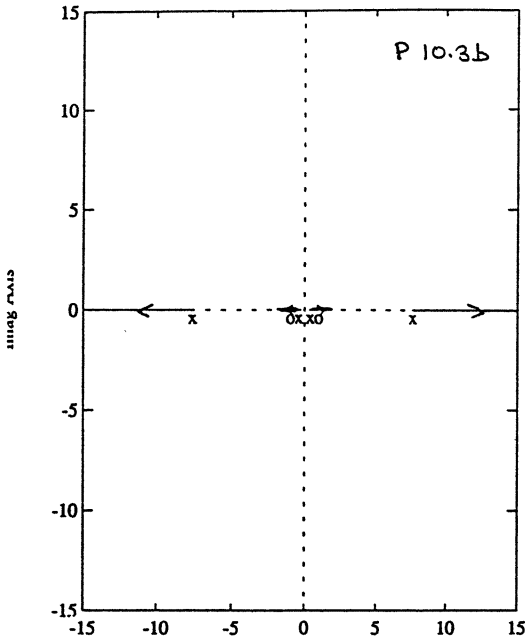
use neg. root locus for stable solution (i.e. no imaginary axis loci).

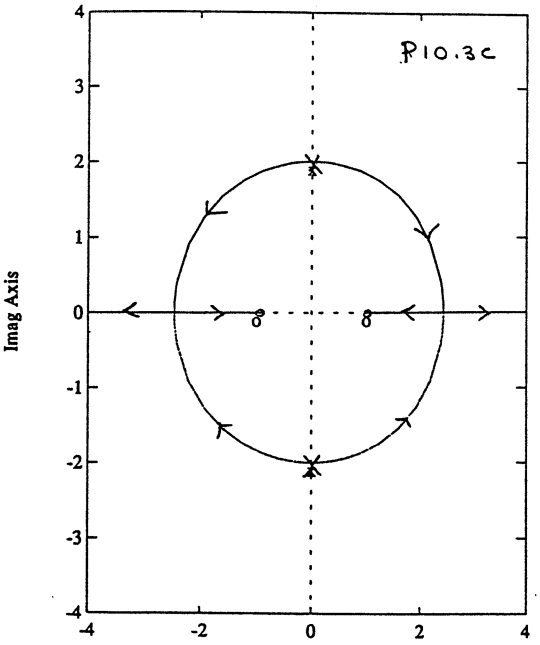

10.2a
Imag Axis
Real Axis


10.2b
Imag Axis
Real Axis


10.2c
Imag Axis
Real Axis


10.3a
Imag Axis
Real Axis


P 10.3b


P10.3c
Imag Axis

(c) $C_q = [1 \quad 0]$, $G_1(s)G_q(-s) = \dfrac{(s+1)(1-s)}{(s^2+4)(s^2+4)}$

Plots are attached

10.4

(a) for

ρ	K_a	
0.1	2.5	0.29
1	0.5	0.06
10	0.06	0.008

b)

ρ	K_b	
0.1	2.5	1.9
1	0.5	0.29
10	0.06	0.03

(c)

ρ	K_c	
0.1	3.01	0.47
1	0.91	0.91
10	0.28	0.06

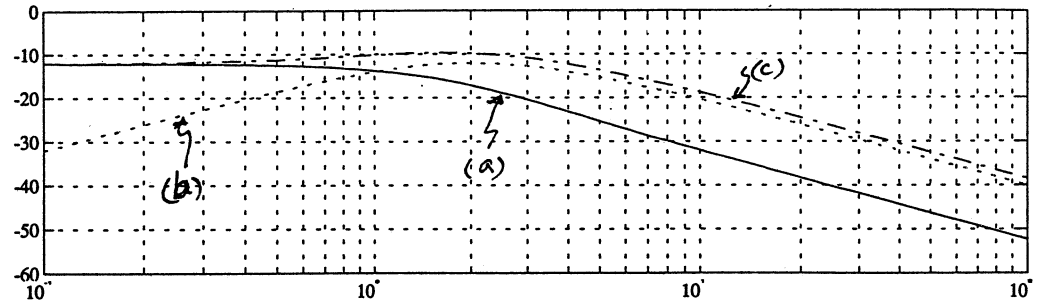
Note that as ρ increases (control cost goes up), values of K decrease. If the eigenvalues are checked, you'll see that in (a) the eigenvalues are pushed farther to the left, compared with (b).

10.5

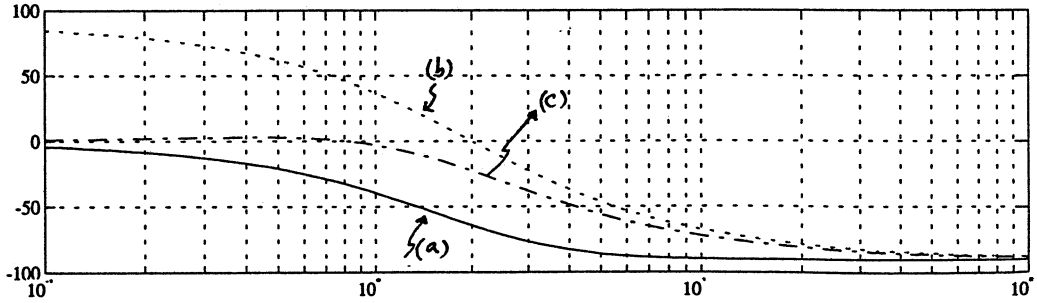
(a) $H(s) = 0.24\,(s + 4.12)$ Closed loop poles $= -2.12 \pm j\,0.707$
GM and PM are not defined (or ∞) as there are no crossover frequencies.

(b) $H(s) = s$ closed loop poles $= -1, -4$
There are no crossover freq. in this case either (∞ margins)

(c) $H(s) = (s + 0.83)\,1.19$ closed loop poles $= -1.2, -3.9$
Margins are infinite.



P 10.5



10.5

10.6

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad Q = \begin{bmatrix} 1 & 0 \\ 0 & q \end{bmatrix}, \quad R = 1$$

solve the Ric. Eq. to get $P = \begin{bmatrix} \sqrt{q+2} & 1 \\ 1 & \sqrt{q+2} \end{bmatrix}$

$$K = \begin{bmatrix} 1 & \sqrt{q+2} \end{bmatrix}$$

$\det[\lambda I - A + BK] =$ optimal char. poly $= \lambda^2 + \lambda\sqrt{q+2} + 1 = 0$

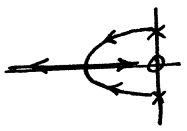
for $q \to 0$, $K \to [1 \quad \sqrt{2}]$, poles $\to (-1,-1) \times 0.7$

In this case, control is expensive, and we try min. energy control, poles are moved from the origin to $(-1,-1) \times (0.7)$

for $q \to \infty$, $K \to [1 \quad \sqrt{q}]$, this is cheap control case. The weight on the first state, $q(1,1) = 1$, is fixed, $K(1) = 1$, but the weight on the second state increases, resulting in strong control action and $K(2) \to \infty$.

The locus of closed loop poles is given by

$$(\lambda^2+1) + \sqrt{q+2}\,\lambda = 0 \quad \Rightarrow \quad 1 + \sqrt{q+2}\,\frac{\lambda}{\lambda^2+1} = 0$$

root locus of $\dfrac{\lambda}{\lambda^2+1}$ is 

Hence, one pole moves to $-\infty$, the other one moves to the origin.

10.7

(a) $K = [395.1 \quad 27.1 \quad -1395.1 \quad -280.8]$

Closed loop poles $= -4.11 \pm j\,9.39$, $-9.94 \pm j\,3.88$

(b) $L = \begin{bmatrix} 16.6 & 283.3 & 4.6 & 6.7 \\ -24 & -541.5 & 4.7 & 19.2 \end{bmatrix}$

filter eigenvalues $= -20.9 \pm j\,20.7$, $-2.3 \pm j\,1.2$

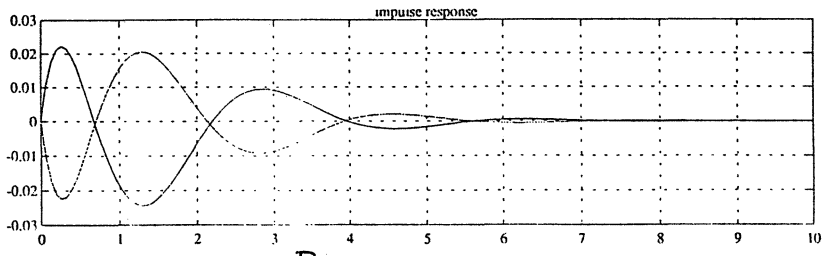
(c) Because There are 2 outputs, the Compensator is a 2-input, 1-output system. $H(s) = [H_1(s) \quad H_2(s)]$

$$H_1(s) = \frac{5.9\times10^{3}\,(s+1)(s-44)(s+22)}{(s+34.5 \pm j\,34.4)(s+2.2 \pm j\,1.2)}$$

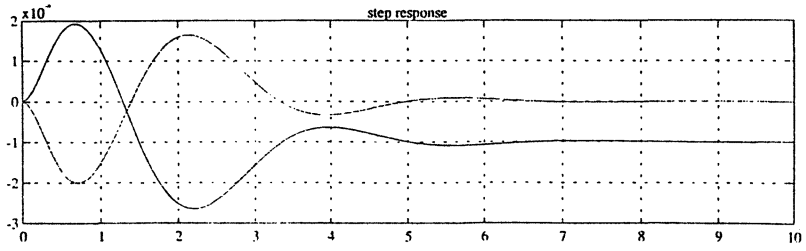
$$H_2(s) = \frac{-3.6\times10^{4}\,(s+3.6)(s+9.5 \pm j\,12.8)}{\text{same poles}}$$

(d) Closed loop poles $= \{-20.9 \pm j\,20.7,\ -4.1 \pm j\,9.4,\ -10 \pm 3.8,\ -2.3 \pm j\,1.2\}$

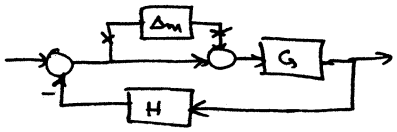
(e)



P10-7e



10-9



(a) $\tilde{G} = (1+\Delta_m)G \Rightarrow \Delta_m = \frac{\tilde{G}}{G} - 1$

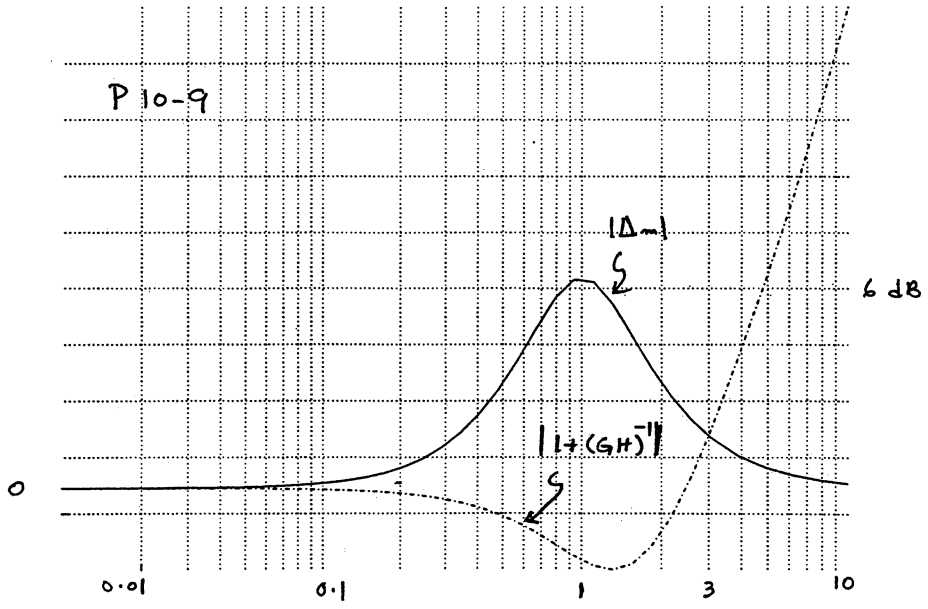
$$\Delta_m = \frac{2(s+1)s^2}{s^2(s^2+s+1)} - 1 = \frac{2s+2-s^2-s-1}{s^2+s+1} = \frac{-s^2+s+1}{s^2+s+1}$$

(b) $$M = \frac{-GH}{1+GH} = \frac{-20(s+1)}{s^2(s+10)+20(s+1)} = \frac{-20(s+1)}{s^3+10s^2+20s+20}$$

(c) $|\Delta_m| < |1+(GH)^{-1}|$

$$1+(GH)^{-1} = 1 + \frac{s^2(s+10)}{20(s+1)} = \frac{s^3+10s^2+20s+20}{20(s+1)}$$

Fig. P10-9 indicates that $|\Delta_m| > |1+(GH)^{-1}|$ for $\omega < 3$. System is not robustly stable. It is also verified by noting that the poles of $\frac{\tilde{G}}{1+\tilde{G}H}$ are in the RHP.



10-10 (a) Equation follow from the block diagram.

(b)

$$\dot{x} = \begin{bmatrix} -14 & 100 & 0 \\ 1 & 0 & 0 \\ 0 & -10 & 0 \end{bmatrix} x + \begin{bmatrix} 0 & 0.1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} y_c \\ d \\ n_1 \\ n_2 \end{bmatrix} + \begin{bmatrix} 100 \\ 0 \\ 0 \end{bmatrix} u$$

$$z = \begin{bmatrix} 0 & 0 & 0.5 \\ 0 & 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} y_c \\ d \\ n_1 \\ n_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 5 & 0 \end{bmatrix} x + \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_c \\ d \\ n_1 \\ n_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} u$$

$$\Rightarrow \begin{bmatrix} \dot{x} \\ z \\ y \end{bmatrix} = \underbrace{\left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{array}\right]}_{P(s)} \begin{bmatrix} x \\ w \\ u \end{bmatrix} \qquad w = \begin{bmatrix} y_c \\ d \\ n_1 \\ n_2 \end{bmatrix} \quad y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

(c) Values obtained using MATLAB and MATRIXx. Also verified by program CC. The answers are in the book.

10-11

$$y = -e = -w - y_1 \qquad \text{where } y_1 = G u \text{ (output of } G)$$

$$z = e = w + y_1 \qquad w = d$$

$$\dot{x}_1 = -2x_1 - 2x_2 - x_3 + u$$

$$\dot{x}_2 = x_1$$

$$\dot{x}_3 = x_2$$

$$y_1 = x_2 - x_3$$

$$z = x_2 - x_3 + w$$

$$y = -x_2 + x_3 - w$$

$$\Rightarrow \begin{bmatrix} \dot{x} \\ z \\ y \end{bmatrix} = \underbrace{\left[\begin{array}{ccc|cc} -2 & -2 & -1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ \hline 0 & 1 & -1 & 1 & 0 \\ 0 & -1 & 1 & -1 & 0 \end{array}\right]}_{P(s)} \begin{bmatrix} x \\ w \\ u \end{bmatrix}$$

$$A = \begin{bmatrix} -2 & -2 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad B_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad B_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$C_1 = [0 \quad 1 \quad -1], \quad D_{11} = 1, \quad D_{12} = 0$$

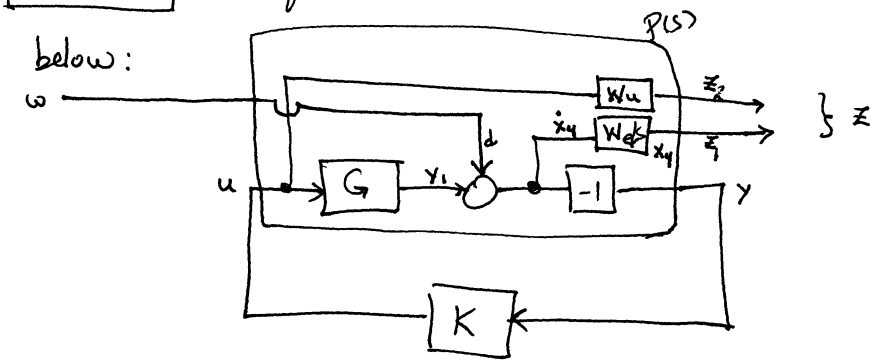
$$C_2 = [0 \quad -1 \quad 1], \quad D_{21} = -1, \quad D_{22} = 0$$

10-11 Cont

The problem as formulated has no stabilizing solution because it violates the H_∞ sol. conditions. In particular $D_{12} = 0$. Looking at P_{ss} and the state equations, it's clear that $D_{12} = 0$ because the control "u" doesn't appear in the "z" equation. Therefore, we have to include "u" as a part of our regulated outputs, i.e., we must "pull it out". We will attach a weight W_u to it for fine tuning.

Another minor problem is that $D_{11} \neq 0$. Although, the most general sol. to the H_∞ problem does not require D_{11} to be zero (Doyle & Glover sol.), the formulas in the book assume that $D_{11} = 0$. Again, note that (from the block diagram) that the disturbance, d, has a direct path to the output (e or z). If we put a dynamic weight on its path (such as $\frac{W_e}{s}$), it will force $D_{11} = 0$. This also makes sense from a control point of view; because the integrator will give us a high gain at low frequencies and help with disturbance rejection.

10-11 Cont. Our final 2-port design block diagram is shown below:



Hence $\begin{cases} z = \begin{bmatrix} W_u \\ \frac{W_e}{s} \end{bmatrix} \begin{bmatrix} u \\ \omega + Gu \end{bmatrix} \\ y = -d - Gu \end{cases}$

Let x_4 = state of the integrator $\triangleq \frac{W_e}{s}$

$$\dot{x}_1 = -2x_1 - 2x_2 - x_3 + u$$
$$\dot{x}_2 = x_1$$
$$\dot{x}_3 = x_2$$
$$\dot{x}_4 = x_2 - x_3 + \omega \quad \Longleftrightarrow \quad z = W_e x_4 \,,\ \dot{x}_4 = \omega + y_1$$

y_1 = plant output.

$$z \begin{cases} z_1 = W_e \cdot x_4 \\ z_2 = W_u \cdot u \end{cases}$$

$$y = -x_2 + x_3 - \omega \quad \Longleftrightarrow \quad y = -(\omega + y_1)$$

$$\therefore \begin{bmatrix} \dot{x} \\ z \\ y \end{bmatrix} = \left[\begin{array}{cccc|cc} -2 & -2 & -1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & W_e & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & W_u \\ \hline 0 & -1 & 1 & 0 & -1 & 0 \end{array}\right] \begin{bmatrix} x \\ \omega \\ u \end{bmatrix} \quad \leftarrow P(s)$$

Note that $D_{12} = \begin{bmatrix} 0 \\ W_u \end{bmatrix}$, $D_{11} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $D_{21} = -1$.

10-11 Cont.

The conditions & assumptions are all satisfied. Once the system matrix $P(s) = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & \cdots & \cdots \\ \vdots & \cdots & \cdots \end{array}\right]$ is available, any software (program CC, MATRIXx, MATLAB) can be used to solve the problem.

We Tried several values for W_u, W_e. It's a good idea to write a program or macro and run it for several values of the weights to see their effects. We settled for:

$$W_u = 0.01 \quad , \quad W_e = 0.2$$

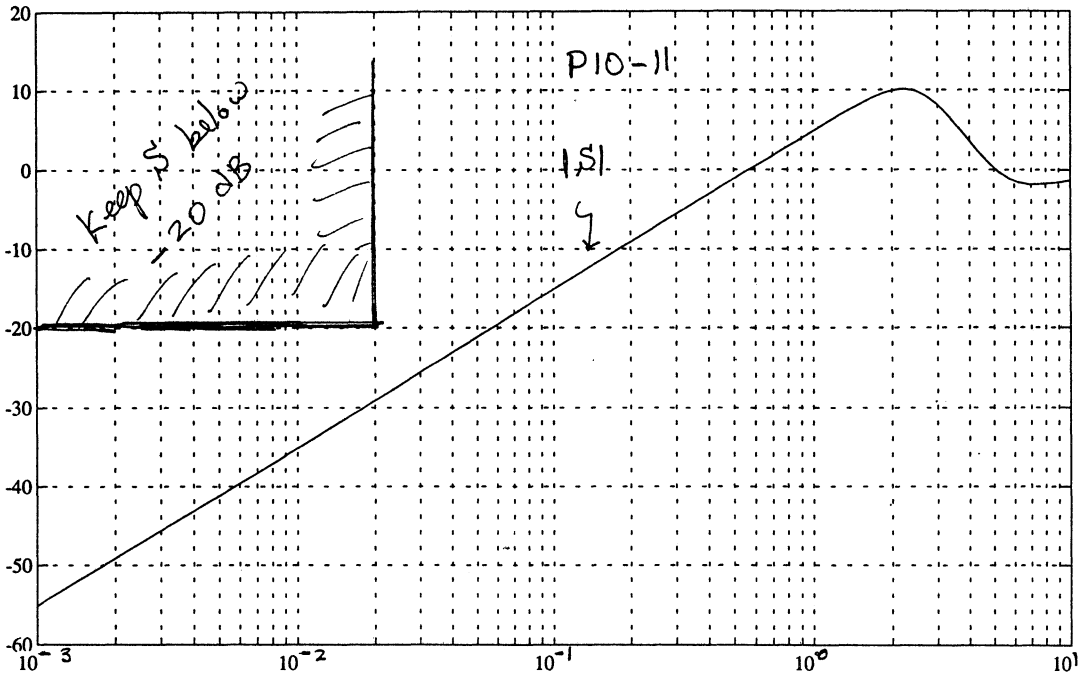
Using γ-iteration, for a value of $\gamma = 5.718$, we get

$$K(s) = \frac{-2766.5\,(s+1)(s^2+s+1)}{s(s+158.3)(s+2.5 \pm j\,4.9)}$$

Closed loop poles $= -158.2, -1.2 \pm j\,2.2, -0.5 \pm j\,0.8, -1, -2.7$

Closed loop zeros $= -0.5 \pm j\,0.8, -1$

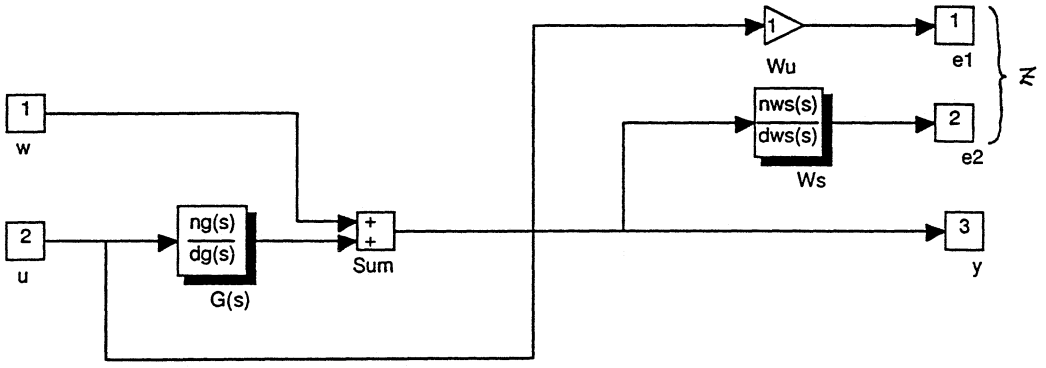
Note that the compensator cancels the stable poles of the system. This usually happens in H_∞ control. The plot of $|S(j\omega)|$ is shown in the next page. Note that $|S(j\omega)| \leq -35$ dB for $\omega \leq 0.01 \Rightarrow |S(j\omega)| < 0.017$ which meets the specs. Actually, S is much smaller than the specs, the price for this is that $|S| > 1$ for $0.6 < \omega < 5$.


P10-11
|S|
Keep S below -20 dB
20
10
0
-10
-20
-30
-40
-50
-60
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}

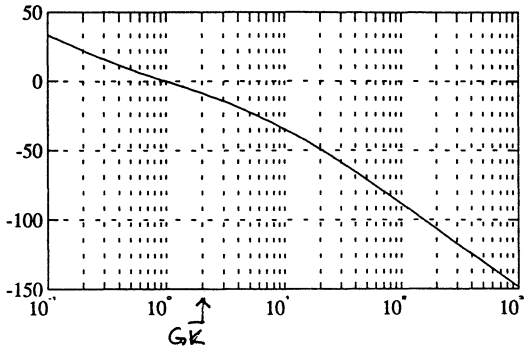
10-12 See the attached plots for the results. Some important features are tabulated.

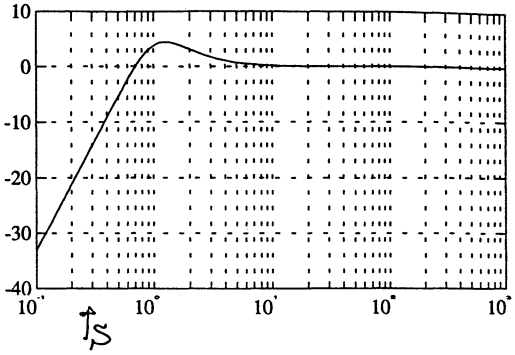
	$\lvert S(j10^{-1})\rvert$	freq. where $\lvert S\rvert \le 1$	freq. where $\lvert S\rvert \ge 1$	$\lVert S\rVert_\infty$	freq. where $\lvert T\rvert \le 1$	$\lVert T\rVert_\infty$	γ
I: $W_u = 1$	-32 dB	$0.1 < \omega < 0.7$	$0.7 < \omega < 6$	1.65	$\omega \ge 1.5$	1.608	2.16
II: $W_u = \frac{10^{-4}}{}$	-65 dB	$0.1 < \omega < 5.5$	$5.5 < \omega < 30$	2.51	$\omega \ge 15$	2.15	1.09

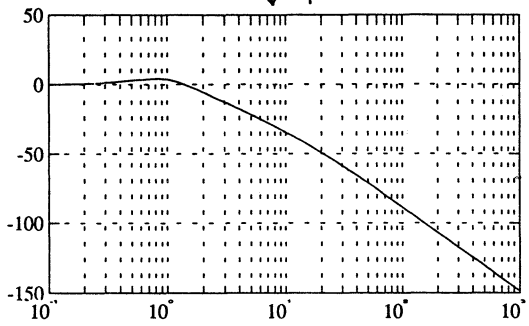
Case II gives better dist. rejection at low freq., signals over a larger freq. range are rejected in case II. This is achieved at the cost of larger peak in T in case II, which means it has smaller $MSM = \frac{1}{\lVert T\rVert_\infty}$. That is, case II is less robust. The gain crossover freq. in case II is higher $\Rightarrow$ higher bandwidth $\Rightarrow$ faster response. But II is more susceptible to higher freq. noise.



Prob. 10-12 Block diagram







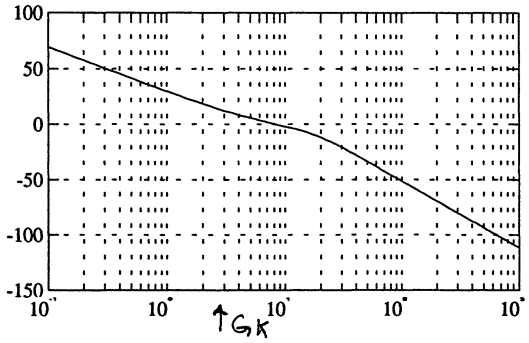
$W_u = 1 \qquad \gamma = 2.1602$

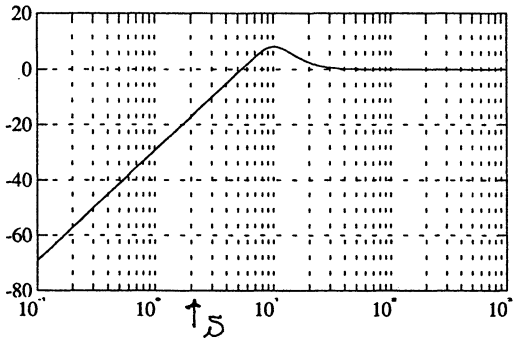
$K(s) = -37.2 \dfrac{(s+0.47)}{(s+2.2)(s+17.08)}$

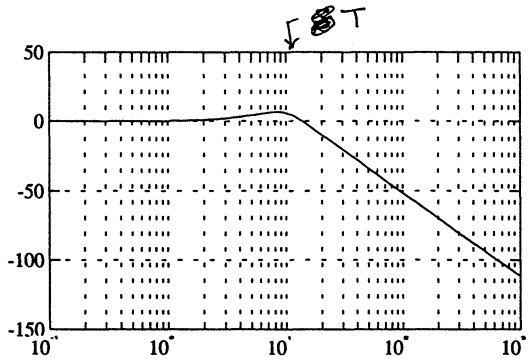
$\|S\|_\infty = 1.6585$

$\|T\|_\infty = 1.6086$

Prob. 10-12







$W_u = 10^{-4} \qquad \gamma = 1.0967$

$K(s) = \dfrac{(s+3.83)(-2607.3)}{(s+13 \pm j13)}$

$\|S\|_\infty = 2.5105$

$\|T\|_\infty = 2.1503$

10-13 See attached plots.

Case I: $\gamma = 1.29$, $K(s) = \dfrac{3.42\times10^{6}(s+0.79)(s+1)}{(s+3.04)(s-6.9\pm j\,107.2)}$

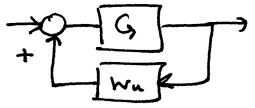
Case II: $\gamma = 3.42$, $K(s) = \dfrac{1.43\times10^{7}(s+958)(s+1)(s+0.41)}{(s+10^{3})(s+500)(s+33.7\pm j\,195)}$

	$\lvert S(j10^{-1})\rvert$	freq. where $\lvert S\rvert\le 1$	freq. where $\lvert S\rvert>1$	$\lVert S\rVert_\infty$	freq. where $\lvert T\rvert\le 1$	$\lVert T\rVert_\infty$	roll off rate at high freq. T
I	-35	0.1 → 0.9	0.9 → 80	91.3	120	90.3	-45 dB/dec.
II	-28 dB	0.1 → 1	1.5 → 70	3.4	180	2.76	-50 dB/dec.

Case II appears to be better overall. Because the system is nonminimum phase (RHP zero), performance improvement is limited. Slightly better dist. rejection at low freq. in case I is paid for by very large peaks in S and T. Hence, Case I is is much less robust than Case II which has a flat freq. response in S and T. The problem shows that appropriate selection of weights has a profound effect on design.

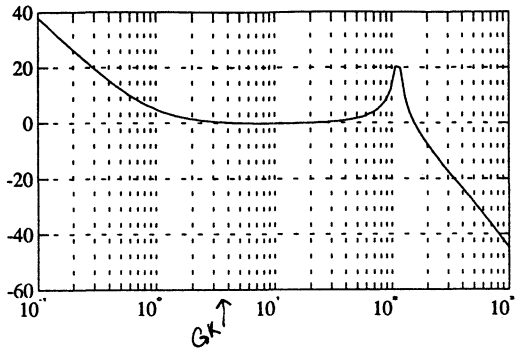
Case III: $\gamma = 1.0024$, $K(s) = 0$.

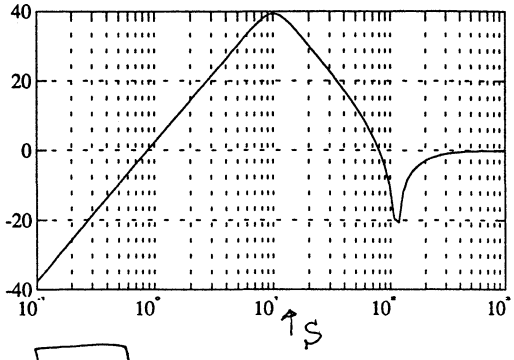
It means that the system is stable for $W_n = 10^{-4}$.

i.e. 

is stable.

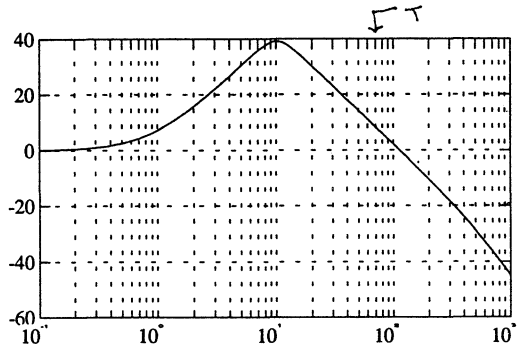
Char. eq. $= s^3 + 1.0026s^2 + 0.0095s + 0.000101$.

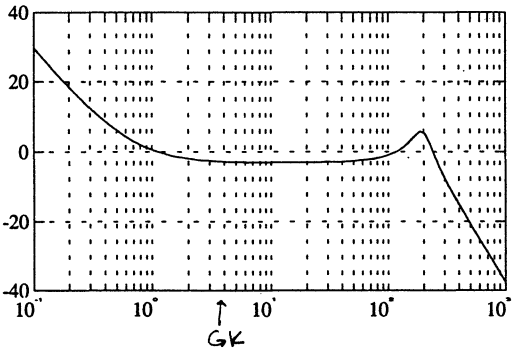


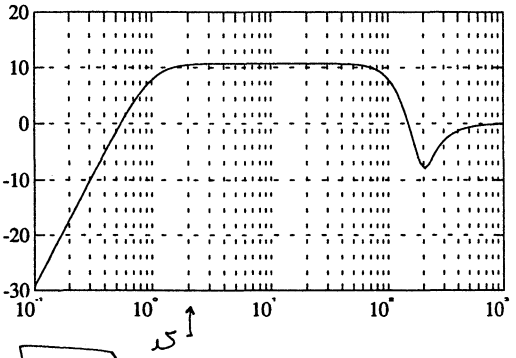


10-13

Case I

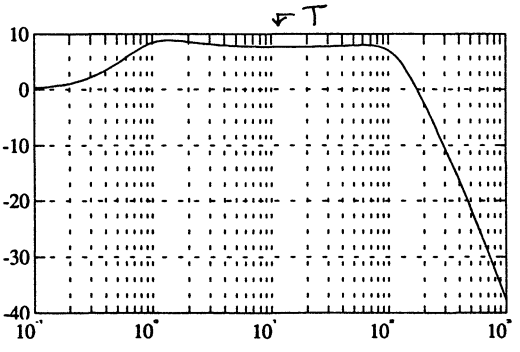




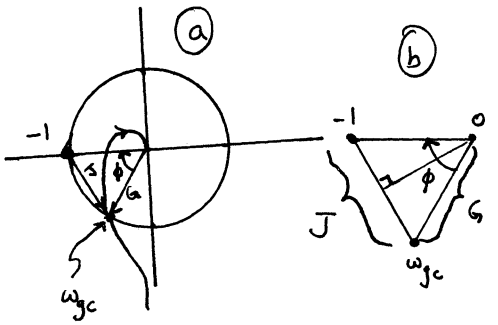


10-13

Case II



10-14



The triangle in (a) is shown in (b). It's an isoscales triangle (because $|G|=1$, at the gain crossover freq.). Phase margin is the angle ϕ.

From the triangle, $\sin\frac{\phi}{2} = \frac{|J(j\omega_{gc})/2|}{1}$

$\therefore \frac{\phi}{2} = \sin^{-1}\left|\frac{J(j\omega_{gc})}{2}\right| \Rightarrow \boxed{\phi = PM = 2\sin^{-1}\left|\frac{J(j\omega_{gc})}{2}\right|}$

Because $S = \frac{1}{J} \Rightarrow \left|\frac{J}{2}\right| = \frac{1}{2|S|} \Rightarrow \boxed{PM = 2\sin^{-1}\frac{1}{2|S(j\omega_{gc})|}}$

Also $J = 1 + G$ and $GM = 20\log\frac{1}{|G|} = -20\log|G|$

$G = J - 1 = \frac{1}{S} - 1 \Rightarrow GM = -20\log\left(1 \pm \frac{1}{|S|}\right)$

```
************  PROBLEM 10-14 ************************************
s=1:.2:3;s=s';gm=-20*log10(1-ones(s)./s);

grm=-20*log10(1+ones(s)./s);
pm=2*asin(ones(s)*.5./s); pm=pm*180/pi; result=[s gm grm pm]
```

|s|	GM	GRM	PM	
1.0000	∞	-6.0206	60.0000	← LQR Stab. margins.
1.2000	15.5630	-5.2648	49.2486	
1.4000	10.8814	-4.6817	41.8497	
1.6000	8.5194	-4.2171	36.4199	
1.8000	7.0437	-3.8377	32.2552	
2.0000	6.0206	-3.5218	28.9550	
2.2000	5.2648	-3.2545	26.2731	
2.4000	4.6817	-3.0254	24.0494	
2.6000	4.2171	-2.8266	22.1750	
2.8000	3.8377	-2.6525	20.5731	
3.0000	3.5218	-2.4988	19.1881	

Chapter 11

11-1

a) refrigerator : temp control,

b) Coffee maker : on-off + temperature control.

c) T.V. : remote control + programming.

d) VCR : timer programming.

e) AC system : on-off control.

f) Home security systems : alarms . etc .

11-2

1) auto : engine control ; sense temp + control cooling fan.
also controls air to fuel ratio, etc.

2) Hotel : central AC system : sense temp + regulate it .

3) Ship : sense heading + direction for navigation.

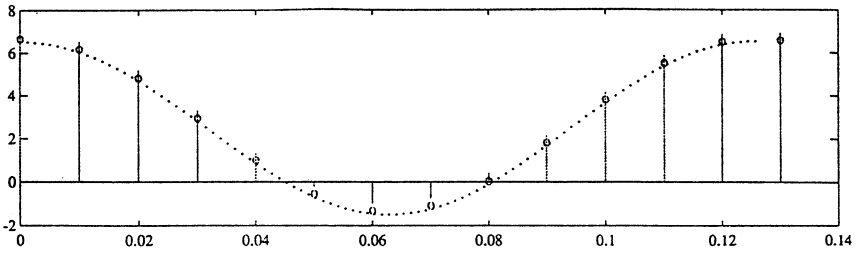
4) Power plant : control of frequency . Control of turbine speed.

5) Operating room : monitor + control blood flow, blood pressure, ...

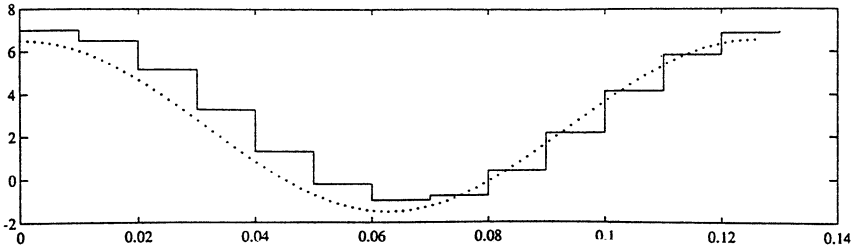
11-3

$f(t) = 3 + 4\cos 50t$ → [A/D] → $f(nT) = 3 + 4\cos 50nT$

for $T = 0.01 \Rightarrow f(n) = 3 + 4\cos\frac{n}{2}$ See attached plot.



P 11-3



11-4 $b_{k+1} = b_k + \frac{i}{12} b_k + d_k = (1 + \frac{i}{12}) b_k + d_k$

$b_{k+1} = (1 + 0.0075) b_k + 65 \qquad b(0) = 1000$

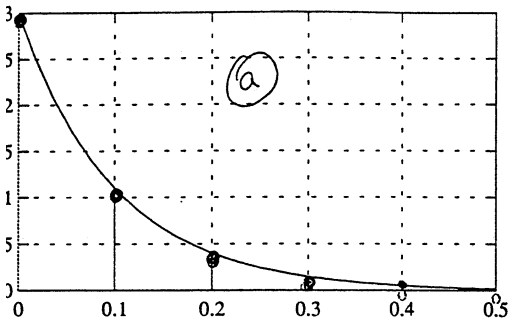
$b_k = (1.0075)^k (1000) + \sum_{j=0}^{k-1} (1.0075)^{k-1-j} \cdot 65$

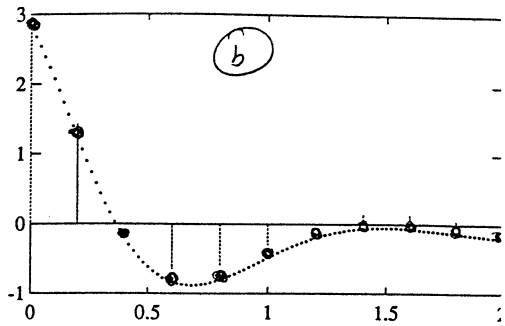
11-5 (a) $f(nT) = 3e^{-10nT} u(nT) = 3e^{-n} \qquad T = 0.1$

(b) $f(n) = 3e^{-0.4n} \cos\left(\frac{\pi}{4} n\right) \qquad T = 0.2$

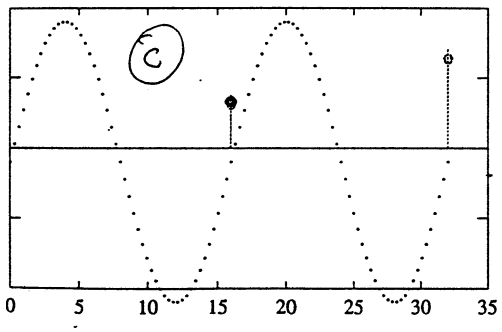
(c) $f(n) = 2 \sin\left(\frac{\pi n}{8}\right) \qquad T = 1$

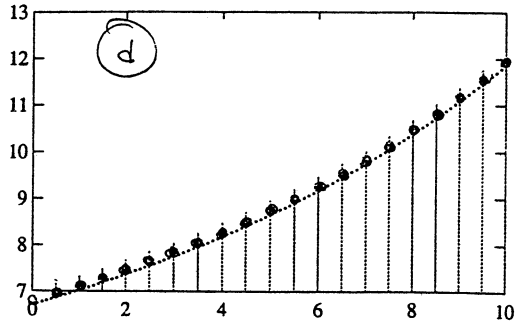
(d) $f(n) = (4 + 3e^{0.05n}) u(n) \qquad T = 0.5$





P11-5





11-6 (a) $F(z) = \frac{z}{z-1} - \frac{z}{z-1} z^{-4} = \frac{z(1-z^{-4})}{z-1}$

(b) $F(z) = 0.1 \frac{z}{(z-1)^2} + \mathcal{Z}\{k\,u(k-4)\,0.1\}$

Note: $k\,u(k-4) = (k-4)u(k-4) + 4\cdot u(k-4)$

$\therefore \mathcal{Z}\{k\,u(k-4)\} = \frac{z}{(z-1)^2} z^{-4} + 4\frac{z}{z-1}\cdot z^{-4}$

$F(z) = 0.1\left\{\frac{z}{(z-1)^2}\left[1+z^{-4}\right] + \frac{z}{z-1}\cdot z^{-4}\right\} = \frac{0.1z}{z-1}\left[\frac{1+z^{-3}}{z-1}\right]$

(c) $f(k) = \frac{\sqrt{2}}{2}\left[\cos(0.1k) - \sin(0.1k)\right]$

$F(z) = \frac{\sqrt{2}}{2}\left[\frac{z(z-\cos(0.1))}{z^2 - 2z\cos(0.1)+1} + \frac{z\sin(0.1)}{z^2-2z\cos(0.1)+1}\right]$

$= \frac{\sqrt{2}}{2}\left[\frac{z^2 - 0.99z + 0.09z}{z^2 - 1.99z + 1}\right] = \frac{\sqrt{2}}{2}\left[\frac{z(z-0.89)}{z^2-1.99z+1}\right]$

(d) $F(z) = \frac{z(-1)\sin(0.5)}{z^2 - 2(-1)\cos(0.5)z + (-1)^2} = \frac{-(0.479)z}{z^2 + 1.88z + 1}$

11-7 (a) $f(n) = 5n(0.5)u(n) = 2.5\,n\,u(n)$

$F(z) = 2.5\frac{z}{(z-1)^2}$

(b) $F(z) = 3\frac{z}{z-1} - 1\frac{z\sin(0.6)}{z^2 - 2z\cos(0.6)+1} = \frac{3z}{z-1} - \frac{2.26z}{z^2 - 1.65z + 1}$

(c) $f(n) = 15(nT)^2 e^{-10nT}u(nT) = 15(16)n^2 e^{-40n}u(n) = 240n^2(e^{-40})^n u(n)$

$F(z) = 240\,e^{-40}\frac{z(z+e^{-40})}{(z-e^{-40})^3}$

(d) $f(n) = 5e^{-3nT}\cos(\pi nT + 45^\circ) = 5e^{-0.3n}\cos\left(\frac{\pi}{10}n + 45^\circ\right)$

$= 5e^{-0.3n}\left[\cos\frac{\pi}{10}n\,\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}\sin\frac{\pi}{10}n\right]$

$F(z) = 5\frac{\sqrt{2}}{2}\left[\frac{z^2 - z(e^{-0.3}\cos\frac{\pi}{10} - e^{-0.3}\sin\pi/10)}{z^2 - 2e^{-0.3}\cos(\pi/10)z + e^{-0.6}}\right]$

11-7-d Cont: $F(z) = 3.53 \dfrac{z^2 - z(0.93)}{z^2 - 1.4z + 0.55}$

11-8 ⓐ $f(n) = \{2, -\frac{5}{2}, \frac{21}{4}, -\frac{31}{8}, \dots\}$

ⓑ $f(n) = \{0, 2, -2, -8, \dots\}$

ⓒ $f(n) = \{0, 0, \frac{10}{3}, \frac{10}{9}, \dots\}$

11-9 ⓐ $f(n) = 12\delta(n) - 12(-1/3)^n u(n)$

ⓑ $f(n) = -0.92(-1/4)^n + 1.66(-1)^n$

ⓒ $f(n) = \frac{1}{4}(n+1)(-1/2)^n$

ⓓ $f(n) = [14(-1/2)^n - 10(-1)^n]u(n) - 4\delta(n)$

ⓔ $f(n) = -4\delta(n-1) + [4(1/2)^n - 4(-1/2)^n]u(n)$

11-10 ⓐ $f(n) = 3e^{-10(0.1)n}u(n) = 3e^{-n}u(n)$

ⓑ $f(n) = 3e^{-0.4n}u(n)\cos\frac{5}{4}\pi\frac{2}{10}n = 3e^{-0.4n}\cos\left(\frac{\pi}{4}n\right)u(n)$

ⓒ $f(n) = 2u(n)\sin\left(\frac{\pi}{8}n\right)$

ⓓ $f(n) = [n^2 10^{-4} e^{-7(0.01)n} + 5e^{-0.06n}\sin(0.1n)]u(n)$

$= [10^{-4} n^2 (0.93)^n + 5(0.94)^n \sin(0.1n)]u(n)$

11-11 ⓐ $F(s) = \dfrac{8(2)}{s^2+4} \Rightarrow F(z) = 8\dfrac{z\sin(0.2)}{z^2 - 2\cos(0.2) + 1}$

$= \dfrac{1.59z}{z^2 - 1.96z + 1}$

ⓑ $F(s) = \frac{1}{s} - \frac{7}{s+4} + \frac{6}{s+5} \Rightarrow F(z) = \dfrac{z}{z-1} - 7\dfrac{z}{z-0.81} + 6\dfrac{z}{z-.78}$

ⓒ $F(z) = 4 \cdot \dfrac{(0.25)z e^{-0.25}}{(z - e^{-0.25})^2} \cdot z^{-2}$ note: $e^{-kST} \Rightarrow z^{-k}$ $e^{-0.5s} = e^{\frac{-.5}{.25}s(.25)}$

$\therefore k = 2 \Rightarrow z^{-2}$ delay.

11-11 (d) $F(z) = 3\frac{z}{z-1}z^{-5} - 4\frac{z}{z-e^{-0.4}} + \frac{z(z-\cos 0.6)}{z^2-2\cos(0.6)z+1}z^{-10}$

$= \frac{3z^{-4}}{z-1} - \frac{4z}{z-0.67} + \frac{z^{-9}(z-0.82)}{z^2-1.65z+1}$

(e) $F(s) = \frac{(1-e^{-2s}-e^{-3s})}{3} \cdot \frac{3}{(s+1)^2+9}$

$\Rightarrow F(z) = \frac{(1-z^{-4}-z^{-6})}{3} \cdot \frac{z e^{-0.5}\sin(1.5)}{z^2-2e^{-0.5}\cos(1.5)z+e^{-1}}$

$F(z) = (1-z^{-4}-z^{-6})\frac{0.2z}{z^2-0.09z+0.37}$.

11-12 (a) $F(z) = \frac{3z^2-2.39z}{z^2+0.83z+1} \Rightarrow F^*(s) = \frac{3e^{2s}-2.39e^{0.5s}}{e^{2s}+0.83e^{0.5s}+1}$

(b) $F(z) = \frac{z}{z+1} - \frac{z}{z-1} = \frac{z^2-z-z^2-z}{z^2-1} = \frac{-2z}{z^2-1}$

$F^*(s) = \frac{-2e^{0.1s}}{e^{-0.2s}-1}$

(c) $F(z) = 6\frac{z(z+0.5\cos 3)}{z^2+\cos 3\, z+0.25} = 6\frac{z(z-0.49)}{z^2-0.98z+0.25}$

$F^*(s) = \frac{6e^{2s}(e^s-0.49)}{e^{2s}-0.98e^s+0.25}$

(d) $f(k) = 10k e^{-0.1k}u(k-3)$

Let $\tilde{f}(k) = 10(k-3)e^{-0.1(k-3)}u(k-3)$

$\tilde{f}(k) = 10k e^{-0.1k}\cdot e^{0.3}u(k-3) - 30e^{-0.1(k-3)}u(k-3)$

$\therefore \tilde{f}(k) = e^{0.3}f(k) - 30e^{-0.1(k-3)}u(k-3)$

$\tilde{F}(z) = 10z^{-3}\frac{e^{-0.1}z}{(z-e^{-0.1})^2} = e^{0.3}F(z) - 30z^{-3}\frac{z}{z-e^{-0.1}}$

$\therefore F(z) = (22.22z^{-1} - 13.4z^{-2})/(z-0.9)^2$

11-12-d Cont: $F^*(s) = \dfrac{22.22\, e^{-0.5s} - 13.4\, e^{-s}}{(e^{+0.5s} - 0.9)^2}$

(e) $F(z) = \dfrac{z}{(z-1)^2} - \left[\dfrac{z}{(z-1)^2} z^{-5} + 5\dfrac{z}{z-1} z^{-5}\right]$

$= \dfrac{z - z^{-4} + 5z^{-4}(z-1)}{(z-1)^2} = \dfrac{z^5 + 5z + 6}{z^4 (z-1)^2} \Rightarrow F^*(s) = \dfrac{e^{1.5s} + 5e^{0.3s} + 6}{e^{1.2s}(e^{0.3s} - 1)^2}$

11-13 $f(n) = 20 - 15\cos(1000\, n\, T) \qquad T = 0.001$

$f^*(t) = \sum_{n=0}^{\infty} f(nT)\,\delta(t - nT) = \sum_{n=0}^{\infty} [20 - 15\cos(10n)]\,\delta\left(t - \frac{n}{100}\right)$

$f^*(t) = 5\,\delta(t) + 32.5\,\delta(t - 0.01) + 13.8\,\delta(t - 0.02) + 17.6\,\delta(t - 0.03) + 30\,\delta(t - 0.04)$

$\tilde{f}(t) = 5[u(t) - u(t - 0.01)] + 32.5[u(t - 0.01) - u(t - 0.02) + \text{etc.}$

$\tilde{f}(t) = \sum_{n=0}^{\infty} [20 - 15\cos(10n)][u(t - n) - u(t - n - 1)]$.

11-14 (a) $\dfrac{Y(z)}{R(z)} = \dfrac{1}{z^2 + 0.2z - 0.5}$ (b) $\dfrac{Y(z)}{R(z)} = \dfrac{4z^2 - 3}{z^3 - 1}$

(c) $\dfrac{Y(z)}{R(z)} = \dfrac{0.75z^3 + 0.25z^2 - 0.25z - 0.75}{z^3}$

11-15 $y(k+2) = 0.5\, y(k) + r(k+1) - 2r(k) \qquad y(0) = 0,\ y(1) = 3$

$r(k) = \left(\frac{1}{2}\right)^k \Rightarrow y(2) = 0.5(0) + \left(\frac{1}{2}\right)^1 - 2\left(\frac{1}{2}\right)^0 = -1.5$

$y(3) = 0.5(3) + \frac{1}{4} - 2\left(\frac{1}{2}\right) = 1.5 + .25 - 1 = 0.75$

$y(4) = 0.5(-1.5) + \frac{1}{8} - 2\left(\frac{1}{4}\right) = -1.125$

$y(5) = 0.5(0.75) + \frac{1}{16} - 2\left(\frac{1}{8}\right) = 0.1875$.

11-16 $R(z) = 2 \Rightarrow Y(z) = R(z)D(z) = \dfrac{2z}{z + \frac{1}{2}}$

(a) $\Rightarrow y(k) = 2(-1/2)^k u(k)$

11-16-b $Y(z) = \frac{1}{z-2} \cdot \frac{3z}{z-1} \Rightarrow \frac{Y(z)}{z} = \frac{3}{(z-1)(z-2)}$

$\Rightarrow \frac{Y(z)}{z} = \frac{-3/2}{z-1} + \frac{3}{z-2} \Rightarrow Y(z) = \frac{-3/2\, z}{z-1} + 3\frac{z}{z-2}$

$y(k) = \left[-\frac{3}{2} + 3(2)^k\right] u(k)$

(c) $Y(z) = \frac{z}{z+1} \cdot \frac{z}{z^2+1} = \frac{z^2}{(z+1)(z^2+1)} \Rightarrow Y = \frac{-\frac{1}{2}z}{z+1} + \frac{\frac{1}{2}z(z+1)}{z^2+1}$

$y(k) = \left[-\frac{1}{2}(-1)^k + \frac{1}{2}\cos\frac{k\pi}{2} + \frac{1}{2}\sin\frac{k\pi}{2}\right] u(k)$

(d) $Y(z) = \frac{1}{z^2 - 0.5z} \cdot \frac{z}{z-0.5} = \frac{z}{z(z-0.5)^2} = \frac{1}{(z-0.5)^2}$

$\frac{Y}{z} = \frac{1}{z(z-.5)^2} = \frac{4}{z} + \frac{2}{(z-0.5)^2} - \frac{4}{(z-0.5)}$

$y(k) = 4\delta(k) + \frac{2}{0.5} k \left(\frac{1}{2}\right)^k - 4(1/2)^k = 4\delta(k) + \left[4(1/2)^k (k-1)\right] u(k)$

(e) $Y(z) = \frac{z^2}{8z^2+6z+1} \cdot \frac{3}{z+\frac{1}{2}}$

$y(k) = -\frac{3}{2}\left(-\frac{1}{4}\right)^k + (1.5)(kn+1)\left(-\frac{1}{2}\right)^k \quad , k \geq 0$

11-17 (a) $\frac{Y(z)}{R(z)} = \frac{z}{z^2+9} \cdot \frac{z^2(0.25)}{z^2 - 0.25}$

(b) $\frac{Y(z)}{R(z)} = \frac{z}{z-1} + \frac{10/(z+.5)}{1 + \frac{10}{z+0.5}} = \frac{z^2 + 20.5z - 10}{(z-1)(z+10.5)}$

(c) $\frac{Y(z)}{R(z)} = \frac{z^2}{1.25z^2 + 1.125z - 0.25}$

11-18 (a) $P_1 = 3, \quad \Delta_1 = 1 - \frac{-2}{z} = \frac{z+2}{z}$

$P_2 = \frac{1}{z^2+4} \cdot \frac{1}{z} \cdot \frac{z}{z-1} = \frac{1}{(z^2+4)(z-1)}, \quad \Delta_2 = 1$

$\Delta = 1 + \frac{2}{z} - \frac{1}{z^3(z^2+4)} = \frac{z^3(z^2+4) + z^2(z^2+4)2 - 1}{z^3(z^2+4)}, \quad \frac{Y}{R} = \frac{\frac{3(z+2)}{z} + P_2}{\Delta}$

11-18-a

$$\frac{Y}{R} = \frac{\frac{3(z+2)}{z} + \frac{1}{(z^2+4)(z-1)}}{\frac{z^3(z^2+4) + z^2(z^2+4)2 - 1}{z^3(z^2+4)}} = \frac{\cdots}{z^2(z^2+4)(z-1)}$$

$$\cdots = \frac{(3z^4 + 3z^3 + 6z^2 + 13z - 24)\, z^2}{(z-1)(z^5 + 2z^4 + 4z^3 + 8z^2 - 1)}.$$

(b) $P_1 = \frac{2}{z}$, $\Delta_1 = 1 - \frac{1}{3z} = \frac{3z-1}{3z}$, $P_2 = \frac{-3}{z}$, $\Delta_2 = \frac{2z+1}{2z}$

$$\Delta = 1 - \left[\frac{-1}{2z} + \frac{1}{3z}\right] + \left[\frac{-1}{2z} \cdot \frac{1}{3z}\right] = 1 + \frac{1}{2z} - \frac{1}{3z} - \frac{1}{6z^2}$$

$$\Delta = \frac{6z^2 + 3z - 2z - 1}{6z^2}$$

$$\frac{Y}{R} = \frac{\frac{2(3z-1)}{3z^2} + \frac{3(2z+1)}{2z^2}}{\frac{\cdots}{6z^2}} = \frac{12z - 4 + 18z + 9}{6z^3 + 3z - 2z - 1} = \frac{30z + 5}{6z^3 + z - 1}$$

(c) $P_1 = \frac{-2}{z^2}$, $\Delta_1 = 1$, $P_2 = \frac{3}{z}$, $\Delta_2 = 1 + \frac{1}{4z} = \frac{4z+1}{4z}$

$$\Delta = 1 - \left[\frac{1}{2z^2} - \frac{1}{4z}\right] = \frac{4z^2 - 2 + z}{4z^2}$$

$$\frac{Y}{R} = \frac{\frac{-2}{z^2} + \frac{3(4z+1)}{4z^2}}{\frac{4z^2 + z - 2}{4z^2}} = \frac{-8 + 12z + 3}{4z^2 + z - 2} = \frac{12z - 5}{4z^2 + z - 2}.$$

11-19

(a) $D(\omega) = \frac{(\omega^2 + 14\omega + 1)(\omega + 1)}{1.98\omega^3 + 3.6\omega^2 + 2.06\omega + .36}$, Stable.

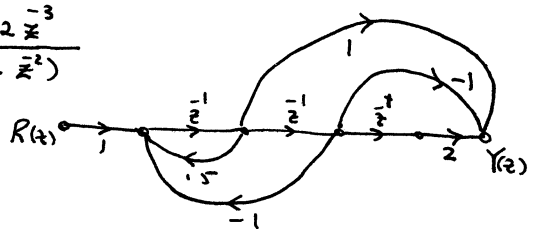
(b) $D(\omega) = \frac{\text{whatever}}{3.96\omega^3 + 4.56\omega^2 + 0.2\omega - 0.72}$ → change of sign, unstable.

(c) $D(\omega) = \frac{\text{whatever}}{5.9\omega^4 + 8.82\omega^3 + 2.58\omega^2 - .98\omega - .36}$ → unstable

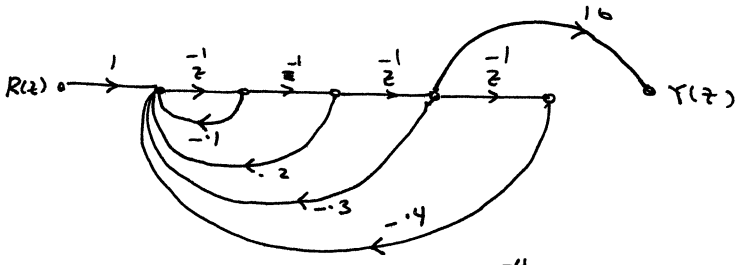
(d) $D(\omega) = \frac{\sim}{0.37\omega^4 + 1.19\omega^3 + 4.89\omega^2 + 6.77\omega + 2.78}$

Stable by R.H. Test.

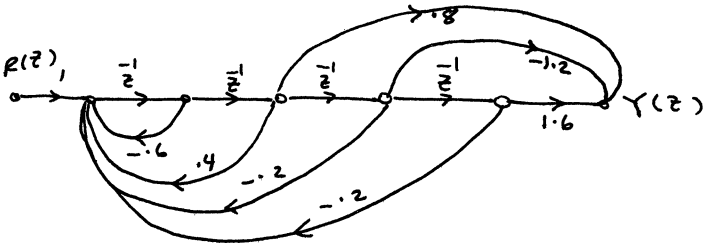
11-20 ⓐ $D(z) = \dfrac{z^{-1} - z^{-2} + 2z^{-3}}{1 - (0.5z^{-1} - z^{-2})}$



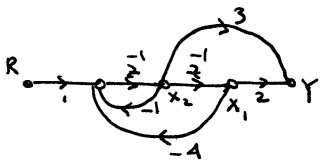
ⓑ $D(z) = \dfrac{10z^{-3}}{1 - (-0.1z^{-1} + 0.2z^{-2} - 0.3z^{-3} - 0.4z^{-4})}$



ⓒ $D(z) = \dfrac{0.8z^{-2} - 1.2z^{-3} + 1.6z^{-4}}{1 - (-0.6z^{-1} + 0.4z^{-2} - 0.2z^{-3} - 0.2z^{-4})}$



11-21 ⓐ $D(z) = \dfrac{3z^{-1} + 2z^{-2}}{1 - (-z^{-1} - 4z^{-2})}$

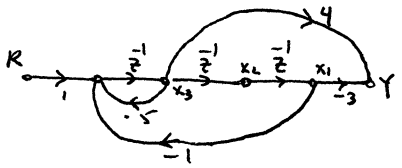


$$\begin{cases} x_1(k+1) = x_2(k) \\ x_2(k+1) = -4x_1(k) - x_2(k) + r(k) \\ y(k) = 2x_1(k) + 3x_2(k) \end{cases} \Rightarrow A = \begin{bmatrix} 0 & 1 \\ -4 & -1 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$C = [2 \quad 3], \; D = 0.$$

ⓑ Answer in the book.

11-21 ⓒ $D(z) = \dfrac{4z^{-1} - 3z^{-3}}{1 - (.5z^{-1} - z^{-3})}$



$$x(k+1) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & .5 \end{bmatrix} x_k + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} r(k)$$

$$y(k) = \begin{bmatrix} -3 & 0 & 4 \end{bmatrix} x_k + [0] \cdot r(k)$$

11-22 $x(1) = Ax(0) = \begin{bmatrix} -4 \\ 4 \end{bmatrix}$, $y(0) = 3x_1(0) = -12$, $y(1) = -12$

ⓐ $x(2) = Ax(1) = \begin{bmatrix} -12 \\ 8 \end{bmatrix}$, $y(2) = -36$

$x(3) = Ax(2) = \begin{bmatrix} -28 \\ 20 \end{bmatrix}$, $y(3) = -84$

ⓑ $x = \begin{bmatrix} 0 & 1 & -2 \\ 1 & -2 & 4 \\ -1 & 4 & -9 \end{bmatrix}$ ($\uparrow k=1$, $\uparrow k=2$, $\uparrow k=3$), $Y = \begin{bmatrix} 0 \\ -4 \\ 19 \\ -42 \end{bmatrix}$ ← $k=0$, ← $k=1$, ← $k=2$, ← $k=3$.

11-23 ⓐ $F_{step}(s) = \dfrac{1}{s} \cdot \dfrac{1}{s} = \dfrac{1}{s^2} \Rightarrow F_{step}(z) = 0.01 \dfrac{z}{(z-1)^2}$

$\Rightarrow D(z) = \dfrac{z-1}{z} \cdot \dfrac{z}{(z-1)^2}(.01) = \dfrac{0.01}{z-1}$.

ⓑ $F_{step}(s) = \dfrac{1}{s+1} \cdot \dfrac{1}{s} = \dfrac{1}{s} - \dfrac{1}{s+1} \Rightarrow F_{step}(z) = \dfrac{z}{z-1} - \dfrac{z}{z-e^{-0.5}}$

$\Rightarrow D(z) = 1 - \dfrac{z-1}{z-e^{-0.5}} = \dfrac{(1-e^{-0.5})}{z-e^{-0.5}} = \dfrac{0.39}{z-0.606}$

ⓒ $F_{step}(s) = \dfrac{s}{s^2+1} \cdot \dfrac{1}{s} = \dfrac{1}{s^2+1} \Rightarrow F_{step}(z) = \dfrac{.02z}{z^2 - 1.99z + 1}$

$D(z) = \dfrac{0.02(z-1)}{z^2 - 1.99z + 1}$.

ⓓ $F_{step}(s) = \dfrac{1}{(s+4)s}\left[1 - e^{-.5s}\right] = \dfrac{1}{4}\left[\dfrac{1}{s} - \dfrac{1}{s+4}\right]\left[1 - e^{-.5s}\right]$

$F_{step}(z) = \dfrac{1}{4}(1 - z^{-5})\left[\dfrac{z}{z-1} - \dfrac{z}{z-0.67}\right]$

$D(z) = \dfrac{1}{4}(1-z^{-5})\left[1 - \dfrac{z-1}{z-.67}\right] = \dfrac{1}{4}(1-z^{-5})\left(\dfrac{0.33}{z-0.67}\right) = \dfrac{0.082(z^5-1)}{z^6 - 0.67z^5}$

11-23 (c) $F_{step}(s) = (1-e^{-0.1s})\dfrac{10}{s^2(s+4)} = (1-e^{-0.1s})\left[\dfrac{2.5}{s^2}+\dfrac{-.625}{s}+\dfrac{.625}{s+4}\right]$

$$F_{stp}(z) = (1-z^{-10})\left[\frac{.025z}{(z-1)^2} - \frac{0.625z}{z-1} + \frac{0.625z}{z-0.96}\right]$$

$$D(z) = \frac{z^{10}-1}{z^{10}}\left[\frac{0.025}{z-1} - 0.625 + \frac{0.625(z-1)}{z-0.96}\right] = \frac{z^{10}-1}{z^{10}}\cdot\frac{4.93\times10^{-4}(z+.98)}{(z-1)(z-0.96)}.$$

11-24 $F_{imp}(s) = \dfrac{10}{s+4} \Rightarrow F_{imp}(z) = 10\cdot\dfrac{z}{z-0.45} = D(z)$.

11-25 (a) $D(s) = (1-e^{-sT})\dfrac{5}{s(s+2)} = (1-e^{-sT})\left[\dfrac{2.5}{s}+\dfrac{-2.5}{s+2}\right]$

$$D(z) = (1-z^{-1})\frac{5}{2}\left[\frac{z}{z-1} - \frac{z}{z-.55}\right] = \frac{5}{2}\left[1-\frac{z-1}{z-.55}\right] = \frac{5}{2}\left[\frac{0.45}{z-.55}\right].$$

(b) Forward path $\Rightarrow (1-e^{-sT})\left[\dfrac{1}{s(s+4)}\right] = (1-e^{-sT})\left[\dfrac{1}{s}+\dfrac{-1}{s+4}\right]$

$$\Rightarrow (1-z^{-1})\left(\frac{z}{z-1} - \frac{z}{z-.3}\right) = 1 - \frac{z-1}{z-.3} = \frac{0.7}{z-0.3}$$

$$\Rightarrow \frac{Y(z)}{F(z)} = \frac{\frac{0.7}{z-0.3}}{1-\frac{z}{z+0.5}\cdot\frac{0.7}{z-0.3}} = \frac{0.7(z+0.5)}{z^2+0.2z-0.15-0.7z}$$

$$= \underline{\frac{0.7(z+0.5)}{z^2-0.5z-0.15}}$$

(c) $D(s) = (1-e^{-sT})\dfrac{4s}{(s^2+4s+4)}\cdot\dfrac{1}{s} = (1-e^{-sT})\dfrac{4}{(s+2)^2}$

$$D(z) = \frac{z-1}{z}\,\frac{0.66z}{(z-0.55)^2} = \frac{0.66(z-1)}{(z-0.55)^2}.$$

11-26 Desired $D(z) = \dfrac{z}{z-1} \Rightarrow$ desired $D(s) = \dfrac{1}{s}$

$$\therefore \frac{1-e^{-sT}}{s}G_c(s) = \frac{1}{s} \Rightarrow G_c(s) = \frac{1}{1-e^{-sT}} = \frac{e^{0.2s}}{1+e^{0.2s}}$$

For the given $T=0.2$, there is no rational $G_c(s)$, only ↑ works.

11-27 (a) $\frac{Y(z)}{F(z)} = (1-z^{-1})\left[\frac{z}{z-1} - \frac{z}{z-e^{-0.6}}\right]\frac{z}{3} = \frac{z}{3}\left[1 - \frac{z-1}{z-e^{-0.6}}\right]$

$= \frac{z}{3}\left[\frac{0.45}{z-0.55}\right] = \frac{0.3}{z-0.55}$, $\frac{Y(z)}{E(z)} = \frac{1}{z-0.5} \cdot \frac{0.3}{z-0.55}$

$\Rightarrow \frac{Y(z)}{R(z)} = \frac{1.2}{z^2-1.05z+1.475}$

(b) $\frac{Y(z)}{F(z)} = (1-z^{-1})\frac{3(0.56)z}{z^2-1.65z+1} = \frac{1.68(z-1)}{z^2-1.65z+1}$

$\frac{Y(z)}{E(z)} = \frac{z-1}{z} \cdot \frac{Y(z)}{F(z)} = \frac{1.68(z-1)^2}{z(z^2-1.65z+1)} \Rightarrow \frac{Y(z)}{R(z)} = \frac{1.68(z-1)^2}{z^3+0.03z^2-2.36z+1.68}$

(c) $\frac{F(z)}{E(z)} = \frac{z(z+2)}{z^2+1.5z+2}$, $\frac{Y(z)}{F(z)} = \frac{(1-z^{-1})}{0.4}\left[\frac{z}{z-1} - \frac{z(z-0.92)}{z^2-1.84z+1}\right]$

$\frac{Y(z)}{F(z)} = \frac{5}{2}\left[\frac{0.08(z+1)}{z^2-1.84z+1}\right] = \frac{0.2(z+1)}{z^2-1.84z+1}$

$\frac{Y(z)}{R(z)} = \frac{0.2z(z+1)(z+2)}{z^4-0.14z^3+0.84z^2-1.78z+2}$.

11-28 (a) Forward $= \frac{z}{z-0.5} \cdot \frac{0.42(z+0.85)}{(z-1)(z-0.6)}$

Closed loop $\frac{Y(z)}{R(z)} = \frac{0.42z(z+0.85)}{(z-0.207)(z-0.73\pm j0.96)}$

The complex poles $= +0.73 \pm j0.96$ have a magnitude of $1.21 \Rightarrow$ unstable

(b) $G(z) = \frac{0.63}{z+0.36}$, forward $= G(z) \cdot \frac{z^{-1}}{1+z^{-1}} = \frac{0.63}{(z+1)(z+0.36)}$

Closed loop $\frac{Y(z)}{R(z)} = \frac{0.631}{(z+1.29)(z-0.33\pm j0.69)}$, → outside unit circle ⇒ unstable.

(c) Forward $= \frac{0.25(z-1)}{z^2-0.65z+0.36}$, $\frac{Y(z)}{R(z)} = \frac{0.25(z-1)}{(z-0.2\pm j0.27)}$ → mag < 1 Stable

11-29

(a) steady state output due to unit step input is given by

$$Y_{ss} = \lim_{z\to 1} (1-z^{-1})\, D(z) \cdot \frac{z}{z-1} = \lim_{z\to 1} D(z) = D(1)$$ assuming system is stable.

$\therefore$ (a) $D(1) = \frac{20}{12+7+1} = \frac{20}{20} = 1 \therefore e_{ss} = 1 - Y_{ss} = 1-1 = 0.$

poles are $= -0.33, -0.25 \Rightarrow$ stable. ✓

(b) poles are $= -1.2, -0.4 \rightarrow$ unstable. ∞ errors.

(c) poles are $= 0.25 \pm j0.66$, mag of poles $= 0.707 \Rightarrow$ stable.

$$D(1) = \frac{2}{2-1+1} = \frac{2}{2} = 1 \Rightarrow e_{ss} = 1 - Y_{ss} = 1-1 = 0.$$

11-30 For deadbeat, all closed loop poles must be at the origin.

$$\frac{Y(z)}{R(z)} = \frac{G}{1+GH} = \frac{\frac{z-.5}{z(z+2)}}{1+\frac{z-.5}{z(z+2)}\cdot\frac{nH}{dH}}$$

$$= \frac{(z-0.5)\,dH}{z(z+2)\,dH + (z-0.5)\,nH} = \frac{(z-0.5)}{z(z+2)+(z-0.5)H(z)}$$

Let $z^2 + 2z + (z-0.5)H(z) = z^2 \Rightarrow \boxed{H(z) = \frac{-2z}{z-0.5}}$

note that $H(z)$ cancels the plant pole & zero. This is O.K. because they are stable (i.e. inside the unit circle). So, $H(z)$ is acceptable.

Appendix - A

A-1 (a) $\begin{bmatrix} 10 & 6 \\ 6 & 4 \end{bmatrix}$ (b) $\begin{bmatrix} 7 \\ 1 \end{bmatrix}$ (c) 50

(d) 5

A-2 (a) $\begin{bmatrix} 29 & 28 \\ 28 & 41 \end{bmatrix}$ (b) $\begin{bmatrix} 14 & 8 \\ 48 & 21 \\ 28 & 16 \end{bmatrix}$ (c) $[35 \quad 31]$

(d) $[326 \quad 167]$

A-3 (a) $(-1)^{1}(2-6) = -4$ (b) $(-1)^{4}(24-24) = 0$

(c) 0 (e) NO, $\det A = 0$

(d) $\begin{bmatrix} 8 & 0 & -4 \\ -28 & 0 & 14 \\ 2 & 0 & -1 \end{bmatrix}$

A-4 (a) -2 (b) 3 (c) 7 (d) $\begin{bmatrix} 3 & 4 & -2 \\ -3 & 3 & 2 \\ 2 & -2 & 1 \end{bmatrix}$

(e) $\frac{1}{7} \times$ cofactor in (d)

A-5 $A^{-1} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & -3/2 & 0 \end{bmatrix}$

A-6 $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} +16 \\ -2 \end{bmatrix} \frac{1}{11}$

A-7 $x = \begin{bmatrix} 1 \\ 2 \\ -1/2 \end{bmatrix}$

A-8 NO. The equations are inconsistent and have no solution.

A-9 $\lambda^2 - 4\lambda + 1$

A-10 for A : $\lambda^2 - \lambda - 2$, for B : $\lambda^2 - 4\lambda + 4$

for AB : $\lambda^2 - 2\lambda - 8$

A-11 $A^2 - 4A + I = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

A-12 $\lambda = 4, -1, 2$

A-13 $\lambda_1 = 2 \longrightarrow v_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\lambda_2 = 1 \longrightarrow v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

A-14 $P = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$, $P^{-1}AP = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$

A-15 $\text{grad}(f) = \begin{bmatrix} 2\cos 2x_1 + 2x_2 e^{2x_1} \\ e^{2x_1} \end{bmatrix}$

A-16 $f = y_1x_1 + y_2x_2 + y_3x_3 = f(x) = y'x$

$\text{grad}(f) = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \Rightarrow \frac{df(x)}{dx} = y$

A-17 $f(x) = 2x_1^2 + 3x_1x_2 + 4x_2^2$

(a) $f(x) = x'Qx = [x_1 \quad x_2] \underbrace{\begin{bmatrix} 2 & 3/2 \\ 3/2 & 4 \end{bmatrix}}_{Q} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

(b) $\frac{df}{dx} = \begin{bmatrix} 4x_1 + 3x_2 \\ 3x_1 + 8x_2 \end{bmatrix} = 2\begin{bmatrix} 2 & 3/2 \\ 3/2 & 4 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 2Qx$

(c) $\lambda(Q) = 1.19, 4.80 > 0 \Rightarrow$ pos. definite.

A-18 $\lambda(Q) = -1, 3, 8 \Rightarrow$ indefinite (both pos. & neg. eigenvalues).

A-19 $A = \begin{bmatrix} 3 & 0 & 2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, $A_s = \frac{1}{2}(A + A') = \begin{bmatrix} 3 & 1/2 & 1 \\ 1/2 & 0 & 1/2 \\ 1 & 1/2 & 0 \end{bmatrix}$

$x'Ax = x'A_sx = 3x_1^2 + x_1x_2 + x_2x_3 + 2x_1x_3$

A-20 $A \xrightarrow{\text{row operation}} \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & 14 \\ 0 & 0 & 0 \end{bmatrix}$ = reduced row echelon form

$\therefore$ rank = 2

A-21 $A \xrightarrow{\text{raw operations}} \begin{bmatrix} \textcircled{1} & 0 & 0 \\ 0 & \textcircled{1} & 0 \\ 0 & 0 & \textcircled{1} \end{bmatrix} \Rightarrow \text{rank} = 3$

A-22 (a) $BA = [13 \quad 7 \quad 16]$ (b) $\begin{bmatrix} 13 \\ 7 \\ 16 \end{bmatrix}$ (c) $\det A = 5$

(d) $A^{-1} = \begin{bmatrix} 4 & 2 & -1 \\ -9 & 3 & 1 \\ 1 & -2 & 1 \end{bmatrix} \frac{1}{5}$

A-23 $A^{-1} = \left[\begin{array}{cc|cc} 3 & 1 & -1 & -4 \\ -6 & -1 & 2 & 8 \\ \hline -4 & 1 & 2 & 5 \\ 5 & 1 & -2 & -6 \end{array}\right]$

